

Residual Fibrations: Exact Reconstruction from Partial Information

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We reconstruct partial information by retaining unresolved structure as typed fibres, kernel pairs and factorization data rather than a scalar error. For an access map q , its conjoint q^* reconstructs q and its tabulation is the free fixed-base split-fibrational completion; a coupled read-out descends exactly when the associated support relation is functional, while the full profunctor retains multiplicity. Point actuality obeys the same descent law: naturally generated targets are classified by kernel inclusion, actualization resources act by kernel refinement, same-datum resources are inert, while canonical joint keys can have incomparable exact implementations.

For a finite family of Set read-outs, the joint image lies canonically between the least common refinement and the rectangular completion over the finest common factor. The resulting reflection separates common invariant, support defect, interaction and multiplicity. On the finite full-support probability face, the conditionally independent law over any common factor is the unique KL projection; the finest common factor uniquely minimizes residual dependence, with total correlation splitting exactly into deterministic common information, support dependence and residual weight dependence. Thus independence is a zero locus of a typed residual profile, not a consequence of Cartesian state-space notation.

Residual fibres compose by Grothendieck–Fubini; support–representability governs terminal composites, while law, occurrence, history and implementation remain distinct. Finite probability laws have no natural point extractor; equivariant actualization exists exactly at the required stabilizer fixed points. Operator-algebraic faces realize Petz sufficiency, BKM geometry, modular reconstruction and relative Cauchy evolution. Standard-subspace reconstruction yields causal homogeneous nets and, on the non-ray face-atomic zero-composition branch, forces the Lorentzian Jordan carrier $M_2(\mathbb{C})_{\text{sa}}$ without assuming dimension or field. The global least-collapse germ reconstructs determined structure while residual fibres retain freedom with no canonical selection.

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I. INTRODUCTION

Partial information determines a quotient and leaves residual fibres. SCO keeps both: kernel pairs encode indistinguishability, fibres multiplicity, factorization categories unresolved composition, and Grothendieck totals their variation over a base. For $q : \mathcal{E} \rightarrow \mathcal{B}$, Yoneda reconstructs q from q^* , tabulation gives the free fixed-base split-fibrational completion, and a read-out a couples by $\tau_{q,a} = a_* \odot q^*$. After corestriction, functional support is exactly descent through q , while the full profunctor retains multiplicity. The same kernel law governs actuality: (GAD) gives naturally generated targets; (DLG) separates point forcedness from non-Dirac law; (RAD/RNB; AKF/ECM) identifies resource refinements and the canonical target-relative key without selecting an implementation.

Composition is fibrewise. Grothendieck–Fubini and (SRTL) control composite residuals and their terminalization; equal local possibility data may still carry distinct transport. Probability adds no missing point: (ANB/SARF/FDSC; OINB) separates invariant law, occurrence and symmetry-breaking implementation, while (CHA/HAG; FIC/HRC/KRPA; ECP/RSP; LPT/ILPT/CLPG) transports law/provenance without manufacturing occurrence. Accordingly (LOPC/PLOC; RLOP/FLV; FLK/BFL; PHQ/PUP) factors occurrence-driven re-entry through the least complete-future quotient; (PLBD/PNB/PBCF) reconstructs future cylinders without a natural branch selector and retains implementation fibres. On the finite commonality face, (CBR/CBIP; LRF/DLCF) further separates canonical common quotient, support/weight dependence, minimal latent complexity and the noncanonically pointed latent-realization moduli.

Extra mathematical structure yields typed residual faces rather than a master scalar: additive/derived, affine–entropic, filtered/Rees, temporal and modular data are connected by explicit maps but remain distinct unless a theorem identifies them. Same-data/different-target witnesses therefore form part of the theorem boundary, separating existence from contractibility, occurrence from selection, endpoint from history and structural closure from ambient realization. The bounded chart $\pi_{\text{Rep}} : \text{Rep}_{<K} \rightarrow \text{SSpec}_{<K}^{\text{disc}}$ realizes this principle: tabulation/Grothendieck reconstruction of its conjoint $\mathfrak{U}_{R,K}$ gives compatible-realization fibres, while Isbell duality gives mutual-determination closure. The standard-Borel/operatorial ladder extends the same access data into a wider σ -finite normal-access perimeter without identifying that ambient atlas with the bounded presentation.

The mathematical-physics branch is a typed consumer. Expected von Neumann subalgebras support Petz sufficiency, BKM tangent geometry and modular reconstruction; algebraic QFT uses relative Cauchy evolution to retain $K/\sim_{\mathcal{A}}$ and the geometric fibre $D_M^{-1}(D_M(K))$; and the

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standard-subspace/Jordan branch yields causal nets and intrinsic Lorentz recognition, forcing $M_2(\mathbb{C})_{\text{sa}}$ and $\mathfrak{so}(1,3) \ltimes \mathbb{R}^{1,3}$ on the non-ray face-atomic zero-composition locus without assuming dimension or field. Yoneda/comma objects, Grothendieck constructions, Isbell duality, Petz recovery, BKM geometry and modular theory are standard inputs at their declared perimeters; the SCO contribution is their typed residual factorization. (RKD/KGC) supplies the least lawful determination quotient and (KGC_{sep}) its invariance under jointly separating presentations, (GAD; RAD/RNB; AKF/ECM/ARF) actuality/implementation, and (LOPC/PLOC; PHQ/PUP; PLBD/PNB/PBCF) predictive law/occurrence/implementation, while non-collapse witnesses retain history, interaction, gauge, clock and ambient realization.

Schematically,

$$q \mapsto q^* \mapsto \text{Tab}(q^*), \quad a \mapsto \top_{q,a} = a_* \odot q^*,$$

followed only by consumer-required attachments. At the endpoint (CGE/CGG/CGT; GDI/ICR/GFC/GGL) forms the cumulative lawful re-entry germ: jointly separating intrinsic Set presentations reconstruct the least-collapse quotient, while fresh inert extensions forbid a final representative. Terminality is invariance under lawful enlargement, not maximal inventory.

Section II constructs the universal residual carrier and bounded structural chart; Section III treats global compatibility; Sections IV–VI local access, sufficiency and residual geometry; Sections VII–VIII effective and temporal realization; Section IX operator-algebraic/modular reconstruction; Section X quantum access, causal geometry and closure; Section XI the relative-global core and ambient fibres. The appendix is a conservative semantic read-out and adds no hypotheses.

II. UNIVERSAL RESIDUALIZATION AND ACCESS REPRESENTATIONS

This section constructs the categorical carrier used throughout the paper. The first result starts from a single map, recovers it from its conjoint, and organizes the failure of exact determination into fibrational and factorization data. Later access, history, closure and realization constructions will be obtained from this carrier by typed restriction, refinement or attachment rather than by introducing unrelated defect notions.

A. Residual Yoneda doctrine and totality

Theorem II.1 (Residual Yoneda object and free fibrational completion). *For $q, r : \mathcal{E} \rightarrow \mathcal{B}$, Yoneda gives (YFF) $\text{Nat}(q, r) \cong \mathbf{Prof}(\mathcal{B}, \mathcal{E})(q^*, r^*)$, so q^* reconstructs q up to unique natural isomorphism. For $H : \mathcal{A}^{\text{op}} \times \mathcal{B} \rightarrow \mathbf{Set}$, $H^1 X(b) = \text{Nat}(X, H(-, b))$ and $H^1 Y(a) = \text{Nat}(Y, H(a, -))$ satisfy (ISB) $\text{Nat}(Y, H^1 X) \cong \text{Nat}(X, H^1 Y)$, with fixed-pair nucleus. Put*

$$[\text{GRM}] \quad \mathbb{T} \downarrow q := \pi_q : (\mathcal{B} \downarrow q) \rightarrow \mathcal{B}, \quad \eta_q(e) = (qe, e, 1_{qe}), \quad \mu_q(b \xrightarrow{v} b' \xrightarrow{u} qe, e) = (b \xrightarrow{uv} qe, e). \quad (1)$$

Squares (r, s) act by $(b, e, \beta) \mapsto (sb, re, s\beta)$. This is the global comma 2-monad; at fixed $\mathcal{B}$ its fibrational restriction is the 2-cell dual of Street's lax-idempotent/Kock–Zöberlein cofibration monad^{1–3}. Thus (RFD) $\mathbb{T}_{\mathcal{B}} q = \text{Tab}(q^) \simeq (\mathcal{B} \downarrow q) \rightarrow \mathcal{B}$; fixed-base pseudoalgebras are cloven fibrations and strict algebras split. (KZP) q is cloven iff there is a normal action $a_q : \mathbb{T}_{\mathcal{B}} q \rightarrow q$ over $\mathcal{B}$ with $\eta_q \dashv a_q$ and $a_q \eta_q = I_q$. $\text{Cart}_{/\mathcal{B}}$ denotes cartesian functors over $\mathcal{B}$; for split $p : \mathcal{F} \rightarrow \mathcal{B}$, (FRF) $\eta_q^* : \text{Cart}_{/\mathcal{B}}(\pi_q, p) \simeq \text{Fun}_{/\mathcal{B}}(q, p)$. For $A : \mathcal{E} \rightarrow \mathcal{Y}$, $\mathcal{Y}$ locally small, whenever the indicated comma limit exists, (RKU) $(\text{Ran}_q A)(b) \cong \{q^*(b, -), A\} \cong \lim_{\leftarrow (b \downarrow q)} A \text{pr}_{\mathcal{E}}^4$ (Ch. X, §§3,5), cf.⁵ (§3.3, Prop. 5.12); hence q^* is the pointwise right-Kan weight. Let $\mathbf{SFib}_{/\mathcal{B}}$ be split fibrations with strict split-cleavage maps and $U_{\mathcal{B}} : \mathbf{SFib}_{/\mathcal{B}} \rightarrow \mathbf{Cat}/\mathcal{B}$ forgetful. Then $\mathbb{T}_{\mathcal{B}}^{\text{fr}} : q \mapsto (\pi_q, \text{can})$ satisfies $\mathbb{T}_{\mathcal{B}}^{\text{fr}} \dashv U_{\mathcal{B}}$ and $U_{\mathcal{B}} \mathbb{T}_{\mathcal{B}}^{\text{fr}} = \mathbb{T}_{\mathcal{B}}$.*

Proof. Yoneda reconstructs q from q^* ; end–Fubini gives Isbell, and comma-monad duality the pseudo/strict and free-adjunction clauses. Kock–Zöberlein is $\text{Hom}(\eta_q e_0, (b, e, \beta)) \cong \text{Hom}(e_0, a_q(b, e, \beta))$, with $a_q \eta_q = I_q$ exactly normality. For split p , $\bar{F}(b, e, \beta) = \beta^* F(e)$ and cartesian uniqueness makes restriction/free extension inverse. Finally $\int q^*(b, -) \cong (b \downarrow q)$ represents the displayed right Kan extension. $\square$

a. Free-completion universality. For fixed $\mathcal{B}$, the adjunction $\mathbb{T}_{\mathcal{B}}^{\text{fr}} \dashv U_{\mathcal{B}}$ makes $(\mathbb{T}_{\mathcal{B}}^{\text{fr}} q, \eta_q)$ the free split-fibrational completion of q : for every split fibration p over $\mathcal{B}$, restriction along η_q is the equivalence

$$[\text{FCU}] \quad \text{Cart}_{/\mathcal{B}}(\mathbb{T}_{\mathcal{B}} q, p) \simeq \text{Fun}_{/\mathcal{B}}(q, p).$$

Thus every same-base split coherent access semantics receiving q factors through $\text{Tab}(q^*)$ up to (FCU). With (YFF), the supplied arrow reconstructs the residual carrier, adding neither a model-specific parameter nor a realization point. *Yoneda self-key.* For small $\mathcal{F}$, full faithfulness of $\gamma_{\mathcal{F}}$ factors every $A : \mathcal{F} \rightarrow \mathcal{Y}$ through its representable image; on discrete F , (YSK) $\delta_F(x)(z) = \mathbf{1}_{\{z=x\}}$ has $R_{\delta_F} = \Delta_F$, while (YFF) reconstructs q from q^* . These keys are complete/target-independent, not minimal/physical. *Generated actuality descent.* Let C be small, let $D, T : C \rightarrow \mathbf{Set}$, fix $c \in C$ and $d \in D(c)$, and write $h_c = C(c, -)$. The Yoneda element map $\hat{d} : h_c \rightarrow D$ is $\hat{d}_x(f) = D(f)d$; factor it objectwise through its image as $h_c \xrightarrow{q_d} \langle d \rangle \hookrightarrow D$. For $z \in T(c)$ define $\bar{z} : h_c \rightarrow T$ analogously and put $R_{\bar{z}}(x) = \{(f, g) : D(f)d = D(g)d \Rightarrow T(f)z = T(g)z\}$ on $C(c, x)$, with relation inclusion below understood objectwise in x . Evaluation at the generating element gives the exact natural-actuality fibre

$$[\text{GAD}] \quad \boxed{\begin{aligned} \text{GenAct}_{D \Rightarrow T}(c, d) &:= \text{Nat}(\langle d \rangle, T) \\ &\cong \{z \in T(c) : R_{\bar{z}} \subseteq R_{\bar{z}}\} \\ &= \{z : (\forall x)(\forall f, g : c \rightarrow x) D(f)d = D(g)d \Rightarrow T(f)z = T(g)z\}. \end{aligned}} \quad (2)$$

The bijection sends $\bar{z}$ to $\bar{z}_c(d)$. Conversely the displayed kernel condition makes $\bar{z}_x(D(f)d) := T(f)z$ well-defined; functoriality makes it natural, and the objectwise epimorphism q_d gives uniqueness. Hence generated actuality is literally a kernel-descent problem, the Yoneda-source form of (RKD), not a second actuality primitive. It composes: for $U : C \rightarrow \mathbf{Set}$,

$$z \in \text{GenAct}_{D \Rightarrow T}(c, d), \quad w \in \text{GenAct}_{T \Rightarrow U}(c, z) \implies w \in \text{GenAct}_{D \Rightarrow U}(c, d),$$

because $R_{\bar{z}} \subseteq R_{\bar{z}} \subseteq R_{\bar{w}}$. Empty, singleton and multiple GenAct therefore give impossible, forced and residually contingent point actuality without choosing a point.

If C is a groupoid, every parallel pair $f, g : c \rightarrow x$ differs by the source automorphism $g^{-1}f$, so the full kernel condition reduces exactly to isotropy:

$$\text{GenAct}_{D \Rightarrow T}(c, d) \cong T(c)^{\text{Stab}_{\text{Aut}_C(c)}(d)}.$$

In particular, for $C = BG$, $\langle d \rangle \cong G/G_d$ and

$$\text{GenAct}_{D \Rightarrow T}(*, d) \cong \text{Hom}_G(G/G_d, T(*)) \cong T(*)^{G_d}.$$

Applied to $d = (s, \xi) \in S \times R$ this is the one-orbit factor $Z^{G_s \cap G_\xi}$ of (SARF), while decomposition into generated orbits gives its full product. On the probability face this computes every law-readable label, not only the uniform obstruction. Over $C = \mathbf{FinBij}_{+, \text{sk}}$, for $p \in \mathbb{D}I$ put $I_r(p) = \{j : p_j = r\}$. Then

$$[\text{LWP}] \quad \boxed{G_p = \prod_{r \in p(I)} \text{Sym}(I_r(p)), \quad \text{GenAct}_{\mathbb{D} \Rightarrow \text{Id}}(I, p) = I^{G_p} = \{i \in I : |I_{p_i}(p)| = 1\}.} \quad (3)$$

Indeed a permutation fixes p exactly when it preserves every weight level, and a label is fixed by that product exactly when its own level is a singleton. Thus u_I has no generated label for $|I| > 1$, recovering the local obstruction behind (ANB), while an asymmetric law may distinguish labels structurally. This readability is support-blind: on $\mathbf{2}$, δ_0 has two singleton weight levels and hence two generated labels although only 0 is its Dirac occurrence. The deterministic-law gate below is therefore strictly stronger. The well-founded equivariant closure is (FDSC).

The groupoidal reduction is sharp: isotropy alone is insufficient on a noninvertible base. Let C have objects c, x and two distinct nonidentity arrows $f, g : c \rightarrow x$, with no nontrivial automorphisms. Take $D(c) = \{d\}$, $D(x) = \{u\}$ and $D(f)d = D(g)d = u$, while $T(c) = T(x) = \mathbf{2}$, $T(f) = I_2$ and $T(g)$ is the transposition. Every automorphism stabilizer is trivial, but the generated kernel contains (f, g) and neither point of $T(c)$ equalizes it; hence $\text{GenAct}_{D \Rightarrow T}(c, d) = \emptyset$. Thus the native obstruction is the full Yoneda kernel, with stabilizer incidence only its groupoidal shadow.

Law and resource gates for point actuality. Let $p : D \Rightarrow \mathbb{D}T$ be a natural finitely supported probability-law map. Put

$$\text{DetLawAct}_p(c, d) := \{z \in T(c) : p_c(d) = \delta_z\}.$$

Naturality and injectivity of the Dirac embedding give the exact gate

$$[\text{DLG}] \quad \boxed{\text{DetLawAct}_p(c, d) \hookrightarrow \text{GenAct}_{D \Rightarrow T}(c, d), \quad |\text{DetLawAct}_p(c, d)| \leq 1,} \quad (4)$$

with the left-hand fibre terminal exactly when $p_c(d)$ is Dirac. Indeed, if $p_c(d) = \delta_z$ and $D(f)d = D(g)d$, naturality gives $\delta_{T(f)z} = p_x(D(f)d) = p_x(D(g)d) = \delta_{T(g)z}$, hence the (GAD) kernel condition. The converse inclusion is false already inside (LWP): take $D = \mathbb{D}$, $T = \text{Id}$, the identity law map and $\varpi = (1/2, 1/4, 1/4) \in \mathbb{D}\mathbf{3}$. Then $\text{GenAct}_{\mathbb{D} \Rightarrow \text{Id}}(\mathbf{3}, \varpi) = \{0\}$ because only ϖ_0 has singleton multiplicity, while $\text{DetLawAct}_{\text{id}_{\mathbb{D}}}(\mathbf{3}, \varpi) = \emptyset$ because ϖ is non-Dirac. Thus a law may force a structurally readable label without making that label an occurrence.

For a second datum functor $R : C \rightarrow \mathbf{Set}$ and $r \in R(c)$, the product Yoneda kernel is the intersection $R_{\widehat{(d,r)}} = R_{\widehat{d}} \cap R_{\widehat{r}}$. Hence (GAD) gives

$$[\text{RAD}] \quad \boxed{\text{GenAct}_{D \times R \Rightarrow T}(c, (d, r)) \cong \{z \in T(c) : R_{\widehat{d}} \cap R_{\widehat{r}} \subseteq R_{\widehat{z}}\}.} \quad (5)$$

A constant singleton resource has universal kernel and changes nothing; a more distinguishing resource can only enlarge the generated-actuality fibre. On a groupoid, (RAD) is $T(c)^{G_d \cap G_r}$ and therefore recovers the local fixed-point law behind (SARF/FDSC). Thus the native actuality resource is kernel refinement, with symmetry breaking only its groupoidal shadow. Neither (DLG) nor (RAD) supplies a realized resource point.

Relative-exogeneity no-bootstrap law. With the preceding notation, the resource is naturally generated from the datum exactly when it is inert for every target:

$$[\text{RNB}] \quad \boxed{r \in \text{GenAct}_{D \Rightarrow R}(c, d) \iff (\forall T : C \rightarrow \mathbf{Set}) \quad \text{GenAct}_{D \times R \Rightarrow T}(c, (d, r)) = \text{GenAct}_{D \Rightarrow T}(c, d) \iff \text{GenAct}_{D \times R \Rightarrow D \times R}(c, (d, r)) = \text{GenAct}_{D \Rightarrow D \times R}(c, d).} \quad (6)$$

Indeed (GAD) identifies the first condition with $R_{\widehat{d}} \subseteq R_{\widehat{r}}$; in which case $R_{\widehat{(d,r)}} = R_{\widehat{d}} \cap R_{\widehat{r}} = R_{\widehat{d}}$ and (RAD) gives equality for every T . Conversely (d, r) always belongs to the left-hand generated actuality fibre for the target $D \times R$; equality for that single target forces $R_{\widehat{d}} \subseteq R_{\widehat{(d,r)}}$, hence $R_{\widehat{d}} \subseteq R_{\widehat{r}}$. Therefore strict enlargement of even one target fibre certifies that r is not naturally generated from d ; if r is not generated, the canonical target $D \times R$ detects the strict enlargement.

The obstruction is closed under iterated self-generation. For a small ordinal λ , choose $R_\alpha : C \rightarrow \mathbf{Set}$ and $r_\alpha \in R_\alpha(c)$, let $D_\alpha := D \times \prod_{\beta < \alpha} R_\beta$ and $d_\alpha = (d, (r_\beta)_{\beta < \alpha})$, and assume $r_\alpha \in \text{GenAct}_{D_\alpha \Rightarrow R_\alpha}(c, d_\alpha)$ for every $\alpha < \lambda$. Transfinite induction gives

$$[\text{RNB}_\lambda] \quad \boxed{R_{\widehat{d_\alpha}} = R_{\widehat{d}} \quad (\alpha \leq \lambda), \quad \text{GenAct}_{D_\lambda \Rightarrow T}(c, d_\lambda) = \text{GenAct}_{D \Rightarrow T}(c, d) \quad (\forall T).} \quad (7)$$

At a successor stage the new kernel contains the accumulated one by (GAD), so its intersection changes nothing; at a limit stage the product kernel is the intersection of the coordinate kernels, all of which contain $R_{\widehat{d}}$. Thus no small transfinite chain of resources generated from its accumulated datum can bootstrap a stricter actuality kernel. On $C = BG$ this says that an internally generated resource cannot reduce the stabilizer of the accumulated datum; (FDSC) is the finite equivariant derivation shadow. This is a relative no-bootstrap theorem, not a denial that an independently supplied boundary, interaction, history or representation-sector datum can refine the kernel; such a datum is precisely outside the first equivalence above. Structural settlement (SRSL) remains distinct because its representable terminal total need not supply such a point-bearing refinement.

Actuality key and exogenous complement fibre. Fix also $z \in T(c)$ and corestrict $\widehat{d} : h_c \rightarrow D$ as $q_d : h_c \twoheadrightarrow \langle d \rangle$. The target-dependent joint key is not a new primitive:

$$[\text{AKF}] \quad \boxed{\text{ActKey}(d; z) := \text{im}(\widehat{d}, \widehat{z}) \cong h_c / (R_{\widehat{d}} \cap R_{\widehat{z}}).} \quad (8)$$

Write $j_{d,z} : h_c \twoheadrightarrow \text{ActKey}(d; z)$ for this corestriction. Images and kernel pairs in $\mathbf{Set}^C$ are objectwise, so this is the functorial assembly of the (RKD) support-key construction, not a new key primitive. For a resource element $r \in R(c)$ let $q_r : h_c \twoheadrightarrow \langle r \rangle$ and let $j_{d,r} : h_c \twoheadrightarrow \text{im}(\widehat{d}, \widehat{r})$

corestrict the joint datum. Then (RAD) and quotient factorization give the exact implementation criterion

$$\begin{aligned} \text{[AKI]} \quad z \in \text{GenAct}_{D \times R \Rightarrow T}(c, (d, r)) &\iff R_{j_{d,r}} \subseteq R_{j_{d,z}} \\ &\iff \exists! \phi_{r,z} : \text{im}(\widehat{d}, \widehat{r}) \rightarrow \text{ActKey}(d; z), \quad j_{d,z} = \phi_{r,z} j_{d,r}. \end{aligned} \quad (9)$$

The first equivalence is (RAD) and the second is the same-source (RKD/KGC_{cons}) quotient universal property, objectwise in $\mathbf{Set}^C$. Thus $\text{ActKey}(d; z)$ is the unique coarsest joint refinement making z naturally admissible. Equality rather than strict refinement defines the exact resource-complement fibre

$$\text{[ECM]} \quad \text{ExComp}(d; z) := \{[q_r] : R_{\widehat{d}} \cap R_{q_r} = R_{\widehat{d}} \cap R_{\widehat{z}}\} = \{[q_r] : j_{d,r} \approx_{h_c} j_{d,z}\}. \quad (10)$$

Its members implement precisely the canonical joint key without joint over-resolution; a successful resource with strict kernel inclusion refines beyond the target key. Here $[q_r]$ denotes the same-source determination class, ordered by the factorization preorder (SDO/SDK). The key (AKF) is a target-relative classifier of what must be jointly determined, not an independently produced physical resource: $\widehat{z}$ occurs in its definition and may not be laundered into the resource input. Hence the required effective distinction is canonical even when its external implementation is not.

That nonuniqueness is irreducible. Let C have objects c, x , identities, and three parallel arrows $u, v, w : c \rightrightarrows x$. Take $D(c) = \{d\}$ with $D(u)d = D(v)d \neq D(w)d$, take $T(c) = \{z\}$ with $T(u)z, T(v)z, T(w)z$ all distinct, and choose resource elements r_1, r_2 whose kernels on $\{u, v, w\}$ have classes $\{u, w\}|\{v\}$ and $\{v, w\}|\{u\}$, respectively. Then $R_{\widehat{d}} \cap R_{\widehat{r}_1} = R_{\widehat{d}} \cap R_{\widehat{r}_2} = \Delta$ for $i = 1, 2$, so both lie in $\text{ExComp}(d; z)$. They are incomparable, while every common coarsening identifies all three arrows and therefore reintroduces $u \sim v$, so it is not successful. Consequently the successful-resource quotient preorder need not have a least external resource even though the joint key (AKF) is unique.

For a finite declared candidate family $(r_j)_{j \in J}$ this residual implementation freedom is exactly the finite frontier

$$\text{[ARF]} \quad \text{ActFront}_J(z \mid d) := \text{Min}_{\subseteq} \left\{ F \subseteq J : R_{\widehat{d}} \cap \bigcap_{j \in F} R_{\widehat{r}_j} \subseteq R_{\widehat{z}} \right\}. \quad (11)$$

It is the Yoneda-source face of (RFF): success is upward under adding resources, but its minimal generators form an antichain rather than a selected cause. In the three-arrow witness $\text{ActFront}_{\{1,2\}}(z \mid d) = \{\{1\}, \{2\}\}$. By (RNB), no member that strictly enlarges the original actuality fibre can itself be generated from d . The theorem therefore separates three levels without scalarization: canonical required joint information, noncanonical exogenous implementation, and an independently supplied realized resource point.

For any bicategorical $L \dashv R$, (URE) $M_{L,R} := R \circ L$, $N_{L,R} := L \circ R$, $\text{Lan}_L a \simeq a \circ R$ and $[a \circ R \Rightarrow H] \cong [a \Rightarrow H \circ L]$. For vertical $q : X \rightarrow Y$ in an equipment write $q_* \dashv q^*$, M_q, N_q ; for $X \xrightarrow{q} Y \xrightarrow{g} Z$, (URE_c) $(gq)^* \simeq q^* \circ g^*$, $M_{gq} \simeq q^* \circ M_g \circ q_*$, $N_{gq} \simeq g_* \circ N_q \circ g^*$, with cells $M_q \Rightarrow M_{gq}$ and $N_{gq} \Rightarrow N_g$. Put $\text{BdRes}(u) := u^*$ for a declared boundary map. Then $X' \xrightarrow{r} X \xrightarrow{q} Y \xrightarrow{s} Y'$ gives $(sqr)^* \simeq \text{BdRes}(r) \circ q^* \circ \text{BdRes}(s)$ and $\text{BdRes}(vu) \simeq \text{BdRes}(u) \circ \text{BdRes}(v)$. Put (TOM) $\tau_{q,a} := a_* \circ q^*$; for $a : X \rightarrow W$, $b : W \rightarrow W'$, $\tau_{gq,ba} \simeq b_* \circ \tau_{q,a} \circ g^*$. Nonempty support sends $(M_q, N_q, \tau_{q,a})$ to $(R_q, \Delta_{\text{im } q}, \text{im}(q, a))$ and preserves $\circ$, companions and conjoints; $\tau_{q,a}^{\text{Prof}}(y, z) \cong \{x : qx = y, ax = z\}$, so $R_q \subseteq R_{gq}$ and $\Delta_{gq}(x) \subseteq \Delta_g(y)$. Thus $\text{supp}_0 \tau_{q,a} := \text{im}(q, a)$; on discrete $\mathbf{Set}$ its corestriction is functional iff $R_q \subseteq R_a$ (equivalently $M_q \subseteq M_a$ there), iff $a = \bar{a}q$, while the Prof-valued object retains fibre multiplicity. Right-to-left horizontal-unit restriction along $f : X' \rightarrow Z'$, $g : Y' \rightarrow Z'$ is $g^* \circ f_*$; hence (RPE) $\text{Tab}(g^* \circ f_*) \simeq (f \downarrow g)$ and $\text{Tab}(M_q) \simeq (q \downarrow q)^5$ (Ex. 5.7, Prop. 5.9). Iteration is Cech/homotopy Cech on $\mathbf{Set}/\mathbf{Gpd}$ with codescent $q(X)$ and $\mathcal{B}|_{V_q}, V_q = \{[b] : N_q(b, b) \neq \emptyset\} = \text{im } \pi_0 q$. On $\mathbf{Set}$, $q = j_q \zeta_q q_{R_q}$, $\check{C}_n(q) = X^{\times_q(X)}(n+1)$, $\check{C}_1(q) = R_q$; its tabulations are (DCC/RKD) below; f^*g is first comma projection. Defined collapses⁶ give the following faces without erasing fixed-access residuals. For $a \leq b$ in a poset P , (RCS) $\text{CmpRes}_P(a, b) := (a \downarrow P) \times_P (P \downarrow b) \cong [a, b]$ and is terminal iff $a = b$. Thus $\mathfrak{A}_{\square}(\alpha; x) = \text{CmpRes}_P(Fx, Gx)$ is the two-sided (RFD) face with (GCS) split zero locus; the 1-cell face $\mathbb{T}c = \text{Tab}(c^*)$ gives (PBC/WCR/LFR) with (HRS/RST) exactness. These dimensions remain typed. *Residual resolution and algebraic contraction*. Put $C_q^{-1} = q$, $C_q^n = \mathbb{T}_{\mathcal{B}}^{n+1} q$. The maps $d_i^n = \mathbb{T}^i \eta_{\mathbb{T}^{n+1-i} q}$ ($0 \leq i \leq n+1$) and $s_i^n = \mathbb{T}^i \mu_{\mathbb{T}^{n-1-i} q}$ ($0 \leq i < n$) satisfy the cosimplicial identities and coaugment $q \rightarrow C_q^*$; the split return is only one such identity. With $\mu_q^{[n]} : \mathbb{T}^n q \rightarrow \mathbb{T} q$, (RIT) $\text{Ob}(\mathbb{T}_{\mathcal{B}}^n q) = \{b_0 \rightarrow \dots \rightarrow b_{n-1} \rightarrow qe\}$, $(\mu_q^{[n]})^{-1}\{(b \xrightarrow{\beta} qe, e)\} \cong \text{Fact}_{\mathcal{B}}^{[n]}(\beta)$. If $\beta : b \rightarrow c$ is noninvertible and $n \geq 2$, $F_- = (1_b, \dots, \beta)$ and $F_+ = (\beta, \dots, 1_c)$ are nonisomorphic (an isomorphism would have first intermediate component β), so (NFP) $\text{Fact}_{\mathcal{B}}^{[n]}(\beta) \neq \mathbf{1}$. The converse fails: in $\mathbf{Set}$, $1 \hookrightarrow [n] \twoheadrightarrow 1$ gives distinct 2-factorizations of 1_1 . For a supplied 2-cell $\eta : F \Rightarrow G$ in a bicategory $\mathcal{K}$ with $\mathcal{K}(X, Y)$ in this size perimeter, set $\text{CellFact}_{\mathcal{K}}^{[n]}(\eta) := \text{Fact}_{\mathcal{K}(X, Y)}^{[n]}(\eta)$. Thus (RIT/YRF) makes CFR the supplied-cell composition fibre; on posetal homs its $n = 2$ face is (RCS/ICP), with the same (NFP) one-way nonterminality boundary. A strict $\mathbb{T}_{\mathcal{B}}$ -algebra $a : \mathbb{T} q \rightarrow q$ supplies extra codegeneracies $\bar{s}_n^a := \mathbb{T}^n a : C_q^n \rightarrow C_q^{n-1}$; $a\eta = I$, $a\mu = a\mathbb{T}a$ are exactly their splitting identities, and degrees 0, 1 recover the algebra laws. Hence by (GRM), (RIT_{alg}) q is a split fibration iff this strict monadic contraction exists. The free algebra μ_q always contracts the tail, so (RIT_{ph}) asks precisely whether contraction extends across $\eta_q : q \rightarrow \mathbb{T} q$: closure is coherence, not exhaustion. For poset $\mathcal{L}$, $q_u(*) = u$, $\ell \leq u$, $n \geq 2$, (RFH) $(\mu_q^{[n]})^{-1}\{(\ell \leq u)\} \cong \{\ell = b_0 \leq \dots \leq b_{n-1} \leq u\}$. Thus $n = 2$ is (RCS); on $[1]$, $|\text{Ob } \mathbb{T}^n q_1| = n + 1$, so no finite stabilization. For discrete $\mathcal{B}$, $\mathbb{T}_{\mathcal{B}} \simeq I_{\text{Cat}/\mathcal{B}}$ because $b \rightarrow qe$ forces $b = qe$ and every $\mathcal{E}$ -arrow is vertical; width is not depth (RCS/CKPF). For $\chi : C \rightarrow [1]$, (POL/LRT) $C \cong_{[1]} \text{Coll}(H_{\chi})$ and $H_{\chi}(0 < 1) \simeq q^*$, $q : C_1 \rightarrow C_0$; $[2]$ has represented $0|2, 0|1|2$, so C alone does not recover χ .

On $\text{Rel}_{\mathcal{E}}$, (REL) $S \circ R = \text{im}(R \times_Y S \rightarrow X \times Z)$, $R^{\circ} = \tau^* R$, $1_X = \Delta_X$, $R_f = \Gamma_f^{\circ} \circ \Gamma_f$, $(f \times g)^* S = \Gamma_g^{\circ} \circ S \circ \Gamma_f$. Cells over (f, g) are $R \subseteq (f \times g)^* S$; in Barr-exact $\mathcal{E}$, effective R is $\Gamma_{q_R}^{\circ} \circ \Gamma_{q_R}$. For chosen $q_R : X \twoheadrightarrow X/R$ and image mono $i : S \hookrightarrow Y$, maps with kernel R and image i are exactly $i\phi_{q_R}$, $\phi \in \text{Iso}_{\mathcal{E}}(X/R, S)$, so fixed kernel/image retain that endpoint-automorphism bitorsor ($n!$ on n -point $\mathbf{Set}$). On $\mathbf{Set}$ let $f_n = f^{\times n}$ and, for $f_n R \subseteq S$, $\text{Sat}_f R := f_n^{-1}(f_n R)$, $\text{RefGap}_f(R, S) := f_n^{-1} S \setminus R = (\text{Sat}_f R \setminus R) \sqcup (f_n^{-1} S \setminus \text{Sat}_f R)$. Then $R \subseteq \Gamma_{f_n}^{\circ} \circ \Gamma_{f_n} \circ R = \text{Sat}_f R \subseteq \Gamma_{f_n}^{\circ} \circ S = f_n^{-1} S$; both grades vanish iff $R = f_n^{-1} S$; $n = 1$ is U0 and finite signatures are componentwise. On the complete same-source face, (REL_{ker}) $\Phi(A) = \bigwedge_{j \in A} \Gamma_{h_j}^{\circ} \circ \Gamma_{h_j}$, $\Phi^f(A) = \Gamma_f^{\circ} \circ \Phi(A) \circ \Gamma_f$, $I_m(R) = \bigwedge_m \Gamma_{T_m}^{\circ} \circ R \circ \Gamma_{T_m}$; hence (DI/KGC/KBC, IOC/BEH, KGC_{pro, res}) are relation readouts.

For $p : \mathcal{E} \rightarrow \mathcal{B}$ put $H_p(f)(e_0, e_1) = \{a : e_0 \rightarrow e_1 : pa = f\}$. Composition makes $H_p : \mathcal{B} \rightarrow \mathbf{Prof}$ normal lax and (LRT)

$\text{Coll}(\mathcal{H}_p) \cong_{/\mathcal{B}} \mathcal{E}$; equivalently $\eta_p : \mathcal{E} \rightarrow \mathbb{T}_{\mathcal{B}} p$ is fully faithful and $\mathcal{H}_p(f)(e_0, e_1) \cong \text{Hom}_{\mathbb{T}_{\mathcal{B}} p}^f(\eta_p e_0, \eta_p e_1)$. For $b_0 \xrightarrow{f} b_1 \xrightarrow{g} b_2$ and a over $gf, \lambda_{g,f}^{-1}(a) \cong \pi_0 \text{Fact}_{g,f}^p(a)$; its fixed-middle slice is (FPF), retaining FRP homotopy gauge/ H^1/Z^0 . Put $\text{Lift}_p^-(f; e) = \int_{e_0 \in \mathcal{E}_{b_0}} \mathcal{H}_p(f)(e_0, e)$ and $\text{Lift}_p^+(f; e) = \int_{e_1 \in \mathcal{E}_{b_1}} \mathcal{H}_p(f)(e, e_1)$. (LYR) identifies these with the two cross-hom element categories; co-Yoneda recovers the slices. The minus/plus slices are (co)representable exactly when they have respectively a terminal/initial universal element, and such elements form a cleavage/opcleavage exactly when $\lambda_{g,f}$ is invertible. Yoneda identifies the represented comparisons with $\kappa_{g,f}^- : f^* g^* e \rightarrow (gf)^* e$ and $\kappa_{g,f}^+ : (gf)_! e \rightarrow g_! f_! e$; invertibility is cloven transport and the identity comparisons split. Thus (KZP/LYR/PLR) makes the KZ phase coherent representability of the minus slices, and (RIT_{ph}) its strict extension gate across $q \rightarrow \mathbb{T}q$, not (R0); $2 \rightarrow 1$ is split with nonterminal fibre. If a represented $\kappa^\pm$ is noninvertible, (NFP) gives $\text{Fact}^{[n]}(\kappa^\pm) \neq \mathbf{1}$ for all $n \geq 2$. For any arrow a of $\mathcal{E}$, (RFG) $\Pi_{p,a}^{[n]} : \text{Fact}_{\mathcal{E}}^{[n]}(a) \rightarrow \text{Fact}_{\mathcal{B}}^{[n]}(pa)$, $(\Pi_{p,a}^{[n]})^{-1}\{\sigma\} \cong \text{Fact}_{p,\sigma}^{[n]}(a)$, $n \geq 2$, where the strict fibre consists of n -step lift-factorizations over σ with vertical intermediate maps; $n = 2$ is LRT. Hence LRT reconstructs the source factorization category as the lax collage of this fibre system: RIT is the base residual and lift factorization its fibre. The split $p : 2 \rightarrow 1$ with nonidentity vertical κ is sharp: the base factorization is terminal while (NFP) keeps every source order nonterminal, so all recursive depth lies in the unique lift fibre. Invertible laxity alone is weaker: $1 \rightarrow \{0 < 1\}$ selecting 1 has no cartesian lift over $0 < 1$ (dually 0). By (RFD/FRF) and Grothendieck correspondence⁷ (Exp. VI, §§6–9), for small $I, F : I^{\text{op}} \rightarrow \mathbf{Cat}$ and cloven $U : \mathcal{E} \rightarrow \mathcal{B}$, with $\text{ResFib}(U)$ its fibre pseudofunctor, (URT) $\text{Tot}_I(F) := (\pi_F : \int_I F \rightarrow I)$, $\text{Circ}(U) := (\pi_U : \int_{\mathcal{B}} \text{ResFib}(U) \rightarrow \mathcal{B})$. Reindexing and Grothendieck–Fubini give, for $\theta : J \rightarrow I$ and compatible $G : (\int_I F)^{\text{op}} \rightarrow \mathbf{Cat}$, $\theta^* \text{Tot}_I(F) \simeq \text{Tot}_J(\theta^* F)$, $\int_{(i,x) \in \int_I F} G(i, x) \simeq \int_{i \in I} \int_{x \in F(i)} G(i, x)$. Thus for composable fibrations $U : \mathcal{E} \rightarrow \mathcal{B}$, $V : \mathcal{B} \rightarrow \mathcal{C}$, and discrete $A : I^{\text{op}} \rightarrow \mathbf{Set}$,

$$[\text{CRF}] \quad \mathcal{E} \simeq_{\mathcal{B}} \text{Circ}(U), \quad \text{Circ}(\pi_U) \simeq_{\mathcal{B}} \text{Circ}(U), \quad (12)$$

$$[\text{RFE}] \quad \text{ResFib}(VU)(c) \simeq \int_{b \in \text{ResFib}(V)(c)} \text{ResFib}(U)(b), \quad \text{Sec}_{\text{cart}}(\text{Tot}_I(A)) \cong \lim_{\leftarrow I} A. \quad (13)$$

Writing $F_U(b) = \text{ResFib}(U)(b)$, (EDO/RFE) gives (ROA) $\text{Poss}_U(b) = [F_U(b) \neq \emptyset]$, $\text{Force}_U(b) = [F_U(b) \simeq \mathbf{1}]$, $\text{Occ}_U(b) \simeq F_U(b)$ and $\text{Occ}_{VU}(c) \simeq \int_{b \in \text{Occ}_V(c)} \text{Occ}_U(b)$; the section identity is its local–global law. Thus (RTD) the single residual spine q^* carries access/composition, free solving/lift fibres, oriented wings and the distinct existence, coherence, exhaustion and return phases; exact-kernel contraction need not imply (R0). Every comparison tabulates as $\text{Tab}(b^*)$. Set, factorization and AB5 are faces of this spine; the KZ sector is reflective with data external only to the local q^* carrier retained as typed stage attachments, and marked closure/re-entry/path data remain distinct. Transgression and persistent return are readouts; $\text{ResFib}(U) \simeq \mathbf{1}_{\mathcal{B}}$ iff $\mathcal{E} \simeq_{\mathcal{B}} \mathcal{B}$.

The support–representability settlement law (SRSL) in Supplementary Material, Sec. S1, is the terminal face of this totalization, not a second actuality primitive. For composable fibrations $\mathcal{E} \xrightarrow{U} \mathcal{B} \xrightarrow{V} \mathcal{C}$ and $c \in \mathcal{C}$, let $R_{U|V,c}$ be the pullback of $\text{ResFib}(U)$ to $\text{Occ}_V(c) = \text{ResFib}(V)(c)$ and let $\text{Occ}_V(c)^+$ be the full support on objects where its fibre is nonempty. Then

$$(\text{SRSL}_{\text{RFE}}) \quad \text{Force}_{VU}(c) = 1 \iff \text{Occ}_V(c)^+ \text{ is groupoidal and } R_{U|V,c}|_{\text{Occ}_V(c)^+} \text{ is representable.}$$

Accordingly occurrence remains the residual fibre itself, while settlement is the stronger terminalization of the compatible total: extinction and coherent symmetry cancellation are two faces of the same criterion and neither supplies a law-to-point selector.

On the effective-groupoid face, relation phase and Grothendieck totalization also recover quotient automorphisms. For effective $\mathcal{R}$ on $\mathcal{Z}$, put $\mathcal{H}_{\mathcal{R}} := \text{Aut}(\mathcal{Z}, \mathcal{R})$ and let $d_{\mathcal{R}} : \mathcal{H}_{\mathcal{R}} \rightarrow \text{Aut}(\mathcal{Q}_{\mathcal{R}})$ be quotient descent. Since $q_{\mathcal{R}} h = d_{\mathcal{R}}(h) q_{\mathcal{R}}$, source reparametrization acts on the REL phase bitorsor through $d_{\mathcal{R}}$, so for replete $\mathcal{S} \subseteq \mathcal{Q}_{\mathcal{R}}$,

$$(\text{QAD}) \quad \text{Phase}(\mathcal{R}, \mathcal{S}) := \text{Eq}(\mathcal{Q}_{\mathcal{R}}, \mathcal{S})_{h\mathcal{H}_{\mathcal{R}}} \simeq \text{hofib}(Bd_{\mathcal{R}} : B\mathcal{H}_{\mathcal{R}} \rightarrow B\text{Aut}(\mathcal{Q}_{\mathcal{R}})).$$

It is contractible iff $d_{\mathcal{R}}$ is an equivalence. The diagonal face is (CRI); on the terminal quotient it is $B\text{Aut}(\mathcal{Z})$. Since $\text{Aut}(BS_3) \simeq *$ but $BS_3 \neq *$, phase principalization does not imply full faithfulness; (QALF/TPGC) remains the typed lift refinement.

On the additive (RWS) face, (UDR) $\mathcal{W}_I(A) = \varprojlim_I A$, $\mathcal{D}\mathcal{W}_I(C) = \mathbf{R} \varprojlim_I C := \mathbf{DRes}_I(C)$ and $H_{-p} \mathcal{D}\mathcal{W}_I(A[0]) \cong \text{Hist}_I^p(A) = R^p \varprojlim_I A$; exactness gives (UDR_Δ) on triangles and the (UAR_±/TRF/PPR) connecting maps. For $u : C \rightarrow \mathcal{D}$, $\text{DKan}_u = \mathbf{R} \text{Ran}_{u^{\text{op}}}$ gives $\mathbf{DRes}_C A \simeq \mathbf{DRes}_{\mathcal{D}}(\text{DKan}_u A)$ and $E_2^{r,s} = \text{Hist}_{\mathcal{D}}^r(R^s \text{Ran}_{u^{\text{op}}} A) \Rightarrow \text{Hist}_C^{r+s}(A)$. With (RFE), put $B_c = \text{ResFib}(V)(c)$, $T_c := \int_{B_c} \text{ResFib}(U) \simeq \text{ResFib}(VU)(c)$ and $\pi_c : T_c \rightarrow B_c$. For $A : T_c^{\text{op}} \rightarrow \mathbf{Ab}$, $E_2^{r,s} = \text{Hist}_{B_c}^r(R^s \text{Ran}_{\pi_c^{\text{op}}} A) \Rightarrow \text{Hist}_{T_c}^{r+s}(A)$; if $R^s \text{Ran}_{\pi_c^{\text{op}}} A = 0$ for $s \neq d$, then $\text{Hist}_{T_c}^{r+d}(A) \cong \text{Hist}_{B_c}^r(R^d \text{Ran}_{\pi_c^{\text{op}}} A)$. Pointwise the coefficient is the derived (RKU) comma limit, not the strict fibre (sharp for $1 \rightarrow [1]$ at 0). The same DKF mechanism on factorization categories is $\mathbf{DRes}_{\text{Fact}_{\mathcal{E}}^{[n]}(a)} A \simeq \mathbf{DRes}_{\text{Fact}_{\mathcal{B}}^{[n]}(pa)} (\text{DKan}_{\Pi_{p,a}^{[n]}} A)$. Whenever the derived Kan adjoints exist, (KAE) $u_! = \mathbf{L} \text{Lan}_{u^{\text{op}}} \dashv u^* \dashv u_* := \text{DKan}_u$, with $K_u^- = \text{Cone}(u_! u^* \rightarrow I)$ and $K_u^+ = \text{Fib}(I \rightarrow u_* u^*)$. Octahedrality gives (KRC) $v_! K_u^-(v^* B) \rightarrow K_{vu}^-(B) \rightarrow K_v^-(B) \rightarrow [1]$ and $K_v^+(B) \rightarrow K_{vu}^+(B) \rightarrow v_* K_u^+(v^* B) \rightarrow [1]$; $u : 1 \rightarrow [1]$ at 0 with $0 \rightarrow \mathbb{Z}$ versus $\mathbb{Z} \xrightarrow{1} \mathbb{Z}$ separates the defects. For fully faithful u , (KDR) $K_u^- \dashv j_u \dashv K_u^+$ on $\mathcal{R}_u = \ker u^*$, $\text{EssIm } K_u^\pm = \mathcal{R}_u$, $(\text{im } u_!, \text{im } u_*) = (\perp \mathcal{R}_u, \mathcal{R}_u^\perp)$ and $\mathcal{D}(\mathcal{D})/\mathcal{R}_u \simeq \mathcal{D}(C)$; fully faithful u, v give the (KRW) recollement $\mathcal{R}_v \rightleftarrows \mathcal{R}_{vu} \rightleftarrows \mathcal{R}_u$ with these triangles. For $i_\alpha : I_\alpha \times I \hookrightarrow \lambda \times I$, (KOF) $\mathcal{R}_\alpha := \ker i_\alpha^*$ satisfies $\mathcal{R}_0 \supseteq \dots \supseteq \mathcal{R}_\lambda = 0$, $\mathcal{R}_\delta = \cap_{\alpha < \delta} \mathcal{R}_\alpha$, $\mathcal{R}_\alpha/\mathcal{R}_{\alpha+1} \simeq \mathcal{D}(I)$. Slice restriction has kernel $\mathcal{R}_{\alpha+1}$ and fully faithful extension-by-zero, so $\mathcal{R}_{\alpha+1} \subsetneq \mathcal{R}_\alpha \iff \mathcal{D}(I) \neq 0 \iff I \neq \emptyset$, with $I = \emptyset$ exactly degenerate. If degree-zero $A : I^{\text{op}} \rightarrow \mathbf{Ab}$ has $\text{Hist}_I^r(A) \neq 0$ for all finite $r \geq 2$, the zero-cross-arrow B_λ with all ordinal slices A has each KOF layer A ; Bernoulli below supplies A , proving ordinal residual depth independent of cohomological depth and excluding transfinite splitting/hosted (ORS) identification. For fully faithful w , (KWF) $\text{Wing}_w(\mathcal{M}) = \ker[\mathcal{D}(\mathcal{E}) \rightarrow \mathcal{D}(\mathcal{M})]$ is a strict residual diagram on $\text{Int}_{\mathcal{E}}(w)^{\text{op}}$; its Grothendieck fibration is (RGQ/CRF), and the [2] skyscraper witness retains the intermediate-base coordinate. On $J = \lambda \times I$, for $B : J^{\text{op}} \rightarrow \mathbf{Ab}$ write $B_\alpha := B|_{\{ \alpha \} \times I}$, let Sk_α be extension-by-zero, and set $\mathcal{A}_{\lambda,I} := (\mathbf{Ab}^{(\lambda \times I)^{\text{op}}})^\simeq$, $\mathcal{G}_{\lambda,I} := (\prod_{\alpha < \lambda} \mathbf{Ab}^{I^{\text{op}}})^\simeq$, $q_{\mathcal{R}}(B) := (B_\alpha)_{\alpha < \lambda}$. (KTR) gives $F_{<\alpha} B \rightarrow B \rightarrow F_{\geq \alpha} B \rightarrow [1]$, $F_{<\alpha} B \rightarrow F_{<\alpha+1} B \rightarrow \text{Sk}_\alpha(B_\alpha) \xrightarrow{\epsilon_\alpha(B)} F_{<\alpha} B[1]$, and $F_{<\delta} B = \varinjlim_{\alpha < \delta} F_{<\alpha} B$ at limits: no limit-stage completion residual, with successor Yoneda class $\epsilon_\alpha(B)$. For fixed $A_\bullet$, (RFD/KTR) gives $\text{Glue}_{\lambda}(A_\bullet) := \text{Fib}_{A_\bullet}(\mathbb{T}_{\mathcal{G}_{\lambda,I}} q_{\mathcal{R}})$ and (KGM) $\text{Glue}_{\alpha+1}(A_{\leq \alpha}) \simeq \int_{F \in \text{Glue}_{\alpha}(A_{<\alpha})} \text{ExtGrp}_{\alpha+1}^1(\text{Sk}_\alpha A_\alpha, F)$, limits gluing by exact colimits; $\text{ExtGrp}_{\alpha+1}^1(X, F)$ is the endpoint-fixed extension groupoid $0 \rightarrow F \rightarrow E \rightarrow X \rightarrow 0$ in $\mathbf{Ab}^{(I_{\alpha+1} \times I)^{\text{op}}}$, with F extended by zero. By⁸, (DKW): Waldhausen S of a small proto-exact category is

2-Segal; (DK_μ) : unital 2-Segal sets carry triangulation-independent fundamental multivalued composition correspondence. Thus (DK_W) yields the fixed-layer face; for unital 2-Segal X , $A_X = X_1 \times_{X_0} X_1$, $p_X = (\partial_0, \partial_2) : X_2 \rightarrow A_X$, $m_X = \partial_1 : X_2 \rightarrow X_1$, and $(R2C/R2F)$ $\text{TriRes}_X(c; u) = \text{Fib}_{(u,c)}(p_X, m_X)$. For $X_n = \text{Fun}([n], C)$ this is (RIT/RFG) ; $S_\bullet(\mathcal{A})$ gives the extension-groupoid face. The $[2]$ intermediate-count and $\mathbb{Z}/2$ split/ $\mathbb{Z}/4$ extension witnesses keep marginals distinct; cyclic versus split all- $\mathbb{Z}/2$ flags realize every finite layer depth. One binary residual generates all finite (KTR/KGM) depths; (KZP) remains a distinct algebraic axis. *Pro-factorization landing completion*. Let $J \neq \emptyset$ be directed, $D : J^{\text{op}} \rightarrow \mathbf{Set}$, $P_D = \varprojlim D_j$, $L \subseteq P_D$, $L_j = p_j L$, $\widehat{L} = \varprojlim L_j$, and $U_x^L = \text{Tot}_J\{\ell \in L : p_j \ell = x_j\}$. Directedness gives

(PFR) $\widehat{L} = \text{FSh}(L) = \{x : \forall F \in J \exists \ell_F \in L, p_j \ell_F = x_j (j \in F)\} = \{x : U_x^L \text{ has (R+)}\}$, $L = \{x : U_x^L \text{ has (RS)}\}$, $\text{LimRes}(L) = \widehat{L} \setminus L$. On the discrete (RFD/URT) face $(\mathbb{T}_{\widehat{L}JL})_x \simeq \text{disc Sect}_{\text{cart}}(U_x^L)$; with $E_j^L = \{(x, \ell) : p_j \ell = x_j\}$, $K_j^L = \ker(\text{pr}_{1\#}|_{E_j^L})$, $(\text{PFR}_{\text{der}})$ gives $\varprojlim K_j^L = 0$ and $\mathbb{Z}^{(\text{LimRes}(L))} = H_0 \mathfrak{R}_\bullet(J_L) \hookrightarrow \text{Hist}_J^1(K_\bullet^L)$. Thus nonlanding is simultaneously empty global Yoneda fibre, $(\text{R+}) \setminus (\text{RS})$ stage fibration and nonzero derived history; block coarsening yields the declared $(\text{R2C}/\text{R2F}, \text{KSL}_\exists/\text{LTX}_K, \text{TCR}/\text{TFC})$ faces. For arbitrary J , topological (P_j) , $P = \prod_j P_j$, $L \subseteq P$,

$$(\text{PEL}_{\text{top}}) \quad \overline{L}^P \cong \varprojlim_{F \in J} \overline{P_F(L)}^{P_F}.$$

This needs no directedness; for directed J with $L \subseteq \varprojlim_j P_j$, PFR is strictly stronger, so closure is not promoted to landing. *Cofinal regime completion-descent and joint-key completion*. For compatible $q_j : \Omega \rightarrow D_j$, $q_i = p_{ij} q_j$, and $a : \Omega \rightarrow A$, set $L_J = (q_j)_j(\Omega) \subseteq P_D$, $q_J : \Omega \twoheadrightarrow L_J$, $\widehat{L}_J = \varprojlim_j q_j(\Omega)$ and $\widehat{q}_J : \Omega \twoheadrightarrow \widehat{L}_J$. Then

$$[\text{RCD}/\text{CRP}] \quad \widehat{L}_J = \text{FSh}(L_J), \quad \text{LimRes}(L_J) = \widehat{L}_J \setminus L_J, \quad \text{CofRes}_{J,a}(x) := a(\widehat{q}_J^{-1}\{x\}), \quad \text{CofRes}_{J,a}(x) = \emptyset \iff x \in \text{LimRes}(L_J).$$

Put $\text{CofKey}_{J,a} := \text{im}(\widehat{q}_J, a) \subseteq \widehat{L}_J \times A$, with corestriction $j_{J,a} : \Omega \twoheadrightarrow \text{CofKey}_{J,a}$ and projection $\pi_{J,a} : \text{CofKey}_{J,a} \rightarrow \widehat{L}_J$. Then $\pi_{J,a} j_{J,a} = \widehat{q}_J$ and $\pi_{J,a}^{-1}\{x\} = \text{CofRes}_{J,a}(x)$. By PFR and $(\text{TOM}/\text{RST}/\text{RPL}/\text{DRX})$,

$$[\text{CRX}] \quad H_0 \mathfrak{R}_\bullet(\pi_{J,a}) \cong \mathbb{Z}^{(\text{LimRes}(L_J))}, \quad H_1 \mathfrak{R}_\bullet(\pi_{J,a}) \cong \text{DepRes}_{q_J}(a), \quad 0 \rightarrow \mathbb{K}(j_{J,a}) \rightarrow \mathbb{K}(q_J) \rightarrow H_1 \mathfrak{R}_\bullet(\pi_{J,a}) \rightarrow 0.$$

The stage keys $\text{JKey}_{j,a} := \text{im}(q_j, a) \subseteq L_j \times A$ have bonds $(u, y) \mapsto (p_{ij} u, y)$. Set $\widehat{\text{JKey}}_{J,a} := \varprojlim_j \text{JKey}_{j,a}$, let $\iota_{J,a} : \text{CofKey}_{J,a} \hookrightarrow \widehat{\text{JKey}}_{J,a}$ be restriction and $\widehat{\pi}_{J,a} : \widehat{\text{JKey}}_{J,a} \rightarrow \widehat{L}_J$ the profile projection. Then

$$[\text{JKC}] \quad \widehat{\text{JKey}}_{J,a} = \text{FSh}(\text{CofKey}_{J,a}), \quad \widehat{\pi}_{J,a}^{-1}\{x\} = \text{StabRes}_{J,a}(x) := \bigcap_j a(q_j^{-1}\{x_j\}), \quad \pi_{J,a} = \widehat{\pi}_{J,a} \iota_{J,a}.$$

Thus $\text{JointLimRes}_{J,a} := \widehat{\text{JKey}}_{J,a} \setminus \text{CofKey}_{J,a}$ consists exactly of profile-target pairs realizable at every finite/cofinal stage but by no single global realization. Since $\iota_{J,a}$ is injective, $(\text{UAR}_\pm/\text{RST}/\text{RPL})$ gives the exact ghost-target sequence

$$[\text{GTS}] \quad 0 \rightarrow \text{DepRes}_{q_J}(a) \rightarrow \mathbb{K}(\widehat{\pi}_{J,a}) \xrightarrow{\partial} \mathbb{Z}^{(\text{JointLimRes}_{J,a})} \rightarrow \mathbb{Z}^{(\text{LimRes}(L_J))} \rightarrow \mathbb{Z}^{(\widehat{L}_J \setminus \text{im } \widehat{\pi}_{J,a})} \rightarrow 0.$$

Here $\mathbb{K}(\widehat{\pi}_{J,a}) \cong \bigoplus_{x \in \text{im } \widehat{\pi}_{J,a}} \mathbb{Z}^{(\text{StabRes}_{J,a}(x))}$, and PFR_{der} applied to the augmented key system injects $\mathbb{Z}^{(\text{JointLimRes}_{J,a})}$ into its Hist_J^1 transgression. Hence CRX is the endpoint quotient of a strictly finer cofinal spine: actual target ambiguity, stagewise-stable target ambiguity, joint profile-target nonlanding, profile nonlanding, and failure of any target value to survive all regimes occur as successive exact terms. Higher Hist^p remains a distinct derived attachment as in (THP) . *Derived joint-key transgression*. Pull each stage key back to the completed profile base,

$$E_{j,a}^\uparrow := \widehat{L}_J \times_{L_j} \text{JKey}_{j,a} = \{(x, y) : y \in a(q_j^{-1}\{x_j\})\}, \quad s_{j,a}(x, y) = x,$$

and put $C_{j,a}^\uparrow := \mathbb{K}(s_{j,a})$ and $A_{j,a}^\uparrow := \mathbb{Z}^{(E_{j,a}^\uparrow)}$. For $i \leq j$ the bond is the inclusion $E_{j,a}^\uparrow \subseteq E_{i,a}^\uparrow$, every $s_{j,a}$ is onto, and $\bigcap_j E_{j,a}^\uparrow = \widehat{\text{JKey}}_{J,a}$. Finite support therefore gives

$$[\text{DJK0}] \quad \varprojlim_j A_{j,a}^\uparrow \cong \mathbb{Z}^{(\widehat{\text{JKey}}_{J,a})}, \quad \varprojlim_j C_{j,a}^\uparrow \cong \mathbb{K}(\widehat{\pi}_{J,a}).$$

Applying $\mathbf{DRes}_J$ to the objectwise exact tower $0 \rightarrow C_{j,a}^\uparrow \rightarrow A_{j,a}^\uparrow \rightarrow \mathbb{Z}^{(\widehat{L}_J)} \rightarrow 0$ gives, since directedness kills the positive-degree history of the constant diagram,

$$[\text{DJK1}] \quad 0 \rightarrow \mathbb{K}(\widehat{\pi}_{J,a}) \rightarrow \mathbb{Z}^{(\widehat{\text{JKey}}_{J,a})} \xrightarrow{\widehat{\pi}_{J,a}^\#} \mathbb{Z}^{(\widehat{L}_J)} \xrightarrow{\tau_{J,a}} \text{Hist}_J^1(C_{j,a}^\uparrow) \rightarrow \text{Hist}_J^1(A_{j,a}^\uparrow) \rightarrow 0,$$

$$[\text{DJK2}] \quad 0 \rightarrow \text{ExtRes}(\widehat{\pi}_{J,a}) \xrightarrow{\bar{\tau}_{J,a}} \text{Hist}_J^1(C_{j,a}^\uparrow) \rightarrow \text{VEx}_{J,a} \rightarrow 0, \quad \text{VEx}_{J,a} := \text{Hist}_J^1(A_{j,a}^\uparrow),$$

while $\text{Hist}_J^p(C_{j,a}^\uparrow) \cong \text{Hist}_J^p(A_{j,a}^\uparrow)$ for $p \geq 2$. Thus stable target ambiguity is exactly the degree-zero inverse limit, the terminal target-extinction term of (GTS) is its canonical degree-one birth, and $\text{VEx}_{J,a}$ is the independent tail excess; no splitting is asserted. This is the base-normalized derived tower: the fibre decompositions of the unpulled $\text{DepRes}_{q_J}(a)$ are indexed by the moving bases L_j and are not identified with it. Derived cofinal invariance is $(\text{MIT}_{\text{cof}})$. At $\Omega = L_J$, $a = I_{L_J}$ this is exactly $(\text{PFR}_{\text{der}})$ and is consumed by RLC/TCR . The $\mathbb{N} \cup \{\infty\}$ identity fixture has bijective $\widehat{\pi}$ but nonstationary $E_n^\uparrow$, hence $\text{VEx} \neq 0$; deleting ∞ gives the independent target-extinction class $[\infty]$ under $\bar{\tau}$. *Two-sided derived residual octahedron*. Put $E_{j,a}^\leftarrow := \widehat{\text{JKey}}_{J,a} \times_{\text{JKey}_{j,a}} \text{CofKey}_{J,a}$, with onto projection $r_{j,a}$ to $\widehat{\text{JKey}}_{J,a}$, and $A_{j,a}^\leftarrow := \mathbb{Z}^{(E_{j,a}^\leftarrow)}$, $C_{j,a}^\leftarrow := \mathbb{K}(r_{j,a})$. The bonds are inclusions and $\bigcap_j E_{j,a}^\leftarrow$ is the graph of $\iota_{J,a}$, so finite support and (UDR_Δ) give

$$[\text{TDO0}] \quad \varprojlim_j A_{j,a}^\leftarrow \cong \mathbb{Z}^{(\text{CofKey}_{J,a})}, \quad \varprojlim_j C_{j,a}^\leftarrow = 0, \quad 0 \rightarrow \mathbb{Z}^{(\text{JointLimRes}_{J,a})} \rightarrow \text{Hist}^1(C_{j,a}^\leftarrow) \rightarrow \text{VEx}_{J,a}^\leftarrow \rightarrow 0,$$

where $\text{VEx}_{J,a}^\leftarrow := \text{Hist}^1(A_{j,a}^\leftarrow)$ and $\text{Hist}^p(C_{j,a}^\leftarrow) \cong \text{Hist}^p(A_{j,a}^\leftarrow)$ for $p \geq 2$. Set $X = \mathbb{Z}^{(\text{CofKey})}[0]$, $M = \mathbb{Z}^{(\widehat{\text{JKey}})}[0]$, $Z = \mathbb{Z}^{(\widehat{L})}[0]$, $Y^- = \mathbf{DRes}(A_{j,a}^\leftarrow)$ and $Y^+ = \mathbf{DRes}(A_{j,a}^\uparrow)$ (all indices J, a suppressed). Graph inclusions and stage projections give

$$X \xrightarrow{\eta} Y^- \xrightarrow{\ell} M \xrightarrow{\sigma} Y^+ \xrightarrow{\rho} Z, \quad \ell\eta = \iota_\#, \quad \rho\sigma = \widehat{\pi}_\#.$$

For $f = \sigma\ell$ and $h = \rho f$, the standard mapping-cone octahedra give the natural morphism

$$\begin{array}{ccccccc} \mathfrak{R}_\bullet(\iota) & \rightarrow & \mathfrak{R}_\bullet(\pi) & \rightarrow & \mathfrak{R}_\bullet(\hat{\pi}) & \rightarrow & [1] \\ \text{[TDO1]} & & \downarrow & & \downarrow & & \downarrow \\ \mathfrak{R}_\bullet(f) & \rightarrow & \mathfrak{R}_\bullet(h) & \rightarrow & \mathfrak{R}_\bullet(\rho) & \rightarrow & [1]. \end{array}$$

Here $\mathfrak{R}_\bullet(\rho) \simeq \mathbf{DRes}(C_\bullet^\uparrow)[1]$, while η, σ are H_0 -isomorphisms with $(H_1, H_0)\mathfrak{R}(\eta) = (H_1, H_0)\mathfrak{R}(\sigma) = (0, 0)$ and $H_{-p}\mathfrak{R}(\eta) \cong \text{Hist}^p(A_\bullet^\uparrow)$, $H_{-p}\mathfrak{R}(\sigma) \cong \text{Hist}^p(A_\bullet^\uparrow)$ for $p \geq 1$. Consequently

$$\text{[TDO2]} \quad 0 \rightarrow \mathbb{Z}^{(\text{JointLimRes})} \rightarrow H_0\mathfrak{R}(f) \rightarrow \text{VEx}^- \rightarrow 0, \quad 0 \rightarrow \mathbb{Z}^{(\text{LimRes } L)} \rightarrow H_0\mathfrak{R}(h) \rightarrow \text{VEx}^- \rightarrow 0, \\ 0 \rightarrow \text{ExtRes}(\hat{\pi}) \rightarrow H_0\mathfrak{R}(\rho) \rightarrow \text{VEx} \rightarrow 0.$$

Naturality identifies the two left tail quotients; their zero-tail kernels and the zero-VEx kernel on the right are exactly the static (GTS) edge. Thus GTS is the degree-0/1 zero-tail edge of one two-sided derived octahedron, not a replacement for higher history. Directed cofinal restriction preserves the whole diagram and both side histories by $(\text{MIT}_{\text{cof}})$. Strictness is simultaneous on $J = \mathbb{N}$, $\Omega = \{0, 1\}^{\mathbb{N}}$, with q_n the length- $(n+1)$ prefix: constant, identity and even-coordinate readouts $a(x) = (x_{2k})_{k \geq 0}$ give respectively $(\text{VEx}^-, \text{VEx}) = (\neq 0, 0), (0, \neq 0), (\neq 0, \neq 0)$ while $L = \hat{L}$ and $\iota, \hat{\pi}$ are bijective. The nonzero cases are precisely the nonstationary countable onto towers and the zero cases stationary, by the tail-excess theorem. Hence the two side-tail vanishing loci are independent of the static GTS spine and of each other; no splitting or scalar exchange is supplied.

Boundary-derived interchange residual. Let $\mathcal{R} = (\Omega, D_\bullet, q_\bullet, a)$ be the preceding regime/readout system, let $r : \Omega' \rightarrow \Omega$, let $s_\bullet : D_\bullet \Rightarrow D'_\bullet$ commute with the bonds, and let $b : A \rightarrow A'$. Write $\mathcal{R}_r = (\Omega', D_\bullet, q_\bullet, r, ar)$, $\mathcal{R}_s = (\Omega, D'_\bullet, s_\bullet, q_\bullet, ba)$ and $\mathcal{R}_{sr} = (\Omega', D'_\bullet, s_\bullet, q_\bullet, r, bar)$. Source restriction gives inclusions of every stage image/key and target postcomposition gives the induced image maps; the resulting squares commute. The same is therefore true after inverse-limit completion, for CofKey and JKey , for both $E^\leftarrow, E^\uparrow$ towers, after $\mathbb{Z}^{(-)}$ and $\mathbf{DRes}_J$, and for the standard mapping cones in (TDO1). If $\mathbf{T}_J(\mathcal{R})$ denotes this full TDO diagram, there is consequently a canonical commutative square

$$\begin{array}{ccc} \mathbf{T}_J(\mathcal{R}_r) & \longrightarrow & \mathbf{T}_J(\mathcal{R}) \\ \downarrow & & \downarrow \\ \mathbf{T}_J(\mathcal{R}_{sr}) & \longrightarrow & \mathbf{T}_J(\mathcal{R}_s). \end{array}$$

Taking the homotopy pullback vertexwise in $\mathcal{D}(\mathbf{Ab})$ gives the comparison and its objectwise residual

$$\text{[BDI]} \quad \chi_{r,s,b} : \mathbf{T}_J(\mathcal{R}_r) \longrightarrow \mathbf{T}_J(\mathcal{R}) \times_{\mathbf{T}_J(\mathcal{R}_s)}^h \mathbf{T}_J(\mathcal{R}_{sr}), \quad \text{BdInt}_J(r; s, b) := \mathfrak{R}_\bullet(\chi_{r,s,b}). \quad (14)$$

Thus $\text{BdInt} = 0$ exactly when boundary change and the full derived residual completion are homotopy Beck–Chevalley; as the mapping-cone cofiber it is the universal exact obstruction to that equivalence. It vanishes identically if $r = I$ or $(s_\bullet, b) = I$, so it is a genuinely mixed boundary coordinate and a (KCM/KRC) face, not a new scalar primitive. Universal vanishing is false already for $J = *$: take $\Omega = \{0, 1\}$, $\Omega' = \{0\}$, constant $q : \Omega \rightarrow \mathbf{1}$, $a = I_\Omega$, $s = I$ and $b : \{0, 1\} \rightarrow \mathbf{1}$. At the joint-key vertex the square is

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{e_0} & \mathbb{Z}^2 \\ \downarrow 1 & & \downarrow (1 \ 1) \\ \mathbb{Z} & \xrightarrow{1} & \mathbb{Z}, \end{array}$$

so the homotopy pullback is $\mathbb{Z}^2$ and $H_0\text{BdInt}^{\text{JKey}} \cong \mathbb{Z}$. This is the exact additive shadow of an excluded source point becoming indistinguishable after target coarse-graining. Hence $(\text{BRX}/\text{TOM}_\circ)$ and TDO commute canonically as a boundary square/correspondence, while lossless interchange is precisely the additional zero condition (BDI); no inverse boundary lift, section or collapse of either TDO side tail is implied.

Boundary-derived pasting law. For a TDO boundary square Q , write $t = (s_\bullet, b)$. Pointwise stability and mapping-cone octahedrality give, for composable source maps r_1, r_2 and target changes t_1, t_2 ,

$$\begin{aligned} \text{BdInt}_J(\mathcal{R}; r; t) &\simeq \text{TotCof}(\mathbf{T}_J(\mathcal{R}; r; t))[-1], \\ \text{BdInt}_J(\mathcal{R}_{r_1}; r_2; t) &\rightarrow \text{BdInt}_J(\mathcal{R}; r_1 r_2; t) \rightarrow \text{BdInt}_J(\mathcal{R}; r_1; t) \rightarrow [1], \\ \text{BdInt}_J(\mathcal{R}; r; t_1) &\rightarrow \text{BdInt}_J(\mathcal{R}; r; t_2 t_1) \rightarrow \text{BdInt}_J(\mathcal{R}_{r_1}; r; t_2) \rightarrow [1]. \end{aligned}$$

Boundary-grid coherence and canonical bigrading. For fixed strings set $r^{[i]} = r_1 \cdots r_i$, $t^{[j]} = t_j \cdots t_1$ (identity at 0) and $\mathcal{F}_{i,j} := \text{BdInt}_J(\mathcal{R}; r^{[i]}; t^{[j]})$. Then (BDP) makes $\mathcal{F} : [m]^{\text{op}} \times [n] \rightarrow \mathcal{D}(\mathbf{Ab})$ a canonical zero-axis rectangle. With $S_{i,j} = \text{Fib}(\mathcal{F}_{i,j} \rightarrow \mathcal{F}_{i-1,j})$ and $T_{i,j} = \text{Cof}(\mathcal{F}_{i,j-1} \rightarrow \mathcal{F}_{i,j})$, the stable 3×3 lemma gives

$$\text{Cof}(S_{i,j-1} \rightarrow S_{i,j}) \simeq \text{Fib}(T_{i,j} \rightarrow T_{i-1,j}) \simeq \text{BdInt}_J(\mathcal{R}_{r^{[i-1]}, t^{[j-1]}}; r_i; t_j) =: \text{BGr}_{i,j}.$$

Thus every recursive BDP resolution of the fixed grid has the same outer object and canonical associated bigrading; cellwise zero implies outer zero, with no converse. Distinct one-dimensional filtrations need not agree or split: (KGM) retains extension gluing. $(\text{MIT}_{\text{cof}})$ preserves the rectangle. Boundary-order choice is presentation, not a new residual primitive; this remains the pasted (KCM/KRC) face.

Cubical residual totalization. For finite nonempty ordered F , $m = |F|$, and a chain-complex cube $X : \mathcal{P}(F) \rightarrow \mathbf{Ch}(\mathbf{Ab})$, viewed in $\mathcal{D}(\mathbf{Ab})$, put

$$\begin{aligned} \text{CubRes}_F(X) &:= \text{Cone}\left(X_\emptyset \rightarrow \text{holim}_{\emptyset \neq A \subseteq F} X_A\right) \simeq \text{TotCof}_F(X)[1-m], \\ \text{[CRT]} \quad F = E \sqcup G, \ E, G \neq \emptyset : \quad &\text{CubRes}_E(A \mapsto \text{CubRes}_G(B \mapsto X_{A \cup B})) \simeq \text{CubRes}_F(X)[1], \\ &\Phi \text{ exact} : \quad \Phi \text{CubRes}_F(X) \simeq \text{CubRes}_F(A \mapsto \Phi X_A), \\ &\Psi \text{ exact contravariant} : \quad \Psi \text{CubRes}_F(X) \simeq \text{CubRes}_F(A \mapsto \Psi X_{F \setminus A})[m-2], \\ &\chi : X \Rightarrow Y : \quad \mathfrak{R}_\bullet(\text{CubRes}_F(X)) \simeq \text{CubRes}_F(A \mapsto \mathfrak{R}_\bullet(X_A)). \end{aligned} \quad (15)$$

Here TotCof_F is the standard iterated chain-level total cofiber. In (CSE), an exact covariant functor preserves this finite total cofiber; an exact contravariant functor exchanges total cofiber with total fiber, reverses suspension, and the complement $A \mapsto F \setminus A$ restores cube orientation, giving the displayed $m-2$ shift. (CRM) is the singleton-arrow block face: every independently supplied chain-level cube of comparison maps is residualized before or after cubical totalization by the same object. Its zero locus is residual descent, not pointwise absence: duplicating the preceding nonzero two-point (BDI) comparison constantly along one spectator cube direction leaves $H_0\mathfrak{R}_\bullet(\chi_A) \cong \mathbb{Z}$ at both vertices but makes their cubical residual zero. It manufactures no cube from isolated Set-level (KCM) data. The Boolean model places X_A in external homological degree $1 - |A|$. Hence $m = 1$ is $\mathfrak{R}_\bullet$ and $m = 2$ on the TDO boundary square is (BDI/BDP₀) objectwise; (BDP/BGC) is the finite

rectangular pasting/bigrading calculus generated by this binary cell. On the (DCE) cube $X_A = Z_b^1(\hat{R}_A; Z)[0]$, $H_{1-p}\text{CubRes}_F(X) \cong \text{BGLu}_Z^p(F)$ ($1 \leq p < m$); for $Y_A = \mathbb{K}(\hat{q}_{F \setminus A})[0]$, after the standard complement sign, $H_{p-m+1}\text{CubRes}_F(Y) \cong \text{DInt}_p(F)$, recovering (UIR/BRMV/DCHD). Thus map residual, mixed boundary cells and finite determination interaction/gluing are typed faces of one stable derived-additive carrier. At $m = 1$, (CSE) with $\Psi = \mathbf{RHom}_{\mathbb{Z}}(-, \mathbb{Z})$ is exactly the (DOD/JOD/MID/DOH) objective-dual route, including its Ext^1/Tor correction. The θ_F classes retain arbitrary finite top order. Upstream (GLT/YRM/KGM/FRP) generation fibres are not determined by the supplied object in $D(\mathbf{Ab})$; (HCF) makes comparison-cell/ICP terminality pre-homotopy; transport, law/isotropy, ambient and occurrence remain outside. Thus CRT supplies neither comparison realization nor scalarization.

Derived integral phase. For bounded finite-free C , the coefficient Bockstein and Pontryagin duality give naturally

$$\begin{array}{c} \text{[DPH]} \quad \text{(DPH)} \quad \begin{array}{c} 0 \rightarrow H_1(C; \mathbb{R})/\text{im } H_1(C; \mathbb{Z}) \rightarrow \text{DPh}(C) := H_1(C \otimes \mathbb{R}/\mathbb{Z}) \xrightarrow{\beta_C} \text{Tor } H_0(C; \mathbb{Z}) \rightarrow 0, \\ \text{Lie DPh}(C) \cong H_1(C; \mathbb{R}), \quad \pi_1 \text{DPh}(C)^0 \cong H_1(C; \mathbb{Z})/\text{Tor } H_1(C; \mathbb{Z}), \quad \pi_0 \text{DPh}(C) \cong \text{Tor } H_0(C; \mathbb{Z}), \\ \overline{\text{DPh}(C)} \cong H^1(\text{Hom}_{\mathbb{Z}}(C, \mathbb{Z})) \cong \text{Obj}_{\mathbb{Z}}^1(C). \end{array} \end{array} \quad (16)$$

Degree-wise exactness of $0 \rightarrow C \rightarrow C \otimes \mathbb{R} \rightarrow C \otimes \mathbb{R}/\mathbb{Z} \rightarrow 0$ gives the first row, since $\ker[H_0(C; \mathbb{Z}) \rightarrow H_0(C; \mathbb{R})] = \text{Tor } H_0(C; \mathbb{Z})$. Cycles are closed and boundaries compact in a finite torus, so $\text{DPh}(C)$ is compact Lie; the full-lattice quotient in the first row is exactly $\text{DPh}(C)^0$. Degree-wise character duality gives the last row, whose dual is the finite-free (DOD) sequence. This standard ABI lands in SCO through the incidence-entropy bridge below.

Exact-functor continuity, coefficient history and prophase. Put $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $\text{Ph}(C) = C \otimes^{\mathbb{L}} \mathbb{T}$. For every J -diagram C in the $\mathbf{DRes}_J$ perimeter and every $E \in \mathcal{D}(\mathbf{Ab})$, the limit cone gives $\chi_E : (\mathbf{DRes}_J C) \otimes^{\mathbb{L}} E \rightarrow \mathbf{DRes}_J(C \otimes^{\mathbb{L}} E)$; set $\text{CoefHist}_J^E(C) := \text{Cofib } \chi_E$. Both sides of χ_E are exact in E (tensor is exact and a stable limit is exact), hence $E \mapsto \text{CoefHist}_J^E(C)$ is exact. Since $\chi_{\mathbb{Z}} = I$, the coefficient triangle $\mathbb{Z} \rightarrow \mathbb{R} \rightarrow \mathbb{T} \rightarrow [1]$ gives

$$\text{[CHI]} \quad \text{(CHI)} \quad \text{CoefHist}_J(C) := \text{CoefHist}_J^{\mathbb{R}}(C) \simeq \text{CoefHist}_J^{\mathbb{T}}(C). \quad (17)$$

Thus realification-history and phase-history are two coefficient faces of one exact residual functor, the derived coefficient instance of (KCM/BDI).

More generally, let $C, \mathcal{D}$ be stable categories with countable products, let $F : C \rightarrow \mathcal{D}$ be exact, and let $X_0 \leftarrow X_1 \leftarrow \dots$ be a tower. Write $P(X) = \prod_{n \geq 0} X_n$ and let T_X be the bond shift. The product projections give the canonical comparison

$$\alpha_F : F(P(X)) \rightarrow P(FX), \quad \text{ProdRes}_F(X) := \text{Cofib } \alpha_F,$$

with induced shift $\tilde{T}_F$ on $\text{ProdRes}_F(X)$. Put $\text{LimInt}_F(X) := \text{Cofib}[F(\mathbf{DRes}_{\mathbb{N}} X) \rightarrow \mathbf{DRes}_{\mathbb{N}}(FX)]$. Since exact F carries the (DLX) fibre sequence for X to the fibre sequence of $1 - F(T_X)$, the product comparison gives a morphism to the (DLX) sequence for FX . Its vertical cofibers are $\text{LimInt}_F(X)$, $\text{ProdRes}_F(X)$, $\text{ProdRes}_F(X)$, so stable 3×3 gives

$$\text{(FCR)} \quad \text{LimInt}_F(X) \simeq \text{Fib}(1 - \tilde{T}_F : \text{ProdRes}_F(X) \rightarrow \text{ProdRes}_F(X)).$$

Thus F commutes with this countable derived limit exactly when $1 - \tilde{T}_F$ is an equivalence on the product residual. Preservation of countable products is sufficient but not necessary. Every exact right adjoint is in the zero-residual class; in particular the fixed-index (KAE/DKF) right-Kan functors $\mathbf{DKan}_u$ and the fixed-index $\mathbf{DRes}_C$ consumers commute with countable $\mathbf{DRes}_{\mathbb{N}}$ wherever typed. Together with (CSE), this isolates the finite/countable boundary for exact transport: finite cubical residuals commute automatically, while countable towers add only product continuity modulo the bond shift.

Specializing (FCR) to $F = - \otimes^{\mathbb{L}} E$ gives $\text{LimInt}_F = \text{CoefHist}_{\mathbb{N}}^E$ and, with

$$\alpha_E : P(C) \otimes^{\mathbb{L}} E \rightarrow P(C \otimes^{\mathbb{L}} E), \quad \text{ProdRes}_E(C) := \text{Cofib } \alpha_E,$$

the coefficient face

$$\text{[CCI]} \quad \text{(CCI)} \quad \text{CoefHist}_{\mathbb{N}}^E(C) \simeq \text{Fib}(1 - \tilde{T}_E : \text{ProdRes}_E(C) \rightarrow \text{ProdRes}_E(C)). \quad (18)$$

Consequently, for every q there is a natural, not canonically split, exact row

$$0 \rightarrow \text{coker}(1 - \tilde{T}_E : H_{q+1}\text{ProdRes}_E \rightarrow H_{q+1}\text{ProdRes}_E) \rightarrow H_q \text{CoefHist}_{\mathbb{N}}^E \rightarrow \ker(1 - \tilde{T}_E : H_q \text{ProdRes}_E \rightarrow H_q \text{ProdRes}_E) \rightarrow 0.$$

If E is perfect (equivalently dualizable in $\mathcal{D}(\mathbb{Z})$), then $- \otimes^{\mathbb{L}} E$ preserves products, so $\text{ProdRes}_E = \text{CoefHist}_{\mathbb{N}}^E = 0$. The converse sufficient condition is false already for $F = - \otimes \mathbb{R}$: on the constant identity tower $C_n = \mathbb{Z}[0]$, $\chi_{\mathbb{R}}$ is an equivalence, whereas $\alpha_{\mathbb{R}} : (\prod_n \mathbb{Z}) \otimes \mathbb{R} \rightarrow \prod_n \mathbb{R}$ is not onto, since every image sequence lies in the finite-dimensional $\mathbb{Q}$ -span of finitely many real coefficients while a sequence of $\mathbb{Q}$ -linearly independent reals does not. Hence $\text{ProdRes}_{\mathbb{R}} \neq 0$ but $1 - \tilde{T}_{\mathbb{R}}$ is an equivalence on it. The history residual is therefore not raw infinite-product failure but its bond-compatible fibre. The coefficient triangle gives (CHI), and (CSE) keeps finite cubical phase transport exact.

Contravariant continuity and limit-born objectives. Let C have countable products, let $\mathcal{D}$ have countable coproducts, and let $G : C^{\text{op}} \rightarrow \mathcal{D}$ be exact; retain the countable tower X . Put $A_G(X) = \bigoplus_{n \geq 0} G(X_n)$ and let S_G be the forward shift induced by the reversed bonds. The product projections give

$$\beta_G : A_G(X) \rightarrow G(P(X)), \quad \text{SumDualRes}_G(X) := \text{Cofib } \beta_G,$$

with induced $\bar{S}_G$. The limit projections give $\delta_G : \text{hocolim}_n G(X_n) \rightarrow G(\mathbf{DRes}_{\mathbb{N}} X)$; set $\text{DualLimInt}_G(X) := \text{Cofib } \delta_G$. Exact contravariance turns the (DLX) fibre sequence into the cofiber sequence of $1 - G(T_X)$, while $\text{hocolim } G(X_n) \simeq \text{Cofib}(1 - S_G)$; the square $G(T_X)\beta_G = \beta_G S_G$ and stable 3×3 therefore give

$$\text{(DFCR)} \quad \text{DualLimInt}_G(X) \simeq \text{Cofib}(1 - \bar{S}_G : \text{SumDualRes}_G(X) \rightarrow \text{SumDualRes}_G(X)).$$

Thus a contravariant exact functor that sends the countable product to the corresponding coproduct has zero dual-limit residual; as on the covariant side, the bond operator is part of the exact criterion and no product/coproduct comparison is itself called history. The sufficient condition is not necessary: in $\mathcal{D}(\mathbb{Q})$ take the constant identity tower $X_n = \mathbb{Q}[0]$ and $G = \mathbf{RHom}_{\mathbb{Q}}(-, \mathbb{Q})$. Then δ_G is an equivalence, whereas $\beta_G : \bigoplus_n \mathbb{Q} \rightarrow \text{Hom}_{\mathbb{Q}}(\mathbb{Q}^{\mathbb{N}}, \mathbb{Q})$ is not onto: $\mathbb{Q}^{\mathbb{N}}/\mathbb{Q}^{(\mathbb{N})} \neq 0$ and a nonzero functional on this quotient extends to a functional outside the finite-coordinate span. Hence $\text{SumDualRes}_G \neq 0$ while $\text{DualLimInt}_G = 0$.

On the derived-Ab face this comparison is index-general. For a small directed I , $X : I^{\text{op}} \rightarrow \mathcal{D}(\mathbf{Ab})$ and exact contravariant G , the limit projections induce $\delta_{G,I} : \text{hocolim}_I G(X_i) \rightarrow G(\mathbf{DRes}_I X)$; put $\text{DualLimInt}_{G,I} = \text{Cofib } \delta_{G,I}$. Since filtered homotopy colimits are homologically exact, with $o_q : \lim_{\rightarrow i} H^q G(X_i) \rightarrow H^q G(\mathbf{DRes}_I X)$ the cofiber LES gives

$$[\text{BDR}] \quad \boxed{(\text{BDR}) \quad 0 \rightarrow \underbrace{\text{coker } o_q}_{\text{Birth}_{G,I}^q} \rightarrow H^q \text{DualLimInt}_{G,I} \rightarrow \underbrace{\text{ker } o_{q+1}}_{\text{Death}_{G,I}^{q+1}} \rightarrow 0 \quad (q \in \mathbb{Z}).} \quad (19)$$

For $I = \mathbb{N}$ this is the derived-Ab specialization of (DFCR). It has genuine higher-history consumers: for $n \geq 1$, (THP/RDXT) with $I = \omega_n$, $K_\alpha = \mathbb{K}(q_\alpha^n)[0]$ and the exact cogenerator dual $G = \mathbf{RHom}(-, \mathbb{Q}/\mathbb{Z})$ gives

$$[\text{HBD}] \quad H^{-n-1} \text{DualLimInt}_{G, \omega_n}(K) \cong \text{Hom}(\text{Hist}_{\omega_n}^{n+1}(\mathbb{K}(q_\bullet^n)), \mathbb{Q}/\mathbb{Z}) \neq 0,$$

because the stage duals are concentrated in degree 0. Thus (DOB) is the degree-one face of an all-degree history carrier.

For the countable incidence tower $C_n = \mathfrak{C}_n^{\text{inc}}$ of (MIB/DIH), put $C_n^\vee = \mathbf{RHom}_{\mathbb{Z}}(C_n, \mathbb{Z})$, $\mathfrak{D}_{\text{inc}} = \mathbf{DRes}_{\mathbb{N}} C_\bullet$, and $\text{Obj}_n = H^1(C_n^\vee)$. Since $H_1(C_n) = \mathbb{I}_n$ is free, (DOD) gives $H^2(C_n^\vee) = 0$, so (BDR) yields

$$[\text{DOB}] \quad \boxed{(\text{DOB}) \quad \text{ObjBirth}_{\text{inc}} := \text{coker} \left[\lim_n \text{Obj}_n \rightarrow \text{Obj}_{\mathbb{Z}}^1(\mathfrak{D}_{\text{inc}}) \right] \cong H^1 \text{DualLimInt}_{(-)^\vee}(C_\bullet).} \quad (20)$$

Here $\text{Death}_{(-)^\vee, \mathbb{N}}^2 = 0$; this is objective information born only after coherent completion, not a selected functional. More precisely, compare the filtered colimit of the stage (DOD) rows with (DOH) and write

$$\begin{aligned} e &: \lim_n \text{Ext}^1(\mathbb{J}_n^{\text{bal}}, \mathbb{Z}) \rightarrow \text{Ext}^1(H_0 \mathfrak{D}_{\text{inc}}, \mathbb{Z}), \\ o &: \lim_n \text{Obj}_n \rightarrow \text{Obj}_{\mathbb{Z}}^1(\mathfrak{D}_{\text{inc}}), \\ h &: \lim_n \text{Hom}(\mathbb{I}_n, \mathbb{Z}) \rightarrow \text{Hom}(\lim_n \mathbb{I}_n, \mathbb{Z}). \end{aligned}$$

Naturality of universal coefficients and Snake give

$$0 \rightarrow \text{ker } e \rightarrow \text{ker } o \rightarrow \text{ker } h \rightarrow \text{coker } e \rightarrow \text{ObjBirth}_{\text{inc}} \rightarrow \text{coker } h \rightarrow 0.$$

Here (DIH) supplies the nontrivial target $H_0 \mathfrak{D}_{\text{inc}}$ through $0 \rightarrow \text{Hist}^1(\mathbb{I}_\bullet) \rightarrow H_0 \mathfrak{D}_{\text{inc}} \rightarrow \text{ker } t_0 \rightarrow 0$. The right flank has an intrinsic two-step resolution. Put $L = \lim_n \mathbb{I}_n$, $L_n = p_n L \subseteq \mathbb{I}_n$, $Q_n = \mathbb{I}_n / L_n$, and give L the prodiscrete topology. Since $\mathbb{Z}$ is discrete and every finite basic neighbourhood is dominated by one tower coordinate, $\text{Hom}_{\text{cts}}(L, \mathbb{Z}) \cong \lim_n \text{Hom}(L_n, \mathbb{Z})$. The bonds carry L_{n+1} onto L_n and induce $Q_{n+1} \rightarrow Q_n$. Restriction $0 \rightarrow \text{Hom}(Q_n, \mathbb{Z}) \rightarrow \text{Hom}(\mathbb{I}_n, \mathbb{Z}) \rightarrow \text{Hom}(L_n, \mathbb{Z}) \rightarrow \text{Ext}^1(Q_n, \mathbb{Z}) \rightarrow 0$ and filtered exactness therefore give

$$[\text{ICG}] \quad \boxed{(\text{ICG}) \quad \begin{aligned} &\text{ker } h \cong \lim_n \text{Hom}(Q_n, \mathbb{Z}), \\ &0 \rightarrow \lim_n \text{Ext}_{\mathbb{Z}}^1(Q_n, \mathbb{Z}) \rightarrow \text{coker } h \rightarrow \text{Hom}(L, \mathbb{Z}) / \text{Hom}_{\text{cts}}(L, \mathbb{Z}) \rightarrow 0. \end{aligned}} \quad (21)$$

Here $\text{Ext}^1(Q_n, \mathbb{Z}) \cong \text{Hom}(\text{Tor } Q_n, \mathbb{Q}/\mathbb{Z})$ and $\text{Tor } Q_n = ((L_n \otimes \mathbb{Q}) \cap \mathbb{I}_n) / L_n$. Thus the character flank records both finite-stage characters lost from the coherent limit and, on the cokernel side, persistent integral nonsaturation plus a possibly noncontinuous algebraic remainder; no splitting is supplied.

The left flank is controlled by the (DIH) extension itself. Write $H = \text{Hist}^1(\mathbb{I}_\bullet)$, $K = \text{ker } t_0$, $E = H_0 \mathfrak{D}_{\text{inc}}$ and let $\kappa_0 \in \text{Ext}^1(K, H)$ be the class of $0 \rightarrow H \rightarrow E \rightarrow K \rightarrow 0$. Let $a : \lim_n \text{Ext}^1(\mathbb{J}_n^{\text{bal}}, \mathbb{Z}) \rightarrow \text{Ext}^1(K, \mathbb{Z})$ be restriction along $K \hookrightarrow \lim_n \mathbb{J}_n^{\text{bal}} \rightarrow \mathbb{J}_n^{\text{bal}}$. Then $e = j a$, where j is pullback along $E \rightarrow K$. Applying $\text{Hom}(-, \mathbb{Z})$ to the (DIH) row and using $\text{Ext}_{\mathbb{Z}}^2 = 0$ gives $\text{ker } j = \text{im } \partial_{\kappa_0}$, $\partial_{\kappa_0}(\lambda) = \lambda_* \kappa_0$, and hence

$$[\text{AHR}] \quad \boxed{(\text{AHR}) \quad \begin{aligned} &\text{ker } e = a^{-1}(\text{im } \partial_{\kappa_0}), \\ &0 \rightarrow \frac{\text{Ext}^1(K, \mathbb{Z})}{\text{im } a + \text{im } \partial_{\kappa_0}} \rightarrow \text{coker } e \rightarrow \text{Ext}^1(H, \mathbb{Z}) \rightarrow 0. \end{aligned}} \quad (22)$$

Thus history acts on the arithmetic flank through pushout of the actual (DIH) extension class; the target arithmetic cannot in general be reconstructed from H and K as an unordered pair. This dependence is sharp already in $\mathbf{Ab}$: for $H = \mathbb{Z}$, $K = \mathbb{Z}/n$, the split extension has $E = \mathbb{Z} \oplus \mathbb{Z}/n$ and $\text{Ext}^1(E, \mathbb{Z}) \cong \mathbb{Z}/n$, whereas the generator extension $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n \rightarrow 0$ has $\partial_{\kappa_0} : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/n$ and $\text{Ext}^1(E, \mathbb{Z}) = 0$. Hence (DOH) is not a new scalar layer: its limit-born part is the degree-one face of the dual continuity residual, with dying interaction characters, arithmetic/history transmutation, persistent saturation and global character generation coupled by (DOB/ICG/AHR), not canonically summed.

These rows have one canonical classification endpoint. Let $\tau_{\text{obj}} : \text{ker } h \rightarrow \text{coker } e$ be the Snake connecting map, $R = \text{im } \tau_{\text{obj}}$, let $A_0 := \text{Ext}^1(K, \mathbb{Z}) / (\text{im } a + \text{im } \partial_{\kappa_0}) \hookrightarrow \text{coker } e$ be the (AHR) kernel term, let $\rho : \text{coker } e \twoheadrightarrow \text{Ext}^1(H, \mathbb{Z})$, and put $R_0 = A_0 \cap R$, $R_1 = \rho(R)$ and $S = \lim_n \text{Ext}^1(Q_n, \mathbb{Z})$. These are gluing/subquotient data, not independent scalar defects; the classification itself is regenerated below by (CEX/ÖBF).

Integral dual-history spectrum. Let I be small directed and let $C_i = \mathfrak{C}_i^{\text{inc}}$ be a diagram of the finite free (MIB/DIH) incidence complexes. Put $\mathfrak{D}_I = \mathbf{DRes}_I C_\bullet$, $\mathfrak{D}_I^\vee = \text{DualLimInt}_{(-)^\vee, I}(C_\bullet)$ and $L_I = \lim_i \mathbb{I}_i$. The stage duals have cohomology only in degrees 0, 1, while (RHTX/DIH) gives $H_q \mathfrak{D}_I = 0$ for $q > 1$. The length-one injective resolution $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ gives for every complex Y the two-column

universal-coefficient row $0 \rightarrow \text{Ext}^1(H_{q-1}Y, \mathbb{Z}) \rightarrow H^q(Y^\vee) \rightarrow \text{Hom}(H_qY, \mathbb{Z}) \rightarrow 0$. Thus (BDR) determines the full amplitude

$$[\text{IDS}] \quad \boxed{\begin{array}{l} H^q \mathfrak{L}_I^\vee = 0 \ (q \geq 3), \quad H^2 \mathfrak{L}_I^\vee \cong \text{Ext}^1(L_I, \mathbb{Z}), \quad H^1 \mathfrak{L}_I^\vee \cong \text{Birth}_{(-)^\vee, I}^1, \\ (\text{IDS}) \quad 0 \rightarrow \text{Birth}_{(-)^\vee, I}^0 \rightarrow H^0 \mathfrak{L}_I^\vee \rightarrow \text{Death}_{(-)^\vee, I}^1 \rightarrow 0, \quad 0 \rightarrow H^{-1}(\mathfrak{D}_I^\vee) \rightarrow H^{-1} \mathfrak{L}_I^\vee \rightarrow \text{Death}_{(-)^\vee, I}^0 \rightarrow 0, \\ H^{-m} \mathfrak{L}_I^\vee \cong H^{-m}(\mathfrak{D}_I^\vee) \quad (m \geq 2). \end{array}} \quad (23)$$

The negative target is itself generated by adjacent (DIH) layers, not by a new defect. For $m \geq 1$ set $A_m = \text{coker } t_{m-1}$, $K_m = \ker t_m$, $E_m = H_{-m} \mathfrak{D}_I$, let $\kappa_m \in \text{Ext}^1(K_m, A_m)$ be the class of $0 \rightarrow A_m \rightarrow E_m \rightarrow K_m \rightarrow 0$, and $\partial_m(\lambda) = \lambda_* \kappa_m$. Dualizing the rows for $m, m+1$ and then applying universal coefficients gives the canonical filtration

$$[\text{HDF}] \quad \boxed{(\text{HDF}) \quad 0 = G_{-1}^{(m)} \subset G_0^{(m)} \subset G_1^{(m)} \subset G_2^{(m)} \subset G_3^{(m)} = H^{-m}(\mathfrak{D}_I^\vee)} \quad (24)$$

with

$$G_0^{(m)} \cong \frac{\text{Ext}^1(K_{m+1}, \mathbb{Z})}{\text{im } \partial_{m+1}}, \quad G_1^{(m)} / G_0^{(m)} \cong \text{Ext}^1(A_{m+1}, \mathbb{Z}), \quad G_2^{(m)} / G_1^{(m)} \cong \text{Hom}(K_m, \mathbb{Z}), \quad G_3^{(m)} / G_2^{(m)} \cong \ker \partial_m.$$

For $m \geq 2$, (IDS) identifies this filtration with $H^{-m} \mathfrak{L}_I^\vee$; for $m = 1$ it is the canonical kernel in the second (IDS) row and the final quotient is Death^0 . At the abstract **Ab** interface, the split/generator extension witness of (AHR) applies to each such short exact row; hence the classes κ_m cannot in general be erased in favor of their endpoint groups. No claim is made that every abstract extension class is realized by an incidence tower.

On $I = \mathbb{N}$, (THP/DLX) kills history above degree 1, so (DIH) gives $H_{-1} \mathfrak{D}_\mathbb{N} \cong \text{Hist}_\mathbb{N}^1(\mathbb{J}_\bullet^{\text{bal}})$ and $H_{-m} = 0$ for $m \geq 2$. Hence the countable incidence residual is supported inside $-1 \leq q \leq 2$, with $H^2 \cong \text{Ext}^1(L, \mathbb{Z})$, $H^1 \cong \text{ObjBirth}_{\text{inc}}$, and

$$[\text{CDS}] \quad \boxed{(\text{CDS}) \quad \begin{array}{l} 0 \rightarrow \text{Birth}_{(-)^\vee, \mathbb{N}}^0 \rightarrow H^0 \text{DualLimInt}_{(-)^\vee} \rightarrow \text{Death}_{(-)^\vee, \mathbb{N}}^1 \rightarrow 0, \\ 0 \rightarrow \text{Hom}(\text{Hist}_\mathbb{N}^1(\mathbb{J}_\bullet^{\text{bal}}), \mathbb{Z}) \rightarrow H^{-1} \text{DualLimInt}_{(-)^\vee} \rightarrow \text{Death}_{(-)^\vee, \mathbb{N}}^0 \rightarrow 0. \end{array}} \quad (25)$$

Thus (OBF) is exactly the internal filtration of the $q = 1$ stratum, not a classification of the whole dual-continuity object; the $q = 0, -1$ seams retain disappearing stage objectives/characters, while higher directed history occupies the (HDF) tail. No direct-sum splitting is supplied.

Curvature transport of continuity. For fixed small directed I , write

$$\mathcal{R}_I(X) := \text{DualLimInt}_{(-)^\vee, I}(X) = \text{Cofib} \left[\text{hocolim}_{i \in I} X_i^\vee \longrightarrow (\mathbf{DRes}_I X)^\vee \right].$$

Both terms are exact contravariant functors of the diagram X , hence so is $\mathcal{R}_I$. The (DIH) two-row Postnikov triangle in $D(\mathbf{Ab}^{I^{\text{op}}})$, $\mathbb{I}_\bullet[1] \rightarrow \mathfrak{C}_\bullet^{\text{inc}} \rightarrow \mathbb{J}_\bullet^{\text{bal}} \xrightarrow{\kappa_{\text{inc}}} \mathbb{I}_\bullet[2]$, therefore yields the distinguished continuity–curvature triangle

$$[\text{CCT}] \quad \boxed{(\text{CCT}) \quad \mathcal{R}_I(\mathbb{I}_\bullet[2]) \xrightarrow{\mathcal{R}_I(\kappa_{\text{inc}})} \mathcal{R}_I(\mathbb{J}_\bullet^{\text{bal}}) \longrightarrow \mathcal{R}_I(\mathfrak{C}_\bullet^{\text{inc}}) \longrightarrow \mathcal{R}_I(\mathbb{I}_\bullet[1])}. \quad (26)$$

Thus the whole (IDS/HDF/CDS) spectrum is generated before taking cohomology by the two one-row continuity residuals plus the transported curvature map; the central $q = 0, -1$ seam is not a new residual primitive. If $\kappa_{\text{inc}} = 0$, the residual triangle splits (noncanonically); the converse is not asserted because $\mathcal{R}_I$ may kill a nonzero curvature class.

Put $R_I = \mathcal{R}_I(\mathfrak{C}_\bullet^{\text{inc}})$. Since $\mathbb{Z}$ has global dimension one, the Postnikov class of $\tau_{[-1,0]} R_I$ lies in $\text{Ext}_\mathbb{Z}^2(H^0 R_I, H^{-1} R_I) = 0$: there is no additional intrinsic cross-degree k -invariant. The load-bearing central data are instead the two canonical Yoneda classes of the adjacent (BDR) rows,

$$[\text{CSM}] \quad \boxed{(\text{CSM}) \quad \varepsilon_{-1} \in \text{Ext}^1(\text{Death}^0, H^{-1}(\mathfrak{D}_I^\vee)), \quad \varepsilon_0 \in \text{Ext}^1(\text{Death}^1, \text{Birth}^0),} \quad (27)$$

together with their common producer $\mathcal{R}_I(\kappa_{\text{inc}})$. Vanishing of Ext^2 gives only a noncanonical formality of the two-degree truncation; it does not split either BDR row. At the abstract **Ab** interface, the split/generator witness of (AHR) shows that fixed endpoint groups with different Ext^1 classes can have different middle groups, so neither ε_{-1} nor ε_0 may be erased. This witness is a sharpness boundary, not a claim that every abstract class is incidence-realizable.

Countable curvature exchange and visibility boundary. Now let $I = \mathbb{N}$ and abbreviate $\mathcal{R}_\mathbb{I} = \mathcal{R}_\mathbb{N}(\mathbb{I}_\bullet)$, $\mathcal{R}_\mathbb{J} = \mathcal{R}_\mathbb{N}(\mathbb{J}_\bullet^{\text{bal}})$ and $\mathcal{R}_{\text{inc}} = \mathcal{R}_\mathbb{N}(\mathfrak{C}_\bullet^{\text{inc}})$. The one-row (BDR/THP) amplitudes lie in $[-1, 1]$. Hence the only ordinary cohomology map of the shifted curvature transport on the overlap is

$$\chi_\kappa := H^1 \mathcal{R}_\mathbb{N}(\kappa_{\text{inc}}) : H^1 \mathcal{R}_\mathbb{N}(\mathbb{I}_\bullet[2]) \cong H^{-1} \mathcal{R}_\mathbb{I} \longrightarrow H^1 \mathcal{R}_\mathbb{J}.$$

The long exact sequence of (CCT) therefore reduces to the exact exchange law

$$[\text{CEX}] \quad \boxed{(\text{CEX}) \quad \begin{array}{l} H^{-1} \mathcal{R}_{\text{inc}} \cong H^{-1} \mathcal{R}_\mathbb{J}, \quad 0 \rightarrow H^0 \mathcal{R}_\mathbb{J} \rightarrow H^0 \mathcal{R}_{\text{inc}} \rightarrow \ker \chi_\kappa \rightarrow 0, \\ 0 \rightarrow \text{coker } \chi_\kappa \rightarrow \text{ObjBirth}_{\text{inc}} \rightarrow H^0 \mathcal{R}_\mathbb{I} \rightarrow 0, \quad H^2 \mathcal{R}_{\text{inc}} \cong H^1 \mathcal{R}_\mathbb{I}. \end{array}} \quad (28)$$

Thus the same visible curvature map sends its kernel to the lower seam and its cokernel to objective birth; the two displayed extensions remain unsplit and retain the (CSM) Yoneda data.

This also regenerates (OBF) from one-row continuity. Let $\sigma_\mathbb{I} : \lim_n \text{Hom}(\mathbb{I}_n, \mathbb{Z}) \rightarrow H^0((\mathbf{DRes}_\mathbb{N} \mathbb{I}_\bullet)^\vee)$ and let $\sigma_\mathbb{I} : \ker h \rightarrow \text{Ext}^1(H, \mathbb{Z})$ be the Snake boundary for the target UCT row, where $H = \text{Hist}^1(\mathbb{I}_\bullet)$ and h is the map in (ICG). Then

$$0 \rightarrow \text{Death}_\mathbb{I}^0 \rightarrow \ker h \xrightarrow{\sigma_\mathbb{I}} \text{Ext}^1(H, \mathbb{Z}) \rightarrow H^0 \mathcal{R}_\mathbb{I} \rightarrow \text{coker } h \rightarrow 0.$$

For a countable tower, (THP) gives $\text{Hist}_\mathbb{N}^2(\mathbb{I}_\bullet) = 0$, hence $t_0 = 0$ and $K = \lim_n \mathbb{J}_n^{\text{bal}}$; therefore $H^1 \mathcal{R}_\mathbb{J} = \text{coker } a$. Naturality of the UCT/Snake 3×3 diagram with (CCT) gives $\text{im}(\rho \tau_{\text{obj}}) = \text{im } \sigma_\mathbb{I}$ (the connecting maps may differ by the standard triangle sign). If $\bar{j} : \text{coker } a \rightarrow \text{coker } e$ is induced by the AHR pullback j , then the restriction of χ_κ to the birth subobject $\text{Hom}(H, \mathbb{Z}) \hookrightarrow H^{-1} \mathcal{R}_\mathbb{I}$ has, up to the same harmless sign convention, the class of ∂_{κ_0} modulo $\text{im } a$ and hence image $\ker \bar{j}$. Thus $\bar{j} \chi_\kappa$ annihilates that birth subobject and descends through the BDR

quotient $H^{-1}\mathcal{R}_I \twoheadrightarrow \text{Death}_I^0 = \ker \sigma_I$; by the same natural Snake diagram its image equals that of $\tau_{\text{obj}}|_{\text{Death}_I^0}$, namely R_0 . Consequently

$$\text{coker } \chi_\kappa \cong A_0/R_0 = F_0, \quad \text{im } \sigma_I = R_1,$$

and $H^0\mathcal{R}_I$ has the three successive grades $\text{Ext}^1(H, \mathbb{Z})/R_1$, S , and $\text{Hom}(L, \mathbb{Z})/\text{Hom}_{\text{cts}}(L, \mathbb{Z})$ by (ICG). Pulling this filtration back along the third (CEX) row gives exactly

$$[\text{OBF}] \quad \boxed{(\text{OBF}) \quad 0 = F_{-1} \subset F_0 \subset F_1 \subset F_2 \subset F_3 = \text{ObjBirth}_{\text{inc}}} \quad (29)$$

with $F_0 \cong A_0/R_0$, $F_1/F_0 \cong \text{Ext}^1(H, \mathbb{Z})/R_1$, $F_2/F_1 \cong S$ and $F_3/F_2 \cong \text{Hom}(L, \mathbb{Z})/\text{Hom}_{\text{cts}}(L, \mathbb{Z})$. Thus OBF is an exact descendant of curvature exchange, not an independent objective primitive. The cross-map remains load-bearing: at the abstract **Ab** interface, with $\ker h = \text{coker } e = \mathbb{Z}$ and $\text{coker } h = 0$, the choices $\tau_{\text{obj}} = \times 2$ and $\times 3$ give objective cokernels $\mathbb{Z}/2$ and $\mathbb{Z}/3$. This is a sharpness witness, not an incidence-realizability claim.

The transport is not faithful to source curvature. If I is finite, finite exactness makes $\mathcal{R}_I(X) = 0$ for every derived-**Ab** diagram X ; the same holds when I admits a finite homotopy-cofinal subcategory, by derived cofinality. Hence $\mathcal{R}_I(\kappa) = 0$ for every curvature class on either perimeter. Nevertheless the finite two-stage (TCP) MIB face has $\text{Ext}_{\mathcal{A}}^2((J \rightarrow 0), (0 \rightarrow I)) \cong \text{Ext}_{\mathbb{Z}}^1(J, I)$ and realizes $\kappa_{\text{inc}} \neq 0$ for, e.g., the generator extension with $I = \mathbb{Z}$, $J = \mathbb{Z}/n$, $n \geq 2$. Therefore

$$[\text{CVB}] \quad \boxed{(\text{CVB}) \quad \kappa_{\text{inc}} = 0 \implies \mathcal{R}_I(\kappa_{\text{inc}}) = 0, \quad \mathcal{R}_I(\kappa_{\text{inc}}) = 0 \not\implies \kappa_{\text{inc}} = 0.} \quad (30)$$

Stage curvature and completion-visible curvature are distinct typed coordinates; genuine visibility requires nontrivial cofinal continuity, not merely nonzero curvature.

Hereditary visibility fibre. Keep $I = \mathbb{N}$ and set

$$A := \mathcal{R}_{\mathbb{N}}(\mathbb{I}_{\bullet}[2]), \quad B := \mathcal{R}_{\mathbb{J}}, \quad U := H^1 A = H^{-1}\mathcal{R}_I, \quad V := H^2 A = H^0\mathcal{R}_I, \quad P := H^0 B, \quad Q := H^1 B.$$

The remaining cohomology of A and B lies outside equal or adjacent degrees. Since $\mathbb{Z}$ has global dimension one, the Postnikov filtration of bounded complexes gives the natural hereditary morphism row

$$[\text{HVS}] \quad \boxed{(\text{HVS}) \quad 0 \rightarrow \text{Ext}^1(U, P) \oplus \text{Ext}^1(V, Q) \rightarrow \text{Hom}_{D(\text{Ab})}(A, B) \xrightarrow{H^1} \text{Hom}(U, Q) \rightarrow 0.} \quad (31)$$

Indeed, filter A by its Postnikov truncations and apply $\text{Hom}_{D(\text{Ab})}(-, B)$. On each layer $H^q A[-q]$ only $\text{Hom}(H^q A, H^q B)$ and $\text{Ext}^1(H^q A, H^{q-1} B)$ can contribute; all higher extension terms vanish. In total degree zero the only surviving diagonal term is $\text{Hom}(U, Q)$ and the only adjacent terms are the two displayed Ext^1 groups. Thus no choice of objectwise splitting is used. In particular the fibre over a fixed ordinary overlap map is a torsor under $\text{OffVis} := \text{Ext}^1(U, P) \oplus \text{Ext}^1(V, Q)$, not a canonically based product.

Let $\text{Curv}_{\mathbb{N}} := \text{Hom}_{D(\text{Ab})^{\text{nop}}}(\mathbb{J}_{\bullet}^{\text{bal}}, \mathbb{I}_{\bullet}[2])$. Exact contravariance gives a homomorphism $\nu_{\mathcal{R}} : \text{Curv}_{\mathbb{N}} \rightarrow \text{Hom}_{D(\text{Ab})}(A, B)$, $\kappa \mapsto \mathcal{R}_{\mathbb{N}}(\kappa)$, and put $\chi := H^1 \nu_{\mathcal{R}}$. On $\ker \chi$, (HVS) canonically identifies $\nu_{\mathcal{R}}(\kappa)$ with a secondary class $\omega_{\mathcal{R}}(\kappa) \in \text{OffVis}$. Put $\text{SecVis}_{\mathbb{N}} := \text{im } \omega_{\mathcal{R}}$. Then

$$[\text{PVS}] \quad \boxed{(\text{PVS}) \quad 0 \rightarrow \ker \nu_{\mathcal{R}} \rightarrow \ker \chi \xrightarrow{\omega_{\mathcal{R}}} \text{SecVis}_{\mathbb{N}} \rightarrow 0, \quad \text{SecVis}_{\mathbb{N}} \hookrightarrow \text{OffVis},} \quad (32)$$

so $\text{SecVis}_{\mathbb{N}} \cong \ker \chi / \ker \nu_{\mathcal{R}}$ is the canonical quotient of source curvature invisible to the primary overlap but visible to the full derived transport, and $\mathcal{R}_{\mathbb{N}}(\kappa) = 0$ iff $\chi(\kappa) = 0$ and $\omega_{\mathcal{R}}(\kappa) = 0$. For the incidence class κ_{inc} , $\chi(\kappa_{\text{inc}}) = \chi_{\kappa}$. If $\chi_{\kappa} = 0$, the two components of $\omega_{\mathcal{R}}(\kappa_{\text{inc}})$ are, up to the standard triangle signs, exactly the Yoneda classes of the two CEX rows $0 \rightarrow P \rightarrow H^0\mathcal{R}_{\text{inc}} \rightarrow U \rightarrow 0$ and $0 \rightarrow Q \rightarrow \text{ObjBirth}_{\text{inc}} \rightarrow V \rightarrow 0$. Thus on the primary-zero locus the full transported curvature vanishes exactly when both exchange rows split. This is a zero-locus theorem, not a canonical decomposition at nonzero χ_{κ} .

The secondary fibre is genuinely necessary at the hereditary target interface: take complexes A_0, B_0 with $H^1 A_0 = \mathbb{Z}/n$, $H^0 B_0 = \mathbb{Z}$ and all other cohomology zero. Then the ordinary overlap group is zero but (HVS) gives $\text{Hom}_{D(\text{Ab})}(A_0, B_0) \cong \text{Ext}^1(\mathbb{Z}/n, \mathbb{Z}) \cong \mathbb{Z}/n$. A generator therefore has zero ordinary cohomology map and nonzero derived morphism, whose cone carries the nonsplit extension $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n \rightarrow 0$. This is an abstract hereditary-interface refuter to any universal replacement of the full transported morphism by χ_{κ} ; no claim is made here that this particular generator is realized by an incidence tower.

For a countable tower of finite-rank free A_n , set $V_n = A_n \otimes \mathbb{R}$, $T_n = A_n \otimes \mathbb{T}$, $A_{\infty} = \varprojlim A_n$, $V_{\infty} = \varprojlim V_n$ and $\text{ProPh}(A_{\bullet}) = \varprojlim T_n$. Images in each finite-dimensional V_n stabilize, hence $\text{Hist}_{\mathbb{N}}^1(V_{\bullet}) = 0$; the coefficient row gives

$$[\text{PPH}] \quad \boxed{(\text{PPH}) \quad 0 \rightarrow A_{\infty} \rightarrow V_{\infty} \xrightarrow{\exp_{\infty}} \text{ProPh}(A_{\bullet}) \rightarrow \text{Hist}_{\mathbb{N}}^1(A_{\bullet}) \rightarrow 0.} \quad (33)$$

Every based path in $\varprojlim T_n$ lifts uniquely and compatibly to $\varprojlim V_n$, while $\ker \exp_{\infty} = A_{\infty}$; hence, algebraically, $\text{ProPh}(A_{\bullet})/\text{ProPh}(A_{\bullet})^{\text{path}, 0} \cong \text{Hist}_{\mathbb{N}}^1(A_{\bullet})$. The path component need not be closed. For (IHRA) $A^{(2)} = (\mathbb{Z} \xleftarrow{2} \mathbb{Z} \leftarrow \dots)$, (PPH/FDE) is $0 \rightarrow \mathbb{R} \rightarrow \Sigma_2 \rightarrow \mathbb{Z}_2^{\text{ad}}/\mathbb{Z} \rightarrow 0$, $\Sigma_2 = \varprojlim (\mathbb{T} \xleftarrow{2} \mathbb{T} \leftarrow \dots)$. Surjective circle bonds make Σ_2 connected; its $\mathbb{R}$ path component is dense because it surjects to every finite coordinate, yet the quotient is nonzero. Thus already this witness carries history inside one connected component, beyond finite Lie direction and ordinary π_0 .

The full derived phase is finer than (PPH). For $C_n = [\mathbb{Z} \xrightarrow{2^n} \mathbb{Z}]_{1,0}$ with bond $(\times 2, 1)$, $H_1 C_n = 0$, $H_0 C_n = \mathbb{Z}/2^n$, while (DPH) gives $H_1 \text{Ph}(C_n) = \mathbb{Z}/2^n$ with reduction bonds; hence $H_1 \mathbf{DRes}_{\mathbb{N}} \text{Ph}(C_{\bullet}) = \mathbb{Z}_2^{\text{ad}}$. But (DLX) gives $\mathbf{DRes} C_{\bullet} \simeq \mathbb{Z}_2^{\text{ad}}[0]$ and $H_1 \text{Ph}(\mathbb{Z}_2^{\text{ad}}[0]) = \text{Tor}_1^{\mathbb{Z}}(\mathbb{Z}_2^{\text{ad}}, \mathbb{T}) = 0$ since $\mathbb{Z}_2^{\text{ad}}$ is torsion-free and therefore $\mathbb{Z}$ -flat. Thus $\text{coker } H_1 \chi_{\mathbb{T}} = \mathbb{Z}_2^{\text{ad}}$: (CHI) may be nonzero even when the (PPH) interaction lattice vanishes stagewise. For incidence towers, finite (DPH/AIPG) and countable (DOH/DIH) can therefore be identified only after the additional zero condition (CHI).

Cubical-history Fubini. Let C be in the (UDR/RHTX) perimeter and let $X : \mathcal{P}(F) \rightarrow \mathbf{Ch}(\mathbf{Ab}^{C^{\text{op}}})$. Taking $\Phi = \mathbf{DRes}_C$ in (CSE) and then (RHTX) gives

$$[\text{CRH}] \quad \begin{aligned} & \mathbf{DRes}_C \text{CubRes}_F(X) \simeq \text{CubRes}_F(A \mapsto \mathbf{DRes}_C X_A), \\ & u : C \rightarrow \mathcal{D} : \mathbf{DRes}_C \text{CubRes}_F(X) \simeq \mathbf{DRes}_{\mathcal{D}} \text{CubRes}_F(A \mapsto \text{DKan}_u X_A), \\ & E_2^{r,q} = \text{Hist}_C^r(H_q \text{CubRes}_F(X)) \implies H_{q-r}(\text{CubRes}_F(A \mapsto \mathbf{DRes}_C X_A)). \end{aligned} \quad (34)$$

The middle line combines (CSE) for the exact functor $\mathbf{DKan}_u$ with (DKF); $u = \pi_c$ is (RFSS). It asserts no Beck–Chevalley law for a varying-base square: mate/interchange failures remain (KCM/BDI), and only independently supplied chain-level comparison cubes enter (CRM). For a C -diagram of complementary (DCHD/CRT) cubes, write $\mathbf{DInt}_p(F)$ for the resulting C^{op} -diagram of pointwise homology groups. Its $q = p - |F| + 1$ row is $E_2^{r,p-|F|+1} \cong \text{Hist}_C^r(\mathbf{DInt}_p(F))$, with total degree $p - |F| + 1 - r$. On each arity- s face A , with $Y_B^A = \mathbb{K}(\widehat{q}_{A \setminus B})[0]$, restoring the DCHD grading gives $\mathbf{D}_\bullet(A) \simeq \text{CubRes}_A(Y^A)[s-1]$; on the full-product face the sole homology is P_F^s/P_F^{s+1} in degree $s-1$, so (RDXT) collapses exactly to (AHT). Thus cubical interaction grade and derived history are the two filtrations of one total object; no equality of the filtrations or splitting is asserted.

For $J = *$, ι is bijective and (GTS) reduces to (RPL/DRX); for constant a , $\widehat{\pi}$ is bijective and it reduces to PFR/SCX. If directed $K \subseteq J$ is cofinal, restriction identifies $\widehat{L}$, $\widehat{\mathbf{JKey}}$, CofKey and the whole (GTS) row with their K -versions. Strictness is witnessed by $D_n = \{0, \dots, n\}$ with truncation bonds, $\Omega = \mathbb{N} \cup \{\infty\}$, $q_n(k) = \min(k, n)$, $q_n(\infty) = n$, $a(k) = 1$, $a(\infty) = 0$: every profile lands and $\pi_{J,a}$ is bijective, but $\text{StabRes}_{J,a}(\infty) = \{0, 1\}$ and $(\infty, 1) \in \text{JointLimRes}_{J,a}$, so ∂ is nonzero although CRX vanishes. Removing ∞ and taking $a(k) = k$ instead gives the independent terminal term $\widehat{L}_J \setminus \text{im } \widehat{\pi}_{J,a} = \{\infty\}$.

For a small abelian category $\mathcal{A}$, $A, B \in \mathcal{A}$ and finite $r \geq 1$, let $\mathbf{YExt}_r(B, A)$ be the Yoneda category of exact r -extensions $0 \rightarrow A \rightarrow E_{r-1} \rightarrow \dots \rightarrow E_0 \rightarrow B \rightarrow 0$ with endpoint-fixed morphisms, and for κ in the Yoneda group $\text{Ext}_{\mathcal{A}}^r(B, A)$ let $\mathbf{YRep}_r(\kappa) \subseteq \mathbf{YExt}_r(B, A)$ be the full subcategory on extensions of Yoneda class κ . For $E \in \mathbf{YRep}_r(\kappa)$, Example 3.2, Theorem 5.2, Proposition 6.1 and Theorem 7.5 give $\pi_0 \mathbf{BYExt}_r \cong \text{Ext}_{\mathcal{A}}^r(B, A)$, $\Omega(\mathbf{BYRep}_r(\kappa), E) \simeq \mathbf{BYExt}_{r-1}$ and $\pi_i \cong \text{Ext}_{\mathcal{A}}^{r-i}(B, A)$ ($1 \leq i \leq r$)⁹ (Ex. 3.2; Thm. 5.2; Prop. 6.1; Thm. 7.5); no large-category passage is used, and Baer translation moves between components without choosing a representative. Hence the split $r = 1$ Hom-torsor does not disappear when the obstruction class vanishes. Filtered realization is the (GLT) carrier/law fibration below: (KGM/YRM) supplies extension-gluing coordinates, while (FRP) supplies the split-carrier differential-law fibre. The (KWF) [2] witness independently varies factorization base; if $B_\alpha \neq 0$, (NFP) keeps every finite raw factorization order nonterminal. Thus (KAE/KRC/KDR/KRW/KOF/KWF/KTR/KGM) is the derived (URE/UAR_±) face with these axes typed separately.

For a countable tower of countable abelian groups $A_0 \leftarrow A_1 \leftarrow \dots$, $I_{n,m} = \text{im}(A_m \rightarrow A_n)$, retain

$$[\text{MPF}] \quad q_{n,m} : A_n \twoheadrightarrow \text{Per}_{n,m}(A) := A_n/I_{n,m}, \quad 0 \rightarrow I_{n,m}/I_{n,\ell} \rightarrow \text{Per}_{n,\ell}(A) \rightarrow \text{Per}_{n,m}(A) \rightarrow 0 \quad (\ell \geq m \geq n). \quad (35)$$

Mittag–Leffler is exactly eventual constancy of these marked quotients for each n , hence $\text{Hist}_{\mathbb{N}}^1 = 0$; for countable A_n the converse is Geoghegan¹⁰. Off ML, the nonstationary image tower and (THP/DLX,FDE) give continuum many branches, hence the 1 versus $2^{\aleph_0}$ dichotomy on this countable abelian face; no converse is asserted for arbitrary groups. Decreasing inclusions give (DIZ/FDE,VST), and $\varprojlim C_n \simeq \text{Fib}(1 - T : \prod C_n \rightarrow \prod C_n)$ gives Milnor/(DLX). For composable fibrations U_r ,

$$[\text{RGS}] \quad \text{ResFib}(U_{[n]})(e_0) \simeq \int_{e_1 \in \text{ResFib}(U_1)(e_0)} \dots \int_{e_{n-1} \in \text{ResFib}(U_{n-1})(e_{n-2})} \text{ResFib}(U_n)(e_{n-1}). \quad (36)$$

Fubini makes every block coarsening equivalent. On corestricted discrete Set, for $p : E \twoheadrightarrow B$, $a : E \rightarrow A$, put $K_{a|p}(b) := a(p^{-1}b)$. Reindexing gives

$$(\text{RGS}_{\text{desc}}) \quad \text{Tot}_{e \in E} H(ae) \cong \text{Tot}_{b \in B} \text{Tot}_{c \in K_{a|p}(b)} ((p, a)^{-1}(b, c) \times H(c)).$$

for $H : A \rightarrow \mathbf{Set}$. Its zero locus/coarsest refinement is (RPL/RKD); quotient/gauge readouts retain it. Let $\mathcal{E}_m \xrightarrow{q_m} \dots \xrightarrow{q_1} \mathcal{E}_0$ be a finite tower of bounded cloven groupoid fibrations, $\Theta_r = \text{ResFib}(q_r)$. Repeated (CRF/RFE) gives

$$\mathcal{E}_m \simeq \int_{x_0 \in \mathcal{E}_0} \int_{x_1 \in \Theta_1(x_0)} \dots \int_{x_{m-1} \in \Theta_{m-1}(x_{m-2})} \Theta_m(x_{m-1}).$$

After a base autoequivalence g_0 , each lift stage is a coherent local-system equivalence $\gamma_r : \Theta_r \simeq \widetilde{\gamma}_{r-1}^* \Theta_r$, and Grothendieck–Fubini gives

$$\text{Aut}_{q_*}(\mathcal{E}_m) \simeq \int_{g_0 \in \text{Aut}(\mathcal{E}_0)} \int_{\gamma_1 \in \text{Eq}(\Theta_1, g_0^* \Theta_1)} \dots \text{Eq}(\Theta_m, \widetilde{\gamma}_{m-1}^* \Theta_m). \quad (\text{IGRT})$$

Thus truncation erases the next lift level exactly when its displayed equivalence fibre is terminal; no extra decomposition primitive is used. A terminal leg cancels, while obstruction vanishing need not: $\text{Ext}^1(\mathbb{Z}, \mathbb{Z}) = 0$ but the split extension retains $\text{Hom}(\mathbb{Z}, \mathbb{Z})$ isotropy. For a groupoid functor F , $\text{HFib}_b(F) \cong (\mathbb{T}_{\mathcal{B}}F)_b$ and (HRS) F is an equivalence iff all these fibres are terminal; (HRF) for $\mathcal{E} \xrightarrow{u} \mathcal{B} \xrightarrow{v} \mathcal{C}$ is $\text{HFib}_c(VU) \simeq \int_{(b, \beta) \in \text{HFib}_b(V)} \text{HFib}_b(U)$. Put $\mathfrak{a}(K) = \text{Cone}(C_\bullet(K; \mathbb{Z}) \rightarrow \mathbb{Z}[0])$ and $\mathfrak{R}_\bullet(u) = [A \xrightarrow{u} B]_{1,0}$.

[HRM] $\mathfrak{H}\mathfrak{R}_b(F) := \mathfrak{a}(\text{NH}\mathfrak{Fib}_b(F))$. For corestricted $a : \mathcal{G} \rightarrow S_{\text{disc}}$,

$$[\text{HCS}] \quad 0 \rightarrow \widetilde{\mathbb{Z}}^{(S)}[0] \rightarrow \bigoplus_s \mathfrak{a}(N\mathcal{G}_s) \rightarrow \mathfrak{a}(N\mathcal{G}) \rightarrow 0,$$

$$[\text{HRD}] \quad \bigoplus_y \mathfrak{H}\mathfrak{R}_y(f) \cong \mathfrak{R}_\bullet(f).$$

Transport is irreducible: over BC_2 with fibre $B\mathbb{Z}$, trivial/inversion monodromy give $H_1 = \mathbb{Z} \oplus C_2$ versus $C_2 \oplus C_2$. HRS implies HRM-acyclicity, not conversely:¹¹ (Cor. 5.6) gives nontrivial acyclic A with $BA \rightarrow *$ HRM-acyclic but nonterminal. Hence (HFF,KSP/RST/RPL/DRX,RFE–RGS/NRF) are exact shadows. *Homotopy comparison firewall*. For every small category $\mathcal{A}$, $\mathcal{A} \simeq \mathbf{1} \Rightarrow N\mathcal{A} \simeq * \Rightarrow \mathfrak{a}(N\mathcal{A}) \simeq 0 \Rightarrow \mathcal{A} \neq \emptyset$; all converses are strict by [1], the nontrivial acyclic group A above with $\pi_1 N(BA) = A$, and $\mathbf{2}_{\text{disc}}$, respectively. Since every (RCS) interval $[a, b]$ has initial object a and terminal object b , every reduced weak-equivalence-invariant E with $E(*) \simeq 0$ has $E(N\text{CmpRes}_p(a, b)) \simeq 0$ for all $a \leq b$, although $\text{CmpRes}_p(a, b) \simeq \mathbf{1} \iff a = b$. Thus $[0, 0]$ versus $[0, 1]$ is an equal-shadow/different-target witness: (RCS/ICP) terminality does not factor through homotopy type, reduced chains or any reduced stable shadow. Any (CSE/DANF) bridge must retain pre-homotopy incidence data; (CFR) is the supplied-cell carrier and the interval its posetal $n = 2$ face. For decreasing onto $q_j : E_j \twoheadrightarrow Y$, $K_j \# \mathfrak{R}_{\mathfrak{K}q_j} q_j \# \mathfrak{H}\mathfrak{R}_J(q_\bullet) := \text{DW}_J(\mathfrak{R}_\bullet(q_\bullet)) \simeq \varprojlim_j K_j[1]$. For finite P with top, $X : P^{\text{op}} \rightarrow \mathbf{Set}$, $d_p : X_\top \rightarrow X_p$, readout t ,

$$[\text{RDA}] \quad \text{Det}_P(t) = \{p : R_{d_p} \subseteq R_t\} = \uparrow \text{Front}_P(t), \quad \text{Front}_P(t) = \text{Min}_P \text{Det}_P(t), \quad \text{Front}_P(\mathcal{T}) = \text{Min}_P \bigcap_{t \in \mathcal{T}} \text{Det}_P(t). \quad (37)$$

By (DCC), membership is exactly $t = \widehat{t}_p d_p$; failure is equal- d_p /different- t . On a chain $D_k = U_{k+1} \cdots U_n$, $D_n = I$,

$$[\text{RDT}] \quad \text{rdepth}_{\mathbf{U}}(t) = \min\{k : R_{d_k} \subseteq R_t\}, \quad \text{rdepth}_{\mathbf{U}}(t) \leq k \iff t = \widehat{t}_k d_k \text{ on } \text{ran } d_k. \quad (38)$$

At $k > 0$ the reachable (RFE) fibre survives; the diamond $\perp < a, b < \top$ with $X_{\top} = \{(0, 0), (1, 1)\}$ and $t = d_a = d_b$ has front $\{a, b\}$, so depth is intrinsically antichain-valued. *Groupoid determination and unbounded residual depth.* For $d : \mathcal{E} \rightarrow \mathcal{B}$, $T : \mathcal{E} \rightarrow \mathcal{Y}$, let $\text{Fac}_d(T) = \text{HFib}_T(d^*)$ and $T_d = (I_{\mathcal{E}}, d)$. (HFC) d is an equivalence iff $\pi_0 \text{Fac}_d(T_d) = *$ (a point gives $Sd \simeq I$, connectedness $dS \simeq I$; HRS gives the converse). For q in Gpd , with image core $\bar{q}$ and off-image $\mathcal{B}_q^{\perp}$,

$$[\text{IFS}] \quad \text{Fac}_q(T) \simeq \text{Fac}_{\bar{q}}(T) \times \text{Fun}(\mathcal{B}_q^{\perp}, \mathcal{Y})^{\simeq}. \quad (39)$$

On Set this is empty unless $R_q \subseteq R_T$, otherwise $\mathcal{Y}^{\mathcal{B} \setminus q(\mathcal{E})}$. For finite-poset groupoid diagrams, (HRT) gives $\text{HDet}_P(T) = \uparrow \text{Min HDet}_P(T)$; (FRT/YRR-/URT/RGQ) totalizes it as $\pi_{P,T} : \text{Tot}_{P^{\text{op}}} \text{Fac}_{\bullet}(T)$ with support/front exactly $\text{HDet}_P(T)$ and its minima. The RDA diamond forbids a canonical single next access. For directed refinement $d_i = p_{ji} d_j$, at a declared limit (LFR) $\text{LimFacRes}_{\lambda}(T) := \mathbb{T}_{\text{Fac}_{d_{\lambda}}(T)} b_{\lambda} = \text{Tab}(b_{\lambda}^*)$, $b_{\lambda} : 2\text{colim}_{i < \lambda} \text{Fac}_{d_i}(T) \rightarrow \text{Fac}_{d_{\lambda}}(T)$, and by (RCS/RFD/HRS/IFS) it is trivial iff b_{λ} is an equivalence. On (EPN/KGC), $b_A : \text{disc } C_{\text{fin}}(A) \hookrightarrow \text{disc } C_{\infty}(A)$ has missing support (SDF) $C_{\infty}(A) \setminus C_{\text{fin}}(A)$; (RSE/SRH) puts every finitary marked-path readout in $C_{\text{fin}}(A)$, so strict (FTL) is limit-born and becomes one (RTO/RRR) residual edge. PFR is inverse-limit landing variance, not a duality. The $\mathbf{1} \rightarrow \mathbf{2}_{\text{disc}}$, $BC_2 \rightarrow B1$, $B1 \rightarrow BC_2$ examples separate support, image core and isotropy; 1-saturation kills only emptiness, while higher-history examples retain unbounded history. *Two-sided Yoneda representation recursion.* For $d : \mathcal{E} \rightarrow \mathcal{B}$, small $\mathcal{Y}$, (YRR-) $\text{RepFib}_{\mathcal{Y}}(d) = \text{Tab}((\text{pre}_d)^*) \simeq (\text{Fun}(\mathcal{E}, \mathcal{Y}) \downarrow \text{pre}_d)$; its fibre at T is $(S, \alpha : T \Rightarrow Sd)$ and its invertible-cell core $\text{Fac}_d(T)$. Dually, for $p : \mathcal{P} \rightarrow I$, (YRR-) is $\text{Tab}((\text{post}_p)^*)$, with homotopy-lift iso locus and noninvertible cells retained. (YFF/LYR/KZP) gives access/pointwise/coherent phases and $\text{pre}_q \dashv \text{Ran}_q$ the (YMR/KCM/RPG) mate face. With $f = \text{pre}_q$, $g = \text{post}_p$, put (YRP) $\text{YPol}_{q,p}^{\rightarrow} = g^* \circ f_*$, $\text{YPol}_{q,p}^{\leftarrow} = f^* \circ g_*$; (RPE) tabulates $(f \downarrow g)$ and $(g \downarrow f)$, with equal iso-comma locus but possibly distinct wings ($\mathbf{1} \rightarrow [1]$ at 0 versus 1 is terminal/empty). For wing cell $\beta : z \rightarrow z'$ and $\delta_n : [1] \rightarrow [n]$, $(\text{pre}_{\delta_n})^{-1}\{\beta\} \cong \text{Fact}^{[n]}(\beta) \cong (\mu_{q,n}^{[n]})^{-1}\{\beta\}$; (NFP/RCS) keeps noninvertible depth, while $BC_2 \rightarrow B1$, the RDA diamond and terminal-moduli/distinct- N_{q_i} fixtures keep kernel, selector and profile coordinates separate. *Source-access coslice.* (SAC) For fixed small $\mathcal{E}$, let $\text{Acc}_{< \kappa}(\mathcal{E})$ be the bounded replete access $(2, 1)$ -coslice with product-closed targets containing $\mathcal{E}$, and $\text{Acc}(d, T) := \text{Fac}_d(T)$. Postcomposition composes, $I_{\mathcal{E}}$ is bi-initial and $\langle d, k \rangle$ product; hence $\text{support}^{\text{op}}$ has $[\langle d, k \rangle] = [d] \vee [k]$, $\top = [I_{\mathcal{E}}]$, and $[d] = \top \iff \text{Fac}_d(I_{\mathcal{E}}) \neq \emptyset \iff (\forall T \in \text{Acc}) \text{Fac}_d(T) \neq \emptyset$. (DCC/RKD/IFS) are its Set/groupoid shadows. *Typed residual totality and strict height.* For small oriented Rld_{res} with reindexings u^* and declared carrier types, let $\mathbf{R} : \text{Rld}_{\text{res}}^{\text{op}} \rightsquigarrow \mathbf{Cat}$. Then

$$[\text{RTO}] \quad \pi_{\mathbf{R}} : \text{RTot}_{\text{res}}(\mathbf{R}) := \int \mathbf{R} \rightarrow \text{Rld}_{\text{res}}, \quad [\text{RGC}] \quad \text{Circ}(\pi_{\mathbf{R}}) \simeq_{\text{Rld}_{\text{res}}} \text{RTot}_{\text{res}}(\mathbf{R}), \quad (40)$$

$$[\text{RGQ}] \quad \text{PsFun}(\text{Rld}_{\text{res}}^{\text{op}}, \mathbf{Cat}) \simeq_{\text{bi}} \mathbf{Fib}_{\text{cl}}(\text{Rld}_{\text{res}}), \quad [\text{RVN}] \quad \text{Hom}_{\int \mathbf{R}}((i, x), (j, y)) \cong \coprod_{u:i \rightarrow j} \text{Hom}_{\mathbf{R}(i)}(x, u^* y). \quad (41)$$

(RKNF) In RVN, u is re-entry and $x \rightarrow u^* y$ vertical repair; (TKN) no total key exists iff every summand is empty, so $u = 1$, failure is same-base only. (AKE/IAK) For bounded $q : \mathcal{E} \rightarrow \mathcal{B}$, $C_q = \mathbb{T}_{\mathcal{B}q} = \text{Tab}(q^*)$ is a (YFF/RFD) descendant and (SCIA) gives $\text{AKE}_q(Q; a, b) = \{f : a \rightarrow b \text{ in } C_q : f \in Q\}$. (PMA/GDM) generates the replete law groupoid of κ -bounded functor packages; after the (DMR) hom-powerclass horizon, (RFD/SCIA) makes $(q, a, b, Q) \mapsto \text{disc AKE}_q(Q; a, b)$ pseudofunctorial. Its (URT) totalization $\text{IAK}_{R, \kappa, \lambda}$ has exactly these fibres; (RVN, EDO/ROA) gives re-entry/repair/occurrence, and the $q = \pi_{\mathbf{R}}$ unit-image face is GAK/KGS with (SKR, DMR/GSRH, BFH/CRFI) read-outs. Realized q, Q /key remain attachments. *Profiled residual Yoneda moduli.* For cloven $p : \mathcal{E} \rightarrow \mathcal{B}$, $\pi : \mathcal{P} \rightarrow I$, small $Q = \text{Prof}_{\pi}(\mathcal{B})$, Grothendieck correspondence/currying gives fibres $\text{Cart}_{\mathcal{B}}(p, p^* \pi)$ and cartesian sections $\text{Cart}_{\mathcal{B} \times Q}(p \times I_Q, \text{Ev}_{\pi, \mathcal{B}})$. Target adjunctions act fibrewise; for split π , (YRR-/FRF) gives $\eta_q^* : \text{Mod}_{\pi}(\mathbb{T}_{\mathcal{B}q}) \simeq \mathbf{Cat}_{\text{Prof}}^{[1]}(q, \pi)$. For $\pi = \pi_{\mathbf{R}}$ this is $\Pi_q^{\mathbf{R}}$, distinct from the q -leg (terminal grammar $q_i : \mathbf{1} \rightarrow \mathbf{2}_{\text{disc}}$ has equal terminal moduli but distinct N_{q_i}), and

$$[\text{RMO}] \quad \text{Mod}_{\pi_{\mathbf{R}}}(\mathbb{T}_{\mathcal{B}q}) \simeq \text{Prof}_{\mathbf{R}}(\mathcal{B}) \times_{\text{Fun}(\mathcal{E}, \text{Rld}_{\text{res}})}^{(2)} \text{Fun}(\mathcal{E}, \text{RTot}_{\text{res}}(\mathbf{R})), \quad (\Pi_q^{\mathbf{R}})^{-1}(\rho) \simeq \text{Sec}_{\mathcal{E}}((\rho q)^* \pi_{\mathbf{R}}). \quad (42)$$

For split $p_{\rho}^{\text{sp}} \simeq \rho^* \pi_{\mathbf{R}}$,

$$[\text{YRC}] \quad (\Pi_q^{\mathbf{R}})^{-1}(\rho) \simeq \text{Cart}_{\mathcal{B}}(\mathbb{T}_{\mathcal{B}q}, p_{\rho}^{\text{sp}}) \simeq \text{Fun}_{\mathcal{B}}(q, p_{\rho}^{\text{sp}}), \quad \text{Sec}_{\text{cart}}(q^* p_{\rho}^{\text{sp}}) \simeq \text{PsNat}(\mathbf{1}_{\mathcal{E}}, (\rho q)^* \pi_{\mathbf{R}}). \quad (43)$$

Profile-restricted base change is (PBC) $\text{HFib}_T(u^* : Q_{\mathcal{B}}^{\simeq} \rightarrow Q_{\mathcal{A}}^{\simeq}) \simeq \text{Fac}_u^Q(T)$, an equivalence iff all these fibres are terminal. The essentially-surjective $\mathbf{1} \rightarrow BC_2$ with (PRH) trivial/regular pair shows MBC does not force classifier Beck–Chevalley; RVN keeps transport and vertical residual separate.

Coherent worlds and closure trichotomy. (RWS) $\text{World}_{\mathcal{B}}(p) = \text{PsNat}(\mathbf{1}_{\mathcal{B}}, \text{ResFib}(p))$; composition is (RWF). On Set, for $\chi_{q,A}^T : T \text{Ran}_q A \Rightarrow \text{Ran}_q TA$, (WCR) $\text{WRes}_{T,q}(A; b) := \mathbb{T}_{(\text{Ran}_q TA)(b)} \chi_{q,A,b}^T = \text{Tab}((\chi_{q,A,b}^T)^*)$, with fibre $\chi^{-1}\{z\}$, and (KCM) $\text{KCM}_{T,q}(A)_b := \mathfrak{R}_{\bullet}(\chi_{q,A,b}^T)$; $\mathbf{R+}/\mathbf{R0}$ are surjectivity/bijection. (YMR/KCM_o) identifies χ^T as the mate of $\text{pre}_q T = T \text{pre}_q$ and composes mates; converse two-out-of-three requires the outer functor to reflect bijections. For $!$, $\text{W}_T(F) := \varinjlim T F$, $\xi_F^T := \chi_{!, F, *}^T$, (LCM) $\text{LCM}_T(F) := \mathfrak{R}_{\bullet}(\xi_F^T)$; α induces (RPG) $\text{PhRes}_{\alpha}(F) = \mathbb{T} \theta_{\alpha, F}$ at every GRM/RIT level. If $I = T_0 \Rightarrow \cdots \Rightarrow T_n$, $T_n \bar{\emptyset} = \emptyset$, then $(\mathbf{R0}) \Rightarrow (\mathbf{RS}) = (\mathbf{RL}_I) \Rightarrow \cdots \Rightarrow (\mathbf{RL}_{T_n}) \Rightarrow (\mathbf{R+})$; (SIC_P/ABE) keeps invariant law distinct from point actualization. (RCE) $\text{Tot}_J(\mathbf{R}|_J) \rightarrow J$ is an equivalence iff $\mathbf{R}|_J \simeq \mathbf{1}_J$; $J = *$ with fibre BC_2 refutes $\text{RGC} \Rightarrow \text{RCE}$. Thus (RCT) separates saturation/fibre completeness/no-exteriority, while (SXE/XRC/EPI; BXT/BOC) keep support/admissibility/output-base axes.

Ordered adjunction phase fibres. For $G : \mathbf{P} \rightleftarrows \mathbf{W} : H$, put $C = HG$, $I = GH$, $p \sim_G p' \iff Gp = Gp'$, $w \sim_H w' \iff Hw = Hw'$. The unit/counit inequalities give

$$(\text{SDR}) \quad GC = G, \quad HI = H, \quad \text{Fix } C = H(W), \quad \text{Fix } I = G(P), \quad p \sim_G p' \iff Cp = Cp', \quad w \sim_H w' \iff Iw = Iw',$$

with $\mathbf{P}/\sim_G \cong \text{Fix } C$, $[p] \mapsto Cp$, and $\mathbf{W}/\sim_H \cong \text{Fix } I$, $[w] \mapsto Iw$. For $c \in \text{Fix } C$, $i \in \text{Fix } I$,

$$(\text{PFX}) \quad C^{-1}(c) = \{p : Gp = Gc\}, \quad c = \max C^{-1}(c); \quad I^{-1}(i) = \{w : Hw = Hi\}, \quad i = \min I^{-1}(i).$$

Indeed $1 \leq C, I \leq 1$ imply idempotence and the displayed fibre laws/extrema. The closure $0 \mapsto 1, 1 \mapsto 1$ on $\{0 < 1\}$ has $C^{-1}(1) = \{0, 1\}$, so the extremal fixed representative does not select the residual fibre.

Exterior residualization. For small groupoid $\mathcal{E}$ and declared $\text{Ext}_{\text{adm}}(\mathcal{E}) \ni I_{\mathcal{E}}$, let $X_{\mathcal{E}}(p, b) = \text{HFib}_b(p)$ and $\pi_{X, \mathcal{E}} : \int X_{\mathcal{E}} \rightarrow \text{Ext}_{\text{adm}}(\mathcal{E}) \times \mathcal{E}$. By (HRS/CRF), (XRC) NEB iff this projection is an equivalence iff every fibre is terminal. If Ext_{eq} is the equivalence subatlas and $\text{Adm}_{\mathcal{E}}$

the replete subatlases containing I_E , then (MTP) $\text{Adm}_E^{\text{NEB}} = \downarrow \text{Ext}_{\text{eq}}$ and $j_E \dashv r_E$, $r_E(\mathbf{C}) = \mathbf{C} \cap \text{Ext}_{\text{eq}}$; this is both the greatest terminal subatlases of $\mathbf{C}$ and, by PFX, the least atlas in its equal-coreflection fibre. For $f : \mathcal{E} \rightarrow \mathcal{B}$ let $\mathbf{D}_{\mathcal{B}}(p) = \{[b] : \text{HFib}_b(p) \neq \mathbf{1}\}$. Then (MBC) $f^\# r_{\mathcal{B}}(\mathbf{C}) \subseteq r_{\mathcal{E}} f^\#(\mathbf{C})$ and $f^\# p$ is an equivalence iff $\mathbf{D}_{\mathcal{B}}(p) \subseteq \pi_0 \mathcal{B}_f^\perp$; reflection therefore means every nonempty defect set meets $\mathbf{V}_f$. With $U = \pi_0 \mathcal{B}$, $\text{LT}_{\mathbf{C}}(V) = \{p : \mathbf{D}_{\mathcal{B}}(p) \cap V = \emptyset\}$ and $\Theta_{\mathbf{C}}(S) = U \setminus \bigcup_{p \in S} \mathbf{D}_{\mathcal{B}}(p)$, $\Theta_{\mathbf{C}} \dashv \text{LT}_{\mathbf{C}}$, $\text{LT}(\bigcup V_i) = \bigcap \text{LT}(V_i)$, $\text{LT}(\emptyset) = \mathbf{C}$, $\text{LT}(U) = r_{\mathcal{B}}(\mathbf{C})$, and $p \in \text{LT}(\mathbf{V}_f)$ iff $f^\# p$ is an equivalence. This is EPN for $[p] \perp b \iff b \notin \mathbf{D}_{\mathcal{B}}(p)$; DCC is the quotient-determination test, finite U the RDA hitting front, and $\mathbb{N}_{\text{disc}}$ with BC_2 defects forbids finite stabilization. If admitted, $p_H : \mathcal{E} \times BH \rightarrow \mathcal{E}$ has fibre BH ; hence any class containing p_{C_2} fails NEB, while (EPI) $\langle I_E \rangle$ versus $\langle I_E, p_{C_2} \rangle$ proves the admissibility perimeter irreducible; (PCL) recovers it from terminal cores/fibres without a selector. *Self-exteriorization and support split.* For $q : \mathcal{E} \rightarrow \mathcal{B}$ in Gpd let $k_q = q^\star q : \text{Tab}(M_q) \rightarrow \mathcal{E}$. Then

$$\begin{aligned} (\text{SXE}) \quad & \text{HFib}_e(k_q) \simeq \text{HFib}_{q(e)}(\bar{q}) \simeq \text{HFib}_{q(e)}(q), \\ & k_q \text{ equivalence} \iff \bar{q} \text{ equivalence} \iff q \text{ fully faithful}, \\ & q \text{ equivalence} \iff k_q \text{ equivalence and } \mathcal{B}_q^\perp = \emptyset \iff k_q \text{ equivalence and } \mathbf{V}_q = \pi_0 \mathcal{B}. \end{aligned}$$

Thus collision/full-faithfulness and essential-image support are independent, and MTP/SXE gives NEB_{self} iff q is fully faithful. For a groupoid, (SIC) $\text{Sect}(\int F \rightarrow \mathcal{G}) \cong \prod_{\lambda} F(c_{\lambda})^{\text{Aut}(c_{\lambda})}$. For finite nonempty fibres put $F_{\lambda} = F(c_{\lambda})$, $\Gamma_{\lambda} = \text{Aut}(c_{\lambda})$, $e_{\lambda} : F_{\lambda} \rightarrow F_{\lambda}/\Gamma_{\lambda}$ and $u_O = \text{Unif}(O)$.

Invariance forces equal point masses on each orbit, hence $e_{\lambda*} : \mathcal{P}(F_{\lambda})^{\Gamma_{\lambda}} \xrightarrow{\text{aff}} \mathcal{P}(F_{\lambda}/\Gamma_{\lambda})$, $e_{\lambda*}^{-1}\{v\} \cap \mathcal{P}(F_{\lambda})^{\Gamma_{\lambda}} = \{\sum_O v(O)u_O\}$. Thus (SIC $_{\mathcal{P}}$) law sections are $\prod_{\lambda} \mathcal{P}(F_{\lambda}/\Gamma_{\lambda})$; extremes are u_O , while Dirac laws are exactly point sections. Hence regular/trivial BG actions separate nonemptiness, point sectionability and terminality. Put $\delta F_{\lambda} = \{\delta_x\}$, $\Gamma_{\lambda, \mu} = \text{Stab}_{\Gamma_{\lambda}}(\mu)$ and $\text{AR}_{\mathcal{G}}(F) = \{a : \mathcal{P}F \Rightarrow F : a\delta_F = 1_F\}$. Then (ABE) $\text{AR}_{\mathcal{G}}(F) \cong \prod_{\lambda} \prod_{[\mu] \in (\mathcal{P}(F_{\lambda}) \setminus \delta F_{\lambda})/\Gamma_{\lambda}} F_{\lambda}^{\Gamma_{\lambda, \mu}}$, and $\text{AR}_{\mathcal{G}}(F) \neq \emptyset$ iff the point-section space is nonempty. For finite $\mathcal{B}$, (PRH) $\widehat{\Gamma}_{\lambda} = 1$ for

every component iff every finite nonempty local system has a point section; a nontrivial finite quotient acting regularly is the sharp converse witness, while uniform law sections always exist.

On Set , for abelian G , (KSP/HRD) gives $\mathbb{K}(f) = H_1 \mathfrak{R}_{\bullet}(f) = \ker f_{\#}$, $\text{ExtRes}(f) = H_0 \mathfrak{R}_{\bullet}(f) = \text{coker } f_{\#}$, $\text{Hom}(\mathbb{K}(f), G) \cong Z^1(R_f; G)$ and $0 \rightarrow G^{f(X)} \rightarrow G^X \rightarrow Z^1(R_f; G) \rightarrow 0$, with higher Čech cohomology zero. *Realization-status decomposition.* $\mathbb{K}(f) \cong \bigoplus_{y \in f(X)} \widetilde{\mathbb{Z}}^{(f^{-1}\{y\})}$, $\text{ExtRes}(f) \cong \bigoplus_{y \notin f(X)} \mathbb{Z}[y]$. Thus $R+$ is $\text{ExtRes} = 0$, $R0$ is $\mathbb{K} = \text{ExtRes} = 0$, and lift/proof failure is no additional status; (RHT/LTX/SLX) separately retains limit nonlanding and multiplicity. For $X \xrightarrow{f} Y \xrightarrow{g} Z$, the cone triangle gives (UAR $_{\perp}$) with $\partial_{g,f} : \mathbb{K}(g) \rightarrow \text{ExtRes}(f)$; the $\{0\} \hookrightarrow \{0, 1\} \rightarrow \{*\}$ case makes ∂ an isomorphism. After corestricting f , (UAR) is $0 \rightarrow \mathbb{K}(f) \rightarrow \mathbb{K}(gf) \rightarrow \mathbb{K}(\bar{g}) \rightarrow 0$ and the dual cocycle exact row, the UDR $_{\Delta}$ /URE $_{\circ}$ composition face. For readout F over Q , $\text{DepRes}_Q(F) = \mathbb{K}(Q)/\mathbb{K}((Q, F)) \cong \mathbb{K}(p_{Q,F})$, so

$$[\text{RPL}] \quad \text{DepRes}_Q(F) \cong H_1 \mathfrak{R}_{\bullet}(p_{Q,F}) \cong \bigoplus_{y \in Q(X)} \widetilde{\mathbb{Z}}^{((\text{supp}_0 \tau_{Q,F})(y))}, \quad \text{supp}_0 \tau_{Q,F} \text{ functional} \iff \text{DepRes}_Q(F) = 0 \iff F = \widehat{F}Q \text{ on } Q(X). \quad (44)$$

Factorization on $Q(X)$ is unique; constants descend, while $\text{DepRes}_Q(I_X) = 0$ iff Q is injective. For $h = (Q, F)$, (DRX) gives the exact row $0 \rightarrow \mathbb{K}(h) \rightarrow \mathbb{K}(Q) \rightarrow \text{DepRes}_Q(F) \rightarrow 0$. If $\mathbb{K}(h) \subseteq \mathbb{K}((Q, G))$, nesting gives $0 \rightarrow \mathbb{K}((Q, G))/\mathbb{K}(h) \rightarrow \text{DepRes}_Q(F) \rightarrow \text{DepRes}_Q(G) \rightarrow 0$. *Bounded cocycle enrichment.* For corestricted $q : X \twoheadrightarrow Y$ and normed real Z , put $d_q F(x, x') = F(x) - F(x')$ and, for $c \in Z^1(R_q; Z)$, $\|c\|_{\infty} := \sup_{R_q} \|c\| \in [0, \infty]$ and $\|c\|_{\text{pr}} := \inf_{d_q H=c} \|H\|_{\infty} \in [0, \infty]$. Let $Z_b^1(R_q; Z) = \{c : \|c\|_{\infty} < \infty\}$, $\omega_q(F) = \|d_q F\|_{\infty}$, and $\mathbf{B}_q(Z) = \{F : \omega_q(F) < \infty\}$. Then

$$[\text{BCR}] \quad 0 \rightarrow Z^Y \xrightarrow{q^*} \mathbf{B}_q(Z) \xrightarrow{d_q} Z_b^1(R_q; Z) \rightarrow 0, \quad Z_b^1(R_q; Z) \subseteq Z^1(R_q; Z_{\text{add}}) \cong \text{Hom}(\mathbb{K}(q), Z_{\text{add}}), \\ \mathfrak{E}_q(F) := \inf_G \sup_x \|F(x) - G(qx)\| = \|d_q F\|_{\text{pr}} = \sup_{y \in Y} \text{rad}_Z F(q^{-1}y), \quad \kappa_q(F) := \frac{1}{2} \omega_q(F), \quad \kappa_q(F) \leq \mathfrak{E}_q(F) \leq 2\kappa_q(F). \quad (45)$$

For corestricted $X \xrightarrow{q} Y \xrightarrow{r} W$, restriction of kernel pairs gives $0 \rightarrow Z_b^1(R_r; Z) \xrightarrow{q^*} Z_b^1(R_{rq}; Z) \xrightarrow{\text{res}} Z_b^1(R_q; Z) \rightarrow 0$. Choose one point in each q -fibre: a bounded c has bounded primitive $c(x, s(qx))$, so the first row is exact; arbitrary primitives differ exactly by q^*G , hence minimization is fibrewise and gives the radius formula in $[0, \infty]$, while radius–diameter gives the norm bounds. Restriction and primitive extension give (BCR $_{\circ}$). Thus (BCR) is the bounded normed enrichment of (KSP/RPL), with the same zero locus but strictly more data: for $q : \{0, 1\} \rightarrow \{*\}$ and $F = (0, 1)$, replacing $|\cdot|$ on $\mathbb{R}$ by $2|\cdot|$ leaves the additive residual unchanged and doubles both quantitative norms. Reference/law/grammar/history/joining/operator-algebraic data remain typed attachments.

On the same corestricted Set face,

$$(\text{FSB}_{\text{core}}) \quad \text{Fac}_c(a) \simeq \mathbf{1} \iff a = \bar{a}c \iff R_c \subseteq R_a \iff [c, a] = [c] \iff \text{DepRes}_c(a) = 0.$$

The support-preorder equality is SAC; strict separation is a genuine key deficit and does not collapse image-core, off-image or attachment fibres. *Determination/kernel descendant.* For corestricted same-source v, w , write $v \preceq_X w$ iff $v = \bar{v}w$. Then

$$v \preceq_X w \iff \text{DepRes}_w(v) = 0 \iff R_w \subseteq R_v, \quad \bar{v} : w(X) \rightarrow v(X) \text{ unique.} \quad (\text{SDO/SDK})$$

For finite same-source F , $R_{\langle F \rangle} = \bigcap_{f \in F} R_f$ (empty intersection X^2) by (REL $_{\text{ker}}$ /RKD/KGC/DCC); hence mutual determination is kernel-pair equality and finite joint readouts are joins in reverse-kernel order. For finite-joint-closed $\mathcal{V}$, put $\mathfrak{D}_X(\mathcal{V}) := \mathcal{V}/\sim$ and $\text{cl}_{\text{det}}(A) := \{\xi : \exists \emptyset \neq F \in A, \xi \preceq_X \vee F\}$. For $A \subseteq B \subseteq C$ and finite G ,

$$\begin{aligned} \text{J}_{\text{det}}(B:A) &= \text{cl}_{\text{det}} B \setminus \text{cl}_{\text{det}} A, & \text{Syn}_{\text{det}}(A, B) &= \text{cl}_{\text{det}}(A \cup B) \setminus (\text{cl}_{\text{det}} A \cup \text{cl}_{\text{det}} B), \\ \text{Front}_G^{\text{det}}(\xi) &= \text{Min}_{\subseteq} \{F \in G : \xi \in \text{cl}_{\text{det}} F\}, & \text{J}_{\text{det}}(C:A) &= \text{J}_{\text{det}}(B:A) \sqcup \text{J}_{\text{det}}(C:B). \end{aligned} \quad (\text{SDC})$$

Let $\text{FinFront}_0(\mathcal{U}) := \text{Min}_{\subseteq}(\mathcal{U} \cap \text{Fin}(I_0))$. For corestricted $q, a, (h_j)_{j \in J} \in \mathcal{V}$, $q : X \twoheadrightarrow Q$, and q_F the corestriction of $\langle q, (h_j)_{j \in J} \rangle$, (RFF) gives $\text{supp}_0 \tau_{q_F, a}$ functional $\iff \text{DepRes}_{q_F}(a) = 0 \iff R_q \cap \bigcap_{j \in J} R_{h_j} \subseteq R_a \iff [a] \in \text{cl}_{\text{det}}(\{[q]\} \cup \{[h_j] : j \in J\})$. Thus $\text{FunFront}_H(a \mid q) := \text{FinFront}_J\{F : \text{supp}_0 \tau_{q_F, a} \text{ functional}\}$ is the minimal functionalizing frontier; refinement only restricts nonempty-support T -fibres, so support is upward and its minima form the finite RDA antichain. This is not law-fibre contractibility: singleton-source functors into $\mathbf{1}$ and BC_2 have the same object/kernel shadow but terminal versus $\text{Aut} = C_2$ law fibre. SDC synergy is the crossed-front case. Moreover $\mathbb{I}_{\{f, g\}} \neq 0$ forces synergy but not conversely: $f = (0, 0, 1)$, $g = (0, 1, 1)$ have injective joint readout and tree $G_{f, g}$, while on $\{0, 1\}^2$

they jointly determine $a = f \oplus g$ and neither does alone. Hence local view, minimal access and ordered route/history remain independently nonunique (RDA; five-point RRCX). If $s = hd$, then $R_d \subseteq R_s$, so $\text{DepRes}_s(a) = 0 \Rightarrow \text{DepRes}_d(a) = 0$, strict for $d = I_{\{0,1\}}$, $s = !$, $a = d$. Also $R_{d\phi} = (\phi \times \phi)^{-1} R_d$, so pullback preserves determination and reflects it for onto ϕ . By the support-key fibre (RKD), nonzero $\text{DepRes}_d(a)$ is same- d /different- a multiplicity, not selection; hence postprocessing cannot create determination. *Banach determination residual functor*. For same-source $(h_j : X \rightarrow Y_j)_{j \in J}$, finite $A \subseteq J$, nonzero Banach Z , let $q_A : X \rightarrow Q_A = X / \cap_{j \in A} R_{h_j}$ and $q_A = \rho_{B,A} q_B$ for $A \subseteq B$. Put $\mathbb{B}_Z(A) = Z_b^1(R_{q_A}; Z)$ and $\mathbb{N}_Z(B:A) = Z_b^1(R_{\rho_{B,A}}; Z)$ with $\|\cdot\|_{\text{pr}}$. Restriction is surjective and

$$[\text{BDRF}] \quad 0 \rightarrow \mathbb{N}_Z(B:A) \rightarrow \mathbb{B}_Z(A) \rightarrow \mathbb{B}_Z(B) \rightarrow 0, \quad 0 \rightarrow \mathbb{N}_Z(B:A) \rightarrow \mathbb{N}_Z(C:A) \rightarrow \mathbb{N}_Z(C:B) \rightarrow 0. \quad (46)$$

for $A \subseteq B \subseteq C$. By (BCR_o) this is the bounded-cochain realization of the same quotient filtration; completeness follows from closed $Z_b^1 \leq \ell_\infty$ and $\frac{1}{2}\|c\|_\infty \leq \|c\|_{\text{pr}} \leq \|c\|_\infty$. It vanishes exactly at $q_A \simeq_X q_B$ and, for finite X , $Z = \mathbb{R}$, has dimension $|Q_B| - |Q_A|$. The five-point (RRCX) routes (1, 3) and (2, 2) share an endpoint, so graded size does not recover route; no canonical splitting/history/BKM identification follows. *Read-out support-key tower and residual-key descent*. On corestricted discrete Set, let $D : \Omega \rightarrow B$, $A : \Omega \rightarrow Y$ and put $\text{Key}_{A|D} := \int_b K_{A|D}(b)$ from (RGS_{desc}); corestrict (D, A) as $j : \Omega \rightarrow \text{Key}_{A|D}$ and write $\pi_D : \text{Key}_{A|D} \rightarrow B$. Then (DCC/RPL/DRX) give

$$[\text{RKD}] \quad \begin{aligned} \text{TabProf } \tau_{D,A} &\cong \Omega \xrightarrow{j} \text{Key}_{A|D} \cong_{B \times Y} \text{TabRel}(\text{supp}_0 \tau_{D,A}) \cong \text{im}(D, A) \cong \Omega / (R_D \cap R_A) \xrightarrow{\pi_D} B, \\ R_j &= R_D \cap R_A, \quad \text{DepRes}_D(A) \cong \mathbb{K}(\pi_D) \cong \mathbb{K}(D)/\mathbb{K}(j), \\ \text{DepRes}_D(A) = 0 &\iff \pi_D \text{ bijective} \iff K_{A|D}(b) \simeq \mathbf{1} \ (b \in B), \quad \text{DepRes}_{(D,k)}(A) = 0 \iff R_D \cap R_k \subseteq R_A. \end{aligned} \quad (47)$$

Write $\alpha_{A|D} := \text{pr}_Y|_{\text{Key}_{A|D}}$, so $D = \pi_D j$ and $A = \alpha_{A|D} j$. Companion/conjoint composition in (URE) gives the exact proarrow-key factorization

$$(\text{PKF}) \quad \tau_{D,A} \simeq (\alpha_{A|D})_* \odot N_j \odot \pi_D^*, \quad N_j(k, k') \cong \begin{cases} j^{-1}\{k\}, & k = k', \\ \emptyset, & k \neq k'. \end{cases}$$

Thus the middle factor is exactly fixed- (D, A) multiplicity, whereas π_D^* carries equal- D /different- A branching. Moreover

$$\text{DepRes}_D(A) = 0 \iff \exists ! \hat{A} : B \rightarrow Y : A = \hat{A}D \text{ and } \tau_{D,A} \simeq \hat{A}_* \odot N_D.$$

Indeed the first display is $(\alpha_{A|D} j)_* \odot (\pi_D j)^*$ with associativity. On the zero locus π_D is bijective by (RKD), $\alpha_{A|D} = \hat{A}\pi_D$, and (URE_o) gives $N_D \simeq (\pi_D)_* \odot N_j \odot \pi_D^*$; conversely the last form has functional support graph $(\hat{A})$, so (TOM/RKD) forces the unique descent. If $D = dq$, $A = aq$ through $q : \Omega \rightarrow Q$, unique $h : Q \rightarrow \text{Key}_{A|D}$ has $j = hq$, so j is the coarsest refinement of D determining A , not a globally least access. Iteration gives $R_{(D,k_1), \dots, k_n} = R_D \cap \cap_i R_{k_i}$. Neither PKF layer selects: (RFD/ROA/YSK) gives a self-key, but an unsupplied resolver k must be produced independently; non-determination is residual freedom, not a chosen hidden alternative. The two-point witnesses $(D \text{ const}, A = I)$ and $(D, A \text{ const})$ make respectively π_D^* nontrivial with singleton N_j , and N_j nontrivial with bijective π_D ; hence neither layer determines the other.

Lossless residual re-entry. On the corestricted same-source Set face let $q_0 : X \rightarrow Q_0$ and readouts $f_1, \dots, f_n$ be given; let q_i corestrict (q_{i-1}, f_i) and put $C_i := \text{DepRes}_{q_{i-1}}(f_i)$. Then (RPL/DRX/RKD) gives

$$[\text{RRCX}] \quad 0 \rightarrow \mathbb{K}(q_i) \rightarrow \mathbb{K}(q_{i-1}) \rightarrow C_i \rightarrow 0, \quad C_i = 0 \iff q_i \simeq_X q_{i-1}, \quad R_{q_n} = R_{q_0} \cap \bigcap_{i=1}^n R_{f_i}. \quad (48)$$

A refinement path $\gamma = (q_0 \preceq_X \dots \preceq_X q_n)$ therefore carries $\mathfrak{R}(\gamma) = \mathbb{K}(q_0)/\mathbb{K}(q_n)$ filtered by $F^i = \mathbb{K}(q_i)/\mathbb{K}(q_n)$, with $F^{i-1}/F^i \cong C_i$. For $a < b < c$,

$$0 \rightarrow C_{b:c} \rightarrow C_{a:c} \rightarrow C_{a:b} \rightarrow 0, \quad C_{a:c} = 0 \iff C_{a:b} = C_{b:c} = 0.$$

Deleting a vertex fuses adjacent grades; concatenation gives this extension. KGC chains have $q_i = q_{A_i}$. Thus endpoint, route and graded/scalar shadows differ; for finite X , $\sum_i \text{rk } C_i = |Q_n| - |Q_0|$. Joint linearization has a distinct interaction face. For finite nonempty F and $H_F = \langle h_i : i \in F \rangle$, (UIR) gives $0 \rightarrow \mathbb{K}(H_F) \rightarrow \cap_i \mathbb{K}(h_i) \rightarrow \mathbb{I}_F \rightarrow 0$, so $\mathbb{I}_F$ is exactly the failure of free residual linearization to commute with joint kernel-pair intersection. For $F = \{f, g\}$ let c corestrict $X/(R_f \vee R_g)$ and $G_{f,g}$ be the bipartite support graph on $f(X) \sqcup g(X)$. Path telescoping gives $\mathbb{K}(c) = \mathbb{K}(f) + \mathbb{K}(g)$ and $\mathbb{I}_{\{f,g\}} \cong H_1(G_{f,g}; \mathbb{Z})$, hence for finite X , $\text{rk } \mathbb{I}_{\{f,g\}} = |c(X)| + |(f, g)(X)| - |f(X)| - |g(X)|$. With

$h = (f, g)$, $\mathbb{K}(h) \xrightarrow{\Delta} \mathbb{K}(f) \oplus \mathbb{K}(g) \xrightarrow{(u,v) \mapsto u-v} \mathbb{K}(c)$ has only $H_1 \cong \mathbb{I}_{\{f,g\}}$; equivalently $0 \rightarrow \mathbb{I}_{\{f,g\}} \rightarrow \text{DepRes}_f(g) \rightarrow \mathbb{K}(c)/\mathbb{K}(g) \rightarrow 0$ and symmetrically. Thus h is the least common refinement, c the finest common factor, and novelty/common invariant/joint interaction form one kernel-pair diamond. The five-point fixture $q_0 = !$, $f = (0, 0, 0, 1, 1)$, $g = (0, 1, 2, 0, 1)$ retains injective endpoint, terminal c , $\mathbb{I}_{\{f,g\}} \cong \mathbb{Z}$ and ordered RRCX ranks (1, 3) versus (2, 2), separating endpoint, route and incidence.

Common-base rectangular reflection and compatibility factorization. Let $\emptyset \neq I$ be a set and let $h_i : X \rightarrow Y_i$ be same-source Set read-outs; replace Y_i by $h_i(X)$ and h_i by its corestriction, so $h_i : X \rightarrow Y_i$. Put $H = \langle h_i \rangle$, $R = \text{im } H \subseteq \prod_i Y_i$, $j : X \rightarrow R$, $R^- := \cap_i R_{h_i} = R_j$, $R^+ := \bigvee_i R_{h_i}$, and $c : X \rightarrow C := X/R^+$. Since $R_{h_i} \subseteq R_c$, unique surjections $u_i : Y_i \rightarrow C$ satisfy $c = u_i h_i$. Define

$$\text{Rect}_C(\mathbf{Y}) := \prod_C Y_i := \{(y_i) \in \prod_i Y_i : u_i(y_i) = u_k(y_k) \ \forall i, k\}, \quad \text{CBGap}(\mathbf{h}) := \text{Rect}_C(\mathbf{Y}) \setminus R.$$

Then

$$[\text{CBR}] \quad X \rightarrow X/R^- \cong R \xrightarrow{\iota_R} \text{Rect}_C(\mathbf{Y}) \subseteq \prod_i Y_i, \quad \text{Rect}_C(\mathbf{Y}) = \min_{\subseteq} \left\{ \prod_D Y_i : R \subseteq \prod_D Y_i \right\}, \quad \iota_R \text{ onto} \iff \text{CBGap}(\mathbf{h}) = \emptyset. \quad (49)$$

Here the minimum ranges over common quotients $v_i : Y_i \rightarrow D$. Since $R_j = R^-$, every h_i factors through j , and $R_q \subseteq R^-$ for any quotient q determining all h_i ; hence j is the coarsest joint-determining quotient. If $R \subseteq \prod_D Y_i$, then $d := v_i h_i$ is independent of i , so $R^+ \subseteq R_d$, $d = \bar{d}c$, and surjectivity of h_i gives $v_i = \bar{d}u_i$; thus c is the finest common factor and proves the minimum in (CBR). For fixed $\mathbf{Y}$ let $\text{Rel}_{\text{surj}}(\mathbf{Y})$ be the inclusion poset of relations with all coordinate projections onto and apply (CBR) to those projections. Its universal minimum gives

$$(\text{CBR}_{\text{cl}}) \quad S \subseteq T \Rightarrow \text{Rect}(S) \subseteq \text{Rect}(T), \quad S \subseteq \text{Rect}(S), \quad \text{Rect}^2(S) = \text{Rect}(S),$$

because $\text{Rect}(T)$ is a common-base overrelation of S and $\text{Rect}(S)$ is itself common-base. Hence Rect reflects onto the common-base fixed points: for fixed T , $\text{Rect}(S) \subseteq T \iff S \subseteq T$. For $I = \{1, 2\}$ these fixed points are exactly the difunctional relations: the three-corner law

makes each bipartite connected component complete, and conversely every $B \times_C Y$ obeys it. Therefore

$$(DFC) \quad R = B \times_C Y \iff R \odot R^\circ \odot R \subseteq R \iff \text{CBGap}(f, g) = \emptyset.$$

For a finite binary fixed point $T = B \times_C Y$, put $\text{RectFib}(T) = \{S \in \text{Rel}_{\text{surj}}(B, Y) : \text{Rect}(S) = T\}$ and $G_S(S) = (B_S \sqcup Y_S, S \cap (B_S \times Y_S))$. Since the join of the projection kernels has the bipartite connected components as classes,

$$(RFS) \quad \begin{aligned} S \in \text{RectFib}(T) &\iff S \subseteq T \text{ \& } G_S(S) \text{ connected } (s \in C), \\ \text{Min}_\subseteq \text{RectFib}(T) &= \{S : G_S(S) \text{ a tree } (s \in C)\}, \quad \max \text{RectFib}(T) = T. \end{aligned}$$

The second line is the minimal-connected-spanning criterion. Thus the fibre is a singleton iff every block has $\min(|B_s|, |Y_s|) = 1$; otherwise it has no least member. No isomorphism-natural minimal-support selector exists: on $K_{2,2}$ the four three-edge minima form one automorphism orbit with no fixed tree. For $S \in \text{RectFib}(T)$ let p_B, p_Y be its two coordinate projections and put $\mathbb{I}_S := \mathbb{I}_{(p_B, p_Y)}$. Euler rank blockwise and (RRCX) give

$$[\text{RCC}] \quad |T \setminus S| + \text{rk } \mathbb{I}_S = \sum_{s \in C} (|B_s| - 1)(|Y_s| - 1). \quad (50)$$

At $S = R = \text{im}(f, g)$ this is the former $|\text{CBGap}| + \text{rk } \mathbb{I}$ law; trees and T are its zero-cycle and zero-gap extremes. The common factor therefore determines a reflection fibre, not a support point. The three relation carriers in (CBR) are discrete (URT/RFE) totalizations of realized-support, common-factor and constant-Cartesian fibres. For finite I, c and j are the empty-face join and full-face meet quotients of (DCE/DCHD); (URE/RKD/PKF) separately retains fixed-output multiplicity N_j .

If $2 \leq |I| = n < \infty$, X is finite nonempty and full-support $\lambda \in \mathcal{P}(X)$, put $\gamma = H_* \lambda$, $\rho = c_* \lambda$, let μ_i^s be the conditional laws on $c = s$, and set $\eta_C = \sum_s \rho(s) \otimes_i \mu_i^s$. Then $\text{supp } \gamma = R$, $\text{supp } \eta_C = \text{Rect}_C(Y)$; with $\eta_C^R = \eta_C(\cdot | R)$,

$$[\text{CBL}] \quad \begin{aligned} \text{TotDep}_C(\gamma) &:= D(\gamma | \eta_C) = \underbrace{-\log \eta_C(R)}_{\text{SuppDep}_C(\gamma)} + \underbrace{D(\gamma | \eta_C^R)}_{\text{WtDep}_C(\gamma)}, \\ \text{SuppDep}_C = 0 &\iff \text{CBGap}(\mathbf{h}) = \emptyset, \quad \text{TotDep}_C = 0 \iff \gamma = \eta_C \iff (\text{SuppDep}_C, \text{WtDep}_C) = (0, 0). \end{aligned} \quad (51)$$

The scalar weight residual has a canonical fibre resolution. Put $\mu_i = (p_i)_* \gamma$, $\mathcal{T}_R(\mu) = \{\pi \in \mathcal{P}(R) : (p_i)_* \pi = \mu_i \forall i\}$, $V_R = \ker[\mathbb{R}^R \xrightarrow{((p_i)_*)_i} \bigoplus_i \mathbb{R}^{Y_i}]$, and let q_R be the unique maximum-entropy point of $\mathcal{T}_R(\mu)$. Full support gives $\eta_C^R > 0$ on R and, by the boundary entropy test against the feasible full-support γ , $q_R \in \mathcal{T}_R^\circ(\mu)$. Equivalently $q_R = \arg \min_{\mathcal{T}_R(\mu)} D(\cdot | \eta_C^R)$: on each common-factor block $\log \eta_C^R$ is, up to a constant, a sum of one-leg potentials. The commutative (EFD) face therefore gives

$$[\text{WFR}] \quad \begin{aligned} (\text{WFR}) \quad \text{WtDep}_C &= \underbrace{D(q_R | \eta_C^R)}_{\text{MargRest}_C} + \underbrace{D(\gamma | q_R)}_{\text{FibDep}_R}, \quad \Lambda_R : \mathcal{T}_R^\circ(\mu) \xrightarrow[\text{an}]{\cong} V_R^*, \quad \Lambda_R(\pi)(z) = \sum_{r \in R} z_r \log \frac{\pi_r}{q_R(r)}. \end{aligned} \quad (52)$$

On this fixed-marginal fibre $d\text{FibDep}_R = \Lambda_R$, $D\Lambda_R(\pi)[u](z) = \sum_r u_r z_r / \pi_r$, and the KL three-point identity gives $\text{TotDep}_C = \text{SuppDep}_C + \text{MargRest}_C + \text{FibDep}_R$. Moreover $\text{MargRest}_C = 0$ iff η_C^R already has the observed marginals, while $\text{FibDep}_R = 0$ iff $\gamma = q_R$; in particular $\text{SuppDep}_C = 0 \implies \text{MargRest}_C = 0$. On the full-support face, supplementary (MIB/TF/MTE) identifies

$$(\text{WFR}_{\text{int}}) \quad V_R = \ker A_{I,\gamma} = \mathbb{I}_{I,\gamma} \otimes \mathbb{R}, \quad \Lambda_R : \mathcal{T}_R^\circ(\mu) \xrightarrow[\text{an}]{\cong} (\mathbb{I}_{I,\gamma} \otimes \mathbb{R})^*.$$

Since $q_R > 0$ on R , a nonzero interaction vector gives $q_R \pm \varepsilon z$ for small ε ; hence

$$\mathbb{I}_{I,\gamma} = 0 \iff \mathcal{T}_R(\mu) = \{q_R\} \iff q_R \in \text{ex } \mathfrak{I}_I(\mu),$$

which is the full-support (MCM) face. Thus the interaction lattice is exactly the linear capacity of law freedom and its real dual is the global EFD coordinate. For the support-corestricted integral incidence complex $C_{I,R} := \mathfrak{R}_\bullet(A_{I,R}^Z)$, (MTE/MSI/DPH) gives

$$[\text{AIPG}] \quad \begin{aligned} (\text{AIPG}) \quad \text{IntPh}_{I,R} &:= \text{DPh}(C_{I,R}), \quad 0 \rightarrow V_R / \mathbb{I}_{I,\gamma} \rightarrow \text{IntPh}_{I,R} \rightarrow \text{SatRes}_I(R) \rightarrow 0, \\ \text{Lie IntPh}_{I,R} &\cong V_R, \quad \pi_1 \text{IntPh}_{I,R}^0 \cong \mathbb{I}_{I,\gamma}, \quad \pi_0 \text{IntPh}_{I,R} \cong \text{SatRes}_I(R), \quad \text{IntPh}_{I,R} \cong \text{Obj}_{I,R}^{\text{inc}}. \end{aligned} \quad (53)$$

Pontryagin duality regenerates supplementary (MID) and the real EFD target:

$$(\text{WFR}_Z) \quad 0 \rightarrow \text{Hom}(\text{SatRes}_I(R), \mathbb{Q}/\mathbb{Z}) \rightarrow \text{Obj}_{I,R}^{\text{inc}} \rightarrow \text{Hom}(\mathbb{I}_{I,\gamma}, \mathbb{Z}) \rightarrow 0, \quad \text{Obj}_{I,R}^{\text{inc}} \otimes \mathbb{R} \cong \text{Lie}(\text{IntPh}_{I,R})^* \cong V_R^*.$$

The phase row is also the $C = \mathbb{R}/\mathbb{Z}$ specialization of (ACS), so coefficient-tensor and objective-dual routes meet on the same carrier. The discrete dual row splits abstractly because its quotient is free. Under $\text{Obj}_{I,R}^{\text{inc}} \otimes \mathbb{R} \cong \mathbb{R}^R / \sum_i p_i^* \mathbb{R}^{Y_i} \cong V_R^*$, the EFD coordinate is exactly $[\log(\pi/q_R)]$. Thus smooth law freedom, the integral interaction lattice, saturation torsion and integral objectives are respectively Lie algebra, fundamental lattice, component group and character group of one compact phase carrier. The old witnesses are its pure faces: $K_{2,2}$ is connected with positive-dimensional Lie algebra, whereas E_n is finite with $\pi_0 \cong \mathbb{Z}/n$. Both can coexist: with alphabets disjoint in all three legs, the union of an embedded $K_{2,2}$ square (constant third leg) and E_n has block-diagonal incidence, hence $\mathbb{I} \cong \mathbb{Z}$, $\text{SatRes} \cong \mathbb{Z}/n$ and $\text{IntPh} \cong (\mathbb{R}/\mathbb{Z}) \times \mathbb{Z}/n$. Its dual row is $\mathbb{Z}/n \hookrightarrow \mathbb{Z} \oplus \mathbb{Z}/n \twoheadrightarrow \mathbb{Z}$; shear automorphisms fixing both ends move its sections, so the extension data select no splitting in this mixed face. Latent realization remains independently noncollapsed by (LRF/DLCF); countable (DOH), higher history and occurrence are not absorbed by this finite phase, with the exact finite/countable boundary supplied by (FCR/CCI/CHI/PPH).

For $I = \{1, 2\}$ let $G_R = (B \sqcup Y, R)$. Then (ICL/HCD/CTH) specializes the interaction lattice graphically and (RFS) identifies the common components, giving

$$[\text{H01}] \quad \begin{aligned} (\text{H01}) \quad C &\cong \pi_0(G_R), \quad \mathbb{R}[C] \cong H_0(G_R; \mathbb{R}), \quad V_R \cong H_1(G_R; \mathbb{R}), \quad V_R^* \cong H^1(G_R; \mathbb{R}), \\ R \in \text{Min RectFib}(B \times_C Y) &\iff H_1(G_R; \mathbb{R}) = 0. \end{aligned} \quad (54)$$

Thus $\dim V_R = |R| - |B| - |Y| + |C|$; the preceding family criterion becomes forest/tree uniqueness. Here (ICL) kills the saturation component group and row-signing gives the sharper phase identification $\text{IntPh}_{\{1,2\},R} \cong H_1(G_R; \mathbb{R}/\mathbb{Z})$, a connected cycle torus with Lie dual $H^1(G_R; \mathbb{R})$. On an alternating cycle the one-leg potentials cancel, so the character/Lie coordinate $[\log(\pi/q_R)]$ evaluates as the logarithmic alternating

product ratio and its differential is Fisher/BKM. Hence common identity, law-freedom capacity, integral cycle lattice and global law coordinate are the $H_0, H_1(\mathbb{R}), H_1(\mathbb{Z}), H^1(\mathbb{R})$ faces of one compatibility graph. Noncollapse is sharp: the uniform law on $\{00, 01, 11\}$ has $H_1 = 0$, $\text{FibDep}_R = 0$ but $\text{WtDep}_C = \text{MargRest}_C = \frac{1}{3} \log(32/27) > 0$; on $K_{2,2}$ with uniform marginals, $\text{SuppDep}_C = \text{MargRest}_C = 0$ while $(1/4 + t, 1/4 - t, 1/4 - t, 1/4 + t)$ has $\text{FibDep}_R > 0$ for $0 < |t| < 1/4$.

This is also variational over common factors. For any $v_i : Y_i \rightarrow D$ with $R \subseteq \prod_D Y_i$, put $d = v_i h_i$, $\rho_D = d_* \lambda$, let $\mu_{i,D}^t$ be the conditional laws, and $\eta_D = \sum_i \rho_D(t) \otimes_i \mu_{i,D}^t$. For every D -conditionally product $v = \sum_i q(t) \otimes_i v_i^t$,

$$\begin{aligned} & D(\gamma \| v) - D(\gamma \| \eta_D) = D(\rho_D \| q) + \sum_{i,t} \rho_D(t) D(\mu_{i,D}^t \| v_i^t), \\ \text{[CBIP/CFIM]} \quad & \text{TC}(\gamma) := D\left(\gamma \left\| \bigotimes_i (h_i)_* \lambda\right.\right) = (n-1) \text{HSh}(\rho_D) + D(\gamma \| \eta_D), \\ & D(\gamma \| \eta_D) - \text{TotDep}_C(\gamma) = (n-1) \text{HSh}(C | D) \geq 0. \end{aligned} \quad (55)$$

The first row is obtained by expanding $\log(\eta_D/v)$ and taking γ -expectation, so η_D is the unique KL minimizer over the D -conditionally independent laws. For the second, D is a function of every Y_i , hence $\text{HSh}(Y_i | D) = \text{HSh}(Y_i) - \text{HSh}(D)$ and likewise for the joint tuple. Finally (CBR) gives $C \rightarrow D$, so subtraction gives the third row; full support makes equality equivalent to $C \simeq D$. Thus C is the unique finest/maximal-entropy common factor and minimizes residual dependence; at $D = C$, (CBL/CBIP/WFR) gives $\text{TC}(\gamma) = (n-1) \text{HSh}(\rho) + \text{SuppDep}_C + \text{WtDep}_C = (n-1) \text{HSh}(\rho) + \text{SuppDep}_C + \text{MargRest}_C + \text{FibDep}_R$. (LRF/DLCF) For a finite two-leg law γ , let $\text{LatRank}(\gamma)$ be the least $|L|$ in $\gamma = \sum_{\ell \in L} \lambda_\ell K_1^\ell \otimes K_2^\ell$, and let rank_+ denote the least number of nonzero nonnegative rank-one summands. Then $\text{LatRank}(\gamma) = \text{rank}_+(M_\gamma)$: latent terms give such summands, and any $M_\gamma = \sum_k a_k b_k^\top$ normalizes by $\lambda_k = \|a_k\|_1 \|b_k\|_1$, $K_1^k = a_k / \|a_k\|_1$, $K_2^k = b_k / \|b_k\|_1$; total mass gives $\sum_k \lambda_k = 1$. For $M_{m,\varepsilon} = (1-\varepsilon)I_m/m + \varepsilon J_m/m^2$, $m \geq 2$, $0 < \varepsilon < 1$, full support and uniform marginals give $(C, \text{SuppDep}) = (1, 0)$, while $\text{rank } M = m \leq \text{rank}_+ M \leq m$ (row decomposition), so $\text{LatRank} = m$; here $R = \mathbf{m}^2$ makes $\text{MargRest}_C = 0$, hence $M_{m,\varepsilon} \neq J_m/m^2$ gives $\text{FibDep}_R = \text{WtDep}_C > 0$. Put $w_m(\varepsilon) := D(\gamma_{m,\varepsilon} \| u_m^{\otimes 2})$; for $0 < \varepsilon < 1$ this is WtDep_C , and its continuous extension has $w_m(0) = \log m$, $w_m(1) = 0$. Thus every fixed $0 < t < \log 2$ occurs for every m : $(C, \text{SuppDep}_C, \text{MargRest}_C, \text{FibDep}_R) = (1, 0, 0, t)$ leaves latent rank unbounded.

For the independent no-least boundary, let $\text{LatCl}(\gamma)$ be latent factorizations modulo relabeling, ordered by deterministic coarsening $r : L \rightarrow L'$ with $r_* \lambda = \lambda'$ and $K_i^\ell = K_i'^{r(\ell)}$. For $0 < \rho < 1$, $\gamma_\rho(a, b) = \frac{1}{4}(1 + \rho ab) = \gamma_{2,1-\rho}$; every $u \in [\rho, 1]$, $v = \rho/u$, gives a uniform latent-bit realization $K_1(a|\ell) = \frac{1}{2}(1 + ua\ell)$, $K_2(b|\ell) = \frac{1}{2}(1 + vb\ell)$ because $uv = \rho$. Both displayed channels vary with ℓ and its only proper latent quotient is one point, hence it is minimal; different u are nonisomorphic by the invariant contrast u . Thus $\text{LatCl}(\gamma_\rho)$ has continuum many minima and no least element. Observable common quotient, minimal latent complexity and realized latent point are distinct. Finally $\gamma = \bigotimes_i \mu_i \Rightarrow R = \prod_i Y_i \iff [C \simeq \mathbf{1} \text{ \& CBGap} = \emptyset] \Rightarrow C \simeq \mathbf{1}$ is strict both ways; constant read-outs isolate N_j , (RFS/RCC) separates support gap from interaction, while a uniform diagonal, the uniform off-diagonal law on $\mathbf{3}^2$, and (LRF/DLCF) isolate respectively $\text{HSh}(\rho)$, $\text{SuppDep} = \log(3/2)$, and unbounded LatRank at fixed positive $\text{FibDep}_R = \text{WtDep}_C$.

Bounded residual Mayer–Vietoris. For nonzero Banach Z put $B_q := Z_b^1(R_q; Z)$. Then

$$\text{[BRMV]} \quad 0 \rightarrow B_c \xrightarrow{\delta^0} B_f \oplus B_g \xrightarrow{\delta^1} B_h \rightarrow 0, \quad \delta^0 \omega = (\omega|_{R_f}, -\omega|_{R_g}), \quad \delta^1(\alpha, \beta) = \alpha|_{R_h} + \beta|_{R_h}, \quad \text{Blnt}_Z(f, g) := H^1, H^0 = H^2 = 0. \quad (56)$$

Its additive comparison is Theorem II A 0 a at $|F| = 2, p = 1$.

Determination Čech envelope. For finite ordered F , $|F| = m \geq 2$, put $\widehat{R}_\emptyset := R_F^+ = \vee_{i \in F} R_{h_i}$ and $\widehat{R}_A := R_A^- = \cap_{i \in A} R_{h_i}$ for $A \neq \emptyset$, and set

$$\text{[DCE]} \quad \mathbf{C}_Z^p(F) := \bigoplus_{|A|=p} Z_b^1(\widehat{R}_A; Z), \quad (\delta^p \omega)_B := \sum_{r=0}^p (-1)^r \omega_{B \setminus \{b_r\}}|_{\widehat{R}_B}, \quad B = \{b_0 < \dots < b_p\}.$$

Then $\delta^{p+1} \delta^p = 0$, $H^0 \mathbf{C}_Z^\bullet(F) = H^m \mathbf{C}_Z^\bullet(F) = 0$ and $\text{BGLu}_Z^p(F) := H^p \mathbf{C}_Z^\bullet(F)$ ($1 \leq p < m$); $m = 2$ is (BRMV) up to the standard sign.

Determination Čech homology duality. Put $\widehat{q}_A : X \rightarrow X/\widehat{R}_A$ and let $\iota_{A,A \setminus a} : \mathbb{K}(\widehat{q}_A) \hookrightarrow \mathbb{K}(\widehat{q}_{A \setminus a})$ be induced by relation inclusion. Set $\mathbf{D}_p(F) = \bigoplus_{|A|=p} \mathbb{K}(\widehat{q}_A)$, $\partial|_A = \sum_{a \in A} (-1)^{\text{pos}_A(a)} \iota_{A,A \setminus a}$, $\text{DInt}_p(F) = H_p \mathbf{D}_\bullet(F)$, and for divisible abelian G , $\mathbf{A}_G^p(F) = \bigoplus_{|A|=p} Z^1(\widehat{R}_A; G)$ with the DCE differential. KSP naturality and injectivity give

$$\text{[DCHD]} \quad H^p \mathbf{A}_G^p(F) \cong \text{Hom}_{\mathbb{Z}}(\text{DInt}_p(F), G), \quad \chi_F^p : \text{BGLu}_Z^p(F) \rightarrow \text{Hom}_{\mathbb{Z}}(\text{DInt}_p(F), Z_{\text{add}}), \quad \text{Clnt}_Z^p(F) = \ker \chi_F^p. \quad (57)$$

Here χ_F^p is induced by $Z_b^1 \subseteq Z^1$ and is an isomorphism for finite X ; $|F| = 2, p = 1$ is UIR/BRMV, $\text{DInt}_1(F) \cong \mathbb{I}_F$ (write $\text{Clnt}_Z = \text{Clnt}_Z^1(F)$). Join generation and (BCR_o) give the bounded endpoint exactness.

On the full-product face $X = \prod_{i \in F} Y_i$, put $M_i = \mathbb{Z}^{(Y_i)}$, $I_i = \ker(M_i \xrightarrow{\varepsilon_i} \mathbb{Z})$, $M_F = \bigotimes_i M_i$ and $P_F^p := \sum_{|A|=p} (\bigotimes_{i \in A} I_i) \otimes (\bigotimes_{j \notin A} M_j)$, $P_F^{|F|+1} = 0$. For marginalization $m_B : M_F \rightarrow \bigotimes_{i \in B} M_i$,

$$\mathbb{I}_F = P_F^2, \quad P_F^p = \bigcap_{|B| < p} \ker m_B, \quad P_F^p / P_F^{p+1} \cong \bigoplus_{|A|=p} \bigotimes_{i \in A} I_i \cong \bigoplus_{|A|=p} \text{DInt}_{p-1}(A).$$

On each finite coordinate face A , $\text{DInt}_{|A|-1}(A) \cong \bigotimes_{i \in A} I_i$. Thus arity is canonical, with rank $\sum_{|A|=p} \prod_{i \in A} (|Y_i| - 1)$; for binary Y_i , $\theta_F = \bigotimes_i ([0] - [1])$ is nonzero in top grade with every proper marginal zero, so bounded alphabets support arbitrarily high joint order. For $Z = \mathbb{R}$, $I_i^{\mathbb{R}} := \ker(\mathbb{R}^{Y_i} \rightarrow \mathbb{R})$, $\text{BGLu}_{\mathbb{R}}^p(F) = 0$ for $p \neq m-1$ and $\text{BGLu}_{\mathbb{R}}^{m-1}(F) \cong (\bigotimes_i I_i^{\mathbb{R}})^*$; KSP and the augmentation decomposition split DCE into cones for $S \neq F$ and the augmented simplex boundary for $S = F$. Hence every full-product interaction grade is facewise determination homology.

The two noncollapse boundaries are explicit. On $X = \{0, 1, 2, 3\}$, $h_1 = (0, 0, 1, 1)$, $h_2 = (0, 1, 0, 1)$, $h_3 = (0, 0, 0, 1)$ have $R_F^- = \Delta$, $R_F^+ = X^2$, $\mathbb{I}_F = 0$, but $u_1 = (0, 0, 0, -1)$, $u_2 = (0, 0, 0, 1)$ give a non-global $(du_1|_{R_{h_1}}, du_2|_{R_{h_2}}, 0) \in \ker \delta^1$: a primitive forces $v_0 = v_1 = v_2$, $v_2 - v_3 = 1$, $v_1 - v_3 = -1$. Thus $\text{BGLu}_{\mathbb{R}}^1(F) \neq 0$, whereas $h_1 = h_2 = !$, $h_3 = I_X$ has the same endpoints and $\mathbb{I}_F = 0$ but acyclic DCE; higher determination gluing is therefore not endpoint/interaction data, and external site/torsor Čech data remain an attachment. For $X = \mathbb{Z}$,

$f(n) = \lfloor n/2 \rfloor$, $g(n) = \lfloor (n+1)/2 \rfloor$, one has $R_h = \Delta$, $R_c = X^2$, $\mathbb{I}_{\{f,g\}} = 0$, while $F(n) = n$ gives $(dF|_{R_f}, -dF|_{R_g}) \in \ker \delta^1 \setminus \text{im } \delta^0$, so $\text{Clnt}_{\mathbb{R}} \neq 0$. Thus bounded/additive collapse fails on infinite supports; no splitting or Hausdorff-completeness claim is made.

(PAC) If $L \vdash i \vdash R$ with i fully faithful, then $iRb \rightarrow b \rightarrow iLb$ and $iRb \simeq b \iff b \simeq iLb \iff b \in \text{EssIm } i$; for an order embedding i , conversely such a triple exists iff $\max\{a : ia \leq b\}$ and $\min\{a : b \leq ia\}$ exist for every b . In a complete lattice $\mathcal{L}$, for $\emptyset \neq S \subseteq \mathcal{L}$ put $\text{Env}_{\mathcal{L}}(S) = [\bigwedge S, \bigvee S]$; this is the least interval containing S , so $\emptyset \neq E \subseteq F$ gives $\bigwedge F \leq \bigwedge E \leq \bigvee E \leq \bigvee F$. If G acts by order automorphisms and $\iota_G : \mathcal{L}^G \hookrightarrow \mathcal{L}$, then $I_G x = \bigwedge_g g x$, $O_G x = \bigvee_g g x$, $\text{Env}_{\mathcal{L}}(Gx) = [I_G x, O_G x]$ and $O_G \vdash \iota_G \vdash I_G$; indeed for $y \in \mathcal{L}^G$, $O_G x \leq y \iff x \leq y$ and $y \leq I_G x \iff y \leq x$.

For a nonempty same-source Set probe family F , use the (CBR) kernels R_F^-, R_F^+ . Then $\text{Env}_{\text{Eq}(X)}\{R_{h_i} : i \in F\} = [R_F^-, R_F^+]$, the least-common-refinement and finest-common-factor kernels; $\emptyset \neq E \subseteq F$ gives $R_F^- \subseteq R_E^- \subseteq R_E^+ \subseteq R_F^+$ and the induced quotient chain. On finite full-support probability shadows, (CBIP/CFIM) makes the common-factor order quantitative: for $C \rightarrow D$, $(n-1)\text{H}_{\text{Sh}}(C \mid D) = \text{TotDep}_D - \text{TotDep}_C$, so refinement converts exactly that deterministic common information into reduced residual dependence. For $G \rightsquigarrow X$ by bijections, $g^*R = (T_g \times T_g)^{-1}R$ gives $I_G(R) = \cap_g g^*R$, $O_G(R) = \vee_g g^*R$ and (IOC) on the right, hence $X \rightarrow X/I_G(R_q) \rightarrow \text{im } q \rightarrow X/O_G(R_q)$. For $I \neq \emptyset$, $X/(\vee_{i \in I} R_{h_i}) \cong \text{colim}_{\emptyset \neq F \subseteq I} X/R_F^+$. Sharpness: on $\{0, 1\}^3$, swapping the last two coordinates with $q(a, b, d) = (a, b)$ gives $(I_G, O_G) = (\Delta, R_a)$; on $\{0, 1\}^2$, coordinate swap with first-coordinate q gives (Δ, X^2) . $j \perp (x, x') \iff h_j x = h_j x'$ gives by (ISB/DCC) $A^\perp = R_A^-, {}^\perp(A^\perp) = C_\infty(A)$, while $\Phi_+(A) = \vee_{j \in A} R_{h_j} \vdash \Psi_+(R) = \{j : R_{h_j} \subseteq R\}$ gives R_A^+ ; for $A \neq \emptyset$ Env is the exact two-sided probe envelope. On same-source Set descendants of (URRS), (RRCX/UIR/DCHD/DCE) retain route, interaction and higher gluing over it; slices without effective kernel-pair joins remain outside this identification. On the derived face (KRC), $K_v^-(B) = 0$ in $v_i K_u^-(v^*B) \rightarrow K_{v_u}^-(B) \rightarrow K_v^-(B) \rightarrow [1]$ gives $K_{v_u}^-(B) \simeq v_i K_u^-(v^*B)$: transport is not annihilation. (KTR/KGM) is the extension/gluing re-entry face, where one binary residual generates every finite depth. Thus complexification is this existing typed filtration/interaction/transport/gluing datum; no scalar primitive is added and the other residual readouts remain nonidentified.

Definition II.2 (Residual access representation and joint read-out). A formal residual-access presentation is the typed pair $\mathfrak{F}_{\text{acc}} = (\mathfrak{P}, \mathfrak{C})$,

$$\mathfrak{P} = \left(\mathcal{G}, \mu, \{(Y_r, w_r, Z_r, q_r, G_r, \text{Imp}_r)\}_{r \in \mathcal{R}}, L, \mathbf{P}, (P_t, \Lambda_t) \right),$$

$$\mathfrak{C} = \left(\mathcal{X}, \mathfrak{C}_{\text{cmp}}, \mathfrak{M}_{\text{cmp}}, \{(\mathcal{F}_r, C_r, i(r), \tau_r, \mathcal{K}_r^{\text{cl}}, \chi_r^{\text{cl}}, \nu_r, \text{Rec}_r, \text{Imp}_r^{\text{ev}})\}_{r \in \mathcal{R}_{\text{adm}}} \right).$$

Here $\mathcal{G}$ is nonempty compact metrizable, μ has full support, $\mathcal{R}$ is countable and $\mathcal{R}_{\text{adm}} \subseteq \mathcal{R}$; Y_r, Z_r are compact metrizable, w_r, q_r Borel, G_r closed, $\text{Imp}_r : Y_r \rightarrow [0, \infty]$ Borel, L continuous, $\mathbf{P}$ a Borel Markov kernel with $\mu \mathbf{P} \sim \mu$, and $P_t, \Lambda_t \geq 0$ locally integrable. The content carrier $\mathcal{X}$ is nonempty and $C_r : \mathcal{X} \rightarrow \mathcal{F}_r$. The comparison spine $\mathfrak{C}_{\text{cmp}} = ((I, \leq), \{\mathcal{T}_i, \Pi_{ij}\}_{i \leq j})$ is a countable directed inverse system of nonempty compact metrizable spaces with continuous bonds; $\mathfrak{M}_{\text{cmp}} = \{d_i\}_{i \in I}$ is optional, with each d_i inducing the topology of $\mathcal{T}_i$. Thus topology/binary extendability belong to $\mathfrak{C}_{\text{cmp}}$, while metric gluing and nearest repair require $\mathfrak{M}_{\text{cmp}}$.

The remaining maps are $i : \mathcal{R}_{\text{adm}} \rightarrow I$, $\tau_r : \mathcal{F}_r \rightarrow \mathcal{T}_{i(r)}$, $\chi_r^{\text{cl}} : \mathcal{F}_r \rightarrow \mathcal{K}_r^{\text{cl}}$, $\nu_r : \mathcal{F}_r \rightarrow [-\infty, 0]$, $\text{Rec}_r : \mathcal{F}_r \rightarrow Z_r$, and $\text{Imp}_r^{\text{ev}} : \mathcal{F}_r \rightarrow [0, \infty]$. A presentation-induced branch declares Borel $\beta_r : \mathcal{T}_{i(r)} \rightarrow Y_r$ and sets $\text{Imp}_r^{\text{ev}} = \text{Imp}_r \beta_r \tau_r$, $\text{Rec}_r = q_r \beta_r \tau_r$; otherwise these read-outs are independent. The family $\mathfrak{B}_{\text{int}} = \{\beta_r\}$ is an interface enrichment external only to the bare pair, retained as a typed stage attachment required by $\mathbf{PEv}^+$ and $\mathbf{PCmp}_c^+$ rather than promoted to a field of $\mathfrak{F}_{\text{acc}}$. The structural generator (P, Λ) produces

$$\text{Sol}_{P, \Lambda}(r)_t = e^{-\int_0^t \Lambda_u du} r + \int_0^t e^{-\int_s^t \Lambda_u du} P_s ds,$$

while $r = R_0$ is the initial-value fiber choice.

For $X \in \mathcal{X}$, $x_r = C_r X$ and

$$\text{JRead}(X; \mathfrak{F}_{\text{acc}}) = \{((r, x_r, \chi_r^{\text{cl}}(x_r), \nu_r(x_r), \text{Rec}_r(x_r), \text{Imp}_r^{\text{ev}}(x_r)))\}_{r \in \mathcal{R}_{\text{adm}}}, \text{Sol}_{P, \Lambda}\}. \quad (58)$$

With $C_{\mathfrak{F}}(X) = (C_r X)_r$, reachable read-outs factor as $\text{JRead} = \Theta_{\mathfrak{F}} C_{\mathfrak{F}}$ and projection to $(x_r)_r$ left-inverts $\Theta_{\mathfrak{F}}$; hence $R_{\text{JRead}} = R_{C_{\mathfrak{F}}} = \bigcap_{r \in \mathcal{R}_{\text{adm}}} R_{C_r}$. Here $\nu_r = -\delta_r$ on the generated inverse-limit specialization. Thus $\mathfrak{F}_{\text{acc}}$ is a typed representation carrier for the doctrine, not the doctrine itself. In particular its nonemptiness and probability hypotheses belong to the post-origin realization domain; they are not premises of the origin calculus. The (SGO/ZBC/SCF) separation separates absence of a sort, a collapsed boundary carrier and a realized nonempty presentation, so no existence assumption in this definition may be used to prove source-free origin.

For the pullback layers below, a typed access arrow is $u = (\rho, \{\eta_{r'}, \zeta_{r'}\})$, with $\rho : \mathcal{R}' \rightarrow \mathcal{R}$, $\rho(\mathcal{R}'_{\text{adm}}) \subseteq \mathcal{R}_{\text{adm}}$, Borel $\eta_{r'} : Y_{\rho(r')} \rightarrow Y'_{r'}$, $\zeta_{r'} : Z_{\rho(r')} \rightarrow Z'_{r'}$, and $q'_{r'} \eta_{r'} = \zeta_{r'} q_{\rho(r')}$, $G_{\rho(r')} \subseteq \eta_{r'}^{-1}(G'_{r'})$, $\text{Imp}'_{r'} \eta_{r'} = \text{Imp}_{\rho(r')}$. These arrows compose componentwise; neutral values cover undeclared attachments.

b. Presented-evaluation pullback and liftability boundary. Put $\text{ev}_r = \beta_r \tau_r C_r$. A $\mathbf{PEv}^+$ arrow over $u = (\rho, \eta, \zeta)$ is $\psi : \mathcal{X} \rightarrow \mathcal{X}'$ with $\eta_{r'} \text{ev}_{\rho(r')} = \text{ev}'_{r'} \psi$, and its cartesian carrier is $\mathcal{X}_u = \{(x', (y_r)_r) : \eta_{r'} y_{\rho(r')} = \text{ev}'_{r'} x' \ \forall r' \in \mathcal{R}'_{\text{adm}}\}$. When nonempty, projection $\mathcal{X}_u \rightarrow \mathcal{X}'$ is the (URE/FRF) pullback; over $u \circ v$ the unique factor is $z \mapsto (hz, (\eta_v^{\text{ev}} \text{ev}_{\rho_v(r)}^{\text{ev}} z)_r)$. With $J_r(u) = \rho^{-1}(r) \cap \mathcal{R}'_{\text{adm}}$ and $\Delta_r^u(y) = (\eta_{r'} y)_{r' \in J_r(u)}$, call u JES when every Δ_r^u is onto. Then $\mathbf{PEv}^+|_{\text{Pres}_{\text{jes}}} \rightarrow \text{Pres}_{\text{jes}}$ is fibered, and JES is maximal for guaranteed nonempty lifts: $\{0\} \rightarrow \{0, 1\}$ with image 0 misses target 1 (empty content is the only extension past this regime). The cartesian arrow transports $\text{Rec}'_{r'}(C'_{r'} \psi X) = \zeta_{r'} \text{Rec}_{\rho(r')}(C_{\rho(r')} X)$, $\text{Imp}'_{r'} \text{ev}_{\rho(r')} (C'_{r'} \psi X) = \text{Imp}_{\rho(r')}^{\text{ev}} (C_{\rho(r')} X)$, and $\text{ev}_{\rho(r')}(X) \in G_{\rho(r')}$ implies $\text{ev}'_{r'}(\psi X) \in G'_{r'}$; record, impact and forward extendability are therefore descendants.

c. Comparison and classicalization lift spine. A comparison lift over $u = (\rho, \eta, \zeta; \psi)$ is $(\kappa, \theta, \lambda)$ with monotone $\kappa : I' \rightarrow I$, continuous $\theta_j : \mathcal{T}_{\kappa(j)} \rightarrow \mathcal{T}_j$, $\lambda_{r'} : \mathcal{F}_{\rho(r')} \rightarrow \mathcal{F}'_{r'}$, and

$$ki'(r') = i\rho(r'), \quad \theta_j \Pi_{\kappa(j)\kappa(k)} = \Pi'_{jk} \theta_k, \quad C'_{r'} \psi = \lambda_{r'} C_{\rho(r')}, \quad \tau'_{r'} \lambda_{r'} = \theta_{i'(r')} \tau_{\rho(r')}, \quad \beta'_{r'} \theta_{i'(r')} = \eta_{r'} \beta_{\rho(r')}. \quad (59)$$

If $B_{r,j}(u) = \{r' : \rho(r') = r, i'(r') = j\}$, every lift equalizes $\tau'_{r'} C'_{r'} \psi$ on $B_{r,j}$; for cartesian $\mathbf{PEv}$ and surjective ψ the targets themselves coincide. The shared $\{0, 1\}$ stage with profiles $(0, 0)$ and $(0, 1)$ proves failure otherwise, so $\text{Pres}_{\text{jes}}^{\text{sep}}$ (admissible ρ injective) is the largest wide presentation-only class removing this collision.

Let $\mathbf{PCmp}_c^+$ retain $(\mathcal{X}, \mathcal{F}, C, \mathfrak{C}_{\text{cmp}}, i, \tau, \beta)$ with continuous bridges, and restrict the base to bijective admissible ρ and continuous η . For $A_j = \{r' : i'(r') \leq j\}$ set $\mathcal{T}_j^u = \{(t, y) : \eta_{r'} y_{r'} = \beta'_{r'} \Pi'_{i'(r'), j} t \ (r' \in A_j)\}$, $\theta_j(t, y) = t$, $\Pi'_{jk}(t, y) = (\Pi'_{jk} t, y|_{A_j})$, let $i^u(\rho r') = i'(r')$,

β_{pr}^u , project to $y_{r'}$, and $\mathcal{F}_{pr'}^u = \mathcal{F}_{r'} \times_{\mathcal{T}_{i'(r')}} \mathcal{T}_{i'(r')}^u$, $X^u = X' \times_{\prod \mathcal{F}_{r'}} \prod \mathcal{F}_{pr'}^u$. Compactness/JES gives nonemptiness and the pullback universal property gives the (URE/FRF) fibration $\mathbf{PCmp}_c^+ \rightarrow \mathbf{Pres}_{\text{jes}}^{\text{bij},c}$. The interface is an attachment: with $X = \mathcal{F} = \{*\}$, $\mathcal{T} = Y = \{0, 1\}$, $\tau(*) = 0$, constant q , Imp , the bridges $\beta_0 \equiv 0$ and $\beta_1 = \text{id}$ have the same bare/reachable data, admit a lift $\beta_0 \rightarrow \beta_1$ via $\theta \equiv 0$, and no reverse lift. A static readout pulls back as $h_{pr'}^u = h_{r'}^u \lambda_{r'}^u$, and replaces h_r exactly when fiber-constant (DI); metric ν retains its Aff_+ lax law.

For $U_{\text{cmp}} : \mathbf{PCmp}_c^+ \rightarrow \mathbf{PEv}^+$, JES makes $\pi_u(x) = (\psi^u x, (e_r^u(x))_r)$ and $Q_u(x', (y_r)) = x'$ surjective. Their (URE) composite is

$$\begin{aligned} \text{(GRT)} \quad X^u &\xrightarrow{\pi_u} X_u \xrightarrow{Q_u} X', \\ \Delta &\subseteq R_u^{\text{cmp}} := R_{\pi_u} \subseteq R_u^{\text{ev}} := R_{Q_u \pi_u}, \\ X^u/R_u^{\text{cmp}} &\cong X_u, \quad X^u/R_u^{\text{ev}} \cong X', \quad R_u^{\text{ev}}/R_u^{\text{cmp}} \cong R_{Q_u}. \end{aligned} \quad (60)$$

Thus $\text{CmpFib}_u(\xi) = \pi_u^{-1}\{\xi\}$ and $Q_u^{-1}\{x'\}$ are successive independent grades, trivial exactly when π_u, Q_u are injective. The $|X_u| = 4$, $|X^u| = 8$ witness gives comparison fibres of size 2, while $\{0, 1\} \rightarrow \{*\}$ gives strict evaluation grade. Compatible actions use (KSP/FRP); other attachments stay outside the tower.

d. Operator-algebraic skeleton, access-presentation arrows and transport. The time-zero measure-algebraic skeleton and its dynamic marked extension are $\mathfrak{S}_0^\# = (\mathcal{A}, \omega, \{Q_r \subseteq \mathcal{N}_r, e_r\}_{r \in \mathcal{R}}, \ell)$, $\mathfrak{S}^\# = (\mathfrak{S}_0^\#, T, (P, \Lambda))$, where $\mathcal{A}$ is abelian, ω faithful normal, $Q_r \subseteq \mathcal{N}_r \subseteq \mathcal{A}$, $e_r \in \text{Proj}(\mathcal{N}_r)$, $\ell = \ell^*$, T normal unital positive with $\omega \circ T \sim \omega$, and (P, Λ) the non-negative residual generator entering the calculus below. Its canonical realization has $\mathcal{A} = L^\infty(\mathcal{G}, \mu)$, $\omega(a) = \int a \, d\mu$, $\mathcal{N}_r = w_r^* L^\infty(Y_r)$, and $Q_r = (q_r \circ w_r)^* L^\infty(Z_r)$, with $e_r = \mathbf{1}_{G_r} \circ w_r$, $\ell = L$, $(Ta)(x) = \int a(y) P_x(dy)$, with residual generator (P, Λ) . Conversely, close a countable weak- $*$ generating self-adjoint set (including the declared marks, ℓ and prescribed countable C^* -data) under T and C^* -operations. Its norm closure is a separable T -invariant core C with $C'' = \mathcal{A}$. Gelfand duality, Riesz representation and faithful ω give the compact-metrizable full-support Feller model, with $\omega \circ T \sim \omega \iff \mu P \sim \mu$; standard-Borel spatiality realizes $\mathcal{N}_r, Q_r$ by w_r, q_r and transports the declared L^0/L^1 attachments, natural extension and type. This supplies the compact Feller representatives of (FTRF) below; it is a realization map, not a second invariant. By (DI), a read-out of $(\mathfrak{P}, \mathfrak{E}) \mapsto \mathfrak{S}^\#$ descends exactly when fiber-constant; the finite typed witness cube $S7$ witnesses properness and the independent evaluation/ambient fibres.

For $f : X \rightarrow Y$ and abelian G , put $\mathcal{R}_G(f)_y := G^{f^{-1}(y)}/\Delta_y G$ ($y \in f(X)$), with $\Delta_y G$ the constants, and $\mathcal{R}_G(f) := (\mathcal{R}_G(f)_y)_y$. An access-presentation morphism is a typed access arrow together with a Borel $\varphi : \mathcal{G} \rightarrow \mathcal{G}'$, modulo μ -a.e. equality, such that $\varphi_* \mu = \mu'$, $w_{r'} \varphi = \eta_{r'} w_{\rho(r')}$, $L' \varphi = L$, $\varphi_* P_x = P'_{\varphi(x)}$, $P' = P$, $\Lambda' = \Lambda$ a.e. (the kernel identity is representative-independent since $\mu P \ll \mu$). The residual equality is essential: without it, $(0, 1)$ and $(1, 1)$ would become isomorphic. Composition is componentwise and undeclared fields retain their neutral values. The induced $j_u(a) = a \circ \varphi : \mathcal{A}' \hookrightarrow \mathcal{A}$ is normal, unital, injective, state preserving and ω -expected, with $j_{vu} = j_u j_v$ and $j_u Q_{r'} \subseteq Q_{\rho(r')} \subseteq \mathcal{N}_{\rho(r')}$, $j_u \mathcal{N}_{r'} \subseteq \mathcal{N}_{\rho(r')}$, $e_{\rho(r')} \leq j_u e_{r'}$, $T j_u = j_u T'$, $j_u \ell' = \ell$, $(P', \Lambda') = (P, \Lambda)$ a.e. The pushforward/intertwining clauses give these identities, while abelianness makes the range expected. *Representative-point descent.* Let $\mathbf{SBP}_{\text{rep}}$ have standard-Borel probability spaces and actual Borel measure-preserving maps and let $Q : \mathbf{SBP}_{\text{rep}} \rightarrow \mathbf{SBP}_{\text{ae}}$ quotient arrows by source-a.e. equality. With $\text{Pnt}(X, \mu) = X$ and $\text{Atm}(X, \mu) = \{x : \mu\{x\} > 0\}$, $(\text{PRF}) \int \text{Pnt} \rightarrow \mathbf{SBP}_{\text{rep}}$ is a discrete opfibration, while $\text{Atm} = \text{Atm}Q$ for a unique $\text{Atm} : \mathbf{SBP}_{\text{ae}} \rightarrow \mathbf{Set}$ and is the largest subfunctor of Pnt descending through Q . For $[f] : (X, \mu) \rightarrow (Y, \nu)$ put $\text{PntFib}_{[f]}(x) = \{g(x) : g \in [f]\}$. Then $(\text{PDB/NPF}) \text{ev}_x : L^\infty(X, \mu) \rightarrow \mathbb{C}$ is well-defined iff $\mu\{x\} > 0$, and $\text{PntFib}_{[f]}(x) = \{f(x)\}$ there, whereas $\text{PntFib}_{[f]}(x) = Y$ when $\mu\{x\} = 0$. If finite G acts measure-preservingly, Z is a finite G -set, $e : X \rightarrow Z$ is equivariant Borel and $K = G_x$, then $(\text{EAF}) \{e'(x) : e' = e \mu\text{-a.e.}, e' \text{ Borel } G\text{-equivariant}\} = \{e(x)\}$ for $\mu\{x\} > 0$ and equals Z^K for $\mu\{x\} = 0$. Indeed atoms are preserved and a.e.-equal maps agree there; a null x admits equal-a.e. maps to $(\{0, 1\}, \delta_0)$ with values 0, 1 at x , and arbitrary null-point modification gives the whole target fiber. On a null finite orbit, every $z \in Z^K$ extends as $gx \mapsto gz$, while equivariance forces the converse. Thus non-atomic pointing and $\mathcal{R}_G$ are representative-level fibers; all a.e.-typed fields above still transport exactly.

Complete-lattice Kan/Galois adjunction. For $f : X \rightarrow Y$ define $(\exists_f a)(y) = \bigvee_{f(x)=y} a(x)$, $(\forall_f a)(y) = \bigwedge_{f(x)=y} a(x)$. The defining order inequalities give $\exists_f \dashv f^* \dashv \forall_f$ directly; a pullback $X' = X \times_Y Y'$ identifies the relevant fibres, hence $b^* \exists_f = \exists_{f'} b^*$ and $b^* \forall_f = \forall_{f'} b^*$, including empty fibres. These are the complete-lattice Kan formulas of⁴ (Chs. IV, X), used here only after the fibrewise proof. If $C \subseteq L$ is order-embedded and $s : C \rightarrow L$ is monotone with $c \leq s(c)$, put $s_!(x) = x \vee \bigvee_{c \leq x} s(c)$, $s_*(x) = \bigwedge_{x \leq c} s(c)$ (with empty join/meet $\perp/\top$). Then $(\text{KEX}) \{F \in \text{PreCl}(L) : F|_C = s\} = [s_!, s_*]_{\text{PreCl}(L)}$, a complete interval: extensivity/monotonicity force $s_! \leq F \leq s_*$ and both endpoints extend s .

Lossless conjugacy of closure genealogies. For an order isomorphism $J : L \rightarrow L'$ of complete lattices set $\Theta_J(F) = JFJ^{-1}$. It is an order isomorphism on preclosures and, for all ordinals α , $(\text{LCG}) (\Theta_J F)^\alpha J = JF^\alpha$, $\overline{\Theta_J F} = \Theta_J(\overline{F})$, $J \text{Fix } F = \text{Fix}(\Theta_J F)$, $\text{clord}_{\Theta_J F}(Jx) = \text{clord}_F(x)$. For $C' = J(C)$ and $s' = JsJ^{-1}$, moreover $\Theta_J(s_!) = s'_!$, $\Theta_J(s_*) = s'_*$ and $\Theta_J([s_!, s_*]_{\text{PreCl}(L)}) = [s'_!, s'_*]_{\text{PreCl}(L')}$. This follows from preservation of arbitrary joins/meets and transfinite induction. Thus chart change transports the whole closure genealogy and its (KEX) law fibre without selecting a law.

e. Reflective genealogy envelope. For complete L , let $\mathcal{W}_\bullet^{F,x} = (F^\alpha x)_{\alpha \leq \kappa_F(x)}$, $\kappa_F(x) = \text{clord}_F(x)$, and corestrict $\text{Orb}_L : (x, F) \mapsto (x, \mathcal{W}_\bullet^{F,x})$ to $\text{GBase}(L)$. For $b = (x, \mathcal{W}_\bullet)$, put $s_{\mathcal{W}}(\mathcal{W}_\alpha) = \mathcal{W}_{\min\{\alpha+1, \kappa\}}$ on $\text{Ch}_{\mathcal{W}} = \text{im } \mathcal{W}_\bullet$ and $(F_{\mathcal{W}}^{\min}, F_{\mathcal{W}}^{\max}) = ((s_{\mathcal{W}})_!, (s_{\mathcal{W}})_*)$. Then (KEX) gives $(\text{GRR}) (F_{\mathcal{W}}^{\min})^\alpha x = \mathcal{W}_\alpha$, $\text{clord}_{F_{\mathcal{W}}^{\min}}(x) = \kappa$, and $(\text{GRI}) \text{Orb}_L^{-1}(b) = \{F : F|_{\text{Ch}_{\mathcal{W}}} = s_{\mathcal{W}}\} = [F_{\mathcal{W}}^{\min}, F_{\mathcal{W}}^{\max}]$. Hence $(\text{URT/RGE}) \text{Tot}_{b \in \text{GBase}(L)} \text{Orb}_L^{-1}(b) \cong L \times \text{PreCl}(L)$. With $\text{End}_L(x, \mathcal{W}) = \mathcal{W}_\kappa$, $\text{PTerm} = \text{End}_L \text{Orb}_L$, $\text{Term}_x(F) := F^{\kappa_F(x)} x$, and $\text{Gen}_x(w) = \{\mathcal{W} : (x, \mathcal{W}) \in \text{GBase}(L), \text{End}_L(x, \mathcal{W}) = w\}$, $\text{PTerm}^{-1}(w) \cong \text{Tot}_{b \in \text{End}_L^{-1}(w)} \text{Orb}_L^{-1}(b)$ by (RFE). A typed family $A = (A_b)$, discrete unless transport is supplied, replaces each fibre by $\text{Orb}_L^{-1}(b) \times A_b$ without changing terminal-fibre law. For order isomorphism $J : L \rightarrow L'$, $(\text{PTG}) \text{Term}_{Jx}(\Theta_J F) = J \text{Term}_x(F)$ and $\Theta_J(\text{Orb}_L^{-1}(b)) = \text{Orb}_{L'}^{-1}(Jb)$. At $x = \perp$, this totalization is $\text{PreCl}(L)$, $\text{Term}_\perp(F) = \overline{F}(\perp) = \min \text{Fix } F$, and the two-point chain keeps the law nonselected. On the Boolean face $a = \mathbf{1}_P : X \rightarrow \mathbf{2}$, (TOM) gives $\text{Svapna}_f(P) := \exists_f \mathbf{1}_P$ and $\text{Catuskoti}_f^{\text{fib}}(P) := \text{Pol}_f(P) = (\text{Svapna}_f(P), \text{Svapna}_f(P^c))$. For $C_{ab} = \text{Pol}_f(P)^{-1}(a, b)$, $C_{11} = f(P) \cap f(P^c)$, $C_{00} = Y \setminus f(X)$, and $C_{11} = \emptyset \iff M_f \subseteq M_a \iff P \in \text{im } f^*$, $C_{00} = \emptyset \iff N_f = \Delta_Y \iff f(X) = Y$; C_{10}, C_{01} are the other support fibres. Beck-Chevalley pulls them back; if $f = pg$, Kan composition gives $\text{Pol}_f(P) = (\exists_p \text{Svapna}_g(P), \exists_p \text{Svapna}_g(P^c))$ and $pC_{11}(g, P) \subseteq C_{11}(f, P)$;

corestriction/descent kill the N_f/M_f defects and give $4 \rightarrow 3 \rightarrow 2$. For $\mathbf{h} = (h_j : X \rightarrow Z_j)_{j \in J}$, $F \subseteq J$, put $H_F = (h_j)_{j \in F}$, $\text{Dep}_F = \text{im } H_F$, $Q_F = H_F : X \rightarrow \text{Dep}_F$ (empty product a point), and for $E \subseteq F$ let $p_{F,E} : \text{Dep}_F \rightarrow \text{Dep}_E$ delete coordinates. Then

$$\begin{aligned} [DI] \quad Q_E &= p_{F,E} Q_F, \quad p_{F,D} = p_{E,D} p_{F,E}, \quad \text{cl}_{DI}(F) := \{j : R_{Q_F} \subseteq R_{h_j}\}, \\ j \in \text{cl}_{DI}(F) &\iff \text{DepRes}_{Q_F}(h_j) = 0 \iff p_{F \cup \{j\}, F} \text{ bijective.} \end{aligned} \quad (61)$$

By first isomorphism $Q_F \simeq_X q_F$ for the (KGC) quotient; (KGC/DCC/RRF) regenerate kernel, tabulation and closure, so $\text{cl}_{DI} = C_\infty$ on Set (not C_{fin} in general), while (RPL) identifies the middle zero with functionality of $\text{supp}_0 T_{Q_F, h_j}$. Thus DI is the concrete image presentation of TOM-neutral closure; $\text{Dep}_{f,h}[y] = h(f^{-1}\{y\})$ is its corestricted nonempty-support TOM fibre, refined/composed by (URE/URT).

f. Monoid saturation, invariant observability, and controlled access. Let M act on a complete lattice L by arbitrary-join endomorphisms $A_e = I$, $A_{mn} = A_m A_n$. With $L^M = \{z : \bigwedge_{m \in M} A_m z \leq z\}$, $\text{MSat}_M(x) = \bigvee_m A_m x = \min\{z \geq x : A_m z \leq z\}$ gives $\overline{\text{MSat}}_M : L \rightarrow L^M$; $L^M \hookrightarrow L$; by (URE/SDR), $\text{MSat}_M^2 = \text{MSat}_M$ and $\text{Fix MSat}_M = L^M$. For a right action $T_{mn} = T_n T_m$ on X , its order-dual equivalence reflector is $I_M(R) = \bigcap_m (T_m \times T_m)^{-1} R = \max\{S \subseteq R : (T_m \times T_m)S \subseteq S\}$, hence $\iota_M \dashv I_M$, $\text{Fix } I_M = \text{Eq}_M(X)$ and $I_M(\bigcap_j R_j) = \bigcap_j I_M(R_j)$. Thus $Q_M(q) : X \rightarrow X/I_M(R_q)$ is coarsest action-carrying, with $Q_M(q) \simeq_X q \iff I_M(R_q) = R_q \iff M \subseteq \text{Stab}(q)$: a cut is dynamically closed iff every action descends; no realization is selected.

Equivariant behavioral pro-envelope. For nonempty I_0 , readouts $q_i : X \rightarrow Y_i$ and a right M -action, put $h_{i,m} = q_i T_m$, apply (KGC_{pro}/YFF) to $A = I_0 \times M$, and write $\beta = H_A$, $D_F = \text{im } H_F$, $Q_{q,M} = q_A$, $B_{q,M} = B_A$ and $\text{BehPro}_{q,M} = (D_F, p_{F,E})_{\text{BEH}}$. Then $R_\beta = \bigcap_{i,m} R_{q_i T_m} = I_M(\bigcap_i R_{q_i})$, $Q_{q,M} \xrightarrow[\beta]{} B_{q,M} \subseteq \prod_{i,m} Y_i$. Let $Q_q = X/(\bigcap_i R_{q_i})$. By (KGC/YFF), $Q_{q,M}$ is its least dynamic refinement; the (RRCX) residual $\mathbb{K}(Q_q)/\mathbb{K}(Q_{q,M})$ vanishes exactly at dynamic closure. Thus (IOC/BEH) is KGC dynamic closure. For corestricted $p : X \rightarrow P$, (BRC) $R_p \subseteq R_\beta \iff Q_{q,M} = c_p p$ and $R_\beta \subseteq R_p \iff p = d_p Q_{q,M}$, uniquely; equality gives inverse factors. If $p T_m = S_m p$, c_p is equivariant; otherwise its fibres are realization distinctions invisible to the orbit profile; $q_i = h_i q$, $q T_m = S_m q$ is included. The induced action satisfies (ELQ) $m \equiv_q n \iff \tilde{T}_m = \tilde{T}_n \iff \sigma_m|_B = \sigma_n|_B$, $M_q^{\text{eff}} = M/\equiv_q$, with faithful quotient action and $\sigma_{mn} = \sigma_n \sigma_m$. Moreover $H_F T_n = H_{nF}$ induces deletion-compatible $D_{nF} \rightarrow D_F$, equivalently $(\sigma_n b)_{i,m} = b_{i,nm}$ and $\tilde{\beta} \tilde{T}_n = \sigma_n \tilde{\beta}$. On the discrete face, $\text{Cpl } \mathcal{U}_\beta(Q_{q,M}) \cong \varprojlim_F D_F \cong \overline{B_{q,M}}^{\text{prod}}$, $\mathcal{U}_\beta = (R_{H_F}/R_\beta)_F$, and $(\tilde{T}_n \times \tilde{T}_n)^{-1} U_F = U_{nF}$ and $m \equiv_q n \iff \tilde{T}_m = \tilde{T}_n$; trivial M is the static joint face and completion changes points, not effective law.

All finite record shells are faces of BEH. For the cyclic action $\langle T \rangle$ and one readout q , set $H_n = O_n = (q, qT, \dots, qT^n)$ and $R_\infty = R_T^\infty(q) = I_{\langle T \rangle}(R_q)$. Then (DHR) $R_{H_n} = \bigcap_{k \leq n} (T^k \times T^k)^{-1} R_q$, $H_{n+1} = (H_n, qT^{n+1})$, and (DRX/RRCX) is the exact next-shell sequence. Constant q makes every shell zero; $q x_0 = q x_1$ but $q T x_0 \neq q T x_1$ gives the (DCC)-least first write; impact and occurrence remain separately typed. On $\mathbb{Z}/4$, $T : x \mapsto x + 1$, $q = \mathbf{1}_{\{0\}}$, the involution $x \mapsto -x$ swaps the first $T^{\pm 1}$ shells while both saturate to Δ_X . A plateau is terminal; (PQTF) gives $X/R_\infty \cong \varprojlim_n \text{im } H_n$, while the one-sided binary shift has $R_\infty = \Delta$ but non-singleton two-sided past fibres. For a right A^* -action, $H_n^A = (q T_w)_{|w| \leq n}$, $L_{n+1}^A = (q T_w)_{|w| \leq n+1}$, so $H_{n+1}^A = (H_n^A, L_{n+1}^A)$, and (COA) $R_{H_n^A} = \bigcap_{|w| \leq n} (T_w \times T_w)^{-1} R_q$, $R_\infty^A(q) = I_{A^*}(R_q)$. Hence $\Pi_n^{\text{fac}} = \{\pi : R_{H_n^A} \subseteq R_\pi\}$ increases and (H2P) $\pi \in \Pi_n^{\text{fac}} \iff \text{DepRes}_{H_n^A}(\pi) = 0$; for $\pi \in \Pi_{n+1}^{\text{fac}}$ the (DRX) row $(Q, F, G) = (H_n^A, L_{n+1}^A, \pi)$ is exact, with nonzero terminal term at first entry, while $\mathfrak{E}_{H_n^A}(F)$ and $\kappa_{H_n^A}(F)$ decrease on the affine record-descent regime. For prefix records $H_w^\partial = (q T_u)_{u \leq w}$, (RRCX) gives $H_{wa}^\partial = (H_w^\partial, q T_{wa})$. On $X = \{0, 1, 2, 3\}$, $q = \mathbf{1}_{\{3\}}$, $T_a = (2\ 3)$, $T_b = (1\ 2\ 3)$, $q T_a = q T_b = \mathbf{1}_{\{2\}}$ but $0 \sim_{H_{ab}^\partial} 1$ and $R_{H_{ba}^\partial} = \Delta_X$; this is the strict (RGS_{desc}) path-over-one-step branch.

On the cyclic BEH face let $B = X/R_\infty$, $\pi : X \rightarrow B$, $\tilde{\pi} \pi = \pi T$. The corestricted identification is (BCF/BCL/BRU):

$$\beta_{q,T}(x) = (q T^k x)_{k \geq 0}, \quad B \xrightarrow[\tilde{\pi}]{} \text{im } \beta_{q,T} \subseteq Q^\mathbb{N}, \quad J([x]) = (q T^k x)_k, \quad J\tilde{T} = \sigma J. \quad (\text{IPD/BRT})$$

For the descended coordinates $\tilde{q}_k$ put $U_N(u) = \bigcap_{k \leq N} \tilde{q}_k^{-1}\{\tilde{q}_k(u)\}$ and $E_N = \bigcap_{k \leq N} R_{\tilde{q}_k}$. Then $\bigcap_N U_N(u) = \{u\}$, $\pi^{-1}(U_N(\pi x)) = [x]_{R_{H_N}}$, and the E_N define the separated non-Archimedean cylinder topology

$$(\text{BRU}) \quad \tau_{\text{beh}} = \min\{\tau : \tilde{q} : (B, \tau) \rightarrow Q_{\text{disc}}, \tilde{T} : (B, \tau) \rightarrow (B, \tau) \text{ continuous}\}.$$

Its intrinsic filtration has gauge $2^{-\ell(u,v)}$; the filtration, not this numerical gauge, is the invariant, $\ell(u,v) = \min\{k : \tilde{q} \tilde{T}^k u \neq \tilde{q} \tilde{T}^k v\}$. For $z \in \text{Fix } \tilde{T}$, (BFH) $\tilde{T}^n u \rightarrow z \iff \exists N \forall n \geq N : \tilde{T}^n u = z$. On this finite-hit locus, with $F = \text{Fix } \tilde{T}$, $\tau_F(u) = \min\{n : \tilde{T}^n u \in F\}$ and $E_{\tilde{T}}(u) = \tilde{T}^{\tau_F(u)} u$, (DAC/EQR) $E_{\tilde{T}}|_F = I$, $E_{\tilde{T}} \tilde{T} = \tilde{T} E_{\tilde{T}} = E_{\tilde{T}}$, $E_{\tilde{T}}^2 = E_{\tilde{T}}$, and $B \xrightarrow[\tilde{T}]{} B \xrightarrow{E_{\tilde{T}}} F$ is the Set coequalizer: every $h \tilde{T} = h$ factors uniquely through F . Multiple fixed basins therefore remain a fibre, not a selected endpoint.

For a convergent sequence $u_m \rightarrow u$ let $\mathfrak{U}_m^{\text{rec}}(u) = \{V \in \mathcal{N}_{\text{rec}}(u) : u_j \in V \ (j \geq m)\}$. They increase and are finite-intersection filters in $\mathcal{N}_{\text{rec}}(u)$, with $\bigcup_m \mathfrak{U}_m^{\text{rec}}(u) = \mathcal{N}_{\text{rec}}(u)$, and with $v_{\text{rec}}(u) = \min\{N : U_N(u) = \{u\}\}$, $v_{\text{rec}}(u) < \infty \iff u$ is isolated in (B, τ_{beh}) . If the rank is infinite, choose $u_N \in U_N(u) \setminus \{u\}$ to get $u_N \rightarrow u$ without exact occurrence; for finite B one horizon satisfies $R_N = R_\infty$. With $D_N = \text{im } O_N \cong X/R_N$,

$$\widehat{B}_{\text{rec}} = \varprojlim_N D_N, \quad \iota_{\text{rec}} : B \hookrightarrow \widehat{B}_{\text{rec}}, \quad \text{LimRes}_{\text{rec}} = \widehat{B}_{\text{rec}} \setminus \iota_{\text{rec}}(B),$$

and every projection of $\iota_{\text{rec}}(B)$ onto D_N is onto; RCD/CRX applied to the induced maps $B \rightarrow D_N$ with $a = I_B$ gives (RLC) $\widehat{B}_{\text{rec}} = \text{FSh}(\iota_{\text{rec}} B)$ and $\text{LimRes}_{\text{rec}} = \emptyset \iff \iota_{\text{rec}}$ is onto. Since ι_{rec} is injective, the associated (CRX) has $H_1 = 0$: record completion is its pure H_0 face. With the identity read-out, $\widehat{\text{Jkey}}_{J, I_B} \cong B$ and ι_{J, I_B} is onto, so RLC is the terminal target-extinction face of (GTS): its CRX H_0 survives while $\text{JointLimRes} = 0$. The same base-normalized tower is (DJK2), so $\text{LimRes}_{\text{rec}}$ is exactly its degree-one target-extinction birth, with tail quotient VEx retained separately. Finite-support binary sequences with shift give the dense-proper-image witness; set-level fibres of ι_{rec} are empty or terminal, so HAH/PRDL isotropy is independent. On the compact exact-record face, compactness makes D_N finite and $\iota_{\text{rec}}(B)$ closed, hence

$$B \text{ compact} \iff [|D_N| < \infty \ (N < \omega) \ \& \ \text{LimRes}_{\text{rec}} = \emptyset].$$

If all orbits hit F there, $|B| < \infty \iff \sup_B \tau_F < \infty \iff \exists N : R_N = R_\infty$. Indeed $A_N = \{u : \tau_F(u) \leq N\}$ injects into finite D_N ; hence the infinite branch is countable with $\tau_F(B) = \mathbb{N}$. For sharpness set $B_\infty = \{b_*\} \sqcup \{x_n : n \geq 1\}$, $x_0 = b_*$, $\tilde{T} b_* = b_*$, $\tilde{T} x_n = x_{n-1}$ ($n \geq 1$), and $\tilde{q}(x_1) = 1$, $\tilde{q}(b_*) = \tilde{q}(x_n) = 0$ ($n \geq 2$); its records are $0^\infty, 0^{n-1} 1 0^\infty$, with no finite horizon.

The discrete record carrier is not a continuum replacement: for a nontrivial unital C^* -algebra, $\mathcal{S}(\mathcal{A})$ is weak- $*$ compact convex and therefore connected, so every continuous discrete readout is constant; a separating $B_0 \subset \mathcal{A}$ instead gives the weak- $*$ embedding $J_{B_0}^{\text{op}} : \mathcal{S}(\mathcal{A}) \rightarrow \mathbb{C}^{B_0}$, $\varphi \mapsto (\varphi(b))_{b \in B_0}$. Thus exact discrete cuts and operatorial continuum evaluation remain typed distinctly.

For finite marked $\mathbf{A}$ and $\nu : \mathbf{A}^* \rightarrow \mathbf{M}_{\mathbf{q}}^{\text{eff}}$ put $\text{Ball}_n = \nu\{w : |w| \leq n\}$, $\text{wl}(g) = \min_{\nu(w)=g} |w|$, $\text{Ball}_m \text{Ball}_n \subseteq \text{Ball}_{m+n}$, $\bigcup_n \text{Ball}_n = \mathbf{M}_{\mathbf{q}}^{\text{eff}}$, $\text{wl}(gh) \leq \text{wl}(g) + \text{wl}(h)$. Writing $q_i = \tilde{q}_i \pi$, (ELQ) makes $h_{i,g} := q_i T_w$ independent of the word w with $\nu(w) = g$. For $F_n^{\mathbf{A}} = I_0 \times \text{Ball}_n$ set $R_n = \bigcap_{(i,g) \in F_n^{\mathbf{A}}} R_{h_{i,g}}$, $D_n \cong X/R_n$, $R_\infty = \bigcap_n R_n$. Then

$$\begin{aligned} \widehat{X}_{I_0 \times \mathbf{A}^*} &\cong \varprojlim_{n < \omega} D_n, & R_\infty &= I_{\mathbf{A}^*} \left(\bigcap_i R_{q_i} \right), & R_{n+1} = R_n &\iff R_n = R_\infty, \\ \text{[KTO]} \quad \delta_{\mathbf{q}, \mathbf{A}}([x], [y]) &= \min\{n : (x, y) \notin R_n\} = \min\{\text{wl}(g) : \exists i, h_{i,g}(x) \neq h_{i,g}(y)\}, \\ d_{\mathbf{q}, \mathbf{A}} &= 2^{-\delta_{\mathbf{q}, \mathbf{A}}}, & d([x], [z]) &\leq \max\{d([x], [y]), d([y], [z])\}, & \min \emptyset &= \infty, & 2^{-\infty} &= 0. \end{aligned} \quad (62)$$

so (IOC) and nested kernels give (KTO/COA), plateau and ultrametricity. On the cyclic face, (RHA) $\widehat{\pi} : \widehat{X}_T \rightarrow \widehat{Q}_T$, $\widehat{\pi}((x_j)) = (\pi x_j)_{j \leq 0}$ and $\text{Past}_{\mathbf{q}, T}(\widehat{z}) = \widehat{\pi}^{-1}\{\widehat{z}\}$. This past fibre is independent: (Q, r, I_Q) and $(Q \times \{0, 1\}^{\mathbb{N}}, r \times \sigma, \text{pr}_Q)$ agree on $(Q, C_3, \text{Ball}_\bullet, \text{KTO})$ and completion but have fibre cardinalities $1, 2^{\aleph_0}$; on the identity shadow $\mathbb{Z}/3$, the trivial and 3-cycle laws agree on orbit kernels/quotient/completion/depth-0 but give $1, C_3$, while regular C_3/S_3 fix the invariant-equivalence reflector. The pair $T_1 = (0, 0, 0, 2), T_2 = (0, 0, 0, 3)$ fixes $01|23 \supset 01|2|3$; the binary-shift identity shadow separates KTO from (SCX/ASL).

Closure reflection, family envelope and word history (RFL/SDR/CWT). Write $\text{Mon}(L)$ for monotone endomaps and put $\text{PreCl}(L) = \{F \in \text{Mon}(L) : I \leq F\}$, $\text{ScPreCl}(L) = \{F \in \text{PreCl}(L) : F \text{ preserves directed joins}\}$, $\text{Cl}(L) = \{F \in \text{PreCl}(L) : F^2 = F\}$, and $\text{ScCl}(L) = \text{ScPreCl}(L) \cap \text{Cl}(L)$. Also $\text{Ret}(F) = I \vee F$; on algebraic L , with compact elements $K(L)$, put $\text{Sc}(G)(x) = \bigvee_{k \in K(L), k \leq x} G(k)$. Algebraicity/compactness makes $\text{Sc}(G)$ the greatest directed-join-preserving map below G ; if $G \in \text{Cl}(L)$, compactness inside the directed family $\{G(k) : k \in K(L), k \leq x\}$ also gives $\text{Sc}(G)^2 = \text{Sc}(G)$, hence $\text{Sc}(G) \in \text{ScCl}(L)$. For $G \in \text{PreCl}(L)$ set $\mathfrak{R}_L G(x) = \bigwedge \{z \geq x : Gz = z\}$ and $\overline{G} = \mathfrak{R}_L G$. Directly (cf. the standard formulation^{12,13}), (RFL) $\mathfrak{R}_L \dashv \iota_{\text{Cl}}$, $\mathfrak{R}_L \iota_{\text{Cl}} = I$, $\text{Fix } \mathfrak{R}_L G = \text{Fix } G$, $\mathfrak{R}_L (\bigvee_i^{\text{pt}} G_i) = \bigvee_i^{\text{Cl}} \mathfrak{R}_L G_i$; for $C \leq D$, $r_{D/C}(y) = Dy : \text{Fix } C \rightarrow \text{Fix } D$ is left adjoint to inclusion, splits it, and $r_{E/D} r_{D/C} = r_{E/C}$. For $C \subseteq \text{Cl}(L)$ put $\downarrow K = \{C : C \leq K\}$ and $E_C = \langle C \rangle = \bigvee^{\text{Cl}} C$. Then $E \dashv \downarrow$, $E \downarrow = I$; its order/max-fibre rows are (SDR). Let $\mathbf{W}_C$ be its pointwise-ordered finite composition monoid (including id), set $\tau := \mathfrak{R}_L|_{\mathbf{W}_C}$, $\text{Att}(C) := \tau(\text{Idem } \mathbf{W}_C) = \text{im } \tau \cap \mathbf{W}_C$, and $\pi_C : \mathbf{W}_C \rightarrow \{E_C\}$. For $w = i_1 \cdots i_n$ write $\text{ev}(w) = C_{i_n} \cdots C_{i_1}$ and $\text{supp } w = \{i_1, \dots, i_n\}$. Then

$$\begin{aligned} \text{[CWT]} \quad \text{Fix}(FG) &= \text{Fix } F \cap \text{Fix } G, & \tau(FG) &= \tau(F) \vee \tau(G) = \tau(GF), \\ \tau(\text{ev } w) &= \langle C_i : i \in \text{supp } w \rangle, & F^2 = F &\iff F = \tau(F) \\ \tau \text{ monotone, } F &\leq D \iff \tau(F) \leq D \ (D \in \text{im } \tau), & F, G \in \tau^{-1}(D) &\implies F, G \leq G \circ F \in \tau^{-1}(D), \\ \max \tau^{-1}(D) &\text{ exists } \iff D \in \text{Att}(C), & D \in \text{Att}(C) &\implies \max \tau^{-1}(D) = D, \\ \tau \dashv s &\iff \text{Att}(C) = \text{im } \tau, & s(D) &= D, & \tau s &= I \\ U, V \in \mathbf{W}_C &\implies U, V \leq V \circ U, & \mathbf{W}_C &\text{ directed,} \\ \max \mathbf{W}_C &\text{ exists } \iff E_C \in \mathbf{W}_C \iff \pi_C \text{ has a right adjoint,} \\ E_C &\notin \mathbf{W}_C \implies \text{Max } \mathbf{W}_C = \emptyset. \end{aligned} \quad (63)$$

More generally, if $\emptyset \neq C \subseteq \text{ScPreCl}(L)$, the pointwise join $F_C(x) := \bigvee_{C \in C} Cx$ is Scott-continuous, extensive, and has the common fixed locus. If C is finite, every word w with $\text{supp } w = C$ has the same properties. Hence $(\text{CWT}_{\text{sc}}) \overline{F}_C(x) = \bigvee_{n < \omega} F_C^n(x)$, $|C| < \infty \implies \text{ev}(w)(x) = \bigvee_{n < \omega} \text{ev}(w)^n(x)$, $C \subseteq \text{ScCl}(L) \implies \overline{F}_C = E_C$. Meet-closure of $\text{Fix } G$ and $G(Kx) = Kx$ for closures $K \geq G$ prove (RFL); its relative row is the adjunction on fixed loci. Extensivity/Scott continuity give (CWT/CWT_{sc}); (RFL/SDR) and generator-fixed maxima give the remaining rows.

On $\mathcal{P}(\{0, 1\})$, $A_0(S) = S \cup \{0\}$ and $B_0(S) = S \cup \{1\}$ have equal τ -shadow $A_0 B_0 = E_{\{A_0, B_0\}}$ but $(B_0 A_0)(\emptyset) = \{1\}$, so evaluation order survives terminalization. On $\mathcal{P}(\mathbb{N}_0)$, $A_\infty(S) = S \cup \{0, 1\} \cup \{2n+1 : 2n \in S\}$ and $B_\infty(S) = S \cup \{2n+2 : 2n+1 \in S\}$ have sole common fixed point $\mathbb{N}_0$ while every finite word sends $\emptyset$ to a finite set. Hence $E = \langle A_\infty, B_\infty \rangle \notin \mathbf{W}_{\{A_\infty, B_\infty\}}$ although $A_\infty, B_\infty \in \text{Att}$, so $E = A_\infty \vee B_\infty \notin \text{Att}$. Adjoining E gives a terminal top, but no finite word in the original $\{A_\infty, B_\infty\}$ alphabet reaches it; thus an exact duration fibre can lack a common upper. This word/duration obstruction is separate from the hosted rule globalization constructed below. No reference-saturation claim is made here because reset generators need not be closures.

For any monoid M , let $\eta_{\text{sl}}(M) = \langle (x^2, x), (xy, yx) : x, y \in M \rangle_{\text{cong}}$ and $q_{\text{sl}}^M : M \rightarrow \text{Sl}(M) := M/\eta_{\text{sl}}(M)$. The quotient is a commutative idempotent monoid, and every monoid map $M \rightarrow S$ to a semilattice kills the generating pairs, hence factors uniquely through q_{sl}^M . For $A = A_\infty$, $B = B_\infty$, $C^+ = \{A, B, E\}$, E is absorbing but no finite A, B -word equals it, and $h : \mathbf{W}_{C^+} \rightarrow (\{0, 1\}, \vee, 0)$, $h(F) = \mathbf{1}_{\{F=E\}}$, is a monoid homomorphism. Put $q_{\text{sl}} = q_{\text{sl}}^{\mathbf{W}_{C^+}}$. By the universal property both τ and h factor, hence (FWQ) $q_{\text{sl}}(E) \neq q_{\text{sl}}(A \circ B)$, $\tau(E) = \tau(A \circ B) = E$, $R_{q_{\text{sl}}} \subseteq R_\tau$, $R_\tau/R_{q_{\text{sl}}} \cong R_{\bar{\tau}}$. The last row is (DI/KSP); the algebraic and terminal quotients coincide exactly when $\bar{\tau}$ is injective, while the intervening fiber remains genuine history.

Primitive-resolution application. For finite C , the marked $\text{BehPro}_{L, \mathbf{W}_C}^C$ has law $\mathbf{W}_C$ and $\lambda_C = \min\{n : E_C \in \text{Ball}_n\}$, with (CWT) $\lambda_C < \infty \iff E_C \in \mathbf{W}_C \iff \max \mathbf{W}_C = E_C$. On $L = \mathcal{P}(\{p, a, b\})$, set $A(S) = S \cup \{a\}$, $B(S) = S \cup \{b\}$, $E = BA$, $A_1 = \{A, B\}$, $A_2 = \{A, B, E\}$. They share effective law/terminal closure, while at $q_b = \mathbf{1}_{\{b \in S\}}$, $x = \{p\}$, $y = \emptyset$, $(\delta_{q_b, A_1}, \lambda_{A_1}) = (2, 2)$ and $(\delta_{q_b, A_2}, \lambda_{A_2}) = (1, 1)$; hence filtration is independent.

B. Bounded structural-realization chart

Proposition II.3 (Structural specification and residual realization). *All clauses below are exact faces of one generated structural-specification fibration; no realization coordinate is promoted into its generator. The five principal groups (A)–(E) are subdivided only where a residual face requires independent navigation.*

(A) Reduction and witness polarity. For complete L and $a : \mathcal{R} \rightarrow L$, $\text{Gen}_a(F) = \bigvee_{r \in F} a_r$ and $\text{Host}_a(\ell) = \{r : a_r \leq \ell\}$ give $\text{Gen}_a \dashv \text{Host}_a$, closure $\text{cl}_a = \text{Host}_a \text{Gen}_a$, and $\text{Gen}_a \text{Host}_a = I_L$ exactly for join-dense $a(\mathcal{R})$; the one-point map into $\mathcal{P}\{0, 1\}$ is the sharp failure. For a polarity $p \perp w$, the subterminal Isbell face is the Galois connection $A^\perp, {}^\perp B$ with nucleus $A = {}^\perp(A^\perp)$. If $W \neq \emptyset$, $\text{Fill}_\perp(A) = A^\perp$ is a presheaf, and on every nonempty downward-closed $\mathcal{B}$,

$$\text{Sec}_{\text{cart}}\left(\int_{\mathcal{B}} \text{Fill}_\perp \rightarrow \mathcal{B}\right) \cong \bigcap_{A \in \mathcal{B}} A^\perp = \left(\bigcup \mathcal{B}\right)^\perp.$$

For $A \subseteq B$, $B^\perp = A^\perp \setminus \text{Kill}_\perp(B/A)$ and zero kill is exactly $B \subseteq \text{cl}_\perp(A)$. A minimal finite $\mathbb{C}$ with $\mathbb{C}^\perp = \emptyset$ has restricted nerve $\partial \Delta^{|\mathbb{C}|-1}$ and top reduced homology $\mathbb{Z}$ without preferred generator (with $\tilde{H}_{-1}(\emptyset) = \mathbb{Z}$ when $|\mathbb{C}| = 1$). Under $i(p) \perp' w' \iff p \perp r(w')$ the extension fibres $\text{ExtFill}_{B/A}(w) = \{w' \in B^\perp : r(w') = w\}$ satisfy $B^\perp \cong \coprod_{w \in A^\perp} \text{ExtFill}_{B/A}(w)$: interface survival is nonempty extension fibre and complete loss is extinction of all of them.

(B) Generated source package and origin/actuality. On the declared R - κ -small structural perimeter, a coherent family $d_x : \mathcal{E} \rightarrow \mathcal{B}_x$, $d_x = p_{yx} d_y$ ($x \leq y$) is exactly a point of

$$\text{CRef}_{R,\kappa}(\mathcal{E}, L) = \int_{\mathcal{B} \in \text{Fun}(L^{\text{op}}, \text{Cat}_{R,\kappa})} \text{Nat}(\Delta \mathcal{E}, \mathcal{B})^\cong.$$

Here $\text{Bas}_{R,\kappa}(L) \subset \text{Cl}(L)^\cong$ is the replete SCIA/PMA R - κ -small closure-state locus and $\text{Key}_{R,\kappa}(L, I) := \text{Bas}_{R,\kappa}(L)^{I_{\text{disc}}}$; RFL supplies finite joins, IAK the distinct admissible-arrow fibration, and PMA/GDM/PEX the bounded functor/natural-transformation package. By (SAC), same-source access keys compose by products, hence by opposite-support joins; SCE is the support-top locus, and no closure-valued key is identified with an access arrow. Therefore

$$\text{SSpec}_{R,\kappa}(L; \mathcal{E}, I) = \int_{C \in \text{Bas}_{R,\kappa}(L)} \int_{K \in \text{Key}_{R,\kappa}(L, I)} \text{CRef}_{R,\kappa}(\mathcal{E}, L) \quad (\text{GSSF})$$

represents bounded SCE packages $(C, (K_k), d)$ naturally. The three source species are universal moduli; occurrence remains an unselected point $1 \rightarrow \text{SSpec}$. Nullary-break/ BC_2 separates local break from cartesian-section residual; regular C_2 , transposition, torsor and framing separate law point, coupling, component and isotropy; provenance weights 1, 2 separate scale.

For well-founded oriented Γ put $\text{ProvHist}(\Gamma) = \text{FreeCat}(\Gamma)$. A positive edge length $\ell : E(\Gamma) \rightarrow \mathbb{R}_{>0}$ gives

$$\tau_\ell(e_m \cdots e_1) = \sum_j \ell(e_j), \quad \tau_\ell(1_x) = 0.$$

The choices $\ell = 1, 2$ fix the unweighted provenance skeleton and give durations $m, 2m$ on nonidentity m -edge paths, so scale is an independent attachment.

(B.1) Structural realization and read-out-profile descent. Fix a regular horizon κ containing the bounded structural coefficients; if their code/law moduli are themselves bounded, choose once a DMR horizon $\lambda > \kappa$. Let $\text{RawAcc}_{<\kappa}$ be the κ -small tuples of the declared residual-access sort and put

$$\text{LawAcc}_{<\kappa} = \mathcal{P}(\text{RawAcc}_{<\kappa}), \quad \text{SatAcc}_{<\kappa} = \{(\ell, F) : F \in \ell\} \rightarrow \text{LawAcc}_{<\kappa},$$

with $\text{Pred}_{\text{acc}} : \text{RawAcc}_{<\kappa} \rightarrow \mathbf{2}$ the characteristic predicate of exactly the declared compactness, measurability, probability, comparison and read-out clauses, and

$$\lambda_{\text{acc}} = \{F \in \text{RawAcc}_{<\kappa} : \text{Pred}_{\text{acc}}(F) = 1\}, \quad \text{FAcc}_{<\kappa}^\cong = (\lambda_{\text{acc}})^\cong.$$

PMA/GDM/PEX make this the unique replete comprehension locus for the declared predicate, but the predicate itself need not be treated as a meta-level choice. Let

$$\text{RLawAcc}_{<\kappa} := \{\ell \in \text{LawAcc}_{<\kappa} : F \in \ell, F' \simeq F \text{ in } \text{RawAcc}_{<\kappa} \Rightarrow F' \in \ell\}.$$

This is set-coded in the already chosen law-moduli horizon λ , and $\lambda_{\text{acc}} \in \text{RLawAcc}_{<\kappa}$. At the declared fibre, arrows preserve every displayed field/read-out, while realization remains an unselected occurrence $1 \rightarrow \text{FAcc}_{<\kappa}^\cong$. Over the declared codes put

$$\text{CRef}_{<\kappa}^{\text{disc}} := \int_{(X, J)} \text{CRef}_{R,\kappa}(\text{disc } X, \text{disc } J), \quad \text{SSpec}_{<\kappa}^{\text{disc}} := \int_{(X, J, I)} \text{SSpec}_{R,\kappa}(\text{disc } J; \text{disc } X, I).$$

The structural leg $U_{\text{str}}(C, (K_i), d) = d$ and the raw formal-access shadow

$$\text{Sh}_{\text{raw}}(F) = (\text{disc } C_r : \text{disc } \mathcal{X} \rightarrow \text{disc } \mathcal{F}_r)_{r \in \mathcal{R}_{\text{adm}}}, \quad \text{Sh} = \text{Sh}_{\text{raw}}|_{\text{FAcc}_{<\kappa}^\cong},$$

land in the same $\text{CRef}_{<\kappa}^{\text{disc}}$. Totalize the admissibility-law point rather than selecting it:

$$\begin{aligned} \text{UAcc}_{<\kappa; \lambda} &:= \int_{\ell \in \text{disc } \text{RLawAcc}_{<\kappa}} \ell^\cong, & p_{\text{law}}(\ell, F) &= \ell, & \text{USh}(\ell, F) &= \text{Sh}_{\text{raw}}(F), \\ \text{[UAL]} \quad \text{URep}_{<\kappa; \lambda} &:= \text{SSpec}_{<\kappa}^{\text{disc}} \times_{\text{CRef}_{<\kappa}^{\text{disc}}}^h \text{UAcc}_{<\kappa; \lambda}, & \Pi_{\text{law}}(s, \ell, F) &= (s, \ell), \\ \text{HFib}_{(s, \ell)}(\Pi_{\text{law}}) &\simeq \text{HFib}_{U_{\text{str}}(s)}(\text{USh}|_{\{\ell\} \times \ell^\cong}), & (\text{pr}_{\text{law}})^{-1}(\lambda_{\text{acc}}) &\simeq \text{Rep}_{<\kappa}. \end{aligned} \quad (64)$$

Here $\text{pr}_{\text{law}} : \text{URep}_{<\kappa; \lambda} \rightarrow \text{disc } \text{RLawAcc}_{<\kappa}$ is the law projection. The last row is the homotopy pullback at the declared law point, so the previous fixed-law chart is an exact fibre of the law-indexed chart, not its source. The identity/flip raw accesses already used above are nonisomorphic; if F_0, F_1 denote them, the replete closures of $\{F_0\}$ and $\{F_0, F_1\}$ are distinct law points while both contain the same F_0 . Hence forgetting ℓ cannot reconstruct the admissibility law from a realization: law/hypothesis provenance is a genuine base coordinate, not a selector generated by the occurrence.

At the declared point λ_{acc} this gives

$$\text{Rep}_{<\kappa} = \text{SSpec}_{<\kappa}^{\text{disc}} \times_{\text{CRef}_{<\kappa}^{\text{disc}}}^h \text{FAcc}_{<\kappa}^\cong,$$

and homotopy base change gives naturally in s

$$(\text{MFAM/SPRF}) \quad \text{PhysFib}_\kappa(s) := \text{HFib}_s(\text{Rep}_{<\kappa} \rightarrow \text{SSpec}_{<\kappa}^{\text{disc}}) \simeq \text{HFib}_{U_{\text{str}}(s)}(\text{Sh}) = \text{RealSh}(U_{\text{str}}(s)).$$

Thus the structural–physical fibre is the full realization profile, not actuality: $R_{J\text{Read}} = \bigcap_r R_{C_r}$ inside each realization, while the model shadow is not the content read-out. The uniform two-point identity/transposition kernels remain nonisomorphic at fixed declared fields and content/readout shadow, so measure, dynamics, metric, interface and operatorial data remain attachments.

For an essentially-surjective bounded groupoid functor $p : \mathcal{E} \rightarrow \mathcal{B}$ (equivalently every $\text{HFib}_b(p)$ inhabited) and bounded groupoidal $F : \mathcal{B} \rightarrow \mathcal{X}$, effective-image factorization/base change give

$$(\text{SSCR}) \quad S_{Fp} \simeq S_F, \quad Q_{Fp} \simeq Q_F, \quad \mathcal{R}_{Fp} \simeq (p \times p)^* \mathcal{R}_F.$$

This is exact for pullback read-outs. With $\text{Ex}(\mathcal{A}) = [\mathcal{A} \neq \emptyset]$, essential surjectivity gives $\text{Ex}(\mathcal{B}) \simeq \mathbf{1} \Rightarrow \text{Ex}(\mathcal{E}) \simeq \mathbf{1}$ (the Poss face), and composition gives finite re-entry towers. Sharpness: $\{0\} \hookrightarrow \{0, 1\}$ changes the identity image, while $B \times \{0, 1\} \rightarrow B$ preserves the pullback quotient but exposes the non-pullback second coordinate; source reparametrization, components and isotropy therefore do not collapse.

Let $\mathcal{A}$ be a bounded model groupoid, Λ a bounded discrete family of replete full $\mathcal{A}_\ell \subseteq \mathcal{A}$, and $K : \mathcal{A} \rightarrow \mathcal{X}$ bounded groupoidal. Put

$$\text{VCons}(\mathcal{X}) := \int_{Z \in \mathbf{Gpd}_{\leq \mathcal{A}}} \text{Fun}(Z, \mathcal{X})^\approx, \quad \text{LawProf}_K(\ell) = [K|_{\mathcal{A}_\ell}],$$

and

$$\ell R_K^{\text{law}} m \iff \text{LawProf}_K(\ell) \simeq \text{LawProf}_K(m), \quad \text{LawFib}_K(\ell) := \text{HFib}_{\text{LawProf}_K(\ell)}(\text{LawProf}_K). \quad (\text{LFCN})$$

For any $F : D_{\text{disc}} \rightarrow \mathcal{Y}$ with $\mathcal{Y}$ a groupoid,

$$(\text{LCPF}) \quad \text{HFib}_{F_d}(F) \simeq \text{disc} \coprod_{e: Fe \simeq Fd} \text{Iso}_{\mathcal{Y}}(Fe, Fd), \quad (\text{LFSF}) \quad \text{HFib}_{F_d}(F) \simeq \mathbf{1} \iff \{e : Fe \simeq Fd\} = \{d\} \ \& \ \text{Aut}(Fd) = 1.$$

The selector-free fibre retains class multiplicity and profile isotropy (nonempty Iso is a bitorsor; its diagonal is Aut), independently sharp for $D = \{d\}$, $F(d) = BC_2$ and constant $\text{disc}\{d, e\} \rightarrow \mathbf{1}$. Postprocessing $K' = hK$ gives $R_K^{\text{law}} \subseteq R_{K'}^{\text{law}}$; for $K = (K_i)_i$,

$$(\text{LFCN/JLFCF}) \quad R_K^{\text{law}} \subseteq \bigcap_i R_{K_i}^{\text{law}}.$$

Diagonal/antidiagonal subsets of $\text{disc}\{00, 01, 10, 11\}$ make this strict for coordinate-bit read-outs, so correlation is genuinely joint. For $F = \text{LawProf}_K$, noncontractibility is multiplicity and/or isotropy, never occurrence; the identity/flip model gives the $K = \text{Sh}$ case.

Supplementary package S1: Bounded structural–realization residual packages (B.2–B.12). The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S1; every use below invokes that exact proof-bearing package, not an additional assumption.

Proposition II.4 (Grammar, proarrows, collapse, and determination quotient). *The material set-theory input is exactly (BDCW): (VBM_{sc}) identifies the theory of the initial $\mathcal{P}_s$ -algebra with RST-provability, not generic grammar completeness. Under $N \rightarrow 1$ and small monomorphisms, $(\text{VDM}_{\text{fin}})$ gives the indexed final $\mathcal{P}_s$ -coalgebra with indexed-initial well-founded core; $(\text{VDM}_{\text{fail}})$ only refutes finality in countable sets with finite-fibre small maps. There $\mathcal{P}_s$ is finite powerset while countable hereditarily finite sets still form the initial algebra, so initiality does not force finality. The empty-signature geometric theory is classified by **Set**¹⁴ (Ch. 11); on the ordinary Set face, **Cat** $\rightarrow$ **Mod** is the tight-Kleisli free cocompletion of **Set** $\rightarrow$ **Span**¹⁵ (Thm. 1.3). For $H : B^{\text{op}} \times E \rightarrow \mathbf{Set}$, $\text{Tab}(H) \rightarrow B$ is the universal local-lift decoder; represented U^* recovers U , coends compose interfaces, and $\text{Fun}(\mathbb{T}_B U, \mathcal{H})^\approx$ remains an occurrence fibre. Terminal-collapse uses the same grammar: $H_1 \mathfrak{R}_\bullet(!_X) = \mathbb{K}(!_X)$ and $H_0 = \text{ExtRes}(!_X)$ separate multiplicity/emptiness; positive syntax has the singleton total-relation model, Boolean collapse excludes it exactly at inconsistency, minimal circuits carry boundary homology without preferred generator, and the extension fibre locates first anti-collapse at common-filler destruction rather than depth. The pointing-free exterior complex $\bigwedge^{n+1} \mathbb{Z}^{(X)}$ resolves $\mathbb{K}(!_X)$ for nonempty X ; provenance codescent $e_A : D_\infty(A) \rightarrow C(A)$ descends exactly on its kernel pair, with unary-loop parity separating provenance from output multiplicity.*

By RFL grammar union is closure join and finite-rule exhaustion is filtered; Scott Proposition 2.5 identifies the directed-join law of Sc with continuity¹⁶ (Prop. 2.5). For complete Barr-exact $\mathcal{E}$ and $\mathbf{h} = (h_j : X \rightarrow Z_j)_{j \in J}$ put $\Phi(A) = \bigcap_{j \in A} R_{h_j}$, $\Psi(R) = \{j : R \subseteq R_{h_j}\}$, $C_\infty = \Psi\Phi$, $C_{\text{fin}} = \text{Sc}(C_\infty) = \bigcup_{F \in A} C_\infty(F)$. EPN/KGC gives $\Phi \dashv \Psi$, $\mathbb{Q}_{\mathbf{h}}(K) = q_K : X \rightarrow D_K = X/\Phi(K)$; put $q_A = q_{\Phi(A)}$, $q \preceq_X r \iff R_r \subseteq R_q$. For every regular-epimorphic $d : X \twoheadrightarrow D$, exactness and the quotient universal property give

$$[\text{KGC}_{\text{cons}}] \quad R_d \subseteq \Phi(A) \iff \exists ! \phi_d : D \rightarrow X/\Phi(A) : q_A = \phi_d d; \quad \mathbf{Set} : (\forall j \in A) \text{DepRes}_d(h_j) = 0 \iff R_d \subseteq \Phi(A). \quad (65)$$

Hence q_A is the unique coarsest effective quotient sufficient for the whole declared consumer family; if I_X is among those consumers, $q_A \simeq_X I_X$. The same object is equivalently the closed distinction–consequence pair $(\Phi(A), C_\infty(A))$: $\Phi(C_\infty(A)) = \Phi(A)$ and $C_\infty(C_\infty(A)) = C_\infty(A)$. Put $\text{Ref}_{\mathbf{h}}(X) := \{[q_K]_X : K \in \text{Fix } C_\infty\}$. For $A \subseteq B \subseteq J$, the inclusion $\Phi(B) \subseteq \Phi(A)$ induces the unique $\rho_{B,A} : D_B \rightarrow D_A$ with $q_A = \rho_{B,A} q_B$, functorially along chains, and

$$[\text{KGC}_{\text{ord}}] \quad \text{Fix } C_\infty \xrightarrow{\sim} \text{Ref}_{\mathbf{h}}(X), \quad K \mapsto [q_K]_X, \quad K \wedge L = K \cap L, \quad K \vee L = C_\infty(K \cup L), \quad \rho_{B,A} \text{ iso} \iff \Phi(B) = \Phi(A) \iff B \subseteq C_\infty(A). \quad (66)$$

Indeed Φ, Ψ restrict to inverse antitone bijections between closed consumer sets and generated kernel relations, while quotient order reverses relation inclusion again; thus reflected objects form a complete lattice. More intrinsically, let $q : X \twoheadrightarrow Q$ be any effective quotient and $(g_i : Q \rightarrow Z_i)_{i \in I}$ a small family; for $h_i = g_i q$, regular-epi descent of subobjects gives

$$[\text{KGC}_{\text{sep}}] \quad q_{\{h_i\}} \simeq_X q \iff \bigcap_{i \in I} R_{g_i} = \Delta_Q \iff (g_i)_{i \in I} \text{ is jointly monic}. \quad (67)$$

Indeed $\bigcap_i R_{g_i q} = (q \times q)^*(\bigcap_i R_{g_i})$, and pullback along the regular epimorphism $q \times q$ reflects subobject equality. Thus every jointly separating lawful presentation of the same quotient has the same consumer reflection: no particular separating vocabulary is ontologically privileged. Moreover $\text{Suff}_A = \{[d]_X : R_d \subseteq \Phi(A)\} \uparrow [q_A]_X$, so q_A is least sufficient and minimizes every isotone quotient valuation. For finite **Set**, cardinality and Shannon entropy are monotone shadows; along $A \subseteq B$ with full-support μ , $\text{Ref}_{\mu}(B:A) = H((q_B)_* \mu) - H((q_A)_* \mu) = H(q_B | q_A)$ is additive and vanishes exactly at $q_B \simeq_X q_A$. A Dirac law hides strict refinement, so full support is necessary. Thus scalar information is an additional valuation shadow along the same (RRCX) refinement chain, not a primitive; code-dependent Kolmogorov complexity is not an

invariant. On the ambient Set read-out class (or any declared class containing q_A), (RFF) applied to target q_A is therefore the minimal finite interface problem for the whole family, so no new interface primitive is required. The three-point witness fixes C_∞ while changing the quotient. For $q[A] = q_{R_q \cap \Phi(A)}$,

$$[\text{DCC}] \quad q \preceq_X q[A], \quad q[A] \preceq_X r \iff [q \preceq_X r \wedge A \subseteq \Psi(R_r)], \quad q[A] \simeq q \iff A \subseteq \Psi(R_q). \quad (68)$$

For $A \subseteq J$ put $K_A = (\{C_\infty(F) : F \in A\}, \subseteq)$, $c_A(F) = C_\infty(F)$, $W_A = \{F \subseteq G : c_A(F) = c_A(G)\}$, $\mathbb{D}_A = (D_{c_A(F)})_{F \in A}$ and $\eta_A = (q_{c_A(F)})_F$. Then

$$[\text{KGC}_{\text{pro}}] \quad \lim_{K \in K_A} \text{Hom}_{\mathcal{E}}(D_K, Z) \cong \{g : X \rightarrow Z : \exists F \in A, \Phi(F) \subseteq R_g\}. \quad (69)$$

On **Set**, $H_A = (h_j)_{j \in A}$, $B_A = \text{im } H_A$ and uniquely $H_A^c = \iota_A q_A$ with $\iota_A : X/\Phi(A) \xrightarrow{\sim} B_A$. For $j \notin A$ the exact refinement trichotomy is

$$[\text{KGC}_{\text{ref}}] \quad \begin{aligned} j \in C_{\text{fin}}(A) &\iff q_{A \cup \{j\}} \simeq_X q_A \text{ and the pro-extension is cofinal,} \\ j \in C_\infty(A) \setminus C_{\text{fin}}(A) &\iff q_{A \cup \{j\}} \simeq_X q_A \text{ and the pro-extension is noncofinal,} \\ j \notin C_\infty(A) &\iff R_{q_{A \cup \{j\}}} \subsetneq R_{q_A}. \end{aligned} \quad (70)$$

Here $R_{q_A} = \Phi(A)$ and, on **Set**, the first two rows are exactly $\text{DepRes}_{q_A}(h_j) = 0$ while the third has nonzero residual. Thus q_A is the consumer reflection, $C_\infty(A)$ its determination closure, and only the third row changes the reflected object; RSC is $C_\infty(A) = J \iff \Phi(A) = \Phi(J) \iff q_A \simeq q_J$. With $\mathcal{U}_j = \{A : j \in C_\infty(A)\}$, $\text{IntSc } \mathcal{U}_j = \{A : j \in C_{\text{fin}}(A)\}$ and $\partial_{\text{Sc}} \mathcal{U}_j = \{A : j \in C_\infty(A) \setminus C_{\text{fin}}(A)\}$; the 0^∞ indicator on $\{0, 1\}^{\mathbb{N}}$ is the sharp nonuniform/nonextendable boundary. On $X = J = A$, $Z_j = Y$, writing $m_A : B_A \hookrightarrow Y^A$, Lawvere¹⁷ (Thm. 1.1) gives fixed-point-free $t : Y \rightarrow Y \Rightarrow m_A$ proper; singleton 2-readouts can have $q_A = I_A$ while m_A is proper, so the image leg remains the independent (REL) coordinate. Indexed base change has

$$[\text{IRF}] \quad \begin{aligned} C_a F(u) \leq F(u) C_b, \quad \text{Nuc}_a = \text{Fix } C_a, \quad \text{IdxRef}_u(A) = [C_a F(u) A, F(u) C_b A], \\ \int_{\mathcal{B}} \text{Nuc} \rightarrow \mathcal{B} \text{ is the } (\text{CTF}_{\text{fib}}) \text{ reflected fibration,} \quad \text{IdxRef}_u(A) \text{ singleton} \iff C_a F(u) A = F(u) C_b A. \end{aligned} \quad (71)$$

with the **Set** equipment⁵ (Ex. 2.12); strict intervals retain nonterminal factorization fibres. Finally

$$[\text{SUP}] \quad \text{IntSc } \mathcal{U} = \bigcup_{F \in \mathcal{U}, F \in I} \uparrow F = \uparrow \text{FinFront}_I(\mathcal{U}), \quad A \in \partial_{\text{Sc}} \mathcal{U} \iff A \in \mathcal{U} \ \& \ (\forall F \in A) F \notin \mathcal{U}. \quad (72)$$

separates finite evidence from infinitary activation.

Proposition II.5 (Closure, local image, law transport, and grammar-world). For $\text{Gr}(L) = \text{Mon}(L, \text{Cl}(L))$, $G_{\mathfrak{Z}}(x) = \mathfrak{Z}(x)(x)$. For $u \in L$ let $u_0 = u$, $u_{\alpha+1} = G_{\mathfrak{Z}} u_\alpha$, $u_\lambda = \bigvee_{\alpha < \lambda} u_\alpha$, κ_u be first stationarity and $C_{\mathfrak{Z}, u} = \{\mathfrak{Z}(u_\alpha) : \alpha < \kappa_u\}$. Before κ_u this is a strict chain; $C \leq D \Rightarrow CD = DC = D$, so $W_{C_{\mathfrak{Z}, u}} = \{I\} \sqcup C_{\mathfrak{Z}, u}$ and (CWT) gives $u_{\kappa_u} = \overline{G_{\mathfrak{Z}}}(u) = E_{C_{\mathfrak{Z}, u}}(u)$. For terminal grammars $\text{TGr}(L) = \{\mathfrak{R} : G_{\mathfrak{R}}^2 = G_{\mathfrak{R}}\}$, the reflector is

$$[\text{GCL}] \quad \text{GrCl}(\mathfrak{Z}) = \bigwedge \{\mathfrak{R} \in \text{TGr}(L) : \mathfrak{Z} \leq \mathfrak{R}\}, \quad \text{GrCl} \dashv \iota_{\text{TGr}}, \quad G_{\text{GrCl}(\mathfrak{Z})} = \mathfrak{R}_L G_{\mathfrak{Z}}, \quad \text{Fix GrCl} = \text{TGr}(L). \quad (73)$$

Set $\delta_L : \text{Gr}(L) \rightarrow \text{Mon}(L, L)$, $\delta_L(\mathfrak{Z}) := G_{\mathfrak{Z}}$, $\bar{\delta}_L := \mathfrak{R}_L \delta_L = \delta_L \text{GrCl}$, $\text{RGW}(L) := \text{Tot}_{\mathfrak{R} \in \text{TGr}(L)} \text{Fix}(G_{\mathfrak{R}})$, and $\mathbb{E}_L^{\text{gr}}(\mathfrak{Z}, x) := (\text{GrCl}(\mathfrak{Z}), \overline{G_{\mathfrak{Z}}}(x))$; by (GCL) it is an extensive idempotent closure with fixed locus $\text{RGW}(L)$, so its corestriction $h_L^{\text{gr}} \dashv \iota_{\text{RGW}}$. The constant grammar embeds ordinary closures. For arbitrary-join $\beta : P \rightarrow Q$, $R_\beta = \ker \beta$, $q \leq \beta v$, set $\Lambda_v(q) = \bigvee \{w \leq v : \beta w \leq q\}$ and $\ell_v(q) = \beta \Lambda_v(q)$. Then

$$[\text{CLD}] \quad \Lambda_v(q) \rightarrow v \text{ cartesian over } q \leq \beta v \iff \ell_v(q) = q, \quad q' \leq q = \ell_v(q) \Rightarrow \Lambda_{\Lambda_v(q)}(q') = \Lambda_v(q'), \quad (74)$$

$$[\text{LCR}] \quad j_v : \text{im}(\beta|_{\downarrow v}) \hookrightarrow \downarrow \beta v \dashv \ell_v, \quad \ell_v^2 = \ell_v, \quad \text{Fix } \ell_v = \text{im}(\beta|_{\downarrow v}), \quad \ell_v(\beta u) = \beta u \iff \Lambda_v(\beta u) R_{\beta u}. \quad (75)$$

Equal kernel with $\beta_a(1) = a$, $\beta_b(1) = b$ gives different local-image fibre. For surjective complete-join $q : L \rightarrow M$, $q \dashv q^*$, put $S_q := q^* q$, $q_{\#}^{\text{law}} G := q G q^*$, $q_{\#}^{\text{law}} F := q^* F q$ and $\text{Man}_q := q_{\#}^{\text{law}} q_{\#}^{\text{law}}$. Then

$$[\text{LMR/PMR}] \quad \begin{aligned} q G \leq F q &\iff q_{\#}^{\text{law}} G \leq F \iff G \leq q_{\#}^{\text{law}} F, \quad q_{\#}^{\text{law}} q_{\#}^{\text{law}} = I, \\ \text{Man}_q G = S_q G S_q &= \max(q_{\#}^{\text{law}})^{-1}(q_{\#}^{\text{law}} G), \quad \mathfrak{A}_{\text{man}}^1 = \mathfrak{A}_{\square}(q G \Rightarrow (q_{\#}^{\text{law}} G) q), \quad \mathfrak{A}_{\text{man}}^\infty = \mathfrak{A}_{\square}(q \overline{G} \Rightarrow \overline{q_{\#}^{\text{law}} G} q), \\ \mathfrak{A}_{\text{man}}^1(q, G; \cdot) &= 0 \iff q G = (q_{\#}^{\text{law}} G) q = q G S_q. \end{aligned} \quad (76)$$

For $S \in \text{Cl}(L)$ put $\Omega_S(C, A) := (C \vee S, (C \vee S)A)$. Then $\Omega_T \Omega_S = \Omega_{S \vee T}$ and $\text{Fix } \Omega_S = \{(C, A) : S \leq C, C(A) = A\}$. Moreover $\text{Cl}(M) \cong \uparrow S_q \subseteq \text{Cl}(L)$ with $T_q C := q(C \vee S_q) q^*$ and $T_r q = T_r T_q$. Put $\mathbb{E}_L := \Omega_I$, $H_L := \text{Fix } \mathbb{E}_L$, $h_L^{\text{cl}} := \mathbb{E}_L|^{H_L}$ and let ι_H be its inclusion; set $d_L(\mathfrak{R}, w) := (G_{\mathfrak{R}}, w) : \text{RGW}(L) \rightarrow H_L$. When L is algebraic put $C_G := \overline{\text{Sc } G}$, $L_G^c := \text{Fix } C_G$, $\mathbb{B}_G := C_G G|_{L_G^c}$, $D_G := \overline{\text{Sc}(\overline{G})}$, $L_G^{\text{sc}} := \text{Fix } D_G$, $R_G := D_G G|_{L_G^{\text{sc}}}$ and $P_G := \text{Sc } L_G^c(\mathbb{B}_G)$. The (TSC) contract is

$$\begin{aligned} C_G &= \bigvee_{n < \omega} (\text{Sc } G)^n, \quad \text{Comp}(L_G^c) = \{C_G k : k \in K(L)\}, \quad \text{Fix } \mathbb{B}_G = \text{Fix } R_G = \text{Fix } G, \\ \overline{\mathbb{B}_G}^{L_G^c}(C_G x) &= \overline{G} x, \quad D_G = \max\{D \leq \overline{G} : D \text{ Scott-continuous closure}\}, \quad \text{Sc } L_G^{\text{sc}} R_G = I, \quad \overline{R_G}^{L_G^{\text{sc}}}(D_G x) = \overline{G} x, \\ R_G = I &\iff D_G = \overline{G}, \quad D_{\mathbb{B}_G} = D_G|_{L_G^c}, \quad R_{\mathbb{B}_G} = R_G, \quad C_G = D_G \iff P_G = I. \end{aligned}$$

On $I = \mathbb{N} \sqcup \{a, b\}$ put $H(S) = S \cup \{n+1 : n \in S \cap \mathbb{N}\}$, $C = \overline{H}$, $G_{\text{seq}}(S) = H(S) \cup (\{a\} : \mathbb{N} \subseteq S) \cup (\{b\} : \mathbb{N} \cup \{a\} \subseteq S)$ and $G_{\text{flat}}(S) = H(S) \cup (\{a, b\} : \mathbb{N} \subseteq S)$. Then $(\text{Sc } G_{\text{seq}}, \overline{G}_{\text{seq}}) = (\text{Sc } G_{\text{flat}}, \overline{G}_{\text{flat}})$ but, for $y = C\{0\} = \mathbb{N}$, $P_{G_{\text{seq}}}(y) = \mathbb{N} \cup \{a\} \subseteq \mathbb{N} \cup \{a, b\} = P_{G_{\text{flat}}}(y)$. Put $\tilde{\mathbf{b}}_L(G) := (\text{Sc } G, \overline{G})$ and $\mathfrak{s}_L(H, K)(x) := K(x) \wedge \bigwedge_{k \in K(L), x \leq k} H(k)$; corestrict $\tilde{\mathbf{b}}_L$ to $\mathbf{b}_L$ on its image, write $c_L := \text{pr}_1 \mathbf{b}_L$, $t_L := \text{pr}_2 \mathbf{b}_L$, and set $\mathfrak{Q}_L := \mathfrak{s}_L \mathbf{b}_L$. The bi-reflection fibre is

$$[\text{BHP}] \quad \begin{aligned} \text{im } \tilde{\mathbf{b}}_L &= \{(H, K) : \overline{\mathfrak{s}_L(H, K)} = K\}, \quad \text{ExtRes}(\tilde{\mathbf{b}}_L) \cong \mathbb{Z}(\{(H, K) : \overline{\mathfrak{s}_L(H, K)} < K\}), \\ \mathbf{b}_L \dashv \mathfrak{s}_L, \quad \mathbf{b}_L \mathfrak{s}_L &= I, \quad \mathbf{b}_L^{-1}(H, K) = \{G : H \leq G \leq \mathfrak{s}_L(H, K), \text{Fix } G = \text{Fix } K\}, \quad \max \mathbf{b}_L^{-1}(H, K) = \mathfrak{s}_L(H, K). \end{aligned} \quad (77)$$

For compact k , $\mathfrak{s}_L(H, K)(k) = H(k)$; every realizer is $\leq \mathfrak{s}_L(H, K)$ and $\overline{\mathfrak{s}_L(H, K)} = K$ gives the converse, proving (BHP). On $L = \{0 < 1\}$, $(H, K) = (I, \top)$ has $\mathfrak{s}_L = I$, so exteriority is nonzero. The inherited $0 < a < 1$ tuple has no cartesian/opcartesian lift, so equality transport remains weaker than unrestricted URE transport.

The grammar-world reflector $\mathbf{RGW}$ satisfies $h_L^{\text{gr}} \dashv \iota_{\mathbf{RGW}}$, $h_L^{\text{cl}} \dashv \iota_H$, $d_L h_L^{\text{gr}} = h_L^{\text{cl}}(\delta_L \times I_L)$, and its exact fibre is

$$[\mathbf{RGF}] \quad (h_L^{\text{gr}})^{-1}(\mathfrak{R}, w) \cong \text{Tot}_{b=(x, w) \in \text{GBase}(L)}^{\text{End}_L(b)=w} \text{Tot}_{F \in \text{Orb}_L^{-1}(b)}^{\overline{F}=G_{\mathfrak{R}}} \{\mathfrak{J} \in \delta_L^{-1}(F) : \text{GrCl}(\mathfrak{J}) = \mathfrak{R}\}. \quad (78)$$

By $(\mathbf{RGE}_A/\mathbf{SAC})$, attachments multiply this fibre and SCE has $Z_{\star} = \{x : \text{Fac}_{d_x}(I_{\mathcal{E}}) \neq \emptyset\} = \cap_T Z_T$, $\mathfrak{J}_{K, \star}(x) = C \vee \bigvee_{F: K_F x \in Z_{\star}} K_F$; $\mathbf{RFL}/\mathbf{GCL}$ fixes exactly $Cx = x$ and active $K_F x = x$. On $\text{Eq}(\Omega)^{\text{op}}$: $Z_{\star} = \{\Delta_{\Omega}\}$ and $R \cap \bigcap_{k \in F} R_k = \Delta_{\Omega}$. $\mathbf{IAK}/\mathbf{SUP}/\mathbf{RGW}/\mathbf{IFS}/\mathbf{TKN}$ retain key closure/boundary/terminal-readout/multiplicity/no-total-key; (KDIS) supplies only standard-Borel conditional laws/disintegration¹⁹ (Thms. 6.3–6.4), not a selector.

Proposition II.6 (Typed residual fibres and quantitative faces). *The realized/corestricted UAR class $\zeta_L(G) = [\mathfrak{Q}_L G] - [G]$ vanishes exactly at the maximal representative. For $\iota_G(x) = \min\{\xi : C_G x \leq G^{\xi} x\}$, $\tau_G(y) = \min\{\xi : \mathbb{B}_G y \leq G^{\xi} y\}$ and $y_{\eta+1} = \mathbb{B}_G y_{\eta}$,*

$$b \leq \text{clord}_G(x) \leq \iota_G(x) + \sum_{\eta < b} \tau_G(y_{\eta}) \leq \omega + \omega b.$$

This combines Theorem II.1, Definition II.2, (SRSL/RAC/PCN) with the (B.8) framing refinement, and Theorems IV.1, IV B 0 b, VIII.1. The BG, duration-1/2, Scott-boundary, local-image, Boolean–Heyting, coinductive-pole, Cantor–Tarski, no-selector and no-finite-key witnesses separate support, quotient realization, law point, provenance, holonomy, scale, transport, temporal occurrence, metric/unitary/clock calibration and physical realization; GSSF plus the displayed fibres regenerates the preceding origin/actualization/proarrow/determination/grammar/closure/terminal coordinates without read-out-visible loss.

Proposition II.7 (Bridge reflection, recurrence, and post-cut unity). *For $F : C \rightarrow \mathcal{D}$ define*

$$\mathbf{BSrc}_F = \coprod_{x, y} \text{Hom}_{C^{\approx}}(x, y), \quad \mathbf{BTrg}_F = \coprod_{x, y} \text{Hom}_{\mathcal{D}^{\approx}}(Fx, Fy), \quad \beta_F(x, y, f) = (x, y, Ff).$$

With $\text{Br}_F(x, y; \alpha) = \{f : Ff = \alpha\}$ and $\Theta_F = \text{ResFib}(\overline{\beta}_F)$, $\mathbf{CRF}/\mathbf{RFE}/\mathbf{IGRT}$ gives

$$[\mathbf{TOB}] \quad \begin{aligned} \text{SuppKill}(\beta_F) &= \{(x, y, \alpha) : \text{Br}_F(x, y; \alpha) = \emptyset\}, & \Theta_F(x, y; \alpha) &\simeq \text{disc Br}_F(x, y; \alpha), \\ F^{\approx} \text{ fully faithful} &\iff \text{SuppKill}(\beta_F) = \emptyset \ \& \ (\forall b \in \mathbf{BTrg}_F^{\text{im}}) \ \Theta_F(b) \simeq \mathbf{1}. \end{aligned} \quad (79)$$

Thus fullness failure is support loss and faithfulness failure vertical multiplicity; $2_{\text{disc}} \rightarrow \mathbf{1}$ and $\text{BG} \rightarrow \mathbf{1}$ separate them. For predicates P, Q , the inclusion $i_{P, Q} : \{x : Q(Fx) \wedge P(x)\} \hookrightarrow \{x : Q(Fx)\}$ has terminal nonempty fibres, so $\text{RefFail}_F(P \Leftarrow Q) = \text{SuppKill}(i_{P, Q}) = \{x : Q(Fx) \wedge \neg P(x)\}$: backward representation is the same residual construction.

Equal semantic labels give only incidence. On the groupoid I of pairs of nonempty sets and bijections, let $\mathbf{H}(X, Y) = \text{Set}(X, Y)$ and $(u, v)_* f = v f u^{-1}$. Then

$$[\mathbf{SNB}] \quad \text{Sec}_{\text{cart}}\left(\int_I \mathbf{H} \rightarrow I\right) \cong \text{Nat}(\mathbf{1}, \mathbf{H}) = \emptyset, \quad (80)$$

because the transposition of $Y = \{0, 1\}$ fixes no $\mathbf{1} \rightarrow Y$; same-label incidence supplies neither bridge data nor coherent selector.

For a dynamical set (X, T) put

$$[\mathbf{RZN}] \quad \text{Res}_n(T) = \text{Eq}(T^n, I_X), \quad \text{Spec}_{\text{res}}(T) = \{n \geq 1 : \text{Res}_n(T) \neq \emptyset\}. \quad (81)$$

If $qT = Sq$, then $E_{q, n} = \text{Res}_n(T) \subseteq B_{q, n} = q^{-1} \text{Res}_n(S)$ and $\text{SuppKill}(E_{q, n} \hookrightarrow B_{q, n})$ is the reflection-failure fibre; it vanishes for injective q and is nonzero for the two-cycle quotient to $\mathbf{1}$. Yet

$$[\mathbf{RZC}] \quad |X| \leq 1 \Rightarrow \text{Res}_n(T) = X \ (n \geq 1), \quad X = \mathbf{1} \Rightarrow \text{Spec}_{\text{res}}(T) = \mathbb{N}_{>0}, \quad (82)$$

so recurrence resonance cannot break collapse; physical/operator resonance needs its own carrier. Finally

$$[\mathbf{POL}] \quad \text{Pol}(X) := \{\sigma \in \text{Aut}(X) : \sigma^2 = I_X, \sigma \neq I_X\}, \quad \text{Pol}(X) \neq \emptyset \iff |X| \geq 2 \iff \mathbb{K}(!_X) \neq 0, \quad (83)$$

so bare polarity is exactly the collapse distinction witness, while for every nonempty connected J ,

$$[\mathbf{UAO}] \quad \text{colim}_J \mathbf{1} \cong \mathbf{1} \cong \lim_J \mathbf{1}, \quad (84)$$

showing universal assembly/endogenous self-readout compatible with collapse. Bridge lifting, reflection, recurrence, polarity and assembly are typed read-outs, not parallel ontological primitives.

C. Derived and affine residual shadows

Supplementary package S2: Derived residual memory and affine residual companion. The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S2; every use below invokes that exact proof-bearing package, not an additional assumption.

a. Residual path envelope, pointed factorization, and quotient flags. For normalized oplax $F : C \rightsquigarrow \mathbf{Pos}$, write $\kappa_w : F_{p(w)} \Rightarrow F_w^{\text{it}}$ and $\mathfrak{M}_w = \mathfrak{A}_{\square}(\kappa_w)$. By (RCS) and local posetality, higher cells are binary pastes and (GCS) $\mathfrak{M}_w = 0 \ \forall w \iff \Gamma_{g, f} = 0 \ \forall (g, f) \iff \int F \rightarrow C$ is a split opfibration; the mixed Beck–Chevalley interval vanishes exactly when the chosen restriction/action lifts commute.

Positive section envelope. For objectwise surjective $q : E \twoheadrightarrow Y$, write $B_Y := \mathbb{Z}^{(Y)}$ and $\xi_q : 0 \rightarrow K \rightarrow \mathbb{Z}^{(E)} \xrightarrow{q\#} B_Y \rightarrow 0$. By freeness, $(-) \dashv \lim_{\leftarrow B}$ and (UDR/DLC), (PSE) $\text{Ext}_{\mathbf{Ab}^{\text{op}}}^n(B_Y, K) \cong \text{Hom}(B_Y, \text{Hist}_{\mathbf{B}}^n(K))$ ($n \geq 0$), $[\xi_q] \mapsto \delta_q$ ($n = 1$), $[\xi_q] = 0 \iff \text{Spl}(q\#) \neq \emptyset$, whereas actual realization requires (PSE₊) $\text{Sect}(q) \cong \text{Spl}_{\mathbf{CMon}}(\mathbb{N}^{(q)}) \cong \text{Spl}_{\star}(q\#) \subseteq \text{Spl}_{\mathbf{Ab}}(q\#)$. For a homomorphism $A : G \rightarrow H$ with source submonoid G_+ put $\text{PosLift}_A(y) = A^{-1}\{y\} \cap G_+$, $\text{Poslm}(A) = A(G_+)$ and, when $A(G_+) \subseteq H_+$, $\text{PosSh}(A) = A(G) \cap H_+$, $\text{PosRes}(A) = \text{PosSh}(A) \setminus \text{Poslm}(A)$. Then (PLI) $\text{PosLift}_A(y) \neq \emptyset \iff y \in \text{Poslm}(A)$ and, for positive composable A, B , (RFE)

$\text{PosLift}_{BA}(z) \cong \int_{y \in \text{PosLift}_B(z)} \text{PosLift}_A(y)$. For $A : \mathbb{Z}^R \rightarrow L$ take $G_+ = \mathbb{N}^R$, $H_+ = L \cap A(\mathbb{R}_+^R)$ and $r_A(y) = \min\{n \geq 1 : \text{PosLift}_A(ny) \neq \emptyset\}$ on $\text{PosSh}(A)$. Rationality gives (FRS) $y \in \text{PosSh}(A) \iff [y \in A\mathbb{Z}^R, r_A(y) < \infty]$, so $\text{PosRes}(A) = \{r_A > 1\}$; (PSE₀/PSE₊) are the shadow/image tests for $D_q : s \mapsto q_\#s$. For finite R and finite-free L , (LCP) $\text{LatProf}_A(U) = \text{coker } A_U$, with quotients for $U \subseteq V$, and $v_A(U) = \text{rk ker}(A_U \otimes \mathbb{Q}) = |U| - \text{rk } L + \text{rk LatProf}_A(U)$, $\text{Tor LatProf}_A(U) = ((\text{im } A_U \otimes \mathbb{Q}) \cap L)/\text{im } A_U$. Since $\text{im } A_U$ is free, (ACS) $0 \rightarrow (\text{ker } A_U) \otimes C \rightarrow \text{ker}(A_U \otimes C) \rightarrow \text{Tor}_1^{\mathbb{Z}}(\text{LatProf}_A(U), C) \rightarrow 0$ naturally in abelian C . Hence rational circuits are the minimal $v_A > 0$ sets, torsion is saturation, $C = \mathbb{R}$ is the characteristic-zero face, and for $v_A(U) = 0$ the $\mathbb{Z}/p^k$ kernel is the p^k -torsion of $\text{LatProf}_A(U)$. For $A_I \otimes \mathbb{Q}$ injective put $d_I = |\text{Tor LatProf}_A(I)|$, $D_A = \max_I d_I$, $N_A = \text{lcm}_I d_I$. (FRB) $r_A(y) \leq D_A$ and $\text{PosLift}_A(N_A y) \neq \emptyset$: a minimal-support conic lift $Ax = y$ has independent I , and $d_I y \in \text{im } A_I$ forces $d_I x \in \mathbb{N}^I$. The family $A_m = (m, m+1)$ has $r_{A_m}(1) = m$, so no A -uniform bound exists. For $q : \coprod_j O_j \rightarrow *$ over BG , $n_j = |O_j|$ and $\text{LatProf}_n(U) \cong \mathbb{Z}/\gcd(n_j : j \in U)$ ($U \neq \emptyset$); hence $\text{Spl}_{\text{Ab}}(q_\#) \neq \emptyset \iff \gcd_j n_j = 1$ and $\text{Sect}(q) \neq \emptyset \iff \text{Spl}_+(q_\#) \neq \emptyset \iff \min_j n_j = 1$, while on $\gcd_j n_j = 1$, $r_n(1) = \min_j n_j$. Also (SIC_P) gives $\mathcal{P}(X)^G \cong_{\text{aff}} \mathcal{P}(\{O_1, \dots, O_r\})$; rows $(2, 3)$, (2) separate replication from signed obstruction. Finally $A_+ = (1, 1)$ and $A_- = (1, -1)$ have identical LatProf but $\text{PosLift}_{A_+}(0) = \{0\}$ and $\text{PosLift}_{A_-}(0) = \mathbb{N}(1, 1)$, so LCP/ACS do not generate positivity, (RCR) stays distinct, and the replication bounds are nonclassifying. For $i : (E, q) \hookrightarrow (E', q')$ over $\underline{Y}$ and induced $i_K : K \rightarrow K'$, $R^1 \lim (i_K) \delta_q = \delta_{q'}$, $\lim K = \lim K' = 0 \Rightarrow \lim (K'/K) \cong \text{ker}(R^1 \lim K \rightarrow R^1 \lim K')$, and (SOI) $\text{Sect}(VU) \cong \coprod_{s \in \text{Sect}(V)} \text{Sect}(s^*U)$; projection gives the preceding obstruction, and on the cloven face this is the object part of (RWF). For $n \geq 1$ let $\mathbf{Q}_n \subseteq \text{Fun}([n]^{\text{op}}, \mathbf{Set})$ be the full subcategory on surjection flags $X_n \rightarrow \dots \rightarrow X_0$ and $V_n = \text{ev}_0$. Terminal pullback preserves surjections, so V_n is a fibration; put $\mathbf{QFlag}_n(Z) := \text{ResFib}(V_n)(Z)$. Splitting off the last arrow gives (URT) (CKPF) $\mathbf{QFlag}_n(Z) \simeq \int_{Y \rightarrow \mathbb{Z}_+} \mathbf{QFlag}_{n-1}(Y)$ ($n \geq 2$); equivalently these are nested kernel-pair flags with terminal marking; adjacent abelian faces are KSP, while activation/law/history remain attachments.

III. GLOBAL COMPATIBILITY AND INVERSE LIMITS

Fix the comparison spine $\mathfrak{C}_{\text{cmp}} = ((I, \leq), \{\mathcal{T}_i, \Pi_{ij}\})$ of Definition II.2, with $\Pi_{ii} = \text{id}$ and $\Pi_{ik} = \Pi_{ij}\Pi_{jk}$; metric repair uses only the declared attachment $\mathfrak{M}_{\text{cmp}} = \{d_i\}$.

a. *Global compatibility datum and closure realization.* $C_*^{\text{rec}} := \varprojlim_i \mathcal{T}_i = \{(z_i) : \Pi_{ij}z_j = z_i \ (i \leq j)\}$, $\text{Obs}(z) := \{(i, j) : \Pi_{ij}z_j \neq z_i\}$. Thus $\text{Obs}(z) = \emptyset \iff z \in C_*^{\text{rec}}; \Pi_{ij}(z) = z$ is shorthand only on a common ambient carrier. Put $X_I = \prod_i \mathcal{T}_i$, $L_I = \mathcal{P}(X_I)^{\text{op}}$ and, for directed $J \neq \emptyset$, $E_J = \{x : \Pi_{ij}x_j = x_i \ (i \leq j \text{ in } J)\}$, $C_J(A) = A \cap E_J \in \text{Cl}(L_I)$. Then

$$\begin{aligned} J \leq J' &\Rightarrow C_J \leq C_{J'}, & E_I &= C_I(X_I) = C_*^{\text{rec}}, \\ \mathcal{D} \text{ directed by inclusion, } \bigcup \mathcal{D} = I &\Rightarrow C_I = E_{\{C_J : J \in \mathcal{D}\}}, & E_I &= \bigcap_{J \in \mathcal{D}} E_J. \end{aligned} \quad (85)$$

Restriction gives $E_J \cong (\varprojlim_{j \in J} \mathcal{T}_j) \times \prod_{i \notin J} \mathcal{T}_i$, with exterior coordinates free until $J = I$. For $\mathcal{R}_I = \{(i, j) : i \leq j\}$ put $\mathcal{O}(A) = \cup_{x \in A} \text{Obs}(x)$, $C_F(A) = \{x \in A : \text{Obs}(x) \cap F = \emptyset\}$, $\mathbf{G}_{\text{cmp}}(F) = C_F(X_I)$ and $\mathbf{H}_{\text{cmp}}(A) = \mathcal{R}_I \setminus \mathcal{O}(A)$. With $a_{ij} = C_{\{(i, j)\}}(X_I)$, this is the common adjunction $\mathbf{G}_{\text{cmp}} \dashv \mathbf{H}_{\text{cmp}}$; monotone $s : L_I \rightarrow \mathcal{P}(\mathcal{R}_I)$ gives the (AFS/CAD) law $G_s(A) = A \vee_{L_I} \mathbf{G}_{\text{cmp}}(s(A)) = C_{s(A)}(A)$, $\mathfrak{F}_s(A) = C_{s(A)}$, with the exact fixed criterion $G_s(A) = A \iff s(A) \cap \mathcal{O}(A) = \emptyset$, and

$$\begin{aligned} \text{[SPB]} \quad s(A) \cap \mathcal{O}(A) = \emptyset \ (\forall A) &\Rightarrow G_s = I, & s(A) \subseteq \mathcal{O}(A) \ (\forall A) &\Rightarrow G_s(X_I) = X_I, \\ & & \mathcal{O}(A) \subseteq s(A) \ (\forall A) &\Rightarrow G_s = C_{\mathcal{R}_I}. \end{aligned} \quad (86)$$

The first row is (AFS); the others use monotonicity and $\mathcal{O}(E_I) = \emptyset$. If H acts equivariantly on X_I , $\mathcal{R}_I$ and $s(hA) = hs(A)$, then $s(X_I) \cap \mathcal{O}(X_I)$ is orbit-saturated; transitivity on nonempty $\mathcal{O}(X_I)$ forces $G_s(X_I) \in \{X_I, E_I\}$. The $1, 2 \leq 3$ projection example is sharp: intermediate staging requires persistent priority or multiple comparison orbits.

Theorem III.1 (Compact inverse-limit exactness). *Let $(\mathcal{T}_i, \Pi_{ij})$ be a nonempty compact-Hausdorff inverse system with continuous bonds. Its reconstructed limit C_*^{rec} is nonempty compact Hausdorff and $G_i = \pi_i(C_*^{\text{rec}}) = \cap_{j \geq i} \Pi_{ij}(\mathcal{T}_j)$. The comparison residual carried by the formal access presentation is*

$$\mathfrak{R}_{\text{cmp}} := (\mathfrak{C}_{\text{cmp}}, C_*^{\text{rec}}, (\pi_i : C_*^{\text{rec}} \rightarrow G_i)_{i \in I}, \mathbf{G}_{\text{cmp}} \dashv \mathbf{H}_{\text{cmp}}), \quad \text{Rta}(\mathfrak{F}_{\text{acc}}) := \mathfrak{R}_{\text{cmp}}.$$

The subdirect map to $\prod_i G_i$ need not be onto; failure of coordinate extension is finite-stage visible. For continuous $\pi : X \rightarrow T$, with X compact and T Hausdorff, $\text{Cat}(\pi) = \{t : \pi^{-1}\{t\} \text{ is not l.s.c. at } t\} \supseteq \partial\pi(X)$, with equality iff π is open over $\text{int } \pi(X)$. If $X \neq \emptyset$, $\text{ProbFib}_\pi(t) := (\pi_*)^{-1}\{\delta_t\} = \mathcal{P}(\pi^{-1}\{t\})$ has $\text{ex ProbFib}_\pi(t) = \delta(\pi^{-1}\{t\})$ and $\text{Cat}_\mathcal{P}(\pi) = \text{Cat}(\pi)$. Onto bonds give onto projections; countable-cofinal compact metrizable abelian systems have $\varprojlim^1 = 0$, while discrete kernel towers retain RPD/R2G; directed iterated limits agree.

For compact X, Y_r and finite F , exact realization of (z_r) , $\cap_r w_r^{-1}(z_r) \neq \emptyset$, a law with Dirac marginals and the separating support inequalities are equivalent. At projective level

$$(PI) \quad \mathcal{P}(C_*^{\text{rec}}) \xrightarrow{\sim} \varprojlim_i \mathcal{P}(\mathcal{T}_i), \quad \mu \mapsto ((\pi_i)_* \mu)_i,$$

an affine weak-* homeomorphism; by (DER) its extreme points are exactly compatible Dirac laws, whose closed convex hull is the whole space (closure unnecessary exactly in the finite case). For $\Omega = \prod_i O_i$ with infinitely many nontrivial finite coordinates, finite marginal families satisfy FMX: their kernel intersection is never the diagonal, whereas all finite marginals jointly are determining. Thus the identity readout lies in $C_\infty \setminus C_{\text{fin}}$ while PI still gives exact point landing: Scott nonfinitarity and limit nonlanding are different.

For $\mu, \nu \in \mathcal{P}(C_*^{\text{rec}})$ put $\mu_i = (\pi_i)_* \mu$, $\nu_i = (\pi_i)_* \nu$, $H_i = L_0^2(\mathcal{T}_i, \nu_i)$ and $\mathbf{CE}_{ij}^\nu f = d(\Pi_{ij})_*(f \nu_j)/d\nu_i$ for $i \leq j$. These conditional-expectation contractions preserve zero mean and compose along the inverse system. Relative entropy and BKM data reconstruct projectively: $D(\mu \parallel \nu) = \sup_i D(\mu_i \parallel \nu_i)$ and

$$(P2) \quad L_0^2(C_*^{\text{rec}}, \nu) \cong \{(f_i) \in \varprojlim_i (H_i, \mathbf{CE}_{ij}^\nu) : \sup_i \|f_i\|_2 < \infty\}, \quad \|f\|_2^2 = \sup_i \|f_i\|_2^2,$$

under $f \mapsto (f_i)$ with $f_i = d(\pi_i)_*(f\nu)/d\nu_i$. For compatible weak-* compact C_i , $C = \varprojlim C_i$, $M = \sup_i \inf_{C_i} D$, $K_i(M) = C_i \cap \{D \leq M\}$, one has

$$\begin{aligned} \text{[PU]} \quad \inf_C D(\cdot \| \nu) &= M, \quad M < \infty \Rightarrow \text{Opt}_{C \rightarrow *}^{D(\cdot \| \nu)}(*) \cong \varprojlim_i K_i(M), \\ D(\mu \| \nu) < \infty : \quad P_i^\nu \mu &= ((d\mu_i/d\nu_i) \circ \pi_i)\nu, \quad D(\mu \| \nu) = D(\mu \| P_i^\nu \mu) + D(\mu_i \| \nu_i). \end{aligned} \quad (87)$$

with $P_i^\nu P_j^\nu = P_i^\nu$ for $i \leq j$; finite entropy gives TV convergence and the BKM tangent projection. The two-stage uniform/Dirac, Bernoulli/Dirac and unbounded partial-sum examples separately show section, weak-only and unbounded- L^2 sharpness. No point or maximal joining is selected.

Proof. Compact FIP yields the limit/images/projections; the compact difference map gives $\varprojlim^1 = 0$. Fibre l.s.c. on the image interior is relative openness, while boundary points have nearby empty fibres. For (DER/PCR), $\pi_*\mu = \delta_t$ iff $\text{supp } \mu \subseteq \pi^{-1}\{t\}$; for non-Dirac μ , choose f nonconstant on $\text{supp } \mu$ and set $h = f - \int f d\mu$; then $(1 \pm \varepsilon h)\mu$ for small $\varepsilon > 0$ split μ , so only Dirac laws are extreme. Forward l.s.c. follows by finite Dirac approximation plus simultaneous support hits; conversely an Urysohn f supported in U makes l.s.c. at δ_x force nearby point fibres to meet U . Finite realization is compact intersection/separation. Compatible marginals give a positive norm-one cylinder functional; Stone–Weierstrass/Riesz²⁰ (Thms. 4.51, 7.2) gives the unique global law and finite-test boxes the closed Dirac hull. Pullback identifies H_i with directed L^2 cylinder subspaces and CE_{ij}^ν with conditional expectations, giving the norm identity. The cylinder entropy formula gives $D(\mu \| \nu) = \sup_i D(\mu_i \| \nu_i)$; compact sublevels give minimizers, conditional-expectation likelihood ratios the Pythagorean/tangent projection, and Pinsker TV convergence. Examples are sharp. $\square$

b. Typed site and gluing obstruction. For an injective frame homomorphism $i : \mathbf{M} \hookrightarrow \mathbf{L}$, $\exists \vdash i \vdash \forall$, and a nucleus j on $\mathbf{L}$, put

$$\text{(SQ)} \quad j_i^\square := \forall ji = \max\{k : ik \leq ji\} \leq j_i^\diamond := \bigwedge_{ji \leq ik} k = \min\{k : ji \leq ik\}, \quad j_i^\square = j_i^\diamond \iff j(i\mathbf{M}) \subseteq i\mathbf{M} \iff \exists ! \bar{j} : i\bar{j} = ji,$$

where the meet/ranges are over nuclei on $\mathbf{M}$. Strictness already occurs for $P = \{c < a, c < b\} \rightarrow Q = \{0 < 1\}$, $q(c) = 0$, $q(a) = q(b) = 1$, $i = q^{-1}$, $j(D) = D \cup \downarrow a$: $j_i^\square(\emptyset) = \{0\} \neq Q = j_i^\diamond(\emptyset)$. For the compact systems of Theorem III.1, landing failure is instead degree zero and finite-stage visible, while the countably cofinal compact metrizable abelian-group face has $\varprojlim^1 = 0$. After a site and a C_2 -local system are supplied, the independent gluing layer is

$$[\epsilon] = 0 \in \check{H}^1(-; C_2) \iff \epsilon_{ab} = s_a s_b^{-1} \text{ for a 0-cochain } (s_a) \iff \text{the local signs glue globally.}$$

The adjunction identities and the Čech coboundary criterion prove the equivalences; site/torsor gluing is therefore attachment data, not hidden degree-zero residual.

For small $D : C \rightarrow \mathbf{Set}$ and cone $(\pi_c : X \rightarrow D(c))_c$, put $X_c = \pi_c(X)$, $\widehat{X}_{\text{sh}} = \varprojlim_c X_c$, $J_X(x) = (\pi_c x)_c$ and $F_p = J_X^{-1}\{p\}$. By (HRD/RST), $\text{SCX}(X, \pi_\bullet) := \bigoplus_{p \in \widehat{X}_{\text{sh}}} \mathfrak{H}_p(J_X) \cong \mathfrak{H}_\bullet(J_X)$, with $(\text{SCX}) H_1 \text{SCX} \cong \bigoplus_{p \in J_X(X)} \widehat{\mathbb{Z}}^{(F_p)}$, $H_0 \text{SCX} \cong \mathbb{Z}^{(\{p : F_p = \emptyset\})}$. Thus $H_1 = 0$ is joint faithfulness and $H_0 = 0$ landing; (URT) transport coherence remains separate. On a directed cone, SCX is the constant-read-out face of (JKC/GTS): $\widehat{\pi}$ is bijective and its H_0 is exactly joint-key nonlanding, whereas the identity-read-out face below puts the same profile nonlanding in target extinction. If D is $\mathbf{Ab}$ -valued and the π_c are homomorphisms, augmentation yields

$$\begin{array}{ccc} \mathbb{Z}(X) & \xrightarrow{(U_{JX})^\#} & \mathbb{Z}(\widehat{X}_{\text{sh}}) \\ \downarrow \varepsilon_X & & \downarrow \varepsilon_{\widehat{X}_{\text{sh}}} \\ X & \xrightarrow{J_X} & \widehat{X}_{\text{sh}} \end{array} \quad \begin{array}{l} H_1 \text{SCX}(UX, U\pi_\bullet) \twoheadrightarrow \ker J_X, \\ H_0 \text{SCX}(UX, U\pi_\bullet) \twoheadrightarrow \text{coker } J_X. \end{array} \quad (88)$$

Representatives are $[r] - [0]$ and target basis classes. Strictness: $2\mathbb{Z} \hookrightarrow \mathbb{Z}$ has raw odd exterior but cokernel $\mathbb{Z}/2$, whereas $\mathbb{Z} \rightarrow \mathbb{Z}/2$ has infinite-rank raw kernel but kernel $2\mathbb{Z}$. Thus the additive interface quotients raw grades. For decreasing $A_i \leq A$, (FDE) gives $\ker J_A = \bigcap_i A_i$, $\text{coker } J_A \cong \text{Hist}_I^1(A_\bullet)$, the quotient read as LimMarg_λ in (RPD) and $\widehat{A}_u/\iota(A)$ in (DIG); on finite-free lattice towers (PPH) identifies the same Hist^1 as the algebraic path-component quotient of the prophase, while bare Set has no such history identification.

Directed non-Archimedean completion boundary. Let I be nonempty directed, $i \leq j \Rightarrow R_j \subseteq R_i$, and put $R_\infty = \bigcap_i R_i$, $Q = X/R_\infty$, $D_i = X/R_i$, $\rho_i : Q \twoheadrightarrow D_i$, $U_i = R_{\rho_i}$ and $\widehat{Q} = \varprojlim_i D_i$. Let $\eta : Q \hookrightarrow \widehat{Q}$ be the canonical joint map. It induces the separated uniformity $\mathcal{U}_R = (U_i)$; (PEL) and completeness give (NCB) $\text{Cpl}_{\mathcal{U}_R}(Q) \cong \widehat{Q}$. By (SCX/HRD), $(H_1, H_0) = (0, \mathbb{Z}^{(\widehat{Q} \setminus \eta(Q))})$. For $X = Q = \mathbb{Z}$ and R_n congruence mod 2^n , one has $D_n = \mathbb{Z}/2^n\mathbb{Z}$, $\widehat{Q} = \mathbb{Z}_2^{\text{ad}}$, and dense proper $\eta(\mathbb{Z})$, proving strictness.

For $R_0 \supseteq R_1 \supseteq \dots$ put $\delta_R(x, y) = \min\{n : \rho_n x \neq \rho_n y\}$, $d_R = 2^{-\delta_R}$ ($2^{-\infty} = 0$), and $\widehat{d}_R(p, p') = 2^{-\min\{n : p_n \neq p'_n\}}$. These metrics generate the preceding uniformities and (UCB) $\text{Cpl}_{d_R}(Q) \cong (\widehat{Q}, \widehat{d}_R)$, with the same (NCB/SCX) boundary; the metric remains a cofinal-resolution coordinate. For mutually cofinal (S_m) with the same terminal quotient set and canonical maps η_R, η_S , put $\alpha(n) = \min\{m : S_m \subseteq R_n\}$, $\beta(m) = \min\{n : R_n \subseteq S_m\}$. Then

$$\begin{aligned} \text{[CFG]} \quad \mathcal{U}_R &= \mathcal{U}_S, \quad \widehat{Q}_R \cong \widehat{Q}_S, \quad \widehat{Q}_R \setminus \eta_R(Q) \cong \widehat{Q}_S \setminus \eta_S(Q), \\ d_S(x, y) < 2^{-\alpha(n)} &\implies d_R(x, y) < 2^{-n}, \quad d_R(x, y) < 2^{-\beta(m)} \implies d_S(x, y) < 2^{-m}. \end{aligned} \quad (89)$$

Sharpness is independent: on $A_{\text{fin}} = \bigoplus_{j \geq 0} \mathbb{Z}/2$, alphabets $\{T\}$ and $\{T, T^2\}$ have depths $k, [k/2]$ but identical completion $\prod_j \mathbb{Z}/2$, raw boundary and (ASL/FDE) history; $(\mathbb{Z}, 2^{n+1}\mathbb{Z})$ versus $(\mathbb{Z}_2^{\text{ad}}, 2^{n+1}\mathbb{Z}_2^{\text{ad}})$ have identical stages/kernel but dense-proper versus onto completion, with $\mathbb{Z}_2^{\text{ad}} \setminus \mathbb{Z} \twoheadrightarrow \mathbb{Z}_2^{\text{ad}}/\mathbb{Z}$. Hence (NCB/CFG/UCB/PQTF/ASL–FDE) remain distinct completion/gauge/metric/compact/additive faces.

Filtered realization-gate law. For $X_m \subseteq \dots \subseteq X_0$, $\lambda : R \rightarrow X_m$ and composite $j : R \rightarrow X_0$, $F^*H_0 = \mathbb{Z}^{(X_r \setminus \lambda(R))}$ and HRD/UAR give $H_1 \mathfrak{H}_\bullet(j) \cong \mathbb{K}(\lambda)$ together with the exact gate grading

$$\text{[RGL]} \quad H_0 \mathfrak{H}_\bullet(j) = \text{ExtRes}(j) \cong F^m H_0 \oplus \bigoplus_{r=1}^m \mathbb{Z}^{(X_{r-1} \setminus X_r)}.$$

Its basis depth is $r, m + 1, +\infty$ on the successive gates, final exterior and realized profiles; inserting an intermediate gate only refines the corresponding grade.

For $p \in \widehat{X}_{\text{sh}}$, $F_c(p) = \{x : \pi_c x = p_c\}$ and $\text{Fail}_X(p) = \{A : \cap_{c \in A} F_c(p) = \emptyset\}$, SUP gives

$$\begin{aligned} \text{Cons}_X(p) &= \{F \in [I_C]^{<\omega} : F \notin \text{Fail}_X(p)\}, & \text{MinObs}_X(p) &= \text{FinFront}_{I_C}(\text{Fail}_X(p)), \\ \text{IntSc Fail}_X(p) &= \uparrow \text{MinObs}_X(p), & \text{ShObs}_X(p) &= (F_p, \text{Fail}_X(p)), \\ \text{FSh}(X) &= \{p : \text{MinObs}_X(p) = \emptyset\}, & \text{FinRes}(X) &= \{p : I_C \in \text{IntSc Fail}_X(p)\}, \\ \text{LimRes}(X) &= \{p : I_C \in \partial_{\text{Sc}} \text{Fail}_X(p)\}. \end{aligned} \quad (\text{FCX}) \quad (90)$$

With $\mathcal{K}_p = \mathcal{P}(I_C) \setminus \text{Fail}_X(p)$ and $\mathcal{K}_p^{\text{dc}} = \{A : [A]^{<\omega} \subseteq \mathcal{K}_p\}$, finite interpolation is exactly Scott closure and

$$[\text{SFC}] \quad \text{IntSc Fail}_X(p) = \mathcal{P}(I_C) \setminus \mathcal{K}_p^{\text{dc}}, \quad \partial_{\text{Sc}} \text{Fail}_X(p) = \mathcal{K}_p^{\text{dc}} \setminus \mathcal{K}_p. \quad (91)$$

Thus KGC gives $\text{FSh}(X) \cong \lim_{\leftarrow F \in I_C} \text{im } \pi_F$; with closed topological finite-coordinate images this is $\overline{J_X(X)}$ and the limit residual is its boundary outside $J_X(X)$.

For decreasing $E_j \subseteq E$ with onto $q_j = q|_{E_j}$, set $E_\infty = \cap_j E_j$, $A_j = \mathbb{Z}^{(E_j)}$, $K_j = \ker q_{j\#}$. LTX/PEL gives $\text{Sect}_{\text{cart}}(U_y) \cong q_\infty^{-1}\{y\}$ and $\mathfrak{o}_q(y) = H_0 \chi_q([y])$, with RS exactly $\mathfrak{o}_q(y) = 0$ and positive locus $q_\infty(E_\infty)$. The independent tail coordinate is $\text{VEx}(q) = \text{coker } H_0 \chi_q \cong \text{im}(\text{Hist}^1 K_\bullet \rightarrow \text{Hist}^1 A_\bullet)$, the $p = 0$ HTF base. On countable onto towers, $\text{VEx}(q) = 0$ iff the tower is eventually constant; every nonstationary tail has at least continuum residual by the split $d_n = [e_n] - [e_n^*]$ retract. The binary-shift tower has surjective limit and nonzero tail excess, while the constant tower has the same limit and zero excess. Hence gate depth, finite obstruction, pure-limit nonlanding, multiplicity and tail excess remain independent; PSE_+ is not absorbed.

Filtered derived residual gate. The relative finite-observation, shadow and first-realization rows are one filtered specialization of the exact functor $\mathbf{DRes}_I = \mathbf{Rlim}_{\leftarrow I}$. Let $X_0 \subseteq \dots \subseteq X_m \subseteq X_{m+1} = R$, let (ρ_i) be jointly faithful, and for finite $F \subseteq I$ put $T = Y_{X_0|R}$, $E_F^{(r)} = \{(t, x) \in T \times X_r : \rho_F x = \rho_F t\}$, $K_F^{(r)} = \ker q_{F\#}^{(r)}$ and $G_F^r = K_F^{(r)} / K_F^{(r-1)}$. Finite realizability makes $q_F^{(r)}$ onto and joint faithfulness gives $\varprojlim K_F^{(r)} = 0$. Exactness of $\mathbf{DRes}$ applied to $K^{(r-1)} \rightarrow K^{(r)} \rightarrow G^r$ yields simultaneously

$$\begin{aligned} (\text{LTX}_{\text{flag}}) \quad \mathbb{Z}^{(T \cap (X_r \setminus X_{r-1}))} &\hookleftarrow \lim_F G_F^r \xrightarrow{\sim} \ker(H_{r-1}^1 \rightarrow H_r^1), \\ (\text{HTF}_{\text{flag}}) \quad 0 \rightarrow \mathbb{Z}^{(T \cap (X_r \setminus X_{r-1}))} &\rightarrow \lim_F G_F^r \rightarrow \ker(V_{r-1} \rightarrow V_r) \rightarrow 0, \\ (\text{RPS}) \quad 0 \rightarrow \text{Birth}_r^p &\rightarrow \text{Hist}^p(G_\bullet^r) \xrightarrow{\partial_r^p} \text{Death}_r^{p+1} \rightarrow 0, \end{aligned}$$

where $H_r^p = \text{Hist}^p(K_\bullet^{(r)})$, $\text{Birth}_r^p = \text{coker}(H_{r-1}^p \rightarrow H_r^p)$ and $\text{Death}_r^{p+1} = \ker(H_{r-1}^{p+1} \rightarrow H_r^{p+1})$. Thus geometric first realization, birth and surviving tail persistence are separate coordinates.

More generally, a finite filtration $0 = F_{-1}K \rightarrow \dots \rightarrow F_m K = K$ with $G_r K = \text{Cone}(F_{r-1}K \rightarrow F_r K)$ has exact couple

$$\mathcal{D}_{r-1}^p \xrightarrow{i_r^p} \mathcal{D}_r^p \xrightarrow{j_r^p} \mathcal{E}_r^p \xrightarrow{k_r^p} \mathcal{D}_{r-1}^{p+1}, \quad \text{im } i = \ker j, \text{ im } j = \ker k, \text{ im } k = \ker i,$$

with $\mathcal{D}_r^p = \text{Hist}^p(F_r K)$, $\mathcal{E}_r^p = \text{Hist}^p(G_r K)$, $E_1^{r,p} = \mathcal{E}_r^p$, $d_1 = j_{r-1}^{p+1} k_r^p$ and $E_\infty^{r,p} \cong \text{gr}_r^F \text{Hist}^p(K)$. This is the common construction of RPS, finite KTR and RHTX. Each page quotient $q_s^v : Z_s^v \twoheadrightarrow E_{s+1}^v$ has

$$(\text{SRP}) \quad \text{PageFib}_{s+1}^v = \text{Tab}((\text{disc } q_s^v)^*) \rightarrow \text{disc } E_{s+1}^v, \quad (\text{PageFib}_{s+1}^v)_e \text{ a } B_s^v\text{-torsor},$$

and $E_s^v / Z_s^v \cong B_s^{v+\delta_s}$. The SUD cosyzygy face gives $\text{Hist}^p(A_{n+1}) \cong \text{Hist}^{p+1}(A_n)$ ($p \geq 1$), so terminal triangle fibres may lie over nonzero bases in every finite degree; carriers are noncanonical while derived groups are invariant.

Naturality gives

$$(\text{LTX}_{\text{van}}) \quad R^1 \varprojlim (i_r) \mathfrak{o}_{r-1} = \mathfrak{o}_r, \quad T \cap X_r = \mathfrak{o}_r^{-1}(0), \quad T \cap (X_r \setminus X_{r-1}) = \{\mathfrak{o}_{r-1} \neq 0, \mathfrak{o}_r = 0\},$$

and, for $Q_\bullet = K_\bullet^{(m+1)} / K_\bullet^{(0)}$,

$$(\text{LTX}_{\text{geom}}) \quad \mathbb{Z}^{(T \setminus X_0)} = \bigoplus_r \mathbb{Z}^{(T \cap (X_r \setminus X_{r-1}))} \hookleftarrow \lim_F Q_F \xrightarrow{\sim} \ker(H_0^1 \rightarrow H_{m+1}^1),$$

whose residual quotient is $\ker(V_0 \rightarrow V_{m+1})$ and may survive unique positive realization. At $m = 0$ retain

$$[\text{LTX}_{\text{rel}}] \quad (\text{LTX}_{\text{flag}})_{|m=0}, \quad R \setminus X = (R \setminus T) \sqcup (T \setminus X).$$

The shadow tower has $H_0 \chi = \bar{\Delta}_X : \mathbb{Z}^{(\text{LimRes } X)} \hookrightarrow \text{Hist}^1(K_\bullet^X)$ and $p \in \text{LimRes}(X) \iff \text{Sect}_{\text{cart}}(U_p) = \emptyset \iff H_0 \chi([p]) \neq 0$. Thus finite obstruction, pure-limit nonlanding and multiplicity remain distinct from Samskara.

The noncollapse examples are unchanged: the two three-point profiles have the same MinObs but different first failure grades; $[\mathbb{N}]^{<\omega}$ and the coinfinite subsets have equal $(F_p, \text{MinObs}, \mathcal{K}_p^{\text{dc}})$ and different failure sets; compact continuous Hausdorff stages have no limit residual. Scott shell and continuation catastrophe vary independently: $[0, 1] \hookrightarrow \mathbb{R}$ has empty shell but $\text{Cat} = \{0, 1\}$ by the compact-fibre catastrophe clause of Theorem III.1, whereas eventually-zero sequences have pure shell with finite-stage open maps, giving CAX. Finally for $P = \varprojlim P_i$, (PEL) identifies $\widehat{L}_{\text{sh}} = \varprojlim L_i$ with R_+ , L with RS and $\text{LimRes}(L) = \widehat{L}_{\text{sh}} \setminus L$; continuous bonds give the closure, SCX/HRD the free H_0 basis, NCB the Cauchy face and LTX_{rel} the zero-multiplicity history lift. FDE/ASL remain independent.

For finite $F : \mathcal{B} \rightarrow \mathbf{Set}$ put $S = P_{\text{coh}} = W_{\text{Id}}(F)$, $\text{CohLaw}(F) = W_{\mathcal{P}}(F)$ and $m_F : \mathcal{P}(S) \rightarrow \text{CohLaw}(F)$. For nonempty finite $\mathcal{B}$, F , $(\text{RL}) = (\text{RL}_{\mathcal{P}})$; Dirac gives $d_F = m_F \delta_S$, $D_F = \text{im } d_F$, and (CPL) yields (CLH) $D_F = \{(\delta_{x_b})_b : x \in S\} \subseteq \text{PtLaw}(F) := \text{im } m_F = \text{conv } D_F \subseteq \text{CohLaw}(F)$. Thus the phase residual splits into non-Dirac point laws and non-point coherent laws; $2/1$ and a regular nontrivial G -set separate the two. Moreover $\text{PtLaw} \neq \emptyset \iff (\text{RS})$, while $\text{CohLaw} = \text{PtLaw}$ is the (FQL/MIS/RPF/ECS) face. With $A = \bigoplus_b (\pi_b)_\#$, (CLF) $m_F^{-1}(a) = A^{-1}(a) \cap \Delta_S$.

For a von Neumann algebra M and continuous real-linear $D : M_{*,h} \rightarrow (Y, \tau)$ write Y' for the continuous dual, $s(x) = \sup \sigma(x)$, $\text{L}_D(y) = \{\varphi \in S(M) : \varphi(D^* \ell) = \ell(y) \forall \ell \in Y'\}$ and $\text{SepRes}_D(y) = \{\ell : \ell(y) > s(D^* \ell)\}$. Hahn–Banach and weak-* compactness give $(\text{PID}_n) \overline{D(S_n)}^\tau = \{y : \text{SepRes}_D(y) = \emptyset\} = \{y : \text{L}_D(y) \neq \emptyset\}$. Let $q_n : M^{**} \twoheadrightarrow M$ be the canonical normal quotient, $\ker q_n = M^{**}(1 - z_n)$.

For $\tilde{\varphi} \in M_*^{**}$ set $\varphi_n = \tilde{\varphi}(z_n \cdot)$, $\varphi_s = \tilde{\varphi}((1 - z_n) \cdot)$. Since positive η is normal iff $s(\tilde{\eta}) \leq z_n$, φ_n is maximal normal and φ_s singular; hence the split is unique (NSD). Thus $\Sigma_s(M) = \{\psi \in M_*^+ : \psi \text{ singular}, \|\psi\| \leq 1\}$; with $e_D(\varphi) = (\varphi(D^*\ell))_\ell$, $j_Y(y) = (\ell(y))_\ell$, $\sigma(\varphi) = \varphi_s$, one has $L_D(y) = e_D^{-1}\{j_Y(y)\}$ and $\text{SingRes}_D(y) = T_{e_D, \sigma}(j_Y y)$. Hence (PID_n/NSR) closure is nonempty residual fibre and exact normal landing is that fibre containing 0. Finite dimension recovers (PID/CLF/TF); for partial traces $D^*(H_C)_C = \sum_C H_C \otimes 1_{F \setminus C}$. For $M = \ell^\infty(\mathbb{N})$, $D(p) = \sum_n p_n/n$ has normal image $(0, 1]$ but shadow $[0, 1]$, realized at 0 only by singular free-ultrafilter states; unequal mixtures of disjoint singular characters have common central support, so the singular coordinate is neither scalar- nor support-determined. Since $m_F = \xi_F^P$, $\text{GlobLawFib}(F) = \text{WRes}_{P, !}(F; *)$ has fibre disc $m_F^{-1}(a)$ and support $\text{PtLaw}(F)$; (R+) removes the second (CLH) gap, (R0) means m_F bijective, and nonterminal fibres are nonunique lifts. (KCM/LCM) is its H_0/H_1 shadow; on full support (AFP)=ker A , so injectivity is $\ker A = 0$ for $S \neq \emptyset$.

Write $\text{Kar}(\mathcal{B})$ for idempotent completion. If $J : \text{Path}(T) \rightarrow \text{Kar}(\mathcal{B})$ is initial with T a finite forest, then for every finite-valued $F : \mathcal{B} \rightarrow \mathbf{Set}$, (CIF) $\text{CohLaw}_{\mathcal{B}}(F) = \text{PtLaw}_{\mathcal{B}}(F)$. Indeed F and $\mathcal{P}F$ extend across the Karoubi envelope because finite sets and finite probability simplices split idempotents, and initiality identifies their limits with those over $\text{Path}(T)$. On each forest component the edge law of $u \xrightarrow{f} v$ is the graph coupling $(\text{id}, Ff)_* \mu_u$; consistent vertex marginals glue recursively along the unique undirected paths to a joint law supported on the point limit. Taking products over components gives a global law with the prescribed coherent marginals. Hence for finite acyclic Q , (FQL) $|Q|$ forest $\iff \text{CohLaw}_{\text{Path}(Q)}(F) = \text{PtLaw}_{\text{Path}(Q)}(F) \forall$ finite-valued F , with cycle counterexample (PRH). The fork $a \rightrightarrows b \rightarrow c$, $pf = pg$, has thin groupoid completion but a coherent uniform law with empty point limit, so (NHR) $(\forall \lambda) \hat{\Gamma}_\lambda = 1 \nRightarrow [\forall F, \text{CohLaw} = \text{PtLaw}]$; its $\mathbb{Z}/2$ face is the nonloop (RCR) 2-cycle for (Ff, Fg) , killed by common postcomposition Fp , while a binary span retains (CLF) multiplicity. For a finite poset Q , choose a Dey–Lesnick initial scaffold $j : P \hookrightarrow Q^{21}$ (Def. 3.2 and proof of Thm. 3.5); its objects are the minima plus essential objects and its nonidentity relations are single minimum-to-essential edges, so $P \cong \text{Path}(\Gamma_P^+)$. Thus (MIS) Γ_P forest $\iff \text{CohLaw}_Q(F) = \text{PtLaw}_Q(F) \forall$ finite-valued $F : Q \rightarrow \mathbf{Set}$. Forward is (CIF). For a cycle choose odd ϵ , put $P_q = P \cap \{x : x \leq q\}$ and $Z_q = \{s : \text{Ob } P_q \rightarrow \mathbb{Z}/2 : s(v) + s(w) = \epsilon_{vw}\}$. The cited scaffold proof gives $P \cap A$ connected for every component A of $\text{dn}(q, Q)$, hence P_q is connected for every q : if $q \notin P$ there is one such component, while if $q \in P$ the scaffold edge from a chosen minimum in each component to q joins those connected pieces through q . Hence every nonempty Z_q is a two-point affine torsor. For $q \leq q'$ with both $Z_q, Z_{q'}$ nonempty, restriction $Z_{q'} \rightarrow Z_q$ is injective because a solution on the connected graph $P_{q'}$ is determined by its value at any vertex, and therefore is bijective; moreover $Z_q = \emptyset$ implies $Z_{q'} = \emptyset$. Thus $F_\epsilon(q) = Z_q$ when nonempty and $F_\epsilon(q) = \{*\}$ otherwise, with inverse restrictions on the nonempty locus and the unique maps into the singleton thereafter, defines a covariant functor. Uniform/Dirac laws are coherent, but the odd cycle forbids a section; if q is first with $Z_q = \emptyset$, its odd cycle lies strictly below q by scaffold minimality. Thus (MIS) is exact and scaffold-independent.

For finite $\mathcal{B}$, F put $K_F(b) = \{x : F\alpha x = F\beta x \forall \alpha \parallel \beta : b \rightarrow c\}$ and $\text{RelRes}_{\mathcal{B}}(F) = \text{cone}(\{(q_b \mu_b)_b : \mu \in \text{CohLaw}(F)\}, q_b : \mathbb{R}^{F(b)} \rightarrow \mathbb{R}^{F(b)} / \mathbb{R}^{K_F(b)}, \text{cone } \emptyset = \{0\})$. K_F is a subfunctor with $K_F \approx \bar{K}_F \pi_{\mathcal{B}}$ through reachability: on K_F parallel arrows coincide, so endomorphisms are identity and mutual arrows inverse. Global points lie in K_F , and $\text{RelRes} = 0$ iff every coherent law is supported there, equivalently all pairwise (RCR) residuals vanish. Therefore (RPF) $\text{RelRes}_{\mathcal{B}}(F) = 0 \Rightarrow (\text{CohLaw}, \text{PtLaw})_{\mathcal{B}}(F) \cong (\text{CohLaw}, \text{PtLaw})_{P_{\mathcal{B}}}(\bar{K}_F)$. Let $\mathbf{FinProb}_+$ be finite full-support probability spaces with deterministic law-preserving maps, $\alpha \equiv_{\text{fp}} \beta$ iff all $S : \mathcal{B} \rightarrow \mathbf{FinProb}_+$ identify them, and $\mathcal{B}_{\text{fp}} = \mathcal{B} / \equiv_{\text{fp}}$. Let $\equiv_0$ be equality and $\equiv_{n+1} = \text{Cg}(\equiv_n \cup \{(\alpha, \beta) : \alpha \parallel \beta, \alpha\gamma \equiv_n \beta\gamma \text{ for some } \gamma \text{ ending at } \text{dom } \alpha\})$; its finite stabilization $\equiv_{\text{epi}}$ gives the universal all-arrow-epic quotient $\mathcal{B}_{\text{epi}}$. Every coherent law restricts to its full-support functor and conversely, so $\mathcal{B}_{\text{fp}}$ is thin iff every coherent law is supported in K_F , equivalently iff all $\text{RelRes}_{\mathcal{B}}(F)$ vanish. Then (FPC/ECS) $\equiv_{\text{fp}} \equiv_{\text{epi}}$, $[\forall F \text{ RelRes}_{\mathcal{B}}(F) = 0] \iff \mathcal{B}_{\text{epi}}$ thin. Surjective $\mathbf{FinProb}_+$ arrows give $\equiv_{\text{epi}} \subseteq \equiv_{\text{fp}}$; if $C = \mathcal{B}_{\text{epi}}$ is nonthin, $H(x) = 2^{\mathbb{I}_y C(x, y)}$ has uniform law-preserving arrows because epicity makes precomposition injective (hence constant-fibre), and separates each distinct parallel pair, giving the reverse inclusion. Thus (RPF/MIS/FPC/ECS) gives universal $\text{CohLaw} = \text{PtLaw}$ iff $\mathcal{B}_{\text{epi}}$ is thin and an initial scaffold of $P_{\mathcal{B}}$ is a forest; (PRH) and (NHR/RCR) are its invertible/noninvertible relation faces.

Let $M_{\mathcal{B}} = \text{Min } P_{\mathcal{B}}$. For $R \subseteq \text{Ob } \mathcal{B}$ put $j_R = (\pi_r)_{r \in R}$ on $\varprojlim F$. Then (MRJ) $[\forall F \text{ finite-valued}, j_R \text{ injective}] \iff M_{\mathcal{B}} \subseteq \{[r] : r \in R\}$. Minimal classes propagate to every object; a missed one is refuted on $P_{\mathcal{B}}$ by $\mathbf{2}$ there and singletons elsewhere, then pulled back. For a transversal R , $S_R = j_R(\varprojlim F)$ and $h_b j_R = \pi_b$ give by (MIB/TF) $\ker A \cong \ker(\bigoplus_b h_{b\#}) = \mathbb{I}_h \otimes \mathbb{R}$, so (AFP/CLF/MRJ) is the minimal-root interaction kernel, canonically conjugate across transversals.

For nonempty small $\mathcal{B}$ with countably many arrows, let $P_{\mathcal{B}}$ be lower-directed when pairs have common lower bounds and $\mathcal{B}_{\text{epi}}$ the universal all-arrow-epic quotient. For standard-Borel-valued Borel F , $S_F = \varprojlim F$ is Borel. Put $m_F^{\text{sb}} : \mathcal{P}(S_F) \rightarrow \varprojlim_b \mathcal{P}(F(b))$ for marginalization and, for $\Gamma \in \mathcal{P}(S_F)$, $\mathcal{N}_b^\Gamma = L^\infty(\sigma(\pi_b), \Gamma)$. Then

$$(\text{DMI}) \quad [\forall F, m_F^{\text{sb}} \text{ injective}] \iff [\forall F, \Gamma, \text{AlgRes}_\Gamma((\mathcal{N}_b^\Gamma)_b) = 0] \iff P_{\mathcal{B}} \text{ lower-directed},$$

$$(\text{CPR}) \quad [\forall F, m_F^{\text{sb}} \text{ bijective}] \iff [P_{\mathcal{B}} \text{ lower-directed}, \mathcal{B}_{\text{epi}} \text{ thin}].$$

For (DMI), $d \rightarrow b$ gives $\mathcal{N}_b^\Gamma \subseteq \mathcal{N}_d^\Gamma$, so (BDS) plus lower-directedness kills AlgRes and predecessors determine cylinders. The two-downset $S_F = 2^2$, uniform Γ_0 , $u = (-1)^{x+y}$ gives distinct equivalent full-support $(1 \pm \varepsilon u)\Gamma_0$ with identical marginals (CPL/CIT). For (CPR), fix a coherent family $(\mu_b)_b$. The induced $(F, \mu) : \mathcal{B} \rightarrow \mathbf{SBP}_{\text{ae}}$ factors through $C := \mathcal{B}_{\text{epi}}$ because every arrow of $\mathbf{SBP}_{\text{ae}}$ is epic. Assume $P_{\mathcal{B}}$ lower-directed and C thin. For finite $J \subseteq \text{Ob } \mathcal{B}$, choose a common predecessor d_J and arrows $\gamma_b : d_J \rightarrow b$ and put

$$\lambda_J := ((F\gamma_b)_{b \in J})_* \mu_{d_J} \in \mathcal{P}\left(\prod_{b \in J} F(b)\right).$$

Thinness of C , after passing two choices to a further common predecessor, makes λ_J choice-independent; these laws are projective. The Daniell–Kolmogorov theorem¹⁹ (Thm. 6.16, p. 115) therefore gives $\lambda \in \mathcal{P}(\prod_b F(b))$ with finite marginals λ_J . For every $\alpha : b \rightarrow c$, the choice $(d, \gamma_b, \gamma_c) = (b, \text{id}_b, \alpha)$ gives $\lambda\{x_c = F(\alpha)x_b\} = 1$. Since $\mathcal{B}$ has countably many arrows, S_F is the Borel intersection of these graph equations and is λ -conull; hence $\lambda|_{S_F}$ is the required lift, unique by (DMI). If C is nonthin, the (FPC/ECS) Bernoulli construction lies in the present standard-Borel class (all coordinate sets are countable) and a distinct parallel pair forces an inverse-limit equality event whose domain marginal has probability 1/2, so no lift exists. Thus the converse follows together with (DMI). For finite $\mathcal{B}$, lower-directedness $\iff |M_{\mathcal{B}}| \leq 1$ gives (UMI/LCM); (NHR/DMI) keep H_0, H_1 independent.

c. *Projective reconstruction and residual-access dynamics bridge.* A tower has thin all-arrow-epic quotient and lower-directed bonding shape, so (CPR) gives set-level existence/uniqueness and (DMI) zero stage-coordinate BKM residual; (PI) strengthens this to affine weak-* reconstruction on its compact branch. On a countable cofinal standard-Borel tower set $\mu_n = (\pi_n)_* \mu$, $\nu_n = (\pi_n)_* \nu$ and $\mathcal{F}_n = \pi_n^{-1} \mathcal{B}(\mathcal{T}_n)$. On the stagewise-normal branch $\mu_n \ll \nu_n$ for every n , $L_n = (d\mu_n/d\nu_n) \circ \pi_n$ is a positive unit-mean ν -martingale and (PN) $\mu \ll \nu \iff (L_n)_n$ is UI. Indeed $\mu \ll \nu$ gives $L_n = \mathbb{E}_\nu[d\mu/d\nu \mid \mathcal{F}_n]$ and hence UI; conversely UI gives an $L^1(\nu)$ limit L , and (DMI) applied to the common stage marginals gives $L\nu = \mu$. If moreover $\mu_n \sim \nu_n$ for every n and $M_n = (d\nu_n/d\mu_n) \circ \pi_n$, then $\mu \sim \nu$ iff both (L_n) and (M_n) are UI; bounded stage KL in both directions suffices by de la Vallée–Poussin. On the compact (PI) regime, Density pullback makes $N_\nu = \mathcal{S}_n(L^\infty(\nu))$ onto closed L^1 stage simplices, and (PEL/PN) gives $\text{LimRes}(N_\nu) = \{\mu : \mu_n \ll \nu_n \forall n, (L_n)_n \text{ non-UI}\}$. Thus topological (PEL) is the product-topology (PID_n) closure, exact normal landing is UI, and Bernoulli laws witness nonempty residual. Compatible Feller kernels on $G_n = \pi_n(C_n^{\text{rec}})$ satisfy

$$\begin{aligned} \text{[PKR]} \quad & \{(\mathbf{P}_n) : (\Pi_{nm})_* \mathbf{P}_m = \mathbf{P}_n \Pi_{nm}\} \cong \{\mathbf{P} : \mathbf{P} \pi_n^* C(G_n) \subseteq \pi_n^* C(G_n) \forall n\}, \\ & \mathbf{P}(g \circ \pi_n) = (\mathbf{P}_n g) \circ \pi_n, \quad (\pi_n)_*(\mu \mathbf{P}) = \mu_n \mathbf{P}_n, \quad [\forall n : \mu_n \mathbf{P}_n = \mu_n] \iff \mu \mathbf{P} = \mu. \end{aligned} \quad (92)$$

If $\mu_n \mathbf{P}_n \sim \mu_n$ for all n , (PN) twice gives $\mu \mathbf{P} \sim \mu$ iff both likelihood-ratio martingales are UI; constant-output Bernoulli kernels make stagewise nonsingularity insufficient. Moreover $(D(\mu \mathbf{P} \parallel \mu), D(\mu \parallel \mu \mathbf{P})) = (\sup_n D(\mu_n \mathbf{P}_n \parallel \mu_n), \sup_n D(\mu_n \parallel \mu_n \mathbf{P}_n))$. The zero pair is invariance and finite two-sided price implies UI; the converse fails on $X = \{0\} \cup \{1/k : k \geq 2\}$ with $\mu\{1/k\} = c_\mu k^{-2}$ and $\nu\{1/k\} = c_\nu / [k(\log k)^2]$: $\mu \sim \nu$, $D(\nu \parallel \mu) = +\infty > D(\mu \parallel \nu)$. For a separating nested quotient tower, (UAR) gives $\bigcap_n \mathbb{K}(\pi_n) = 0$ (finite supports separate at one stage), whereas the Bernoulli (PN) boundary still permits equivalent finite-stage laws with singular global limits. The independent (RPD) axis is sharp: its countable witness has kernel exhaustion with $\text{LimMarg}_\omega \neq 0$ even when $\mu = \nu$ kills the analytic defect, whereas the Bernoulli (PN) boundary with a constant same-source read-out has $A = K_0$ and $\text{LimMarg} = 0$. Thus marginal exhaustion, marginal-limit continuity and normal/analytic completion are pairwise non-collapsing over (PQTF/PI/P2/PUI–PN/PKR/PCT/RPD).

IV. LOCAL ACCESS AND CONTROLLED RESPONSE

A. Extendability and operator-algebraic access fibers

Let $G_i := \pi_i(C_i^{\text{rec}})$ be the compact extendable locus from Theorem III.1. Its intrinsic set-level read-out is $\text{Ext}_i^{\text{bin}} = \mathbf{1}_{G_i}$; a quantitative gluing attachment is any $\delta_i : \mathcal{T}_i \rightarrow [0, +\infty]$, $\delta_i^{-1}(0) = G_i$, $\nu_i = -\delta_i$. Metric regimes may take $\delta_i = d_i(-, G_i)$ and $\text{Ext}_i^{\text{gr}} = e^{-\delta_i}$; otherwise use 0 on G_i , $+\infty$ off it. On this countable compact-metrizable spine coherent metrics exist: choose topology-compatible $0 \leq \rho_i \leq 1$ and $a_i > 0$ with $\sum_i a_i < \infty$, and set (CMR) $d_j(x, y) = \sum_{i \leq j} a_i \rho_i(\Pi_{ij} x, \Pi_{ij} y)$. Then the sum is uniform, $d_j \geq a_j \rho_j$, so d_j induces the original topology; the cocycle makes every Π_{ij} 1-Lipschitz, hence $G_i = \Pi_{ij}(G_j)$ gives $\delta_i(\Pi_{ij} z) \leq \delta_j(z)$. Put $\delta_r = \delta_{i(r)} \tau_r$, $\nu_r = -\delta_r$; (DI) descends fiber-constant scores and monotone zero-set-preserving scores are reparametrizations. Rescaling δ preserves extendability but rescales response, so calibration remains attachment data; (MRK/PCRG) below shows fixed distance-to-goal need not determine repair continuation. The inverse-limit formula excludes every nonextendable point at finite stage; nontrivial binary extendability requires non-surjective π_i .

The same U0 Boolean fiber is the access-opacity carrier. A restriction $\text{Res}_{O,t} : \mathcal{G} \rightarrow \mathcal{C}_{\text{loc}}^{O,t}$ is opaque at c exactly when $|\text{Res}_{O,t}^{-1}\{c\}| > 1$; for a unital inclusion $\iota_O : \mathcal{A}_O \hookrightarrow \mathcal{A}$ this is the state restriction fiber $K_O(\rho_O) = \{\rho : \rho|_{\mathcal{A}_O} = \rho_O\}$. Input opacity and subsequent record loss are therefore the same universal fiber mechanism on different typed arrows, not the same fiber. For a weak-* closed self-adjoint unital operator system $V \subseteq \mathcal{M}$, let $r_V^n : \mathcal{S}_n(\mathcal{M}) \rightarrow \mathcal{S}_n(V)$ be normal-state restriction, $r_V^n(\rho) = \rho|_V$, and put $R_V^n = R_{r_V^n} = \{(\rho, \sigma) \in \mathcal{S}_n(\mathcal{M})^2 : \rho|_V = \sigma|_V\}$ and, for a relation R on normal states, $\text{Const}_n(R) = \{x \in \mathcal{M} : \rho(x) = \sigma(x) \forall (\rho, \sigma) \in R\}$. The closed (EPN) face $x \perp (\rho, \sigma) \iff \rho(x) = \sigma(x)$ plus normal Hahn–Banach/Jordan gives (NAT) $V \subseteq \text{Const}_n(R) \iff R \subseteq R_V^n$, $\text{Const}_n(R_V^n) = V$, $R_{V_i V_i}^n = \bigcap_i R_{V_i}^n$, for arbitrary families in the operator-system lattice. Thus (GEP), for $V_I := \bigvee_i V_i$, $R_{V_I}^n = \bigcap_i R_{V_i}^n$ and $V_I = \mathcal{M} \iff \bigcap_i R_{V_i}^n = \Delta$: generated observable content and extinction of common opacity are the same operation in opposite order. Hence $V \mapsto R_V^n$ is a lossless opposite chart and $R \mapsto R_{\text{Const}_n(R)}^n$ its least-realizable closure. For every von Neumann subalgebra $\mathcal{N} \subseteq \mathcal{M}$, this specializes to $r_{\mathcal{N}}^n : \mathcal{S}_n(\mathcal{M}) \rightarrow \mathcal{S}_n(\mathcal{N})$, $R_{\mathcal{N}}^n = R_{r_{\mathcal{N}}^n}$, $\mathbb{K}_{\mathcal{N}}^n := \mathbb{K}(r_{\mathcal{N}}^n)$. (RRP) If $J : L \hookrightarrow P$ is an antitone order embedding and $\rho \mapsto \iota : \mathfrak{F} \hookrightarrow L$ a reflector, then $J\rho x = \max\{Jf : f \in \mathfrak{F}, Jf \leq Jx\}$; for bounded L , $(J\rho \perp, J\rho \top) = (\max J(\mathfrak{F}), \min J(\mathfrak{F}))$. This is least-content/greatest-residual polarity read through J , and (DCC) is the kernel-pair face. By (YFF/RFD), ρ^* reconstructs ρ , while $\mathbb{T}_{\mathfrak{F}} \rho = \text{Tab}(\rho^*)$ is its comma residual fibration; hence ρ^* determines the reflective fixed locus and every strict fibre without a selector. On an expected flag $\mathcal{N} \subseteq \mathcal{K}$, Jordan decomposition and (KAR/PST) give $\ker(E_{\mathcal{N}^*} |_{\mathcal{M}_{s,h}}) = \text{span}_{\mathbb{R}}\{\rho - \sigma : (\rho, \sigma) \in R_{\mathcal{N}}^n\}$, $\ker(E_{\mathcal{N}^*} |_{\mathcal{M}_{s,h}}) / \ker(E_{\mathcal{K}^*} |_{\mathcal{M}_{s,h}}) \cong V_{\mathcal{K}:\mathcal{N}}$. For a restricted normal-state family Ω , put $J_\Omega(V) = R_V^\Omega := R_V^n \cap \Omega^2$ and $\text{Const}_\Omega(R) = \{x : \rho(x) = \sigma(x) \text{ } (\rho, \sigma) \in R\}$. Then $J_\Omega(V) \leq_{\text{Eq}(\Omega)^{\text{op}}} R \iff V \subseteq \text{Const}_\Omega(R)$, so $J_\Omega \dashv \text{Const}_\Omega$ on the operator-system lattice, with closure $\text{Const}_\Omega J_\Omega$. If $\Omega \subseteq \mathcal{S}_{\text{fn}}(\mathcal{M})$, put $H_\Omega(\mathcal{N}) = \Omega \cap \text{Host}_\omega(\mathcal{N})$, $\text{l}_\Omega(\mathcal{N}) = \text{Gen}_\omega H_\Omega(\mathcal{N})$ and $\Sigma_\Omega(\mathcal{N}) = (R_{\mathcal{N}}^\Omega, H_\Omega(\mathcal{N}))$. Restricting (MSA) gives $\text{Gen}_\omega |_{\mathcal{P}(\Omega)} \dashv H_\Omega$, so l_Ω is idempotent and $\text{l}_\Omega(\mathcal{N}) \subseteq \mathcal{N} \subseteq \mathcal{C}_\Omega(\mathcal{N}) := \text{Const}_\Omega(R_{\mathcal{N}}^\Omega)$. Here $\mathcal{C}_\Omega$ is operator-system-valued; a von Neumann-algebra-valued right adjoint need not exist. For $\Omega = \mathcal{S}_{\text{fn}}(\mathcal{M})$, (MSA/NAT) gives $\text{l}_\Omega(\mathcal{N}) = \mathcal{N} = \mathcal{C}_\Omega(\mathcal{N})$.

a. *Restricted-state double-closure normal form.* Put $\mathbf{B}_\Omega(\mathcal{N}) := (\text{l}_\Omega(\mathcal{N}), \mathcal{C}_\Omega(\mathcal{N}))$. The equality and fiber-intersection laws for Σ_Ω are exactly the componentwise (SDR) quotient for $(\text{Gen}_\omega, H_\Omega)$ and $(J_\Omega, \text{Const}_\Omega)$. The sector-specific identification is

$$\text{[SBC]} \quad \overline{\mathbf{B}}_\Omega : \text{im } \Sigma_\Omega \xrightarrow{\cong} \text{im } \mathbf{B}_\Omega, \quad (R, H) \mapsto (\text{Gen}_\omega H, \text{Const}_\Omega R), \quad \overline{\mathbf{B}}_\Omega^{-1}(I, C) = (J_\Omega(C), H_\Omega(I)). \quad (93)$$

Here $H_\Omega \text{l}_\Omega = H_\Omega$ and $J_\Omega \mathcal{C}_\Omega = J_\Omega$ give the inverse. The two coordinates are independently necessary already for $\mathcal{M} = \mathbb{C}^3$, uniform ω , $\Omega = \{\rho = (1, 1, 6)/8, \sigma = (1, 2, 5)/8\}$: for block-constant algebras, $H_\Omega(\mathcal{N}_{0|1}) = H_\Omega(\mathcal{N}_{0|12}) = \emptyset$ but $(R_{\mathcal{N}_{0|1}}^\Omega, R_{\mathcal{N}_{0|12}}^\Omega) = (\Delta, \Omega^2)$, whereas $(R_{\mathcal{N}_{0|12}}^\Omega, R_{\mathcal{N}_{0|1}}^\Omega) = (\Delta, \Delta)$ but $(H_\Omega(\mathcal{N}_{0|12}), H_\Omega(\mathcal{N}_{0|1})) = (\Omega, \{\rho\})$. Hence neither one-coordinate subprofile determines Σ_Ω . Restricted Ω is also sharply nonfaithful: in tracial M_2 , with $\Omega = \{\rho_\pm = (1 \pm \epsilon \sigma_y)/2, 0 < \epsilon < 1\}$, both $\mathcal{N} = \mathcal{D}_x, \mathcal{D}_z$ have $H_\Omega(\mathcal{N}) = \emptyset$, $R_{\mathcal{N}}^\Omega = \Omega^2$, $\text{l}_\Omega(\mathcal{N}) = \mathbb{C}1$, $\mathcal{C}_\Omega(\mathcal{N}) = \text{span}\{1, \sigma_x, \sigma_z\}$ and equal Σ_Ω , although $\mathcal{D}_x \neq \mathcal{D}_z$ and $R_{\mathcal{D}_x \vee \mathcal{D}_z}^\Omega = \Delta$. Thus the common signature/double-closure fiber is nonprincipal

and (JCR/JCP) is nonzero/strict. More generally, for expected $\mathcal{N}_i$ put $V_I = \overline{\sum_i \mathcal{N}_i}^{w^*}$, $\mathcal{N}_V = W^*(\bigcup_i \mathcal{N}_i)$ and $W^\perp = \{\eta \in \mathcal{M}_{*,h} : \eta|_W = 0\}$. The full multiplicative-correlation residual and its finite-energy tangent are

$$\begin{aligned} 0 \rightarrow V_I \rightarrow \mathcal{N}_V \rightarrow \text{MulCor}_I &:= \mathcal{N}_V/V_I \rightarrow 0, & \text{MulCor}_{I,*} &:= V_I^\perp/\mathcal{N}_V^\perp, \\ \text{[MCR]} \quad R_{\mathcal{N}_V}^n \subseteq R_{V_I}^n = \bigcap_i R_{\mathcal{N}_i}^n, & \text{MulCor}_I = 0 \iff V_I = \mathcal{N}_V \iff R_{\mathcal{N}_V}^n = R_{V_I}^n. \end{aligned} \quad (94)$$

b. Restricted multiplicative-correlation fiber. For $\Omega \subseteq S_n(\mathcal{M})$ corestrict $Q_{I,\Omega} := (r_{\mathcal{N}_i}^n|_\Omega)_i$ and $r_{V,\Omega} := r_{\mathcal{N}_V}^n|_\Omega$. Restriction transitivity gives the unique surjection $q_{I,\Omega} : r_{V,\Omega}(\Omega) \rightarrow Q_{I,\Omega}(\Omega)$ with $q_{I,\Omega} r_{V,\Omega} = Q_{I,\Omega}$; write $\text{MulFib}_{I,\Omega}(Q_{I,\Omega}\rho) := \{r_{V,\Omega}\sigma : Q_{I,\Omega}\sigma = Q_{I,\Omega}\rho\}$. For a read-out $\pi : \Omega \rightarrow Z$ with $\pi = \bar{\pi}_V r_{V,\Omega}$ put $R_\pi = \{(\rho, \sigma) : \pi(\rho) = \pi(\sigma)\}$. Then

$$\begin{aligned} \text{[MCF]} \quad [\forall y \mid \text{MulFib}_{I,\Omega}(y) = 1] &\iff R_{\mathcal{N}_V}^\Omega = R_{Q_{I,\Omega}} = \bigcap_i R_{\mathcal{N}_i}^\Omega, \\ [\exists \bar{\pi}_Q : \pi = \bar{\pi}_Q Q_{I,\Omega}] &\iff [\bar{\pi}_V \text{ is constant on every } \text{MulFib}_{I,\Omega}] \iff R_{Q_{I,\Omega}} \subseteq R_\pi. \end{aligned} \quad (95)$$

At $\Omega = S_n(\mathcal{M})$, (MCR) turns the first row into $\text{MulCor}_I = 0$ iff all fibers are singletons; with $\pi = r_{V,\Omega}$ the second row regenerates the full-state factorization clause below. The restricted converse to the full-state criterion is false: in tracial M_2 , $(\mathcal{N}_1, \mathcal{N}_2) = (\mathcal{D}_x, \mathcal{D}_z)$ has $\text{MulCor}_I \cong \mathbb{R}\sigma_y \neq 0$, while for $0 < \epsilon < 1$ the family $\Omega_x = \{(1 \pm \epsilon\sigma_x)/2\}$ has singleton fibers and $\Omega_y = \{(1 \pm \epsilon\sigma_y)/2\}$ has a two-point fiber. Thus $\text{MulFib}_{I,\Omega}$ is the exact Ω -visible multiplicative-correlation coordinate, not the full quotient. On each faithful finite-dimensional generated-algebra fibre the quotient gives a global chart. For finite I , define $Q_I^\vee : S_n(\mathcal{N}_V) \rightarrow \prod_i S_n(\mathcal{N}_i)$ by restriction. Let τ be a faithful finite trace on $\mathcal{N}_V$, $w > 0$, $\tau(w) = 1$, and let y have a faithful $Q_I^\vee$ -fibre; write $\rho_d = \tau(d \cdot)$ and let d_y^0 be the unique minimizer of $D(d||w)$ in that density fibre. Since the fibre tangent is $V_{I,\text{sa}}^\perp$ and its trace dual is $(\mathcal{N}_{V,\text{sa}})/V_{I,\text{sa}}$, (EFD) gives

$$\text{[MGC]} \quad \boxed{\text{(MGC)} \quad \text{LogMul}_{I,y}^w : \{d > 0 : \tau(d) = 1, Q_I^\vee \rho_d = y\} \xrightarrow[\text{an}]{\cong} (\text{MulCor}_I)_{\text{sa}}, \quad d \mapsto [\log d - \log d_y^0]_{V_{I,\text{sa}}}} \quad (96)$$

with differential the dual BKM Riesz map. This does not absorb ambient extension freedom, which remains the (LAR/KCE) attachment. Normal Hahn–Banach/Jordan duality identifies $\text{MulCor}_{I,*}$ with the linearized opacity-kernel shell. Corestrict $Q_I = (r_{\mathcal{N}_i}^n)_i$ to its effective image. Then (NAT/DI/TOM/MCR/MCF) gives

$$\text{[LGS]} \quad \text{Const}_n(R_{Q_I}) = V_I, \quad \mathcal{N}_V = W^*(V_I), \quad \text{Tab } \tau_{Q_I, r_{\mathcal{N}_V}^n} \cong S_n(\mathcal{N}_V). \quad (97)$$

At full normal-state scope, (MCF/MCR) also gives $[r_{\mathcal{N}_V}^n = \bar{r} Q_I \text{ for some } \bar{r}] \iff \text{MulCor}_I = 0$. If $(\mathcal{N}_i)_i$ is directed, $V_I = \mathcal{N}_V$, so $R_{\mathcal{N}_V}^n = \bigcap_i R_{\mathcal{N}_i}^n$ and $\mathcal{N}_V = \mathcal{M} \iff \bigcap_i R_{\mathcal{N}_i}^n = \Delta$. Thus kernel data reconstruct the generated subalgebra; (MCF) isolates its restricted-state visible part. The M_2 (LAR) example is strict; (BDS/KCE) is only the finite-energy/kernel-completion face. Entropy and temporal/residual coordinates remain typed.

B. Proper localization forces access noninjectivity

a. Positive affine fibers, normal chord tangents, and finite unique-extension locus. Let $A : \mathcal{M}_{*,h} \rightarrow Y$ be bounded real linear on a von Neumann algebra, $K_y = \{\sigma \in \mathcal{M}_{*,+} : A\sigma = y\}$, $\rho \in K_y$, and $p = s(\rho)$. Then $(\text{PES}_n) \mathcal{T}_\rho^\pm K_y = \{\eta \in \mathcal{M}_{*,h} : A\eta = 0, -c\rho \leq \eta \leq c\rho \ (c < \infty)\}$, $\rho \in \text{Ext } K_y \iff \mathcal{T}_\rho^\pm K_y = \{0\}$. Indeed $\rho \pm \varepsilon\eta \in K_y$ is equivalent to the displayed kernel/order condition for some $\varepsilon > 0$; extremality is absence of a nonzero symmetric chord. Moreover $K_y \cap \{s(\sigma) \leq p\} = \arg \min_{K_y} \sigma(1 - p)$. Under finite-dimensional trace identification every $q = s(X)$ -supported hermitian direction is X -dominated; hence for $K_y = \{X \geq 0 : LX = y\}$, $F_X = K_y \cap (q\mathcal{D}q)_+$ has aff $F_X = X + (\ker L \cap (q\mathcal{D}q)_{\text{sa}})$ and $X \in \text{Ext } K_y \iff \ker L \cap (q\mathcal{D}q)_{\text{sa}} = 0 \iff X$ is uniquely linearly exposed (PES_n) ; faithful finite τ exposes F_X by $\tau((1 - q) \cdot)$; it is not a selector. Relative interior gives the affine-face clause above. On a finite commutative face $K_y = \{p \in \mathbb{R}_+^E : Lp = y\}$ put $E_p = \text{supp } p$, $L_p = L|_{\mathbb{R}^{E_p}}$ and let $\mathfrak{M}_{L,p}$ be its real column matroid. Then $(\text{PCM}) \mathcal{T}_p^\pm K_y = \ker L_p$, $p \in \text{ex } K_y \iff \mathfrak{M}_{L,p}$ has no circuit, and the circuit lines span $\ker L_p$. On the integral PCM face, the real matroid is the $C = \mathbb{R}$ face of (LCP/ACS), while $K_y = \text{PosLift}_L(y)$ makes (PLI) the existence/fibre layer and PCM its symmetric-chord geometry; LCP does not determine that positive fibre. For a Banach space X and closed $Y \subseteq X$, Hahn–Banach gives (BDT) $X^*/Y^\perp \cong Y^*$, $J : X/Y \hookrightarrow (Y^\perp)^*$, J onto $\iff X/Y$ reflexive. Support-stratified local-global affine-fiber kernel profile. For the positive affine fiber K_y and $\rho \in K_y$, put $p = s(\rho)$, $\mathfrak{F}_p(y) = K_y \cap \{\sigma : s(\sigma) \leq p\}$, $K_{\text{loc}} = \overline{\mathcal{T}_p^\pm K_y}$, $K_{\text{face}} = \text{span}_{\mathbb{R}}(\mathfrak{F}_p(y) - \mathfrak{F}_p(y))$, $K_{\text{sec}} = \text{span}_{\mathbb{R}}(K_y - K_y)$ and $K_{\text{alg}} = \ker A$ (norm closures). Then the positive-affine residual is the single four-stage (KSP) flag

$$\begin{aligned} \text{[AFP]} \quad \mathbf{A}_{\rho,y}(A) &:= (K_{\text{loc}} \subseteq K_{\text{face}} \subseteq K_{\text{sec}} \subseteq K_{\text{alg}}), \\ K_{\text{alg}}^\perp &\subseteq K_{\text{sec}}^\perp \subseteq K_{\text{face}}^\perp \subseteq K_{\text{loc}}^\perp, \quad \mathcal{M}_{\text{sa}}/K_\bullet^\perp \cong K_\bullet^*, \\ \dim \mathcal{M} < \infty &\implies K_{\text{loc}} = K_{\text{face}} = \ker A \cap (pMp)_{*,h}, \\ \dim \mathcal{M} < \infty, \quad p = 1 &\implies K_{\text{loc}} = K_{\text{face}} = K_{\text{sec}} = K_{\text{alg}} = \ker A. \end{aligned} \quad (98)$$

Its three grades $\mathfrak{D}_{\text{loc}} = K_{\text{face}}/K_{\text{loc}}$, $\mathfrak{D}_{\text{supp}} = K_{\text{sec}}/K_{\text{face}}$, $\mathfrak{D}_{\text{pos}} = K_{\text{alg}}/K_{\text{sec}}$ are local saturation, support escape and positive realizability. For $K_0 = K_{\text{loc}}$, $K_1 = K_{\text{face}}$, $K_2 = K_{\text{sec}}$, $K_3 = K_{\text{alg}}$, Hahn–Banach gives the whole dual spine at once:

$$\begin{aligned} 0 \rightarrow (K_{i+1}/K_i)^* &\rightarrow K_{i+1}^* \rightarrow K_i^* \rightarrow 0 \quad (i = 0, 1, 2), \\ K_2^* &\cong Z_{\text{aff}}^1(R_{q_y}), \quad 0 \rightarrow Z_{\text{aff}}^1(R_{q_y}) \rightarrow Z^1(R_{q_y}; \mathbb{R}) \rightarrow \text{NAFib}(q_y) \rightarrow 0. \end{aligned} \quad (99)$$

Here $q_y : K_y \rightarrow \{*\}$ and (KAR) gives $V_{q_y} = K_2$; hence $\text{NAFib}(q_y) = 0$ exactly for a singleton fiber. (PES_n) gives $K_0 \subseteq K_1$, the face/fiber inclusions give the rest, and (BDT) gives the dual flag. In finite dimension faithfulness on pMp makes every element of $\ker A \cap (pMp)_{*,h}$ locally order-bounded, proving the collapse clauses.

Independence is sharp: (OEL) has $K_0 = 0 \subseteq K_1 = K_2 \subseteq K_3$, while M_2 , $A(\sigma) = \sigma(1)$, pure $\rho = e_{00}$ has $K_0 = K_1 = 0 \subseteq K_2 = K_3$; on the faithful face $K_1 = K_2$.

We use the following standard-form facts throughout: (HSTD) denotes Haagerup’s standard form, positive-vector representation, canonical implementation and continuity results²² (Def. 2.1, Lem. 2.10, Thm. 3.2 and Cor. 3.6); thus $JMJ = \mathcal{M}'$, positive normal functionals have

unique natural-cone vector representatives, and every σ -weakly continuous locally compact group action has a strongly continuous canonical unitary implementation preserving J and the cone.

For the faithful order-tame specialization, let ω be faithful normal and R a bounded real-linear idempotent on $\mathcal{M}_{*,h}$ with $R^*1 = 0$ and $\pm\zeta \leq c\omega \Rightarrow \pm R\zeta \leq \kappa_R c\omega$. Put $\mathfrak{F}_R(\omega) = (\omega + \text{Ran } R) \cap \mathcal{S}_{\text{fin}}(\mathcal{M})$ and $q_R : \mathfrak{F}_R(\omega) \rightarrow \{*\}$. Then $\mathcal{T}_\omega^\pm \mathfrak{F}_R = \{\eta \in \text{Ran } R : \exists c < \infty, -c\omega \leq \eta \leq c\omega\}$. By (HSTD), write $\phi = \omega_{\xi_\phi} \geq 0$; faithfulness makes ξ_ω cyclic for $\mathcal{M}'$, so choose $a'_n \in \mathcal{M}'$ with $a'_n \xi_\omega \rightarrow \xi_\phi$. Then $\omega_{a'_n \xi_\omega} \leq \|a'_n\|^2 \omega$ and converges in norm. Thus Jordan decomposition gives order-bounded $\zeta_n \rightarrow \eta \in \text{Ran } R$; since $R\eta = \eta$, boundedness gives $R\zeta_n \rightarrow \eta$, and order-tameness makes every $R\zeta_n$ order-bounded. Hence (AFP/KAR/BDT) is exactly

$$\begin{aligned} \mathbf{A}_{\omega,*}(I - R) &= (\text{Ran } R \subseteq \text{Ran } R \subseteq \text{Ran } R \subseteq \text{Ran } R), \\ V_{q_R} &= \text{Ran } R, \quad (\text{Ran } R)^\perp = \ker R^*, \quad \mathcal{M}_{\text{sa}}/\ker R^* \cong (\text{Ran } R)^*, \quad \text{NAFib}(q_R) = 0 \iff \text{Ran } R = 0. \end{aligned} \quad (100)$$

(NPS) If positive mass-preserving idempotents P, Q obey $PQ = QP = P$ and fix faithful ψ , then $R = Q - P$ has $R^2 = R$, $R^*1 = 0$ and $\pm\zeta \leq c\psi \Rightarrow \pm R\zeta \leq 2c\psi$; hence (AFP) applies. The L^∞ singleton-fiber witness above keeps the order-tame boundary sharp.

Let $(I, \leq)$ be directed and let (P_α) be positive mass-preserving idempotents on $\mathcal{M}_{*,h}$, fixing faithful ω , with $P_\alpha P_\beta = P_\alpha = P_\beta P_\alpha$ for $\alpha \leq \beta$ and strong predual limit P_∞ . Put $Q_{\beta:\alpha} = P_\beta - P_\alpha$. For $\alpha_0 \in I$, let Coh_{α_0} be the norm-convergent nets $(\psi_\alpha \in \mathcal{S}_{\text{fin}})_{\alpha \geq \alpha_0}$ with $\psi_{\alpha_0} = \omega$, $P_\alpha \psi_\beta = \psi_\alpha$ for $\alpha_0 \leq \alpha \leq \beta$, and faithful limit ψ_∞ .

$$\begin{aligned} Q_{\beta:\alpha}^2 &= Q_{\beta:\alpha}, \quad \text{Ran } Q_{\beta:\alpha} = \text{Ran } P_\beta \cap \ker P_\alpha, \\ Q_{\gamma:\alpha} &= Q_{\gamma:\beta} + Q_{\beta:\alpha}, \quad Q_{\gamma:\beta} Q_{\beta:\alpha} = Q_{\beta:\alpha} Q_{\gamma:\beta} = 0, \\ \text{[PST]} \quad Q_{\beta:\alpha} \eta &\xrightarrow[\beta \uparrow]{\|\cdot\|_*} (P_\infty - P_\alpha)\eta, \quad \{\phi \in \mathcal{S}_{\text{fin}} : P_\infty \phi = \phi, P_{\alpha_0} \phi = \omega\} \cong \text{Coh}_{\alpha_0}. \end{aligned} \quad (101)$$

Nesting and the strong limit give the interval identities; (NPS) supplies the (AFP) shell and quotient dual for each $Q_{\beta:\alpha}$. The map $\phi \mapsto (P_\alpha \phi)$ identifies P_∞ -fixed states with norm-convergent coherent nets (inverse lim), and (PN) is the countable classical UI case. For $P_\infty = I$, a cofinal chain gives $Q_i Q_j = \delta_{ij} Q_j$, the ordered norm series and shell tower. Uniform Δ_3° with constants $\subset \{1\} \cup \{2, 3\}$ and last radius $(1 - p_1)/2$ makes it non-Cartesian; dyadic averages of normalized $\text{dist}(\cdot, C)$ for a Smith–Volterra–Cantor set give faithful finite stages with nonfaithful limit.

For the finite split face of (PST) put $P_0 \leq \dots \leq P_m = I$, $Q_j = P_j - P_{j-1}$, $V_j = \text{Ran } Q_j$, $\mathfrak{F}_j^P = \{\rho \in \mathcal{S}_{\text{fin}} : P_j \rho = \rho, P_0 \rho = \omega\}$ and $p_j^P = P_{j-1}|_{\mathfrak{F}_j^P}$. By (KSP), $P_j - P_0 = \sum_{k \leq j} Q_k$ with pairwise annihilating shells; hence p_j^P has the inclusion section and, for $\sigma \in \mathfrak{F}_{j-1}^P$,

$$\begin{aligned} \mathfrak{F}_j^P &= \left(\omega + \bigoplus_{1 \leq k \leq j} V_k \right) \cap \mathcal{S}_{\text{fin}}, \quad (p_j^P)^{-1}\{\sigma\} = (\sigma + V_j) \cap \mathcal{S}_{\text{fin}}, \\ \overline{\mathcal{T}_\sigma^\pm (p_j^P)^{-1}\{\sigma\}}^{\|\cdot\|_*} &= V_j, \quad V_j^\perp = \ker Q_j^*, \quad \mathcal{M}_{\text{sa}}/\ker Q_j^* \cong V_j^*. \end{aligned} \quad (102)$$

The last row is (AFP/NPS). If positive mass-preserving a intertwines the tower, then $a\omega = \omega$, $aV_j \subseteq V_j$ and $a(\omega + \sum_{j>0} \eta_j) = \omega + \sum_{j>0} (Q_j a Q_j) \eta_j$, so off-diagonal residual blocks vanish. Equivariant inclusion sections lie on the first (FRP) zero locus, so $[\tau]_{\mathcal{G}_F} = 0$ and $E_1 = E_\infty$; strict (FRP) arrows block the converses and (KSP) gives the same relation flag. The (PST) non-Cartesian witness prevents reconstruction of the positive tower from its grades. Entropic data remain independent: $(0, 1) \rightarrow \{*\}$ with Bernoulli references $1/2, 1/3$ has, at $r = 1/2$, fiber entropies 0 and $\frac{1}{2} \log(9/8)$. For the free word monoid I^* , put $\text{ev}_P(\alpha_1 \dots \alpha_k) = P_{\alpha_k} \dots P_{\alpha_1}$, $\text{Word}(P) = \text{ev}_P(I^*)$, and $\text{Hist}(P) = \overline{\text{Word}(P)}^s$ in point- $\|\cdot\|_*$ strong topology. Positivity and mass survive strong limits and multiplication is jointly strong-continuous on contractions, so this is the least strong-closed composition carrier; Jordan decomposition gives $S_\lambda \rightarrow S \iff S_\lambda \rho \rightarrow S \rho$ for every normal state ρ . For $S \in \text{Hist}(P)$, its adjoint S^* is normal unital positive; put $\Lambda_S(q) := 1 - s(S^*(1 - q))$ on $\text{Proj } \mathcal{M}$. Support faithfulness gives the unique left adjoint Σ_S and

$$\begin{aligned} s(S\rho) \leq q &\iff s(\rho) \leq \Lambda_S(q), \quad \Sigma_S + \Lambda_S, \quad s(S\rho) = \Sigma_S(s(\rho)), \\ \text{[HSD]} \quad c\Sigma_S &= \Sigma_S^Z c \text{ for some } \Sigma_S^Z \iff \Lambda_S(\text{Proj } Z(\mathcal{M})) \subseteq \text{Proj } Z(\mathcal{M}), \\ &\iff s(S^*z) \in \text{Proj } Z(\mathcal{M}) \quad (z \in \text{Proj } Z(\mathcal{M})), \\ P &= E_{N^*}, \quad E_N \omega = \omega \implies \text{Proj } N = \text{Fix } \Lambda_P, \quad N = W^*(\text{Fix } \Lambda_P). \end{aligned} \quad (103)$$

The first row is $S\rho(1 - q) = \rho(S^*(1 - q))$. For central z , $c\Sigma_S(p) \leq z \iff p \leq \Lambda_S(z)$, and $p = \Lambda_S(z)$ gives necessity. For $P = E_{N^*}$, $q \in \text{Fix } \Lambda_P \iff s(E_N(1 - q)) = 1 - q \iff q \in N$. Thus $\text{Hist}_Z(P) := \{S \in \text{Hist}(P) : c\Sigma_S = \Sigma_S^Z c \text{ for some } \Sigma_S^Z\}$ is a submonoid; a fixed orbit may descend further.

Applied to $(q_P, \Sigma, T_\rho, sT_\rho, csT_\rho)$, (DI) makes the per-state orbit flag and flat/support quotient below the chain and joint-coordinate faces of one history read-out system; (HSD) is the stability criterion for its central face. For fixed ρ , the orbit tower is $I^* \rightarrow \text{Word}(P) \rightarrow \text{Hist}(P) \xrightarrow{T_\rho} \mathcal{O}_\rho \xrightarrow{s} \mathcal{P}_\rho \xrightarrow{c} \mathcal{Z}_\rho$, with ambient central cover c (not base- A^{**} support without the (GC) interface). Put $R_\rho^0 := \ker T_\rho \subseteq R_\rho^1 := \ker(sT_\rho) \subseteq R_\rho^2 := \ker(csT_\rho)$. On $Y = \text{Hist}(P)$, let the one-object category $\text{Hist}_Z(P)^{\text{op}}$ act on the right by $S \cdot R := RS$. Then $\mathbf{R}_\rho^H := (\Delta \subseteq R_\rho^0 \subseteq R_\rho^1 \subseteq R_\rho^2) \in N_3 \text{Eq}_{\text{Hist}_Z(P)^{\text{op}}}(\text{Hist}(P))$. Indeed endpoint equality is preserved by left composition, support equality by $s(R\eta) = \Sigma_R(s(\eta))$, and central-support equality precisely by $c\Sigma_R = \Sigma_R^Z c$. Hence, for every abelian coefficient group G , (KSP/FRP) applies functorially to $\mathbf{R}_\rho^H$: its three relation grades are the history-to-endpoint, endpoint-to-support and support-to-central-support residuals, and its Maurer–Cartan gauge class is the exact continuation-gluing class under central-descending left composition at those fixed grades. On the full $\text{Hist}(P)^{\text{op}}$ -action only $\Delta \subseteq R_\rho^0 \subseteq R_\rho^1$ is automatic; no central continuation class is asserted without the (HSD) descent criterion. The sharp central-support witness below shows this restriction cannot be dropped. For the trivial action, effective-range corestrictions recover the (KSP) history/state/support flag; the (FRP) spectral row governs finite-history lifting, while all-state tomography makes the represented $\text{Hist}(P)^{\text{op}}$ -action faithful.

Put $q_P := q_{\text{sl}}^{\text{Hist}(P)} : \text{Hist}(P) \twoheadrightarrow \text{Flat}(P) := \text{Sl}(\text{Hist}(P))$. This is the same universal semilattice reflection used in (FWQ); no quotient topology is asserted, and fixed-locus quotients remain distinct. Its universal property and (KSP) give $Z^1(R_{q_P}; \mathbb{R}) = 0 \iff q_P$ injective $\iff \text{Hist}(P)$ semilattice $\iff [P_\alpha, P_\beta] = 0 \ (\forall \alpha, \beta)$. The other global monoid read-out is $\Sigma : \text{Hist}(P) \rightarrow \text{End}_V(\text{Proj } \mathcal{M})$. (DI) therefore supplies, without a new primitive, the dependent-image quotient $Q_{q_P, \Sigma} : \text{Hist}(P) \twoheadrightarrow \text{Dep}_{q_P, \Sigma}$, $\ker Q_{q_P, \Sigma} = R_{q_P} \cap R_\Sigma$. By (KSP), $R_{q_P}/(R_{q_P} \cap R_\Sigma)$ is exactly the support-propagation attachment grade over the flat shadow; it vanishes exactly when Σ descends through q_P . This least flat/support

quotient is irreducibly two-coordinate and still lossy. On uniform $\mathbb{C}^3$, partition expectations $A = 12|3$, $B = 1|23$ have $q_P(AB) = q_P(BA)$ but $\Sigma_{AB}(e_1) = e_1 + e_2 \neq 1 = \Sigma_{BA}(e_1)$. Conversely, for tracial M_2 with distinct MASA expectations P_z, P_n , $0 < |z \cdot n| < 1$, the scalar expectation P_0 is the strong alternating-word limit and $\Sigma_{P_z P_n} = \Sigma_{P_0}$, whereas $q_P(P_z P_n) \neq q_P(P_0)$: every non- P_0 Bloch block is a nonzero multiple of $|u\rangle\langle v|$, $u, v \in \{z, n\}$, so $h(P_0) = 0$, $h(S \neq P_0) = 1$ is a semilattice morphism. Moreover $Q_{q_P, \Sigma}(P_z P_n) = Q_{q_P, \Sigma}(P_n P_z)$ although the maps differ. For $G = \mathcal{L}(\mathcal{M}_*)_{\text{add}}$, $u(S) = S$, (KSP) reads the joint fiber by $d_{Q_{q_P, \Sigma}} u(S, S') = S - S'$; its (FRP) $(1, 1)$ face is the nonzero continuation curvature. Thus flat shadow, support propagation and residual history are one dependent profile. The flat branch is (RIF/ISM/AFP/BRB). On the finite-dimensional orthogonal two-generator face put $H_{ij} = \ker(P - iI) \cap \ker(Q - jI)$. Off the four corners, Halmos' form²³ (Thm. 2) on $K \oplus K$ is

$$P = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} A & \sqrt{A(I-A)} \\ \sqrt{A(I-A)} & I-A \end{pmatrix}, \quad 0 < A < I.$$

Hence (H2A) $\text{Ang}(P, Q) = ((\dim H_{ij})_{i,j \in \{0,1\}}, [A]_u)$ is a complete simultaneous-unitary invariant; alternating returns recover powers of A , whereas $[P, Q]^* [P, Q]$ sees only two copies of $A(I-A)$. Thus $\theta = \pi/6, \pi/3$ in tracial M_2 have equal nonzero commutator but $A = 3/4, 1/4$. More generally, for faithful normal tracial (M, τ) , (N, σ) and projection families $\mathbf{P} = (P_i)_{i \in I}$, $\mathbf{Q} = (Q_i)_{i \in I}$ put $\text{TW}_\tau(\mathbf{P}) = (\tau(w(\mathbf{P})))_{w \in \mathcal{W}_I}$. Then (TWR) $\text{TW}_\tau(\mathbf{P}) = \text{TW}_\sigma(\mathbf{Q}) \iff \exists! \theta : W^*(\mathbf{P}) \xrightarrow{\sim} W^*(\mathbf{Q})$ normal trace-preserving with $\theta(P_i) = Q_i$. Indeed word traces determine polynomial L^2 -norms and evaluation kernels; $p(\mathbf{P}) \mapsto p(\mathbf{Q})$ induces an L^2 unitary intertwining left multiplication, so strong closure gives the unique normal trace-preserving θ . In M_d , with $\text{TW} = d \text{TW}_{\text{Tr}/d}$, trace multiplicities make θ spatial, so for arbitrary I , (TWC) $\mathbf{P} \simeq_u \mathbf{Q} \iff \text{TW}(\mathbf{P}) = \text{TW}(\mathbf{Q})$. With $r_0 = 0$, $r_\xi = |\xi| \langle \xi | / \|\xi\|^2$ and $\text{TWM}(\xi, \mathbf{P}) = (\|\xi\|, \text{TW}(r_\xi, \mathbf{P}))$, augmented (TWC) and phase correction give (TWM) $\exists U[U\xi = \eta, U P_i U^* = Q_i] \iff \text{TWM}(\xi, \mathbf{P}) = \text{TWM}(\eta, \mathbf{Q})$. The normal-quotient law used here and below is the following. If $N \triangleleft G$ acts on X , $\pi : G \rightarrow G/N$, and the transporter from $[x]_N$ to $[y]_N$ is nonempty, then for any transporter $\pi(g_0)$,

$$\begin{aligned} \text{(NQT)} \quad & \text{Trans}_{G/N}([x]_N, [y]_N) = \pi(g_0)\pi(G_x), \\ & \text{Stab}_{G/N}([x]_N) = \pi(G_x) \cong G_x/(G_x \cap N), \\ & (G/N) \cdot [x]_N \cong G/(NG_x). \end{aligned}$$

Indeed gN sends $[x]_N$ to $[y]_N$ iff $g_0^{-1}g \in NG_x$. If a group L acts on X commuting with G , write $q_N : X \rightarrow X/N$, $q_G : X \rightarrow X/G$ and $\bar{q} : X/N \rightarrow X/G$. Then (NQT_{ex}) $(X/N)/(G/N) \cong_L X/G$, $q_G = \bar{q}q_N$, $0 \rightarrow \mathbb{K}(q_N) \rightarrow \mathbb{K}(q_G) \xrightarrow{(q_N)^\#} \mathbb{K}(\bar{q}) \rightarrow 0$, $\text{Hom}_L(X/G, Y) \cong \text{Hom}_L(X/N, Y)^{G/N}$, and a right G -torsor descends to a right G/N -torsor. The quotient orbits coincide; surjectivity of $(q_N)^\#$ gives the exact kernel row, and G/N -invariant maps factor uniquely through $\bar{q}$. On a common finite tracial ambient (M, τ) with equal trace words let $\theta = \theta_{\mathbf{P}, \mathbf{Q}} : A_{\mathbf{P}} = W^*(\mathbf{P}) \rightarrow A_{\mathbf{Q}} = W^*(\mathbf{Q})$ be (TWR), $H_{\mathbf{P}} = \{\beta \in \text{Aut}_\tau(M) : \beta|_{A_{\mathbf{P}}} = \text{id}\}$ and $\pi : \text{Aut}_\tau(M) \rightarrow \text{Out}_\tau(M)$. Then (AOR) $\text{AOut}_M(\mathbf{P}, \mathbf{Q}) := \pi\{\alpha \in \text{Aut}_\tau(M) : \alpha|_{A_{\mathbf{P}}} = \theta\}$, $\mathbf{P} \simeq_u \mathbf{Q} \iff 1 \in \text{AOut}_M(\mathbf{P}, \mathbf{Q})$.

Applying (NQT) to postcomposition on the trace-preserving embeddings of $A_{\mathbf{P}}$ into M , an inner mismatch at the target is corrected inside the same outer class; hence this quotient transporter is exactly AOut_M , and whenever one extension α_0 exists, $\text{AOut}_M = \pi(\alpha_0)\pi(H_{\mathbf{P}})$. The displayed equivalence is innerness of an extension. Thus $A_{\mathbf{P}} = A_{\mathbf{Q}} = M$ gives $\{\theta\}$ and M_d gives $\{1\}$. Empty and outer-only fibers both occur: in $\mathbb{C}^4$ with weights $(1, 2, 3, 4)/10$, $P = 1_{\{1,4\}}$, $Q = 1_{\{2,3\}}$ have equal trace words but $\text{Aut}_\tau(M) = 1$; a nontrivial trace-preserving automorphism of diffuse abelian $L^\infty(\mathbb{T})$ gives a full-generation residual missing 1. The (TWM) face is obtained by adjoining r_ξ . Central descent is exactly (HSD) and is sharp: the tracial expectation on $M_2 \oplus M_2$ onto $\text{span}\{(P, 0), (Q, P), (0, Q)\}$, $Q = 1 - P$, has $s(\bar{E}(1, 0)) = (1, P)$ noncentral, while $(0, Q)$ and $(0, \frac{1}{2}1)$ have equal central support but image central supports $(0, 1)$, $(1, 1)$.

For finite-dimensional convex $K \subset E$, $V = \text{span}(K - K)$ gives $E^*/V^\perp \cong V^*$, $N_K(x) = \{\ell \in V^* : \ell(K - x) \geq 0\}$, and $x \in \text{relint } K \implies N_K(x) = 0$. For a closed convex cone $C \subset H$ and linear $D : H \rightarrow \mathbb{R}^m$, $\lambda \in (DC)^* \iff D^\top \lambda \in C^*$ directly by pairing. Writing $K_0 = \overline{DC}$, so $K_0^* = (DC)^*$, strong separation of K_0 from the compact negative simplex $\{-p : p \geq 0, \mathbf{1}^\top p = 1\}$ gives $K_0 \cap (-\mathbb{R}_+^m) = \{0\} \iff K_0^* \cap \mathbb{R}_+^m \neq \emptyset$: conicity forces the separator into K_0^* , strictness gives $\min_i \lambda_i > 0$, and the reverse implication follows by pairing.

For a state $\varphi = (\sigma_k)_k$ on $B = \bigoplus_k (M_{n_k} \otimes \mathbf{1}_{m_k}) \subset M_d$, $\sigma_k \geq 0$, $\sum_k \text{Tr } \sigma_k = 1$, write $A = \{k : \sigma_k \neq 0\}$, $r_k = \text{rank } \sigma_k$, $p = \bigoplus_A s(\sigma_k) \otimes \mathbf{1}_{m_k}$, $N = \sum_A r_k m_k$. Then $\rho^\circ = \bigoplus_A \sigma_k \otimes \mathbf{1}_{m_k}/m_k$, $\dim \text{aff Ext}_B(\varphi) = N^2 - \sum_A r_k^2$, and $|\text{Ext}_B(\varphi)| = 1 \iff pBp = pM_d p \iff pB'p = \mathbb{C}p \iff A = \{k\}, m_k = 1$. Thus $\text{UEP}(B \subset M_d) \cong \bigsqcup_{k:m_k=1} S(M_{n_k})$ and restriction on $pM_d p$ has real rank $\sum_A r_k^2$. Hence $M_{d_0} \otimes \mathbf{1}_{d_E}$, $d_E \geq 2$, has no UEP state and rank- r fiber dimension $r^2(d_E^2 - 1)$, whereas the diagonal of M_2 has only its two characters. Every proper unital closed $B \subsetneq A$ is globally noninjective by normalized Jordan parts of a nonzero hermitian functional annihilating B (predually for W^* -algebras).

Data processing lets restriction erase positive relative entropy; reference-preserving automorphisms conserve it and ω -invariant MLSI semigroups contract it. (BRO/TSB) separate orientation, dissipation and structural terminality; Stinespring dilation does not identify access loss, modular clock or dissipation.

b. Same-system response witness. If admissible actions contain an ω -invariant MLSI step P_τ and modular flow σ_s^ω , then for finite-action $\rho \neq \omega$ $\Delta_{P_\tau} D(\rho||\omega) \leq (e^{-2\lambda\tau} - 1)D(\rho||\omega) < 0$, $\Delta_{\sigma_s^\omega} D(\rho||\omega) = 0$. Thus the same system contains a strict descent and a neutral action; any positive entropic response under an ω -preserving action is excluded by data processing and therefore requires reference motion.

C. Controlled response and descent

a. Action read-out coordinates. For a declared null action a_0 , the response path is the typed composite

$$\begin{aligned} a & \mapsto x_a(t^+) = \Phi_a(x(t), H_t) \mapsto x_a^{\text{fut}} = C_{\text{future}}(x_a(t^+)) \\ & \mapsto z_a = \tau_{\text{future}}(x_a^{\text{fut}}) \mapsto \delta_{\text{future}}(a) = d(z_a, G_{i(\text{future})}). \end{aligned} \tag{104}$$

Put $\Delta_a \delta := \delta_{\text{future}}(a) - \delta_{\text{future}}(a_0)$, ND $\iff a \mapsto \delta_{\text{future}}(a)$ is nonconstant. Descent means $\Delta_a \delta < 0$ for some admissible a . This chain is only the source/target typing of the response coordinate: the scalar defect is the already declared global-compatibility read-out, while intrinsic descent through visible data is decided first by the following fiber theorem.

b. *Normal-world access spine.* Let $\mathbf{NW}$ have objects the σ -finite von Neumann algebras with faithful normal state and arrows $\theta : (M, \omega) \rightarrow (M', \omega')$ the normal $*$ -isomorphisms with $\omega' \circ \theta = \omega$. Put

$$\begin{aligned} \text{ExpLat} : \mathbf{NW} &\longrightarrow \mathbf{Pos}, \\ (M, \omega) &\longmapsto \text{Lat}_\omega(M) := \{N \subseteq M \mid \forall t \in \mathbb{R} : \sigma_t^\omega(N) = N\}, \\ \text{ExpLat}(\theta)(N) &= \theta N. \end{aligned}$$

Since $\theta \sigma_t^\omega = \sigma_t^{\omega'} \theta$, this is strict. Set $\mathbf{PRes} := \int \text{ExpLat}$, with core $\mathbf{PAcc} := \mathbf{PRes}^\times = \int \text{Disc ExpLat}$.

Theorem IV.1 (Residual calculus and future descent). *For affine $Q : S \rightarrow R$, let $\bar{Q} : S \rightarrow \text{ran } Q$ be its corestriction. For $F : S \rightarrow Z$, (RPL) gives the unique affine descended map $\hat{F} : \text{ran } Q \rightarrow Z$ on the zero locus. On the normed branch, (BCR) gives $\mathfrak{E}_{\bar{Q}}(F) = \|d_{\bar{Q}}F\|_{\text{pr}} = \sup_{\tau \in \text{ran } Q} \text{rad}_Z F(Q^{-1}\{\tau\})$, $\kappa_{\bar{Q}}(F) = \frac{1}{2} \|d_{\bar{Q}}F\|_\infty$, $\kappa_{\bar{Q}} \leq \mathfrak{E}_{\bar{Q}} \leq 2\kappa_{\bar{Q}}$. Both vanish exactly at descent; $\mathfrak{E}_{\bar{Q}} = \kappa_{\bar{Q}}$ when every fiber image has radius half its diameter. (BAF) separates bounded from affine cocycle directions on the real bounded-affine face. (QAF) For corestricted $q : X \twoheadrightarrow Y$, (IOC/BEH) gives $\text{Stab}(q) = \{a : (a \times a)R_q \subseteq R_q\}$ and unique monoid descent $\text{Desc}_q(a)q = qa$. For an M -action β on Y , $\text{Lift}_M(q; \beta) = \{\alpha : q\alpha_m = \beta_m q \ \forall m\}$ is the possibly-empty realization fibre. For an affine semigroup $T_t : S \rightarrow S$ ($t \geq 0$), put $\mathcal{R}_t(\tau) = \{QT_t\rho : Q\rho = \tau\}$, $\kappa_t := \kappa_{\bar{Q}}(QT_t)$ and $\mathfrak{E}_t := \mathfrak{E}_{\bar{Q}}(QT_t)$. By (QAF/BCR), $T_t \in \text{Stab}(\bar{Q}) \iff \kappa_t = \mathfrak{E}_t = 0$, and all-time descent gives the semigroup $\hat{T}_t = \text{Desc}_{\bar{Q}}(T_t)$. Semigroup structure is essential: for $Q = I_{\mathbb{R}}$, $T_t = (1+t)I_{\mathbb{R}}$ descends pointwise but is not a semigroup. On the quantum-state branch let Q, T be CPTP and, whenever the displayed entropies are finite, put $d_Q(\rho\|\sigma) := D(\rho\|\sigma) - D(Q\rho\|Q\sigma)$ and likewise for QT . Then $D(QT\rho\|QT\sigma) - D(Q\rho\|Q\sigma) = d_Q - d_{QT}$; for $Q\rho = Q\sigma$ and finite $D(\rho\|\sigma)$, $QT\rho = QT\sigma \iff d_{QT}(\rho\|\sigma) = d_Q(\rho\|\sigma)$ (Pinsker). On the normal-UCP restriction face, (QAF/NAT) identifies $T = \mathcal{T}_*$ in $\text{Stab}(Q)$ for $Q = \iota_*$ with $\mathcal{T}(Q) \subseteq Q$; the unique normal UCP Heisenberg descent is $\mathcal{T}|_Q$. On the compact commutative Feller branch this is Feller lumpability. For commuting idempotents E_i let $\mathcal{L}_E := \langle E_i \rangle_{\wedge, 1}$ be their finite unital meet semilattice ($\wedge \emptyset = I$) and let μ_E be the inverse of its zeta matrix $\zeta(P, Q) = \mathbf{1}_{P \preceq Q}$. Put $A_P := \sum_{Q \preceq P} \mu_E(Q, P)Q$, $G_P := \text{im } A_P$, $j_E((v_P)_P) := \sum_P v_P$. Finite triangular inversion gives $P = \sum_{Q \preceq P} A_Q$; equivalently $A_P = P - \sum_{Q \prec P} A_Q$, and induction yields $RA_P = \mathbf{1}_{P \preceq R} A_P$ and hence $A_R A_P = \delta_{RP} A_P$. Represented-incidence frame normal form.*

$$\begin{aligned} A_P A_R &= \delta_{PR} A_P, & \sum_P A_P &= \text{id}, & Q &= \sum_{P \preceq Q} A_P, & j_E \text{ is an isomorphism,} \\ \text{[RIF]} & & j_E^{-1} Q j_E &= \bigoplus_{P \in \mathcal{L}_E} \chi_P(Q) I_{G_P}, & & \\ & & j_E^{-1} f(E_1, \dots, E_n) j_E &= \bigoplus_{P \in \mathcal{L}_E} f(\chi_P(E_1), \dots, \chi_P(E_n)) I_{G_P} \quad (f \in \mathbb{R}\langle z_1, \dots, z_n \rangle). \end{aligned} \quad (105)$$

Conversely, finite meet-semilattice L with top, atoms (G_P) and $j : \bigoplus_P G_P \rightarrow V$ gives this representation, faithful iff $Q \neq R \implies \exists P : G_P \neq 0$, $\mathbf{1}_{P \preceq Q} \neq \mathbf{1}_{P \preceq R}$. On this locus, for $G = \bigoplus_P G_P$, $D_L = \prod_P GL(G_P)$ and $B_L = \text{Stab}(G|_Q)$, (TRC) gives $B_L = \mathfrak{T}_L^\times = (1 + \mathfrak{n}_L) \rtimes D_L$; put $U_L := 1 + \mathfrak{n}_L$, so every $b \in B_L$ factors uniquely as $b = ud$ with $(u, d) \in U_L \times D_L$. $\text{Rep}_{L, G}(V) \cong \text{Iso}(G, V)/D_L \rightarrow \text{Fil}_{L, G}(V) \cong \text{Iso}(G, V)/B_L$, $\text{Fib}_{\text{model}} \cong B_L/D_L \cong U_L$. For $L = \{0 < 1\}$, $G_0 = G_1 = \mathbb{R}$, $u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ fixes the flag but moves $\text{diag}(1, 0)$ for $t \neq 0$, whereas block-diagonal gauges fix it. For disjoint $S, T \subseteq [n]$, $\prod_{i \in S} E_i \prod_{j \in T} (I - E_j) = \sum_{P \in \Lambda_{S, T}} A_P$, $\Lambda_{S, T} = \{P : P \preceq E_i \ (i \in S), P \not\preceq E_j \ (j \in T)\}$. If V is Hilbert and the E_i are orthogonal projections, j_E is unitary, the A_P are orthogonal, and $\|f(E_1, \dots, E_n)v\|^2 = \sum_P |f(\chi_P(E_1), \dots, \chi_P(E_n))|^2 \|A_P v\|^2$. Thus orthogonal incidence is the frame-rigid (KSP) locus; Boolean atoms are the nonzero P_E and $A_{PR} = A_P A_R$ multiplies as blocks. Abstract incidence does not determine the representation (rank-one/rank-two two-flags), while typed nested expectations identify atoms with $E_j - E_{j-1}$. For a g -orthogonal nested filtration,

$$\begin{aligned} A(\text{ran } E_r) \subseteq \text{ran } E_r &\iff P_i A_P j = 0 \quad (i > r, j \leq r), \\ q_r(v) &= \|(1 - E_r)v\|_g^2 = \sum_{j > r} \|P_j v\|_g^2, \quad q_{-1}(v) = \|v\|_g^2, \quad q_{r-1} - q_r = \|P_r v\|_g^2. \end{aligned} \quad (106)$$

c. *Derivation/causal boundary.* For real-linear $v \mapsto \delta_v$ on $\mathcal{K} \subseteq \mathcal{A}_0 \subseteq M$, set $\Xi_{\mathcal{K}, V}(v)(a) = q_{\mathcal{K}}^{\text{alg}}(\delta_v a)$, $\mathfrak{R}_{\mathcal{K}, V}^\partial = V/\ker \Xi$. Then (DBC) $[v]_\partial = 0 \iff \delta_v(\mathcal{K}) \subseteq \mathcal{K}$, $\mathfrak{R}_{\mathcal{K}, V}^\partial \cong \text{im } \Xi$. This quotient is universal for Ξ ; a declared projection E realizes it by $(1 - E)\delta_v|_{\mathcal{K}}$. For $G \curvearrowright M$, let $F_{\mathcal{K}}, H_{\mathcal{K}}$ be the action/orbit stabilizers. The (GOR/UAR) causal quotient is

$$\begin{aligned} \text{Star}_{\mathcal{K}} \mathbf{Caus}_G(\mathcal{K}) &\cong G/F_{\mathcal{K}} \longrightarrow \mathcal{O}_{\mathcal{K}} \cong G/H_{\mathcal{K}}, & \text{Aut}_{\mathbf{Caus}_G}(\mathcal{K}) &\cong H_{\mathcal{K}}/F_{\mathcal{K}}, \\ \text{[FCB]} & & 0 \longrightarrow I_{F_{\mathcal{K}}}^r \mathbb{Z}^{(G)} \longrightarrow I_{H_{\mathcal{K}}}^r \mathbb{Z}^{(G)} \longrightarrow I_{H_{\mathcal{K}}/F_{\mathcal{K}}}^r \mathbb{Z}^{(G/F_{\mathcal{K}})} \longrightarrow 0. \end{aligned} \quad (107)$$

Here $\mathbf{Caus}_G(\mathcal{K})$ is the restricted-action image groupoid; right multiplication supplies its action. Write $\text{Karma}_{\mathcal{K}, G}^{\text{caus}, \text{fin}}(g) = gF_{\mathcal{K}}$; only $F_{\mathcal{K}} \triangleleft H_{\mathcal{K}}$ is used. The exact augmentation row is (NQT_{ex}) for the commuting left- G /right- $H_{\mathcal{K}}$ actions on G with normal subgroup $F_{\mathcal{K}}$. By (CDR), orbit descent iff $F_{\mathcal{K}} = H_{\mathcal{K}}$; the S_3 witness shows this need not imply normality in G . (KTL) $\mathfrak{f}_{\mathcal{K}} = \{X : \delta_X|_{\mathcal{K}} = 0\} \subseteq \mathfrak{h}_{\mathcal{K}} = \{X : \delta_X \mathcal{K} \subseteq \mathcal{K}\}$, $0 \rightarrow \mathfrak{h}_{\mathcal{K}}/\mathfrak{f}_{\mathcal{K}} \rightarrow \mathfrak{g}/\mathfrak{f}_{\mathcal{K}} \rightarrow \mathfrak{g}/\mathfrak{h}_{\mathcal{K}} \cong \mathfrak{R}_{\mathcal{K}, \mathfrak{g}}^\partial \rightarrow 0$, For $a \perp g \iff \alpha_g(a) = a$, with group subsets ordered by reverse inclusion, (SDR) gives (CIG) $S \subseteq \text{Inv}_G(\mathcal{L}) \iff \mathcal{L} \subseteq \text{Fix}_G(S)$, $\text{Sat}_G(\mathcal{L}) := \text{Fix}_G \text{Inv}_G(\mathcal{L})$. Hence $\text{Inv}_G(\mathcal{L}) = \text{Inv}_G(\mathcal{K}) \iff \text{Sat}_G(\mathcal{L}) = \text{Sat}_G(\mathcal{K})$. It is sharp: for M_2 , $G = \{e, \text{Ad } \sigma_x\}$ and $\mathcal{D}_z = W^*(\sigma_z)$, both $\mathcal{D}_z$ and M_2 have fixer $\{e\}$, so $\text{Sat}_G(\mathcal{D}_z) = M_2$. Put $G_M^{\text{rec}} := \langle \text{rcem}[h] : h \in H(M) \rangle \leq \text{Aut}(\mathcal{A}(M))$ and, for compact $K \subset M$, $G_{K^\perp}^{\text{rec}} := \langle \text{rcem}[h] : h \in H(M; K^\perp) \rangle \leq G_M^{\text{rec}}$. Then (DLR) $\mathcal{A}^\bullet(M; K) = \text{Fix}_{G_M^{\text{rec}}} (G_{K^\perp}^{\text{rec}})$, $\text{Sat}_G^{\text{rec}}(\mathcal{A}^\bullet(M; K)) = \mathcal{A}^\bullet(M; K)$.

d. *Relative Cauchy evolution and operational geometry.* We use Fewster-Verch's results²⁴ (Prop. 3.5; §5.1, Eq. (5.3), Thm. 5.2(a), Def. 6.2; Thm. 6.13) in three distinct roles: (FVDL_{dyn}) is the RCE-fixed/dynamically-local net and its isotony on the concrete $\mathbf{Alg}$ face, (FVDL_{nat}) is the natural-state no-go at the printed hypotheses, and (FVDL_{rce}) is the local relative-Cauchy-evolution invariance input of Prop. 3.5. For a dynamically local theory and nonempty O , (FVDL_{dyn}) gives, on the concrete algebraic face, $\mathcal{A}^{\text{kin}}(M; O) = \mathcal{A}^{\text{dyn}}(M; O) = \bigvee_{K \in \mathcal{K}(M; O)} \mathcal{A}^\bullet(M; K)$, with subobject isomorphism represented by equality; Thm. 5.2(a) supplies the isotony below. On (DLR), set $D_M(K) := \mathcal{A}^\bullet(M; K) = \text{Fix}_{G_M^{\text{rec}}} (G_{K^\perp}^{\text{rec}})$, $K \sim_{\mathcal{A}} L \iff D_M(K) = D_M(L)$, $\text{OpReg}_{\mathcal{A}}(M) := \text{im } D_M$, $\text{GeoFib}_{\mathcal{A}}(X) := D_M^{-1}(X)$, and $[K] \preceq_{\mathcal{A}} [L] \iff D_M(K) \subseteq D_M(L)$. Then (ROG) $\mathcal{K}(M)/\sim_{\mathcal{A}} \cong \text{OpReg}_{\mathcal{A}}(M)$, $K \subseteq L \implies D_M(K) \subseteq D_M(L)$. With $X \perp_{\text{op}} Y \iff [X, Y] = 0$, Einstein causality gives $K \perp L \implies D_M(K) \perp_{\text{op}} D_M(L)$. Bare-region recovery is exactly reflection of inclusion

and disjointness; the constant abelian locally covariant theory refutes both and has $\text{GeoFib}_{\mathcal{A}}(\mathbb{C}^2) = \mathcal{K}(M)$. Thus RCE determines the operational quotient, while the typed GeoFib is the sharp (CIG/DLR) non-compression boundary. For $M = B(\mathcal{H}_X \otimes \mathcal{H}_E)$, $N = B(\mathcal{H}_X) \otimes 1$ and $\delta_H = i[H, \cdot]$,

$$\begin{aligned} \mathfrak{R}_N^{\text{caus}} &= \mathcal{M}_{\text{sa}} / (\mathcal{N}_{\text{sa}} + \mathcal{N}'_{\text{sa}}), & \Xi_N([H]) &= (1 - E_N)\delta_H E_N, \\ \text{[HCB/HIR/HID]} \quad H_X &= H - E_N H - E_{N'} H + \tau(H)\mathbf{1}, & \mathfrak{R}_N^{\text{caus}} &\cong (\ker \tau_X)_{\text{sa}} \otimes_{\mathbb{R}} (\ker \tau_E)_{\text{sa}}, \\ \dim_{\mathbb{R}} \mathfrak{R}_N^{\text{caus}} &= (d_X^2 - 1)(d_E^2 - 1). \end{aligned} \quad (108)$$

Here $H_X = (I - E_N)(I - E_{N'})H$ is the top (RIF) atom and unique Hilbert–Schmidt representative; Ξ_N is quotient-injective with full-diagonal image the zero-marginal interaction space (UIR/TF). For $\alpha_t = \text{Ad}(e^{itH})$,

$$\begin{aligned} [H] &= 0 \iff \delta_H(N) \subseteq N \iff \alpha_t(N) = N \quad (\forall t), \\ [H] \neq 0 &\iff \exists \rho \in \text{Host}_{\tau}(N), \eta > 0 : \alpha_{t+\rho} \notin \text{Host}_{\tau}(N) \quad (0 < |t| < \eta). \end{aligned} \quad (109)$$

For $h, k \in N_{\text{sa}}$ put $\mathfrak{A}_{N,H}(h, k) := \tau(\Xi_N([H])h \Xi_N([H])k)$, while $q_N^{\tau}(x) = \|(1 - E_N)x\|_{2,\tau}^2$. $q_N^{\tau}(\alpha_t h) = t^2 \mathfrak{A}_{N,H}(h, h) + O(t^3)$, $\mathfrak{A}_{N,H} \equiv 0 \iff [H] = 0$, $\mathfrak{A}_{N,H} = \mathfrak{A}_{N,-H}$. For the Hilbert (KSP) factorization of $\Xi_H := \Xi_N([H])$, put $H_{\mathfrak{A}_H} := N_{\text{sa}} / \ker \mathfrak{A}_{N,H}$, let $\iota_H : N_{\text{sa}} \rightarrow H_{\mathfrak{A}_H}$ be the quotient map, and define the residual boundary frame $j_H : H_{\mathfrak{A}_H} \xrightarrow{\cong} \overline{\text{Ran } \Xi_H}$ by $j_H \iota_H(h) = \Xi_H h$. Thus $\Xi_H = j_H \iota_H$.

$$\begin{aligned} \text{[BFE]} \quad \mathfrak{A}_{N,H} &= \mathfrak{A}_{N,K} \iff \exists! U_{K \leftarrow H} : \overline{\text{Ran } \Xi_H} \xrightarrow{\cong} \overline{\text{Ran } \Xi_K}, \quad U_{K \leftarrow H} \text{ isometric, } \Xi_K = U_{K \leftarrow H} \Xi_H, \\ [H] &\mapsto (\mathfrak{A}_{N,H}, j_H) \text{ is injective on } \mathfrak{R}_N^{\text{caus}}. \end{aligned} \quad (110)$$

Factor-boundary moduli. For qubits, invertible $K_H : V_X \rightarrow V_E$ fixes the relative-orientation line while H_{θ} gives continuum many classes with the same $\mathfrak{A}$. For M_d , $d = mn$, let $\mathfrak{F}_{m,n}$ be the type- I_m factor boundaries: $(\text{BMF}) \mathfrak{F}_{m,n}(\mathcal{M}) \cong \mathcal{U}(mn) / ((\mathcal{U}(m) \times \mathcal{U}(n)) / \mathbb{T})$, $\dim_{\mathbb{R}} \mathfrak{F}_{m,n} = (m^2 - 1)(n^2 - 1)$. If $\min(m, n) = 1$, then $\mathfrak{F}_{m,n} \simeq \mathbf{1}$, so $T\mathfrak{F}_{m,n} = 0$, every V_H vanishes, and every minimization over the boundary space has its unique point as minimizer; this is the exact degenerate face and its associated minimizer bundle has the trivial section. All uses of (BGT–BIF) below, including the HNS/PLP injectivity, nonselection, properness and separation conclusions that invoke them, are henceforth on the nontrivial factor-boundary regime $m, n \geq 2$. There the local stabilizer is proper, so the $\text{PU}(mn)$ -action on $\mathfrak{F}_{m,n}$ is nontrivial; simplicity and centerlessness of $\text{PU}(mn)$ make it effective. With $\mathfrak{R}_{m,n}^{\text{bd}} := \coprod_N \mathfrak{R}_N^{\text{caus}}$, the (HCB/HIR/HID) quotient

identifies the tangent bundle losslessly: $T\mathfrak{F}_{m,n} \xrightarrow{\cong} \mathfrak{R}_{m,n}^{\text{bd}}$, $\partial_{N_N}([iA]) = [A]$. For $H = H^*$ put $V_H(N) = \frac{d}{dt}|_0 \text{Ad}(e^{itH})N$. (BIV) $\partial(V_H)(N) = [H]_N$, $V_H(N) = 0 \iff [H]_N = 0 \iff \text{Ad}(e^{itH})N = N \quad (t \in \mathbb{R})$, and the Hilbert–Schmidt representative is exactly the boundary representative H_X^N constructed above. (BGT) $V_H = V_K$ on $\mathfrak{F}_{m,n}(\mathcal{M}) \iff H - K \in \mathbb{R}\mathbf{1}$.

e. Projective law and pointing. Put $(\mathcal{M}_{\text{sa}})^{\text{simp}} := \{H = H^* : H \text{ has simple spectrum}\}$, $\mathfrak{h}_d^{\text{simp}} := (\mathcal{M}_{\text{sa}})^{\text{simp}} / \mathbb{R}\mathbf{1}$, and $\mathbf{PH}_d^{\text{simp}} := \mathfrak{h}_d^{\text{simp}} / \text{PU}(d)$. For $[H] \in \mathfrak{h}_d^{\text{simp}}$ set $V_{[H]} := V_H$, $I_{[H]}(N) := \|H_X^N\|_{2,\tau}^2$ and $(\text{HNS/OMF}) \text{BndFib}_{m,n}([H]) := \text{pr}_2 \text{Opt}_{\text{pr}_1}^I([H])$ on $\mathfrak{h}_d^{\text{simp}} \times \mathfrak{F}_{m,n}$. By (BGT) these are representative-independent and $[H] \mapsto V_{[H]}$ is injective; equivalently, H, K have the same full boundary-vector-field read-out iff $H - K \in \mathbb{R}\mathbf{1}$. For simple $[H]$ (equivalently any representative H), torus-rank comparison gives a non-singleton fiber and, for every $N \in \text{BndFib}_{m,n}([H])$, (BIO) $\text{Stab}(H)N \subseteq \text{BndFib}_{m,n}([H])$, $\dim(\text{Stab}(H)N) \geq mn - (m + n - 1) = (m - 1)(n - 1) > 0$. By (BIO/IAE), boundary laws have no Borel actualizer; covariance/(SIC) give (BIF) $\int \text{BndFib}_{m,n} \rightarrow \mathbf{PH}_d^{\text{simp}}$, $\text{Sect}(\int \text{BndFib}_{m,n} \rightarrow \mathbf{PH}_d^{\text{simp}}) = \emptyset$. Also $\min I_{[H]} = 0$ iff some factor boundary is H -invariant; the zero locus is proper since for $m = n = 2$, $[\text{diag}(0, 1, 2, 4)]$ has positive minimum. Put $N_t := \alpha_t(N)$, $\theta_t := \alpha_t|_N : N \xrightarrow{\sim} N_t$, and let $\alpha_{t\#} := (\alpha_t^{-1})_*$ and $\theta_{t\#}(\varphi) := \varphi \circ \theta_t^{-1}$ denote the corresponding state push-forwards. Then (BCA) $r_{N_t} \alpha_{t\#} = \theta_{t\#} r_N$. Hence $(\alpha_t, =)$ is the horizontal $\mathbf{PAcc}$ leg of (PRG), while $N_t \subseteq \mathcal{K}$ is vertical, with the unique factorization $(\alpha_t, N_t \subseteq \mathcal{K}) = (\text{id}, N_t \subseteq \mathcal{K}) \circ (\alpha_t, =)$. The (BMF) tangent identification and its Lie quotient give (BIV/BGT), torus-rank comparison gives (BIO), (SIC) gives (BIF), and covariance gives (BCA/PRG). Let $J : \mathbf{PH}_d^{\text{simp}} \rightarrow \mathbf{NW}$ be $J([H]) = (\mathcal{M}, \tau)$ on objects and $J([U]) = \text{Ad } U$ on arrows. $\text{BndFib}_{m,n} \hookrightarrow (\text{Disc ExpLat}) \circ J$. Grothendieck integration gives the fully faithful functor (PLP) $\iota_{m,n} : \int \text{BndFib}_{m,n} \hookrightarrow J^* \mathbf{PAcc} \hookrightarrow J^* \mathbf{PRes}$, with minimizer-pointing image. The inclusion is proper ($\mathbb{C}\mathbf{1}, M_d$ are not $m \times n$ boundaries), and $\text{BndFib}([H]) = \text{BndFib}([-H])$ while (BGT) separates the laws, so constrained pointing does not determine projective dynamics. Local unitary actions satisfying $\alpha_a(M_X) \subseteq M_{v_a(X)}$ obey (CC) $\alpha_{a_n} \cdots \alpha_{a_1}(M_X) \subseteq M_{v_{a_n} \circ \cdots \circ v_{a_1}(X)}$. For radius $\leq R$, n actions generate local algebras only inside $B_{nR}(X)$; this is the exact finite-speed local-algebra regime, with no continuum-limit claim. For affine idempotent state projection E and affine a , let $\bar{E}$ be the corestriction $E : S \twoheadrightarrow \text{im } E$ and put $\mathcal{E}_E^{\text{fut}}(a) := \mathcal{E}_E(Ea)$. Then $\text{supp}_0 T_{E,Ea} \text{functional} \iff \text{DepRes}_E(Ea) = 0 \iff Ea = EaE \iff \mathcal{E}_E^{\text{fut}}(a) = 0 \iff \text{visible future factors through visible present}$. These zeros are composition-stable. If E , a extend linearly to signed states, put $L_E(a) := Ea(1 - E) = P_0 a P_1$; it is the (TRC) lower-cut read-out, with the same zero locus and block-product law.

Proof. (RPL/BCR) gives exact/quantitative descent and fibre minimax/norm bounds; quotient universality gives (QAF) action fibres. Möbius inversion gives the represented-incidence normal form and causal/Lie faces; tensoriality, affine-kernel exactness and projective covariance give subsystem clauses. DPI/Pinsker plus normal-state separation gives unique UCP descent; Petz/Pythagoras the entropic clauses; examples and word induction give strictness/history. $\square$

For the cartesian PEv projection $Q_u : X_u \rightarrow X'$, future descent is fiber constancy, equivalently $\mathfrak{E}_{Q_u} = \kappa_{Q_u} = 0$ on the convex-affine branch; the two-point surjection is therefore a comparison-cartesian nonzero-future-fiber witness.

For typed access-presentation transport put $N_0 = j_u N'_r \subseteq N_1 = N_{\rho(r')} \subseteq \mathcal{A}$ and $(P_0, P_1, P_2) = (E_0, E_1 - E_0, \text{id} - E_1)$. The filtered calculus gives target descent iff $T_{10} = T_{20} = 0$ and source descent iff $T_{20} = T_{21} = 0$; they coincide when $N_0 = N_1$, but neither implies the other under strict inclusion. On four equiprobable points with target 12|34 and source 1|2|34, the strictly positive kernels $K_+ = 20^{-1}[(8, 2, 5, 5), (2, 8, 5, 5), (6, 4, 5, 5), (4, 6, 5, 5)]$ and $K_- = 10^{-1}[(4, 4, 1, 1), (1, 1, 4, 4)^{\times 3}]$ are respectively target-only and source-only lumpable. Thus P_1 carries both directional future-leakage quantities while the ambient top is fixed.

f. Dependent action geometry. For an expected visible projection E_r , define

$$\mathcal{D}_{\text{desc}}(a) := (\text{DepRes}_{E_r}(E_r a), \mathcal{E}_{E_r}^{\text{fut}}(a)).$$

By (QAF/KSP/BCR), its zero locus is exactly intrinsic quotient-level response; on the signed-linear branch $\mathcal{L}_E(a)$ is the (TRC) read-out of its first leg. There define $\text{Pot} := \sup_a (-\Delta_a \delta)_+$, $\text{Expr} := (-\Delta_{a_t} \delta)_+$, $\text{Mis} := \text{Pot} - \text{Expr}$. In a differentiable finite-dimensional control regime with $0 \in U$, for a local response chart $u \mapsto x(u)$ differentiable at 0, write $x := x(0)$, $B := D_0[u \mapsto x(u)]$, and put

$$\begin{aligned} T_0 U &= \{v : \exists t_n \downarrow 0, u_n \in U, u_n/t_n \rightarrow v\}, \\ C_{\tan}(x) &= \{(B^* \nabla \delta(x), v) : v \in T_0 U\}. \end{aligned} \quad (111)$$

and $\text{Desc}^{(1)} := \mathbf{1}_{\{C_{\tan}(x) \cap (-\infty, 0) \neq \emptyset\}}$; write $\text{Desc} := \mathbf{1}_{\{\exists a : \Delta_a \delta < 0\}}$.

Theorem IV.2 (Action–response hierarchy). *For $F \in C^1(\Omega)$, $\Omega = A_{\text{att}} \sqcup A_{\text{plat}} \sqcup A_{\text{sing}}$, where $A_{\text{att}} = \{dF \neq 0\}$, $A_{\text{plat}} = \{F \text{ locally constant}\}$, and A_{sing} is the remainder. If moreover F is real analytic on connected analytic Ω , then F is constant or its critical set is a proper locally null analytic subset, with no open plateau and local Łojasiewicz gradient inequality²⁵ (§18, Prop. 1). On bounded open definable charts, Kurdyka²⁶ (Thm. 1) gives the local definable C^1 gradient inequality; the C^∞ singular stratum may nevertheless have positive measure.*

On $\mathcal{D}_{\text{desc}} = 0$, let $\mathcal{R}_{\text{avail}} = (\text{ND}, C_{\tan}, \text{Desc}^{(1)}, \text{Desc}, \text{Pot})$ and $\mathcal{R}_{\text{sel}} = (\text{Expr}, \text{Mis})$, with tangent entries only in their differentiable regime. Then $\text{Desc}^{(1)} > 0 \Rightarrow \text{Desc} > 0 \iff \text{Pot} > 0 \Rightarrow \text{ND}, 0 \leq \text{Expr} \leq \text{Pot}$, and $\text{Mis} = \text{Pot} - \text{Expr}$. Thus on the common source carrying the available response data and the selected action, the combined profile has the affine-graph normal form $(\text{Pot}, \text{Expr}) \mapsto (\text{Pot}, \text{Expr}, \text{Pot} - \text{Expr})$, $0 \leq \text{Expr} \leq \text{Pot}$, with projection to the first two coordinates as inverse on the realized image. Hence $R(\mathcal{R}_{\text{avail}}, \mathcal{R}_{\text{sel}}) = R(\mathcal{R}_{\text{avail}}, \text{Expr})$, $\text{Mis} \in \text{cl}_{\text{DI}}(\{\text{Pot}, \text{Expr}\})$. The mismatch remains a semantically useful selection-gap read-out, but it is not an additional determination generator once potential and expressed response are retained. If $x(u) = x + Bu + o(\|u\|)$, first-order descent is exactly $\langle B^* \nabla \delta(x), v \rangle < 0$ for some $v \in T_0 U$, hence gives actual local descent. Symmetric $T_0 U$ gives both signs; if $0 \in \text{int } U$, then $T_0 U = V$ and nonzero first-order response is equivalent to $B^* \nabla \delta(x) \neq 0$.

On an attained objective branch $\mathcal{F}_t^O = \mathcal{A}_t$, put $d_a = \mathcal{A}_{\text{vert}}(a) = \delta_{\text{future}}(a)$, $D_t = \text{Argmin}_a d_a$, and let $q : \mathcal{A}_t \rightarrow \{*\}$. Then (RFP) gives $V_q^d(a_t) = d_{a_t} - \min d = \text{Mis} + (d_{a_t} - d_{a_0})_+$, $a_t \in D_t \iff V_q^d(a_t) = 0$. Thus $\text{Mis} = V_q^d(a_t)$ exactly for $d_{a_t} \leq d_{a_0}$: the vertical residual, rather than a new selection scalar, completes the selected-response layer.

For the missing local converse, let U be a semianalytic germ at 0 and $\phi(u) = \delta(x(u)) - \delta(x)$ real analytic. For each nonconstant analytic $\gamma : [0, \varepsilon) \rightarrow U$ from zero with $\phi \circ \gamma \neq 0$, write $\phi(\gamma(t)) = c_\gamma t^{k_\gamma} + O(t^{k_\gamma+1})$, $c_\gamma \neq 0$, and set $C_{\text{arc}}^{\text{an}}(x) = \{(k_\gamma, c_\gamma)\}_\gamma$. Then (AJ) $0 \in \{u \in U : \phi(u) < 0\} \iff C_{\text{arc}}^{\text{an}}(x) \cap (\mathbb{N} \times (-\infty, 0)) \neq \emptyset$. The sign of c_γ is invariant under positive analytic reparameterization. Degree one recovers the first-order rate on analytic arcs with nonzero velocity, while higher degrees are necessary: $\delta(x) = (1 - x^2)^2$ at zero gives $(k, c) = (2, -2)$, and $\phi(u, v) = (v - u^2)^2 - u^6$ has positive leading coefficient on every nonzero line through zero but $\phi(t, t^2) = -t^6$. Also $\delta(a) = 1 + a^2$ gives ND without descent; on the objective branch, $d_{a_0} = 0 < d_{a_1} = 1$ with selected a_1 gives $\text{Pot} = \text{Mis} = 0$ but $V_q^d(a_1) = 1$, whereas selecting the baseline a_0 in the same objective gives $\text{Mis} = V_q^d(a_0) = 0$. Thus the affine determination of Mis from $(\text{Pot}, \text{Expr})$ does not collapse the vertical residual: these two selections are an exact same-Mis/different- V_q^d obstruction outside the favorable branch $d_{a_t} \leq d_{a_0}$. Analyticity is sharp: the smooth $\phi(u) = -e^{-1/u^2}$ for $u \neq 0$, $\phi(0) = 0$, descends locally while all finite Taylor jets vanish.

Proof. If analytic F is nonconstant, some analytic partial derivative is nonzero, so $\text{Crit } F$ lies in its proper null zero set and no plateau is open. For smooth sharpness let $K \subset (0, 1)$ be a fat Cantor set with complementary intervals I_n ; choose $b_n \in C^\infty$, $\{b_n > 0\} = I_n$, and $\varepsilon_n > 0$ with $\sum_n \varepsilon_n \|b_n\|_{C^k} < \infty$ for every k . Then $F(x) = \int_0^x \sum_n \varepsilon_n b_n$ has $A_{\text{sing}} = K$. First variation and $0 \leq \text{Expr} \leq \text{Pot}$ give the tangent and selection-mismatch formulas. For (AJ), Ruiz²⁷ (§7, (7.2)) gives the required analytic γ in $\{\phi < 0\}$; Taylor sign gives the stated equivalence. The displayed examples give all nonconverses. $\square$

For the same-system MLSI/modular witness above, $\text{Pot} \geq (1 - e^{-2\lambda\tau})D(\rho\|\omega) > 0$ off the reference. Thus forcing and gluing response remain type-distinct without that attachment. Closed-set distance is 1-Lipschitz, so an L -Lipschitz action with $\Delta_a \delta \leq -\eta$ obeys $\Delta_{\tilde{a}} \delta \leq -\eta/2$ throughout $|\tilde{a} - a| \leq \eta/(2L)$, with the positive-margin branch identical; this is a derived robustness bound, not a coordinate.

g. *Objective attachment.* For $q : \mathcal{F}_t^O \rightarrow \{*\}$ with finite objective infimum, (OMF/AMF) gives $D_t = \text{Opt}_q^{\mathcal{A}_{\text{vert}}}(*)$. Compact/l.s.c. data kill the (AOL) H_0 grade and strict convexity its H_1 grade; otherwise nonattainment is its LimRes class, while attained pullbacks use the RF-completed clause of Theorem IV.2.

V. SUFFICIENCY, RECOVERY, AND MULTI-ACCESS CLOSURE

A. Normal state-preserving access calculus

Write $\mathfrak{S}_\omega$ for the category whose objects are $(\mathcal{A}', \omega|_{\mathcal{A}'})$ and whose arrows are unital completely positive Heisenberg maps $\alpha : \mathcal{A}'' \rightarrow \mathcal{A}'$ satisfying $\omega' \alpha = \omega''$; α^* always denotes the Schrödinger predual. For a normal state channel $\Phi : (\mathcal{A}, \eta)_* \rightarrow (\mathcal{B}, \nu)_*$ put $\text{Karma}_\eta^{\text{ref}}(\Phi) := [\Phi]_{R_{\Psi \mapsto \Psi\eta}} \cong \Phi\eta$, $\text{Karma}_\eta^{\text{forc}}(\Phi) := P_\Phi^{\eta \rightarrow \nu} := D(\Phi\eta\|\nu)$, with Araki D for arbitrary normal states²⁸ (Def. 3.1, Thm. 3.6(1)); hence $P_\Phi^{\eta \rightarrow \nu} = 0 \iff \Phi\eta = \nu$. Binary AF restriction at a Jordan projection²⁸ (Thm. 3.8(2 γ)) plus classical Pinsker gives $D(\rho\|\sigma) \geq \frac{1}{2}\|\rho - \sigma\|_1^2$. On (M_2, τ) replacements to $\rho_\pm = \text{diag}(\frac{1}{2} \pm \delta, \frac{1}{2} \mp \delta)$ have equal $D(\rho_\pm\|\tau)$ and distinct $\text{Karma}_\tau^{\text{ref}}$, so forcing is nonclassifying. The causal U0 faces (FCB/DBC/KTL) remain type-distinct. On one pointed object write $P_\Phi := P_\Phi^{\omega \rightarrow \omega}$.

B. Relative-entropy defect and sufficiency maps

a. *Relative-entropy defect cocycle map.* For reference-preserving UCP $\alpha : (\mathcal{A}'', \omega'') \rightarrow (\mathcal{A}', \omega')$, Uhlmann monotonicity²⁹ (Prop. 18) gives the global (RFP) cell $\text{Cell}_\alpha(\rho\|\sigma) = [D(\alpha^* \rho\|\alpha^* \sigma), D(\rho\|\sigma)]$ and, on its finite upper face, $d_\alpha(\rho\|\sigma) = D(\rho\|\sigma) -$

$D(\alpha^* \rho \| \alpha^* \sigma) \geq 0$. Corestrict $q = \alpha^*$ to its image and put $A = D(\cdot \| \sigma)$, $b = D(\cdot \| \alpha^* \sigma)$. This is $\mathfrak{A}_\square$ pasting; for the right action $(\rho, \sigma) \cdot \alpha = (\alpha^* \rho, \alpha^* \sigma)$ one has $d_\alpha = \delta(-D)$, so (RFP/FRP) kills every composable access-path history. Relative-Tomita factorization proves $D(\rho_1 \otimes \rho_2 \| \sigma_1 \otimes \sigma_2) = D(\rho_1 \| \sigma_1) + D(\rho_2 \| \sigma_2)$ on product pairs (extended-real by support/spectral truncation); hence, when finite,

$$d_{\alpha \otimes \beta}(\rho_1 \otimes \rho_2 \| \sigma_1 \otimes \sigma_2) = d_\alpha(\rho_1 \| \sigma_1) + d_\beta(\rho_2 \| \sigma_2), \quad \sum_k \delta_k = D(\rho_0 \| \sigma_0) - D(\rho_n \| \sigma_n), \quad P_\Phi = D(\eta_n \| \omega_n) + \sum_k \delta_k^\Phi.$$

Hence closed pair-cycles have all $\delta_k = 0$; if $\rho_n = \sigma_n$, then $\max_k \delta_k \geq D(\rho_0 \| \sigma_0)/n \geq \|\rho_0 - \sigma_0\|_1^2/(2n)$, and on $\eta_n = \omega_n$, $\max_k \delta_k^\Phi \geq P_\Phi/n$. A declared reference-preserving MLSI(λ) step of duration τ contracts visible relative entropy by $e^{-2\lambda\tau}$; this introduces no second current.

C. Sufficiency and Petz recovery

a. Petz sufficiency interface. Put $\text{Suff}(\alpha) = \{\rho : D(\rho \| \omega') < \infty, D(\alpha^* \rho \| \omega'') = D(\rho \| \omega')\}$. For faithful normal references and normal unital 2-positive access let $\mathcal{R}_\alpha^{\omega'} = (\alpha_\omega^+)^*$ be the reference/KMS dual. By Jenčová–Petz³⁰ (Thms. 1–2, post-Thm. 2 remark, Appendix), write $(\text{JP}_{\text{suff}})$ for Thm. 2 + that remark + Appendix and (JP_{br}) for Thm. 1 + its minimal modular cocycle algebra; then $(\text{JP}_{\text{suff}})$ gives

$$\rho \in \text{Suff}(\alpha) \iff \exists \beta : \beta^* \alpha^* \rho = \rho, \beta^* \alpha^* \omega' = \omega' \iff \mathcal{R}_\alpha^{\omega'} \alpha^* \rho = \rho,$$

with β normal unital 2-positive and equivalently $\alpha([D(\rho \circ \alpha) : D(\omega' \circ \alpha)]_t) = [D\rho : D\omega']_t$; in finite dimension $\mathcal{R}_\alpha^{\omega'}(\sigma) = \omega'^{1/2} \alpha(\omega''^{-1/2} \sigma \omega''^{-1/2}) \omega'^{1/2}$. KMS adjointness gives reverse-order recovery composition. On the commutative deterministic finite-entropy face $w_* \eta = \tau$,

$$D(\eta \| \mu) = D(\tau \| w_* \mu) + D(\eta \| \eta^*), \quad \frac{d\eta^*}{d\mu} = \left(\frac{d\tau}{d(w_* \mu)} \right) \circ w,$$

so η^* is the unique minimum-entropy lift; deterministic sufficiency is fibre saturation, mutual Blackwell sufficiency is generated-algebra equality modulo null sets, and invertible natural extension kills the information defect while reference preservation remains $\Theta_\Xi = 0$.

For $\Delta_\alpha(\rho) = d_\alpha(\rho \| \omega')$ set Avidya(α, ρ) = $\Delta_\alpha(\rho)$ and Pramana(α, ρ) $\iff \Delta_\alpha(\rho) = 0$. The averaged rotated-Petz predual³¹ obeys

$$F(\rho, \bar{\mathcal{R}}_\alpha^{\omega'}(\alpha^* \rho)) \geq e^{-\Delta_\alpha(\rho)/2}, \quad \|\rho - \bar{\mathcal{R}}_\alpha^{\omega'}(\alpha^* \rho)\|_1 \leq 2\sqrt{1 - e^{-\Delta_\alpha(\rho)}} \leq 2\sqrt{\Delta_\alpha(\rho)}.$$

If $\alpha^* \rho = \alpha^* \sigma = \tau$, every common recovery R obeys $\max\{\|\rho - R\tau\|_1, \|\sigma - R\tau\|_1\} \geq \frac{1}{2}\|\rho - \sigma\|_1$. Thus exact, approximate and obstructed recovery are respectively the zero-defect, stability and terminal-fibre-diameter faces of this access coordinate.

D. Bi-closure absorption and normal entropic resolution

The Petz map VC0a treats one access at a time; a family requires the interaction of the U1 collective section with the common expected-sufficiency meet. Fix a nonempty finite or countable F , σ -finite $\mathcal{M}, \mathcal{N}_r$, faithful normal ω , and normal UCP $\alpha_r : \mathcal{N}_r \rightarrow \mathcal{M}$, $\omega_r := \omega \circ \alpha_r$, $\mathcal{N}_r := \alpha_r^*$. We support-reduce once: with $s_r = s(\omega_r)$, faithfulness of ω and CP Schwarz give $\alpha_r(1 - s_r) = 0$ and $\mathcal{N}_r \eta = s_r(\mathcal{N}_r \eta)s_r$; replacing $\mathcal{N}_r$ by $s_r \mathcal{N}_r s_r$ preserves every record/fiber and makes ω_r faithful. Write $\mathcal{S}_{\omega, c}^\infty = \{\rho : D(\rho \| \omega) < \infty\}$ and $K_F(\rho) = \{\eta : \mathcal{N}_r \eta = \mathcal{N}_r \rho \ \forall r \in F\}$. For the purely lattice clauses use $\mathcal{V}_\partial = \mathbb{R}1$, $\mathcal{A}_\partial^\Delta = \mathcal{M}$; no P_∂^Δ is introduced, so all variational clauses keep $F \neq \emptyset$.

a. Normal-cluster compactness map. If normal UCP $\lambda_i : \mathcal{B} \rightarrow \mathcal{C}$ satisfy $\varphi \lambda_i \rightarrow \sigma_*$ pointwise with φ faithful normal and σ_* normal, product compactness gives a point-ultraweak UCP cluster λ with $\varphi \lambda = \sigma_*$. For $0 \leq x_j \uparrow x$, normality gives $\varphi(\sup_j \lambda x_j) = \sup_j \varphi \lambda(x_j) = \varphi \lambda(x) = \sigma_*(x)$; faithfulness yields $\sup_j \lambda x_j = \lambda x$, so λ is normal. Bind once Jenčová's type-distinct interfaces ($\text{JEO}_{\text{mix}}, \text{JEO}_{\text{cmp}}, \text{JEO}_{\text{gib}}, \text{JEO}_{\text{ort}}, \text{JEO}_{\text{ch}}, \text{JEO}_p$) to Eq. (4), Prop. 4(i),(iii), Thm. 1, Prop. 7 plus its following commutative paragraph, Props. 8 and 1, respectively³².

Theorem V.1 (Sufficiency and tangent loss). *Let $\mathcal{M}$ be σ -finite with faithful normal ω , $\emptyset \neq C \subseteq \mathcal{S}_n(\mathcal{M})$ convex and $\sigma(\mathcal{M}_*, \mathcal{M})$ -closed, Y Hausdorff locally convex, and $Q : C \rightarrow Y$ continuous affine. Put $D_\omega := D(\cdot \| \omega)|_{\mathcal{S}_n(\mathcal{M})}$, $\mathcal{S}_{\omega, c} := \{D_\omega \leq c\}$, $\mathcal{J} := Q!(D_\omega|_C)$, $G_c := Q(C \cap \mathcal{S}_{\omega, c})$, $Y_{\text{fin}} := \{\mathcal{J} < \infty\}$, $C_{\text{fin}} := Q^{-1}Y_{\text{fin}}$. Since $F^\omega = D_\omega - 1$ on states, $(\text{JEO}_{\text{cmp}})$ gives compact sublevels and strict convexity; (TGF/RFP) gives, for $c < \infty$, $\{\mathcal{J} \leq c\} = G_c$, $s_{Q, C}^\omega(y) = \arg \min_{Q\eta=y} D(\eta \| \omega)$, $P := P_{Q, C}^\omega = s_{Q, C}^\omega Q|_{C_{\text{fin}}} : C_{\text{fin}} \rightarrow C_{\text{fin}}$, with $QP = Q$, $P^2 = P$ on C_{fin} and all displayed support/Legendre, visible-penalty, selector-tower, penalty-hard and zero-at-infinity rows of (TGF/RFP), and $s(Q\omega) = \omega$ when $\omega \in C$. For a normal state channel $J : \mathcal{M}_* \rightarrow \mathcal{B}_*$, faithful $v = J\omega$, and $\kappa_y = s_{Q, C}^\omega(y)$, (SIM) $\mathcal{J}(y) = P_J^{\kappa_y \rightarrow y} + \text{Avidya}(J^*, \kappa_y)$ by (RFP/Petz), and the loss vanishes iff $\kappa_y \in \text{Suff}(J^*)$. On uniform $\mathbb{C}^2$, $Q = \text{id}$, $\kappa = \delta_0$, identity/depolarizing J give $P_J = \log 2, 0$ at the same $\mathcal{J} = \log 2$, so J remains attachment data. Thus (TGF) identifies finite- β Gibbs selection, its infimal push-forward tower and the $\beta \rightarrow \infty$ hard KSP gauge; the two-point witness keeps the soft attachment irreducible.*

For bounded measurable L on $L^\infty(K, \mu)$, $d\kappa_\beta = Z_\beta^{-1} e^{-\beta L} d\mu$, $Z_\beta = \int e^{-\beta L} d\mu$, and $\frac{d}{d\beta} D(\kappa_\beta \| \mu) = \beta \text{Var}_{\kappa_\beta}(L) \geq 0$. If $\text{supp } \mu$ is compact, L continuous, $m_\mu = \min L$ on the support and $M_\mu = \{L = m_\mu\} \cap \text{supp } \mu$, then $\mu(M_\mu) > 0 \implies (\kappa_\beta, D(\kappa_\beta \| \mu)) \rightarrow (\mu(\cdot|_{M_\mu}), -\log \mu(M_\mu))$, $\mu(M_\mu) = 0 \implies D(\kappa_\beta \| \mu) \rightarrow \infty$. For Borel A , $\inf_{\kappa(A)=1} D(\kappa \| \mu) = -\log \mu(A)$, with unique finite minimizer $\mu(\cdot|_A)$ when $\mu(A) > 0$; if $\mu(A) = 0$ the value is $+\infty$ and no finite-entropy feasible state exists. If m_A is the essential minimum on A , positive mass occurs arbitrarily near m_A , and $m_{A^c} \geq m_A + \eta$, then for $0 < \varepsilon < \eta$, $-\beta^{-1} \log \int_A e^{-\beta L} d\mu \rightarrow m_A$, $\kappa_\beta(A^c) \leq \frac{\mu(A^c) e^{-\beta(\eta - \varepsilon)}}{\mu(A \cap \{L < m_A + \varepsilon\})}$.

These are (TGF) entropy read-outs and add no closure coordinate. For finite-dimensional products with faithful ω_i and affine Q_i , $(Q_1 \boxplus Q_2)(\eta) = (Q_1 \eta_1, Q_2 \eta_2)$ gives by the relative-entropy chain rule, rather than bare U0, $\mathcal{J}_{Q_1 \boxplus Q_2}^{\omega_1 \otimes \omega_2} = \mathcal{J}_{Q_1}^{\omega_1} + \mathcal{J}_{Q_2}^{\omega_2}$, $s_{Q_1 \boxplus Q_2}^{\omega_1 \otimes \omega_2} = s_{Q_1}^{\omega_1} \otimes s_{Q_2}^{\omega_2}$.

For the finite affine face, let $\mathcal{D}$ be unital finite-dimensional with faithful finite trace τ , $w > 0$, $\tau(w) = 1$, and $R : \mathcal{D}_{\text{sa}} \rightarrow W$ real linear; corestrict R to its finite-dimensional image (which does not change its fibres), and let $K_y = \{d \geq 0 : \tau(d) = 1, Rd = y\}$ contain a faithful

density. Put $K_y^\circ = \{d \in K_y : d > 0\}$, $V_y = \{v = v^* : \tau(v) = 0, Rv = 0\}$, let $R_\tau^* : W^* \rightarrow \mathcal{D}_{\text{sa}}$ be the trace-pairing adjoint, and write d_0 for the minimizer of $D(d\|w)$ on K_y . Then $d_0 > 0$, and the tangent law globalizes: with

$$(EFD) \quad F_y(d) := D(d\|d_0), \quad \Lambda_y(d)(v) := \tau(v(\log d - \log d_0)),$$

$$\Lambda_y : K_y^\circ \xrightarrow[\text{an}]{\cong} V_y^*, \quad dF_y = \Lambda_y, \quad D\Lambda_y(d)[u](v) = \tau(vD \log_d[u]) = g_d^\vee(u, v).$$

For $h, k \in \mathcal{D}_{\text{sa}}$ let d_h minimize $D(d\|w) + \tau(hd)$, put $w_h = Z_h^{-1}e^{\log w - h}$, $Z_h = \tau(e^{\log w - h})$, $d_* = d_h$ and $\dot{d}_k = \frac{d}{d\varepsilon}\big|_0 d_{h+\varepsilon k}$. Then $d_h \in K_y^\circ$ is the faithful selector for reference w_h , and

$$\Lambda_y(d_h) = -\tau(h \cdot)|_{V_y}, \quad \log d_h - \log w + h = R_\tau^* \lambda + c\mathbf{1}, \quad d_h = \frac{e^{\log w - h + R_\tau^* \lambda}}{\tau(e^{\log w - h + R_\tau^* \lambda})},$$

for some (λ, c) , and

$$D(d\|w) = D(d_0\|w) + D(d\|d_0) \quad (d \in K_y), \quad g_{d_*}^\vee(\dot{d}_k, v) = -\tau(kv) \quad (v \in V_y).$$

Thus feasibility plus the KKT row characterizes d_h , $\dot{d}_k = 0 \iff k \in \text{Ran } R_\tau^* + \mathbb{R}\mathbf{1}$, and $D^2\mathcal{A}_y(h)[k, k] = -g_{d_*}^\vee(\dot{d}_k, \dot{d}_k) \leq 0$ for $\mathcal{A}_y(h) = \min_{d \in K_y} \{D(d\|w) + \tau(hd)\}$. Trace pairing and (KAR) give $\mathcal{D}_{\text{sa}}/(\text{Ran } R_\tau^* + \mathbb{R}\mathbf{1}) \cong V_y^* \cong Z_{\text{aff}}^1(R_{q_y})$ for $q_y : K_y \rightarrow \{*\}$, the finite-dimensional affine face of Theorem S2; thus EFD globally integrates the affine residual shadow and the former BKM quotient is its differential. On this face (E2S) is $g_{d_*}^\vee$ on V_y and $+\infty$ off it; $R = 0$ is Gibbs–Araki and marginal restriction is the tensor endpoint of (LAR).

Proof. Petz plus the residual profile gives sufficiency; Gibbs–Laplace/entropy chain rule gives hard-event, gap and product clauses. For EFD, the same boundary test first shows $d_0 > 0$: toward any faithful feasible σ , each zero eigenvalue of a singular candidate acquires positive first-order mass, so the one-sided derivative of $\tau(d \log d)$ is $-\infty$ while the remaining terms are finite. Now minimize $F_y - \ell$ on compact K_y ; the same test makes its unique minimizer faithful, and its first-order equation is $\Lambda_y(d) = \ell$. Thus Λ_y is onto, while strict convexity gives injectivity. The integral formula for $D \log_d$ makes $\tau(uD \log_d[u]) > 0$ for $u \neq 0$, so $D\Lambda_y$ is invertible and the analytic inverse theorem gives the global analytic diffeomorphism. KKT subtraction yields the d_h /BKM rows, and $\tau((d - d_0)(\log d_0 - \log w)) = 0$ gives Pythagoras; envelope differentiation and trace duality give the remaining finite-face clauses. $\square$

b. Normal-record specialization. For nonempty finite or countable F write

$$N_F : \mathcal{M}_{*,h} \rightarrow \prod_{r \in F} (N_r)_{*,h}, \quad N_F \eta = (N_r \eta)_{r \in F}, \quad \mathcal{R}_F := N_F(\mathcal{S}_n(\mathcal{M})).$$

For $\emptyset \neq E \subseteq F$, let $\pi_{F \rightarrow E} : \prod_{r \in F} \mathcal{S}_n(N_r) \rightarrow \prod_{r \in E} \mathcal{S}_n(N_r)$ be coordinate restriction; the same symbol acts on the linear products, so $N_E = \pi_{F \rightarrow E} N_F$ and $\pi_{G \rightarrow E} = \pi_{F \rightarrow E} \pi_{G \rightarrow F}$. For $y = (y_r)_{r \in F} \in \prod_{r \in F} \mathcal{S}_n(N_r)$ put

$$\begin{aligned} \text{Comp}_F &= \mathbf{1}_{\mathcal{R}_F}, & K_F(y) &= N_F^{-1}\{y\}, & J_F^\omega &= (N_F)_! D_\omega, \\ \mathcal{S}_{\text{fin}}^\omega &:= \mathcal{S}_\omega^{<\infty}(\mathcal{M}), & Y_{F,\text{fin}}^\omega &:= \{J_F^\omega < \infty\}, & \mathcal{D}_F^\omega &:= \mathcal{S}_n(\mathcal{M}) \cap N_F^{-1} Y_{F,\text{fin}}^\omega \supseteq \mathcal{S}_{\text{fin}}^\omega. \end{aligned}$$

The entropic U0 chart gives $\{J_F^\omega \leq c\} = N_F(\mathcal{S}_{\omega,c})$ and

$$\begin{aligned} s_F : Y_{F,\text{fin}}^\omega &\rightarrow \mathcal{S}_{\text{fin}}^\omega, & P_F^\vee &:= s_F N_F|_{\mathcal{D}_F^\omega} : \mathcal{D}_F^\omega \rightarrow \mathcal{D}_F^\omega, \\ N_F P_F^\vee &= N_F|_{\mathcal{D}_F^\omega}, & (P_F^\vee)^2 &= P_F^\vee, & P_F^\vee \omega &= \omega. \end{aligned}$$

If $F \subseteq G$, $J_F^\omega \pi_{G \rightarrow F} \leq J_G^\omega$, hence $\mathcal{D}_G^\omega \subseteq \mathcal{D}_F^\omega$ and $P_F^\vee P_G^\vee = P_F^\vee|_{\mathcal{D}_G^\omega}$. Since $J_F^\omega = +\infty$ conflates empty and compatible entropy-singular fibers, Comp_F is the least Boolean emptiness coordinate and K_F the full section carrier; no selector is asserted off $Y_{F,\text{fin}}^\omega$. Layers, support/pressure and soft/hard read-outs are representations of $(\text{Comp}_F, J_F^\omega)$, not additional coordinates.

c. Donald–Petz entropy splitting and entropic RFP face. For a normal faithful conditional expectation $H : \mathcal{M} \rightarrow \mathcal{A}$ put $Q\eta = \eta|_{\mathcal{A}}$, $K\beta = \beta \circ H$ ($QK = \text{id}$); for commuting normal faithful conditional expectations E, F put $G = EF = FE$. With $q_j \geq 0$, $\sum_j q_j = 1$, $\bar{\eta} = \sum_j q_j \eta_j$. The finite positive-functional mixture identity (JEO_{mix}), expected Pythagoras³³ (Thm. 5.15), and (RFP) give

$$\begin{aligned} \sum_j q_j D(\eta_j \| \omega) &= D(\bar{\eta} \| \omega) + \sum_j q_j D(\eta_j \| \bar{\eta}), \\ D_{\mathcal{M}}(\eta \| K\beta) &= D_{\mathcal{M}}(\eta \| KQ\eta) + D_{\mathcal{A}}(Q\eta \| \beta), & D_{\mathcal{M}}(K\alpha \| K\beta) &= D_{\mathcal{A}}(\alpha \| \beta), \\ \mathfrak{J}_{E,F}(\rho) &:= D(\rho \| \rho E) - D(\rho F \| \rho G) \geq 0, \\ D(\rho \| \rho G) &= D(\rho E \| \rho G) + D(\rho F \| \rho G) + \mathfrak{J}_{E,F}(\rho). \end{aligned} \tag{112}$$

The isometric row also follows from DPI in both directions because $QK = \text{id}$. Donald’s row extends to countable finite-action mixtures by finite truncation: joint convexity controls the tail against ω , while $\bar{\eta} \geq r_n \eta_{>n}$ gives $r_n D(\eta_{>n} \| \bar{\eta}) \leq -r_n \log r_n \rightarrow 0$. If $D(\rho \| \rho E) < \infty$, the third is exactly the finite-upper-face length of the global (RFP) cell $\text{Cell}_F(\rho \| \rho E) = [D(\rho F \| \rho G), D(\rho \| \rho E)]$; expected-split Pythagoras for E plus that length gives the fourth. If also $D(\rho \| \rho F) < \infty$, swapping E, F gives $\mathfrak{J}_{E,F} = \mathfrak{J}_{F,E}$; globally retain both ordered cells, without subtraction at infinite upper endpoints. For expected $N \subseteq \mathcal{K}$ put $\delta_N(\rho) := D(\rho \| \rho E_N)$ and $A_\rho(\mathcal{K} : N) := D(\rho E_{\mathcal{K}} \| \rho E_N)$. The same split gives, in $[0, \infty]$ and without subtraction, (EAP) $\delta_N(\rho) = \delta_{\mathcal{K}}(\rho) + A_\rho(\mathcal{K} : N)$, $A_\rho(\mathcal{L} : N) = A_\rho(\mathcal{L} : \mathcal{K}) + A_\rho(\mathcal{K} : N)$, for $N \subseteq \mathcal{K} \subseteq \mathcal{L}$. If moreover H is ω -preserving, put $\omega_{\mathcal{A}} := Q\omega$, so $\omega = K\omega_{\mathcal{A}}$. For any normal-state readout $R = \bar{R}Q$, expected Pythagoras and $QK = \text{id}$ give the exact variational descent

$$(EVD) \quad R_! D_\omega = \bar{R}_! D_{\omega_{\mathcal{A}}}, \quad \text{Opt}_R^{D_\omega}(y) = K \text{Opt}_{\bar{R}}^{D_{\omega_{\mathcal{A}}}}(y) \quad \text{when the common infimum is finite.}$$

Indeed $D_{\mathcal{M}}(\eta \| \omega) = D_{\mathcal{M}}(\eta \| KQ\eta) + D_{\mathcal{A}}(Q\eta \| \omega_{\mathcal{A}})$; taking fibre infima and using the K-lift gives the first equality, while finite equality forces the vertical term to vanish and gives the minimizer identity. Thus fixed-state Avidya is an antitone exact potential. Extended scalarization is sharp on $\ell^\infty(\mathbb{N}^2)$ for $p_n = q_n = 2^{-n}$, $s_n = 6/(\pi^2 n^2)$, coordinate expectations E_q, F_p , and $\rho = p \otimes s$: $D(s \| q) = +\infty$, so the first cell is $[+\infty, +\infty]$ with no scalar readout while the swapped finite-face defect is 0. The product balance is the expected-split identity applied successively, on faithful supports, to the σ_A - and η_B -preserving one-leg expectations; support reduction gives the extended-value form. With

$I_\eta(A:B) := D(\eta\|\eta_A \otimes \eta_B)$, the two applications are

$$\begin{aligned} D(\eta\|\sigma_A \otimes \sigma_B) &= D(\eta\|\sigma_A \otimes \eta_B) + D(\eta_B\|\sigma_B) \\ &= I_\eta(A:B) + D(\eta_A\|\sigma_A) + D(\eta_B\|\sigma_B). \end{aligned} \quad (113)$$

Thus, if $\eta_B = \gamma$, $D(\eta\|\sigma_A \otimes \tau) = D(\eta\|\sigma_A \otimes \gamma) + D(\gamma\|\tau)$. For relative entropy, the vertical residual of Proposition S2 admits the following exact normal-profile refinement. For $\rho \in C \cap S_\omega^{\leq \infty}$, put $y = Q\rho$, $P = P_{Q,C}^\omega$, $\sigma = P\rho = s_{Q,C}^\omega(y)$, $\sigma_t = (1-t)\sigma + t\rho$, and define the *entropic normal defect* $\mathfrak{n}_P^\omega(\rho) := D(\rho\|\omega) - D(\rho\|P\rho) - D(P\rho\|\omega)$. Corestrict $Q_{<\infty} := Q|_{C \cap S_\omega^{\leq \infty}} : C \cap S_\omega^{\leq \infty} \rightarrow Y_{\text{fin}}$ and put $A_{<\infty} := D_\omega|_{C \cap S_\omega^{\leq \infty}}$. Then $A_{<\infty} \in \mathcal{B}_{Q_{<\infty}}$ and $(Q_{<\infty})!A_{<\infty} = \mathcal{J}|_{Y_{\text{fin}}}$; hence on this finite-action domain, $V_Q^{D_\omega}(\rho) = V_{Q_{<\infty}}^{A_{<\infty}}(\rho) = \mathcal{N}_{Q_{<\infty}}^{Q_{<\infty}}(d_{Q_{<\infty}}A_{<\infty})(\rho)$; the normal profile is the further split (FN) $V_Q^{D_\omega}(\rho) = D(\rho\|P\rho) + \mathfrak{n}_P^\omega(\rho)$, $\mathfrak{n}_P^\omega(\rho) = \partial_+ D_\omega(\sigma; \rho - \sigma) = \lim_{t \downarrow 0} \frac{D(\sigma_t\|\omega) - D(\sigma\|\omega)}{t} \geq 0$. Apply (JEO_{mix}) to $\sigma_t = (1-t)\sigma + t\rho$. With reference ω , selector minimality gives $D(\rho\|\sigma_t) \leq \Delta := D(\rho\|\omega) - D(\sigma\|\omega)$, hence $D(\rho\|\sigma) \leq \Delta$ by lower semicontinuity; joint convexity gives $D(\rho\|\sigma_t) \leq (1-t)D(\rho\|\sigma)$, hence convergence. With reference σ , the same identity gives $0 \leq (1-t)t^{-1}D(\sigma\|\sigma_t) \leq D(\rho\|\sigma) - D(\rho\|\sigma_t) \rightarrow 0$; substitution in the ω identity proves (FN). The domain is sharp: on $\ell^\infty(\mathbb{N})$ take $C = S_n$ and the constant quotient, $\omega_n = 2^{-n}$ and $\rho_n = 6/(\pi^2 n^2)$ ($n \geq 1$); then $P\rho = \omega$, $D(\rho\|\omega) = +\infty$, and the displayed difference is $+\infty - +\infty$. Hence flatness is $\mathfrak{n}_P^\omega = 0$ on this domain; on the faithful finite-affine tilt branch the Bregman identity and stationarity give exact Pythagoras and hence flatness, while at full σ -finite ambient scope finite-entropy normals can still leave the perturbative cotangent. For $x_0, \dots, x_m, \zeta$ with all displayed relative entropies finite, put $\delta_\zeta(x, y) := D(x\|\zeta) - D(x\|y) - D(y\|\zeta)$; then $D(x_0\|\zeta) = \sum_{j=1}^m [D(x_{j-1}\|x_j) + \delta_\zeta(x_{j-1}, x_j)] + D(x_m\|\zeta)$. For $y = Px$, $\zeta = \omega$, $\delta_\omega(x, y) = \mathfrak{n}_P^\omega(x)$; exact Pythagoras is exactly $\delta_\zeta = 0$. On $\ell^\infty(\mathbb{N})$ take $\sigma_n = 2^{-n}$, $\omega_n = 3 \cdot 4^{-n}$, $C = \{p : p_n \geq \sigma_n \ (n \geq 2)\}$, $g_N = \min\{\log(\sigma/\omega), N\}$. Then $p(g_N) \geq \sigma(g_N)$ gives the Gibbs minimizer σ and $\mathfrak{R}_{D_\omega, 1}(\sigma; e_n - e_1) = (n-1)\log 2$. For $g_n = n \log 2 - \log 3$, (JEO_{ori}) gives $E_{\text{exp}} = B_\Phi = E^\phi$, $L_{\log} = B_{\Phi_\omega}^*$, $B_{\Phi_\omega}^{**} = L^\phi$ (equivalent norms), $\phi = \cosh - 1$; $Z_g(t) = 3^{1-t} \sum_{n \geq 1} (2^t/4)^n$ has radius 2, so $[g] \in L^\phi/\mathbb{R}1 \setminus E^\phi/\mathbb{R}1$. Thus E_{exp} is proper and sublinear normals remain profiles.

d. *Expected BKM spectral representation.* Put $t_\omega^b = M_{\text{sa}}/\mathbb{R}1$ and $\text{Tan}_\omega(\mathcal{N}) = [\mathcal{N}_{\text{sa}}]$. By Takesaki³⁴, (TEXP) $\mathcal{N} \in \text{Lat}_\omega(\mathcal{M}) \iff \exists!$ normal ω -preserving expectation $E_N : \mathcal{M} \rightarrow \mathcal{N}$; GNS: $E_N(x)\Omega = e_{N,x}\Omega$, $[e_N, \Delta_\omega] = 0$. With g_ω ³⁵ (Sec. 14), $\Pi_{\omega, \mathcal{N}}^{\text{BKM}}[h] = [E_N h]$ and e_N commutes with $T_\Omega = ((\Delta_\omega - 1)/\log \Delta_\omega)^{1/2}$, so $(\Pi_{\omega, \mathcal{N}}^{\text{BKM}})^2 = \Pi_{\omega, \mathcal{N}}^{\text{BKM}}$, $\text{Ran } \Pi_{\omega, \mathcal{N}}^{\text{BKM}} = \text{Tan}_\omega(\mathcal{N})$, $g_\omega(\Pi_{\omega, \mathcal{N}}^{\text{BKM}} v, w) = g_\omega(v, \Pi_{\omega, \mathcal{N}}^{\text{BKM}} w)$. Let $\widehat{\Pi}_{\omega, \mathcal{N}}^{\text{BKM}}$ be its orthogonal extension to $\mathcal{H}_\omega^{\text{BKM}}$, with range $\overline{\text{Tan}_\omega(\mathcal{N})}^{g_\omega}$, and define $\mathfrak{B}_\omega : \text{Lat}_\omega(\mathcal{M}) \rightarrow \text{Proj}(\mathcal{H}_\omega^{\text{BKM}})$, $\mathcal{N} \mapsto \widehat{\Pi}_{\omega, \mathcal{N}}^{\text{BKM}}$. Its source is $\mathcal{N}$, never E_N . Restricting projection order to bounded tangents and using $\omega E = \omega$ to kill quotient scalars gives $\mathfrak{B}_\omega(\mathcal{N}) \leq \mathfrak{B}_\omega(\mathcal{K}) \iff E_{\mathcal{K}} E_N = E_N \iff \mathcal{N} \subseteq \mathcal{K}$, so $\mathfrak{B}_\omega$ is lossless and order-faithful. For expected $\mathcal{N} \subseteq \mathcal{K}$ define the common interval-shell pair $\mathfrak{b}_{\mathcal{N}:\mathcal{K}} := \widehat{\Pi}_{\omega, \mathcal{K}}^{\text{BKM}} - \widehat{\Pi}_{\omega, \mathcal{N}}^{\text{BKM}} \in \text{Proj}(\mathcal{H}_\omega^{\text{BKM}})$, $\mathfrak{e}_{\mathcal{N}:\mathcal{K}}(\rho) := D(\rho E_{\mathcal{K}}\|\rho E_N) = A_\rho(\mathcal{K}:\mathcal{N})$. Nested orthogonal projections and (EAP) give, for $\mathcal{N} \subseteq \mathcal{K} \subseteq \mathcal{L}$, $\mathfrak{b}_{\mathcal{N}:\mathcal{K}} = \mathfrak{b}_{\mathcal{N}:\mathcal{K}} + \mathfrak{b}_{\mathcal{K}:\mathcal{L}}$, $\mathfrak{e}_{\mathcal{N}:\mathcal{L}} = \mathfrak{e}_{\mathcal{N}:\mathcal{K}} + \mathfrak{e}_{\mathcal{K}:\mathcal{L}}$, $\delta_{\mathcal{N}} = \delta_{\mathcal{K}} + \mathfrak{e}_{\mathcal{N}:\mathcal{K}}$. (BES) For every expected $\mathcal{N} \subseteq \mathcal{K}$ and bounded $h = h^*$, the Araki arc $\rho_t = \omega^{th}$ satisfies

$$\frac{d^2}{dt^2} \Big|_0 \mathfrak{e}_{\mathcal{N}:\mathcal{K}}(\rho_t) = \|\mathfrak{b}_{\mathcal{N}:\mathcal{K}}[h]\|_{g_\omega}^2.$$

Hence $\mathfrak{b}_{\mathcal{N}:\mathcal{K}} = 0$ iff $\mathfrak{e}_{\mathcal{N}:\mathcal{K}}(\rho) = 0$ for every finite-action normal ρ , iff the displayed Hessian vanishes for every bounded h . Given $\mathcal{K}$, the shell reconstructs $\mathcal{N}$ via $\widehat{\Pi}_{\omega, \mathcal{N}}^{\text{BKM}} = \widehat{\Pi}_{\omega, \mathcal{K}}^{\text{BKM}} - \mathfrak{b}_{\mathcal{N}:\mathcal{K}}$ and order-faithfulness, hence (TEXP) reconstructs E_N and the full entropic shell profile. Thus the nonlinear all-state shell and its reference-local BKM germ carry the same expected-interval information at this typed base. For any increasing expected family $(\mathcal{K}_i)$ abbreviate $\mathfrak{b}_{s:t} = \mathfrak{b}_{\mathcal{K}_s:\mathcal{K}_t}$ and $\mathfrak{e}_{s:t} = \mathfrak{e}_{\mathcal{K}_s:\mathcal{K}_t}$. More sharply, for $(\mathcal{N}_i)_{i \in I} \subseteq \text{Lat}_\omega(\mathcal{M})$, $\mathcal{N}_\wedge = \cap_i \mathcal{N}_i$ and $\mathcal{N}_\vee = \overline{\text{span}} \cup_i \mathcal{N}_i$, $\mathfrak{B}_\omega(\mathcal{N}_\wedge) = \bigwedge_i \mathfrak{B}_\omega(\mathcal{N}_i)$, $(\mathcal{N}_i) \text{ directed} \implies \mathfrak{B}_\omega(\mathcal{N}_\vee) = \bigvee_i \mathfrak{B}_\omega(\mathcal{N}_i)$. For arbitrary families set $\text{AlgRes}_\omega(\mathcal{N}_\bullet) := \mathfrak{B}_\omega(\mathcal{N}_\vee) - \bigvee_i \mathfrak{B}_\omega(\mathcal{N}_i) \geq 0$ and $\mathcal{H}_I^{\text{lin}} := \sum_i \text{Ran } \widehat{\Pi}_{\mathcal{N}_i}^{\text{BKM}}^{g_\omega}$. On the tensor-tracial face, let F be finite, $(\mathcal{A}_i, \tau_i)$ finite von Neumann algebras with faithful normal traces, $\tau_F = \bigotimes_i \tau_i$, and $\mathcal{C}$ a cover of F . Put $\mathcal{N}_C = (\bigotimes_{i \in C} \mathcal{A}_i) \bar{\otimes} (\bigotimes_{j \notin C} \mathbb{C}1_j)$ and $H_i^0 = L^2(\mathcal{A}_i, \tau_i)_{\text{sa}} \ominus \mathbb{R}1$. Tensor orthogonality gives

$$(\text{CIT}_{\mathcal{C}}) \quad \text{Ran AlgRes}_{\tau_F}((\mathcal{N}_C)_C) = \bigoplus_{A \in F: A \not\subseteq C \ \forall C \in \mathcal{C}} \bigotimes_{i \in A} H_i^0,$$

with scalar factors on $F \setminus A$. (TEXP) makes $\mathcal{N}_\wedge, \mathcal{N}_\vee$ expected. For finite F , $a_F = |F|^{-1} \sum_{i \in F} e_{\mathcal{N}_i}$ has $a_F^n \rightarrow p_F = \bigwedge_i e_{\mathcal{N}_i}$. Since a_F implements $\Phi_F = |F|^{-1} \sum_i E_{\mathcal{N}_i}$, any ultraweak cluster y of $\Phi_F^n x$ has $y\Omega = \overline{p_F x \Omega}$ and $E_{\mathcal{N}_i} y = y$, hence $p_F = e_{\cap_i \mathcal{N}_i}$; finite-subset descent gives arbitrary meets. Directed joins follow from Kaplansky $\mathcal{N}_\vee \Omega = \overline{\cup_i \mathcal{N}_i \Omega}$; commutation with T_Ω transports both identities to BKM and proves (BDS). Also $Y = \sum_i \mathcal{N}_i$ is modular-invariant and weak-* dense in V_I , and normal separation plus Gaussian modular smoothing makes $Y\Omega$ a T_Ω -core. Hence $\mathcal{H}_I^{\text{lin}} = \overline{[V_I]_{\text{sa}}}^{g_\omega}$ and $\text{Ran AlgRes}_\omega = \text{Ran } \mathfrak{B}_\omega(\mathcal{N}_\vee) \ominus \mathcal{H}_I^{\text{lin}}$, the (MCR) BKM tangent face; for finite families it is the first (LAR) summand, strict in M_2 . Thus arbitrary joins fail exactly by AlgRes_ω . On M^* put $E_N^* v = v \circ E_N$ (normal restriction E_{N^*}) and, for $v = v^*$, $v(1) = 0$, $g_\omega^\vee(v) = \sup_{h=h^*} \{2v(h) - g_\omega([h], [h])\}$. If this is finite, optimization along $t[h]$ gives $|v(h)|^2 \leq g_\omega^\vee(v)g_\omega([h], [h])$, hence a unique Riesz vector $\mathcal{R}_\omega v$ with $v(h) = g_\omega(\mathcal{R}_\omega v, [h])$ and $g_\omega^\vee(v) = \|\mathcal{R}_\omega v\|_{g_\omega}^2$. For faithful ω set $M_{*,h}^0 = \{v = v^* : v(1) = 0\}$, $\text{Ord}_\omega = \{v \in M_{*,h}^0 : -c\omega \leq v \leq c\omega \text{ for some } c\}$ and $\text{En}_\omega = \{v \in M_{*,h}^0 : g_\omega^\vee(v) < \infty\}$. With Haagerup density h_ω , Radon–Nikodym writes $v = h_\omega^{1/2} x h_\omega^{1/2}$ on Ord_ω ; logarithmic mean $\geq$ geometric mean gives $g_\omega^\vee(v) \leq \|h_\omega^{1/4} x h_\omega^{1/4}\|_2^2 \leq \|x\|^2$. Thus, for bounded A with $\ker A \subseteq \overline{M_{*,h}^0}$, (OEL) $\mathcal{T}_\omega^\pm K_{A\omega} = \ker A \cap \text{Ord}_\omega \subseteq \ker A \cap \text{En}_\omega \subseteq \ker A$, with equality in finite dimension. Define (KCE) $\text{KEx}_\omega(A) := \ker A / \overline{\ker A \cap \text{En}_\omega}$, with predual-norm closure. For strictness on $\ell^\infty(\mathbb{N})$ take $\omega_n = 2^{-n}$, $f_n = n - 2$, $g_n = (-1)^n 2^{n/2} + \sqrt{2} - 1$, $V = \text{span}\{f\omega, g\omega\}$ and quotient A . Then $f \in L^2 \setminus L^\infty$, $g \in L^1 \setminus L^2$ with unbounded alternating signs, $K_{A\omega} = \{(1+af)\omega : 0 \leq a \leq 1\}$, and $0 = \mathcal{T}_\omega^\pm K_{A\omega} = \ker A \cap \text{Ord}_\omega \subsetneq \ker A \cap \text{En}_\omega = \mathbb{R}f\omega = K_{\text{face}} = K_{\text{sec}} \subsetneq \ker A = V$ ($\text{KEx}_\omega(A) \cong \mathbb{R}$). On the commutative faithful face, with $D_\omega f = f\omega$, $\phi = \cosh - 1$ and subscript 0 centered, (JEO_{ori}) plus classical BKM gives (EOL) $\text{Ord}_\omega = D_\omega L_0^\infty \subset D_\omega E_0^\phi \subset D_\omega L_0^\phi \subset \text{En}_\omega = D_\omega L_0^2 \subset M_{*,h}^0 = D_\omega L_0^1$, where E^ϕ/L^ϕ are the all/local exponential-moment classes. For $\omega_n = 3 \cdot 4^{-n}$ above, centered $\sqrt{n}, g, 2^{n/2}, 2^n$ separate the four inclusions. Thus OEL/KCE/EOL are pre-closure and (AFP) norm-closed.

Therefore, for every real noncommutative polynomial p ,

$$\begin{aligned} \mathcal{R}_\omega(p(E_{N_1}^*, \dots, E_{N_m}^*)v) &= p(\widehat{\Pi}_{\omega, N_1}^{\text{BKM}}, \dots, \widehat{\Pi}_{\omega, N_m}^{\text{BKM}})\mathcal{R}_\omega v, \\ \text{[BRB]} \quad g_\omega^\vee(p(E_{N_1}^*, \dots, E_{N_m}^*)v) &= \|p(\widehat{\Pi}_{\omega, N_1}^{\text{BKM}}, \dots, \widehat{\Pi}_{\omega, N_m}^{\text{BKM}})\mathcal{R}_\omega v\|_{g_\omega}^2. \end{aligned} \quad (114)$$

The linear case is $v(E_N h) = g_\omega(\mathcal{R}_\omega v, \Pi_{\omega, N}^{\text{BKM}}[h])$; composition gives (BRB), with strict kernel boundary (OEL/KCE). For two projections (H2A/BRB) gives the complete pairwise corner/angle carrier; in M_d , (TWC/TWM) gives the corresponding all-state trace-word/marker carrier.

For a finite commuting expected family, let $\mathcal{L}_E = \langle E_i \rangle_{i=1}^\infty$ be the distinct expectation products (empty product I), with Möbius function μ_E , let $\widehat{A}_P$ be the (RIF) atoms of the commuting $\widehat{\Pi}_i^{\text{BKM}}$, and set $A_P^\vee = \sum_{Q \leq P} \mu_E(Q, P)Q^*$. Then (BRB) gives

$$\mathcal{R}_\omega(A_P^\vee v) = \widehat{A}_P \mathcal{R}_\omega v, \quad g_\omega^\vee(f(E_1^*, \dots, E_n^*)v) = \sum_P |f(\chi_P(E_1), \dots, \chi_P(E_n))|^2 \|\widehat{A}_P \mathcal{R}_\omega v\|_{g_\omega}^2,$$

with Parseval at $f = 1$. Thus the commuting spectral measure, Boolean faces and nested shells are one represented incidence calculus. *Incidence spectral measure and residual exactness.* For $B \subseteq \mathcal{L}_E$ put $A(B) = \sum_{P \in B} \widehat{A}_P$, $\varepsilon_B(v, w) = g_\omega(A(B)v, A(B)w)$ and $\widehat{q}_Q(v, w) = g_\omega((I - \widehat{Q})v, (I - \widehat{Q})w)$. The orthogonal RIF atoms give

$$\begin{aligned} \widehat{Q} &= A(\downarrow Q), \quad \widehat{q}_Q = \varepsilon_{\mathcal{L}_E \setminus \downarrow Q}, \quad Q \leq R : \widehat{q}_Q - \widehat{q}_R = \varepsilon_{\downarrow R \setminus \downarrow Q} \geq 0, \\ \varepsilon_{\{P\}} &= \sum_{Q \leq P} \mu_E(Q, P)(g_\omega - \widehat{q}_Q), \\ \text{[ISM]} \quad \prod_{i \in S} \widehat{\Pi}_i^{\text{BKM}} \prod_{j \in T} (I - \widehat{\Pi}_j^{\text{BKM}}) &= A(\Lambda_{S,T}) \quad (S \cap T = \emptyset), \\ Z_Q &:= Z_{\text{tr}}^1(R_{\widehat{Q}}) \cong_\perp A(\mathcal{L}_E \setminus \downarrow Q) \mathcal{H}_\omega^{\text{BKM}}, \quad \ker(\text{res}_{Q,R}) \cong_\perp A(\downarrow R \setminus \downarrow Q) \mathcal{H}_\omega^{\text{BKM}}. \end{aligned} \quad (115)$$

For $S \cap T = \emptyset, T \neq \emptyset$, set

$$E_{S^*} = \prod_{j \in S} E_{j^*}, \quad B_{S,T}^\vee = E_{S^*} \prod_{i \in T} (I - E_{i^*}), \quad B_{S,T} = \prod_{j \in S} \Pi_j^{\text{BKM}} \prod_{i \in T} (I - \Pi_i^{\text{BKM}}), \quad \widehat{B}_{S,T} = A(\Lambda_{S,T}),$$

$\mathfrak{F}_{S,T} = \{\varphi \in \mathcal{S}_{\text{fin}} : \varphi E_j = \varphi (j \in S), \varphi E_i = \omega (i \in T)\}$. Commutativity, (AFP) and (BRB) give $\mathfrak{F}_{S,T} = (\omega + \text{Ran } B_{S,T}^\vee) \cap \mathcal{S}_{\text{fin}}$, $\mathcal{R}_\omega(B_{S,T}^\vee v) = \widehat{B}_{S,T} \mathcal{R}_\omega v$, $\mathfrak{F}_{S,T} = \{\omega\} \iff B_{S,T}^\vee = 0 \iff B_{S,T} = 0 \iff \widehat{B}_{S,T} = 0$, and $g_\omega^\vee(B_{S,T}^\vee v) = \|\widehat{B}_{S,T} \mathcal{R}_\omega v\|^2$. With the conflict-zero signed-union product $a \star b$ on $\mathfrak{C}_n = \{(S, T) : S \cap T = \emptyset\} \cup \{\perp\}$, $B_a^\vee B_b^\vee = B_{a \star b}^\vee$, $B_a B_b = B_{a \star b}$, $\widehat{B}_a \widehat{B}_b = \widehat{B}_{a \star b}$, $\text{Ran } B_{a \star b}^\vee = \text{Ran } B_a^\vee \cap \text{Ran } B_b^\vee$, $\mathfrak{F}_{a \star b} = \mathfrak{F}_a \cap \mathfrak{F}_b$. If $U_0 = [n] \setminus (S \cup T)$, the three carriers decompose over all terminal sign choices $U \subseteq U_0$ (orthogonal sums for $B, \widehat{B}$ and an ordinary sum for $B^\vee$). Hence the predual cube, sign bands, $\mathfrak{A}_\square$ telescope, (KSP/BKT) intervals and kernel-Čech grades are exact read-outs of (ISM), not new coordinates. Writing $\widehat{P}_* = \prod_i \widehat{\Pi}_i^{\text{BKM}}$, $V = \text{Ran } \widehat{P}_*$, $Z_* = Z_{\text{tr}}^1(R_{\widehat{P}_*})$ and $\mathcal{K}_Q^{\text{inc}} = \ker(\text{res}_{\widehat{P}_*, \widehat{Q}})$, $C_{O(\mathcal{H}_\omega^{\text{BKM}})}(\widehat{\mathcal{L}}_E) \cong O(V) \times O_{\text{inc}}(Z_*)$, $O_{\text{inc}}(Z_*) = \{T \in O(Z_*) : T \mathcal{K}_Q^{\text{inc}} = \mathcal{K}_Q^{\text{inc}} \forall Q\}$. This is exactly preservation of the incidence kernel flag, and descends compatibly to every Z_Q . Sharpness is unchanged: noncommuting (LAR) has no common PVM, positive Gram shadows retain (BFE), and the static PVM does not determine (FRP/TAN).

Second entropic blow-up and concentration profile. Assume $\omega \in C$. Use weak-star tangents in $\mathcal{M}_{\text{sa},0}^* = \{v = v^* : v(\mathbf{1}) = 0\}$ and product weak-star record tangents; Pinsker gives (E2) compactness. With the BKM form g_ω , set $\mathfrak{d}_{\omega,C}^{(2)} = (D_\omega|_C)^{(2)}$, $\mathfrak{d}_{F,C}^{(2)} = (\mathcal{J}_{N_F,C}^\omega)_{N_F \omega}^{(2)}$, and $L_F^\vee v = (v \circ \alpha_r)_r$. Then

$$\begin{aligned} \text{[E2S]} \quad \mathfrak{d}_{\omega,C}^{(2)} &\geq g_\omega^\vee, & \mathfrak{d}_{\omega, \mathcal{S}_n}^{(2)} &= g_\omega^\vee, \\ \mathfrak{d}_{F,C}^{(2)} &= (L_F^\vee)_{\mathfrak{d}_{\omega,C}^{(2)}}, & \dim \mathcal{M} < \infty : \mathfrak{d}_{\omega,C}^{(2)} &= g_\omega^\vee + I_{T_\omega C}. \end{aligned} \quad (116)$$

Here I_T is the closed-contingent-cone indicator. By (BRB), finite-energy tangents are represented in the BKM completion. On $g_\omega^\vee < \infty$ put $\mathfrak{c}_{\omega,C}^{(2)} = \mathfrak{d}_{\omega,C}^{(2)} - g_\omega^\vee$. If $N \in \text{Lat}_\omega(\mathcal{M})$ and $b = (b_j) \subset N_{\text{sa}}$ is finite, then $\mathfrak{d}_{\omega, C_{N,b}}^{(2)} = g_\omega^\vee + I_{\{v: v E_N = v, v(b_j) = 0\}}$. Indeed $k = \mathcal{R}_\omega v$ lies in $\text{Ran } \widehat{\Pi}_{\omega, N}^{\text{BKM}} \cap V_b^{\perp g}$; bounded approximants in N_{sa} can be Gram-projected to $V_b^{\perp g}$, and the moment map of $\omega^{tk_n + \sum a \lambda_a c_a}$ has invertible $D_\lambda = G_c$ and $D_t = 0$, so IFT gives exact-moment arcs with $\lambda(t) = O(t^2)$. Expected sufficiency and a diagonal limit give recovery. For $N = \mathcal{M}$, this yields the image formula $u^\top G_b^\dagger u + I_{\text{Ran } G_b}(u)$. The residual is genuinely second order. For $1 < \alpha < 2$, set $r_\alpha = (\alpha - 1)/\alpha$, $dv_\alpha = \alpha r_\alpha^\alpha r^{-\alpha-1} \mathbf{1}_{[r_\alpha, \infty)} dr$, $w = r - 1$, attach a fair sign $u = \varepsilon$, and let $C_{\alpha, \lambda} = \{(1 + xu + yw)\omega_\alpha : 0 \leq x, y \leq \delta, y \geq \lambda x^{2/\alpha}\}$. These convex faithful families share their closed-contingent/radial first-order and ambient BKM data (and zero terminal FN profile on the radial cone), whereas (E2X) $\mathfrak{d}_{\omega, C_{\alpha, \lambda}}^{(2)}(u) = 1 + 2\kappa_\alpha \lambda^\alpha$, $\kappa_\alpha = -\frac{\pi(\alpha-1)^{\alpha-1}}{\alpha^\alpha \sin(\pi\alpha)} > 0$. Thus $\mathfrak{c}^{(2)}$ has continuum residual; $\alpha = 3/2$ gives $1 + 4\pi(3\sqrt{3})^{-1} \lambda^{3/2}$. (JEO_{gib}) and the BKM Hessian³⁵ (Sec. 14) give $\mathfrak{d}^{(2)} \geq g_\omega^\vee$; bounded Riesz approximation along these perturbation arcs gives equality on all normal states, and (E2) gives the record and finite-dimensional cone rows. Finally $D((1 + yw)v_\alpha \| v_\alpha) = \kappa_\alpha y^\alpha + o(y^\alpha)$ from $\int_0^\infty \phi(z) z^{-\alpha-1} dz = -\pi/[\alpha(\alpha-1) \sin(\pi\alpha)]$, while sign averaging contributes $x^2/2 + o(x^2)$; the boundary $y = \lambda x^{2/\alpha}$ therefore gives and attains (E2X).

On smooth pointed tangent charts, relative-entropy chain rule gives $\text{Hess}(D_1 \circ T) = g_1(A \cdot, A \cdot)$ and $q_A = g_0 - g_1(A \cdot, A \cdot) \geq 0$, with (RFP) composition $q_{BA} = q_A + q_B(A \cdot, A \cdot)$. Same-chart contractive idempotents are BKM-orthogonal. On a finite constant-rank chart let $L_Q[h] = \frac{d}{dt} \big|_0 Q(\omega^{th})$, $A_Q = \text{proj}_{(\ker L_Q)^\perp}^{\mathcal{H}_\omega}$, $\text{Hess}_{Q\omega} \mathcal{J}_{Q,C}^\omega(u, u) = \min_{L_Q h = u} g_\omega(h, h)$. For records, $A_F = \bigvee_{r \in F} A_{N_r}$ and $\text{Hess}_0[J_F^\omega(N_F \omega^h)](v, w) = g_\omega(A_F v, w)$, $A_F - A_E \geq 0$. For two records $K_{12} = A_{\{1,2\}} - A_1 - A_2$ has on each generic block

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}, \quad K_\lambda = I - P - Q, \quad K_\lambda^2 = \lambda I \quad (0 < \lambda < 1).$$

Hence $K_{12} \leq 0 \iff [A_1, A_2] = 0 \iff (A_1 A_2 A_1)^2 = A_1 A_2 A_1 \iff \text{spec}(A_1 A_2 A_1 |_{\text{Ran } A_1}) \subset \{0, 1\}; K_{12} = 0 \iff A_1 A_2 = 0$, generic

blocks contribute $\{\pm\sqrt{\lambda}\}$, and at commutation $K_{12} = -A_1A_2$, while CSQ is $(1 - A_1)(1 - A_2)$ with the $\lambda = 1/4$ three-point witness. With $\mathcal{V}_F$ from Theorem V.2, duality gives $\ker L_F = [\mathcal{V}_F]^{\perp_{g\omega}}$, $\text{Ran } A_F = [\mathcal{V}_F]$ and $\text{Alg}_\omega(\text{Ran } A_F) = \mathcal{G}_F^\omega$.

For a finite canonical U1 chain $P_j P_{j-1} = P_j$ on (C, ω) , $\sigma_j = P_j \rho$ satisfies $\delta_\omega(\sigma_{j-1}, \sigma_j) = \mathfrak{n}_{P_j}^\omega(\sigma_{j-1}) \geq 0$; refinement is the bare (KSP) / weighted (RFP) telescope. If instead $P_i P_j = P_{\max(i,j)}$ on a finite-action quotient, each P_j has exact Pythagoras, and $\text{QK} = \text{id}$, put $\widehat{P}_0 = \text{KQ}$, $\widehat{P}_j = \text{KP}_j \text{Q}$. Then, for $\beta \in \text{Ran } P_m$,

$$D_{\mathcal{M}}(\eta \| \text{K}\beta) = D(\eta \| \widehat{P}_0 \eta) + \sum_{j=1}^m D(\widehat{P}_{j-1} \eta \| \widehat{P}_j \eta) + D(\widehat{P}_m \eta \| \text{K}\beta)$$

in $[0, \infty]$. On the all-finite locus this is the zero- $\delta_{\text{K}\beta}$ specialization above; the lifted $\widehat{P}_j$ are the reverse-KL retractions and obey the same absorption law. At a smooth common fixed point, (E2S) differentiates it to nested BKM projections and their shell differences; the nonlinear exact-flat identity remains independent.

For commuting ω -preserving $E_1, \dots, E_n$ and finite-action ρ , let $E_S = \prod_{i \in S} E_i$, $\Phi(S) = D(\rho E_S \| \omega)$, $\Delta_i \Phi(S) = \Phi(S) - \Phi(S \cup \{i\})$. Expected splitting gives $\Phi(S) - \Phi(T) = D(\rho E_S \| \rho E_T) \geq 0$ for $S \subseteq T$. Thus $r(S) = \Phi(\emptyset) - \Phi(S)$, $J_{ij}(S) = \Delta_i \Delta_j \Phi(S) = \mathfrak{J}_{E_i, E_j}(\rho E_S) \geq 0$ is a normalized monotone polymatroid rank; its base-polytope greedy/support theorem is Edmonds³⁶, while restriction (reduction) and contraction are the submodular-system minors of Fujishige³⁷ (Sec. 2). Expected conditioning realizes contraction: $(r/E)(T) = D(\rho E_E \| \rho E_{E \cup T})$. Wall weights are $J_{ij}(S)$, and equality of distributional support Hessians is equivalent to translation of $B(r)$, hence to a modular rank difference. At $\rho = \omega$, (ISM) gives for disjoint S, T the tangent sign projector $B_{S,T} = \prod_S^{\text{BKM}} \prod_{i \in T} (1 - \Pi_i^{\text{BKM}})$ and $\text{Hess}_\omega \Delta_T \Phi(S)(v, w) = g_\omega(B_{S,T} v, B_{S,T} w)$, $\widehat{B}_{S,T} = A(\Lambda_{S,T})$, so every tangent finite difference is positive semidefinite and, by the zero locus above, $\text{Hess}_\omega \Delta_T \Phi(S) \equiv 0 \iff \mathfrak{F}_{S,T} = \{\omega\}$. The exact nonlinear carrier is $r/B(r)$; this Hessian is only its (E2S/ISM) Boolean-atom tangent. The distinction is sharp: for coordinate averages with uniform reference and $p = (5, 1, 1, 1, 1, 1, 1, 5)/16$, $\Delta_{123} \Phi(\emptyset) = \frac{1}{8} \log(2^{16} 5^5 / 3^{18}) < 0$.

e. Local-ambient factorization. For finite $N_i \subseteq \mathcal{K}$ in $\text{Lat}_\omega(\mathcal{M})$ put $N_V = W^*(\cup_i N_i)$ and let $\widehat{\Pi}_{\text{loc}}$ project onto $\overline{\sum_i \text{Ran } \widehat{\Pi}_i}^{g_\omega}$. From $\widehat{\Pi}_{\text{loc}} \leq \widehat{\Pi}_V \leq \widehat{\Pi}_{\mathcal{K}}$, (KSP) gives (LAR) $\widehat{\Pi}_{\mathcal{K}} - \widehat{\Pi}_{\text{loc}} = (\widehat{\Pi}_V - \widehat{\Pi}_{\text{loc}}) \oplus (\widehat{\Pi}_{\mathcal{K}} - \widehat{\Pi}_V)$. The tracial M_2 example $(N_1, N_2) = (\mathcal{D}_x, \mathcal{D}_z)$ has zero (RLE) shell but $N_V = M_2$, internal residual $\mathbb{R}[\sigma_y]$ and nonzero idempotent curvature, so LAR survives non-Boolean atomization. If the expectations commute, $\widehat{\Pi}_{\text{loc}} = I - \prod_i (I - \widehat{\Pi}_i)$ and, for $Q_I = \prod_i (I - E_{i*})$, (BRB/AFP) gives $E_{\mathcal{K}*} Q_I = E_{N_V*} Q_I \oplus (E_{\mathcal{K}*} - E_{N_V*})$. The faithful map $\varphi \mapsto \varphi E_{N_V}$ has fiber $(\psi + \text{Ran}(E_{\mathcal{K}*} - E_{N_V*})) \cap \mathcal{S}_{\text{fin}}$; hence by (MCR/BDS) the two LAR summands are the finite-energy multiplicative-correlation face and ambient extension, with (ISM) supplying the Boolean predual/Hilbert and tangent faces.

(Alg) *Local-global RFE reconstruction flag.* Put $\mathcal{G}_F^{\text{alg}} := W^*(\mathcal{V}_F)$ and $\mathcal{K}_F^{\text{alg}} := ((\mathcal{G}_F^{\text{alg}})_{\text{sa}})^\perp \subseteq \mathcal{M}_{*,h}$. The preceding annihilator identities give

$$(\mathcal{A}_F^\wedge)_{\text{sa}} \subseteq \mathcal{V}_F \subseteq (\mathcal{G}_F^{\text{alg}})_{\text{sa}} \subseteq \mathcal{M}_{\text{sa}}, \quad 0 \subseteq \mathcal{K}_F^{\text{alg}} \subseteq \mathcal{K}_F^{\text{vis}} \subseteq \mathcal{K}_F^{\text{rec}}.$$

Set $\mathfrak{L}\mathcal{G}_F := \mathcal{M}_{\text{sa}} / (\mathcal{A}_F^\wedge)_{\text{sa}}$, $\mathfrak{L}_F := (\mathcal{G}_F^{\text{alg}})_{\text{sa}} / (\mathcal{A}_F^\wedge)_{\text{sa}}$, $\mathfrak{C}_F^{\text{alg}} := (\mathcal{G}_F^{\text{alg}})_{\text{sa}} / \mathcal{V}_F$, and $\mathfrak{D}_F^{\text{amb}} := \mathcal{M}_{\text{sa}} / (\mathcal{G}_F^{\text{alg}})_{\text{sa}}$. Normal Hahn-Banach/Jordan duality gives

$$\begin{aligned} 0 \subseteq \mathfrak{R}_F \subseteq \mathfrak{L}_F \subseteq \mathfrak{L}\mathcal{G}_F, \quad 0 \rightarrow \mathfrak{R}_F \rightarrow \mathfrak{L}_F \rightarrow \mathfrak{C}_F^{\text{alg}} \rightarrow 0, \quad 0 \rightarrow \mathfrak{L}_F \rightarrow \mathfrak{L}\mathcal{G}_F \rightarrow \mathfrak{D}_F^{\text{amb}} \rightarrow 0, \\ \text{[LGR]} \quad \mathfrak{C}_F^{\text{alg}} \cong (\mathcal{K}_F^{\text{vis}} / \mathcal{K}_F^{\text{alg}})^*, \quad \mathfrak{D}_F^{\text{amb}} \cong (\mathcal{K}_F^{\text{alg}})^*, \quad 0 \rightarrow \mathcal{K}_F^{\text{vis}} / \mathcal{K}_F^{\text{alg}} \rightarrow \mathcal{K}_F^{\text{rec}} / \mathcal{K}_F^{\text{alg}} \rightarrow \mathcal{K}_F^{\text{rec}} / \mathcal{K}_F^{\text{vis}} \rightarrow 0, \\ \text{gr } \mathfrak{L}\mathcal{G}_F \cong \mathfrak{R}_F \oplus \mathfrak{C}_F^{\text{alg}} \oplus \mathfrak{D}_F^{\text{amb}}. \end{aligned} \quad (117)$$

The direct sum is only associated graded, not a splitting: this is the linear (KSP) face of the same filtered-residual architecture on the discrete (RFE) tower. On $\Omega = \mathcal{S}_n(\mathcal{M})$ put $Y_F^{\text{alg}} = r_{\mathcal{G}_F^{\text{alg}}}^n(\Omega)$, $Y_F^{\text{loc}} = N_F(\Omega)$, $Y_F^{\text{rec}} = \Pi_F^\wedge(\Omega)$ and corestrict

$$q_F^{\text{corr}}(r_{\mathcal{G}_F^{\text{alg}}}^n \rho) = N_F \rho, \quad q_F^{\text{rec}}(N_F \rho) = \Pi_F^\wedge \rho.$$

They are well-defined by $\alpha_r(\mathcal{N}_r) \subseteq \mathcal{G}_F^{\text{alg}}$ and $\mathcal{K}_F^{\text{vis}} \subseteq \mathcal{K}_F^{\text{rec}}$, and

$$\Pi_F^\wedge|_\Omega = q_F^{\text{rec}} q_F^{\text{corr}} r_{\mathcal{G}_F^{\text{alg}}}^n, \quad \text{Fib}_z(\Pi_F^\wedge|_\Omega) \cong \coprod_{y \in \text{Fib}_z(q_F^{\text{rec}})} \coprod_{u \in \text{Fib}_y(q_F^{\text{corr}})} \text{Fib}_u(r_{\mathcal{G}_F^{\text{alg}}}^n).$$

Thus ambient coverage, algebraic/correlation completion and common recoverability are successive fibre grades. By (NAT/PBE/VRC) plus Jordan decomposition their singleton zero loci are exactly $\mathfrak{D}_F^{\text{amb}} = 0$, $\mathfrak{C}_F^{\text{alg}} = 0$, and $\mathfrak{R}_F = 0$. For expected inclusions $\alpha_i : \mathcal{N}_i \hookrightarrow \mathcal{M}$, $\mathcal{V}_F = (V_I)_{\text{sa}}$ and $\mathcal{G}_F^{\text{alg}} = \mathcal{N}_V$; hence $\mathfrak{C}_F^{\text{alg}}$ is the self-adjoint (MCR) face, q_F^{corr} is (MCF), and (LGS/BDS/AlgRes/LAR) are typed descendants rather than parallel primitives.

The zero loci are ordered but not equivalent. Since $\mathcal{A}_F^\wedge$ is a von Neumann algebra,

$$\mathfrak{R}_F = 0 \implies \mathcal{V}_F = (\mathcal{A}_F^\wedge)_{\text{sa}} \implies \mathcal{G}_F^{\text{alg}} = \mathcal{A}_F^\wedge \implies \mathfrak{C}_F^{\text{alg}} = 0.$$

For $\emptyset \neq E \subseteq F$, the displayed four-kernel chain gives $\mathfrak{R}_F = 0 \implies \mathfrak{R}_E = 0$; equivalently a nonzero common-recovery defect is upward persistent under access addition. Thus access accumulation is not a monotone “coherence improvement.” By contrast,

$$\mathcal{K}_F^{\text{vis}} = 0 \iff N_F \text{ is injective on normal states} \implies \mathcal{V}_F = \mathcal{M}_{\text{sa}} \implies \mathfrak{D}_F^{\text{amb}} = \mathfrak{C}_F^{\text{alg}} = 0.$$

If additionally some $\mathcal{A}_r^{\text{sat}} \neq \mathcal{M}$, then $\mathcal{A}_F^\wedge \neq \mathcal{M}$ and

$$\boxed{\mathcal{K}_F^{\text{vis}} = 0 \ \& \ (\exists r \in F : \mathcal{A}_r^{\text{sat}} \neq \mathcal{M}) \implies \mathfrak{R}_F \neq 0.}$$

Hence exact tomography by genuinely proper accesses need not, and generally cannot, coincide with common Petz recoverability. On its finite-action face $P_F^\vee \rho = \rho$ and $M_F(\rho) = D(\rho \| \Pi_F^\wedge \rho)$, so the Petz/BKM interval measures this collective grade, not a physical energy.

The tracial qubit is sharp. For expected MASAs $\mathcal{D}_z$, then $\{\mathcal{D}_x, \mathcal{D}_z\}$, then $\{\mathcal{D}_x, \mathcal{D}_y, \mathcal{D}_z\}$,

$$(\dim \mathfrak{D}_F^{\text{amb}}, \dim \mathfrak{C}_F^{\text{alg}}, \dim \mathfrak{R}_F) = (2, 0, 0), \ (0, 1, 2), \ (0, 0, 3).$$

Every singleton MASA has $\mathfrak{R}_{\{r\}} = 0$, while the transverse pair already has $\mathfrak{R}_F \neq 0$: the recovery grade is genuinely collective. The triple is tomographically complete with common sufficient algebra C1. In finite dimension the tomography transition is exactly $r_v(F) = \dim_{\mathbb{R}} M_{sa} - 1$; minimal saturating families form its antichain frontier. Whenever (OOP) is realized by such access spans, $0 \rightarrow \mathbb{C}_F^{\text{alg}} \rightarrow M_{sa}/\mathcal{V}_F \rightarrow \mathfrak{D}_F^{\text{amb}} \rightarrow 0$, so finite-horizon opacity extinction kills ambient and correlation opacity, not the independent recovery grade.

This theorem is post-cut and fixed-world. Modular saturation $\mathcal{G}_F^\omega$, the c_F world/ambient-realization fibre, support/placement and dynamics remain attachments; (CSL/ETL/VJT) are other (RFE) gluing realizations, (QGR) keeps zero state opacity compatible with nonzero information-geometric residual, and (SCF/BPR) still forbids source-free first distinction.

For $\text{QK} = I$ and $\text{QC}\rho = \text{Q}\rho$, DPI gives $D_{\mathcal{M}}(\text{C}\rho \parallel \text{U}\rho) \geq D_{\mathcal{A}}(\text{Q}\rho \parallel \text{Q}\text{U}\rho)$. If also $\text{U} = \text{KR}$, $\text{QC} = \text{A}$, Donald–Petz gives $D_{\mathcal{M}}(\text{C}\rho \parallel \text{U}\rho) = D_{\mathcal{M}}(\text{C}\rho \parallel \text{KA}\rho) + D_{\mathcal{A}}(\text{A}\rho \parallel \text{R}\rho)$, so the finite-faithful infimum is the last term, uniquely at $\text{KA}\rho$. If $\hat{\rho}_t = L_t^0 \rho_t + B_t \rho_t$ and $\text{Q}\rho_t$ is constant, $\text{QB}_t \rho_t = -\text{QL}_t^0 \rho_t$, $\|\text{B}_t \rho_t\|_1 \geq \|\text{QL}_t^0 \rho_t\|_1$, and Pinsker gives $\liminf_{h \downarrow 0} h^{-2} D(\text{C}_{t,h} \rho_t \parallel \text{U}_{t,h} \rho_t) \geq \frac{1}{2} \|\text{QL}_t^0 \rho_t\|_1^2$; this remains a differential shadow.

Theorem V.2 (Closure, recovery, and entropic resolution). *Let $\mathcal{R}$ be the declared regime set.*

(A) Residual interval. For $F \subseteq \mathcal{R}$ put $\mathcal{V}_F := \mathbb{R}1 + \overline{\text{span}_{\mathbb{R}} \cup_{r \in F} \alpha_r(N_{r,sa})}^{\sigma\text{-weak}}$ and $\mathcal{A}_F^\wedge := \cap_{r \in F} \mathcal{A}_r^{\text{sat}}$, where $\mathcal{A}_r^{\text{sat}} = \text{Fix}((R_r^\omega N_r)^*)$; use $\mathcal{V}_\emptyset = \mathbb{R}1$, $\mathcal{A}_\emptyset^\wedge = \mathcal{M}$. Let $\text{cl}_v = H_v G_v$, $G_v(F) = \mathcal{V}_F$, $H_v(V) = \{r : \alpha_r(N_{r,sa}) \subseteq V\}$, and $\text{cl}_\wedge(F) = \{r : \mathcal{A}_r^\wedge \subseteq \mathcal{A}_F^\wedge\}$ in reverse inclusion order. For $F \neq \emptyset$ put $\mathcal{K}_F^{\text{vis}} := \cap_{r \in F} \ker N_r$, $\Pi_F^\wedge := (E_F^\wedge)_*$ and $\mathcal{K}_F^{\text{rec}} := \ker \Pi_F^\wedge$; all interval, biflag and variational clauses below have this scope.

$$\begin{aligned} \mathcal{V}_F &= (\mathcal{K}_F^{\text{vis}})^\perp, \quad (\mathcal{A}_F^\wedge)_{sa} = (\mathcal{K}_F^{\text{rec}})^\perp, \quad \mathcal{K}_F^{\text{vis}} \subseteq \mathcal{K}_F^{\text{rec}}, \\ r \in \text{cl}_v(F) &\iff \mathcal{K}_F^{\text{vis}} = \mathcal{K}_{F \cup \{r\}}^{\text{vis}} \iff [N_s \rho = N_s \eta \ (s \in F) \Rightarrow N_r \rho = N_r \eta] \iff K_F(\rho) = K_{F \cup \{r\}}(\rho) \ \forall \rho, \\ r \in \text{cl}_\wedge(F) &\iff \mathcal{K}_F^{\text{rec}} = \mathcal{K}_{F \cup \{r\}}^{\text{rec}} \iff \Pi_{F \cup \{r\}}^\wedge = \Pi_F^\wedge, \quad F \subseteq G \Rightarrow E_F^\wedge E_G^\wedge = E_G^\wedge E_F^\wedge = E_G^\wedge, \\ \text{Ran } \Pi_F^\wedge \cap \mathcal{S}_\omega^{\leq \infty} &= \cap_{r \in F} \text{Suff}(\alpha_r), \quad \text{Fix}_f(\Pi_F^\wedge) = \text{Host}_\omega(\mathcal{A}_F^\wedge), \quad \mathcal{A}_F^\wedge = \text{Gen}_\omega \text{Fix}_f(\Pi_F^\wedge). \end{aligned}$$

$r \in \text{cl}_v(F)$ implies $(\mathcal{D}_{F \cup \{r\}}^\omega, P_{F \cup \{r\}}^\vee) = (\mathcal{D}_F^\omega, P_F^\vee)$. On the residual-kernel domain $\emptyset \neq F \subseteq \mathcal{R}$, set $\text{BiRes}_\omega(F) = (\mathcal{K}_F^{\text{vis}} \subseteq \mathcal{K}_F^{\text{rec}})$. For $\emptyset \neq E \subseteq F$,

$$\begin{aligned} \mathcal{K}_F^{\text{vis}} &\subseteq \mathcal{K}_E^{\text{vis}} \subseteq \mathcal{K}_E^{\text{rec}} \subseteq \mathcal{K}_F^{\text{rec}}, \quad (\mathcal{A}_F^\wedge)_{sa} \subseteq (\mathcal{A}_E^\wedge)_{sa} \subseteq \mathcal{V}_E \subseteq \mathcal{V}_F, \\ (\text{PBE}) \quad \mathbf{P}_F^\omega &= [\mathcal{M}_{*,h} \xrightarrow{\tilde{N}_F} N_F(\mathcal{M}_{*,h}) \xrightarrow{\tilde{\Pi}_F^\wedge} \text{Ran } \Pi_F^\wedge], \quad \Pi_F^\wedge = \tilde{\Pi}_F^\wedge \tilde{N}_F, \\ 0 \rightarrow \mathcal{K}_F^{\text{rec}} / \mathcal{K}_F^{\text{vis}} &\rightarrow \mathcal{M}_{*,h} / \mathcal{K}_F^{\text{vis}} \rightarrow \mathcal{M}_{*,h} / \mathcal{K}_F^{\text{rec}} \rightarrow 0, \quad (\text{VRI}) \quad \mathfrak{R}_F := \mathcal{V}_F / (\mathcal{A}_F^\wedge)_{sa} \cong (\mathcal{K}_F^{\text{rec}} / \mathcal{K}_F^{\text{vis}})^*. \end{aligned}$$

U1/(TEXP) reconstructs $P_F^\vee, \Pi_F^\wedge$ from $\text{BiRes}_\omega(F)$, hence $M_{sa}/\mathcal{V}_F \cong (\mathcal{K}_F^{\text{vis}})^*$, $\mathcal{G}_F^\omega = W^*(\sigma_t^\omega(\mathcal{V}_F) : t \in \mathbb{R})$ and $\mathfrak{S}_F = [\mathcal{A}_F^\wedge, \mathcal{G}_F^\omega]_{\text{Lat}_\omega(\mathcal{M})}$. If

$\theta_g \alpha_r = \alpha_{gr} \kappa_{g,r}$ for an ω -preserving normal G -action, then (BRG) $(\theta_g)_\# \text{BiRes}_\omega(F) = \text{BiRes}_\omega(gF)$ and $\mathcal{R}_{E:F} \xrightarrow{\theta_g} \mathcal{R}_{gE:gF}$. For G -stable F this is (ASP); **NW** is the normal-lift subfiber and (FIT) keeps set-stability $\Rightarrow$ lift. In $(A, V) \preceq (B, W) \iff A \supseteq B, V \subseteq W$, put $\alpha_r = (\mathcal{A}_r^{\text{sat}}, \mathcal{V}_{\{r\}})$; the common adjunction gives (BCG) $\bigvee_{r \in F} \alpha_r = (\mathcal{A}_F^\wedge, \mathcal{V}_F) \preceq (A, V) \iff F \subseteq \{r : A \subseteq \mathcal{A}_r^{\text{sat}}, \mathcal{V}_{\{r\}} \subseteq V\}$, with fixed set $\text{cl}_v(F) \cap \text{cl}_\wedge(F)$. For finite F , finite-dimensional visible span gives the integer polymatroid $r_v(S) = \dim_{\mathbb{R}} \mathcal{V}_S - 1$ with $r \in \text{cl}_v(F) \iff r_v(F \cup \{r\}) = r_v(F)$; commuting saturated expectations give, for every $\rho \in \mathcal{S}_\omega^{\leq \infty}$, the polymatroid $r_{\wedge, \rho}(S) = D(\rho \parallel \omega) - D(\rho E_S^\wedge \parallel \omega)$. With $\mathcal{K} = \mathcal{A}_{F \cup \{r\}}^\wedge \subseteq \mathcal{N} = \mathcal{A}_F^\wedge$, (EAP/BES) gives $r_{\wedge, \rho}(F \cup \{r\}) - r_{\wedge, \rho}(F) = \mathbf{e}_{\mathcal{K}; \mathcal{N}}(\rho)$ and $r \in \text{cl}_\wedge(F) \iff \mathbf{b}_{\mathcal{K}; \mathcal{N}} = 0 \iff \text{Hess}_0 \mathbf{e}_{\mathcal{K}; \mathcal{N}}(\omega^{th}) = 0$ for every bounded h , equivalently iff the rank increment vanishes for every finite-action ρ . Thus the state-separating rank profile is the nonlinear read-out of the pre-existing BKM shell, not an additional closure coordinate. The visible integer rank and recoverability shell remain type-distinct instances of the ordered adjunction construction (SDR) above^{36,37}.

(B) Entropic realization. For finite-action ρ and nonempty finite/countable F , $P_F^\vee \Pi_F^\wedge = \Pi_F^\wedge P_F^\vee = \Pi_F^\wedge$. Put $\rho_F^\vee = P_F^\vee \rho$, $\rho_F^\wedge = \Pi_F^\wedge \rho$, $B_F = \mathbf{n}_{P_F^\vee}^\omega(\rho) = \mathfrak{R}_{D_\omega, N_F}(\rho_F^\vee; \rho - \rho_F^\vee) \geq 0$, and $H_F = D(\rho \parallel \rho_F^\vee)$, $M_F = D(\rho_F^\vee \parallel \rho_F^\wedge)$, $C_F = D(\rho_F^\wedge \parallel \omega)$. Then $D(\rho \parallel \omega) = H_F + B_F + M_F + C_F$, $V_{N_F}^{D_\omega} = H_F + B_F$, $J_F^\omega = M_F + C_F$, with zeros respectively $\rho = \rho_F^\vee$, fiber Pythagoras, $\rho_F^\vee = \rho_F^\wedge$, $\rho_F^\wedge = \omega$. On a faithful finite-dimensional constant-rank Araki chart, $A = dP_F^\vee|_\omega$, $C = d\Pi_F^\wedge|_\omega$ satisfy $AC = CA = C$ and $(\text{Hess } H_F, \text{Hess } B_F, \text{Hess } M_F, \text{Hess } C_F)_\omega(v, v) = (\|(1-A)v\|_g^2, 0, \|(A-C)v\|_g^2, \|Cv\|_g^2)$. Hence B_F is quadratically invisible, $\text{Ran } C = \text{Tan}_\omega(\mathcal{A}_F^\wedge)$ is JLCl_ω -fixed, $\text{Alg}_\omega(\text{Ran } A) = \mathcal{G}_F^\omega$, and $A - C$ is the BKM interval shell; Petz-flatness remains stronger than U1 flatness by (E).

(C) Exact channelization. For $p_r > 0$, $\sum_r p_r = 1$, set $N_{F,p} \rho = \oplus_r p_r N_r \rho$; finite action gives

$$\begin{aligned} [\text{VRC}] \quad R_{F,p} N_{F,p} |_{\mathcal{S}_{\text{fin}}^\omega} &= P_F^\vee |_{\mathcal{S}_{\text{fin}}^\omega} \text{ for a normal state channel } R_{F,p} \iff P_F^\vee |_{\mathcal{S}_{\text{fin}}^\omega} = \Pi_F^\wedge |_{\mathcal{S}_{\text{fin}}^\omega} \iff \mathcal{K}_F^{\text{vis}} = \mathcal{K}_F^{\text{rec}} \iff \mathfrak{R}_F = 0, \\ &\iff \mathcal{V}_F = (\mathcal{A}_F^\wedge)_{sa} \iff \mathcal{G}_F^\omega = \mathcal{A}_F^\wedge \iff |\mathfrak{S}_F| = 1. \end{aligned} \quad (118)$$

Equivalently $P_F^\vee(\mathcal{S}_\omega^{\leq \infty}) \subseteq \cap_r \text{Suff}(\alpha_r)$. For $r \in F$, $J_{\{r\}}^\omega \pi_F|_{\{r\}} \leq J_F^\omega$ and a local finite minimizer supplies a joint finite lift, so $\pi_r^{\text{fin}} := \pi_F|_{\{r\}}|_{Y_{F,\text{fin}}^\omega} : Y_{F,\text{fin}}^\omega \rightarrow Y_{\{r\},\text{fin}}^\omega$. The equivalent finite-face RFP/length criterion is the simultaneous singleton-leg vanishing in the normal-record refinement cocycle below. Its DepRes is distinct from the global full-product DepRes below, whose additional domain-saturation grade is separated by the duplicate-identity witness. Thus channelization is exactly collapse of the sole interval grade; off collapse, any exact normal totalizer lies in the positive reference-displacement sector below.

(D) Bifiltration. For $\emptyset \neq E \subseteq F$ put $\pi = \pi_{F \rightarrow E}$ and retain the global RFP carrier $\text{Cell}_\pi^{J_F^\omega | J_E^\omega} = [J_E^\omega \pi, J_F^\omega]$; on $Y_{F,\text{fin}}^\omega$ put $A_{F:E}(y) := J_F^\omega(y) - J_E^\omega(\pi y) \geq 0$. For finite-action ρ ,

$$\begin{aligned} [\text{BFP}] \quad M_F(\rho) - M_E(\rho) &= A_{F:E}(N_F \rho) + D(\Pi_E^\wedge \rho \parallel \Pi_F^\wedge \rho), \quad M_F(\rho) = D(P_F^\vee \rho \parallel \Pi_F^\wedge \rho), \\ (\text{VAG}) \quad \mathcal{R}_{E:F}/\mathbf{t} &\cong (\mathcal{A}_E^\wedge)_{sa} / (\mathcal{A}_F^\wedge)_{sa} \oplus \mathfrak{R}_E \oplus \mathcal{V}_F / \mathcal{V}_E, \quad \mathcal{R}_{E:F}/(\mathbf{t} - 1) \cong \mathfrak{R}_F. \end{aligned} \quad (119)$$

$\mathcal{R}_{E:F}$ is the Rees module of $0 \subset (\mathcal{A}_E^\wedge)_{sa} / (\mathcal{A}_F^\wedge)_{sa} \subset \mathcal{V}_E / (\mathcal{A}_F^\wedge)_{sa} \subset \mathfrak{R}_F$; no global split. Its grades are dual to $\mathcal{K}_F^{\text{rec}} / \mathcal{K}_E^{\text{rec}}$, $\mathcal{K}_E^{\text{rec}} / \mathcal{K}_E^{\text{vis}}$, $\mathcal{K}_E^{\text{vis}} / \mathcal{K}_F^{\text{vis}}$. On the Araki chart, $C_F \leq C_E \leq A_E \leq A_F$ and $A_F - C_F = (A_F - A_E) + (A_E - C_E) + (C_E - C_F)$ orthogonally, with $\text{Hess}_\omega(M_F - M_E)(v, v) = \|(A_F - A_E)v\|_g^2 + \|(C_E - C_F)v\|_g^2$. Moreover $\text{gr}_{E:F}^{\text{BKM}} := \text{Ran } C_E / \text{Ran } C_F \oplus \text{Ran } A_E / \text{Ran } C_E \oplus \text{Ran } A_F / \text{Ran } A_E \cong \perp \text{Ran } (A_F - C_F)$; hence $\mathbf{t} = 0$ is the BKM graded fiber, $\mathbf{t} = 1$ the unsplit extension, and (BFP) leaves the middle grade. For finite nonempty

$F_n \uparrow F$, $D(P_{F_n}^\vee \rho \| \omega) \uparrow D(P_F^\vee \rho \| \omega)$, $\|P_{F_n}^\vee \rho - P_F^\vee \rho\|_1 \rightarrow 0$, $\|\Pi_{F_n}^\wedge \rho - \Pi_F^\wedge \rho\|_1 \rightarrow 0$ and $E_{F_n}^{\wedge, (2)} \rightarrow E_F^{\wedge, (2)}$ strongly; the compact projective (PI/UI) clause supplies the degree-zero step.

(E) Sharpness. On tracial M_2 , with $D_\lambda(x) = \lambda x + (1 - \lambda)\tau(x)1$ ($0 < \lambda < 1$), $(\mathcal{V}_I, \mathcal{A}_I^{\text{sat}}) = (M_{2, \text{sa}}, M_2)$, $(\mathcal{V}_{D_\lambda}, \mathcal{A}_{D_\lambda}^{\text{sat}}) = (M_{2, \text{sa}}, \mathbb{C}1)$, $(\mathcal{V}_T, \mathcal{A}_T^{\text{sat}}) = (\mathbb{R}1, \mathbb{C}1)$, so either endpoint varies alone. For $F = \{I, T\}$, $\rho \neq \tau$, the $E = T, I$ (BFP) pairs are $(D(\rho \| \tau), 0)$ and $(0, D(\rho \| \tau))$. For finite-dimensional product systems and nonempty marginal families F, G , $P_{F \sqcup G}^\vee(\rho_1 \otimes \rho_2) = P_F^\vee \rho_1 \otimes P_G^\vee \rho_2$ but common marginal sufficiency is $\{\omega_1 \otimes \omega_2\}$, so $\Pi_{F \sqcup G}^\wedge$ collapses and additivity fails. In classical $A - C - B$, $D(p_{ACB} \| \sigma_A \sigma_C \sigma_B) = I_p(A:B | C) + \sum_c p_C(c) [D(p_{A|c} \| \sigma_A) + D(p_{B|c} \| \sigma_B)] + D(p_C \| \sigma_C)$ with $B_F = 0$ while non-reference Markov conditionals give $H_F = 0 < M_F$. Hence endpoints/increments/monoidality are independent; covariance is (BRG) and deterministic Blackwell equivalence.

Proof. Banach duality, quotient identities, Petz projection and Takesaki expectedness give visible/recovered fibres and the predual exact sequence. Collective section, finite-action entropy and BKM second variation give entropic splitting; nested expected Pythagoras and Araki-arc separation identify $\text{cl}_\wedge$ with equality of the full state-indexed rank profile; DPI equality gives channelization. Pythagoras/Rees gives the bifiltration, KMS–Petz its covariance without equating set-stability and action lift. Compact entropy sublevels/Pinsker give $P^\vee$ exhaustion; Tsukada³⁸ (Thm. 3) gives the $\Pi^\wedge$ norm limit and strong expectation convergence. $\square$

f. *Normal-record refinement cocycle.* For $\emptyset \neq E \subseteq F$ and $\pi = \pi_{F \rightarrow E}$, use the cell and finite-face length from (D). RFP gives $J_E^\omega = \pi_! J_F^\omega$ and $\text{DepRes}_\pi(J_F^\omega) = 0 \iff J_F^\omega = J_E^\omega \pi \iff [Y_{F, \text{fin}}^\omega = \pi^{-1}(Y_{E, \text{fin}}^\omega) \ \& \ s_F = s_E \pi \text{ on } Y_{F, \text{fin}}^\omega]$, where the inverse image is inside the F -record carrier. Indeed, on the finite locus the two values are the relative entropies of the displayed unique selectors, while domain equality matches their $+\infty$ loci. The domain clause is necessary already for two labeled identity accesses on a finite classical simplex: $Y_{F, \text{fin}}^\omega$ is the diagonal record locus and the selectors agree there, whereas $\pi^{-1}(Y_{E, \text{fin}}^\omega)$ also contains incompatible off-diagonal pairs. For $y \in Y_{F, \text{fin}}^\omega$, $A_{F:E}(y) = D(s_F(y) \| s_E(\pi y)) + n_{P_E^\vee}^\omega(s_F(y)) \geq 0$, $A_{F:E}(y) = 0 \iff s_F(y) = s_E(\pi y)$. For $E \subseteq F \subseteq G$ and $y \in Y_{G, \text{fin}}^\omega$, RFP cell pasting reads $A_{G:E}(y) = A_{G:F}(y) + A_{F:E}(\pi_{G \rightarrow F} y)$. For $r \in F$, put $\pi_r = \pi_{F \rightarrow \{r\}}$, $D_r = D(\cdot \| \omega_r)$, $c_r = H_{\pi_r}^\omega | D_r$ and $o_{F,r}(y) := A_{F:\{r\}}(y) + c_r(\pi_r y)$ on $Y_{F, \text{fin}}^\omega$. There $o_{F,r}(y) = 0 \iff A_{F:\{r\}}(y) = 0 = c_r(\pi_r y) \iff s_F(y) \in \text{Suff}(\alpha_r)$; at $y = N_F \rho$ with $\rho \in S_{\text{fin}}^\omega$ this is $P_F^\vee \rho = \Pi_F^\wedge \rho$. Hence (VRC) is equivalently $o_{F,r} \equiv 0$ for every $r \in F$. Thus $O_F = (\text{Comp}_F, J_F, (o_{F,r} : Y_{F, \text{fin}}^\omega \rightarrow [0, \infty))_r$; the $A > 0 = c$ and $A = 0 < c$ examples separate the finite-face grades, with (E2S) their tangent shell.

g. *Bounded-layer quantitative map.* For finite F , $c < \infty$, $\lambda_r > 0$, set $\psi_{F,c}^\omega(h) = \sup_{J_F^\omega(z) \leq c} \langle h, z \rangle$, $q_\lambda(h) = 2 \sum_r \|h_r\|^2 / \lambda_r$, $\tau_{F,c}^{\omega, \lambda}(y) = \min_{J_F^\omega(z) \leq c} \sum_r \lambda_r D(y_r \| z_r)$. U1/Hahn–Banach/Cauchy–Schwarz/Pinsker give $\tau_{F,c}^{\omega, \lambda}(y) \geq \sup_{h \neq 0} [\langle h, y \rangle - \psi_{F,c}^\omega(h)]_+^2 / q_\lambda(h)$ and $\tau_{F,c}^{\omega, \lambda}(y) = 0 \iff J_F^\omega(y) \leq c$, hence $J_F^\omega(y) = \inf\{c : \tau_{F,c}^{\omega, \lambda}(y) = 0\}$. The floor $m_F^\lambda(y) = \inf_\eta A_y(\eta)$, $A_y = \sum_r \lambda_r D(y_r \| N_r \eta)$, splits incompatible tuples into $m_F^\lambda > 0$ and zero-floor nonattainment. In the latter (AMF) has nonempty tolerance fibres but section fibre $\{A_y = 0\} = K_F(y) = \emptyset$; Pinsker lands their images in the (PID_n) norm closure but not the normal image. Both statements are independent of positive reweighting.

h. *Escaping-entropy witness and modular-generation consequence.* If $K_F(y) = \emptyset$, $m_F^\lambda(y) = 0$ forces minimizing nets out of every finite entropy layer, so a uniform entropy bound excludes the zero-floor unattained branch; in finite dimension $D(\eta \| \omega) \leq \log \|\omega^{-1}\|$. For sharpness take $\mu(A_n) = 2^{-n}$, $(\alpha f)|_{A_n} = (1 - 1/n)f(0) + n^{-1}f(n)$, whence $(N\eta)_0 = \sum_n p_n(1 - 1/n)$, $(N\eta)_n = p_n/n$. With $y = \delta_0$, $K_{\{N\}}(y) = \emptyset$, $\eta_n = \mu(\cdot | A_n)$ and $\inf_\eta D(y \| N\eta) = 0$, $D(y \| N\eta_n) = -\log(1 - 1/n) \rightarrow 0$, $D(\eta_n \| \mu) = n \log 2 \rightarrow \infty$; indeed $N\eta = y$ forces all $p_n = 0$. Take $X = \mathbb{N}_{\geq 1}$ atomically; on a non-atomic standard probability space an admitted Krieger realization changes temporal type without changing N .

For $\beta > 0$ the unique soft selector is $p_n^{(\beta)} = Z_\beta^{-1} 2^{-n} e^{-c_\beta/n}$, $Z_\beta = \sum_m 2^{-m} e^{-c_\beta/m}$, $c_\beta = \beta/q_0^{(\beta)}$, $q_0^{(\beta)} = \sum_n p_n^{(\beta)}(1 - 1/n)$. If $0 < |t| < 2\pi/c_\beta$, then $[Dp^{(\beta)} : D\omega]_t(n) = Z_\beta^{-it} e^{-itc_\beta/n}$ has pairwise distinct values, since no phase difference reaches $2\pi\mathbb{Z}$; its Borel spectral calculus therefore generates $\ell^\infty(\mathbb{N}_{\geq 1})$. Thus full modular generation coexists with an unattained hard target.

E. Exact normal totalization and terminal recovery

Let $\alpha : (\mathcal{A}'', \omega'') \rightarrow (\mathcal{A}', \omega')$ be normal UCP between σ -finite von Neumann algebras with faithful normal references and $\omega'' = \alpha^* \omega'$. For nonempty $\mathcal{E} \subset S_{\omega'}^{\leq \infty}(\mathcal{A}')$, let $\mathfrak{T}_{\alpha, \mathcal{E}}$ be the normal channels $\Lambda : (\mathcal{A}'')_* \rightarrow (\mathcal{A}')_*$ satisfying $\Lambda \alpha^* \rho = \rho$ for every $\rho \in \mathcal{E}$, and put $\varepsilon_{\alpha, \mathcal{E}}^{\omega'} := \inf_{\Lambda \in \mathfrak{T}_{\alpha, \mathcal{E}}} P_\Lambda^{\omega'' \rightarrow \omega'}$, $\inf \emptyset := +\infty$. Set $C_{\alpha, \mathcal{E}} := \{\Lambda \omega'' : \Lambda \in \mathfrak{T}_{\alpha, \mathcal{E}}\}$. It is convex and weak-predual closed (possibly empty), since normal-cluster map $\vee D 0$ preserves UCP and exact recovery. When nonempty it is the terminal-map U1 chart of the U0 weighted quotient, $\varepsilon_{\alpha, \mathcal{E}}^{\omega'} = \mathcal{J}_{!, C_{\alpha, \mathcal{E}}}^{\omega'}(*)$.

a. *Terminal exact-recovery specialization.*

For nonempty $\mathcal{E}$ put $\text{Upadana}(\alpha, \mathcal{E}) = \mathfrak{T}_{\alpha, \mathcal{E}}$, $\text{Rec}_\alpha(\Lambda) = \{\rho : \Lambda \alpha^* \rho = \rho\}$ and $\text{Fix}_\alpha(\mathcal{L}) = \bigcap_{\Lambda \in \mathcal{L}} \text{Rec}_\alpha(\Lambda)$, with empty intersection the full finite-entropy set. The (EPN) face gives $\mathcal{L} \subseteq \mathfrak{T}_{\alpha, \mathcal{E}} \iff \mathcal{E} \subseteq \text{Fix}_\alpha(\mathcal{L})$. Thus $\text{RecCl}_\alpha \mathcal{E} := \text{Fix}_\alpha(\mathfrak{T}_{\alpha, \mathcal{E}})$ is a closure with $\mathfrak{T}_{\alpha, \text{RecCl}_\alpha \mathcal{E}} = \mathfrak{T}_{\alpha, \mathcal{E}}$ and $\mathfrak{T}_{\alpha, \mathcal{E}} = \mathfrak{T}_{\alpha, \mathcal{F}} \iff \text{RecCl}_\alpha \mathcal{E} = \text{RecCl}_\alpha \mathcal{F}$; it preserves $C_{\alpha, \mathcal{E}}$ and $\varepsilon_{\alpha, \mathcal{E}}^{\omega'}$; target inclusion reverses totalizer inclusion and increases the tariff, and by (PFX) is the greatest target family in its equal-totalizer fibre. It is strictly stronger than convexification: (AKN) with $0 < p < r < 1$ has the unique decoder $0 \mapsto \delta_\alpha$, $1 \mapsto \delta_c$ and $\text{RecCl}_\alpha\{\rho_p, \rho_r\} = \{(t, 0, 1 - t) : 0 \leq t \leq 1\} \supseteq \text{conv}\{\rho_p, \rho_r\}$.

On the nonempty totalizer locus,

$$\varepsilon_{\alpha, \mathcal{E}}^{\omega'} = \inf_{\Lambda \in \text{Upadana}(\alpha, \mathcal{E})} \text{Karma}_{\omega'' \rightarrow \omega'}^{\text{forc}}(\Lambda), \quad \varepsilon_{\alpha, \mathcal{E}}^{\omega'} = 0 \iff \mathcal{E} \subseteq \text{Suff}(\alpha), \quad \varepsilon_{\alpha, \mathcal{E}}^{\omega'} \geq \frac{1}{2}(\delta_{\alpha, \mathcal{E}}^{\omega'})^2,$$

where $\delta_{\alpha, \mathcal{E}}^{\omega'} := \inf_{\Lambda \in \mathfrak{T}_{\alpha, \mathcal{E}}} \|\Lambda \omega'' - \omega'\|_1$. Closedness of $C_{\alpha, \mathcal{E}}$ makes $\delta = 0$ attained; Petz identifies it with sufficiency and Pinsker gives the bound, so $\delta_{\alpha, \mathcal{E}} > 0$ off sufficiency. When the U1 tariff is finite, compact entropy sublevels and strict convexity (JEO_{cmp}) give a unique optimal reference output (the channel need not be unique); the forward access is forcing-neutral, while every off-sufficiency exact reverse

has forcing-Karma at least ε . For a singleton ρ , $0 \leq \varepsilon_{\alpha,\varepsilon}^{\omega'} \leq D(\rho\|\omega')$, with strict positivity exactly off sufficiency. In finite dimension, UO support duality and Gibbs variationality reduce the tariff to the compact-convex output set $C_{\alpha,\varepsilon}$; Sion's one-compact-side minimax theorem³⁹ (Cor. 3.3) then gives

$$\varepsilon_{\alpha,\varepsilon}^{\omega'} = \sup_{h=h^*} \left\{ \min_{\Lambda \in \mathfrak{T}_{\alpha,\varepsilon}} (\Lambda\omega'')(h) - \log \text{Tr } e^{\log d_{\omega'} + h} \right\}.$$

b. Faithful-image support shadow and noncalibration. Let $\mathcal{E}_f = \{\rho \in \mathcal{E} : \alpha^* \rho \text{ faithful}\}$, $p_{\alpha,\varepsilon} = \bigwedge_{\rho \in \mathcal{E}_f} s(\rho)$, with $\bigwedge \emptyset = \mathbf{1}$, and put $\text{SuppProf}_\alpha(\mathcal{E}) := \{s(\rho) : \rho \in \mathcal{E}_f\}$. For any normal state channel Λ and faithful normal states ξ, ζ on its input algebra, $s(\Lambda\xi) = s(\Lambda\zeta)$: for every projection e , $(\Lambda\xi)(e) = 0 \iff \xi(\Lambda^*e) = 0 \iff \Lambda^*e = 0 \iff \zeta(\Lambda^*e) = 0 \iff (\Lambda\zeta)(e) = 0$. Denote this faithful-input output support by q_Λ . Consequently

$$\mathfrak{T}_{\alpha,\varepsilon} \neq \emptyset, \mathcal{E}_f \neq \emptyset \implies \text{SuppProf}_\alpha(\mathcal{E}) = \{p_{\alpha,\varepsilon}\}, \quad q_\Lambda = s(\Lambda\omega'') = p_{\alpha,\varepsilon} \ (\Lambda \in \mathfrak{T}_{\alpha,\varepsilon}), \quad p_{\alpha,\text{RecCl}_\alpha \mathcal{E}} = p_{\alpha,\varepsilon}.$$

Indeed $\alpha^* \rho$ and ω'' are faithful and $\Lambda\alpha^* \rho = \rho$; applying the support identity first fixes every $s(\rho)$ and then every reachable reference-output support. Since $\text{RecCl}_\alpha \mathcal{E}$ has the same totalizer fibre and contains $\mathcal{E}$, the last equality follows by the same argument. Thus $|\text{SuppProf}_\alpha(\mathcal{E})| > 1$ already obstructs common exact totalization. Faithfulness is load-bearing: the identity channel on a two-point algebra sends the nonfaithful states δ_0, δ_1 to outputs with different supports. In all cases

$$C_{\alpha,\varepsilon} \subseteq \{\sigma : s(\sigma) \leq p_{\alpha,\varepsilon}\}, \quad \varepsilon_{\alpha,\varepsilon}^{\omega'} \geq -\log \omega'(p_{\alpha,\varepsilon}),$$

when $\mathfrak{T}_{\alpha,\varepsilon} \neq \emptyset$ and $\mathcal{E}_f \neq \emptyset$ by (TSR), while an empty totalizer fibre makes the inclusion vacuous and $\mathcal{E}_f = \emptyset$ gives $p_{\alpha,\varepsilon} = \mathbf{1}$; two-outcome DPI then gives the bound. Hence $p = 0$ obstructs common exact totalization and $p < 1$ forces positive tariff when the fibre is nonempty.

c. Support-face entropy envelope. For a projection $p \in \mathcal{A}'$ put $\mathcal{F}_p := \{\sigma \in \mathcal{S}_n(\mathcal{A}') : s(\sigma) \leq p\} = \{\sigma : \sigma(p) = 1\}$ and, with $\inf \emptyset = +\infty$,

$$\text{SFE}_{\omega'}(p) := \inf_{\sigma \in \mathcal{F}_p} D(\sigma\|\omega').$$

This is the $Q = !$, $C = \mathcal{F}_p$ specialization of the entropic UO quotient above, not a new entropy primitive. The face is convex and weak-predual closed, so finite $\text{SFE}_{\omega'}(p)$ is attained at a unique selector by (JEO_{cmp}). Moreover

$$p \leq q \implies \text{SFE}_{\omega'}(p) \geq \text{SFE}_{\omega'}(q), \quad \text{SFE}_{\omega'}(\mathbf{1}) = 0, \quad \text{SFE}_{\omega'}(0) = +\infty, \quad \text{SFE}_{\omega'}(p) \geq -\log \omega'(p).$$

The last inequality is DPI for the two-outcome channel $\sigma \mapsto (\sigma(p), \sigma(1-p))$. If some finite-entropy τ has $s(\tau) = p$, then convexly mixing τ into any finite point of $\mathcal{F}_p$ shows $\text{SFE}_{\omega'}(p) = \inf_{s(\sigma)=p} D(\sigma\|\omega')$; hence on the finite-tariff (TSR) branch this is the largest lower bound obtainable from the reference and the forced output support alone. In particular, when $\mathcal{E}_f \neq \emptyset$ and $\varepsilon_{\alpha,\varepsilon}^{\omega'} < \infty$, with $p = p_{\alpha,\varepsilon} \neq 0$,

$$\varepsilon_{\alpha,\varepsilon}^{\omega'} = -\log \omega'(p) + \text{FaceEx}_{\omega'}(p) + \text{TotEx}_{\alpha,\varepsilon}^{\omega'}, \quad \text{FaceEx}_{\omega'}(p) := \text{SFE}_{\omega'}(p) + \log \omega'(p) \geq 0, \quad \text{TotEx}_{\alpha,\varepsilon}^{\omega'} := \varepsilon_{\alpha,\varepsilon}^{\omega'} - \text{SFE}_{\omega'}(p) \geq 0.$$

Thus UIS is the mass shadow, SFE is the exact support-face envelope, and TotEx is the remaining target/recovery grade; no subtraction is used when the tariff is infinite.

On a finite-dimensional tracial realization $(\mathcal{D}, \tau)$ with faithful density $w > 0$, $\tau(w) = 1$, Gibbs variationality on the corner $p\mathcal{D}p$ gives the closed form

$$Z_{w,p} := \tau(\exp_{p\mathcal{D}p}(p(\log w)p)), \quad \text{SFE}_w(p) = -\log Z_{w,p}, \quad d_{w,p}^* = Z_{w,p}^{-1} \exp_{p\mathcal{D}p}(p(\log w)p).$$

If $[p, w] = 0$, then $Z_{w,p} = \tau(pw)$ and $\text{FaceEx}_w(p) = 0$; the classical hard-event identity above is exactly this commuting specialization. The mass shadow is nevertheless not complete. With ordinary trace on M_3 , take $w = \text{diag}(1/2, 1/3, 1/6)$, $p_0 = |e_2\rangle\langle e_2|$ and $p_1 = |v\rangle\langle v|$, $v = (e_1 + e_3)/\sqrt{2}$. Then $\tau(wp_0) = \tau(wp_1) = 1/3$ while $\text{SFE}_w(p_0) = \log 3$ and $\text{SFE}_w(p_1) = \log(2\sqrt{3}) > \log 3$, so equal support mass can carry different hard-face entropy.

The upper grade is independently strict. In (AKN) at $q = 1/2$, let $\omega = (1/4, 1/4, 1/2)$ and $p = 1_{\{a,c\}}$. For $\rho_A = (2/5, 0, 3/5)$ the decoder $0 \mapsto \delta_c$, $1 \mapsto (2/3, 0, 1/3)$ has reference output $(1/3, 0, 2/3) = d_{\omega,p}^*$, so $\varepsilon_{\alpha,\{\rho_A\}}^{\omega} = \text{SFE}_\omega(p) = \log(4/3)$ and $\text{TotEx} = 0$. For $\rho_B = (1/2, 0, 1/2)$ one has $\alpha^* \rho_B = \omega'' = (1/2, 1/2)$, so every exact totalizer satisfies $\Lambda\omega'' = \rho_B$ and $\varepsilon_{\alpha,\{\rho_B\}}^{\omega} = D(\rho_B\|\omega) = \frac{1}{2} \log 2 > \log(4/3)$, hence $\text{TotEx} = \log 3 - \frac{3}{2} \log 2 > 0$ at the same forced support. The following Choi fibre still varies at fixed optimal output, so neither support-face rigidity nor the two entropy grades collapse implementation multiplicity.

d. Support-face interpolation and Pythagorean zero locus. Assume $\mathcal{E}_f \neq \emptyset$ and $\varepsilon_{\alpha,\varepsilon}^{\omega'} < \infty$, put $p = p_{\alpha,\varepsilon}$, let $\sigma_{\omega',p}^{\text{sf}}$ be the unique $\text{SFE}_{\omega'}(p)$ selector, and let $\sigma_{\alpha,\varepsilon}^*$ be the unique optimal reference output in $C_{\alpha,\varepsilon}$. Since (TSR) gives $C_{\alpha,\varepsilon} \subseteq \mathcal{F}_p$, define the support-face interpolation fibre

$$\mathfrak{Z}_{\alpha,\varepsilon}^{\text{sf}} := \{\Lambda \in \mathfrak{T}_{\alpha,\varepsilon} : \Lambda\omega'' = \sigma_{\omega',p}^{\text{sf}}\}.$$

Strict convexity of $D(\cdot\|\omega')$ on the finite face gives the exact zero locus

$$\text{TotEx}_{\alpha,\varepsilon}^{\omega'} = 0 \iff \sigma_{\alpha,\varepsilon}^* = \sigma_{\omega',p}^{\text{sf}} \iff \sigma_{\omega',p}^{\text{sf}} \in C_{\alpha,\varepsilon} \iff \mathfrak{Z}_{\alpha,\varepsilon}^{\text{sf}} \neq \emptyset.$$

Thus the upper grade is exactly a compatibility obstruction between the canonical support-face entropy selector and the affine exact-recovery fibre; it is not another support invariant.

On the finite-dimensional tracial branch the obstruction has an exact relative-entropy realization. For every density d with $s(d) \leq p$, the corner Gibbs formula above gives

$$D(d\|w) = \text{SFE}_w(p) + D(d\|d_{w,p}^*).$$

Indeed $d = p\mathcal{D}p$ implies $\tau(d \log w) = \tau(d \log(w)p)$ and $\log d_{w,p}^* = p(\log w)p - (\log Z_{w,p})p$ on $p\mathcal{D}p$. Hence for the optimal totalizer output

$$\text{TotEx}_{\alpha,\varepsilon}^w = D(\sigma_{\alpha,\varepsilon}^* \| d_{w,p}^*) \geq \frac{1}{2} \|\sigma_{\alpha,\varepsilon}^* - d_{w,p}^*\|_1^2,$$

and (SFZ) is the exact equality locus. In finite matrices, $\mathfrak{Z}_{\alpha,\varepsilon}^{\text{sf}} \neq \emptyset$ is precisely feasibility of the already typed Choi face obtained below by setting $\sigma_* = d_{w,p}^*$, so the zero grade still leaves the support-corrected recovery-channel fibre free.

Record and support do not determine this compatibility. Let $X = \{a, b, c, d, e, f\} \rightarrow Y = \{0, 1\}$ send $a, b, c \mapsto 0$ and $d, e, f \mapsto 1$, take the uniform reference $w = (1/6, 1/6, 1/6, 1/6, 1/6, 1/6)$ and $p = 1_{\{a,b,d,e\}}$. Then $w'' = (1/2, 1/2)$ and

$$\rho_0 = (1/4, 1/4, 0, 1/4, 1/4, 0) = d_{w,p}^*, \quad \rho_1 = (3/8, 1/8, 0, 1/8, 3/8, 0)$$

have the same faithful record w'' and the same exact support p . For either singleton target, the replacement channel proves totalizability, while every exact totalizer must send the reference record w'' to that target. Therefore

$$\text{TotEx}_{\alpha, \{\rho_0\}}^w = 0, \quad \text{TotEx}_{\alpha, \{\rho_1\}}^w = D(\rho_1 \| \rho_0) = \frac{3}{4} \log 3 - \log 2 > 0.$$

Thus fixed record plus fixed support can still cross the (SFZ) boundary; the canonical positive compatibility object at this interface is the interpolation fibre $\mathfrak{S}^{\text{sf}}$ (or its finite Choi realization), and no scalar function of record and support alone determines its nonemptiness.

The meet $p_{\alpha, \varepsilon}$ alone does not retain this obstruction. On the finite classical access $X = \{a, b, c, d\} \xrightarrow{f} Y = \{0, 1\}$ with $a, b \mapsto 0, c, d \mapsto 1$ and uniform faithful reference, write tuples in the order (a, b, c, d) and set

$$\mathcal{E}_0 = \{(2/5, 0, 3/5, 0), (3/5, 0, 2/5, 0)\}, \\ \mathcal{E}_1 = \{(1/5, 1/5, 3/5, 0), (3/5, 0, 1/5, 1/5)\}.$$

Then $\alpha^* \mathcal{E}_0 = \alpha^* \mathcal{E}_1 = \{(2/5, 3/5), (3/5, 2/5)\}$ and $p_{\alpha, \mathcal{E}_0} = p_{\alpha, \mathcal{E}_1} = 1_{\{a,c\}}$. The decoder $0 \mapsto \delta_a, 1 \mapsto \delta_c$ recovers $\mathcal{E}_0$, whereas the two states in $\mathcal{E}_1$ have faithful records but distinct supports $\{a, b, c\}$ and $\{a, c, d\}$, so (TSR) gives $\mathfrak{T}_{\alpha, \mathcal{E}_1} = \emptyset$. Thus equal record data and equal meet support can have different exact-totalizer existence; the faithful-image support profile is a genuine positive residual coordinate, not another scalar tariff. This support coordinate is not calibrated by Δ_α . On (AKN), $X = \{a, b, c\} \xrightarrow{f} \{0, 1\}$ with $f(a) = f(b) = 0, f(c) = 1$, $\omega_q = (q/2, q/2, 1 - q)$ and $\rho_p = (p, 0, 1 - p)$ gives

$$\Delta_\alpha(\rho_p) = p \log 2, \quad -\log(1 - q/2) \leq \varepsilon_{\alpha, \rho_p}^{\omega_q} \leq q \log 2.$$

The decoder $0 \mapsto \delta_a, 1 \mapsto \delta_c$ gives the upper bound, so either coordinate can tend to zero with the other fixed positive. Fiberwise, for $0 < q < 1/2$, $\varepsilon_{\alpha, \rho_q}^{\omega_q} = q \log 2$; choosing $u \in (1/2, 1)$ with $D((u, 1 - u) \| (1/2, 1/2)) = \frac{1}{2} \log 2$ and $\tilde{\rho} = (2qu, 2q(1 - u), 1 - 2q)$ yields $\Delta_\alpha(\tilde{\rho}) = \Delta_\alpha(\rho_q) = q \log 2$ but $0 < \varepsilon_{\alpha, \tilde{\rho}}^{\omega_q} \leq \frac{q}{2} \log 2$. Hence $\varepsilon = 0 \iff \Delta_\alpha = 0$ does not extend to an Avidya–Karma scalar exchange law: support geometry is an independent exact-totalization coordinate.

In finite matrices $M_m \leftarrow M_n$, U1 fixes σ_* but generally not the channel. Put $S = \text{span}_{\mathbb{R}}(\{\omega''\} \cup \alpha^* \mathcal{E})$ and $\Gamma_\xi(J) = \text{Tr}_n[J(\mathbf{1}_m \otimes \xi^T)]$. Then $\mathfrak{R}_* := \{J \geq 0 : \text{Tr}_m J = \mathbf{1}_n, \Gamma_{\omega''} J = \sigma_*, \Gamma_{\alpha^* \rho} J = \rho (\rho \in \mathcal{E})\}$ and $\mathcal{V}_* := \{H = H^* : \text{Tr}_m H = 0, \Gamma_\xi H = 0 (\xi \in S)\}$. For $J^\circ \in \text{relint } \mathfrak{R}_*$, $p = s(J^\circ)$, the relative-interior affine-face clause and the displayed positivity, partial-trace and Γ constraints give $\text{aff } \mathfrak{R}_* = J^\circ + (\mathcal{V}_* \cap p(M_m \otimes M_n)_{\text{sa}} p)$, $\dim \mathcal{V}_* = (m^2 - 1)(n^2 - \dim S)$. Under trace identification this is the finite-dimensional support-face branch of (AFP): $K_{\text{loc}} = K_{\text{face}} = \mathcal{V}_* \cap p(M_m \otimes M_n)_{\text{sa}} p$, while the unconstrained raw kernel $\mathcal{V}_*$ may be larger. Uniqueness is therefore the face-corrected-kernel zero locus and, for $p = \mathbf{1}$, the displayed dimension formula gives exactly

$$|\mathfrak{R}_*| = 1 \iff [m = 1 \text{ or } S = (M_n)_{\text{sa}}].$$

For nontrivial output $m \geq 2$ this recovers the full-span criterion. The scalar-output boundary is forced by trace preservation itself: when $m = 1$, $\text{Tr}_m J = \mathbf{1}_n$ gives the unique $J = \mathbf{1}_n$ even if $S \subsetneq (M_n)_{\text{sa}}$. Sharpness already occurs at $m = n = 2$: for $\tau = 1/2$, completely depolarizing α^* , $\mathcal{E} = \{\text{diag}(3/4, 1/4)\}$ and $Z = \text{diag}(1, -1)$, preparing $\rho \pm \varepsilon Z$ after the computational-basis measurement ($|\varepsilon| < 1/4$) gives distinct $J > 0$ optima in the nine-dimensional affine fiber; for pure $\rho = |0\rangle\langle 0|$ the same raw kernel meets the support face in the unique replacement channel. Thus U1 fixes output/tariff while U0/AFP retains the support-corrected channel fiber.

VI. RESIDUAL GEOMETRY AND TRANSPORT

A. Reference-conditioned entropic geometry

Relative entropy has three type-distinct quadratic regimes here: absolute reference conditioning controls finite Type I, relative Loewner control alone has no reverse constant, and the commutative unbounded chart is $L^2/L \log L$ -Orlicz. In the commutative branch put $h = d\rho/d\omega$, $f = h - 1 = f^+ - f^-$, $0 \leq f^- \leq 1$, $\Phi_1(u) = (1 + u) \log(1 + u) - u$ for $u \geq -1$, with $0 \log 0 = 0$, and

$$\|\rho\|_{\Phi_1} := \inf\{\lambda > 0 : \int \Phi_1(g/\lambda) d\omega \leq 1\}, \quad N_+ = \|f^+\|_{\Phi_1}, \\ D_+ = \int_{\{h>1\}} \Phi_1(f^+) d\omega, \quad D_- = \int_{\{h<1\}} \Phi_1(-f^-) d\omega. \quad (120)$$

These are exact type-distinct metric regimes of the U1 potential, not a second entropy coordinate.

(I) *Finite Type-I absolute-reference chart.* For faithful density ω let $\chi_\omega^2(\rho) = \text{Tr}((\rho - \omega)\omega^{-1}(\rho - \omega))$. Then $\frac{1}{2}\|\rho - \omega\|_1^2 \leq D(\rho\|\omega) \leq \log(1 + \chi_\omega^2) \leq \chi_\omega^2$, $\omega \geq \varepsilon \mathbf{I} \Rightarrow \chi_\omega^2 \leq (2/\varepsilon)D(\rho\|\omega)$. The second box also factors through $\chi_\omega^2 \leq \varepsilon^{-1}\|\rho - \omega\|_2^2 \leq \varepsilon^{-1}\|\rho - \omega\|_1^2$.

(II) *Relative-sandwich non-determination.* For $0 < b < 1$,

$$\omega_\varepsilon = \begin{pmatrix} \varepsilon & 0 \\ 0 & 1 - \varepsilon \end{pmatrix}, \quad X = \begin{pmatrix} 1 & b \\ b & 1 \end{pmatrix}, \quad \rho_\varepsilon = \omega_\varepsilon^{1/2} X \omega_\varepsilon^{1/2}, \quad (121)$$

satisfy $(1 - b)\omega_\varepsilon \leq \rho_\varepsilon \leq (1 + b)\omega_\varepsilon$, $\chi_{\omega_\varepsilon}^2(\rho_\varepsilon) = b^2$, but $\|\rho_\varepsilon - \omega_\varepsilon\|_1 = 2b\sqrt{\varepsilon(1 - \varepsilon)} \rightarrow 0$, $D(\rho_\varepsilon\|\omega_\varepsilon) \rightarrow 0$. Hence no reverse χ^2 /entropy or uniform χ^2 /trace-distance constant depends only on the relative sandwich; absolute reference conditioning is an independent coordinate.

(III) *Commutative unbounded Orlicz chart.* $D(\rho\|\omega) = D_+ + D_-$, $\frac{1}{2}\|f^-\|_2^2 \leq D_- \leq \|f^-\|_2^2$, $N_+ \leq 1 : N_+^2 \leq D_+ \leq N_+$, $N_+ \geq 1 : N_+ \leq D_+ \leq N_+ + \log N_+$, with $D_+ = 1$ at $N_+ = 1$. Sharpness: on $[0, 1]$, take $h_n = e^{n^2}$ on A_n with $\omega(A_n) = (ne^{n^2})^{-1}$ and constant c_n off

A_n , normalized. Then $D(\rho_n\|\omega) = n + o(1)$, $D(\omega\|\rho_n) \rightarrow 0$, $\|\rho_n - \omega\|_1 \rightarrow 0$, $\frac{\chi^2(\rho_n\|\omega)}{D(\rho_n\|\omega)} \sim \frac{e^{n^2}}{n^2} \rightarrow \infty$. If $h \leq M$, the bounded-density subchart is $\kappa(M)\chi^2(\rho\|\omega) \leq D(\rho\|\omega) \leq \chi^2(\rho\|\omega)$, $\kappa(M) = \Phi_1(M-1)/(M-1)^2$. Thus no branch is a universal σ -finite Petz metric. The concentration profile constructed above is the constrained second-order refinement: it vanishes on finite-dimensional tangent-feasible directions but can be positive at fixed BKM germ in the unbounded σ -finite chart.

a. Verification. For (I), first suppose ρ is faithful. On Hilbert–Schmidt space let $\Delta_{\rho|\omega} = L_{\rho}R_{\omega^{-1}}$. Since $\|\rho^{1/2}\|_2 = 1$ and $\log \Delta = L_{\log \rho} - R_{\log \omega}$, operator Jensen gives $D(\rho\|\omega) \leq \log \langle \rho^{1/2}, \Delta_{\rho|\omega} \rho^{1/2} \rangle = \log \text{Tr}(\rho \omega^{-1}) = \log(1 + \chi^2_{\omega})$, the $f(t) = t^2$ Petz quasi-entropy face⁴⁰ (§1). For arbitrary ρ , apply this bound to $\rho_{\eta} = (1-\eta)\rho + \eta\omega$ and let $\eta \downarrow 0$; finite-dimensional continuity of $D(\cdot\|\omega)$ for faithful ω and of χ^2_{ω} gives the same inequality. Pinsker, $\log(1+x) \leq x$ and $\omega \geq \varepsilon I$ give the other bounds. In (II) direct calculation gives the sandwich, $\chi^2 = b^2$ and the displayed trace norm; equal diagonals give $D(\rho_{\varepsilon}\|\omega_{\varepsilon}) = S(\omega_{\varepsilon}) - S(\rho_{\varepsilon}) \rightarrow 0$. For (III), $D = \int \Phi_1(f)$, $\frac{1}{2}u^2 \leq \Phi_1(u) \leq u^2$ on $[-1, 0]$, and strict monotonicity gives $\int \Phi_1(f^+/N_+) = 1$ when $f^+ \neq 0$; the inequalities $\lambda^2 \Phi_1(u) \leq \Phi_1(\lambda u) \leq \lambda \Phi_1(u)$ for $0 < \lambda \leq 1$ and $\lambda \Phi_1(u) \leq \Phi_1(\lambda u) \leq \lambda \Phi_1(u) + \lambda u \log \lambda$ for $\lambda \geq 1$ give the two regimes; substitution gives the sharp sequence and monotonicity of $\Phi_1(u)/u^2$ the bounded-density chart.

B. Functoriality under regime changes

For each regime set $\mathfrak{A}_r = (\mathcal{F}(r), C_r, \tau_r, \delta_r, \nu_r, \text{Rec}_r, \text{Imp}_r^{\text{ev}}, T_r)$. Restrictions $\rho_{rs} : \mathcal{F}(s) \rightarrow \mathcal{F}(r)$ form $\mathbf{Reg}_{\text{adm}}$ with $\rho_{rt} = \rho_{rs}\rho_{st}$. On realized states, $(r, x) \preceq (s, y)$ iff $x = \rho_{rs}(y)$ and $\delta_s(y) \leq \delta_r(x)$; quotienting mutual reachability gives a partial order. A strict comparison $M_{rs} : \mathcal{F}(r) \rightarrow \mathcal{F}(s)$ requires $i(s) \leq i(r)$ unless a comparison span is declared and satisfies $\tau_s M_{rs} = \Pi_{i(s)i(r)} \tau_r$; its ε -lax version replaces equality by $d_{i(s)}(\tau_s M_{rs} x, \Pi_{i(s)i(r)} \tau_r x) \leq \varepsilon$.

a. Positive affine distortion map. All quantitative laxity uses $\text{Aff}_+ = \mathbb{R}_{\geq 0}^2$, $(L_1, \varepsilon_1) \star (L_2, \varepsilon_2) = (L_1 L_2, L_1 \varepsilon_2 + \varepsilon_1)$, unit $(1, 0)$, action $(L, \varepsilon)d = Ld + \varepsilon$; hence $\delta_s M_{rs} \leq L\delta_r + \varepsilon$. Non-expanding is $(L, 0)$, $L \leq 1$, sound logical transport $(1, 0)$; (CMR) realizes all structural bonds there. If $G_i = \Pi_{ij}(G_j)$ and θ_j is L -Lipschitz with $\sup_{g \in G_{s(j)}} d'_j(\theta_j g, G'_j) \leq \varepsilon$, then $\delta'_{r'} \lambda_{r'} \leq L\delta_{\rho(r')} + \varepsilon$, $\nu'_{r'} \lambda_{r'} \geq L\nu_{\rho(r')} - \varepsilon$, with composition $\star$. On bounded defects put $e_f(L) = \sup_x (\delta_Y(fx) - L\delta_X(x))_+$. Then $\delta_Y f \leq L\delta_X + \varepsilon \iff \varepsilon \geq e_f(L)$ and $e_{gf}(L_g L_f) \leq L_g e_f(L_f) + e_g(L_g)$. Hence exact affine transport factors through (δ_X, δ_Y, f) ; full metrics only certify. For $\delta = (0, 1)$, $f = \text{id}$, $e_f(L) = (1-L)_+$; its epigraph has no product-order least point. It also grades $X \vdash_r^{\varepsilon} Y \iff \delta_r Y \leq \delta_r X + \varepsilon$ and $X \vdash_{r \rightarrow s}^{(L, \varepsilon)} Y \iff \delta_s Y \leq L\delta_r X + \varepsilon$.

Finally, fiber-constancy of a content update a for every C_r implies fiber-constancy for $C_{\mathfrak{F}}$, hence for $\mathbf{JRead}$. The stronger componentwise hypothesis is retained because each induced $\bar{a}_r$ is used below; (DI) gives $\bar{a}_r(C_r X) = C_r(aX)$ uniquely. Since $R_{\mathbf{JRead}} = \bigcap_r R_{C_r}$, there is likewise a unique map $\bar{a}_{\text{sp}}$ on $\mathbf{JRead}(-; \mathfrak{F}_{\text{acc}})$ with $\bar{a}_{\text{sp}} \mathbf{JRead}(X; \mathfrak{F}_{\text{acc}}) = \mathbf{JRead}(aX; \mathfrak{F}_{\text{acc}})$; for such updates a, b , $\bar{b} \circ \bar{a}_{\text{sp}} = \bar{b}_{\text{sp}} \bar{a}_{\text{sp}}$ and $\bar{\text{id}}_{\text{sp}} = \text{id}$. The solution operator remains fixed by (P, Λ) , with R_0 selecting a trajectory; extensions of either induced map beyond the reachable image are presentation data.

C. Record algebra and geometric attachments

a. Topological, modular, and Dirichlet structure. Gelfand duality gives $A_{\text{rec}}^{\text{cl}} \cong C(M_{\text{phys}})$ with M_{phys} compact Hausdorff, while faithful normal modular flow is trivial on an abelian algebra: in its L^2 -GNS model the Tomita operator is conjugation, hence $\Delta = \mathbf{1}$. By (GTR), any independently produced separating record subalgebra already reconstructs this Gelfand topology; the Dirichlet flag below therefore begins strictly after, and does not regenerate, the record-topological carrier. For a regular strongly local symmetric Dirichlet form $\mathcal{E}$ on $L^2(X, \mu)$, writing $P_t^{\mathcal{E}} = e^{-tA_{\mathcal{E}}}$, spectral calculus and Sturm's intrinsic pseudometric⁴¹ (Eq. (1.1)) give (DG) $\mathcal{E}(f, f) = \lim_{t \downarrow 0} t^{-1} \langle f - P_t^{\mathcal{E}} f, f \rangle$, $d_{\mathcal{E}}(x, y) = \sup\{u(x) - u(y) : u \in \mathcal{D}_{\text{loc}}(\mathcal{E}) \cap C(X), d\Gamma(u, u) \leq d\mu\}$. Let $\mathbf{RDir}(X)$ be the full-support probability/regular strongly-local symmetric-form pairs and $\mathbf{IMet}(X) = \{d_{\mathcal{E}}\}$. Their forgetting flag is

$$[\mathbf{RDF}] \quad \mathbf{RDir}(X) \xrightarrow{q_d} \mathbf{IMet}(X) \xrightarrow{q_X} \{X\}, \quad q_d(\mu, \mathcal{E}) = d_{\mathcal{E}}, \quad q_X(d) = X, \quad \mathbf{DirFib}_X(d) = q_d^{-1}(d). \quad (122)$$

On fixed L^2 , the semigroup determines $\mathcal{E}$, whereas $d_{\mathcal{E}}$ is only its intrinsic extended-pseudometric shadow. For bounded mean-zero f , $\frac{1}{2} \partial_t q_{P_t}(f, f)|_0 = \mathcal{E}(f, f)$, so $e^{-2\kappa t}$ entropy decay implies $\mathcal{E}(f, f) \geq \kappa \|f\|_2^2$ and identifies the Fisher/BKM tangent. If a μ -preserving E_N preserves $\mathcal{D}(\mathcal{E})$, then (DL) $\mathcal{E}(E_N f, (1 - E_N)g) = -\lim_{t \downarrow 0} t^{-1} \langle E_N f, E_N P_t(1 - E_N)g \rangle$. Together with orthogonality, vanishing of this off-diagonal form plus domain invariance is equivalent to $P_t \mathcal{N} \subseteq \mathcal{N}$, equivalently $[E_N, P_t] = 0$; by (QAF) this is the quotient-descent locus, and then $\mathcal{E} = \mathcal{E}|_{E_N} \oplus \mathcal{E}|_{1-E_N}$; where defined, $L = -A_{\mathcal{E}}$ and $\Gamma_2(u) = \frac{1}{2}(L\Gamma(u) - 2\Gamma(u, Lu))$ stay in this carrier. For $V \in C^2([0, 1])$, let $d\mu_V = Z^{-1} e^{-V} dx$ and let $\mathcal{E}_V$ be the Neumann form on $H^1([0, 1], \mu_V)$, $\mathcal{E}_V(f, f) = \int_0^1 |f'|^2 d\mu_V$, $(d_{\mathcal{E}_V}, L_V, \Gamma_2) = (|x - y|, u'' - V'u', (u'')^2 + V''(u')^2)$, and $C^2([0, 1])/\mathbb{R} \hookrightarrow \mathbf{DirFib}_{[0, 1]}(|\cdot|)$, $[V] \mapsto (\mu_V, \mathcal{E}_V)$. Equality of normalized densities is exactly $V - W \in \mathbb{R}$, so this is an infinite-dimensional fixed-metric fibre. Zero/Neumann energy give distinct $\mathbf{IMet}$ points over one Gelfand topology; $V = 0$ and $V = \kappa x^2/2$ have the same metric but different dynamics. If $V'' \geq \kappa > 0$, then $\Gamma_2 \geq \kappa \Gamma$; Bakry–Émery⁴² (Cor. 2) with $\lambda = 2\kappa$, plus entropy dissipation, gives $D(\rho_t\|\mu_V) \leq e^{-2\kappa t} D(\rho_0\|\mu_V)$ for probability densities ρ_0 and $\rho_t = P_t^{\mathcal{E}_V} \rho_0$.

D. Residual fibers and independent attachments

By (DI/U0), total fibres, target shadows, kernel pairs, deterministic refinement and the measure/support ladder are already generated; only the following attachments remain. For $z \in \mathcal{T}_i$ put $\text{ExtFib}_i(z) = \{c \in C_{*}^{\text{rec}} : c_i = z\}$. With inclusion $J_i : G_i \hookrightarrow \mathcal{T}_i$ and $\bar{\pi}_i = J_i \pi_i$,

$\text{ExtFib}_i(z) = \tilde{\pi}_i^{-1}\{z\}$ and Boolean Kan gives $\text{Svapna}_{\tilde{\pi}_i}(C_*^{\text{rec}}) = \mathbf{1}_{G_i} = \text{Ext}_i^{\text{bin}}$. Hence

$$\begin{aligned} \text{Satya}_r(X) &\iff \text{Svapna}_{\tilde{\pi}_{i(r)}}(C_*^{\text{rec}})(\tau_r(C_r X)) = 1 \iff \text{ExtFib}_{i(r)}(\tau_r(C_r X)) \neq \emptyset \iff \delta_r(C_r X) = 0, \\ \text{Maya}_r(X) &\iff \text{Svapna}_{\tilde{\pi}_{i(r)}}(C_*^{\text{rec}})(\tau_r(C_r X)) = 0 \text{ and } \text{Imp}_r^{\text{ev}}(C_r X) > 0. \end{aligned} \quad (123)$$

Empty/singleton/multiple encode non-extendability/uniqueness/opacity; Maya is positive-impact empty, distinct from zero-impact. (DI) makes the binary profile lossless; scalarization is lossless iff its factor is injective, and two regimes ($\mathcal{T}_i = \{0, 1\}$, $G_i = \{1\}$) with identity evaluation realize all four profiles. For closed $G_i \subseteq \mathcal{T}_i$, $a \setminus b = \overline{a \cap b^c}$, $\sim a = \overline{\mathcal{T}_i \setminus a}$ and $a \wedge \sim a = \partial a$; hence proper nonempty G_i in connected $\mathcal{T}_i$ has the U0 fragile boundary/null-representative witness. On (RGL/SCX), $\mathbb{K}(\pi_i)$ sums reduced free modules on nonempty $\text{ExtFib}_i(z)$ and $\text{ExtRes}(\pi_i)$ is free on empty fibres. A compact metric adds $\text{Rep}_i(z) = \arg \min_{w \in G_i} d_i(z, w)$ with u.s.c. compact fibres. Theorem III.1/(PCR) gives : nonempty compact-valued $K : T \rightrightarrows X$ (X compact Hausdorff) and $t \mapsto \mathcal{P}(K(t))$ have the same l.s.c. locus. Thus randomization preserves nearest-repair catastrophe; $G = \{-1, 1\} \subset [-1, 1]$ is strict at 0, while excess instability is failure of relative openness.

For compact-metrizable (T, τ) and $\emptyset \neq G \subseteq T$ closed, write Ω_τ for compatible metrics, $D_G(d) = (\min_{w \in G} d(z, w))_z$, $A_G(d) = (\text{pr}_2 \text{Opt}_{\text{pr}_1}^{d|_{T \times G}}(z))_z$ by (OMF), corestricting D_G . Compactness kills the (AOL) H_0 grade; $T = [-1, 1]$, $G = \{-1, 1\}$, $z = 0$ has rank-one H_1 . By (RKD), $\text{RepKey}_{\tau, G} := \text{Key}_{A_G|D_G} \cong \Omega_\tau / (R_{D_G} \cap R_{A_G})$ and $\text{DepRes}_{D_G}(A_G) = 0 \iff A_G$ descends through D_G . On the (FTRF) identity point $T = \{a, b, c, x_n : n \geq 2\}$, $G = \{a, b\}$, take Euclidean pullbacks with $a = (-1, 0)$, $b = (1, 0)$, $c = 0$, $x_n = u_n^j$, $u_n^0 = (0, n^{-1})$, $u_n^1 = (-n^{-2}, \sqrt{3n^{-2} - n^{-4}})$. Both metrize τ and share D_G : $D_G(x_n)^2 = 1 + n^{-2}$, while $A_0(x_n) = G$, $A_1(x_n) = \{a\}$ and $A_j(c) = G$. A_0 is l.s.c.; A_1 fails only at c ; (PCRG) preserves the loci. Thus (FTRF) and D_G miss continuation-relevant repair. Under (RKD) let $q_M = \langle D_G, A_G \rangle$, $\pi_D q_M = D_G$. (URT/RFE) gives $\Omega_\tau \cong \int_{u \in \text{im } D_G} \int_{k \in \pi_D^{-1}\{u\}} q_M^{-1}\{k\}$. Each metric leg of the affine frontier factors through $u = D_G$, repair through k . For $T = \{g, a, b\}$, $G = \{g\}$, metric triples $(1, 2, 3/2)$, $(1, 2, 5/2)$ in (d_{ga}, d_{gb}, d_{ab}) share D_G, A_G but are nonisometric; already on one-stage (CMR) ($a_i = 3, \rho_i = d/3$). The inner fibre is nonterminal; a singleton ε -lax comparison has sharp $\varepsilon = d(a, b)$. Of these consumers, only metric gluing sees it.

For standard-Borel $e : \mathcal{G} \rightarrow E$ and $\lambda \in \mathcal{P}(E)$, (CDNF) at empty constraint family gives (LPA) $\text{Lift}_e(\lambda) := \{\nu : e_* \nu = \lambda\} \cong_{\text{aff}} \text{LKer}_e(\lambda) := \{[\kappa]_\lambda : \kappa_y \in \mathcal{P}(e^{-1}\{y\}) \text{ } \lambda\text{-a.e.}, \nu = \int \kappa_y d\lambda(y)\}$. Put $\text{Sec}_e(\lambda) = \{[s]_\lambda : s : E \rightarrow \mathcal{G} \text{ Borel, } e_* s = 1_E \text{ } \lambda\text{-a.e.}\}$. Then $\text{ex Lift}_e(\lambda) \cong \text{Sec}_e(\lambda)$ by $[s] \mapsto \int \delta_{s(y)} d\lambda(y)$; (DER/PCR) is the compact-metrizable $\lambda = \delta_y$ face.

For composable Borel maps between standard-Borel spaces $X \xrightarrow{U} Y \xrightarrow{V} Z$, (RFE) applied to U_*, V_* has the affine graph face

$$\text{LiftTot}_{U, V}(\lambda) := \{(\eta, \nu) : V_* \eta = \lambda, U_* \nu = \eta\} \cong_{\text{aff}} \text{Lift}_{VU}(\lambda), \quad \nu \leftrightarrow (U_* \nu, \nu).$$

At $\lambda = \delta_z$ the outer/inner fibres are $\mathcal{P}((VU)^{-1}\{z\})$, $\mathcal{P}(V^{-1}\{z\})$ and $y \mapsto \mathcal{P}(U^{-1}\{y\})$; (ROA/LPA/ELR) therefore makes only the extreme boundary deterministic. Dependence is strict for $V : \{0, 1\} \rightarrow *$, $U^{-1}(0) = \{a, b\}$, $U^{-1}(1) = \{c\}$: the two inner fibres over δ_0, δ_1 are a segment and a point, so no product decomposition or occurrence selector follows.

For finite $G \curvearrowright F$, (CDNF/LPA/OIL) gives the OIL singleton and with (ELR/ABE) yields $\text{ex}(\mathcal{P}(F))^G = \{\text{Unif}(O) : O \in F/G\}$ and $(\text{ex } \mathcal{P}(F))^G = \delta(F^G)$, equal iff the action is trivial. For second-countable G , equivariant e and invariant λ , uniqueness of disintegration gives $(g\kappa)_y = g_* \kappa_{g^{-1}y}$ and $(\text{LPA}_G) \text{Lift}_e(\lambda)^G \cong_{\text{aff}} \text{LKer}_e(\lambda)^G$; fixed Dirac classes are fixed sections, and compact G Haar-averages any existing lift to an invariant one.

The same kernel proves (DD): for measurable $u : X \rightarrow Y$, $v : X \rightarrow Z$ on (X, μ) with Y, Z standard Borel,

$$v = f \circ u \text{ } \mu\text{-a.e. for some Borel } f : Y \rightarrow Z \iff \sigma(v) \subseteq \sigma(u) \text{ mod } \mu.$$

For (ELR), if κ_y is non-Dirac on positive λ -mass, a countable separating algebra supplies one A with $p_y = \kappa_y(A) \in (0, 1)$ there; $dk_y^\pm = (1 \pm \varepsilon(\mathbf{1}_A - p_y))dk_y$ ($0 < \varepsilon < 1$) are distinct admissible kernels with midpoint κ . Conversely a.e.-Dirac kernels are exactly $\text{Sec}_e(\lambda)$ and are pointwise extreme, with uniqueness from disintegration. For (DD), disintegrate $(u, v)_* \mu$ over u and take a countable separating algebra (A_n) of Z ; the assumed σ -inclusion gives $B_n \subseteq Y$ with $\mathbf{1}_{\{v \in A_n\}} = \mathbf{1}_{\{u \in B_n\}}$ a.e., hence on one conull set every $\kappa_y(A_n) \in \{0, 1\}$, so $\kappa_y = \delta_{f(y)}$ by separation and f is Borel with $f^{-1}(A_n) = B_n$ there. Extend off that set; the converse is immediate. For history $\mathbf{x}_n = (x_0, \dots, x_n)$ set $K_n = \bigcap_{k \leq n} G_{x_k}$, $O_n = G\mathbf{x}_n$. Then (HSB) $\text{Stab}_G(\delta_{\mathbf{x}_n}) = K_n$, $O_n \cong G/K_n$, $\text{Hom}_G(O_n, Z) \cong Z^{K_n}$, so $K_{n+1} \leq K_n$ and $Z^{K_n} \subseteq Z^{K_{n+1}}$. For transitive $Z \cong G/H$, (HSB_{act}) $\text{Hom}_G(G/K, Z) \neq \emptyset \iff Z^K \neq \emptyset \iff K \leq gHg^{-1}$ for some g ; hence $O_\emptyset := G/G = *$ admits a deterministic equivariant first point iff $Z^G \neq \emptyset$. (SKR) Read the history orbit as an admissible-key filtration:

$$\text{Key}_n^G(Z) := \text{Hom}_G(O_n, Z) \cong Z^{K_n}, \quad \text{Key}_n^G(Z) \hookrightarrow \text{Key}_{n+1}^G(Z), \quad Z = G/H \implies [\text{Key}_n^G(Z) \neq \emptyset \iff K_n \leq gHg^{-1} \text{ for some } g].$$

Precomposition proves monotonicity; $K_\emptyset = G$ is the fixed-point gate and $\text{Hom}_G(G, Z) \cong Z$ makes $G/1$ the complete symmetry key. SKR tests resource admissibility below YSK and manufactures none. After $BK \rightarrow BG$, (ABE) gives the same existence criterion, while (PRF/EAf) makes Z^K the exact null-representative outcome fiber before the a.e. operator-algebraic quotient. For finite G let $C_H = \{gHg^{-1} : g \in G\}$, $Z_L = \{z \in Z : G_z = L\}$ and $N_G(L) = \{g : gLg^{-1} = L\}$. Evaluation at eK and exact-stabilizer partition give (IAR) $\text{Hom}_G(G/K, G/H) \cong (G/H)^K = \bigsqcup_{\substack{L \in C_H \\ K \leq L}} Z_L$, with each Z_L an $N_G(L)/L$ -torsor; hence $gH \mapsto gHg^{-1}$ is $q_H : G/H \twoheadrightarrow G/N_G(H)$ with fiber

$N_G(H)/H$. Put $W_K := \text{Aut}_G(G/K) \cong N_G(K)/K$. Its right action has orbit map q_K ; (NQT_{ex}) for the commuting left- G /right- $N_G(K)$ actions on G gives (UOQ) $(G/K)/W_K \cong_G G/N_G(K)$, $\text{Hom}_G(G/N_G(K), Y) \cong \text{Hom}_G(G/K, Y)^{W_K}$. For $Y = G/H$ this is $(G/H)^{N_G(K)}$, nonempty iff $N_G(K) \leq gHg^{-1}$; thus q_H is point-injective, equivalently splits G -equivariantly, iff $N_G(H) = H$, and $[G : H] = [G : N_G(H)][N_G(H) : H]$. On nonempty $\text{Hom}_G(G/K, G/H)$, (ABE) makes the same criterion equivalent to natural actualizability of invariant implementation laws. $r : O_n \rightarrow R$ equivariant and $L_n = G_{r(\mathbf{x}_n)}$ imply $K_n \leq L_n$ and $Z^{L_n} \subseteq Z^{K_n}$. For finite G set $u_n = \text{Unif}(O_n)$, $D_n = D(\delta_{\mathbf{x}_n} \| u_n)$, $D'_n = D(\delta_{r(\mathbf{x}_n)} \| r_* u_n)$; then $D_n = \log[G : K_n]$, $D_{n+1} - D_n = \log[K_n : K_{n+1}]$, $D_n - D'_n = \log[L_n : K_n]$. For a diagonal G -equivariant deterministic implementation $A : S \times R \rightarrow Z$, equivariance gives the pointwise resource law $G_s \cap G_\xi \leq G_{A(s, \xi)}$. If G is finite, put $\mathfrak{d}_G(y) = \log[G : G_y]$; then

$$\begin{aligned} \mathfrak{d}_G(A(s, \xi)) &\leq \log[G : G_s \cap G_\xi] \leq \mathfrak{d}_G(s) + \mathfrak{d}_G(\xi), \quad s \in S^G \implies \mathfrak{d}_G(A(s, \xi)) \leq \mathfrak{d}_G(\xi), \\ \text{[SAB]} \quad x_{n+1} &= A(r(\mathbf{x}_n), \xi_{n+1}) \implies 0 \leq D_{n+1} - D_n \leq \log[K_n : K_n \cap G_{\xi_{n+1}}] \leq \mathfrak{d}_G(\xi_{n+1}), \quad D_m - D_0 \leq \sum_{j=1}^m \mathfrak{d}_G(\xi_j). \end{aligned} \quad (124)$$

Indeed $K_n \leq L_n = G_{r(x_n)}$ and $G_{x_{n+1}} \supseteq L_n \cap G_{\xi_{n+1}}$ imply $K_{n+1} \supseteq K_n \cap G_{\xi_{n+1}}$; the index bounds are the injective coset maps induced by subgroup intersection. Thus a trivial-action auxiliary seed has zero symmetry resource and cannot create a new stabilizer drop from a symmetric input. More sharply, for $s \in S^G$ and $Z \cong G/H$, a resource orbit G/K can produce a point exactly on the $(\text{HSB}_{\text{act}})$ locus $K \leq gHg^{-1}$; $[G : K] \geq [G : H]$ is only necessary. It is not sufficient: for $G = C_2 \times C_2$, $H = \langle a \rangle$ and $K = \langle b \rangle$ have the same index two but $\text{Hom}_G(G/K, G/H) = \emptyset$, whereas $K = H$ works. Hence subgroup incidence, not its scalar index, is the complete transitive feasibility resource; (IAR/UOQ) leaves the point torsor after forgetting the orbit basepoint. The regular action $G/1$ ($G \neq 1$) is strict by (OIL). If R, Z are measurable G -spaces, $s \in S^G$ and $\lambda \in \mathcal{P}(R)^G$, then $A(s, \cdot)_* \lambda \in \mathcal{P}(Z)^G$: an invariant stochastic law does not remove the point-resource requirement of a deterministic realization. Base-coset evaluation and the W_K -orbit fibers prove (IAR/UOQ); exact- L points form the displayed $N_G(L)/L$ -torsor, and orbit maps prove the KL symmetry-resource order.

The pointwise resource inequality has an exact residual-fibre refinement. For finite G -sets S, R, Z , choose one representative (s_O, ξ_O) from each diagonal orbit $O \in G \backslash (S \times R)$. Evaluation on orbit representatives gives

$$[\text{SARF}] \quad \boxed{\text{Hom}_G(S \times R, Z) \cong \prod_{O \in G \backslash (S \times R)} Z^{G_{s_O} \cap G_{\xi_O}}.} \quad (125)$$

The displayed product is independent of the representative choices up to the canonical conjugacy bijections. Indeed a G -map on $O \cong G/(G_{s_O} \cap G_{\xi_O})$ is determined exactly by the image of its basepoint, and that image must be fixed by the orbit stabilizer. Consequently, for transitive $Z \cong G/H$ an equivariant deterministic implementation exists exactly when, for every diagonal orbit O , there is $g_O \in G$ with $G_{s_O} \cap G_{\xi_O} \leq g_O H g_O^{-1}$. Thus (SAB) is the scalar shadow of (SARF); the full fixed-point product retains implementation multiplicity and torsor data.

There is no resource-free probabilistic escape. Let $\mathbb{D}$ be the finitely supported probability-distribution monad on **Set**. Then

$$[\text{ANB}] \quad \boxed{\text{Nat}(\mathbb{D}, \text{Id}_{\text{Set}}) = \emptyset.} \quad (126)$$

For the two-point set, the uniform law is fixed by the transposition; naturality would force its image point to be fixed by the same transposition, which is impossible. This strengthens the law-occurrence no-retraction boundary: even a non-retractive natural law-to-point transformation is impossible. Hence a law can constrain an occurrence but cannot universally generate it. (SARF) identifies the positive missing object: an independently available resource whose stabilizer incidence makes the target point feasible. A realized resource point remains an occurrence input; replacing it by an invariant resource law does not remove this boundary, as (ERD/PSC/OCV) already shows.

Realized-seed landing in the actuality complement. The preceding abstract resource has an existing target-blind realization. Let I be finite, $G = \text{Sym}(I)$, let u_I be the uniform law, and let $\mathcal{J}_I^{\text{sym}} = |I|^{-1} \text{id}_M$ be the symmetric normal instrument. Give $\Omega_I = (0, \infty)^I$ the coordinate-permutation action and, on the unique-minimum locus, put $i = \mathbf{R}_I(u_I, \xi) = \arg \min_j \xi_j$ as in (ERD). Then

$$[\text{RSL}] \quad \boxed{G_{\mathcal{J}^{\text{sym}}} = G, \quad G_\xi \leq G_i, \quad [q_\xi] \in \text{ExComp}(\mathcal{J}^{\text{sym}}; i) \iff G_\xi = G_i.} \quad (127)$$

Indeed the unique minimum forces every stabilizer of ξ to fix i . Hence the groupoid face of (AKI) makes the joint instrument-seed datum successful, while (ECM) is exactly $G_{\mathcal{J}^{\text{sym}}} \cap G_\xi = G_{\mathcal{J}^{\text{sym}}} \cap G_i$. On the unique-minimum locus, $G_\xi = G_i$ iff all coordinates ξ_j , $j \neq i$, are equal. Therefore for the product exponential seed law ν_I , the nontrivial binary case $|I| = 2$ lands in the exact complement almost surely, whereas for $|I| \geq 3$ a generic seed has trivial stabilizer and lands successfully but strictly over-resolves the target key; the exact-complement locus has ν_I -measure zero. The map $\mathbf{R}_I$ is permutation-natural on this locus and receives no target label as input, so this is a genuine positive consumer of (AKF/ECM), not target laundering. It still does not generate the realized seed from the instrument: (RNB) forces that point-bearing input to remain relatively exogenous on the symmetric witness.

Projective race key. The raw exponential seed contains one universally invisible coordinate. Put $\Delta_I^\circ := \{p \in \Delta_I : p_i > 0 \forall i\}$, $\mathcal{P}_+(I) := \{A \subseteq I : A \neq \emptyset\}$ and

$$W_I(p, \xi) := \arg \min_{i \in I} \frac{\xi_i}{p_i} \in \mathcal{P}_+(I), \quad \text{RaceProf}_I(\xi) := (W_I(p, \xi))_{p \in \Delta_I^\circ}.$$

Then

$$[\text{PRK}] \quad \boxed{\text{RaceProf}_I(\xi) = \text{RaceProf}_I(\eta) \iff \eta = t\xi \text{ for a unique } t > 0, \quad \text{RaceKey}_I := \text{im RaceProf}_I \cong \Omega_I / \mathbb{R}_{>0} \cong \Delta_I^\circ,} \quad (128)$$

where the last isomorphism is $[\xi] \mapsto \xi / (\sum_i \xi_i)$. Since the full-support family $(W_I(p, -))_p$ is jointly separating on RaceKey_I , $(\text{KGC}_{\text{sep}}/\text{KGC}_{\text{cons}})$ makes it the coarsest target-blind seed quotient sufficient for the whole ERD family; it removes exactly common positive scale.

Scaling invariance is immediate. Conversely, if ξ, η are not proportional, choose i, j with $\xi_i/\xi_j \neq \eta_i/\eta_j$ and choose r strictly between these two ratios. Take positive unnormalized law weights with $p_i : p_j = r : 1$ and all remaining p_k sufficiently small. The remaining ratios are then too large to win, while i and j win on opposite sides of the two distinct thresholds, so $W_I(p, \xi) \neq W_I(p, \eta)$ with unique winners. This proves the first equivalence; universal factorization is $(\text{KGC}_{\text{sep}}/\text{KGC}_{\text{cons}})$.

In logarithmic coordinates $s_i = \log \xi_i$ the carrier is $\mathbb{R}^I / \mathbb{R} \mathbf{1}$, and for fixed p the unique-outcome chambers are

$$[\text{PRC}] \quad \boxed{C_i(p) = \{[s] : s_i - \log p_i < s_j - \log p_j \ (j \neq i)\}, \quad s_i - s_j = \log(p_i/p_j) \text{ on the walls.}} \quad (129)$$

Thus the law translates a type- $A_{|I|-1}$ braid arrangement on one fixed projective seed carrier rather than changing the carrier itself. For $\xi_i \stackrel{\text{iid}}{\sim} \text{Exp}(1)$, write $\xi = tv$ with $v \in \Delta_I^\circ$. The Jacobian is $t^{|I|-1}$ and the joint density factorizes as $e^{-t} t^{|I|-1} dt$ times constant simplex volume, so t is independent of v , $t \sim \text{Gamma}(|I|, 1)$ and $v \sim \text{Dirichlet}(1, \dots, 1)$. By (ERD), therefore,

$$\mathbb{P}\{[\log v] \in C_i(p)\} = p_i.$$

The Born/instrument face only substitutes $p_i = \text{Tr } \mathcal{I}_i(p)$ in this identity. The projective compression is a realization/key theorem, not an origin theorem: (RNB) still forbids deriving the realized projective seed from the same pre-event datum when it strictly enlarges actuality.

Outcome-label interface firewall. Let $\mathbf{K} : \mathbf{FinBij}_{+, \text{sk}} \rightarrow \mathbf{Set}$ be the projective-race-key functor $\mathbf{K}(I) = \text{RaceKey}_I \cong \Delta_I^\circ$, with bijections acting by coordinate push-forward. For every set A , viewed as the constant functor $\underline{A}$,

$$[\text{OLIF}] \quad \boxed{\text{Nat}(\underline{A}, \mathbf{K}) \simeq \mathbf{1}, \quad \kappa_I(a) = u_I := (|I|^{-1})_{i \in I}.} \quad (130)$$

Indeed naturality at the automorphism group $G_I = \text{Sym}(I)$ forces $\kappa_I(a) \in (\Delta_I^{\circ})^{G_I}$, and this fixed locus is the singleton $\{u_I\}$; the uniform assignment is natural. More generally, for any $P : \mathbf{FinBij}_{+,sk} \rightarrow \mathbf{Set}$, natural $\kappa : P \Rightarrow K$ and $x \in P(I)$,

$$\text{Stab}_{G_I}(x) \leq \text{Stab}_{G_I}(\kappa_I(x)).$$

Thus a G_I -fixed pre-event datum can reach only u_I . Applied to $P = \underline{A} \times \text{Nlnstr}_{\mathcal{M}}$ at $(a, \mathcal{J}^{\text{sym}})$, the whole pre-event pair is G_I -fixed and therefore has the same conclusion. At the symmetric law itself,

$$[\text{OLW}] \quad \boxed{W_I(u_I, u_I) = I, \quad u_I \notin \bigcup_{i \in I} C_i(u_I) \quad (|I| > 1),} \quad (131)$$

so the unique label-blind projective key lies on all pairwise race walls and supplies no unique outcome. Consequently, adjoining any fixed normal state or any fixed datum from an outcome-unindexed physical carrier as a constant pre-event factor cannot repair the symmetric (OINB) witness through the race key. In particular the channel-only dilation carrier (DGF) and the marked time-envelope carrier MEnv, at their declared types, carry no outcome-label coordinate; a fixed point of either, pulled back along $\mathbf{FinBij}_{+,sk}$, is such a constant factor. A successful physical producer must therefore supply an independently realized outcome-indexed datum $x \in P(I)$ together with a natural map $P \Rightarrow K$ whose value leaves the symmetric wall and enters an outcome chamber. This is a type/perimeter obstruction, not a claim that no such physical interface can exist; adding that interface is additional physical data rather than a consequence of the outcome-unindexed carriers above.

The same obstruction has a law-level form. For a finite group G , a finite G -set X , $x \in X$ and $\mu \in \mathbb{D}X$, write $G_x = \{g : gx = x\}$ and $G_\mu = \{g : g*\mu = \mu\}$. If $K : X \rightarrow \mathbb{D}Y$ is equivariant and $K^\#$ its Kleisli extension, then

$$[\text{FDSC}] \quad \boxed{G_x \leq G_{K(x)}, \quad G_\mu \leq G_{K^\#\mu}, \quad K^\#\mu = \delta_y \implies G_\mu \leq G_y.} \quad (132)$$

Indeed $g*K^\#\mu = K^\#g*\mu$; taking $\mu = \delta_x$ gives the point face. Thus an equivariant Kleisli chain has $G_{\mu_0} \leq \dots \leq G_{\mu_n}$: invariant input stays invariant, and a Dirac endpoint must be fixed by every symmetry of the input law.

A deterministic well-founded finite-arity typed tree composes to one equivariant map from the product of its external realized leaves to the root; nullary generated vertices are G -fixed. Hence the old tree law is an immediate face:

$$[\text{FDSC}_{\text{tree}}] \quad \boxed{K_T := \bigcap_{\ell} G_{x_\ell} \leq G_{x_{\text{root}}}} \quad (133)$$

(with $K_T = G$ if there are no external leaves). Thus a nonfixed root needs a nonfixed external leaf; equivalently, with boundary X_s^G , every equivariant rule has empty boundary-changing ignition in $(\text{BPR}_{\text{ign}})$. For G/H its occurrence requires $K_T \leq gHg^{-1}$, the $(\text{HSB}_{\text{act}}/\text{SAB})$ gate. This is the finite groupoidal shadow of (GAD/RNB) , not a second source theorem.

At the predictive law–occurrence seam, the next-law map and positive-branch conditioning in $(\text{PLBD}/\text{PBCF})$ are equivariant, so

$$[\text{FDSC}_{\text{br}}] \quad \boxed{G_s \leq G_{p_s}, \quad G_s \cap G_a \leq G_{C_a s} = G_{U_a(s)}.} \quad (134)$$

The second inclusion is sharp: for $s = \frac{1}{2}(\delta_{0^\infty} + \delta_{1^\infty})$, the binary transposition fixes s , whereas $C_0 s = \delta_{0^\infty}$ has trivial stabilizer and $G_s \cap G_0 = 1$. Hence unconditioned natural/Kleisli evolution cannot create symmetry loss; conditional symmetry loss enters through the occurred branch. (PNB) forbids generating that branch naturally from the full path law, and the full-invariance specialization is exactly the stochastic source-conservation face consumed by (QSSC). Stabilizer monotonicity is elementary equivariance; the SCO content is its typed bridge between (RNB), law dynamics and (PLBD/PBCF). The scope is finite- G /finite-carrier; no thermodynamic-limit or spontaneous-symmetry-breaking claim is made.

For compact metrizable X , let $\text{AR}_G^{\text{Bor}}(X)$ be Borel equivariant $a : \mathcal{P}(X) \rightarrow X$ with $a\delta = 1_X$. Then $(\text{IAE}) \mathcal{P}(X)^G \neq \emptyset \implies [\text{AR}_G^{\text{Bor}}(X) \neq \emptyset \iff X^G \neq \emptyset]$: an actualizer sends invariant laws to fixed points, while fixed x_0 defines the Borel retraction $a(\delta_x) = x$, $a = x_0$ off the closed Dirac locus. Compact G supplies orbit laws; on transitive X Haar averaging makes that invariant law unique, while nontriviality gives no actualizer. The hypothesis is sharp because $\text{PSL}_2(\mathbb{R}) \curvearrowright S^1$ has none (rotations force Lebesgue, a hyperbolic Möbius map breaks it). Finite BG is $(\text{SIC}_{\mathcal{P}})$. At $\lambda = e_*\mu$, cost/Bayes are $D(\kappa_y \| \mu_y)$ and $a_y^* \in \arg \min_a \int \ell(g, a) d\mu_y(g)$; implementation is Theorem V.1; finite-normal selectors come from U0/Gibbs. Diagonal $E : M_2 \rightarrow D_2$: $A_0 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} > 0$, $A_0 E(A_0)^{-1} = \begin{pmatrix} 1 & 1 \\ 1/2 & 1 \end{pmatrix} \notin M_2^{\text{sa}}$, so right division fails.

Deterministic Blackwell refinement is the (DI) factorization $w_s = f \circ w_r$; randomized comparison remains the Blackwell–Sherman–Stein decision import⁴³. On a full-support commutative realization, $M_{\text{cl}} = L^\infty(\mathcal{G}, \mu)$ and $\mathcal{N}_r = w_r^* L^\infty(Y_r, w_{r*}\mu)$, disintegration is conditional expectation, U0/U1 are the modal/entropic quotients, and (DD) gives $w_s = f \circ w_r$ μ -a.e. $\iff \sigma(w_s) \subseteq \sigma(w_r) \bmod \mu \iff \mathcal{N}_s \subseteq \mathcal{N}_r$.

a. Residual Blackwell shadow. With $\alpha_r : \mathcal{N}_r \hookrightarrow M_{\text{cl}}$ and $\mathcal{S}_b^\times = \{p\omega_\mu : 0 < \text{ess inf } p \leq \text{ess sup } p < \infty\}$, (DD/EIO) identifies deterministic Blackwell refinement with both $d_{\alpha_r} \leq d_{\alpha_s}$ on $\mathcal{S}_b^\times$ and $\text{Suff}(\alpha_s) \subseteq \text{Suff}(\alpha_r)$ there; call this commutative specialization (RBO). It is the measure-algebra zero-determination order of (CDR)/Samskara, while pointwise CDR keeps the strict null-representative boundary; if factorization fails, $p = 1 + \varepsilon g$ with real mean-zero $g \in \mathcal{N}_s \setminus \mathcal{N}_r$ is the bounded faithful witness. The canonical mixed-fiber certificate uses $\mathcal{T}_n = [0, 1]$, $\Pi_{12} = \text{clamp}_{[1/4, 3/4]}$, $\Pi_{n,n+1} = \text{id}$ ($n \geq 2$), Lebesgue u , $L(u) = |u - 1/2|$ and $w = \pi_1$, one has $G_1 = [1/4, 3/4]$, boundary fibers $[0, 1/4]$, $[3/4, 1]$, point-collapse cost $+\infty$ and half-fiber tariff $\log 2$; on $A = [1/8, 1/4]$ the support-relative minima differ by $1/8$. Since $\pi_1 = \text{clamp} \circ \pi_2$, this single example also realizes the exponential barrier, strict deterministic Blackwell refinement and Theorem III.1's boundary instability.

VII. EFFECTIVE REPRESENTATION AND REFLECTIVE REDUCTION

A. Category of residual-access presentations

An access presentation is the leg $\mathfrak{P}$ of Definition II.2, with the typed access morphisms defined above. Ambient Y_r , reachable support $S_r = \text{supp}(w_{r*}\mu)$, and marked G_r remain distinct; inverse-limit gluing is asserted only when $G_r = w_r(\mathcal{G})$. (DI) supplies the effective read-out fibers; U0 and the attachments supply the modal and reflector structure of this category.

B. Effective access and intrinsic reduction

a. Effective-data scope. On a standard Borel probability space (X, μ) , an effective input is the typed family $D = (\mathcal{E}_D, \mathcal{Q}_D, G_D, L_D, I_D, P_D, P, \Lambda)$, where $\mathcal{E}_D \subseteq \mathcal{B}(X)$ is countably generated, $\mathcal{Q}_D \subseteq \mathcal{E}_D$ is a retained countably generated sub- σ -algebra, $G_D \in \mathcal{E}_D$, L_D is bounded Borel, $I_D \geq 0$ is $\mathcal{E}_D$ -measurable when declared, $\mu P_D \sim \mu$, and $P, \Lambda \geq 0$ are locally integrable. Undeclared fields take respectively the trivial retained algebra, full event, zero cost/impact, identity kernel and zero production. Effectivity means precisely that the declared distinctions factor through $\mathcal{E}_D$; if all fields are null, only $\{\emptyset, X\}$ is generated.

A *standard-Borel presentation leg* is the presentation leg of Definition II.2 with $\mathcal{G}, Y_r, Z_r$ standard Borel, μ a Borel probability measure, $w_r, q_r, \text{Imp}_r, L$ Borel (with L bounded), G_r Borel, and P a Borel Markov kernel satisfying $\mu P \sim \mu$; $P_t, \Lambda_t \geq 0$ remain locally integrable.

Theorem VII.1 (Realization and observational reflection). *Let D be effective on a standard Borel probability space and let $\mathfrak{A}$ be a standard-Borel presentation leg as above.*

1. (DD/DI) gives standard-Borel $X \xrightarrow{w_D} Y_D \xrightarrow{q_D} Z_D$ with $\mathcal{E}_D = w_D^{-1} \mathcal{B}(Y_D)$ and $\mathcal{Q}_D = (q_D w_D)^{-1} \mathcal{B}(Z_D)$; every declared $\mathcal{E}_D$ -read-out factors through w_D , and effective-range/null normalization makes the quotient generator-independent up to the declared measure-algebra isomorphism.
2. For countable families $\mathcal{L}_{\text{aux}}$ (bounded Borel costs) and $\mathcal{R}_{\text{aux}} = \{(P_t^b, \Lambda_t^b)\}_{b \in B}$ (non-negative locally integrable residual generators), Theorem II A 0 d on a T -invariant core containing $\mathcal{L}_{\text{aux}}$ transports access/disintegration, marks, visible/record algebras, cost/impact, dynamics/temporal envelope and $\mathcal{R}_{\text{aux}}$ in native measurable classes, preserving Krieger–Connes type on the ergodic nonatomic stratum. The active reduct removes exactly these families without scalarization, preserves marked morphisms/residual Sol-families, and forgetting marks recovers $\mathfrak{A}$ (empty family conservative).
3. (PI) gives inverse-limit realization for countable compact inverse-system presentations; (PUI/P2) is the finite-entropy Radon branch, and the constant system is the converse base. Hence standard-Borel effectivity is equivalent to access presentability with the same effective quotient and the typed complement of Theorem S7.
4. **Operator-algebraic ladder.** The skeleton sends $\mathfrak{S}$ to the exact abelian time-zero normal-access/record object $\mathfrak{S}_0^\#$; evaluation coordinates enter only after (DI) descent, and ambient/evaluation data remain attachments. For every typed access morphism $u = (\rho, \eta, \zeta; \varphi)$, (OR1)

$$j_u \alpha_{r'}^{\text{ab}} = \alpha_{\rho(r')}^{\text{ab}} \eta_{r'}^*, \quad \eta_{r'}^* \iota_{r'}^{\text{rec}} = \iota_{\rho(r')}^{\text{rec}} \zeta_{r'}^*.$$

The marked temporal extension is $\text{MEnv}(P, \mu)$ of Theorem VIII.1, with $\mathcal{A}_0 \subseteq \widehat{\mathcal{A}} \subseteq \mathcal{M}_{\text{time}}$ expected, abelian locus exactly $P = I$, and modular flow fixing $\widehat{\mathcal{A}}$ pointwise. On any unital C^* state-space face whose independently produced observables separate states, (GTR) reconstructs only the weak- $*$ topology from evaluations, not a state, dynamics or occurrence. A general sigma-finite normal-access $(\mathcal{M}, \omega, (\alpha_r)_r)$ is not generated by $\mathfrak{S}_0^\#$; it is external only to that local skeleton and is retained by the global typed realization/reference fibres. For $\mathcal{K} \subseteq \mathcal{M}$, put $C_\omega \mathcal{K} := \text{Exp}_\omega(\mathcal{K}) = W^*(\sigma_t^\omega(\mathcal{K}) : t \in \mathbb{R})$, the least expected algebra containing $\mathcal{K}$. By modular generation and BKM order-faithfulness, (RLE) $C_\omega(\bigvee_i \mathcal{K}_i) = \bigvee_i C_\omega \mathcal{K}_i$, $C_\omega(\cap_i \mathcal{K}_i) \subseteq \cap_i C_\omega \mathcal{K}_i$, and $\mathbf{b}_{C_\omega(\cap_i \mathcal{K}_i) : \cap_i C_\omega \mathcal{K}_i} = 0$ iff equality; (TEXP) supplies the unique ω -preserving normal expectations.

b. Reference fibration and modular transport. For $\mathcal{A} : \mathbf{Loc}_{\text{sep}} \rightarrow \mathbf{C}^* \mathbf{Alg}_{\text{inj}}$ with separable unital controls, a pointed reference is $r = (\mathcal{H}_r, \pi_r, \omega_r)$, π_r faithful on separable $\mathcal{H}_r$, $\mathcal{M}_r = \pi_r(\mathcal{A}(M))''$, and ω_r faithful normal. Arrows are unitary marked intertwiners $u : r \rightarrow r'$ with $u\pi_r(a)u^* = \pi_{r'}(a)$ and $\omega_{r'} \circ \text{Ad } u = \omega_r$. Thus (NRF) $\mathbf{NRef}_{\mathcal{A}} : \mathbf{Loc}_{\text{sep}}^{\text{op}} \rightarrow \mathbf{Gpd}$, with $\mathbf{Ext}_\psi(r) = \{(s, u) : u : r \cong \psi^* s\}$, has RVN/RTO Grothendieck realization and

$$(\text{HRF/CRF}) \quad \mathbf{Ext}_{\phi\psi}(r) \simeq \int_{(s,u) \in \mathbf{Ext}_\psi(r)} \mathbf{Ext}_\phi(s),$$

without a chosen section. Under the Fewster–Verch no-go hypotheses (dynamical locality, natural state, a noncompact-Cauchy test spacetime with faithful Reeh–Schlieder GNS representation, and extended locality), a natural marked-state section would force triviality; a nontrivial theory in this regime has no such preferred section by (FVDL_{nat}).

Let $\mathbf{ModBFlag}_n$ be the groupoid of pointed raw/reflected length- n flags and $\mathbf{ModInt}$ its length-one truncation. For $e = (s, u)$ set $\mathcal{M}_e = \pi_s(\mathcal{A}(N))''$, $\mathcal{K}_e = (\pi_s \mathcal{A}(\psi)(\mathcal{A}(M)))''$, and $\widehat{\mathcal{K}}_e = \text{Exp}_{\omega_s}(\mathcal{K}_e)$; for a finite chain ψ ending at t , write $\mathcal{K}_i^t$ for the raw image flag and $\mathcal{H}_i^t = \text{Exp}_{\omega_t}(\mathcal{K}_i^t)$. Modular covariance yields $\mathcal{M}_{\psi, r} : \mathbf{Ext}_{\psi_n \cdots \psi_1}(r) \rightarrow \mathbf{ModBFlag}_n$ and $\mathcal{R}_{\psi, r}(e) = (\mathcal{M}_e, \omega_s; \mathcal{K}_e \subseteq \widehat{\mathcal{K}}_e)$, with

$$\mathcal{H}_0^t \subseteq \cdots \subseteq \mathcal{H}_n^t = \mathcal{M}_t, \quad \text{Exp}_{\omega_t|_{\mathcal{H}_i^t}}^{\mathcal{H}_j^t}(\mathcal{K}_i^t) = \mathcal{H}_i^t \quad (i \leq j).$$

These are functorial under extension isomorphisms, with inner faces/degeneracies from deleting/repeating intermediate pairs. The (RLE) meet can be strict: for $\tau = \text{Tr}/2$, $0 < |c| < 1$ and $\omega_h = \tau((1 + c\sigma_x) \cdot)$, $\mathcal{D}_z \cap \mathcal{D}_y = \mathbb{C}1$ while $C_\omega \mathcal{D}_z = C_\omega \mathcal{D}_y = \mathcal{M}_2$; its RLE shell is nonzero but the reflected-family $\mathbf{AlgRes}$ vanishes.

c. Closure–transport factorization. For closure operators $C_x : \mathcal{L}_x \rightarrow \mathcal{L}_x$ on posets and monotone raw transports $j_e : \mathcal{L}_x \rightarrow \mathcal{L}_y$ with $j_{fe} X \leq j_f j_e X$, put $\Phi_e = C_y j_e$ on $\text{Fix } C_x$. Then (CTF) $C_z j_{fe} X \leq C_z j_f j_e X \leq C_z j_f C_y j_e X = \Phi_f \Phi_e X$, the $n = 2$ (RCS/RFH) two-cut profile; strict raw composition kills only the first component. For strict $F : \mathcal{B}^{\text{op}} \rightarrow \mathbf{Pos}$ set $\mathbf{ClFib}_C(f; y) = [C_a F(f)y, F(f)C_b y]$ and $\widetilde{C}(b, x) = (b, C_b x)$. Then

$$\begin{aligned} \mathbf{ClFib}_C(f; y) \neq \emptyset \ (\forall f, y) &\iff C_a F(f) \leq F(f)C_b \ (\forall f) \\ &\iff F(f) \text{Fix } C_b \subseteq \text{Fix } C_a \\ (\text{CTF}_{\text{fib}}) &\iff \widetilde{C} \dashv \iota_C \text{ over } \mathcal{B}, \quad \text{Fix } \widetilde{C} \simeq_{\mathcal{B}} \int_{\mathcal{B}} \text{Fix } C. \end{aligned} \tag{135}$$

For $\ell = C_a F(f)y$, $u = F(f)C_b y$, this fibre is $[\ell, u] = \{x : \text{Fact}_{1,1}(\ell \rightarrow u; x) \neq \emptyset\}$: fixed-midpoint (FPF) fibres are singleton, $\ell \not\leq u$ gives none, and all are singleton iff $C_a F(f) = F(f)C_b$. Composition gives $C_a F(f)F(g)y \leq F(f)C_b F(g)y \leq F(f)F(g)C_c y$. The chain $0 < 1 < 2$ with $F(f) = I$ and $(C_a, C_b) = (T_\tau, I), (I, T_\tau), (J, J), Jx = x \vee 1$, realizes empty/strict/singleton fibres. Singleton fibres make the adjunction cartesian; its $p = I_B$ fibre is the (RWS) world reflector and selects no section. At $C = I$, the C_2 (PRH) face has two/no point-worlds but one uniform (SIC_P) law-world under swap, with $\mathbf{W}_{BC_2}(\mathbb{Z}[2]) \cong \mathbb{Z}$ and $H_0 \text{LCM}_P \neq 0$ despite exact transport. Nonempty intervals only stabilize fixed transport. In one fixed fibre $\mathcal{L}^C = \text{Fix } C$, for monotone endomaps F, G ,

$$\begin{aligned} \text{Ret}(FG) &\leq \text{Ret}(F) \text{Ret}(G), \\ \text{Ret}(Cj_{fe}) &\leq \text{Ret}(Cj_f j_e) \leq \text{Ret}(\Phi_f \Phi_e) \leq \text{Ret}(\Phi_f) \text{Ret}(\Phi_e), \end{aligned} \quad (136)$$

where the first inequality is $FGX \leq F(\text{Ret}(G)X)$ and the final grade is defined only after all transports land in one fibre.

Let $\mathbf{RefPath}_{\mathcal{A}}$ have objects (M, r) and arrows $e = (\psi, u) : (M, r) \rightarrow (N, s)$, and let $\mathbf{Sup}$ be the locally posetal 2-category of complete lattices and complete-join maps. Set $U_e = u$, $j_e = \text{Ad } U_e : \mathcal{M}_r \xrightarrow{\sim} \mathcal{K}_e \hookrightarrow \mathcal{M}_s$ and $\Phi_e(\mathcal{L}) = \text{Exp}_{\omega_s}(j_e \mathcal{L})$. Then $U_{f \star e} = U_f U_e$, $j_{f \star e} = j_f j_e$, $\Phi_{\text{id}} = I$ and (RLE) preserves joins, hence (MWO) $\Phi : \mathbf{RefPath}_{\mathcal{A}} \rightsquigarrow \mathbf{Sup}$, $\Phi_{f \star e}(\mathcal{L}) \subseteq \Phi_f \Phi_e(\mathcal{L})$. For $\mathfrak{A}_{\text{mod}}(f, e; \mathcal{L}) = \mathfrak{A}_{\square}(\Phi_{f \star e} \Rightarrow \Phi_f \Phi_e; \mathcal{L})$, (CTF/GCS) pastes the binary cells along every chain, with $\Phi_S \mathcal{L} \subseteq \Phi_T \mathcal{L}$ for $S \leq T$, from $\Phi_{\emptyset} = \Phi_{e_{n,0}}$ to $\Phi_{\{1, \dots, n-1\}} = \Phi_{e_n} \cdots \Phi_{e_1}$; modular-flat outer image kills the corresponding edge. Strictness is the explicit $\mathcal{D}_2 \subset M_2 \rightarrow M_2 \oplus M_2$ witness $A \mapsto A \oplus A$, with $h = (\frac{2}{1/2} \frac{1/2}{1})$, $V = \text{diag}(1, i)$, $h_1 = VhV^*$ and target density $\propto h \oplus h_1$: (MAW) $\Phi_{\phi\psi}(\mathcal{D}_2) = \{A \oplus VAV^* : A \in M_2\} \subseteq M_2 \oplus M_2 = \Phi_{\phi} \Phi_{\psi}(\mathcal{D}_2)$.

For separable unital $\mathcal{A}$, countable declared $G \leq \text{Aut}(\mathcal{A})$ and faithful separable π_0 , the regular covariant representation $\pi_G(a)(\delta_g \otimes \xi) = \delta_g \otimes \pi_0(g^{-1}a)\xi$, $U_h(\delta_g \otimes \xi) = \delta_{hg} \otimes \xi$ is faithful/covariant; thus $M_G = \pi_G(\mathcal{A})''$ is σ -finite with separable predual, G acts normally, and every faithful normal ω is admissible but none preferred.

For an ω -preserving expected inclusion $N \subseteq \mathcal{A}$ with ω faithful, let $E_N : \mathcal{A} \rightarrow N$ be the ω -preserving normal expectation. Then

$$\diamond_N p = s(E_N p), \quad \square_N p = 1 - s(E_N(1 - p)).$$

On point presentations these are $\mu_y(p) > 0$ and $\mu_y(p) = 1$; S5, normal join preservation and $\square p \leq p \leq \diamond p$ are inherited from $U0$, not new modal primitives. On separable-predual non-Type-I factors the injective-factor classification below gives the exact injective range and a Cartan; access still requires a realization/read-out, with ambient freedom a typed fibre.

After effective-range/null normalization let $\mathbf{Acc}_{\text{sb}}$ be the resulting standard-Borel presentations. For $\mathfrak{S} \in \mathbf{Acc}_{\text{sb}}$ with kernel P , define

$$\partial P_x = P_x - \delta_x, \quad \kappa_{\mathfrak{S}}(x) = ((w_r(x))_r, L(x), \partial P_x).$$

For a countable determining Boolean algebra $C = \{C_n\}$ put $d_C(x) = (P_x(C_n) - \mathbf{1}_{C_n}(x))_n$ and $\widehat{\kappa}_{\mathfrak{S}, C}(x) = ((w_r(x))_r, L(x), d_C(x))$, and let $\pi_{\mathfrak{S}} : \mathcal{G} \rightarrow \mathcal{G}_{\mathfrak{S}}^{\text{red}}$ be the effective-range/null quotient of $\sigma(\widehat{\kappa}_{\mathfrak{S}, C})$. Since C determines finite signed measures, its fibres agree mod null sets with equality of $\kappa_{\mathfrak{S}}$, so

$$R_{\pi_{\mathfrak{S}}} = \left(\bigcap_r R_{w_r} \right) \cap R_L \cap R_{\partial P},$$

independently, up to the declared standard-Borel/measure-algebra isomorphism, of C .

Centered rigidity. If $\partial P_x = \partial P_y$, then either $x = y$ or $P_x = \delta_x$ and $P_y = \delta_y$; hence the centered kernel is constant exactly for identity dynamics, and every determining d_C generates the same centered-carrier σ -algebra.

Descent. Record, extendability and impact are $q_r w_r$, $\mathbf{1}_{G_r} w_r$ and $\text{Imp}_r w_r$, while (P, Λ) is a global typed attachment. Thus the quotient carries every state-dependent field; centered rigidity makes $P_{\pi_{\mathfrak{S}}(x)}^{\text{red}} = (\pi_{\mathfrak{S}})_* P_x$ well defined and $(\pi_{\mathfrak{S}})_* \mu^{\text{red}} \sim (\pi_{\mathfrak{S}})_* \mu$. Hence $\text{Red}(\mathfrak{S}) \in \mathbf{Acc}_{\text{sb}}$, with compact realization from clause (2).

Reflector law. $U0$ fixed-locus factorization makes Red idempotent up to measure isomorphism and left adjoint to inclusion of reduced presentations: every $\varphi : \mathfrak{S} \rightarrow \mathfrak{I}$ with reduced $\mathfrak{I}$ factors uniquely through $\pi_{\mathfrak{S}}$. Clause (2) transports $\kappa_{\mathfrak{S}}$, so Red is canonical up to standard-Borel/measure-algebra isomorphism.

Topological/Feller realization residual. Let $U : \mathbf{Pol} \rightarrow \mathbf{SB}$ forget topology and $\text{TopFib}(e) = \text{HFib}_e((U^{[1]})^{\approx})$. For $S = \mathbb{N} \cup \{\infty\}$, the same Borel arrow $e = 1_S$ has realizations $S_{\text{disc}} \rightarrow S_{\text{disc}}$ with l.s.c. singleton fibres and $S_{\text{disc}} \rightarrow S_{\text{conv}}$ whose singleton-fibre map fails l.s.c. exactly at ∞ ; both Borel structures are $\mathcal{P}(S)$ and the targets are nonhomeomorphic. Hence continuation is a genuine realization coordinate. Let $V_{\text{Fel}} : \mathbf{FelPres} \rightarrow \mathbf{Acc}_{\text{sb}}$ forget the compact-metrizable full-support topology from Feller Markov presentations (arrows are marked homeomorphic Markov conjugacies), and put

$$\text{FelFib}(\mathfrak{S}) := \text{HFib}_{\mathfrak{S}}(V_{\text{Fel}}^{\approx}).$$

The commutative Markov representation makes this fibre inhabited on the present perimeter; forgetting the Feller/kernel mark gives the underlying (TPR) topology point on the chosen standard-Borel representative. In general it is neither terminal nor the whole topology fibre. Indeed let $S = \{a, b\} \sqcup \{e_n, o_n : n \geq 1\}$, $\mu(a) = \mu(b) = 1/4$ and $\mu(e_n) = \mu(o_n) = 2^{-n-2}$. In τ_1 the interleaved sequence $e_1, o_1, e_2, o_2, \dots$ converges to a and b is isolated; in τ_2 , $e_n \rightarrow a$ and $o_n \rightarrow b$. Both are compact metrizable, have Borel algebra $\mathcal{P}(S)$ and full-support μ , and the identity kernel is Feller for both, but they are nonhomeomorphic (one versus two nonisolated points). For the same Borel involution $R(a) = b, R(b) = a, R(e_n) = o_n, R(o_n) = e_n$, R is a τ_2 -homeomorphism but is not τ_1 -continuous; hence its deterministic Markov kernel is Feller only on τ_2 . Thus Feller admissibility is a topology-sensitive support locus, not a measurable invariant or selector. For morphisms there is a complementary rigidity: (FML) V_{Fel} is faithful. Equivalently, for fixed points $\mathfrak{S} \in \text{FelFib}(\mathfrak{S})$ and $\mathfrak{I} \in \text{FelFib}(\mathfrak{I})$ and $u : \mathfrak{S} \rightarrow \mathfrak{I}$, the lift fibre $\{\tilde{u} : \mathfrak{S} \rightarrow \mathfrak{I} : V_{\text{Fel}}(\tilde{u}) = u\}$ is empty or singleton. Indeed two such lifts have continuous underlying maps equal μ -a.e.; their disagreement set is open, so full support forces it empty. On the constant-mark zero-residual identity-dynamics presentation on $([0, 1], \lambda)$, (NPF) gives many literal representatives of $[1]$ by null-point modification, while (FML) leaves the identity as its unique continuous lift. Fullness fails on the same presentation: the measure-preserving Borel rotation $r(x) = x + 1/2$ for $x < 1/2$ and $r(x) = x - 1/2$ for $x \geq 1/2$ is a legal measurable access arrow, but no member of its a.e. class is continuous in the interval topology, since the two one-sided limits at $1/2$ are 1 and 0. Thus (PRF/NPF) representative multiplicity collapses on a fixed full-support Feller morphism locus to a pure existence obstruction, while the object-level (FTRF) topology fibre remains nonterminal. On compact-Hausdorff points (PCRG) preserves continuation under probability convexification, while (LPA/ELR) is topology-forgetting invariant.

Neutral stratum. *Reduction is observationally inert exactly when all w_r are constant, $L = 0$ a.e., reachable extendability is full, $P = \text{id}$ a.e., $P = \Lambda = 0$, and every reachable impact vanishes. The static (PI/P2) tower cannot absorb dynamics: on the constant inverse system of (X, μ) all μ -preserving deterministic kernels $P_x^S = \delta_{Sx}$ have the same projective data, while Red reads $\delta_{Sx} - \delta_x$; identity is neutral and nontrivial S is not. Thus the centered transition field is the missing positive carrier. Any nonzero constant cost or other non-neutral typed attachment likewise keeps reduction non-inert.*

Proof. Clauses (1)–(3) are (DD/DI), the commutative Markov representation and (PI). Clause (4) uses the skeleton, Theorem VIII.1, residual normal form, (RVN/NRF–HRF/CRF, TEXT, CTF); Fewster–Verch excludes a natural section and the displayed M_2 witnesses make reflection/oplaxity strict. Boundary-kernel equality and (U0) give centered rigidity and the reflector law; determination/reducedness plus pushforward give descent and nonsingularity. For (TPR), both displayed topologies have Borel algebra $\mathcal{P}(S)$, whereas $\{\infty\}$ is target-open only in the discrete realization, proving the l.s.c. separation. For (FTRF), the C^* -core/Gelfand construction of clause (2) supplies a compact-metrizable full-support Feller representative. In the displayed S , all subsets are Borel and every point has positive μ -mass; τ_1 has one nonisolated point and τ_2 two. The involution R preserves μ and swaps the two convergent components in τ_2 , while $e_n \rightarrow a$ but $R(e_n) = o_n \rightarrow a \neq b = R(a)$ in τ_1 , proving the Feller-support separation. For (FML), equality a.e. of continuous maps into a Hausdorff target has closed equality locus; full support forbids a nonempty open complement. The displayed half-rotation preserves Lebesgue measure, and any continuous representative equal to it a.e. would agree with the two continuous branches on dense subsets of the two half-intervals, forcing incompatible one-sided limits at $1/2$. $\square$

For an effective datum D on its declared standard-Borel probability base (X, ν) , put

$$x R_{\mathcal{E}_D} x' \iff \mathbf{1}_A(x) = \mathbf{1}_A(x') \quad (A \in \mathcal{E}_D), \quad R_D^{\text{eff}} := R_{\mathcal{E}_D} \cap R_{L_D} \cap R_{\partial \mathcal{P}_D}, \quad \partial \mathcal{P}_{D,x} := \mathcal{P}_{D,x} - \delta_x,$$

and let $\text{Red}(D)$ be the normalized standard-Borel effective-range quotient by R_D^{eff} . Since $\mathcal{Q}_D \subseteq \mathcal{E}_D$, while G_D, I_D factor through $\mathcal{E}_D$ and (P, Λ) are global typed attachments, clause (1) and centered rigidity identify this intrinsic quotient with the reduction of any standard-Borel access presentation of D supplied by clauses (1)–(4); no presentation selector is asserted. Hence write $D \sim_{\text{obs}} E$ iff $\text{Red}(D) \cong \text{Red}(E)$. Theorem II A 0 d, (PI/PUI/P2), and the typed complement of Theorem S7 make the construction independent of the determining algebra. Product completion keeps canonicity presentation-relative: dynamics may activate a residual fiber. The classification restricts to the non-inert stratum; inert effective data share the neutral one-point reduction. It classifies objects, not an undeclared morphism category.

VIII. TIME, MEMORY, AND MODULAR ENVELOPES

For $\mathfrak{S} \in \mathbf{Acc}$, write $\mathcal{N}_r := w_r^* L^\infty(Y_r, w_r \mu)$, $\mathcal{Q}_r := (q_r \circ w_r)^* L^\infty(Z_r, (q_r \circ w_r)_* \mu)$, $e_r := \mathbf{1}_{G_r} \circ w_r$.

A. Natural extension of the dynamics

For the Borel kernel P with $\mu P \sim \mu$, let $(\mathbb{P}_\mu, \theta)$ be its one-sided nonsingular path system. The shift θ is a nonsingular transformation of its probability path space, hence Markovian for its Radon–Nikodym cocycle; the inverse-limit natural-extension theorem is then canonical⁴⁴ (Def. 5.8, Ex. 1.3(a), Thm. 5.9 and Rem. 5.10(b)) and yields $\text{Smrti}(\mathfrak{S}) = (\widehat{\mathcal{G}}, \widehat{\mu}, \widehat{S})$. Since the time-zero pushforward is $\mathbb{P}_\mu$, $\widehat{\mu}$ is a probability; deterministic noninvertible/invertible maps give inverse limits/themselves. The bilateral measure is the nonsingular extension measure, and temporal crossed products use $\widehat{S}$, not P .

a. Finite-history innovation. Let $r_0 : \widehat{\mathcal{G}} \rightarrow \mathcal{G}$ be the time-zero factor (identity, inverse-limit projection, or path-space zero coordinate), $\mathcal{F}_n = \sigma(r_0 \widehat{S}^k : |k| \leq n)$, $\mathcal{A}_n = L^\infty(\mathcal{F}_n, \widehat{\mu})$ and $\widetilde{E}_n = E_n^{\text{hist}} E : \mathcal{M}_{\text{time}} \rightarrow \mathcal{A}_n$, with Fourier expectation E and conditional expectation E_n^{hist} ; write $\omega_{\mathfrak{S}}(x) := \int E(x) d\widehat{\mu}$. Natural-extension minimality and modular triviality give (HIF) $\mathcal{A}_n \in \text{Lat}_{\omega_{\mathfrak{S}}}(\mathcal{M}_{\text{time}})$, $\mathcal{A}_n \nearrow \widehat{\mathcal{A}}$, $\bigvee_{n < \omega} \mathcal{A}_n = \widehat{\mathcal{A}}$. For $m, N < \omega$ put $\text{MObs}_{m,N} := \overline{\text{span}}^{w^*} \{u^{-r} a u^s : a \in \mathcal{A}_m, |r|, |s| \leq N\}$, a weak- $*$ closed operator system. HIF, crossed-product Fourier density and (NAT/GEP) give (MGO) $\bigvee_{m,N} \text{MObs}_{m,N} = \mathcal{M}_{\text{time}}$ and $\bigcap_{m,N} R_{\text{MObs}_{m,N}}^n = \Delta$. No finite Fourier horizon exists: in $\mathbb{C} \rtimes \mathbb{Z} \cong L^\infty(\mathbb{T})$, 1 and $1 + \varepsilon \cos((N+1)\theta)$ agree for $|k| \leq N$. With $\mathcal{A}_\omega = \widehat{\mathcal{A}}$, (PBG) gives the horizontal Rd ERHist($\mathcal{A}_\omega$) read-out while $\widehat{S}$ stays attached; BKM= L^2 gives $B_n^{\text{hist}} = b_{n-1:n}$ and $\|f - \widehat{\Pi}_n f\|_2^2 = \sum_{j>n} \|B_j^{\text{hist}} f\|_2^2$, with the displayed e/RLH/GRR–SRF tails and independent IHC/HKC clock/flag fibre. For $\kappa = \min\{\alpha : \mathcal{A}_\alpha = \widehat{\mathcal{A}}\}$, any plateau is terminal by natural-extension minimality and PGF retains the law fibre. The RHA uniform/Bernoulli witness has $\kappa = 0/\omega$ with nonzero shells at every positive Bernoulli depth, retaining the relative-past attachment.

B. Continuous-time realization fibers

a. Temporal lift fiber and rigidity loci. For a declared realization category $\mathcal{C}$, the time-one map $F_1 : \text{Flow}_{\mathcal{C}} \rightarrow \text{End}_{\mathcal{C}}$ has fiber $\text{TimeFib}_{\mathcal{C}}(T) = F_1^{-1}(T)$. In a strong-closed contraction monoid $\mathcal{E} \subset B(X)$, for $n \geq 1$ let $\text{TimeFib}_{\mathcal{E}}^0(T)$ be the C_0 -semigroup fibre at $T_1 = T$ and put $\mathcal{R}_n(T) = \{S \in \mathcal{E} : S^{n!} = T\}$, $\pi_{n+1,n}(S) = S^{n+1}$, $\text{Root}_{\mathcal{E}}(T) = \varprojlim_n \mathcal{R}_n(T)$. Write $\mathbf{BE}$ for the one-object category of $\mathcal{E}$ and $\text{DiagFact}_{\mathbf{BE}}^{\mathcal{E}}(T) := \{(S, \dots, S) \in \text{Ob Fact}_{\mathbf{BE}}^{[m]}(T)\}$. Then the output/decomposition face of (R2C/R2F) gives (TFC) $\mathcal{R}_n(T) \cong \text{DiagFact}_{\mathbf{BE}}^{\mathcal{E}}(T)$, $\pi_{n+1,n} \leftrightarrow \text{blk}_{n+1}$, $\text{Root}_{\mathcal{E}}(T) \cong \varprojlim_n \text{DiagFact}_{\mathbf{BE}}^{\mathcal{E}}(T)$, where blk_{n+1} groups consecutive blocks of $n+1$ equal factors. For $r = (S_n)$ set $U_{k/n!}^r = S_n^k$ and $\text{Land}_{\mathcal{E}}(T) = \{r : U_q^r \xrightarrow[q \downarrow 0]{s} I\}$. Then (TRP) $\text{TimeFib}_{\mathcal{E}}^0(T) \cong \text{Land}_{\mathcal{E}}(T) \subseteq \text{Root}_{\mathcal{E}}(T)$. Restriction to $1/n!$ and rational-semigroup extension prove (TRP), while $\|U_q x - U_p x\| \leq \|U_{q-p} x - x\|$ identifies landing with the C_0

condition. If X is Banach, $K \subset X$ is closed and $q_K : X \twoheadrightarrow X/K$, then $\mathcal{E}_K := \{S \in \mathcal{E} : SK \subseteq K\} = \mathcal{E} \cap \text{Stab}(q_K)$ is strong-closed; applying (TRP/QAF/BCR) gives (TDF) $\text{TimeFib}_{\mathcal{E}_K}^0(T) \cong \{(S_n) \in \text{Land}_{\mathcal{E}}(T) : S_n K \subseteq K \forall n\} = \{(S_n) \in \text{Land}_{\mathcal{E}}(T) : \mathfrak{E}_{q_K}(q_K S_n) = 0 \forall n\}$. Closedness of K promotes factorial-rational invariance to all t , so these and only these lifts descend to C_0 contraction semigroups on X/K . For the full contraction monoid, $T = I$: on $\mathbb{R}^2$ with $K = \mathbb{R}e_2$, planar rotations show the inclusion in $\text{TimeFib}_{\mathcal{E}}^0(I)$ can be strict, while on $\mathbb{R} \oplus \mathbb{R}^2$ with $K = 0 \oplus \mathbb{R}^2$, the lifts $1 \oplus R_{2\pi m t}$ are distinct and compatible with the same quotient flow. Compact nonempty root levels reduce existence to landing by Theorem III.1, with (PQTF) when the bonds are onto. Put $L_T = \text{Land}_{\mathcal{E}}(T)$, let p_n be the root coordinates and $\widehat{L}_T^{\text{fin}} = \lim_{\leftarrow n} p_n(L_T)$, $\text{TempRes}_{\mathcal{E}}(T) = \widehat{L}_T^{\text{fin}} \setminus L_T$. By (RCD/CRX/PFR_{der}/PEL/SCX/TRP), (TCR) $\text{TimeFib}_{\mathcal{E}}^0(T) \cong L_T$, $\text{TempRes}_{\mathcal{E}}(T) = \text{LimRes}(L_T)$, $\text{ExtRes}(J_{L_T}) = H_0 \text{SCX}(L_T, p_\bullet) \cong \mathbb{Z}^{(\text{TempRes}_{\mathcal{E}}(T))} \hookrightarrow \text{Hist}_{\mathbb{N}}^1(K_\bullet^{L_T})$. This is the $a = I_{L_T}$ pure- H_0 face of (CRX); its H_1 vanishes because the full root sequence already separates points of L_T . Moreover $\widehat{\text{JKey}}_{J, I_{L_T}} \cong L_T$ and $\iota_{J, I_{L_T}}$ is onto, so (GTS) identifies $\text{TempRes}_{\mathcal{E}}(T)$ with its terminal target-extinction term, not with a joint-key ghost; (DJK2) identifies the same term with the first derived birth and keeps the tail excess as the quotient. The two-state flip witnesses missing finite roots, while $U_t^{(m)} f(z) = f(e^{-2\pi i m t} z)$ on $L^2(S^1)$ gives distinct landed lifts. For a genuinely pro-only defect, on S^1 with Haar law put $R_\theta f(z) = f(e^{-2\pi i \theta} z)$, $\mathcal{E}_{\text{Fel}} = \{A \in B(C(S^1)) : A \geq 0, A1 = 1\}$, $S_n = R_{\theta_n}$, $\theta_1 = 0$ and $\theta_{n+1} = (\theta_n + \lfloor (n+1)/2 \rfloor)/(n+1)$. Then $S_{n+1}^{n+1} = S_n$, $S_n^1 = I$ and $1/2 \leq \theta_n \leq 5/8$ for $n \geq 2$. With $m_N = N! \theta_N \in \mathbb{Z}$, $U_t^{(N)} = R_{m_N t}$ lies in $\mathcal{E}_{\text{Fel}}$, has $U_1^{(N)} = I$ and $U_{1/n!}^{(N)} = S_n$ for all $n \leq N$, but for $f_1(z) = z$, $\|S_n f_1 - f_1\|_\infty = |e^{-2\pi i \theta_n} - 1| \not\rightarrow 0$. Hence $(S_n)_n \in \text{TempRes}_{\mathcal{E}_{\text{Fel}}}(I)$ although every finite truncation is realized by a genuine flow: no finite diagonal factorization depth detects C_0 landing. Coherence, finite satisfiability, realization and continuity remain distinct.

On the positive spectral/OS locus, $T \geq 0$ self-adjoint selects the unique nonnegative-self-adjoint semigroup on $s(T)\mathcal{H}$: with $A = -\log(T|_{s(T)\mathcal{H}})$, $\text{OSFib}(T) := \{(e^{-tB})_{t \geq 0} : B = B^* \geq 0, e^{-B} = T|_{s(T)\mathcal{H}}\} = \{(e^{-tA})_{t \geq 0}\}$. A full-space nonnegative-self-adjoint $T = e^{-A}$ exists iff $\ker T = 0$. This spectral selector does not collapse the ambient TimeFib : for $T = I$ the rotation groups $U^{(m)}$ above remain distinct temporal lifts, while only $m = 0$ lies in $\text{OSFib}(I)$.

For a reversible Markov kernel with two-sided path law and half-step reflection, let $\mathcal{H}_+ = L^2(\sigma(X_k : k \geq 1), \mathbb{P})$, let G_1 be conditional expectation at X_1 , and let $\iota_1 h = h(X_1)$. Markovity and reversibility give $\mathbb{E}[\overline{F} \circ \theta_{1/2} F] = \langle G_1 F, T G_1 F \rangle$, so reflection positivity on $\mathcal{H}_+$ is equivalent to $T \geq 0$. Since $G_1 \iota_1 = I$, $\mathcal{N}_{\text{OS}} = \ker G_1 \oplus \iota_1(\ker T)$, $\mathcal{H}_{\text{OS}} \cong s(T)L^2(\mu)$, $T|_{s(T)L^2(\mu)} = e^{-H}$, $H \geq 0$. The future restriction is essential: for the independent two-point kernel, $F = f(X_0) - f(X_1)$ has $F \circ \theta_{1/2} = -F$ on the bilateral space. Future prediction and one-time spectral null therefore remain distinct.

Positivity still does not imply Markov embeddability. The symmetric stochastic matrix $T = 32^{-1} \begin{pmatrix} 19 & 3 & 10 \\ 3 & 19 & 10 \\ 10 & 10 & 12 \end{pmatrix}$ has spectrum $\{1, 1/2, 1/16\}$ but $(T^{1/4})_{12} = 5/12 - 2^{-5/4} < 0$ and $(\log T)_{12} = -(\log 2)/6 < 0$; simple positive spectrum makes the principal logarithm its only real logarithm. For a finite-state μ -reversible Markov matrix, let $\text{TimeFib}_{\text{revM}}(T)$ be the reversible continuous-time Markov semigroups with time-one map T . Then the set-level (RST) fiber is exactly

$$[\text{FRM}] \quad \text{TimeFib}_{\text{revM}}(T) = \begin{cases} \{(e^{t \log T})_{t \geq 0}\}, & T > 0 \text{ on } L^2(\mu) \text{ and } (\log T)_{ij} \geq 0 \ (i \neq j), \\ \emptyset, & \text{otherwise.} \end{cases} \quad (137)$$

Indeed a reversible generator is self-adjoint, so $e^G = T$ forces $G = \log T$; conversely the displayed off-diagonal condition, $Q1 = 0$ and detailed balance make $Q = \log T$ the reversible rate matrix. A kernel of T forbids any matrix-exponential lift. Hence the OS-selected lift is either the unique reversible Markov lift or none exists; for simple positive spectrum the same criterion decides unrestricted real finite-state Markov embeddability. Full field-theoretic OS reconstruction uses the Euclidean Green-function hierarchy, not T alone; deterministic constant fields $\phi \equiv c$ have $T_t = I$ for every $c \in \mathbb{R}$ but $S_n \equiv c^n$, so one-time/semigroup data do not determine field correlators⁴⁵.

Finally, for a standard pointed pair $(\mathcal{M}, \Omega)$ let $\text{Bor}_+(\mathcal{M}, \Omega)$ consist of $U(s) = e^{isP}$, $P \geq 0$, fixing Ω and satisfying $U(s)MU(-s) \subseteq \mathcal{M}$ for $s \geq 0$. Its half-sided rigidity is the (BW/BO) read-out below; this locus remains type-distinct from path/OS positivity and from the Radon–Nikodym modular clock.

C. Temporal envelope, modular clock, and gauge dissipation

a. Reference-asymmetry cocycle. The two-sided history carrier already generates the clock. Use $\widehat{\mu} \circ \widehat{S}^n := (\widehat{S}^{-n})_* \widehat{\mu}$. For $n \in \mathbb{Z}$ put $c_{\widehat{\mathfrak{S}}}(n) := \log \frac{d\widehat{\mu} \circ \widehat{S}^n}{d\widehat{\mu}}$. The Radon–Nikodym chain rule gives $c_{\widehat{\mathfrak{S}}}(n+m) = c_{\widehat{\mathfrak{S}}}(n) + c_{\widehat{\mathfrak{S}}}(m) \circ \widehat{S}^n$, $c_{\widehat{\mathfrak{S}}}(1) = \Theta_{\widehat{\mathfrak{S}}} = \text{Kala}(\widehat{\mathfrak{S}})$. Thus Kala is Smrti's generator-one cocycle read-out.

For $\mathfrak{S} = (\text{Smrti}(\widehat{\mathfrak{S}}), r_0, \pi)$, Smrti determines $\text{Kala}(\widehat{\mathfrak{S}})$ by the Radon–Nikodym chain rule, whereas $\text{Samskara}_{r_0}(\pi)$ also uses the access mark. Neither read-out factors through the other: the bilateral fair Bernoulli shift has $\text{Kala} = 0$ but, for $\pi^0 = x_0, \pi^- = x_{-1}$, $\text{Samskara}_{r_0}(\pi^0) = 0 \neq \text{Samskara}_{r_0}(\pi^-)$ by (CDR), while on $(\{0, 1\}, \mu_p)$, $p \neq 1/2$, identity and flip with $\pi = r_0$ both have zero Samskara but the distinct Kala values displayed below. Hence the lossless BPR/PFP provenance datum does not generate the required access/dynamic marks: those marks remain typed attachments. Any generic origin-to-clock theorem therefore needs declared marked nonsingular realization data sufficient to determine the cocycle; Smrti is the sufficient carrier, where Kala is derived and needs no independent clock primitive.

Invertible-face history–clock obstruction. On the invertible deterministic face the natural extension is the system itself and $r_0 = \text{id}$. Hence the first HIF algebra is already the whole history algebra: (IHC) S invertible $\implies \sigma(r_0) = \mathcal{B}(\widehat{\mathfrak{G}})$, $\mathcal{A}_0 = \widehat{\mathcal{A}}$, $\kappa = 0$. Thus finite-window history resolution is blind to the invertible temporal action, while its cocycle and conjugacy class can still vary. The three non-implications are sharp on two points. Let $\mathcal{G} = \{0, 1\}$ and $\mu_p(0) = p, \mu_p(1) = 1 - p$ with $0 < p < 1$. The identity and flip $S_{\mathfrak{H}}(x) = 1 - x$ have the same terminal HIF flag by (IHC). For $p \neq 1/2$, however, $\text{Kala}(S_{\text{id}}) = 0$, $\text{Kala}(S_{\mathfrak{H}})(0) = \log \frac{1-p}{p}$, $\text{Kala}(S_{\mathfrak{H}})(1) = \log \frac{p}{1-p}$, so HIF does not determine Kala. Conversely, at $p = 1/2$ the identity and the reset kernel $P_x = \mu_{1/2}$ both have $\text{Kala} = 0$, whereas the identity has $\kappa = 0$ and the bilateral Bernoulli natural extension of the reset has the strict finite-window flag $\mathcal{F}_n = \sigma(X_k : |k| \leq n)$ for every finite n , hence $\kappa = \omega$. Finally, at $p = 1/2$ the identity and flip have the same HIF flag and the same $\text{Kala} = 0$ but are not conjugate: the identity

fixes both atoms and the flip fixes none. Therefore (HKC) $\text{HIF} \not\Rightarrow \text{Kala}$, $\text{Kala} \not\Rightarrow \text{HIF}$, $(\text{HIF}, \text{Kala}) \not\Rightarrow [\widehat{S}]_{\text{conjugacy}}$. The residual is temporal action/realization data already retained by (PGF/TLO), not a further scalar clock coordinate.

Theorem VIII.1 derives the modular action, reference-preservation zero locus and one-step entropy price from $\text{MEnv}(\mathbf{P}, \mu)$. It is non-vacuous: for $S(x) = x/(2-x)$ on $[0, 1]$, $d(dx \circ S)/dx = 2/(2-x)^2 \neq 1$ on positive measure. Its ratio-set/type remains a separate invariant of the unmarked envelope.

D. Classification of temporal envelopes

a. Unmarked-envelope classification. Krieger type and associated-flow classification. For an ergodic nonsingular natural extension on a standard non-atomic space, write $r(\widehat{S}) := r_{\text{ext}}(\widehat{S}) \cap [0, \infty)$ for the finite part of Krieger's extended ratio set (thus suppressing $+\infty$). The Krieger–Connes–Takesaki classification gives

$$\begin{array}{l|l} r(\widehat{S}) = \{1\} & \text{semifinite,} \\ r(\widehat{S}) = \{0, 1\} & \text{III}_0, \\ r(\widehat{S}) = \{0\} \cup \lambda\mathbb{Z}, 0 < \lambda < 1 & \text{III}_\lambda, \\ r(\widehat{S}) = [0, \infty) & \text{III}_1. \end{array} \quad (138)$$

For temporal classification we use^{46–49} and⁵⁰ (Secs. 1–2, Thms. 3.2, 4.3 and Sec. 5): $(\text{KCT}_{\text{type}})$ is the displayed finite extended-ratio-set/type table; $(\text{KCT}_{\text{mod}})$ identifies the associated flow of an essentially free ergodic $\mathbb{Z}$ -action with the Connes–Takesaki flow of weights, complete for orbit equivalence on III_0 and periodic/trivial on $\text{III}_\lambda/\text{III}_1$; $(\text{KCT}_{\text{real}})$ realizes every properly ergodic flow by an ergodic nonsingular III_0 $\mathbb{Z}$ -action. Realization is not identification of an action with its flow, and ratio-set type does not determine the III_0 flow class. Takesaki duality⁵¹ (Thm. 4.5) gives $\mathcal{M}_{\text{time}} \rtimes_{\delta} \mathbb{T} \cong L^{\infty}(\widehat{\mathcal{G}}, \widehat{\mu}) \otimes \mathcal{B}(\ell^2\mathbb{Z})$, dual averaging onto the base, and a two-periodic duality orbit up to Type-I amplification; type/flow is a crossed-product, not amplified-base, readout. *Injective factor range.* Haagerup's injective–hyperfinite theorem, the Connes/Haagerup classification, and Connes's single-automorphism theorem^{52–55}, together with the atomic shift face below, give

$$\begin{aligned} & \mathcal{M} \text{ a separable-predual factor:} \\ & \mathcal{M} \text{ injective and } \mathcal{M} \not\cong M_n(\mathbb{C}) \ (1 \leq n < \infty) \\ \iff & \exists (X, \mu, S) \text{ standard probability, } S \text{ ergodic, essentially free and nonsingular,} \\ & \mathcal{M} \cong L^{\infty}(X, \mu) \rtimes_S \mathbb{Z}. \end{aligned} \quad (139)$$

For the range statement, Haagerup's injective–hyperfinite theorem⁵² makes an injective separable-predual factor hyperfinite. Choose an increasing finite-dimensional weakly dense approximation (N_k) ; each finite-dimensional unital von Neumann subalgebra is the range of a normal conditional expectation (choose matrix units in each summand and normal corner states). Thus the hypothesis in Connes' single-automorphism criterion⁵⁴ is met, and every infinite such factor is $A \rtimes_{\alpha} \mathbb{Z}$ with A abelian. Separability identifies A with $L^{\infty}(X, \mu)$ for a standard probability space and α with a nonsingular automorphism S . Factoriality forces ergodicity. Moreover $\text{Fix}(S^n)$ is S -invariant for every $n \neq 0$; if it had positive measure, ergodicity would make $S^n = \text{id}$ a.e., so the canonical u^n would be a nonscalar central unitary. Hence S is essentially free. Conversely Connes' criterion gives an expected finite-dimensional exhaustion for every factor $L^{\infty}(X, \mu) \rtimes_S \mathbb{Z}$ of this form, hence hyperfiniteness and therefore injectivity by Haagerup. The Connes/Haagerup classification^{53,55} identifies the resulting II/III isomorphism classes; the III_0 crossed-product realization is already explicit in⁵⁴. For $A = L^{\infty}(X)$ the Fourier expectation is faithful, $uAu^* = A$, $W^*(A, u) = A \rtimes_S \mathbb{Z}$, and $x = \sum_n a_n u^n \in A'$ forces a_n onto $\text{Fix}(S^n)$, hence $a_n = 0$ for $n \neq 0$: A is Cartan (FZC). The shift on $X = \mathbb{Z}$ gives the Type I_{∞} boundary $B(\ell^2\mathbb{Z})$ via matrix units $p_i u^{i-j} p_j$. A finite matrix factor cannot occur: essential freeness is impossible on a finite ergodic atomic $\mathbb{Z}$ -space, while every infinite standard base yields an infinite-dimensional crossed product. Thus precisely the finite matrix factors are excluded.

By $(\text{KCT}_{\text{real}})$, every properly ergodic flow (every $\mathbb{R}$ -orbit null) is realized by an ergodic nonsingular $\mathbb{Z}$ -action of type III_0 ; $(\text{KCT}_{\text{mod}})$ gives the periodic and trivial associated-flow boundaries for III_λ and III_1 , while finite Type-I and non-injective factors lie outside the range above.

By the finite typed witness cube S7 and DI/U0, a skeleton field is removable exactly on its determination closure; no external-encoding claim is made. On the ergodic non-atomic stratum put $\text{Stat}(\mathfrak{G}^{\#}) = [(\mathcal{A}, \omega; \{Q_r \subseteq \mathcal{N}_r, e_r\}_r; \ell)]_{\cong}$ (state-preserving), $\text{Mod}(\mathfrak{G}^{\#}) = [\text{AssocFlow}(\widehat{S})]_{\cong}$ and $\text{KCEnv}(\mathfrak{G}^{\#}) = (\text{Stat}(\mathfrak{G}^{\#}); \text{type}(\mathcal{M}_{\text{time}}), \text{Mod}(\mathfrak{G}^{\#}); (P, \Lambda))$. Here $(\text{KCT}_{\text{type}}/\text{KCT}_{\text{mod}})$ is exactly the orbit-equivalence datum and Feldman–Moore⁵⁶ (Thm. 1) reconstructs the regular-MASA/expectation Cartan datum. On the free- $\mathbb{Z}$ leg every normalized $L^{\infty}(X, \mathbb{T})$ -valued 2-cycle is a coboundary: $b_0 = 1$, $b_{n+1} = b_n \sigma(n, 1)^{-1}$ gives $\sigma(n, m) = b_n \alpha_n(b_m) b_{n+m}^{-1}$ (FMC). Thus, for componentwise KC equality, $[\mathfrak{G}^{\#}]_{\cong} \twoheadrightarrow [\mathfrak{G}^{\#}]_{\sim_{\text{KC}}} \xrightarrow{\sim} \text{im}(\text{KCEnv})$, with no imposed compatibility between static isomorphism and orbit equivalence.

For $c \in \text{im}(\text{KCEnv})$ put $\text{KCFib}(c) = \{[\mathfrak{G}^{\#}]_{\cong} : \text{KCEnv}(\mathfrak{G}^{\#}) = c\}$. Choose a standard measured-relation representative $(Y, \nu, \mathcal{R}_c)$ and let $\text{RAut}(\mathcal{R}_c) \leq \text{Aut}_{\text{ns}}(Y, [\nu])$ be the subgroup preserving $\mathcal{R}_c$. Let $\text{FrPlac}(c)$ consist of $(\eta, S, \mathcal{B}^{\#})$ with $\eta \sim \nu$, $S \in \text{Aut}_{\text{ns}}(Y, [\nu])$, $\mathcal{R}_S = \mathcal{R}_c$, and $\mathcal{B}^{\#} \subset L^{\infty}(Y, \eta)$ a state-preserving marked copy of the static component with (P, Λ) fixed, satisfying for $\alpha_S(f) = f \circ S$, $T_{\mathcal{B}} = E_{\mathcal{B}}^{\eta} \alpha_S|_{\mathcal{B}}$,

$$L^{\infty}(Y, \eta) = W^*(\alpha_S^n \mathcal{B} : n \in \mathbb{Z}), \quad \eta(a_0 \alpha_S(a_1) \cdots \alpha_S^n(a_n)) = \eta|_{\mathcal{B}}(a_0 T_{\mathcal{B}}(a_1 T_{\mathcal{B}}(\cdots T_{\mathcal{B}}(a_n) \cdots))).$$

for every $n \geq 0$ and $a_j \in \mathcal{B}$.

Simultaneous push-forward/conjugation gives $(\text{KFP}) \text{KCFib}(c) \cong \text{FrPlac}(c)/\text{RAut}(\mathcal{R}_c)$, $(\text{KFG}) \text{KFG}(c) := \text{RAut}(\mathcal{R}_c) \rtimes \text{FrPlac}(c)$, $\pi_0 \text{KFG}(c) \cong \text{KCFib}(c)$. Its object isotropy is the exact relation-automorphism stabilizer, so by (HRS) the set classifier is groupoid-lossless iff all stabilizers vanish. Natural extension supplies the frame; conversely the moments reconstruct every finite path law, bilateral generation the minimal natural extension, and the marked copy plus (P, Λ) the skeleton, with precisely the RAut orbits. For ergodic S and $\eta = h\nu$, $c_{\eta}^S(n) - c_{\nu}^S(n) = \log h \circ S^n - \log h$, $c_{\rho}^S(n) = \log d(\rho \circ S^n)/d\rho$; equality makes h invariant up to scale, so normalization recovers it. Reference density is therefore the one-leg (JRN) coordinate.

The time-zero mark is irreducible: for a bilateral Bernoulli shift, $\mathcal{B}_+ = L^{\infty}(\sigma(x_k : k \geq 0))$ with $\alpha_S(\mathcal{B}_+) \subsetneq \mathcal{B}_+$ yields noninvertible $T = \alpha_S|_{\mathcal{B}_+}$, while the full algebra yields invertible $T = \alpha_S$; static probability algebra, minimal bilateral extension, orbit-equivalence datum and trivial residual data agree; since invertibility of T is a skeleton-isomorphism invariant, generator placement alone does not classify KCFib .

For the automorphic full-time-zero face put $\text{KCFib}_{\text{det}}(c) = \{[\mathfrak{G}^{\#}] \in \text{KCFib}(c) : T \text{ automorphic}\}$, where automorphic means induced by a nonsingular transformation. Choose a static representative $\mathfrak{B}_c$ on (X, μ) and a relation representative $(Y, [\nu], \mathcal{R}_c)$, set $H_c = \text{Aut}(\mathfrak{B}_c)$,

$G_c = \text{Aut}(\mathcal{R}_c)$, $l_c = \text{Iso}_{\text{mc}}((Y, [\nu]), (X, [\mu]))$ and $\text{Gen}_{\mathbb{Z}}^{\text{ns}}(\mathcal{R}_c) = \{S : \mathcal{R}_S = \mathcal{R}_c\}$ with $(\theta, S)g = (\theta g^{-1}, g S g^{-1})$. Then

$$\begin{aligned} \text{(KPB)} \quad \text{DynPlac}(c) &:= (l_c \times \text{Gen}_{\mathbb{Z}}^{\text{ns}}(\mathcal{R}_c)) / G_c \xrightarrow{[\theta, S] \mapsto \theta S \theta^{-1}} \{T : [\mathcal{R}_T]_{\cong} = [\mathcal{R}_c]_{\cong}\}, \\ \text{KCFib}_{\text{det}}(c) &\cong H_c \backslash \text{DynPlac}(c). \end{aligned}$$

(KPG) $\text{KPGpd}(c) := H_c \ltimes \text{DynPlac}(c)$ has $\pi_0 \cong \text{KCFib}_{\text{det}}(c)$, $\text{Aut}(T) = H_c \cap C_{\text{Aut}_{\text{ns}}(X, [\mu])}(T)$, and is the choice-independent full-time-zero KFG subgroupoid. With $G_0 = \text{Aut}(X, \mu)$, (KLP) $\lambda_c^{-1}([T]_{G_0}) \cong H_c \backslash G_0 / C_{G_0}(T)$, singleton iff $G_0 = H_c C_{G_0}(T)$. This label-placement residual is independent of outer-relation isotropy and the contractible (KMG) implementer/cocycle fibre; trivial static data erase it, while $H_c = 1$ retains the full conjugacy orbit.

On the trivial-static/residual p.m.p. hyperfinite ergodic face, (OGF) $\text{KCFib}_{\text{det}}(c_{\text{pmp}}) \cong \text{Gen}_{\mathbb{Z}}(\mathcal{R}_{\text{hyp}}) / \sim_{\text{Ad}}$, where $S \sim_{\text{Ad}} T$ iff $T = \phi S \phi^{-1}$ for $\phi \in \text{Aut}(\mathcal{R}_{\text{hyp}})$. Put $G_{\mathcal{R}} := \text{Aut}(\mathcal{R})$, $\text{InnGen}(\mathcal{R}) = \text{Gen}_{\mathbb{Z}}(\mathcal{R}) / \sim_{\text{Ad}[\mathcal{R}]}$, and $\text{Pllso}_{\mathcal{R}}(S) = C_{G_{\mathcal{R}}}(S) / C_{[\mathcal{R}]}(S)$. Since $[\mathcal{R}] \triangleleft G_{\mathcal{R}}$, (NQT) gives (OGL) $\text{KCFib}_{\text{det}}(c_{\text{pmp}}) \cong \text{InnGen}(\mathcal{R}_{\text{hyp}}) / \text{Out}(\mathcal{R}_{\text{hyp}})$, $\text{Pllso}_{\mathcal{R}}(S) \cong \text{Stab}_{\text{Out}(\mathcal{R})}([S]_{\text{in}})$ and $p^{-1}p([S]_{\text{in}}) \cong \text{Out}(\mathcal{R}) / \text{Pllso}_{\mathcal{R}}(S)$. Equivalently $\text{OGpd}(\mathcal{R}) := \text{Out}(\mathcal{R}) \ltimes \text{InnGen}(\mathcal{R})$ has $\pi_0 \cong \text{KCFib}_{\text{det}}(c_{\text{pmp}})$ and isotropy $\text{Pllso}_{\mathcal{R}}(S)$: the inner-reduced p.m.p. KFG/KPG face. For a nontrivial Bernoulli base, $\text{Aut}(A, \kappa) \hookrightarrow \text{Pllso}_{\mathcal{R}_S}(S)$; for $\theta \neq \text{id}$, $J_{\theta}x = S^n x$ is null for all n , so $J_{\theta} \notin [\mathcal{R}_S]$, giving cardinal $2^{\aleph_0}$ on non-atomic bases. By (NQT_{ex}), the orbit-equivalence torsor descends from $\text{Aut}(\mathcal{R}_{\hat{T}})$ to the (GOQ) $\text{Out}(\mathcal{R}_{\hat{T}})$ -torsor; GOQ and OGL share the outer group, not the action. Different-entropy Bernoulli shifts separate placement orbits. FRW reduce ill-founded trees continuously to self-inverse ergodic p.m.p. transformations. Their footnote 12 states that the tree-dependent prime support in E1 is used only to make the coding strongly one-to-one through Kronecker factors; Claims 10–11 use the remaining E/Q/A/J specifications, Lemma 17 and the joining analysis⁵⁷ (Thm. 7; Claims 10–11; footnote 12; Secs. 6–8). Thus fixed-relation hardness needs a uniform bridge rather than parameterwise (DOE), but need not preserve that fingerprint.

Set $N_0 = 2$, $\ell_0 = 1$. For $n \geq 1$ and a legal rooted prefix $S \subseteq \{\sigma_i : i \leq n\}$ set $d_n(S) = \max_{\sigma_i \in S, i \leq n} \text{lh}(\sigma_i)$ and $D_n = \max_{i \leq n} \text{lh}(\sigma_i)$. Choose a_n divisible by $1, \dots, D_n + 1$ so that $k_n(S) = a_n / (d_n(S) + 1)$ meets all finitely many FRW k -bounds. Since Q_0^n is one class, Q6 and the terminal class size give $|W_n(S)| = 2^{a_n} =: N_n$. Choose common ϵ_n and a prime p_n larger than every previous prime factor, satisfying the stronger adjacent-level forms of (18)–(21) and all Sec. 8 substitution/finishing bounds; finitely many prefixes permit successive maxima⁵⁷ (Sec. 7.1; Prop. 44; Lem. 45; Sec. 8.3).

Call a word level *neutral* when it is obtained from FRW Case 1 through the word-construction step immediately before adjoining the final $\mathbb{Z}_2$ generator, and all existing G_s are then carried unchanged. Neutral insertion is a stuttering refinement of the reduction proof: the Case 1 construction preserves E2–E3, Q4–Q6 and A7–A9 and supplies J10–J11 on the new adjacent pair (together with the equal-length/power-of-two part of E1); Q5/product transport and A9/skew-diagonal transport propagate the old factors/actions, while the abstract depth system $(G_s(T), \rho_{t,s})$ is literally unchanged. Hence Lemma 17 and the proofs of Claims 10–11/Corollary 38 are invariant under the present locally finite neutral insertion; only the discarded tree-dependent prime-support fingerprint can change. Build every global level from $n - 1$ and use such a neutral level exactly when $\sigma_n \notin S$. Then $|W_n| = N_n$, $\ell_n = p_n^2 N_{n-1} \ell_{n-1}$ and $\#_u(w) = p_n^2$ for $u \in W_{n-1}$, $w \in W_n$, tree-independently, and the padded tower still satisfies T ill-founded iff $F_*(T) \cong F_*(T)^{-1}$.

Let B have root $V_{-1} = \{*\}$, $V_n = \{1, \dots, N_n\}$, one root edge to each V_0 , and p_n^2 edges between every pair in $V_{n-1} \times V_n$. If h_n is the number of root paths to a fixed $v \in V_n$, then $h_0 = 1$ and $h_n = N_{n-1} p_n^2 h_{n-1} = \ell_n$; hence the central measure gives each length- n path cylinder mass $(N_n \ell_n)^{-1}$, and the simple tail relation $\mathcal{R}_B$ is fixed hyperfinite ergodic p.m.p. Lexicographically identify $W_n(T)$ with V_n and order incoming edges by parent-word occurrence position. E2–E3 give a Vershik map V_T measure-conjugate to $F_*(T)$. Maximal/minimal n -prefix unions have mass $\ell_n^{-1} \rightarrow 0$, so off their null tail saturations $\mathcal{R}_{V_T} = \mathcal{R}_B$. Prefix agreement through n fixes the edge orders through n and hence the successor off the all-maximal n -prefix set; thus $T \mapsto V_T$ is Borel. Choose once by $\Psi : \mathcal{R}_B \cong \mathcal{R}_{\text{hyp}}$ and put $S_T = \Psi V_T \Psi^{-1}$. Then $T \mapsto S_T \in \text{Gen}_{\mathbb{Z}}(\mathcal{R}_{\text{hyp}})$ is Borel and T is ill-founded iff $S_T \cong S_T^{-1}$. For fixed-relation generators ordinary conjugacy equals the $\text{Aut}(\mathcal{R}_{\text{hyp}})$ orbit relation, since $U = \phi S \phi^{-1}$ implies $\phi \mathcal{R}_{\text{hyp}} \phi^{-1} = \mathcal{R}_U = \mathcal{R}_{\text{hyp}}$. Hence $T \mapsto (S_T, S_T^{-1})$ reduces ill-foundedness to (OGF). Finally $\text{Gen}_{\mathbb{Z}}(\mathcal{R}_{T_0})$ is Borel for fixed T_0 : $R_S \subseteq R_{T_0}$ iff $\mu\{x : Sx \in \{T_0^k x : |k| \leq N\}\} \rightarrow 1$, and $R_{T_0} \subseteq R_S$ is the symmetric condition with S, T_0 exchanged; each finite-exhaustion mass is Borel in the standard weak MPT coding. Thus (OGF) is a Borel restriction of analytic MPT conjugacy, hence complete analytic and non-Borel. Thus type/flow, Markov time-zero placement, orbit equivalence and labeled conjugacy remain distinct.

Finite-dimensional peripheral decomposition. For CPTP Φ with faithful invariant ω_* write $\mathcal{N}_{\text{fix}} = \ker(\Phi^{\dagger} - I)$, $\mathcal{N}_{\text{per}} = \text{span}\{x : \Phi^{\dagger}x = \lambda x, |\lambda| = 1\}$, $E_{\text{fix}} = \lim_{N \rightarrow \infty} N^{-1} \sum_{k < N} (\Phi^{\dagger})^k$, and $\text{Mix}(\Phi)$ for $\|\Phi^n(\rho) - \omega_*\|_1 \rightarrow 0$ for every state ρ . Since $\Phi^{\dagger}$ is a unital CP contraction, power-boundedness puts its spectrum in $\overline{\mathbb{D}}$ and excludes nontrivial Jordan blocks on the unit circle. Finite-dimensional spectral decomposition therefore gives $\mathcal{N}_{\text{per}} = \mathcal{N}_{\text{fix}} \iff (\Phi^{\dagger})^n \rightarrow E_{\text{fix}}$. Kernel duality plus faithful fixed-state perturbations gives $\mathcal{N}_{\text{fix}} = \mathbb{C}1 \iff \omega_*$ is unique; on this face $E_{\text{fix}}(x) = \omega_*(x)1$, so duality identifies power convergence with trace-norm mixing. Hence $\mathcal{N}_{\text{fix}} = \mathbb{C}1 \iff \omega_*$ unique; $\mathcal{N}_{\text{per}} = \mathcal{N}_{\text{fix}} \iff (\Phi^{\dagger})^n \rightarrow E_{\text{fix}}$; $\mathcal{N}_{\text{per}} = \mathbb{C}1 \iff \omega_*$ unique and $\text{Mix}(\Phi)$. Strictness already occurs on M_2 : for trace τ , diagonal expectation D and Pauli X, Z , the pairs $(\mathcal{N}_{\text{fix}}, \mathcal{N}_{\text{per}})$ are $(\mathbb{C}1, \text{span}\{1, Z\})$, $(\text{span}\{1, Z\}, \text{span}\{1, Z\})$ and $(\mathbb{C}1, \mathbb{C}1)$ for $\text{Ad}_X \circ D$, D and $x \mapsto \tau(x)1$, respectively; only the last is mixing. Hence equilibrium uniqueness, fixed-algebra convergence and mixing are distinct, and no identification with temporal-lift or modular-clock fibres is asserted.

Theorem VIII.1 (Temporal envelope and gauge dissipation). *Let $(\widehat{\mathcal{G}}, \widehat{\mu}, \widehat{S})$ be the canonical natural extension of a dynamic kernel $\mathbf{P}$ with $\mu \mathbf{P} \sim \mu$, with the canonical time-zero factor r_0 . Put*

$$\text{MEnv}(\mathbf{P}, \mu) = (\mathcal{A}_0 \subseteq \widehat{\mathcal{A}} \subseteq \mathcal{M}_{\text{time}}, u, \widehat{\omega}), \quad \mathcal{A}_0 = r_0^* L^{\infty}(\mathcal{G}, \mu), \quad \widehat{\mathcal{A}} = L^{\infty}(\widehat{\mathcal{G}}, \widehat{\mu}), \quad \widehat{\omega}(a) = \int a d\widehat{\mu},$$

with $u^ f u = f \circ \widehat{S}$ and $\widehat{\mathcal{A}} = W^*(u^{-n} \mathcal{A}_0 u^n : n \in \mathbb{Z})$. The gauge and heat are $\delta_{\theta}(f u^n) = e^{in\theta} f u^n$ and $T_1(f u^n) = e^{-tn^2} f u^n$; their closed L^2 form is $\mathcal{E}_N(x) = \sum_n n^2 \|x_n\|_{\omega_{\widehat{\mathcal{A}}}}^2$ (and $+\infty$ outside its domain), with fixed algebra $\widehat{\mathcal{A}}$ and gap 1 on its orthogonal complement.*

A normal covariant expected realization is a faithful normal unital $$ -monomorphism $\iota : \widehat{\mathcal{A}} \rightarrow \mathcal{M}$, a unitary $U \in \mathcal{M}$ with $U^* \iota(f) U = \iota(f \circ \widehat{S})$, and a faithful normal expectation $E' : W^*(\iota(\widehat{\mathcal{A}}), U) \rightarrow \iota(\widehat{\mathcal{A}})$ satisfying $E'(\iota(f) U^n) = \mathbf{1}_{n=0} \iota(f)$. It receives a unique normal isomorphism*

$$\Phi : \mathcal{M}_{\text{time}} \xrightarrow{\sim} W^*(\iota(\widehat{\mathcal{A}}), U), \quad \Phi(f) = \iota(f), \quad \Phi(u) = U, \quad E' \Phi = \iota E,$$

where E is the Fourier expectation. This is crossed-product infrastructure, not a reconstruction invariant. With $\omega_\Xi = \widehat{\omega}E$,

$$\Theta_\Xi = \log \frac{d\widehat{\mu} \circ \widehat{S}}{d\widehat{\mu}}, \quad \sigma_t^{\omega_\Xi}(u) = ue^{it\Theta_\Xi}, \quad \sigma_t^{\omega_\Xi}(f) = f,$$

and (MC) $\Theta_\Xi = 0 \iff \mu P = \mu \iff \sigma^{\omega_\Xi}$ is trivial, $D(\theta_* \mathbb{P}_\mu \| \mathbb{P}_\mu) = D(\mu P \| \mu) \geq \frac{1}{2} \|\mu P - \mu\|_1^2$. The modular clock is therefore derived from MEnv; continuous-time realization remains the independent fibre of Proposition VIII B 0 a, and (TDF) is its all-time descent locus along the declared Banach quotient, not a clock coordinate.

The marks determine the unique $\widehat{\omega}$ -preserving $E_0 : \widehat{\mathcal{A}} \rightarrow \mathcal{A}_0$. With $j_0(f) = f \circ r_0$,

$$[\text{NC}] \quad T_0(j_0 f) := E_0(u^* j_0(f) u) = j_0(T f), \quad T f(x) = \int f(y) P_x(dy), \quad E_0(u^* j_0(f) u - j_0(f)) = j_0((T - I)f). \quad (140)$$

Hence $\mathcal{M}_{\text{time}}$ is abelian iff $[u, \mathcal{A}_0] = 0$ iff $T = I$ iff $P_x = \delta_x$ a.e.; (MGO) adds no primitive beyond this centered transition. The stationary two-point flip has zero clock but nonabelian envelope.

On the stationary branch let $T^\#$ be the reverse Markov operator and let P_0, V implement $E_0, \text{Ad}(u^*)$ on $L^2(\widehat{\mu})$. For $L_0 = P_0 V(1 - P_0)$,

$$\begin{aligned} L_0 = 0 &\iff TT^\# = I \\ [\text{RDL}] \quad &\iff T^\# \text{ is normal unital }^*\text{-homomorphic} \\ &\iff \exists R \text{ m.p. : } T^\# f = f \circ R, \quad P_x(R^{-1}\{x\}) = 1 \text{ } \mu\text{-a.e.} \end{aligned} \quad (141)$$

This strictly contains $T = I$: Haar doubling with its equal-weight two-root kernel has $L_0 = 0 \neq T - I$, while the independent two-point kernel has $L_0 \neq 0$; both have zero clock. If $L_0 = 0$ and $T = T^\#$, then R is a measure-preserving involution, and intersection with the positive spectral/OS locus of Proposition VIII B 0 a forces $T = I$. Thus leakage, centered transition and clock are distinct.

Finite stationary mixing versus reverse leakage. For a finite nontrivial Markov kernel K with faithful invariant law π ,

$$\delta(K) < 1 \implies \Theta_\Xi = 0 \quad \text{and} \quad L_0 \neq 0.$$

Indeed stationarity gives the first assertion by (MC), whereas (RDL) with $L_0 = 0$ forces distinct rows to have disjoint supports and hence $\delta(K) = 1$. The DLA/STA kernel K_ε , $0 < \varepsilon < 1/2$, has faithful uniform stationary law and $\delta(K_\varepsilon) = 1 - 2\varepsilon < 1$. Thus reverse leakage can coexist with zero modular clock; by STA this is law-level and creates no event.

For standard probability spaces with nonsingular kernels, modulo a.e. equality and measure-preserving kernel isomorphism,

$$(\mathcal{G}_1, \mu_1, P_1) \cong (\mathcal{G}_2, \mu_2, P_2) \iff \text{MEnv}(P_1, \mu_1) \cong_{\text{mark}} \text{MEnv}(P_2, \mu_2),$$

where a marked Ψ is a normal * -isomorphism preserving $(\widehat{\mathcal{A}}, \mathcal{A}_0, u, \widehat{\omega})$; E, E_0 are then derived. Forgetting the marks is sharp: the uniform kernels $P_a^1(b) = 1/2$ and $P_{(a,b)}^2(b, c) = 1/2$ (zero otherwise) have the same bilateral Bernoulli extension but 2- versus 4-atom $\mathcal{A}_0$, while the two-point identity kernel with weights $(1/3, 2/3)$ and $(1/4, 3/4)$ has the same algebraic marks but non-isomorphic states.

Gauge permanence. Brannan–Gao–Junge⁵⁸ (Def. 2.5, Prop. 2.4 and Thm. 4.4) supplies $c_{S^1} > 0$ uniformly for finite von Neumann coefficients, with

$$D(\rho(H_t \otimes \text{id}) \| \rho(\tau_{S^1} \otimes \text{id})) \leq e^{-2c_{S^1} t} D(\rho \| \rho(\tau_{S^1} \otimes \text{id})).$$

For a sigma-finite coefficient algebra $\mathcal{R}$, choose a faithful normal state and its Haagerup reduction $\widehat{\mathcal{R}} \supset \mathcal{R}$, expectation $E_{\mathcal{R}}$ and finite expected stages $\mathcal{R}_n$ from⁵⁹ (Thm. B.1). With $\mathcal{A} = L^\infty(S^1) \widehat{\otimes} \widehat{\mathcal{R}}$, $\widehat{\mathcal{A}} = L^\infty(S^1) \widehat{\otimes} \widehat{\widehat{\mathcal{R}}}$, $\mathcal{A}_n = L^\infty(S^1) \widehat{\otimes} \mathcal{R}_n$, $G_t = H_t \otimes \text{id}$ and $P = \tau_{S^1} \otimes \text{id}$, product modular factorization makes the stages finite, expected, strongly dense and G_t, P -invariant. For $\widehat{\rho} = \rho \circ (\text{id} \otimes E_{\mathcal{R}})$ and $\widehat{\rho}_n = \widehat{\rho}|_{\mathcal{A}_n}$, two-sided DPI and D_1 martingale convergence⁵⁹ (Lem. B.2) give

$$\begin{aligned} D_{\mathcal{A}}(\rho G_t \| \rho P) &= \lim_n D_{\mathcal{A}_n}(\widehat{\rho}_n G_t \| \widehat{\rho}_n P), \\ D_{\mathcal{A}}(\rho \| \rho P) &= \lim_n D_{\mathcal{A}_n}(\widehat{\rho}_n \| \widehat{\rho}_n P), \end{aligned} \quad (142)$$

so the same c_{S^1} holds for every sigma-finite $\mathcal{R}$. More generally, if $\iota : \mathcal{A} \rightarrow \mathcal{B}$ is normal unital * -monic and $r : \mathcal{B} \rightarrow \mathcal{A}$ is normal UCP with $r\iota = I$, $rS_t = T_t r$, $rF = Er$, two-sided DPI transfers any complete relative-entropy decay estimate from (S_t, F) to (T_t, E) with the same constant at every matrix amplification.

For $\mathcal{A} = \mathcal{M}_{\text{time}}$ take $z(\vartheta) = e^{i\vartheta}$, $J_\delta(x)(\vartheta) = \delta_\vartheta(x)$, $\Gamma_\phi F(\vartheta) = \delta_\phi(F(\vartheta - \phi))$, $P_\Gamma = \int \Gamma_\phi d\phi$ and $r_\delta = J_\delta^{-1} P_\Gamma$. Then r_δ is normal UCP, $\text{Fix } \Gamma = J_\delta(\mathcal{M}_{\text{time}})$, $r_\delta J_\delta = I$, and (GST) $J_\delta(fu^n) = z^n \otimes fu^n$, $r_\delta(H_t \otimes \text{id}) = T_t r_\delta$, $r_\delta(\tau_{S^1} \otimes \text{id}) = Er_\delta$. Consequently, for every k and normal φ on $\mathcal{M}_{\text{time}} \widehat{\otimes} M_k$,

$$D(\varphi(T_t \otimes \text{id}_k) \| \varphi(E \otimes \text{id}_k)) \leq e^{-2c_{S^1} t} D(\varphi \| \varphi(E \otimes \text{id}_k)).$$

The constant is independent of Θ_Ξ , k and the access presentation; faithful $\widehat{\omega}E$ makes $\mathcal{M}_{\text{time}}$ sigma-finite. At $k = 1$, $\mathcal{R}_t^\Xi(\varphi) := D(\varphi T_t \| \varphi E) \leq e^{-2c_{S^1} t} \mathcal{R}_0^\Xi \rightarrow 0$; if $\omega_\varphi = \varphi E$ is faithful and $\rho_t = \varphi T_t$, then $\rho_t E = \omega_\varphi$ and $\mathcal{R}_t^\Xi = D(\rho_t \| \omega_\varphi)$, the fixed-reference current of Paragraph X B 0 c, not a new primitive.

Proof. For a normal covariant expected realization, Fourier covariance gives the * -homomorphism $\Phi_0(\sum f_n u^n) = \sum \iota(f_n) U^n$ and $E'\Phi_0 = \iota E$; the state $\omega' = \widehat{\omega} \iota^{-1} E'$ makes its polynomial GNS map isometric, so faithful normal closure yields the displayed unique Φ . For (RDL), $P_0 V P_0 = T$, $P_0 V^* P_0 = T^\#$ give $L_0 L_0^* = I - TT^\#$. Hence $L_0 = 0$ iff $T^\#$ is L^2 -isometric; Kadison/faithfulness gives full multiplicative domain and stationarity the converse. Standard-Borel spatiality gives $T^\# f = f \circ R$ and the displayed adjoint-kernel support. If also $T = T^\#$, positive spectral/OS support removes -1 from the involutive Koopman spectrum, so $T = I$. Path intertwining gives (NC); Radon–Nikodym/KMS covariance plus Pinsker gives (MC); the kernels give the strict boundaries. Marked expectations recover the natural extension and crossed product, and the two forgetting examples show sharpness. For gauge permanence, BGJ⁵⁸ (Def. 2.5, Prop. 2.4 and Thm. 4.4) applies on each finite Haagerup stage; product modular factorization, two-sided DPI and the displayed D_1 limits pass the same constant to the sigma-finite coefficient algebra, while (GST) is the split transference interface. Fourier monomials give the fixed algebra and gap. $\square$

b. Projective temporal consequence. On the projectively adapted compact-Radon Feller branch, Paragraph III 0 c and (MC) give $[\forall n : \mu_n P_n = \mu_n] \iff \mu P = \mu \iff \Theta_\Xi = 0$. Here the UI criterion above controls MEnv entry and (PCT) its bidirectional entropy-price

shadow; the compact-countable witness separates entry from finite price. MEnv is therefore the corresponding operator-algebraic realization, not another projective invariant.

c. *Expectation/extension boundary.* For MEnv, modular invariance of the base gives the unique $\omega_{\otimes}$ -preserving normal expectation onto $\widehat{\mathcal{A}}$, namely the Fourier expectation already reconstructed above, and hence the canonical extension $\widehat{\omega} \circ E$. The expectation/covariance hypothesis is essential: in $\mathbb{C} \rtimes \mathbb{Z} = L^\infty(\mathbb{T})$, infinitely many normal probability densities restrict to the same scalar base state, while point evaluations may integrate at the C^* -level without defining normal states of $L^\infty(\mathbb{T})$. Thus extension is canonical only relative to the expected/covariant structure; forgetting it leaves a genuinely non-singleton extension fiber.

IX. OPERATOR-ALGEBRAIC REALIZATION AND MODULAR RECONSTRUCTION

A. Modular completion and normal-access rigidity

Retain the normal access regime of Subsection VD: $\mathcal{M}$ and the $\mathcal{N}_r$ are σ -finite von Neumann algebras, ω is a faithful normal state, F is non-empty and finite or countable, and $\alpha_r : \mathcal{N}_r \rightarrow \mathcal{M}$ is normal, unital, and completely positive, with $N_r = \alpha_r^*$ and faithful reference record $\omega_r = \omega \circ \alpha_r$. All tuples below have finite support when F is countable. For $h = (h_r) \in \bigoplus_{r \in F} \mathcal{N}_{r,sa}$ write $H_F(h) := \sum_{r \in F} \alpha_r(h_r)$, and retain $\mathcal{V}_F = G_\vee(F)$ from Theorem V.2.

a. *Bounded perturbation, pressure, and expansionals.* For faithful normal φ , bounded $a = a^*$, and every normal state $\rho \in S_n(\mathcal{M})$, (JEO_{gib}) gives the unique faithful positive $\tilde{\varphi}^a$ satisfying $\rho(a) + D(\rho||\tilde{\varphi}^a) = D(\rho||\varphi)$. With $Z_\varphi(a) = \tilde{\varphi}^a(\mathbf{1})$ and $\varphi^a = Z_\varphi(a)^{-1}\tilde{\varphi}^a$ this is $\rho(a) - D(\rho||\varphi) = \log Z_\varphi(a) - D(\rho||\varphi^a)$; hence $\mathcal{P}_\varphi(a) = \log Z_\varphi(a)$, uniquely maximized by φ^a , and $\tilde{\varphi}^{a+b} = (\tilde{\varphi}^a)^b$. This identity fixes $h(\varphi^a/\varphi) = \mathcal{P}_\varphi(a)\mathbf{1} - a$ in Araki's sign convention;⁶⁰ (Eq. (1.8)) gives (ARH) $[D\varphi^a : D\varphi]_t$ as the ordered cocycle generated by $a - \mathcal{P}_\varphi(a)\mathbf{1}$. Araki's Banach-algebra expansional⁶¹ (§2; §4, Thm. 2) supplies the absolutely convergent norm-series and its cocycle solution. Araki's multiple-time KMS theorems⁶² (Thms. 3.1, 3.3) supply a function holomorphic on the simplicial tube, continuous and bounded on its closure, with cyclic distinguished-boundary values. For bounded $x_1, \dots, x_q$ let $C_\varphi(x_1, \dots, x_q; t)$ denote that imaginary-time multiple-KMS boundary function on Δ_{q-1} , with $C_\varphi(x_1, \dots, x_q; t) = \varphi(x_1 \sigma_{-it_2}^\varphi(x_2) \cdots \sigma_{-i(t_2+\dots+t_q)}^\varphi(x_q))$ on Tomita-analytic inputs and $\text{Tr}(\rho^{t_1} x_1 \cdots \rho^{t_q} x_q)$ in finite Type I. Polarization of the ordered term on $\Sigma_q = \{s_a \geq 0 : \sum_{a=0}^q s_a = 1\}$, $ds = ds_1 \cdots ds_q$, $s_0 = 1 - \sum_{j=1}^q s_j$, gives (AD) $D^q Z_\varphi(0)[x_1, \dots, x_q] = \sum_{\pi \in S_q} \int_{\Sigma_q} C_\varphi(x_{\pi(1)}, \dots, x_{\pi(q)}; s_0 + s_q, s_1, \dots, s_{q-1}) ds$. For finitely supported h on $E \subseteq F$, U0 duality and $\langle h, N_F \rho \rangle = \rho(H_F(h))$ give $\Lambda_F^\omega(h) = \log Z_\omega(H_F(h))$, $Q_F^\omega(h) = \omega^{H_F(h)}$, $\rho(H_F(h)) - D(\rho||\omega) = \Lambda_F^\omega(h) - D(\rho||Q_F^\omega(h))$. By (ARH), $h(Q_F^\omega(h)/\omega) = \Lambda_F^\omega(h)\mathbf{1} - H_F(h)$ and $u_{h,t}^F = [DQ_F^\omega(h) : D\omega]_t$ is its ordered cocycle;⁶¹ (§2; §4, Thm. 2) gives the norm-convergent series in modular translates of $H_F(h) - \Lambda_F^\omega(h)\mathbf{1}$. Thus finite cocycle words are absolutely convergent combinations of C_ω correlations, so C_ω determines χ_F^ω . For the Orlicz channel maps $N_r : L_{\log}(\mathcal{M}, \omega) \rightarrow L_{\log}(\mathcal{N}_r, \omega_r)$ and $\alpha_r : E_{\exp}(\mathcal{N}_r, \omega_r) \rightarrow E_{\exp}(\mathcal{M}, \omega)$ put $X = E_{\exp}(\mathcal{M}, \omega)$, $V = \mathcal{V}_F^{\exp} := \overline{\sum_{r \in F} \alpha_r^{\exp}(E_{\exp}(\mathcal{N}_r, \omega_r))}^{\|\cdot\|_{\exp, \omega}}$ and $K = \mathcal{K}_F^{\log} := \bigcap_r \ker N_r$; then (BDT) gives (VQ) $V^\perp = K$, $(X/V)^* \cong K$, $K^* \cong X^{**}/V^{\perp\perp}$. The preceding finite row and $Y = \mathbb{R}\mathbf{1}$ witness allow $X/V \subseteq K^*$. On the bounded finite-action U1 chart, if $\rho \in \mathcal{C} \cap S_{\omega}^{\leq \infty}$, $\sigma = \omega^h$, $\delta = \rho - \sigma \in K$, then (FN) gives $\mathfrak{n}_{\rho, \omega}^\omega(\rho) = \langle \delta, [h] \rangle$; hence $X/V = 0$ forces flatness while $B_F > 0$ may remain cotangent-invisible.

b. *Modular sufficiency (Takesaki–Petz theory).* For σ -finite $(\mathcal{M}, \omega)$ with faithful normal ω , a faithful normal-state family Φ , $N \in \text{Lat}_\omega(\mathcal{M})$ and self-adjoint operator system $V \subseteq \mathcal{M}$, set

$$\begin{aligned} \text{Gen}_\omega(\Phi) &= W^*([D\varphi : D\omega]_t : \varphi \in \Phi, t \in \mathbb{R}), & \text{Host}_\omega(N) &= \{\varphi \in S_{\text{fin}}(\mathcal{M}) : [D\varphi : D\omega]_t \in N \forall t\}, \\ \Gamma_\omega(V, \Phi) &= W^*(\sigma_s^\omega(V), [D\varphi : D\omega]_t : s, t \in \mathbb{R}, \varphi \in \Phi). \end{aligned} \quad (143)$$

Connes' cocycle law⁴⁸ (§1.2), (TEXP/JP_{suff}) and Araki expansional/differentiation give the exact host–generator interface

$$\begin{aligned} \text{[MSA]} \quad \text{Gen}_\omega(\Phi) \subseteq N &\iff \Phi \subseteq \text{Host}_\omega(N), & \text{Gen}_\omega \text{Host}_\omega(N) &= N, \\ \Gamma_\omega(V, \Phi) \subseteq N &\iff V \subseteq N \text{ and } \Phi \subseteq \text{Host}_\omega(N). \end{aligned} \quad (144)$$

Indeed $u_t = [D\varphi : D\omega]_t$ satisfies (CRN) $u_{s+t} = u_s \sigma_s^\omega(u_t)$ and $\sigma_t^\varphi = \text{Ad}(u_t) \sigma_t^\omega$; for $h \in N_{sa}$, ω^h is hosted and cocycle differentiation recovers h modulo scalars, so hosted cocycles generate N . For $r_N : S_{\text{fin}}(\mathcal{M}) \rightarrow S_{\text{fin}}(N)$, $r_N(\rho) = \rho|_N$, and $s_N : S_{\text{fin}}(N) \rightarrow S_{\text{fin}}(\mathcal{M})$, $s_N(\eta) = \eta \circ E_N$, $r_N s_N = I$, $\text{im } s_N = \text{Host}_\omega(N) = \{\rho : E_N \rho = \rho\}$. Thus, with $\widehat{\rho}_N = E_N \rho$ and $u_N(\rho) = \rho - \widehat{\rho}_N$, $r_N^{-1}\{r_N \rho\} = (\widehat{\rho}_N + \ker(E_N|_{\mathcal{M}_{*,h}})) \cap S_{\text{fin}}(\mathcal{M})$, $u_N(\rho) = 0 \iff \rho \in \text{Host}_\omega(N)$. For $N \subseteq \mathcal{K}$, $u_N(\rho) = u_{\mathcal{K}}(\rho) + (E_{\mathcal{K}^*} - E_N)\rho$ with the second term in $V_{\mathcal{K},N}$. Put $\text{Cl}_\omega = \text{Host}_\omega \text{Gen}_\omega$, $\text{Alg}_\omega(S) = W^*(\sigma_t^\omega(h) : h = h^*, [h] \in S, t \in \mathbb{R})$ and $\text{JLCl}_\omega = \text{Tan}_\omega \text{Alg}_\omega$. For $H \subset \mathcal{M}_{sa}$ and nonzero s_h , (ST) $\text{Gen}_\omega(\text{Host}_\omega(N) \cup \{\omega^{s_h} : h \in H\}) = \text{Alg}_\omega(\text{Tan}_\omega(N) + \text{span}_{\mathbb{R}}\{[h] : h \in H\})$. All maps are natural under normal state-preserving $*$ -isomorphisms; on the finite-dimensional fixed locus this is Jordan–Lie saturation.

Theorem IX.1 (Hosted closure and BKM geometry). *Fix $N \subseteq \mathcal{K} \subseteq \mathcal{K}_*$ in $\text{Lat}_\omega(\mathcal{M})$. Put $\text{Ext}_\omega(N) = \{\varphi : \varphi|_N = \omega|_N\}$, $H_N(\mathcal{K}) = \text{Host}_\omega(\mathcal{K}) \cap \text{Ext}_\omega(N)$, $G_N(\Phi) = \Gamma_\omega(N, \Phi)$ and $V_{\mathcal{K},N} = \text{Ran}(E_{\mathcal{K}^*} - E_N)|_{\mathcal{M}_{*,h}}$. The (AFP/SDR) split gives the fixed-local adjunction (FLG) $G_N \dashv H_N$, $G_N H_N = I$, and $\mathcal{K}_1 \subseteq \mathcal{K}_2 \iff V_{\mathcal{K}_1,N} \subseteq V_{\mathcal{K}_2,N}$; equality of generated algebras, closure values and shell spaces are equivalent. For $N \subseteq \mathcal{K} \subseteq \mathcal{L}$, nested expectations split $V_{\mathcal{L},N} = V_{\mathcal{L},\mathcal{K}} \oplus V_{\mathcal{K},N}$, killing the topological affine obstruction and giving (AFC). With $p_{\mathcal{L},\mathcal{K}} = E_{\mathcal{K}^*}|_{H_N(\mathcal{L})}$ and $N_{\mathcal{K}} = \text{NAFib}(p_{\mathcal{K},N})$,*

$$\begin{aligned} 0 \rightarrow N_{\mathcal{K}} \rightarrow N_{\mathcal{L}} \rightarrow \text{NAFib}(p_{\mathcal{L},\mathcal{K}}) \rightarrow 0, & \quad 0 \rightarrow V_{\mathcal{L},\mathcal{K}}^* \rightarrow Z^1(R_{p_{\mathcal{L},\mathcal{K}}}; \mathbb{R}) \rightarrow \text{NAFib}(p_{\mathcal{L},\mathcal{K}}) \rightarrow 0, \\ \text{[FLR]} \quad \text{gr}_{\mathcal{F}} N_{\mathcal{K}_*} \cong \bigoplus_{j=1}^m \text{NAFib}(p_{j,j-1}). \end{aligned} \quad (145)$$

Rees exactness recovers the filtered extension. For nested $E_{\mathcal{K}^*}$, $E_{\mathcal{L}^*}$, (NPS/AFP/SDR/HSF) gives shell fibre $(\psi + V_{\mathcal{L},\mathcal{K}}) \cap S_{\text{fin}}$ and $\mathcal{L} = \mathcal{K} \iff p$ injective $\iff V = 0 \iff \mathcal{A}(R_p)^0 = \mathcal{N}(R_p)^0 = 0$; otherwise each faithful fibre contains an interval. Quadratic, four-point and tracial- M_2

witnesses make scalar/set shadows nonfaithful. With BKM projection Π_N and $q_N = g_\omega((I - \Pi_N) \cdot, (I - \Pi_N) \cdot)$,

$$\begin{aligned} \text{[BKT]} \quad \text{Lat}_\omega(\mathcal{M}) &\cong \text{Fix}(\text{Cl}_\omega) \cong \text{Fix}(\text{JLCl}_\omega), \\ N &= \text{Alg}_\omega(\text{Ran } \Pi_N) = \text{Alg}_\omega(\ker q_N), \quad N \subseteq \mathcal{K} \iff q_N \geq q_{\mathcal{K}}. \end{aligned} \quad (146)$$

Full-state chart conjugacy. For $N \in \text{Lat}_\omega(\mathcal{M})$ put $J_H N = \text{Host}_\omega(N)$, $J_R N = R_N^n$ and $J_Q N = q_N$, with reverse relation/Loewner orders on the last two images. Corestricted to their images, (MSA/NAT/BKT)Agives $(J_H^{-1}, J_R^{-1}, J_Q^{-1}) = (\text{Gen}_\omega, \text{Const}_n, q \mapsto \text{Alg}_\omega(\ker q))$, $(J_H \mathcal{M}, J_R \mathcal{M}, J_Q \mathcal{M}) = (\mathcal{S}_m(\mathcal{M}), \Delta, 0)$. Thus (LCG) transports every preclosure genealogy through each J_X ; J_R also determines $\mathbb{K}_N^n$. By (SBC), restricted Ω is sharp: neither relation nor Host coordinate alone determines Σ_Ω . For an increasing expected flag put $K_i := \mathbb{K}_{N_i}^n$, $P_i := E_{N_i^*} |_{\mathcal{M}_{s,h}}$, $L_i := \ker P_i$, $\beta_i := \beta_{r_{N_i}^n} : K_i \rightarrow L_i$, $C_i := \ker \beta_i$, $\widehat{\Pi}_i := \mathfrak{B}_\omega(N_i)$, $H_i := \ker \widehat{\Pi}_i$. (NAT/KAR) gives β_i onto; $g_\omega([h], [h]) \leq 4\|h\|^2$ gives $\|J_\omega \xi\|_* \leq 2\|\xi\|_{g_\omega}$, so $J_\omega : \mathcal{H}_\omega^{\text{BKM}} \hookrightarrow \mathcal{M}_{s,h}^0$, $J_\omega \xi(h) := g_\omega(\xi, [h])$ is injective with norm-dense range. Riesz and (BRB) give $J_\omega(H_i) = L_i \cap \text{En}_\omega$; moreover $J_\omega(I - \widehat{\Pi}_i)\xi_n = (I - P_i)J_\omega \xi_n \rightarrow v$ whenever $J_\omega \xi_n \rightarrow v \in L_i$. Hence (KCE) vanishes for every expected P_i and $J_i := J_\omega|_{H_i}$ is norm-dense. Put $\text{ERHist}_\omega(N_\bullet) := \mathfrak{R}_\bullet(\beta - J)$ and $\text{JRes}_i := K_i \times_{L_i} H_i$; then

$$\text{[JRC]} \quad 0 \rightarrow \text{JRes}_\bullet \rightarrow K_\bullet \oplus H_\bullet \xrightarrow{\beta-J} L_\bullet \rightarrow 0, \quad 0 \rightarrow C_\bullet \rightarrow \text{JRes}_\bullet \rightarrow H_\bullet \rightarrow 0. \quad (147)$$

Both rows are exact by β_i onto and $J_i(H_i) \subseteq L_i$. For small directed I put $\text{NFI}_I := (\int_{\text{NW}} \text{Fun}(I, \text{ExpLat}(-)))^\times$ and $\text{NFI}_m := \text{NFI}_{[m]}$. Naturality gives

$$\text{[DRLS]} \quad \text{ER}_I(F) = [K_\bullet \oplus H_\bullet \xrightarrow{\beta-J} L_\bullet]_{1,0}, \quad \text{DHist}_I = \text{DRes}_I \text{ER}_I : \text{NFI}_I \rightarrow D(\text{Ab})^\cong. \quad (148)$$

(UDR $_\Delta$ /DLX) on (JRC) gives the coefficient long-exact face. For $I = [m]$, corestrict $\mathcal{M}_{\psi,r}$ to $\widehat{\mathcal{M}}_{\psi,r} : \text{Ext}_{\psi_m \cdots \psi_1}(r) \rightarrow \text{NFI}_m$; (NRF/DRF) gives $\text{DHist}_{[m]} \widehat{\mathcal{M}}_{\psi,r}$ with no preferred section; joining/pointing remain attachments. For $r \leq i \leq j$ put $\mathfrak{H}_{j|r} := \text{Host}_{\omega, N_r}(N_j)$, $p_{ji}^{(r)} := P_i|_{\mathfrak{H}_{j|r}}$, $\mathbf{b}_{i:j} := \mathbf{b}_{N_i:N_j}$, $\mathbf{e}_{i:j} := \mathbf{e}_{N_i:N_j}$. (MSA/FLG/HSF) gives the split Host ladder, (BFS/ERF) its BKM/entropy Pythagoras readouts.

For a directed I with least element and a reveal map $\tau : U \rightarrow I$, let $\mathcal{M}_U^{\text{Ber}} = L^\infty(\{-1, 1\}^{U \times \{0,1\}}, \varpi)$ carry the Bernoulli product state and coordinate signs (X_u, Y_u) . Put

$$\begin{aligned} \mathcal{N}_i^{\text{Ber}} &:= W^*(X_u (u \in U), Y_u (\tau(u) \leq i)), \quad \frac{d\rho_u^\pm}{d\varpi} := (1 + X_u/4)(1 \pm Y_u/3), \\ K_i^{\text{Ber}} &:= \mathbb{K}_{\mathcal{N}_i^{\text{Ber}}}^n \text{ and } P_i^{\text{Ber}} := \bigoplus_{\tau(u) \leq i} \mathbb{Z}g_u. \text{ With } \chi_u(\rho) := \mathbf{1}_{\{\rho|_{\mathcal{N}_{\tau(u)}^{\text{Ber}}} = \rho_u^\pm|_{\mathcal{N}_{\tau(u)}^{\text{Ber}}}\}} \text{ define} \\ \text{[BER]} \quad e_i(g_u) &= [\rho_u^+] - [\rho_u^-], \quad \pi_i \left(\sum_\rho n_\rho [\rho] \right) = \sum_{\tau(u) \leq i} \left(\sum_\rho n_\rho \chi_u(\rho) \right) g_u, \quad \pi_i e_i = I_{P_i^{\text{Ber}}}. \end{aligned} \quad (149)$$

If $\tau(u) \not\leq i$, invisible Y_u gives $[\rho_u^+] - [\rho_u^-] \in K_i^{\text{Ber}}$; visible X_v gives $\chi_u(\rho_v^+) - \chi_u(\rho_v^-) = \delta_{uv}$, proving (BER). Once $\tau(u) \leq j$, χ_u factors through $r_{\mathcal{N}_j}^n$, so its coefficient vanishes on K_j^{Ber} when the u -coordinate has disappeared from P_j^{Ber} . Hence $e : P_\bullet^{\text{Ber}} \hookrightarrow K_\bullet^{\text{Ber}}$ and π are natural, with e split monic; coordinate expectations make the flag ϖ -expected and successor-valued τ limit-continuous.

For $I = \omega_n$, $U_n = [\omega_n]^{n+2}$ and $\tau_n(u) = \min(u) + 1$, write $(\mathcal{M}_n, \varpi_n, \mathcal{N}_\alpha^{\text{Ber},n}, K_\alpha^{\text{Ber},n}, P_\alpha^{\text{Ber},n})$. Since unrevealed coordinates are $[\alpha, \omega_n]^{n+2}$, $g_u \mapsto [(u, 1)] - [(u, 0)]$ gives $P_\alpha^{\text{Ber},n} \cong \mathbb{K}(q_\alpha^n)$, so the split retract and (THP) give

$$\text{(EHR)} \quad \text{Hist}_{\omega_n}^{n+1}(K_\bullet^{\text{Ber},n}) \neq 0, \quad \text{cd}(\omega_n) = n + 1$$

by (MIT $_{\text{cd}}$). For simultaneous depth take $I_\infty = \prod_{m \geq 1} \omega_m$, $U_\infty = \bigsqcup_{m \geq 1} [\omega_m]^{m+2}$, with $\tau_\infty(m, u)$ equal to $\min(u) + 1$ in coordinate m and 0 elsewhere; write $K_\bullet^{\text{Ber},\infty}, P_\bullet^{\text{Ber},\infty}$ for this specialization. Then $P_\alpha^{\text{Ber},\infty} = \bigoplus_m P_{\alpha_m}^{\text{Ber},m}$ and each $\text{pr}_m^* P_{\alpha_m}^{\text{Ber},m}$ is a natural split summand. With $s_m(\alpha)_j = \alpha$ for $j = m$ and 0 otherwise, $s_m + \text{pr}_m$; (MIT $_{\text{cof}}$) gives $R^{m+1} \varprojlim_{I_\infty} \text{pr}_m^* P_{\alpha_m}^{\text{Ber},m} \cong R^{m+1} \varprojlim_{\omega_m} P_{\alpha_m}^{\text{Ber},m} \neq 0$. Functoriality preserves this summand and (BER) retract without commuting derived limit with the infinite sum; hence one σ -finite expected Bernoulli system satisfies

$$\text{(SUD)} \quad \text{Hist}_{I_\infty}^p(K_\bullet^{\text{Ber},\infty}) \neq 0 \quad (p \geq 2).$$

With $A = K_\bullet^{\text{Ber},\infty}$, (KOF) makes the ordinal/cohomological witness unconditional; higher derived depth is not a scalar stationarity coordinate.

For $N \in \text{Lat}_\omega(\mathcal{M})$ put

$$Z_N := \{P \subseteq S_n(\mathcal{M}) : \text{Samskara}_{r_N^n}(\mathbf{1}_P) = 0\}, \quad Z_r := Z_{N_r}, \quad Z_s := Z_{N_s}.$$

The fixed-local order is simultaneously modal, BKM, Host, entropy, sufficiency and zero-Samskara order:

$$\begin{aligned} \text{[EIO]} \quad N_s \subseteq N_r &\iff \diamond_r p \leq \diamond_s p \iff \square_s p \leq \square_r p \iff \widehat{\Pi}_s \leq \widehat{\Pi}_r \\ &\iff \text{Host}_\omega(N_s) \subseteq \text{Host}_\omega(N_r) \iff \Delta_r \leq \Delta_s \text{ on } \mathcal{E}_b^\omega \\ &\iff \text{Suff}(t_s) \cap \mathcal{E}_b^\omega \subseteq \text{Suff}(t_r) \cap \mathcal{E}_b^\omega \iff Z_s \subseteq Z_r. \end{aligned} \quad (150)$$

A bounded Araki perturbation makes the first failure strict. For $N \subseteq \mathcal{K}$, with predual shell $Q^\vee$ and BKM shell B ,

$$\begin{aligned} \text{[RSR]} \quad \text{DepRes}_{r_N^n}(r_{\mathcal{K}}^n) &\cong \mathbb{K}(p), \quad 0 \rightarrow \ker \beta_p \rightarrow \mathbb{K}(p) \xrightarrow{J\kappa\beta_p} \text{Ran } Q^\vee \rightarrow 0, \\ \mathcal{R}_\omega(Q^\vee v) &= \widehat{B} \mathcal{R}_\omega v, \quad g_\omega^\vee(Q^\vee v) = \|\widehat{B} \mathcal{R}_\omega v\|_{g_\omega}^2, \quad \text{Hess}_\omega(\Delta_N - \Delta_{\mathcal{K}})(u, w) = g_\omega(Bu, Bw). \end{aligned} \quad (151)$$

For $N \subsetneq \mathcal{K}$, same-fibre states make $\ker \beta_p$ and (EIO) strict; (LCG) conjugates all closure iterates/ordinals across $n/\mathbb{K}/q$. For continuous $B_\alpha = \widehat{\Pi}_{\alpha+1} - \widehat{\Pi}_\alpha$, (BFS) gives orthogonal shells, $\widehat{\Pi}_\beta = \widehat{\Pi}_0 + \sum_{\alpha < \beta} B_\alpha$, bijectively recovering the flag. For countable flags $W_n := \text{Ran } B_n$, (BHC) gives $\widehat{H}_{\text{BKM}} \cong \prod_n W_n$ and $\partial_{\text{hist}} H_\bullet \cong (\prod_n W_n) / (\bigoplus_{2,n} W_n)$.

With $N_\infty = (\cup_j N_j)''$ and $\rho_j = P_j \rho$,

$$\begin{aligned} \text{[ERF]} \quad \rho_n &\rightarrow \rho_\infty \text{ in } \|\cdot\|_1, \quad \overline{\bigoplus_{j \geq 1} \text{Ran}(P_j - P_{j-1})}^{\|\cdot\|_1} = \text{Ran}(P_\infty - P_0), \\ \delta_{N_r}(\rho) &= D(\rho \| \rho_\infty) + \sum_{j > r} D(\rho_j \| \rho_{j-1}). \end{aligned} \quad (152)$$

Because the base is a groupoid both projections are discrete bifibrations. $\text{Law}(z, x, e) := e_* \eta$ descends functorially through Q_{ae} since $(e \circ r^{-1})_*(r_* \eta) = e_* \eta$; $\text{Out}(z, x, e) := e(x)$ has fibre

$$[\text{ROF}] \quad \{\text{Out}(z, x, e') : e' \in \text{BOut}_G(Y, Z), e' \sim_\eta e\} = \begin{cases} \{e(x)\}, & \eta\{x\} > 0, \\ Z^{G_x}, & \eta\{x\} = 0. \end{cases} \quad (159)$$

Thus (PRF/EAF) controls the null-point fibre, (URT/CRF) representative actualization over placement, with (KDL) pulled from the nonsingular base and (MPB) only p.m.p. On objects, (RPO) gives the forgetting chain $X_3 \rightarrow X_2 \rightarrow X_1 \rightarrow X_0$ with $X_3 = \text{Ob RPO}$, $X_2 = \text{Ob ARPO}$, $X_1 = \text{Ob KFG}_{\text{ns}}^G(c)$ and $X_0 = \{c\}$. Let $\text{KCFib}_{\text{ns}}^G(c)$ be the image of X_1 in $\text{KCFib}(c)$ and write $\text{Pl} = [z]_{\text{KFG}}$ for the placement read-out. Then (RDPL) gives $\text{rdepth}(\text{Pl}) = \mathbf{1}_{\{|\text{KCFib}_{\text{ns}}^G(c)| > 1\}}$, $\text{rdepth}(\text{Law}) \leq 2$, $\text{rdepth}(\text{Out}) = 3 \iff \exists(z, x, [e]_\eta) : |Z^{G_x}| > 1$. The p.m.p. binary witness/(RDT) realize strict depths 1, 2, 3; (NFR) forbids uniform finite derived height.

Put $C = \partial_{\text{hist}} C_\bullet$, $L = \partial_{\text{hist}} L_\bullet$. Warfield–Huber⁶⁵ (p. 434) makes $C = \varprojlim^1 C_n$ cotorsion; quotient models make L, E_ω real vector spaces, hence L $\mathbb{Z}$ -flat and E_ω injective. Thus $\text{Ext}_{\mathbb{Z}}^1(L, C \oplus E_\omega) \cong \mathbb{S}[\mathbb{S}]$ and $\text{JHist}_\omega \cong L \oplus C \oplus E_\omega$. Derived chart descent. By (JHP/JSP/HSC), $C = E_\omega = 0 \iff \partial_{\text{hist}} K_\bullet \xrightarrow{\sim} L \xleftarrow{\sim} \partial_{\text{hist}} H_\bullet \iff \text{JHist}_\omega \rightarrow L$ bijective, and eventual stationarity iff $L = C = E_\omega = 0$. Thus (EAC) is stagewise lossless, with independent first-history defects C, E_ω and noncanonical splitting witnessed by torsors/dyadic shears. For normal faithful finite-index $E : \mathcal{B} \rightarrow \mathcal{A}$ put

$$I_E^{-1} := \sup\{\lambda \geq 0 : E(x) \geq \lambda x \ (x \geq 0)\} > 0, \quad H_E = \log I_E, \quad h_E = H_E/2, \quad c_E(\rho) = D_{\mathcal{B}}(\rho \| \rho E), \quad a_E = c_E - h_E, \\ \text{Ext}_E^\circ(\nu) = \{\eta \in \mathcal{S}_\circ(\mathcal{B}) : \eta|_{\mathcal{A}} = \nu\}, \quad \text{Cap}_E(\nu) = \sup_{\text{Ext}_E^n(\nu)} c_E, \quad \omega_\nu = \nu E \quad (\nu \text{ faithful}).$$

I_E is the scalar Pimsner–Popa order index, not the finite-center minimal-index invariant outside its typed subfactor face. From $\eta E = \omega_\nu \geq I_E^{-1} \eta$, Araki monotonicity²⁸ (Thm. 3.6(1),(3)–(4)), convex interpolation and l.s.c. give

$$\text{CertProf}_E(\nu) := a_E(\text{Ext}_E^{\text{fn}}(\nu)) = [-h_E, \text{Cap}_E(\nu) - h_E], \quad T_{\omega_\nu} \text{Ext}_E^{\text{fn}}(\nu) = \ker \Pi_{\omega_\nu, \mathcal{A}}^{\text{BKM}} (\text{BRB/RFP}).$$

Hiai⁶⁶ (Thm. 1)/Hollands⁶⁷ (Cor. 1) give factor inputs; Giorgetti–Longo⁶⁸ (Thm. 5.2, Cor. 5.4, Prop. 5.6, Def. 7.25, Rem. 7.26 and Props. 7.46–7.47) gives finite-center: $(\text{GLM}_{\text{dim}})$ additive on orthogonal subarrows on fixed central axes, conjugate-transpose compatible, $d_X = \|D_X\|_2$, connected-block additive and compatible-center multiplicative; $(\text{GLM}_{\text{id}_X})$ has $c_3 \leq c_1 c_2$, with equality/composition of minimal expectations on the matched Perron–Frobenius locus, in particular through a factor. Finite-center minimal index is not generally multiplicative. Thus (EAP/GLM_{id_X}) gives additive two-step certainty/Bellman exactly on the factor-intermediate or matched-PF boundary and the corresponding submultiplicative bound otherwise; (FRP) is independent. For normal automorphism β and the EHB bridge,

$$[\text{EHB}] \quad \begin{aligned} \beta(\text{Exp}_\omega \mathcal{K}) &= \text{Exp}_{\omega^\beta}(\beta \mathcal{K}), & \mathfrak{T}_{\omega, N}(\beta) &= W^*(N, \beta N, u_t^\beta : t \in \mathbb{R}), \\ \mathfrak{T}_{gh}^\omega(N) &\subseteq N \vee \text{Exp}_\omega(J_g^\omega J_h^\omega N) \subseteq N \vee A_g^\omega A_h^\omega(N) \subseteq \mathfrak{T}_g^\omega(\mathfrak{T}_h^\omega(N)), \\ \widehat{\mathcal{K}}_e &= \mathcal{K}_e \iff \sigma_t^{\omega_s}(\mathcal{K}_e) = \mathcal{K}_e \iff \exists! E_e : \mathcal{M}_e \rightarrow \mathcal{K}_e \text{ normal conditional expectation with } \omega_s E_e = \omega_s \iff \text{Lift}_{\omega_s}(\mathcal{K}_e) = \{E_{e*}\}. \end{aligned} \quad (160)$$

The M_2 modular-orbit witness makes the reference fibre strict. At faithful ω , (JP_{br}) makes $\text{Br}_{\omega, N}(\varphi) = W^*(N, [D\varphi : D\omega]_t)$ least expected sufficient over N ; equal-spectrum z/x states keep equal entropy scalars but bridge algebras $\mathcal{D}, M_2$.

Derivation. State-preserving expectations/Petz plus expected-flag residual exactness give FLG/BKM/EAC and higher history; placement/polar bridges give PHB/KDL/MPB, and the finite-index/modular-covariance inputs give certainty and reference transport. The displayed witnesses give the stated strictness.

Theorem IX.2 (Generation and reflective closure). Let $(\mathcal{M}, \omega)$ be σ -finite with faithful normal ω and T normal UCP, $\omega T = \omega$. Retain the preceding raw/hosted response data of $\mathfrak{B}_{\text{ctrl}} = (\Omega, A, (\Phi_a), \Pi)$, in particular $C_\pi^r, C_\pi^h, \Pi_N^{\text{fac}}$ and Out_N^ε . For weak-* closed (V_i) put $V_0 = \bigvee_i V_i$, $V_\infty = \text{Fut}_T(V_0) = \overline{\text{span}}_{\eta < \omega}^{w*} T^n V_0$, $\text{MAlg}_\omega(V) = \text{Exp}_\omega W^*(V)$, and let Pred_T be the Scott reflector onto expected T -invariant algebras. Then $(\text{NAT}/\text{BEH}/\text{CWT}_{\text{sc}})$ gives

$$[\text{PCE}] \quad \begin{aligned} R_{Q_{\mathfrak{q}, (T_*)}} &= I_{\langle T_* \rangle}(R_{V_0}^n) = R_{V_\infty}^n, & Q_{\mathfrak{q}, (T_*)} &\cong r_{V_\infty}^n \text{ corestricted}, \\ R_{\text{Pred}_T(V_0)}^n &\subseteq R_{V_\infty}^n, & R_{\text{Pred}_T(V_0)}^n &= R_{V_\infty}^n \iff \text{Pred}_T(V_0) = V_\infty \iff V_\infty \in \text{Lat}_\omega(\mathcal{M}). \end{aligned} \quad (161)$$

Put $J_n(V) := R_V^n$ and $V_\perp := \{\eta \in \mathcal{M}_* : \eta|_V = 0\}$. By (NAT), J_n is lossless on weak-* closed operator systems, so the first (PCE) row is the exact observability–opacity conjugacy

$$[\text{OOP}] \quad J_n \text{Fut}_T = I_{\langle T_* \rangle} J_n, \quad (V_\infty)_\perp = \bigcap_{n \geq 0} (T_*^n)^{-1}(V_0)_\perp, \quad V_\infty = \mathcal{M} \iff R_{V_\infty}^n = \Delta \iff (V_\infty)_\perp = 0. \quad (162)$$

The middle term is the greatest T_* -invariant opacity in $(V_0)_\perp$; the full-state (BRC) fibre is trivial exactly on the (OOP) zero locus. In finite dimension $V_n = \text{span}_{k \leq n} T^k V_0$ and $V_{n+1} = V_n \Rightarrow V_n = V_\infty$, hence

$$V_\infty = \mathcal{M} \iff \exists N \leq \dim_{\mathbb{C}} \mathcal{M} - \dim_{\mathbb{C}} V_0 : V_N = \mathcal{M}, \quad R_{V_N}^n = R_{V_\infty}^n = \Delta. \quad (\text{OOP}_{\text{fin}})$$

$\text{Pred}_T(V_0)$ is the least expected invariant algebra, its relation the greatest realizable invariant opacity in $R_{V_0}^n$; equality with raw future closure is the (PCE) expectedness condition. (PAD) and the tracial M_2 witness keep predictive shells strict.

Put $\text{PredLat}_{\omega, T}(\mathcal{M}) := \text{Fix Pred}_T = \{\mathcal{K} \in \text{Lat}_\omega(\mathcal{M}) : T(\mathcal{K}) \subseteq \mathcal{K}\}$. Set $\mathcal{P}_T(\Phi) = \text{Pred}_T \text{Gen}_\omega \Phi$, $\mathcal{J}_\mathcal{P}(N) = \text{Pred}_T(N \vee \mathcal{P})$ and $\text{H}_T = \text{Host}_\omega|_{\text{PredLat}_{\omega, T}(\mathcal{M})}$. Then

$$[\text{RCF}] \quad C_\Phi = \text{Pred}_T \Gamma_\omega(-, \Phi) = \mathcal{J}_{\mathcal{P}_T(\Phi)}, \quad W_{\text{ctrl}}^\varepsilon(N) = \mathcal{P}_T(\text{Out}_N^\varepsilon), \quad \mathbb{G}^\varepsilon(N) = \mathcal{J}_{W_{\text{ctrl}}^\varepsilon(N)}(N),$$

and the fixed-local adjunction specializes to $\mathcal{P}_T \dashv \text{H}_T$, $\mathcal{P}_T \text{H}_T = I$. Policy factorization is $(\text{HPS}) \Pi_N^{\text{fac}} \iff R_N^\Omega \subseteq R_\pi$; raw/hosted control factors through restricted opacity/(SBC), with join-factorable quotient (MCF).

Put $F = \text{Pratityasamutpada}_{T, \Pi} = \mathbb{G}_{T, \Pi}^h$ and $\text{Sunyata}_{T, \Pi} = \overline{F}$. The (PGA/FRF/CTF_{fib}/SDR) face of (RGW/CAD/GCL) gives (HGL) and

$$\text{Fix } \mathbb{G}_{T, \Pi}^\varepsilon = \{N : T N \subseteq N, \text{Out}_N^\varepsilon \subseteq \text{Host}_\omega(N)\}, \quad \text{Fix Sunyata} = \text{Fix } \mathbb{G}^h.$$

For $\mathfrak{F}_{N_0} = \{\mathcal{K} \supseteq N_0 : \mathcal{K} \in \text{Fix}(\mathbb{G}^h)\}$, $N_\infty = \text{Sunyata}(N_0) = \bigwedge \mathfrak{F}_{N_0}$, $R_{N_\infty}^n = \max_{\subseteq} \{R_{\mathcal{K}}^n : \mathcal{K} \in \mathfrak{F}_{N_0}\}$. By (EAC), $N_\infty = \mathcal{M} \iff (J_H N_\infty, J_R N_\infty, J_Q N_\infty) = (S_{\text{fin}}(\mathcal{M}), \Delta, 0)$; below top residuals persist, and hosted policies descend to $r_{N_\infty}(H_{N_\infty}^h)$ by rFs. For normal UCP Θ , (QAF/NAT/ECL) gives $\Theta N_\infty \subseteq N_\infty \iff \Theta_*$ descends uniquely $\iff R_{N_\infty}^n$ is $\Theta_*^{\times 2}$ -invariant; (RAP/RTG/RRD/FIT) keeps exchange-action lift stricter. At $\mathbb{C}1$, (ZSB) gives $\mathbb{G}^r = \mathbb{G}^h = \text{Pratityasamutpada} = \text{Sunyata} = I$. The polarity $N \perp_{\text{Sun}} \mathcal{K} \iff N \subseteq \mathcal{K}$ ($\mathcal{K}$ fixed) reconstructs the fixed-world nucleus and N_∞ ; $(\{\mathbb{C}1\}, \text{Fix Sunyata})$ is the zero/maximal duality, not a seed. The M_2 and M_4 examples retain (HGN) equal generated law with different local closures and the independent control profile

$$[\text{SCN}] \quad W_0^h(\mathcal{K}) = \mathbb{C}1, \quad W_1^h(\mathcal{K}) = \begin{cases} \mathcal{D}, & \mathcal{D} \subseteq \mathcal{K}, \\ \mathbb{C}1, & \mathcal{D} \not\subseteq \mathcal{K}, \end{cases} \quad \mathcal{K} \in \text{Lat}_{\tau_4}(M_4). \quad (162)$$

Hence primitive spine, generated law and local closure stay distinct; state-preserving isomorphisms transport fibrewise closure, while grammar/reference remain attachments.

For a declared normal action $\beta : G \rightsquigarrow \mathcal{M}$, with (EHB/GCS), put $\text{CSat}_{G,\omega} = E_{\{\text{Sat}_G, \text{Exp}_\omega\}}$ and $\text{RefSat}_{G,\omega} = E_{\{\mathfrak{T}_g^\omega : g \in G\}}$. Then

$$\text{RefSat}_{G,\omega}(N) = W^*(\beta_g N, u_t^g : g \in G, t \in \mathbb{R}), \quad \text{RefSat}_{G,\omega}(\mathcal{K}) = \mathcal{K} \iff [\beta_g \mathcal{K} = \mathcal{K}, R_g^\omega \subseteq \mathcal{K}].$$

The intrinsic seed is $N_{\text{seed}} = N_{\text{str}} \vee N_{\text{ref}}$ with $N_{\text{str}} = \text{CSat}_{G,\omega}(\mathbb{C}1)$, $N_{\text{ref}} = \text{RefSat}_{G,\omega}(\mathbb{C}1)$ and $N_{\text{ref}} = W^*(u_t^g)$, $N_{\text{seed}} = \mathbb{C}1$ iff both seeds are trivial. If expected, $\text{Fix}_G(G)$ is the least structural fixed algebra (DLR) when applicable; structural/reference seeds stay independent, and (NAT/BKT) gives the exact constant/lossy/injective restriction trichotomy $\mathbb{C}1 \subseteq N \subseteq \mathcal{M}$. On the full $\mathbf{Pres}_{d,\tau} \subset \mathbf{Pres}$ of (M_d, τ, N) , $\mathfrak{D}N = N'$, $\mathfrak{D}(\theta, \theta N \subseteq \mathcal{K}) = (\theta^{-1}, \theta^{-1}\mathcal{K}' \subseteq N')$ is involutive contravariant: (PSN) flags reverse; fixed vertices are MASAs; terminalization is preserved iff the original starts at $\mathbb{C}1$; an m -flag is fixed iff $N_i = N'_{m-i}$; the M_2 z/x table keeps EAP non-self-dual. On the finite-factor (BMF/PLP) boundary, $\partial N = N'$ and $G_{\text{out}}(N) = \{\text{Ad } v : v \in \mathcal{U}(N')\}$, $\text{Fix } G_{\text{out}} = N = N_{\text{seed}}$, $N_{\text{ref}} = \mathbb{C}1$; Haar gives, (FAS) the tensor-factor face, and (BIF/IAE) forbids law-only selection. For $H \leq G$, put $\mathcal{K}_H = \text{RefSat}_{H,\omega} N$ and $\mathcal{P}_{gH|H}^\omega = \mathfrak{T}_g^\omega \mathcal{K}_H$. Then $\mathcal{P}_{gH|H}^\omega$ is well-defined on G/H , $\text{RefSat}_{G,\omega}(N) = \bigvee_{gH \in G/H} \mathcal{P}_{gH|H}^\omega$, and $gH \stackrel{\sim}{\sim}_\omega g'H \iff \mathcal{P}_{gH|H}^\omega = \mathcal{P}_{g'H|H}^\omega$. Put $\text{CosEff}_{G,\omega}(\mathcal{K}) := \{\mathfrak{T}_g^\omega(\mathcal{K}) : g \in G\}$. Then

$$[\text{ISF}] \quad \begin{aligned} \text{Abs}_\omega(\mathcal{K}) &= \{g : \beta_g \mathcal{K} \subseteq \mathcal{K}, R_g^\omega \subseteq \mathcal{K}\} \text{ is a submonoid,} \\ H_{\mathcal{K}}^\omega &= \{g : \beta_g \mathcal{K} = \mathcal{K}, R_g^\omega \subseteq \mathcal{K}\} = \max\{H \leq G : \text{RefSat}_{H,\omega}(\mathcal{K}) = \mathcal{K}\}, \\ \text{RefSat}_{G,\omega}(\mathcal{K}) &= \bigvee \text{CosEff}_{G,\omega}(\mathcal{K}), \quad |\text{CosEff}_{G,\omega}(\mathcal{K})| = 1 \iff \text{RefSat}_{G,\omega}(\mathcal{K}) = \mathcal{K} \iff H_{\mathcal{K}}^\omega = G. \end{aligned} \quad (163)$$

The coset quotient $G \twoheadrightarrow G/H_{\mathcal{K}}^\omega$ retains (GOR); $\chi_{\mathcal{K}} : G/H_{\mathcal{K}}^\omega \twoheadrightarrow \text{CosEff}_{G,\omega}(\mathcal{K})$, $gH \mapsto \mathfrak{T}_g^\omega \mathcal{K}$ has the separate (KSP) collision fibre; wordwise $\mathfrak{T}$ joins recover RefSat .

On smooth finite-matrix control, $\mathcal{D}_{g,\omega}(\mathcal{K}) = \text{Exp}_\omega W^*(\mathcal{K}, \delta_X \mathcal{K}, \eta_X^\omega : X \in \mathfrak{g})$ reaches $\text{RefSat}_{G,\omega} N$ within $d^2 - \dim_{\mathbb{C}} N$ strict steps (at trace $\eta_X^\tau = 0$); Hadamard/rotation separate structural/reference/effect levels. For σ -weakly continuous connected actions, (HSTD)/Stone⁶⁹ (Thm. B) and Kato C_1 – C_3 ⁷⁰ (Thms. 2–3) mutually reconstruct standard generators, resolvent history and propagator; $D(L_X) = \text{Ran } \mathcal{R}_\beta(X)$ and $L_X = iI + \mathcal{R}_\beta(X)^{-1}$. For $I = [a, b]$, let KHist_I be common-domain self-adjoint L_t histories with $A(t) = -iL_t$, standard frozen groups and $B(t, s) = (I - A(t))(I - A(s))^{-1}$ satisfying C_2 – C_3 (C_1 : resolvent contraction), and set $\mathcal{R}_A(t) = (L_t - iI)^{-1}$. Kato gives $U_A(t, s)$; uniqueness for $-A$ gives $U_A(t, s)^{-1} = U_A(s, t)$ and standardness the canonical normalizer. Hence

$$\beta_{t,s}^A := \text{Ad } U_A(t, s)|_{\mathcal{M}}, \quad \Gamma_I(A) := (\beta_{t,a}^A)_{t \in I}, \quad \text{KTr}_I := \Gamma_I(\text{KHist}_I).$$

For $\gamma = \Gamma_I(A)$, the canonical standard implementer's strong right derivative satisfies $A(t) \subseteq G_t^\gamma$; unitarity/maximal skew-adjointness give

$$D(G_t^\gamma) = D(A(t)), \quad G_t^\gamma = A(t).$$

Thus $A_t^\gamma := G_t^\gamma$, $Q_A(t) := (I - A(t))^{-1}$ and $Q_\gamma(t) := (I - G_t^\gamma)^{-1} = Q_A(t)$ on $[a, b]$, and $Q_A(t) = Q_A(s)B(s, t)$ uniquely extends the endpoint and inverts Γ_I . For strongly dense unital $\ast$ -algebra $\mathcal{A}_c \subset \mathcal{M}$ carrying the initial frozen norm generator, set

$$R_A(A) := \mathcal{R}_A(A), \quad E_I(A) := \beta_{b,a}^A, \quad d_c(R) := \left. \frac{d}{dt} \right|_0 \alpha_t^R|_{\mathcal{A}_c}, \quad \text{ResEnd}_I := (R_A, E_I), \quad \text{CoreEnd}_I := (d_c R_A, E_I),$$

where Stone reconstructs α^R from R . (KTS) therefore gives a lossless $\Gamma_I : \text{KHist}_I \xrightarrow{\cong} \text{KTr}_I$ (the autonomous factor face quotients only scalar Hamiltonians), with strict quotient flag

$$[\text{KTS}_1] \quad \begin{aligned} \Gamma_I : \text{ResEnd}_I^{-1}(R, \beta) &\xrightarrow{\cong} \{\gamma \in \text{KTr}_I : (iA_a^\gamma - iI)^{-1} = R, \gamma_b = \beta\}, \\ \text{CoreEnd}_I^{-1}(\delta_c, \beta) &\cong \coprod_{\substack{R \in \text{im } R_A \\ d_c R = \delta_c}} \text{ResEnd}_I^{-1}(R, \beta), \quad E_I^{-1}(\beta) \cong \coprod_{\delta_c \in d_c(\text{im } R_A)} \coprod_{\substack{R \in \text{im } R_A \\ d_c R = \delta_c}} \text{ResEnd}_I^{-1}(R, \beta), \\ R_{\text{CoreEnd}_I} / R_{\text{ResEnd}_I} &\cong R_{d_c \times I}, \quad R_{E_I} / R_{\text{CoreEnd}_I} \cong R_{\text{pr}_2}. \end{aligned} \quad (164)$$

A bounded M_2 loop, P_θ and scalar loops witness all three strict quotients. Piecewise Kato histories embed in the Moore path monoid with additive positive duration, reversal, exact seams and no positive-duration inverse; two-step (TRC) cocycle extension stays separate.

For a rule system $\mathcal{R} \subseteq \mathcal{P}(I) \times I$ put $\mathcal{A}_{\mathcal{R}}(S) = S \cup \{j : (A, j) \in \mathcal{R}, A \subseteq S\}$. On $\mathcal{M}_I^H = L^\infty(\{-1, 1\}^{\{*\} \sqcup (I \times \{0, 1\})}, \varpi)$ let $\iota(S) = W^*(Z, X_i (i \in I), Y_i (i \in S))$, $d\sigma^\pm/d\varpi = 1 \pm Z/5$, $d\rho_i^\pm/d\varpi = (1 + X_i/4)(1 \pm Y_i/3)$ and $\Omega_I^H = \{\sigma^\pm, \rho_i^\pm : i \in I\}$. Then $\rho_i^+|_{\iota(S)} = \rho_i^-|_{\iota(S)} \iff i \notin S$, all other marks are separated, $\sigma^\pm$ are always hosted, and $\rho_i^\pm$ is hosted iff $i \in S$. For $(A, j) \in \mathcal{R}$, choose the policy $\pi_{A,j}$ sending σ^+ to the ρ_j^+ -replacer and the remaining states to the σ^- -replacer, with distinct labels on $\rho_i^\pm$ exactly for $i \in A$. It factors through $\iota(S)$ iff $A \subseteq S$ and then adjoins exactly Y_j . Hence, for identity dynamics,

$$(\text{WHP}) \quad \mathbb{G}^h \iota(S) = \mathbb{G} \iota(S) = \iota(\mathcal{A}_{\mathcal{R}}(S)), \quad \overline{\mathbb{G}^h} \iota(S) = \overline{\mathbb{G}} \iota(S) = \iota(\overline{\mathcal{A}_{\mathcal{R}}(S)}).$$

Every monotone extensive $F : \mathcal{P}(I) \rightarrow \mathcal{P}(I)$ is $\mathcal{A}_{\mathcal{R}_F}$ for $\mathcal{R}_F = \{(A, j) : j \in F(A)\}$; (HUR) transports (GRI/TSC/LHF/RFI) law/history/duration/terminal invariants. On a prescribed host, (HRA) $\text{HReal}_T = \{F \in \text{PreCl}(L) : F(\mathbb{C}1) = \mathbb{C}1, \text{Pred}_T F = F\}$. For $b = (N_0, \mathbf{W}_\bullet) \in \text{GBase}(L)$ of closure ordinal κ , (HLF) gives $\text{HostLawFib}_T(b) = \text{Orb}_L^{-1}(b) \cap \text{HReal}_T = [F_{\mathcal{W}}^{\min}, F_{\mathcal{W}}^{\max}] \cap \text{HReal}_T$, nonempty exactly for $\kappa = 0$ with $\text{Pred}_T N_0 = N_0$, or $\kappa > 0$, $N_0 \neq \mathbb{C}1$ with every successor $\mathcal{W}_{\alpha+1}$ Pred_T -fixed; its least law is $\text{Pred}_T F_{\mathcal{W}}^{\min}$.

Presentation remains a separate fibre. For control $\mathfrak{B}$,

$$(HGL) \quad \mathfrak{Z}_{\mathfrak{B}}^h(\mathcal{N}) = \mathcal{J}_{\text{ctrl}, \mathfrak{B}}^h(\mathcal{N}), \quad G_{\mathfrak{Z}_{\mathfrak{B}}^h} = \mathbb{G}_{\mathfrak{B}}^h, \quad G_{\text{GrCl}(\mathfrak{Z}_{\mathfrak{B}}^h)} = \overline{\mathbb{G}_{\mathfrak{B}}^h}.$$

For $F \in \text{HReal}_T$ let $\text{CtrlGrFib}_T(F)$ be controls of hosted law F modulo equality of $\mathfrak{Z}^h$ and define

$$\text{SeedProf}_T(F) = \{W \in \text{Mon}(L, \text{Fix Pred}_T) : W(\mathbb{C}1) = \mathbb{C}1, \text{Pred}_T(\mathcal{N} \vee W(\mathcal{N})) = F(\mathcal{N}) \ (N \in L)\}.$$

(ZSB/RCF,HRA/HPS/PGA) sends every hosted control of law F to its seed profile and, conversely, the HRA/HPS threshold construction applied to $W \in \text{SeedProf}_T(F)$ realizes a hosted control with seed W . Equality of hosted generators recovers the seed pointwise because $\mathcal{J}_{W(\mathcal{N})}(\mathbb{C}1) = \text{Pred}_T(W(\mathcal{N})) = W(\mathcal{N})$ for $W(\mathcal{N}) \in \text{Fix Pred}_T$. Hence $(\text{HSP}) \text{CtrlGrFib}_T(F) \xrightarrow{\sim} \text{SeedProf}_T(F)$, $[\mathfrak{B}] \mapsto W_{\text{ctrl}, \mathfrak{B}}^h$ with $\max \text{SeedProf}_T(F) = F$; indeed the seed equation gives $W \leq F$, while $W = F$ is admissible since $F \in \text{HReal}_T$ is extensive and $\text{Pred}_T F = F$. For $N_0, w \in L$ put $\text{CtrlEndFib}_{T, N_0}(w) := \text{Tot}_{F \in \text{HReal}_T, \text{Term}_{N_0}(F)=w} \text{CtrlGrFib}_T(F)$; then (TFD/RFE/HLF/HSP) gives

$$(\text{EHF}) \quad \text{CtrlEndFib}_{T, N_0}(w) \simeq \text{Tot}_{W_* \in \text{Gen}_{N_0}(w)} \text{Tot}_{F \in \text{HostLawFib}_T((N_0, W_*))} \text{SeedProf}_T(F).$$

No reverse selector follows: on (M_2, τ, I) , I and $F_T(\mathbb{C}1) = \mathbb{C}1$, $F_T(\mathcal{N}) = M_2$ otherwise, share the $\mathbb{C}1$ endpoint. Disjoint used inert labels a_j with $\Phi_{a_j} = I$, $\pi_j \equiv a_j$ leave $(\text{HPI}) (\Pi_N^{\varepsilon, J}, \text{Out}_N^{\varepsilon, J}, \mathfrak{Z}^h, \mathbb{G}^h) = (\Pi_N^{\varepsilon}, \text{Out}_N^{\varepsilon}, \mathfrak{Z}^h, \mathbb{G}^h)$ ($\varepsilon = r, h$). At fixed Ω , after disjoint action relabeling write $\mathfrak{B} \sqcup_{\Omega} \mathfrak{C}$ for union. For $s \in \{\text{gr}, \text{law}\}$ let $\sim_{\text{ext}}^s$ be the equivalence relation generated by $\mathfrak{B} \leftrightarrow \mathfrak{B} \sqcup_{\Omega} \mathfrak{C}$ whenever the corresponding statistic $S_{\text{gr}}(\mathfrak{B}) := \mathfrak{Z}_{\mathfrak{B}}^h$ or $S_{\text{law}}(\mathfrak{B}) := \mathbb{G}_{\mathfrak{B}}^h$ is preserved. Then

$$[\text{HCI}] \quad \mathfrak{B} \sim_{\text{ext}}^{\text{gr}} \mathfrak{B}' \iff \mathfrak{Z}_{\mathfrak{B}}^h = \mathfrak{Z}_{\mathfrak{B}'}^h, \quad \mathfrak{B} \sim_{\text{ext}}^{\text{law}} \mathfrak{B}' \iff \mathbb{G}_{\mathfrak{B}}^h = \mathbb{G}_{\mathfrak{B}'}^h. \quad (165)$$

The forward implications are invariant by construction. Conversely, after disjoint relabeling, equal generator statistic has union seed $W \vee W = W$; equal hosted law F has union payload still hosted in F and containing either original realization, hence again law F . Joining through $\mathfrak{B} \sqcup_{\Omega} \mathfrak{B}'$ proves both reverse implications. For $n \geq 1$ set $\kappa = \omega_n$, $U_n = [\kappa]^{n+2}$, $S_{\alpha}^n = \{u : \min u < \alpha\}$ and $\iota_n(S) = W^*(X_u \ (u \in U_n), Y_u \ (u \in S))$, so $\iota_n(S_{\alpha}^n) = N_{\alpha}^{\text{Ber}, n}$. Let $\mathcal{R}_n = \{(S_{\alpha}^n, u) : \min u = \alpha < \kappa\}$, $F_n = A_{\mathcal{R}_n}$ and choose $u_* \in U_n$. With $d\sigma_{\kappa, n}^{\pm}/d\omega_n = 1 \pm X_{u_*}/5$, $d\rho_u^{\pm}/d\omega_n = (1 + X_u/4)(1 \pm Y_u/3)$ and $\Omega_n = \{\sigma_{\kappa, n}^{\pm}, \rho_u^{\pm} : u \in U_n\}$, (WHP/HUR) gives a closed-loop $\mathfrak{B}_n$ with

$$\mathbb{G}_{I_{M_n}, \Pi_n}^h \iota_n(S) = \iota_n(F_n(S)).$$

Since $F_n(S_{\alpha}^n) = S_{\alpha+1}^n$ and limits are unions, transfinite induction gives

$$(\mathbb{G}_{I_{M_n}, \Pi_n}^h)^{\alpha} \iota_n(\emptyset) = \iota_n(S_{\alpha}^n), \quad \text{clord}_{I_{M_n}, \Pi_n}(\iota_n(\emptyset)) = \omega_n, \quad \text{Hist}_{\omega_n}^{n+1}(K_{\bullet}^{\text{Ber}, n}) \neq 0.$$

Thus closure time ω_n and derived depth $n+1$ coexist on one endogenous realization and remain distinct. On $L = \mathcal{P}(I)$ fix Scott-continuous H , put $C = \overline{H}$, and for $F \in \text{Sc}^{-1}(H)$ set $q_F = (H, \overline{F})$, $\Sigma_F = \{\mathbb{B}_G : G \leq F, \text{Sc } G = H\}$, $P_F = \text{Sc}_{\text{Fix } C}(\max \Sigma_F)$; at fixed $q = (H, K)$ put $R_F = D_F F|_{\text{Fix } D_F}$. For monotone extensive F put

$$\mathcal{U}_i^F = \{S \subseteq I \setminus \{i\} : i \in F(S)\}, \quad \partial_{\text{Sc}}^i F = \mathcal{U}_i^F \setminus \text{IntSc } \mathcal{U}_i^F, \quad F(S) = S \cup \{i \notin S : S \in \mathcal{U}_i^F\}.$$

By (SUP), $\text{Sc } F = H \iff \text{IntSc } \mathcal{U}_i^F = \mathcal{U}_i^H$; $F \mapsto (\partial_{\text{Sc}}^i F)_i$ is lossless, with image the upward $\mathcal{U}_i^H \cup V_i$ satisfying $V_i \cap \mathcal{U}_i^H = \emptyset$, $\text{IntSc}(\mathcal{U}_i^H \cup V_i) = \mathcal{U}_i^H$, inverse $F_V(S) = S \cup \{i \notin S : S \in \mathcal{U}_i^H \cup V_i\}$ and componentwise joins. The local-image/micro/base flag is

$$(\text{LHF}) \quad \text{L}_H^{\text{lift}} = \text{im}(F \mapsto (q_F, \Sigma_F, R_F)), \quad \text{L}_H^{\text{micro}} = \text{im}(F \mapsto (q_F, P_F, R_F)), \quad \text{B}_H = \text{im}(F \mapsto q_F), \\ \text{LawFlag}_H : \text{Sc}^{-1}(H) \rightarrow \text{L}_H^{\text{lift}} \rightarrow \text{L}_H^{\text{micro}} \rightarrow \text{B}_H.$$

At fixed $q = (H, K)$, put $\mathcal{F}_q = \mathfrak{b}_L^{-1}(q)$, $\text{HistSh}_q(F) = (P_F, R_F)$, $r_{\text{lift}}^q(F) = (\Sigma_F, R_F)$, $\text{MicroFib}_q(P, R) = \text{HistSh}_q^{-1}(P, R)$ and $\text{LiftHistFib}_q(\Sigma, R) = (r_{\text{lift}}^q)^{-1}(\Sigma, R)$. Image-corestricted join preservation gives maxima

$$\mathfrak{s}_q^{\text{micro}}(z) = \bigvee \{F \in \mathcal{F}_q : \text{HistSh}_q(F) \leq z\}, \quad \mathfrak{s}_q^{\text{lift}}(z) = \bigvee \{F \in \mathcal{F}_q : r_{\text{lift}}^q(F) \leq z\},$$

so nonempty micro/lift fibres are directed with maxima. For countably infinite A , $I = A \sqcup \{p, q\}$, take $H_0(S) = S \cup (\{q\} : p \in S) \cup (\{p\} : q \in S)$, $F_p(S) = H_0(S) \cup (\{p\} : A \subseteq S)$, $F_q(S) = H_0(S) \cup (\{q\} : A \subseteq S)$ and $F_+ = F_p \vee F_q$ have common $(\text{Sc}, \tau, \mathbb{B}, P, R)$ and $\Sigma_{F_p} = \Sigma_{F_q}$ but $F_p \neq F_q$. Define $\text{ClHit}_F(S) = \min\{n < \omega : F^n(S) = \overline{F}(S)\}$, $\min \emptyset = \infty$. Then $\text{ClHit}_{F_p}(A) = \text{ClHit}_{F_q}(A) = 2$ and $\text{ClHit}_{F_+}(A) = 1$. With $q_0 = (H_0, K)$ put $K_{\text{dur}} = \{G \in \text{LiftHistFib}_{q_0}(\Sigma_{F_p}, R_{F_p}) : \text{ClHit}_G = \text{ClHit}_{F_p}\}$; which is nondirected. Put $\text{Sect}(X) = (\mathbf{1}_{\{\max X \text{ exists}\}}, \mathbf{1}_{\{X \text{ directed}\}})$; with the (CWT) witness $E = \langle A_{\infty}, B_{\infty} \rangle \notin W_{\{A_{\infty}, B_{\infty}\}}$, $\text{Sect}(\text{MicroFib}_q)$, $\text{Sect}(\text{LiftHistFib}_q)$, $\text{Sect}(\tau^{-1}(E))$, $\text{Sect}(K_{\text{dur}}) = ((1, 1), (1, 1), (0, 1), (0, 0))$. (HLC) For $J \neq \emptyset$ and $0 < \alpha \neq \beta$, $|\alpha|, |\beta| \leq |J|$, write $T_X^{\top}(S) = X$ and choose nonempty-block partitions $J = \bigsqcup_{\xi < \gamma} B_{\xi}^{\gamma}$ with rules $\bigcup_{\eta < \xi} B_{\eta}^{\gamma} \Rightarrow j \ (j \in B_{\xi}^{\gamma})$; then

$$(G_{\gamma})^{\xi} \emptyset = \bigcup_{\eta < \xi} B_{\eta}^{\gamma} \ (\xi \leq \gamma), \quad \overline{G}_{\gamma} = T_J^{\top}, \quad \text{clord}_{G_{\gamma}}(\emptyset) = \gamma.$$

On the countable singleton-block face, $\overline{G}_{\alpha}^{\text{seq}} = \overline{G}_{\alpha}^{\text{flat}} = T_{\alpha}^{\top}$ with depths $\alpha, 1$. More generally take disjoint $K = \{k_n\}$, $P = \{p_n\}$, $A_{\gamma} = \{a_{\gamma, n}\}$ ($\gamma < b$), $J = K \cup P$, common rules $p_n \Rightarrow p_{n+1}$, $a_{\gamma, n} \Rightarrow a_{\gamma, n+1}$, and for $\mathbf{d} = (d_{\gamma}) \in (\mathbb{N}_{\geq 1} \cup \{\omega\})^b$ put $D_{\gamma} = J \cup \bigcup_{\delta < \gamma} A_{\delta}$, $Q_{\gamma, m} = \{a_{\gamma, r} : m \mid r\}$, $Q_{\gamma, \omega} = \{a_{\gamma, 0}\}$ with rules (D_{γ}, q) for $q \in Q_{\gamma, d_{\gamma}}$. The $G_{\mathbf{d}}$ share (TSC) boundary/history and terminal closure but

$$\text{clord}_{G_{\mathbf{d}}}(J) = \sum_{\gamma < b} d_{\gamma}, \quad \text{clord}_{G_{\mathbf{d}}}(K \cup \{p_0\}) = \omega + \sum_{\gamma < b} d_{\gamma},$$

and $\tau_{G_{\mathbf{d}}}$ recovers $\mathbf{d}$. HUR transports the witnesses: outer (EHF) is strict and boundary profile, sectionability, duration, ordinal and return time remain distinct. On the hosted $\omega + 1$ face,

$$\mathbb{G}\left(\bigvee_{n < \omega} \mathcal{N}^{(n)}\right) = \mathcal{N}^{(\omega+1)} \supseteq \mathcal{N}^{(\omega)} = \bigvee_{n < \omega} \mathbb{G}(\mathcal{N}^{(n)}).$$

For feedback $F(x) = x \vee \bigvee_{\gamma: x \in A_\gamma} o_\gamma$, activation/landing and output ranks satisfy

$$\begin{aligned} h_\gamma &\leq a_\gamma + 1, & x_{\alpha+1} > x_\alpha &\iff \exists \gamma : (a_\gamma, h_\gamma) = (\alpha, \alpha + 1), \\ x_{\alpha+1} &= x_\alpha \vee \bigvee_{b_e \leq \alpha} o_e, & c_e &\leq b_e + 1, \\ \text{[GBR]} \quad \text{clord}_F(x_0) &= \sup_{a_\gamma < \infty, h_\gamma = a_\gamma + 1} (a_\gamma + 1) = \sup_{b_e < \infty, c_e = b_e + 1} (b_e + 1). \end{aligned} \quad (166)$$

For predictive N , hosted states are the zero-entropy/contained-point locus and (PFL) vanishes exactly at fixed points; Theorem IV.1/(JCP) gives same-policy future residual/seams, (RDO/RHC) expected chains, (GKF/BKT/EIO) lossless orbit/history, and (ECD) first-history defect; $M_2/(\text{RLH/RSR/BHC/EHC})$ forbid scalarization.

The HUR image of $G_\kappa^\beta \emptyset = \beta \cap \kappa$ has identity T_κ and (ORS) productive ranks/closure ordinal κ , so no small ordinal uniformly bounds hosted productivity; at nonzero limits (RFI/RPD/DIH/DHF) separates successor extinction, limit history and equal-profile Postnikov continuation. For every nonzero limit λ , (ZI) $N^{(\lambda)} = \mathcal{M} \iff K_\lambda = 0 \iff \cap_{\beta < \lambda} K_\beta = 0$. At ω , under $N^{(\omega)} = \mathcal{M}$,

$$\begin{aligned} \text{[ZIH]} \quad \exists N < \omega : N^{(N)} = \mathcal{M} &\iff \partial_{\text{hist}} K_\bullet = \partial_{\text{hist}} L_\bullet = \partial_{\text{hist}} H_\bullet = 0, \\ \forall N < \omega : N^{(N)} \neq \mathcal{M} &\iff |\partial_{\text{hist}} K_\bullet|, |\partial_{\text{hist}} L_\bullet|, |\partial_{\text{hist}} H_\bullet| \geq 2^{\aleph_0}. \end{aligned} \quad (167)$$

On the type-I factor face, equivalently $\cap_{\beta < \lambda} N'_\beta = \mathbb{C}1$, sharp at ω by the diagonal-density flag. The (RGH/UAF/FRP/DLX) and (R2G/THP/BEC) coordinates remain distinct; productive effect classes give the standing Hilbert/policy/output cardinal bound. On the HLF genealogy face, $F \in \text{Orb}_L^{-1}(b)$ gives

$$\text{[HXR]} \quad \left(\text{Term}_{N_0}(F), \bigcap_{\alpha \leq \kappa} R_{W_\alpha}^n \right) = (\mathcal{W}_\kappa, R_{W_\kappa}^n). \quad (168)$$

For $F = \mathbb{G}_{T, \Pi}^h$, (NAT/HXR) yields

$$\text{(HOP)} \quad \text{Sunity}_{T, \Pi}(N_0) = \mathcal{M} \iff \mathcal{W}_\kappa = \mathcal{M} \iff R_{W_\kappa}^n = \Delta \iff (\mathcal{W}_\kappa)_\perp = 0.$$

Completeness is terminal-endpoint completeness while (EHF) retains genealogy/law/control. At (ZSB) bottom with $\mathcal{M} \neq \mathbb{C}1$, (HOP) leaves terminal opacity; (PFL/SRF/HGN/SCN) retains productivity, zero-Samskara and presentation fibres. For every state-preserving normal *-isomorphism θ , with transported data (T^θ, Π^θ) , (PTG/AGR/RGW) gives the fibred covariance

$$\text{(CFP)} \quad \theta(\mathbb{G}_{T, \Pi}^h(N)) = \mathbb{G}_{T^\theta, \Pi^\theta}^h(\theta(N)), \quad \theta(\text{Sunity}_{T, \Pi}(N)) = \text{Sunity}_{T^\theta, \Pi^\theta}(\theta(N)).$$

Declared normal RCE is the causal-star arrow plus least expected Host algebra, local by (FVDL_{rce}) with Klein–Gordon stress response⁷¹ (Thm. 4.3); no lattice-to-continuum limit is asserted.

Derivation. The hosted-closure theorem (Theorem IX.1) plus restriction-kernel/Scott reflection gives future/hosted-control reflection; iteration, reference saturation and RCE give closure/covariance; the retained M_2/M_4 witnesses give non-collapse. *Supplementary package S3: Finite graph and Perron–Frobenius certainty packages.* The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S3; every use below invokes that exact proof-bearing package, not an additional assumption.

c. Orientation fiber and tracial forgetting. For the finite-dimensional JBW algebras used below set $Z_J = Z(\text{ODer}(J))$, $\widehat{\text{ODer}}(J) = \text{ODer}(J)/Z_J$, and let $\text{Or}(J)$ be the Connes orientations of this quotient. Alfsen–Shultz identify these with dynamical correspondences and Jordan-compatible von Neumann products⁷² (Thm. 1 and Cor. 2)⁷³; after existence, fixing one compatible factor product leaves exactly the opposite pair $z = \pm 1$ and product/opposite⁷² (Cor. 2 and Thm. 6). For the simple rank-two spin factor $E = \mathbb{R}e \oplus V$, $n = \dim V \geq 2$, their order-derivation decomposition is $\text{ODer}(E) = \mathbb{R} \text{id}_E \oplus B_V \oplus \mathfrak{so}(V)$, $(a \text{id} + B_u + R)^* = a \text{id} + B_u - R$, with $B_u = L_{(0, u)}$ and $\text{Der}(E) = \mathfrak{so}(V)$. Commuting with all boosts and rotations leaves center $\mathbb{R} \text{id}$, hence $\widehat{\text{ODer}}(E) \cong V \oplus \mathfrak{so}(V)$. A Connes orientation exchanges the $+1$ boost and -1 rotation eigenspaces and squares to -1 ⁷², so existence forces $n = n(n-1)/2$, i.e. $n = 3$; conversely $E \cong M_2(\mathbb{C})_{\text{sa}}$ then supplies exactly two orientations. Intrinsically put $\text{Ax}(V) = \{\kappa \in \text{Hom}_{SO(V)}(\Lambda^2 V, V) : \|\kappa\|_{\text{op}} = 1\}$. It is empty for $n \neq 3$; for $n \geq 4$ the stabilizer of a unit bivector forces its image into both its 2-plane and its orthogonal complement (decomposable bivectors span), while $n = 2$ has no fixed vector. For $n = 3$, $\text{Hom}_{SO(V)}(\Lambda^2 V, V) = \mathbb{R} * v$, hence $\text{Ax}(V) = \{\pm * v\}$. Under the Pauli identification, $\frac{1}{2}[u \cdot \sigma, v \cdot \sigma] = -(u \times v) \cdot \sigma$ matches product/opposite with $\kappa = -*v, +*v$; equivalently $I_\kappa(B_u) = R_u^\kappa$, $I_\kappa(R_u^\kappa) = -B_u$, $R_u^\kappa v = \kappa(u \wedge v)$. Therefore (ORI) $\text{Or}(E) \cong \text{Ax}(V)$, $\text{Or}(E) \neq \emptyset \iff \dim_{\mathbb{R}} E = 4 \iff E \cong M_2(\mathbb{C})_{\text{sa}}$, $|\text{Or}(E)| = 2$. Via $\mathfrak{so}(Q_J) \cong \Lambda^2 E$, the two Connes orientations are the two Lorentzian Hodge signs $\pm * Q_J$. For $n \geq 3$ fix $W_0 \in \text{Gr}_3(V)$ and $I_0 \in \text{Or}(\mathbb{R}e \oplus W_0)$. With $\text{Fr}_V = SO(V)$, $H_{W_0} = \text{Stab}(W_0)$, $q_{W_0}(g) = gW_0$, define $\Theta_{I_0}(g) = (gW_0, g_* I_0)$. The residual fiber action is $h_* I_0 = \chi_{W_0}(h) I_0$, $\chi_{W_0}(h) = \det(h|_{W_0})$. Hence (DI) gives Θ_{I_0} descends through $q_{W_0} \iff \chi_{W_0} = 1 \iff n = 3 \iff \text{Or}(E) \neq \emptyset$. For $n > 3$ a paired reflection in W_0 and $W_0^\perp$ gives $\chi = -1$; for $n = 3$, $W_0 = V$. Put $P_{\text{or}} := \text{Fr}_V \times_{H_{W_0}} \{\pm 1\} \cong \text{Gr}_3^+(V) \rightarrow B := \text{Gr}_3(V)$, $o_{\text{or}} := [P_{\text{or}}] \in \check{H}^1(B; C_2)$. Torsor descent gives $o_{\text{or}} = 0$ exactly when the local signs glue. For $n > 3$, choose orthonormal $e_1, \dots, e_4$ and let $W(t) = \text{span}(e_1, e_2, \cos(\pi t)e_3 + \sin(\pi t)e_4)$, $0 \leq t \leq 1$. This is a loop in B whose oriented lift runs from (e_1, e_2, e_3) to $(e_1, e_2, -e_3)$, so it changes sheet and $o_{\text{or}} \neq 0$; for $n = 3$, $B = *$ and $o_{\text{or}} = 0$. Thus o_{or} is the degree-one Čech instance of clause III 0b: four-dimensionality is exactly global $SO(V)$ -frame descent of the local orientation, while higher-dimensional local 4-spin faces retain the nonzero C_2 residual. Rank two alone is still insufficient: $\text{Sym}_2(\mathbb{R})$ has $n = 2$ and empty orientation fiber. By (RHG/RHO, RAP/RTG) this is the C_2 local-system face of the same stabilizer parity χ_{W_0} : the orbit base is canonical, orientation remains vertical, and no affine midpoint is asserted. The fixed-tracial ancillary slice can quotient this sign fiber, as the following strict example shows. Let

$$A = \text{diag}(1/5, 2/5, 3/5), \quad \rho = \begin{pmatrix} 1/3 & r & r e^{-i\theta} \\ r & 1/3 & r \\ r e^{i\theta} & r & 1/3 \end{pmatrix}, \quad r = 1/12, \quad \theta = \pi/4, \quad (169)$$

put $A = (A, 1 - A)$, $\omega(X) = \text{Tr}(\rho X)$, and $\omega^{\text{op}}(X) = \text{Tr}(\rho^\top X)$. Strict diagonal dominance makes both references faithful. For $n \geq 1$ put $\mathfrak{X}_n^{\text{tr}} := \{\eta \in \mathcal{S}(M_3 \otimes M_n) : \eta(1 \otimes b) = \text{Tr}(b)/n \forall b \in M_n\}$. Write $\tau_{A, n, c}^\omega$ and $\mathcal{A}_{A, n, \beta}^\omega$ for the hard and soft matricial entropic prices at target

$(Y, 1 - Y)$ on $\mathfrak{X}_n^{\text{tr}}$. Full matrix transpose preserves positivity, unit trace and the displayed marginal (partial trace commutes with transpose), and preserves relative entropy (functional calculus commutes with transpose and trace is invariant), while $\mathcal{R}_{A,n}(\hat{\rho}^\top) = \mathcal{R}_{A,n}(\hat{\rho})^\top$. Since each Hermitian Y is unitarily equivalent to $Y^\top$, ancillary conjugation preserves the section, reference, record and entropy terms and gives (TR) $\tau_{A,n,c}^\omega(Y) = \tau_{A,n,c}^{\omega^\text{op}}(Y)$, $\mathcal{A}_{A,n,\beta}^\omega(Y) = \mathcal{A}_{A,n,\beta}^{\omega^\text{op}}(Y)$. Nevertheless no reference-preserving UCP access simulation exists, even modulo output automorphisms. Indeed $\omega(A) = 2/5$, so the output swap is excluded. If $\varphi : M_3 \rightarrow M_3$ were UCP with $\varphi(A) = A$ and $\omega\varphi = \omega^\text{op}$, then $\Delta = \varphi(A^2) - A^2 \geq 0$ by Kadison–Schwarz and $\omega(\Delta) = \omega^\text{op}(A^2) - \omega(A^2) = 0$; faithfulness gives $\Delta = 0$. The simple spectrum of A therefore puts all E_{ii} in the multiplicative domain and forces $\varphi(E_{ij}) = c_{ij}E_{ij}$ for a correlation matrix $C \geq 0$, $c_{ii} = 1$. The reference condition gives $c_{ij} = \rho_{ij}/\rho_{ji}$, so every off-diagonal entry has modulus one; positivity then forces rank one and $c_{12}c_{23}c_{31} = 1$, whereas here $c_{12}c_{23}c_{31} = e^{2i\theta} = i$. On $E = M_2(\mathbb{C})_{\text{sa}}$ with trace τ , identity access in the two opposite (ORI) realizations $M_\pm$ has $\text{Suff}(\text{id}) = \mathcal{S}_n(M_\pm)$, $q_{\text{id}}^{\text{Av}} = 0$ and $(N_{\text{rev}})_{\text{sa}} = E$ by (PCH/PAH). After the canonical Jordan identification the PAH data coincide while the Connes orientations are distinct; their exact orientation fiber is $\text{Or}(E) \cong \{\pm *V\}$, so the sign is not determined. The strict (TR) example is the faithful price-level/no-UCP witness of the same boundary, without identifying the remaining Jordan fibers.

B. Marginal resolution and complete modular reconstruction

The missing coordinate of the tracial-ancilla invariant is the full family of finite ancillary base changes with faithful marginal retained, not a scalar. For $n \geq 1$, $\gamma \in M_n^+$, $\text{Tr } \gamma = 1$, put $\mathfrak{X}_{n,\gamma} := \{\eta \in \mathcal{S}_n(M \otimes M_n) : \eta(\mathbf{1} \otimes b) = \text{Tr}(\gamma b) \forall b \in M_n\}$ and let $\mathcal{Q}_{F,n}$ be restriction to $(\mathcal{V}_F \odot M_n)_{\text{sa}}$. The normal ancillary completion is

$$\text{Anc}_F^\omega := \mathfrak{F}_{F,\text{nmr}}^\omega := \left\{ \mathcal{J}_{F,n}^{\omega,\gamma} : n \geq 1, \gamma > 0, \text{Tr } \gamma = 1 \right\}, \quad \mathcal{J}_{F,n}^{\omega,\gamma}(\ell) := \inf_{\substack{\eta \in \mathfrak{X}_{n,\gamma} \\ \mathcal{Q}_{F,n}\eta = \ell}} D(\eta \| \omega \otimes \gamma). \quad (170)$$

Its support representation $\mathfrak{F}_{F,\text{nmr}}^\omega$ is the tower $\psi_{F,n,c}^{\omega,\gamma}(V) := \sup_{\eta \in \mathfrak{X}_{n,\gamma}, D(\eta \| \omega \otimes \gamma) \leq c} \eta(V) = \sup_{\mathcal{J}_{F,n}^{\omega,\gamma}(\ell) \leq c} \ell(V)$. Thus U0 support duality gives $\mathfrak{F}_{F,\text{nmr}}^\omega \iff \mathfrak{F}_{F,\text{nmr}}^\omega$ at every base change; $n = 1$ is $\mathcal{J}_F^\omega$, and in finite Type I $\mathfrak{X}_{n,\gamma} = \{\eta : \text{Tr}_{M_d} \eta = \gamma\}$, the density-matrix tower rather than a parallel construction.

a. Marginal-disintegration map. For $\Phi_{F,n}^{\omega,\gamma}(V) := \sup_{\eta \in \mathfrak{X}_{n,\gamma}} \{\eta(V) - D(\eta \| \omega \otimes \gamma)\}$, U0 support duality gives $\Phi_{F,n}^{\omega,\gamma}(\beta V) = \sup_{c \geq 0} \{\beta \psi_{F,n,c}^{\omega,\gamma}(V) - c\}$ for $\beta > 0$, with inverse concave Legendre transform. Put $\mathcal{P}_{\omega \otimes \tau_n}(K) := \sup_{\eta} \{\eta(K) - D(\eta \| \omega \otimes \tau_n)\}$, $\tau_n = \mathbf{1}_n/n$. The bipartite chain rule (113) gives $D(\eta \| \omega \otimes \tau_n) = D(\eta \| \omega \otimes \gamma) + D(\gamma \| \tau_n)$. Thus, on the full density simplex, for $g_{F,n}^\omega(\gamma; V) := \sup_{\eta \in \mathfrak{X}_{n,\gamma}} \{\eta(V) - D(\eta \| \omega \otimes \tau_n)\}$, $g_{F,n}^\omega(\gamma; V) = \Phi_{F,n}^{\omega,\gamma}(V) - D(\gamma \| \tau_n)$ on the faithful stratum, while grouping by the marginal yields $\mathcal{P}_{\omega \otimes \tau_n}(V + \mathbf{1} \otimes D) = \sup_{\gamma \geq 0, \text{Tr } \gamma = 1} \{g_{F,n}^\omega(\gamma; V) + \text{Tr}(\gamma D)\}$ for $D = D^*$. Mixing with τ_n makes faithful densities cofinal and fiber mixing makes g concave. Since $\omega \otimes \gamma$ gives $g(\gamma; V) \geq -\|V\| - \log n$, every 1-maximizer lies in $D(\cdot \| \omega \otimes \tau_n) \leq 2\|V\| + \log n + 1$; this sublevel is weak-* compact by (JEO_{cmp}). Continuity of the marginal map and upper semicontinuity of $\eta(V) - D(\eta \| \omega \otimes \tau_n)$ therefore make g upper semicontinuous. Finite-dimensional concave biconjugacy gives, for faithful γ , $\Phi_{F,n}^{\omega,\gamma}(V) - D(\gamma \| \tau_n) = \inf_{D=D^*} \{\mathcal{P}_{\omega \otimes \tau_n}(V + \mathbf{1} \otimes D) - \text{Tr}(\gamma D)\}$. Hence marginal and ancillary-Hamiltonian resolution are exact coordinates of one completion. In finite Type I, $\mathcal{P}(K) = \log \text{Tr} \exp(\log(\rho \otimes \tau_n) + K)$, with the same Fenchel pair.

All pressure, analyticity, KMS-boundary and mixed-derivative data below are those of Paragraph IX A 0 a. For each q write $D_{\mathbb{C}}^q Z_\varphi(0)$ for the unique complex-multilinear extension of the real Fréchet jet $D^q Z_\varphi(0)$ from M_{sa}^q to M^q ; (AD) extends to this jet by complex multilinearity. No value of $Z_\varphi(a)$ for non-self-adjoint a is introduced, and no second free-energy, multi-time or Duhamel object is used.

b. Normal labeled-cycle separation map. The preceding resolved pressures determine the full multiple-KMS correlation of every position-labeled word. The perturbation chain law gives $Z_{\omega \otimes \gamma_D}(V) = \exp(\mathcal{P}_{\omega \otimes \tau_q}(V + \mathbf{1} \otimes D) - \mathcal{P}_{\omega \otimes \tau_q}(\mathbf{1} \otimes D))$, $D = \text{diag}(d_1, \dots, d_q)$ real, $\gamma_D = e^D / \text{Tr } e^D$. For $x = (x_1, \dots, x_q)$ set $B_j = E_{j,j+1}$ cyclically. By the complexified (AD) jet, $D_{\mathbb{C}}^q Z_{\omega \otimes \gamma_D}(0)[x_1 \otimes B_1, \dots, x_q \otimes B_q]$ kills every noncyclic matrix-unit ordering. For the cyclic start r , the unit-Jacobian change $t_r = s_0 + s_q$, $t_{r+a} = s_a$ ($1 \leq a < q$), $u = s_0$, has fiber $0 \leq u \leq t_r$; hence its ancillary factor is $(\text{Tr } e^D)^{-1} e^{(d,t)}$ and its system factor is $C_r(t) := C_\omega(x_r, x_{r+1}, \dots, x_{r+q-1}; t_r, t_{r+1}, \dots, t_{r+q-1})$. Writing $M_x(d) := (\text{Tr } e^D) D_{\mathbb{C}}^q Z_{\omega \otimes \gamma_D}(0)[x_1 \otimes B_1, \dots, x_q \otimes B_q]$, the multiple-KMS cyclic boundary theorem⁶² gives

$$C_r(t) = C_1(t) \quad (1 \leq r \leq q),$$

$$\stackrel{[\text{LC}]}{M_x(d)} = \int_{\Delta_{q-1}} e^{(d,t)} \sum_{r=1}^q t_r C_r(t) dt = \int_{\Delta_{q-1}} e^{(d,t)} C_1(t) dt. \quad (171)$$

Polynomial moments make this compact-simplex Laplace transform injective; continuity recovers C_1 pointwise, and uniqueness on the connected multiple-KMS tube recovers its canonical boundary values and cocycle expansionals. Position labels allow repeated letters. The directed cycle preserves orientation; only the cyclic basepoint is quotiented.

X. QUANTUM ACCESS, CAUSAL GEOMETRY, AND CLOSURE

A. Quantum access and reconstruction boundaries

The quantum branch is the general normal-access realization of Theorem VII.1(4): a σ -finite von Neumann algebra $\mathcal{M}$ with faithful normal ω , $\mathcal{Q}_r \subseteq \mathcal{N}_r \subseteq \mathcal{M}$, record projections, bounded self-adjoint cost, normal positive dynamics and typed residuals. The abelian skeleton and MEnv are exact subrealizations; arbitrary $\mathcal{M}$ is declared realization data. Effects may replace projections for positivity/membership/expectation claims, with fiber-modal refinement; no second quantum skeleton is introduced.

a. *Operatorial instrument no-bootstrap.* Let $\mathbf{FinBij}_{+,sk}$ be a small skeleton of the groupoid of nonempty finite sets and bijections and fix a unital von Neumann algebra $\mathcal{M}$. For $I \in \mathbf{FinBij}_{+,sk}$ let $\mathbf{Nlnstr}_{\mathcal{M}}(I)$ be the set of finite normal Heisenberg instruments $\mathcal{J} = (\mathcal{J}_i)_{i \in I}$, where every $\mathcal{J}_i : \mathcal{M} \rightarrow \mathcal{M}$ is normal completely positive and $\sum_i \mathcal{J}_i$ is unital. Relabeling by $\sigma : I \rightarrow J$ is $(\sigma_* \mathcal{J})_j = \mathcal{J}_{\sigma^{-1}j}$, giving a functor $\mathbf{Nlnstr}_{\mathcal{M}} : \mathbf{FinBij}_{+,sk} \rightarrow \mathbf{Set}$; write $U_{\text{set}} : \mathbf{FinBij}_{+,sk} \rightarrow \mathbf{Set}$ for the underlying-set functor, $U_{\text{set}}(I) = I$. With $G_{\mathcal{J}} = \{\sigma \in \text{Sym}(I) : \sigma_* \mathcal{J} = \mathcal{J}\}$, the groupoid face of (GAD) gives

$$[\text{OIGF}] \quad \boxed{\text{GenAct}_{\mathbf{Nlnstr}_{\mathcal{M}} \Rightarrow U_{\text{set}}}(I, \mathcal{J}) \cong I^{G_{\mathcal{J}}}.} \quad (172)$$

Nevertheless there is no universal natural outcome extractor:

$$[\text{OINB}] \quad \boxed{\text{Nat}(\mathbf{Nlnstr}_{\mathcal{M}}, U_{\text{set}}) = \emptyset.} \quad (173)$$

Indeed on $I = \mathbf{2}$ the normal instrument $\mathcal{J}_0 = \mathcal{J}_1 = \frac{1}{2} \text{id}_{\mathcal{M}}$ is fixed by transposition, while $\mathbf{2}$ has no transposition-fixed point. The obstruction therefore survives knowledge of the full selective CP maps, not merely their Born probabilities. If ω is any fixed normal state, $p_{\omega, \mathcal{J}}(i) = \omega(\mathcal{J}_i(1))$ is natural under relabeling; (DLG) then shows that a deterministic law-compatible outcome can occur from the same unpointed datum only on its Dirac-law locus. In particular the symmetric binary witness has uniform Born law. More generally, for every nonempty set A viewed as a constant functor, $\text{Nat}(A \times \mathbf{Nlnstr}_{\mathcal{M}}, U_{\text{set}}) = \emptyset$ by the same binary witness. Hence pre-event fields that are constant under outcome relabeling—in particular a fixed normal state or fixed label-blind ambient data—do not remove the obstruction; a field that does remove it is additional kernel-refinement data in the sense of (RAD). By (RNB), adjoining any small well-ordered chain of auxiliary resources naturally generated from the accumulated full-instrument datum leaves its Yoneda kernel unchanged. In the symmetric binary witness the outcome generated-actuality fibre therefore remains empty throughout such a chain. Any datum that enables an outcome there is necessarily relatively exogenous to the pre-event instrument datum in the exact (RNB) sense. For a chosen label i , (AKF) then supplies the target-relative quotient $\text{ActKey}(\mathcal{J}; i)$ that every successful instrument–resource joint datum must determine; (ECM) is its exact-complement fibre. This key is a requirement classifier after i is specified, not an extractor of i from $\mathcal{J}$, so (OINB) is not bypassed. Since any natural extractor on the unskeletonized finite-bijection groupoid would restrict to the chosen skeleton, (OINB) also blocks that wider presentation.

b. *Unpointed measurement-dilation firewall.* Fix a unital von Neumann algebra $\mathcal{M}$. For finite nonempty I , let $\mathbf{MMod}_{\mathcal{M}}(I)$ contain finite-dimensional environment-framed tuples $m = (K, \sigma, U, (P_i)_{i \in I})$, where σ is a normal state on $B(K)$, $U \in \mathcal{M} \otimes B(K)$ is unitary and (P_i) is a PVM. Put

$$\Pi_I(m)_i(a) := (\text{id}_{\mathcal{M}} \bar{\otimes} \sigma)(U^*(a \otimes P_i)U).$$

These are normal CP maps and $\sum_i \Pi_I(m)_i$ is unital. Let $\mathbf{MMod}_{\mathcal{M}}(I)$ be the quotient by environment-frame unitary equivalence $W : K \rightarrow K'$: $\sigma'(x) = \sigma(W^*xW)$, $U' = (1 \otimes W)U(1 \otimes W^*)$, and $P'_i = WP_iW^*$. Outcome relabeling sends P_i to $P_{\tau^{-1}j}$, so both the framed and quotient families are functorial on $\mathbf{FinBij}_{+,sk}$ and $\Pi : \mathbf{MMod}_{\mathcal{M}} \Rightarrow \mathbf{Nlnstr}_{\mathcal{M}}$ is natural.

For $n = |I| > 1$ take $K_I = \mathbb{C}^I$, $\sigma_I = \text{tr}_I/n$, $U_I = 1$ and $P_i = |i\rangle\langle i|$. Then

$$[\text{MDNB}] \quad \boxed{\Pi_I([m_I])_i = n^{-1} \text{id}_{\mathcal{M}}, \quad \text{Stab}_{\text{Sym}(I)}([m_I]) = \text{Sym}(I), \quad \text{Nat}(\mathbf{MMod}_{\mathcal{M}}, U_{\text{set}}) = \emptyset.} \quad (174)$$

Indeed every relabeling of m_I is frame-equivalent to it by the corresponding permutation unitary on K_I , whereas I has no $\text{Sym}(I)$ -fixed point. Thus even the full unpointed finite measurement model—environment state, interaction unitary and pointer PVM—has a symmetric object above the symmetric (OINB) instrument and admits no universal natural outcome extractor. By (OLIF), every natural $\kappa : \mathbf{MMod}_{\mathcal{M}} \Rightarrow \text{RaceKey}$ sends this class to u_I , so the projective-race route also remains on the total tie wall. No dilation-existence or surjectivity theorem is used here: the no-bootstrap conclusion needs only the explicit symmetric object above, and no claim that Π is onto is made.

The frame quotient is load-bearing rather than cosmetic. Before quotienting by environment-frame unitaries, the relabeling orbit of the same concrete tuple m_I is free:

$$[\text{MFR}] \quad \boxed{\text{Stab}_{\text{Sym}(I)}(m_I) = 1, \quad \text{Hom}_{\text{Sym}(I)}(\text{Sym}(I) \cdot m_I, I) \cong I.} \quad (175)$$

The first statement is literal equality of the labelled raw PVM tuple; the second is (SARF/HSB_{act}) for the regular resource orbit. Hence retaining an independently physicalized pointer frame changes the deterministic-actualization fibre from empty to nonempty, but leaves exactly the outcome multiplicity rather than selecting one. Merely refusing to quotient bookkeeping labels is not a physical mechanism: the framed tuple counts as a resource only when that frame is independently realized in the declared physical interface. A realized pointer value or some further frame-relative asymmetry is still additional occurrence data. This is the measurement-dilation version of the DGF frame firewall, not a claim that unitary dilation is dynamically incapable of correlating system and apparatus.

Pointer-weight landing and residual dilation fibre. On the full-support locus $\mathbf{MMod}_{\mathcal{M}}^+(I) := \{[K, \sigma, U, (P_i)] : \sigma(P_i) > 0 \forall i\}$ put

$$[\text{PWK}] \quad \boxed{\Lambda_I([K, \sigma, U, (P_i)]) := (\sigma(P_i))_{i \in I} \in \Delta_I^{\circ} \cong \text{RaceKey}_I.} \quad (176)$$

This is invariant under environment-frame equivalence and natural under outcome relabeling, so $\Lambda : \mathbf{MMod}_{\mathcal{M}}^+ \Rightarrow \text{RaceKey}$ is a target-blind physical read-out. It is not the Born law: for a normal system state ω the latter is $p_i = \omega(\Pi_I(m)_i(1))$ and depends on the interaction as well as the pointer state.

The separation is maximal already over one fixed observed instrument. Take the scalar system $\mathcal{M} = \mathbb{C}$ and let u_I denote the uniform scalar instrument $u_I(i)(a) = |I|^{-1}a$. For $\text{UFib}(I) := \Pi_I^{-1}\{u_I\} \cap \mathbf{MMod}_{\mathbb{C}}^+(I)$,

$$[\text{UDF}] \quad \boxed{\Lambda_I(\text{UFib}(I)) = \Delta_I^{\circ} \cong \text{RaceKey}_I.} \quad (177)$$

Indeed, for arbitrary $v \in \Delta_I^{\circ}$, choose an identification of I with $\mathbb{Z}/n\mathbb{Z}$, $K = \mathbb{C}^I$, $\sigma_v(x) := \sum_j v_j \langle j, xj \rangle$, the standard pointer $P_i = |i\rangle\langle i|$, and a Fourier unitary F with $|F_{ij}|^2 = 1/n$. Then

$$\Pi_I(m_v)_i(a) = a \sigma_v(F^* P_i F) = a/n, \quad \Lambda_I([m_v]) = v.$$

No natural section $v \mapsto [m_v]$ is asserted: the auxiliary identification/Fourier frame is used only to prove surjectivity of the fibre read-out. Since Λ descends through the frame quotient, distinct v give distinct unpointed dilation classes. Thus the same symmetric instrument admits the full projective actuality-key continuum in its physical dilation fibre; the instrument determines none of it. On the unique-minimum locus the declared race read-out $W_I(u_I, \Lambda_I(m))$ is a singleton and is permutation-natural, whereas the symmetric class of (MDNB) has $\Lambda = u_I$ and remains the total tie. This supplies a positive outcome-indexed physical carrier for (OLIF) without contradicting (OINB): the deterministic race

read-out depends on which dilation point is independently realized, not on the instrument law alone. Identifying that read-out with a physical occurrence still requires the declared occurrence interface; no such identification is made here.

Universal race calibration. Let $|I| = n \geq 2$ and let μ be a Borel probability measure on $\text{RaceKey}_I \cong \Delta_I^\circ$. With the open chambers $C_i(p)$ of (PRC), the following are equivalent:

$$[\text{URC}] \quad \left[(\forall p \in \Delta_I^\circ) (\forall i \in I) \mu(C_i(p)) = p_i \right] \iff \mu = \text{Dirichlet}(1, \dots, 1). \quad (178)$$

Thus the uniform Dirichlet law is not merely one seed law realizing every finite full-support law through the race: it is the unique target-blind projective-key law with that universal calibration.

For the converse direction, Dirichlet sufficiency is (PRK/ERD). For uniqueness, fix $i \in I$ and arbitrary $r_j > 0$ for $j \neq i$, and choose $p_i = (1 + \sum_{j \neq i} r_j)^{-1}$, $p_j = r_j p_i$. Then

$$C_i(p) = \{v \in \Delta_I^\circ : v_j > r_j v_i \forall j \neq i\}.$$

Hence, for $Y_j = v_j/v_i$, the assumed calibration determines the full joint survival function

$$[\text{URS}] \quad \mu\{Y_j > r_j \forall j \neq i\} = \frac{1}{1 + \sum_{j \neq i} r_j}. \quad (179)$$

The upper orthants generate the Borel sigma-algebra of $(0, \infty)^{n-1}$, so this determines the joint law of Y . If Ξ_j are iid $\text{Exp}(1)$ and $V_j = \Xi_j / \sum_k \Xi_k$, then

$$\mathbb{P}\{V_j/V_i > r_j \forall j \neq i\} = \int_0^\infty e^{-t} \prod_{j \neq i} e^{-r_j t} dt = \frac{1}{1 + \sum_{j \neq i} r_j}.$$

The ratio map $v \mapsto (v_j/v_i)_{j \neq i}$ is a Borel bijection $\Delta_I^\circ \rightarrow (0, \infty)^{n-1}$, so μ equals the law of V , namely $\text{Dirichlet}(1, \dots, 1)$.

Consequently, if an independently realized physical carrier has a target-blind read-out $V \in \text{RaceKey}_I$ whose law is independent of the supplied full-support law p , then the declared race output has law p for every p exactly when V has this Dirichlet law. On the quantum face one substitutes $p_i = \omega(\mathcal{F}_i(1))$. This is a calibration theorem for the finite classical outcome read-out, not a noncontextual hidden-state model: it neither produces the realized V from its law nor assigns values across incompatible quantum contexts; (ANB), (RNB), and (QRS) remain unchanged.

Fixed-instrument compensation orbit and Haar key law. Fix a framed measurement model $m = (K, \sigma_\psi, U, (P_i)_{i \in I}) \in \widetilde{\text{MMod}}_{\mathcal{M}}(I)$ with $|I| = n \geq 2$, $\sigma_\psi(x) = \langle \psi, x\psi \rangle$ pure and every $P_i \neq 0$. For $W \in \mathcal{U}(K)$ put

$$\sigma_\psi^W(x) := \sigma_\psi(WxW^*), \quad U^W := U(1 \otimes W), \quad m[W] := (K, \sigma_\psi^W, U^W, (P_i)_i),$$

and $\lambda_m(W) := (\sigma_\psi^W(P_i))_{i \in I} \in \Delta_I$. Then

$$[\text{FIC}] \quad \Pi_I(m[W]) = \Pi_I(m) \quad (W \in \mathcal{U}(K)), \quad \lambda_m(W)_i = \|P_i W^* \psi\|^2, \quad \lambda_m(\mathcal{U}(K)) = \Delta_I. \quad (180)$$

Thus $W \mapsto m[W]$ is an explicit family inside one fixed observed-instrument fibre. Its pointer-weight read-out sweeps the closed simplex Δ_I ; on the full-support locus it corestricts onto $\Delta_I^\circ \cong \text{RaceKey}_I$. Indeed, for $Y = U^*(a \otimes P_i)U$,

$$(\text{id} \otimes \sigma_\psi^W)((1 \otimes W^*)Y(1 \otimes W)) = (\text{id} \otimes \sigma_\psi)(Y),$$

and the second identity is immediate. For $v \in \Delta_I$, choose unit $e_i \in P_i K$ and $x_v = \sum_i \sqrt{v_i} e_i$; transitivity of $\mathcal{U}(K)$ on the unit sphere gives $W^* \psi = x_v$, proving surjectivity. The raw weight tuple is itself invariant under environment-frame equivalence, since $\sigma'(P'_i) = \sigma(P_i)$; hence distinct values of λ_m cannot be identified by that quotient. On the full-support locus this is exactly the invariance used in (PWK).

Let $r_i = \text{rank } P_i$ and $N = \dim K = \sum_i r_i$. If W is Haar on $\mathcal{U}(K)$, then $W^* \psi$ is Haar on the unit sphere and

$$[\text{HRC}] \quad \lambda_m(W) \sim \text{Dirichlet}(r_i)_{i \in I}, \quad \lambda_m(W) \in \Delta_I^\circ \text{ a.s.} \quad (181)$$

Indeed write a Haar unit vector as $Z/\|Z\|$ in a basis adapted to the orthogonal blocks $P_i K$, with independent standard complex Gaussians as coordinates. The block energies $S_i = \|P_i Z\|^2$ are independent Gamma($r_i, 1$) variables, hence $(S_i/\sum_j S_j)_i$ has the displayed Dirichlet law. Combining (HRC) with (URC) gives the sharp Haar-calibration boundary

$$[\text{HRC}_0] \quad \left[(\forall p \in \Delta_I^\circ) (\forall i) \mathbb{P}\{W_I(p, \lambda_m(W)) = \{i\}\} = p_i \right] \iff r_i = 1 \text{ for every } i. \quad (182)$$

For the forward direction (URC) forces $\text{Dirichlet}(r_i) = \text{Dirichlet}(1, \dots, 1)$; the i th marginal is Beta($r_i, N - r_i$), so comparison with Beta($1, n - 1$) at the endpoint 0 forces $r_i = 1$ for every i . The reverse direction is (URC). Hence a pure rank-one pointer realization turns Haar compensation inside a fixed observed instrument into the unique universally calibrated race-key law. This is realization-relative, not instrument-determined: already for the uniform scalar instrument u_I , any positive rank vector (r_i) occurs by taking $K = \bigoplus_i \mathbb{C}^{r_i}$, a pure ψ , the block PVM (P_i) , and a unitary U with $U\psi = n^{-1/2} \sum_i e_i$ for unit $e_i \in P_i K$. Then $\Pi_I(m)_i(a) = a/n$ for all i , while its Haar compensation law is Dirichlet(r_i). Thus the observed instrument does not select a Haar realization law. Nor does the Haar law select a realized W : (ANB/RNB) still types the run-level point as occurrence input.

Marked-Poisson partition race and restart. For standard-Borel X and $\mu \in \mathcal{P}(X)$, the iid marked-interarrival law $\Pi_X(\mu) = (\text{Exp}(1) \otimes \mu)^{\otimes \mathbb{N}}$ has $\text{First}_X((E_m, A_m)_m) := A_1$ as its natural first-mark map. For a finite Borel $q : X \rightarrow J$, writing $\bar{\mu} = q_* \mu$ and $\mu_j = \mu(\cdot \cap q^{-1}j)/\bar{\mu}(j)$ on active cells, the marked-Poisson mapping, restriction, marking and thinning theorems¹⁸ (Thms. 5.1, 5.2, 5.6, 5.8, 5.10, 5.11) give $\text{Law}((M_j, Y_j)_j) = \bigotimes_j (\text{Exp}(\bar{\mu}(j)) \otimes \mu_j)$, and under a further finite coarsening $\text{Exp}(\bar{\mu}) \rightarrow K$, $M_k^{sq} = \min_{g(j)=k} M_j^q$, $Y_k^{sq} = Y_j^q$ almost surely. Borel naturality and iid head-tail splitting give independent restart. Thus first-mark push-forward is μ , but a natural process/mark selector would contradict (ANB) already on the uniform binary face. On finite active support, singleton cells recover (CAL/SRTK/SRCT/ECRL); on the full-support chart they recover (RTK/RCT), and orbit cells recover the PORS race. The Poisson facts are STANDARD_ABI; only this common actuality-spine landing is MIXED_BRIDGE.

Resolution-natural race-time transport. Let $\mathbf{FinSurj}_+$ be the category of nonempty finite sets and surjections. Write $\text{TimeKey}_I := (0, \infty)^I / \mathbb{R}_{>0}$ as a typed copy of the projective positive cone. For $f : I \twoheadrightarrow J$ define

$$[\text{RTK}] \quad M_f([t]) := \left[\left(\min_{i \in f^{-1}(j)} t_i \right)_{j \in J} \right]. \quad (183)$$

Then $M_{gf} = M_g M_f$ and $M_{1_I} = 1$, so TimeKey is a covariant $\mathbf{FinSurj}_+$ -functor. Its occurrence read-out is exactly resolution-natural:

$$[\text{RTW}] \quad \text{argmin}_{j \in J} (M_f[t])_j = f \left(\text{argmin}_{i \in I} t_i \right), \quad (184)$$

with both sides interpreted as nonempty subsets, so ties require no exceptional convention.

For $p \in \Delta_I^\circ$ put $\Theta_p : \text{RaceKey}_I \rightarrow \text{TimeKey}_I$, $\Theta_p([\xi]) = [(\xi_i/p_i)_i]$. If $q = f_*p$, conjugating (RTK) back to the law-independent seed chart gives

$$[\text{RCT}] \quad C_f(p, [\xi]) := \Theta_q^{-1} M_f \Theta_p([\xi]) = \left[\left(q_j \min_{i \in f^{-1}(j)} \frac{\xi_i}{p_i} \right)_{j \in J} \right]. \quad (185)$$

It is projectively well defined and satisfies

$$C_{gf}(p, [\xi]) = C_g(f_*p, C_f(p, [\xi])), \quad f(W_I(p, [\xi])) = W_J(f_*p, C_f(p, [\xi])).$$

By the active-singleton finite face of (MPRR/MPRS), the block minima have rates q_j and are independent; after q_j -rescaling the coarse seed coordinates are again iid $\text{Exp}(1)$. Hence

$$[\text{RCT}_\mu] \quad (C_f(p, -))_* \text{Dirichlet}_I(1, \dots, 1) = \text{Dirichlet}_J(1, \dots, 1). \quad (186)$$

Thus the calibrated actualization law is stable under arbitrary finite surjective resolution change, but only on the coupled law–resource carrier: in the time chart the transport is the law-independent minimum operation, while in the seed chart the same transport is necessarily p -dependent.

That dependence is irreducible. For a surjection $f : I \twoheadrightarrow J$, a map $\kappa_f : \text{RaceKey}_I \rightarrow \text{RaceKey}_J$ satisfying

$$f(W_I(p, v)) = W_J(f_*p, \kappa_f(v)) \quad (p \in \Delta_I^\circ, v \in \text{RaceKey}_I)$$

exists exactly when f is bijective or $|J| = 1$:

$$[\text{SKF}] \quad |J| > 1 \text{ and } f \text{ noninjective} \implies \text{no law-independent exact race-key transport } \kappa_f. \quad (187)$$

The positive cases are relabeling and the terminal one-point target. For the converse choose $a \neq b$ in one fibre of f and c in another. Fix one projective seed with $\xi_a = \xi_b = \xi_c = 1$ and make all unused coordinates sufficiently large. Choose two full-support laws with the same coarse law, putting total mass 0.6 on the a, b fibre and 0.35 on the c fibre (the remaining mass is spread over any other fibres): in the first law concentrate almost all 0.6 on a , while in the second split it almost equally between a, b . Unused fine labels receive sufficiently small positive mass. The first fine race has its unique winner in the a, b fibre and the second in the c fibre, although f_*p, ξ and hence the hypothetical coarse input are identical. This contradicts the displayed intertwining law.

The physical pointer coarse-graining is a different map: $S_f(v)_j = \sum_{i \in f^{-1}(j)} v_i$. If $V \sim \text{Dirichlet}_I(1, \dots, 1)$, Gamma aggregation gives

$$[\text{SKF}_\mu] \quad S_f(V) \sim \text{Dirichlet}(|f^{-1}(j)|)_{j \in J}, \quad C_f(p, V) \sim \text{Dirichlet}_J(1, \dots, 1). \quad (188)$$

For a genuine merge with $|J| > 1$ these laws differ. Hence pointer-weight summation and resolution-natural actuality transport are distinct typed maps; identifying them would erase a real interface residual. The positive replacement is the law–seed/time change of chart (RTK/RCT), not a hidden resolution-independent selector.

On the quantum instrument face this is a genuine resolution consumer. Let $\mathcal{F} = \Pi_I(m)$ be fixed, let ω be a normal system state with $p_i = \omega(\mathcal{F}_i(1)) > 0$, and for $f : I \twoheadrightarrow J$ define the coarse instrument $(f_*\mathcal{F})_j := \sum_{i \in f^{-1}(j)} \mathcal{F}_i$, whose Born law is $q = f_*p$. For every realized projective key v ,

$$[\text{QRC}] \quad f(W_I(p, v)) = W_J(q, C_f(p, v)). \quad (189)$$

Thus the fine race record and its coarse race record are the same set-valued actualization read-out at two resolutions; on the unique-minimum locus they are the corresponding singleton occurred records. If m is on the rank-one pure-pointer locus of (HRC_0) and its compensation variable is Haar, the fine key law is universally calibrated and (RCT_μ) transports it to the universally calibrated coarse key law. The effective coarse key is generally not the raw weight of the merged pointer projector $P'_j = \sum_{i \in f^{-1}(j)} P_i$, whose law is the first law in (SKF_μ) . Hence exact record compatibility is carried by the law–time transport, not by identifying pointer weights across resolutions. This remains one instrument/context at a time and supplies no context-independent spectral valuation; (QRS/OINB) are therefore unchanged.

Contextual actuality fibration and global law–point gap. Let $\mathcal{M}$ be a unital von Neumann algebra. Let $\text{PVM}_f(\mathcal{M})$ have as objects labelled finite nonzero PVMs $P = (P_i)_{i \in I_P}$ and as arrows $f : P \rightarrow Q$ the surjections $f : I_P \twoheadrightarrow I_Q$ satisfying $Q_j = \sum_{i \in f^{-1}(j)} P_i$. Put

$$\mathcal{O}_\mathcal{M}(P) := I_P, \quad \mathcal{T}_\mathcal{M}(P) := \text{TimeKey}_{I_P}, \quad \mathcal{T}_\mathcal{M}(f) := M_f,$$

and let $\mathcal{T}_\mathcal{M}^{\text{one}}(P)$ be the locus on which the minimum is unique. By (RTK/RTW) this is a subfunctor. For $i \in I_P$ let $\tau_P^i \in (0, \infty)^{I_P}$ have value 1 at i and 2 elsewhere. Unique minimization and this two-level embedding are natural and give

$$[\text{QCAF}] \quad \mathcal{O}_\mathcal{M} \xrightarrow{\iota} \mathcal{T}_\mathcal{M}^{\text{one}} \xrightarrow{w} \mathcal{O}_\mathcal{M}, \quad \iota_P(i) = [\tau_P^i], \quad w_P([t]) = \text{argmin } t, \quad wt = 1. \quad (190)$$

Indeed $M_f[\tau_P^i] = [\tau_Q^{f(i)}]$: the block containing i has minimum 1 and every other block minimum 2. Hence

$$\text{Nat}(1, \mathcal{O}_\mathcal{M}) \neq \emptyset \iff \text{Nat}(1, \mathcal{T}_\mathcal{M}^{\text{one}}) \neq \emptyset.$$

Moreover there is a canonical bijection

$$[\text{QSG}] \quad \text{Nat}(1, \mathcal{O}_\mathcal{M}) \cong \lim_{\leftarrow C \in \mathcal{C}(\mathcal{M})} \text{Spec}(C), \quad (191)$$

where $C(\mathcal{M})$ is the poset of unital abelian von Neumann subalgebras and transition maps are spectral restriction. To see this, a coherent PVM outcome family defines $v(e) \in \{0, 1\}$ for every projection e by the binary PVM $(e, 1 - e)$, with the trivial $e = 0, 1$ cases understood. Coarse-graining coherence says that exactly one projection in every finite nonzero orthogonal partition of 1 has value 1. On each abelian C , this ultrafilter on its projections defines a character first on finite spectral sums and then uniquely on C by uniform spectral approximation; common refinements make the definition independent of the chosen spectral sum and compatible under inclusion. Conversely a compatible family of characters selects the unique P_i on which the character is 1. The constructions are inverse.

Consequently the (CSL) no- I_1/I_2 perimeter gives the sharp contextual settlement gap

$$[\text{QSG}_0] \quad \boxed{\mathcal{M} \text{ has no type } I_1/I_2 \text{ summand} \implies \text{Nat}(\underline{1}, \mathcal{T}_{\mathcal{M}}^{\text{one}}) = \emptyset, \quad \text{Nat}(\underline{1}, \mathcal{T}_{\mathcal{M}}) \neq \emptyset.} \quad (192)$$

The second assertion is witnessed by the total-tie section $P \mapsto [(1)_{i \in I_P}]$, preserved by every block-minimum map. Thus every local unique-minimum fibre is inhabited, and the full residual family even has a coherent global section, while no coherent global singleton occurrence exists on this perimeter. The obstruction is gluing, not local absence.

If now ρ is faithful normal, put $p_P^\rho(i) = \rho(P_i) \in \Delta_{I_P}^\circ$ and define the state-indexed race-key functor

$$R_\rho(P) := \text{RaceKey}_{I_P}, \quad R_\rho(f) := C_f(p_P^\rho, -).$$

Since $p_Q^\rho = f_* p_P^\rho$, (RCT) makes this a functor and the charts $\Theta_{p_P^\rho}$ assemble to a natural isomorphism

$$[\text{BTS}] \quad \boxed{\Theta^\rho : R_\rho \xrightarrow{\sim} \mathcal{T}_{\mathcal{M}}, \quad b_\rho(P) := [p_P^\rho], \quad R_\rho(f)b_\rho(P) = b_\rho(Q), \quad W_{I_P}(p_P^\rho, b_\rho(P)) = I_P.} \quad (193)$$

Thus the Born law determines, by the displayed typed coordinate map, a canonical globally coherent *tie* section of the actualization-key carrier: p_P^ρ remains a probability law while $b_\rho(P) = [p_P^\rho]$ is its distinct RaceKey-typed image. In race-time coordinates it is the constant section $[(1)_i]$, so it contains no preferred local outcome. Let $R_\rho^{\text{one}} := (\Theta^\rho)^{-1} \mathcal{T}_{\mathcal{M}}^{\text{one}}$. Then

$$[\text{QALG}] \quad \boxed{\mu_P := \text{Dirichlet}_{I_P}(1, \dots, 1), \quad (R_\rho(f))_* \mu_P = \mu_Q, \quad \mathcal{M} \text{ no } I_1/I_2 \implies \text{Nat}(\underline{1}, R_\rho^{\text{one}}) = \emptyset.} \quad (194)$$

Here $\mu_P(R_\rho^{\text{one}}(P)) = 1$, and the first equality is precisely (RCT) $_\mu$. Hence the calibrated local actualizer laws glue under every compatible finite resolution change although their realized unique-outcome points do not glue globally. A compatible family of fibre laws is not a probability law on a nonexistent global point-section space. Thus the first separation in (QRS) acquires an explicit positive carrier rather than a hidden-variable completion: contextuality is preserved, not bypassed. In particular local occurrence may be real and resolution-coherent without assigning pre-existing outcomes to incompatible contexts. No identification with an arbitrary external contextuality cover is asserted here; the comparison is only through the declared PVM-context category above.

Support-stratified race transport and sequential quantum restart. The full-support hypothesis in (RCT/QCAF) is not stable under selective quantum update, so retain the zero-probability boundary rather than regularizing it. For a nonempty finite set I put

$$\text{STime}_I := \coprod_{\emptyset \neq S \subseteq I} \text{TimeKey}_S.$$

For a surjection $f : I \twoheadrightarrow J$ define

$$[\text{SRTK}] \quad \boxed{\text{SM}_f(S, [t]) := (f(S), M_{f|_S}[t]), \quad \text{SM}_{gf} = \text{SM}_g \text{SM}_f.} \quad (195)$$

This is well defined because $f|_S : S \twoheadrightarrow f(S)$ is surjective, and functoriality is exactly (RTK) on the restricted supports.

For $p \in \Delta_I$ let $S_p = \{i : p_i > 0\}$. Regard the effective seed key as RaceKey_{S_p} and put $\Theta_p^s([\xi]) = [(\xi_i/p_i)_{i \in S_p}] \in \text{TimeKey}_{S_p}$. If $q = f_* p$, then $S_q = f(S_p)$ and the support-restricted seed transport is

$$[\text{SRCT}] \quad \boxed{C_f^s(p, [\xi]) := \left[\left(q_j \min_{\substack{i \in S_p \\ f(i)=j}} \frac{\xi_i}{p_i} \right)_{j \in S_q} \right] \in \text{RaceKey}_{S_q}.} \quad (196)$$

The square with Θ_p^s, Θ_q^s and SM_f commutes, hence

$$C_{gf}^s(p, -) = C_g^s(f_* p, -) C_f^s(p, -), \quad f(W_{S_p}(p|_{S_p}, v)) = W_{S_q}(q|_{S_q}, C_f^s(p, v)).$$

Here $p|_{S_p}$ denotes the same probability vector with its zero coordinates deleted. If $V \sim \text{Dirichlet}_{S_p}(1)$, the exponential-minimum proof of (RCT) $_\mu$ on the restricted support gives

$$[\text{SRCT}_\mu] \quad \boxed{(C_f^s(p, -))_* \text{Dirichlet}_{S_p}(1) = \text{Dirichlet}_{S_q}(1).} \quad (197)$$

Thus zero Born weights remove impossible outcomes from the active fibre; they do not require an ε -regularization or an arbitrary tie convention.

This support stratification removes the faithfulness restriction from the state-indexed part of (QCAF). For any normal state ρ and finite nonzero PVM $P = (P_i)$ put $S_{\rho, P} = \{i : \rho(P_i) > 0\}$ and define

$$R_\rho^s(P) := \text{RaceKey}_{S_{\rho, P}}, \quad T_\rho^s(P) := \text{TimeKey}_{S_{\rho, P}}.$$

On a PVM coarse-graining $f : P \rightarrow Q$, use respectively $C_f^s(p_P^\rho, -)$ and $M_{f|_{S_{\rho, P}}}$. The preceding identities make both assignments functors and $\Theta_{p_P^\rho}^s$ a natural isomorphism $R_\rho^s \simeq T_\rho^s$. The typed Born image $[p_P^\rho|_{S_{\rho, P}}]$ is again the total-tie section, and the Dirichlet laws in (SRCT) $_\mu$ glue under resolution. Its unique-minimum subfunctor still maps naturally to $\mathcal{O}_{\mathcal{M}}$, so on the no- I_1/I_2 perimeter (QSG/CSL) still forbids a global singleton section. Loss of faithfulness after an occurrence therefore changes the active support stratum but does not remove the contextual law-point gap.

For a finite-dimensional system this gives an exact sequential consumer. Let a finite-depth history-dependent family of Schrödinger instruments $\mathcal{I}^h = (\mathcal{I}_i^h)_{i \in I_h}$ be assigned to each reachable history h , with $\sum_i \mathcal{I}_i^h$ trace preserving. Starting from a density matrix $\rho_\emptyset$, recursively put

$$p_h(i) = \text{Tr } \mathcal{I}_i^h(\rho_h), \quad S_h = \text{supp } p_h, \quad i_h = W_{S_h}(p_h|_{S_h}, V_h), \quad \rho_{hi} = \frac{\mathcal{I}_i^h(\rho_h)}{p_h(i)} \quad (i \in S_h),$$

where, at each visited history, V_h is an independently realized $\text{Dirichlet}_{S_h}(1)$ key. Then for every reachable path $\mathbf{i} = (i_0, \dots, i_{m-1})$, writing $h_r = (i_0, \dots, i_{r-1})$,

$$[\text{SQAR}] \quad \mathbb{P}(\mathbf{i}) = \prod_{r \leq m} p_{h_r}(i_r) = \text{Tr} \left[I_{i_{m-1}}^{h_{m-1}} \circ \dots \circ I_{i_0}^{\varnothing}(\rho_{\varnothing}) \right], \quad \rho_{\mathbf{i}} = \frac{I_{i_{m-1}}^{h_{m-1}} \circ \dots \circ I_{i_0}^{\varnothing}(\rho_{\varnothing})}{\mathbb{P}(\mathbf{i})}. \quad (198)$$

The first equality is the conditional Dirichlet race calibration on the active support; the second and the state formula follow by induction from the definition of selective instruments. This is the support-complete history-dependent form of (IRO/KLC): occurrence updates the state, the updated state determines the next law and support fibre, and a new realized key actualizes the next branch. The recursion therefore composes law, support, occurrence and update without identifying them. It does not generate the realized V_h from ρ_h or from the instrument; (RNB/OINB) still forbids that bootstrap at the corresponding naturality perimeter, while (QIF/QOF/QRS) keep post-outcome update and global quantum extension as independent residual coordinates.

Law-actuality Grothendieck carrier. The preceding law, support and actualizer transports are the charts of one dependent object rather than separate primitives. Let $\mathbf{A} : \mathbf{FinSurj}_+ \rightarrow \mathbf{Set}$ send a nonempty finite set I to its probability simplex Δ_I and a surjection f to pushforward f_* . Write

$$\text{Law}_{\text{fin}} := \int_{I \in \mathbf{FinSurj}_+} \mathbf{A}(I),$$

so an arrow $f : (I, p) \rightarrow (J, q)$ means $f : I \rightarrow J$ and $q = f_*p$. For $S_p = \text{supp } p$ define

$$\mathbf{K}_{\text{act}}(I, p) := \text{RaceKey}_{S_p}, \quad \mathbf{K}_{\text{time}}(I, p) := \text{TimeKey}_{S_p},$$

with arrow maps $C_f^s(p, -)$ and $M_{f|_{S_p}}$, respectively. By (SRTK/SRCT) these are covariant functors on Law_{fin} , and the support charts assemble to a natural isomorphism. Thus

$$[\text{LAC}] \quad \pi_{\text{act}} : \mathbf{Act}_{\text{fin}} := \int_{(I, p) \in \text{Law}_{\text{fin}}} \mathbf{K}_{\text{act}}(I, p) \longrightarrow \text{Law}_{\text{fin}} \text{ is a discrete opfibration,} \quad \Theta^s : \mathbf{K}_{\text{act}} \xrightarrow{\sim} \mathbf{K}_{\text{time}}. \quad (199)$$

Explicitly, for every base arrow $f : (I, p) \rightarrow (J, q)$ and every $v \in \mathbf{K}_{\text{act}}(I, p)$ there is a unique arrow of $\mathbf{Act}_{\text{fin}}$ over f with source $((I, p), v)$, namely the arrow to $((J, q), C_f^s(p, v))$; this is the discrete-opfibration lifting property. In the time chart the transport is law-independent blockwise minimum; in the seed chart the same opfibration is the law-dependent transport C_f^s . The dependence is therefore a change of chart on one carrier, not an extra interaction primitive.

Let $\text{Win}(I, p) = \mathcal{P}_+(S_p)$, with direct-image transport under $f|_{S_p}$. The race winner defines a natural transformation

$$w : \mathbf{K}_{\text{act}} \Rightarrow \text{Win}, \quad w_{I, p}(v) = W_{S_p}(p|_{S_p}, v),$$

by (SRCT). Let $\mathbf{K}_{\text{act}}^{\text{one}}$ be its unique-winner subfunctor (unique winners remain unique after surjective coarse-graining), and $w_{I, p}^{\text{one}} : \mathbf{K}_{\text{act}}^{\text{one}}(I, p) \rightarrow S_p$ the singleton winner. There is a canonical natural section of the unpointed key carrier,

$$[\text{LNB}] \quad b_{I, p} := [p|_{S_p}] \in \mathbf{K}_{\text{act}}(I, p), \quad \mathbf{K}_{\text{act}}(f)b_{I, p} = b_{J, f_*p}, \quad w_{I, p}(b_{I, p}) = S_p, \quad (200)$$

but no natural section of the unique-winner carrier:

$$\text{Nat}_{\text{Law}_{\text{fin}}}(\underline{1}, \mathbf{K}_{\text{act}}^{\text{one}}) = \emptyset.$$

Indeed at the binary uniform object $(2, u_2)$ the transposition is an automorphism of Law_{fin} ; a natural section would have to choose a transposition-fixed unique-winner key, while the only fixed projective key is the total tie. Thus a law has a canonical coherent residual key but not a canonical occurred point.

The carrier also has a canonical calibrated fibre law. Put $\mu_{I, p} := \text{Dirichlet}_{S_p}(1, \dots, 1)$, with the singleton convention when $|S_p| = 1$. Then (SRCT $_{\mu}$ /URC) gives

$$[\text{CAL}] \quad (C_f^s(p, -))_* \mu_{I, p} = \mu_{J, f_*p}, \quad \mu_{I, p}(\mathbf{K}_{\text{act}}^{\text{one}}(I, p)) = 1, \quad (w_{I, p}^{\text{one}})_* \mu_{I, p} = p|_{S_p}. \quad (201)$$

Moreover this law is unique under the exact target-blind calibration hypothesis: for each fixed nonempty support S , if a Borel probability ν_S on RaceKey_S is independent of $p \in \Delta_S^{\circ}$ and satisfies $(w_{S, p}^{\text{one}})_* \nu_S = p$ for every such p , then

$$[\text{CAL}_0] \quad \nu_S = \text{Dirichlet}_S(1, \dots, 1). \quad (202)$$

For $|S| = 1$ this is tautological; for $|S| \geq 2$ it is exactly (URC) on the support stratum. The law (SRCT $_{\mu}$) supplies the compatibility between strata and resolutions. Thus the law determines a calibrated law on the actualizer fibre under the stated target-blind universality condition, but not its realized point.

The physical and quantum consumers are now pullbacks/realizations of the same carrier. On the pure rank-one pointer locus of (FIC/HRC), Haar compensation has projective coordinates represented by iid exponentials, so restriction to any active support S_p has precisely the fibre law $\mu_{I, p}$ in (CAL). For a normal state ρ , the Born assignment

$$[\text{BPF}] \quad B_{\rho} : \text{PVM}_f(\mathcal{M}) \longrightarrow \text{Law}_{\text{fin}}, \quad P \longmapsto (I_P, p_P^{\rho}), \quad f \longmapsto f \quad (203)$$

is a functor, and $B_{\rho}^* \mathbf{K}_{\text{act}} = \mathbf{R}_{\rho}^s$, $B_{\rho}^* \mathbf{K}_{\text{time}} = \mathbf{T}_{\rho}^s$ exactly. Hence (QRC) is the one-arrow pullback of (LAC), the state-indexed part of (QCAF) is its context pullback, and (SQAR) repeatedly re-enters its law base after selective update. The latter update is an attachment, not a morphism forced by $\mathbf{FinSurj}_+$: different post-outcome quantum states can occur at fixed outcome law by (QIF/QOF). Thus the carrier absorbs law/support/key/resolution transport without erasing update, context choice, history or the realized occurrence coordinate. More precisely, (PRK/ERD) is its realized projective-seed presentation followed by w , (URC) is the fixed-support uniqueness clause (CAL $_0$), (RTK/SRCT) are its two transport charts, (QRC/QCAF) are pullbacks along resolution/context maps, and (SQAR) is repeated re-entry after the independent update attachment. These are derivability maps, not identifications of law, fibre point, occurrence and update.

Occurrence-settlement firewall and event-conditioned residual law. The preceding carrier also makes precise why occurrence cannot be identified with (SRS�) terminalization. The two notions are incomparable already inside $\mathbf{Law}_{\text{fin}}$. First, pull K_{act} back along the arrow $(2, \delta_0) \longrightarrow (1, \delta_*)$.

Both active supports are singletons, so the pulled-back key functor is constantly $\mathbf{1}$ and has a unique section, while its Grothendieck total is $[1] \neq \mathbf{1}$; by (SRS�) the noninvertible surviving base arrow prevents settlement. Conversely, at the binary uniform law let τ exchange the two labels and choose any non-tie key v . Its C_2 -orbit $O_v = \{v, \tau v\} \subset K_{\text{act}}^{\text{one}}(2, u_2)$ is free and transitive, so

$$[\text{OSF}] \quad \boxed{\text{Nat}_{[1]}(\underline{1}, \mathbf{1}) \simeq \mathbf{1} \text{ but } \int_{[1]} \mathbf{1} \simeq [1] \neq \mathbf{1}, \quad \int_{BC_2} O_v \simeq EC_2 \simeq \mathbf{1} \text{ but } \text{Nat}_{BC_2}(\underline{1}, O_v) = \emptyset.} \quad (204)$$

Thus coherent settlement of a residual total and a realized/natural point section cannot be compressed to one predicate. This is the (PRA/SRS�/TSF) distinction instantiated on the law-actuality carrier, not a new actuality primitive.

The positive replacement is the residual left after an occurrence. Fix (I, p) , put $S = S_p$, $\mu_p = \text{Dirichlet}_S(1)$, and for $i \in S$ let

$$C_i(p) := \{v \in \text{RaceKey}_S : W_S(p|_S, v) = \{i\}\}, \quad \mu_p^i(A) := \frac{\mu_p(A \cap C_i(p))}{p_i}.$$

The chambers are disjoint and cover a μ_p -conull set by (CAL), hence

$$[\text{ECRL}] \quad \boxed{\mu_p = \sum_{i \in S} p_i \mu_p^i, \quad \mu_p^i(C_i(p)) = 1, \quad |S| > 1 \implies \mu_p^i \text{ is non-Dirac.}} \quad (205)$$

By the singleton-cell splitting and restart clauses of (MPRR/MPRS), conditional on winner i , if $M = T_i$, then

$$[\text{ECRL}_i] \quad \boxed{M \sim \text{Exp}(1), \quad T_k = M + E_k \ (k \neq i), \quad (E_k)_{k \neq i} \text{ independent, } E_k \sim \text{Exp}(p_k),} \quad (206)$$

with M independent of the residuals. Projectivization gives $(\Theta_p^S)_* \mu_p^i$; for $|S| > 1$, $T_k/T_i = 1 + E_k/M$ has nondegenerate continuous law, proving that occurrence fixes the read-out but leaves a nonterminal actualizer continuum. For every $f : (I, p) \rightarrow (J, q)$, the MPRS product law also gives internal-winner/block-minimum independence and its refinement/restart clauses give

$$[\text{ECRL}_o] \quad \boxed{(C_f^S(p, -))_* \mu_p^i = \mu_q^{f(i)}}. \quad (207)$$

Thus CAL disintegration commutes with finite resolution change. On the quantum pullback B_p , a recorded outcome terminalizes only its outcome coordinate; the conditioned actualizer fibre remains residual, while (QIF/QOF) independently retain post-outcome update and global extension fibres. Hence neither (SRS�) settlement nor occurrence launders the remaining quantum coordinates into a selected hidden world.

Kraus-refined physical actualization. The rank-one calibration face is not a restriction on the observed finite instrument. Let H be finite-dimensional and let $\mathcal{J} = (\mathcal{J}_i)_{i \in I}$ be a finite Heisenberg instrument on $B(H)$, after corestricting away identically zero outcome maps, with $\sum_i \mathcal{J}_i$ unital. Choose a finite Kraus refinement

$$\mathcal{J}_i(a) = \sum_{\alpha \in A_i} K_{i\alpha}^* a K_{i\alpha}, \quad A := \coprod_{i \in I} A_i, \quad f : A \twoheadrightarrow I,$$

with all $K_{i\alpha} \neq 0$. Writing $K_a = K_{i\alpha}$, the map

$$V : H \longrightarrow H \otimes \mathbb{C}^A, \quad Vx = \sum_{a \in A} K_a x \otimes e_a$$

is an isometry because $\sum_a K_a^* K_a = 1$. Choose a unit vector $\psi_0 \in \mathbb{C}^A$ and extend the isometry $x \otimes \psi_0 \mapsto Vx$ to a unitary U on $H \otimes \mathbb{C}^A$. With $P_a = |e_a\rangle\langle e_a|$ and $\sigma_0 = |\psi_0\rangle\langle\psi_0|$, the framed model $m = (\mathbb{C}^A, \sigma_0, U, (P_a)_{a \in A})$ satisfies

$$[\text{KRPA}_0] \quad \boxed{\Pi_A(m)_a(x) = K_a^* x K_a, \quad f_* \Pi_A(m) = \mathcal{J}.} \quad (208)$$

Thus the usual finite Kraus dilation supplies a pure rank-one pointer refinement; no novelty is claimed for this dilation step itself. If $|A| = 1$ all key/outcome laws below are singleton and the conclusion is tautological, so the nontrivial case has $|A| \geq 2$ and lies in the stated (FIC/HRC) perimeter.

For a density matrix ρ , put

$$r_a = \text{Tr}(\rho K_a^* K_a), \quad p_i = \sum_{a \in f^{-1}(i)} r_a,$$

so p is the observed physical Born law and $f(S_r) = S_p$. Apply the fixed-instrument compensation orbit (FIC) to the refined model. If W is Haar on $\mathcal{U}(\mathbb{C}^A)$, then (HRC) gives $\lambda_m(W) \sim \text{Dirichlet}_A(1)$. Using its iid-exponential projective representative, restriction to S_r again has law $\text{Dirichlet}_{S_r}(1)$; hence the support transport

$$V_{\text{obs}}(W) := C_f^S(r, [\lambda_m(W)|_{S_r}]) \in \text{RaceKey}_{S_p}$$

satisfies, by (SRCT $_{\mu}$ /CAL),

$$[\text{KRPA}] \quad \boxed{\Pi_A(m[W]) = \Pi_A(m), \quad \text{Law}(V_{\text{obs}}(W)) = \text{Dirichlet}_{S_p}(1), \quad \mathbb{P}[W_{S_p}(p|_{S_p}, V_{\text{obs}}(W)) = \{i\}] = p_i.} \quad (209)$$

Moreover winner naturality gives

$$W_{S_p}(p|_{S_p}, V_{\text{obs}}(W)) = f(W_{S_r}(r|_{S_r}, [\lambda_m(W)|_{S_r}])) \quad \text{a.s.}$$

The fine Kraus-label winner on the right is only an internal read-out used to prove coarse naturality. It is not declared to be a physically recorded sub-outcome: unless that refinement is separately measured, the selective state update remains the given coarse instrument update $\mathcal{I}_i(\rho) = \sum_{\alpha \in A_i} K_{i\alpha} \rho K_{i\alpha}^*$, where $\mathcal{I}_i$ is the Schrödinger predual of $\mathcal{J}_i$. Therefore every finite-dimensional finite-outcome instrument admits a pure rank-one refined physical realization whose fixed-refined-instrument Haar compensation orbit realizes the canonical (CAL) law at the observed outcome resolution after the canonical (SRCT) transport. Raw coarse pointer weights are not substituted for this map; their Dirichlet block-rank law remains the distinct (SKF $_{\mu}$) interface.

Different Kraus refinements need not identify realized keys or compensation points, but they induce the same observed fibre law $\text{Dirichlet}_{S_p}(1)$ and the same Born winner law p . Thus Kraus/Stinespring presentation is erased at the calibrated-law/output interface while its realization-level

multiplicity remains a typed fibre. For a finite-depth family of selective Schrödinger instruments, choosing such a refinement at each reachable history and independent Haar compensation points W_h replaces the abstract V_h in (SQAR) by the displayed physical read-out and reproduces its exact path law and posterior states. The theorem still does not generate the realized W_h from the instrument: (MDNB/RNB/OINB) is the exact no-bootstrap boundary. The new content here is the bridge from the standard rank-one refinement through (FIC/HRC/SRCT) to the law–actuality carrier, not the standard dilation theorem.

Compensation-covariant convolution dynamics and Haar-attractor boundary. The preceding Haar law can be dynamically selected as a law only after an additional mixing input; compensation symmetry by itself supplies invariance but not attraction. Let G be a compact group with normalized Haar probability h_G , and let P be a Borel Markov kernel on G satisfying full left covariance

$$P(gW, gA) = P(W, A) \quad (g, W \in G, A \subseteq G \text{ Borel}).$$

Put $\nu = P(e, \cdot)$, write ν^{*k} for the k -fold convolution under group multiplication, and use the total-variation convention of (DLA). Then, for every $k \geq 1$,

$$[\text{CHA}] \quad \boxed{P(W, A) = \nu(W^{-1}A), \quad P^k(W, \cdot) = (L_W)_* \nu^{*k}, \quad h_G P = h_G,} \quad (210)$$

and therefore

$$[\text{CHA}_\delta] \quad \boxed{\delta(P^k) := \sup_{W, W'} \|P^k(W, \cdot) - P^k(W', \cdot)\|_{\text{TV}} = \sup_{g \in G} \|\nu^{*k} - (L_g)_* \nu^{*k}\|_{\text{TV}}.} \quad (211)$$

Indeed covariance with $g = W^{-1}$ gives the first identity and induction gives the convolution power. For fixed V , normalized Haar is right invariant on the compact group, hence

$$\int_G P(W, A) dh_G(W) = \int_G \int_G \mathbf{1}_A(WV) d\nu(V) dh_G(W) = h_G(A).$$

Left-translation invariance of total variation gives (CHA $_\delta$).

If for some $\ell \geq 1$ $d := \delta(P^\ell) < 1$, then h_G is the unique P -invariant probability and, for every probability η on G ,

$$[\text{HAG}] \quad \boxed{\|\eta P^n - h_G\|_{\text{TV}} \leq d^{\lfloor n/\ell \rfloor} \|\eta - h_G\|_{\text{TV}} \quad (n \geq 0).} \quad (212)$$

For completeness, if $Q = P^\ell$, α, β are probabilities and $\sigma = \alpha - \beta$, the Hahn decomposition writes $\sigma = \|\sigma\|_{\text{TV}}(\alpha_0 - \beta_0)$ when $\sigma \neq 0$, with α_0, β_0 probabilities. Since

$$\alpha_0 Q - \beta_0 Q = \int_{G \times G} (Q(x, \cdot) - Q(y, \cdot)) d\alpha_0(x) d\beta_0(y),$$

convexity of total variation gives $\|\alpha_0 Q - \beta_0 Q\|_{\text{TV}} \leq d\|\alpha - \beta\|_{\text{TV}}$. Iterating and using Markov nonexpansiveness for the final remainder proves (HAG); uniqueness follows by applying the same contraction to two invariant laws. A concrete sufficient mixing input is the Doeblin minorization

$$[\text{HAG}_0] \quad \boxed{\nu^{*\ell} \geq \varepsilon h_G \quad (\varepsilon > 0) \implies \delta(P^\ell) \leq 1 - \varepsilon.} \quad (213)$$

This is an input on the increment law, not a consequence of compensation symmetry.

The deterministic boundary is exact. If $P(W, \cdot) = \delta_{T(W)}$ and the same covariance holds, then $T(gW) = gT(W)$; putting $a = T(e)$ gives

$$[\text{CHA}_0] \quad \boxed{T(W) = Wa, \quad P^n(W, \cdot) = \delta_{W a^n}, \quad \delta(P^n) = 1, \quad \|\eta P^n - h_G\|_{\text{TV}} = \|\eta - h_G\|_{\text{TV}}} \quad (214)$$

for every $n \geq 1$ whenever G has more than one point. More generally, a finite composition of possibly time-dependent deterministic left-covariant steps remains a right translation and preserves total-variation distance from Haar. Thus deterministic full compensation covariance cannot generate Haar attraction of one-time laws; in particular, symmetry plus longer evolution time does not by itself produce the Haar law. This does not rule out Cesaro/time-average equidistribution, which is a different actualization notion from the one-time law convergence used here.

Now take $G = \mathcal{U}(K)$ in the fixed-instrument orbit (FIC) and let $\lambda_m : G \rightarrow \Delta_I$ be its pointer-weight map. Data processing for total variation and (HRC) give, for $k_n = \lfloor n/\ell \rfloor$,

$$[\text{HAG}_{\text{FIC}}] \quad \boxed{\|(\lambda_m)_*(\eta P^n) - \text{Dirichlet}(r_i)_{i \in I}\|_{\text{TV}} \leq d^{k_n} \|\eta - h_G\|_{\text{TV}}.} \quad (215)$$

On the rank-one pointer locus this limit is exactly the calibrated Dirichlet $_I(1)$ law of (CAL). For full-support p , if $A_i(p) = \{W : W_I(p, \lambda_m(W)) = \{i\}\}$ and $A_{\text{tie}}(p)$ is the complementary tie set, (URC/HRC $_0$) therefore yield

$$[\text{HAG}_{\text{Born}}] \quad \boxed{|\eta P^n(A_i(p)) - p_i| \leq d^{k_n} \|\eta - h_G\|_{\text{TV}}, \quad \eta P^n(A_{\text{tie}}(p)) \leq d^{k_n} \|\eta - h_G\|_{\text{TV}}.} \quad (216)$$

For the rank-one Kraus refinement in (KRPA), with observed-key map $V_{\text{obs}} : G \rightarrow \text{RaceKey}_{S_p}$, put

$$B_i(p) := \{W : W_{S_p}(p|_{S_p}, V_{\text{obs}}(W)) = \{i\}\}, \quad B_{\text{tie}}(p) := G \setminus \bigcup_{i \in S_p} B_i(p).$$

The same contraction survives support restriction and resolution transport, while occurrence is kept relation-valued on the transient tie locus:

$$[\text{HAG}_{\text{KRPA}}] \quad \boxed{\begin{aligned} & \| (V_{\text{obs}})_*(\eta P^n) - \text{Dirichlet}_{S_p}(1) \|_{\text{TV}} \leq d^{k_n} \|\eta - h_G\|_{\text{TV}}, \\ & |\eta P^n(B_i(p)) - p_i| \leq d^{k_n} \|\eta - h_G\|_{\text{TV}} \quad (i \in S_p), \quad \eta P^n(B_{\text{tie}}(p)) \leq d^{k_n} \|\eta - h_G\|_{\text{TV}}. \end{aligned}} \quad (217)$$

Thus a supplied compensation-covariant mixing dynamics inside one fixed refined-instrument fibre selects the canonical CAL/Haar law asymptotically and transports it to the exact Born law at the observed resolution. The pure compact-group/Haar and Dobrushin ingredients are standard; the SCO content is their typed bridge through (FIC/HRC/KRPA) and the exact deterministic no-mixing boundary. No claim is made that the instrument itself determines P , that a microscopic apparatus dynamics satisfies (HAG) or (HAG $_0$), or that convergence of laws supplies a realized Markov path. A realized compensation trajectory/noise remains point-bearing occurrence data, so (MDNB/RNB/OINB) is unchanged.

Compensation-dynamics nonidentifiability. The mixing input in (HAG) is a genuine residual coordinate even after the complete static fixed-instrument compensation carrier has been specified. Let G be any nontrivial compact group. For $a \in G$ and normalized Haar h_G , define

two left-covariant Markov kernels

$$P_a(W, \cdot) := \delta_{W_a}, \quad P_H(W, \cdot) := h_G.$$

Both obey the covariance hypothesis of (CHA), have invariant law h_G , and live on exactly the same state carrier G , but

$$[\text{CDNI}] \quad \boxed{\delta(P_a^n) = 1 \quad (n \geq 1), \quad \delta(P_H) = 0, \quad \eta P_H = h_G \quad (\forall \eta).} \quad (218)$$

Thus compensation symmetry, its Haar measure, and its full static point space do not determine the dynamical mixing class. This remains true on one fixed measurement model. If $m = (K, \sigma_\psi, U, (P_i))$ is in the pure FIC perimeter and $G = \mathcal{U}(K)$, both kernels evolve only the same compensation coordinate $W \mapsto m[W]$, while

$$\Pi_I(m[W]) = \Pi_I(m) \quad (W \in G)$$

for every state reached by either dynamics. Hence the observed instrument, the compensation orbit, the pointer-weight map λ_m , and the Haar/calibration law are all held fixed while the one-time approach to Haar varies from exact nonconvergence in total variation to one-step mixing. In particular no function of those static data can recover the actual compensation kernel or its Dobrushin class.

This is also the exact interface with the existing temporal environment construction. The (MEnv) carrier takes a Markov kernel P as input and reconstructs/classifies its marked temporal envelope; on the fixed pointed skeleton the current theorem gives $R_m = R_c = \Delta$ for $m(P) = [\text{MEnv}(P, \mu)]$ and $c(P) = T_P - I$. Therefore MEnv can distinguish P_a from P_H once the dynamics is supplied, but cannot be used in the reverse direction to infer which kernel is physically realized from the static measurement instrument. Doing so would promote an input coordinate into its own producer.

Consequently (CHA/HAG) and (CDNI) give a sharp two-sided boundary:

$$\begin{aligned} \text{static compensation symmetry} &\implies h_G \text{ is invariant,} \\ \text{static compensation data} &\not\Rightarrow \text{mixing class,} \\ \delta(P^\ell) < 1 &\implies \eta P^n \longrightarrow h_G \text{ geometrically in TV.} \end{aligned}$$

The last line is a sufficient dynamical gate, not a classification of every possible mechanism of Haar convergence. The physical source of that dynamical datum is therefore a separate attachment. This is an anti-compression obstruction, not a new noise primitive: P_a versus P_H is an equal-static-data/different-dynamics witness. Neither branch supplies a realized trajectory from its law.

Equivariant environment reduction and stochastic-provenance fibre. The preceding dynamical attachment has an exact provenance normal form on the classical standard-Borel reduction perimeter. Let G be a compact metrizable group, let E be a standard Borel space with probability law λ , and let $A : G \times E \rightarrow G$ be Borel and left equivariant in the compensation coordinate,

$$A(gW, e) = gA(W, e) \quad (g, W \in G, e \in E).$$

Put $a(e) := A(1, e)$, $\nu := a_*\lambda$, and define the reduced kernel obtained by forgetting the environment point,

$$P_\lambda(W, B) := \int_E \mathbf{1}_B(A(W, e)) d\lambda(e).$$

Then equivariance is equivalent to $A(W, e) = Wa(e)$, and hence

$$[\text{ECP}] \quad \boxed{P_\lambda(W, B) = \nu(W^{-1}B), \quad \nu = a_*\lambda.} \quad (219)$$

Thus every such deterministic environment coupling reduces exactly to the convolution face of (CHA), and its complete one-step stochastic input is the push-forward law $a_*\lambda$, not an additional hidden coordinate. Conversely every convolution kernel $P(W, B) = \nu(W^{-1}B)$ has such a dilation, for example with $E = G$, $a = I_G$, $\lambda = \nu$, and $A(W, e) = We$. This converse is a realization of a supplied law, not a producer of that law.

For fixed $a : E \rightarrow G$ and fixed reduced increment law ν , the full environment-law implementation fibre is precisely the existing probability lift

$$[\text{RSP}] \quad \boxed{\text{EnvLift}_a(\nu) := \{\lambda \in \mathcal{P}(E) : a_*\lambda = \nu\} = \text{Lift}_a(\nu) \cong_{\text{aff}} \text{LKer}_a(\nu).} \quad (220)$$

Hence (LPA/ELR) supplies its disintegration and extreme boundary; no new realization primitive is introduced. The fibre can be nonterminal even when the reduced kernel is fixed: for $E = G \times \{0, 1\}$, $a(g, j) = g$, the distinct laws $\nu \otimes \delta_0$ and $\nu \otimes \delta_1$ induce the same P . Therefore neither the reduced dynamics nor its mixing class reconstructs the ambient implementation law without an additional section/identification.

The probabilistic provenance boundary is exact. If $\lambda = \delta_e$, then $\nu = \delta_{a(e)}$ and P_λ is the deterministic right translation of (CHA₀). Consequently, for nontrivial G ,

$$[\text{ECP}_0] \quad \boxed{\delta(P_\lambda^\ell) < 1 \text{ for some } \ell \implies a_*\lambda \text{ is non-Dirac} \implies \lambda \text{ is non-Dirac.}} \quad (221)$$

The converses are false: a non-Dirac environment law may collapse to a Dirac increment under constant a , and a non-Dirac increment need not satisfy the mixing gate of (HAG). If instead $(a_*\lambda)^{\ast\ell} \geq \varepsilon h_G$, then (HAG) applies and

$$[\text{ECP}_{\text{mix}}] \quad \boxed{\|\eta P_\lambda^n - h_G\|_{\text{TV}} \leq (1 - \varepsilon)^{\lfloor n/\ell \rfloor} \|\eta - h_G\|_{\text{TV}.} \quad (222)$$

On $G = \mathcal{U}(K)$, the (FIC/HRC/KRPA) push-forwards then give the corresponding Dirichlet/CAL and Born winner-law convergence of (HAG_{FIC}/HAG_{KRPA}).

For repeated fresh environment uses, an independently supplied product law $\lambda^{\otimes \mathbb{N}}$ makes the increments $a(e_n)$ iid with law ν and produces the Markov kernel above at every step. A realized sequence (e_n) , however, gives a realized compensation trajectory, whereas $\lambda^{\otimes \mathbb{N}}$ gives only its law. Marginalization therefore transports a probabilistic environment input into reduced stochastic dynamics; it does not provide a law-to-point selector. This theorem is deliberately classical/standard-Borel: it makes no claim that a mixed quantum environment state is an epistemic classical mixture, nor that unitary system-environment dynamics by itself solves occurrence. The (RNB/OINB/MDNB) point-bearing firewall remains intact.

Law-occurrence tower factorization and settlement. For standard-Borel X put $\text{LawOcc}(X) := \mathcal{P}(X) \times X$ and, for Borel $e : X \rightarrow Y$, $\text{LawOcc}(e)(\nu, x) := (e_*\nu, e(x))$. This is functorial, with exact fibre

$$[\text{LOPC}] \quad \text{LOLift}_e(\lambda, y) := \text{LawOcc}(e)^{-1}\{(\lambda, y)\} = \text{Lift}_e(\lambda) \times e^{-1}\{y\}. \quad (223)$$

The product records two lifting constraints, not statistical independence. For a finite tower of Borel maps between standard-Borel spaces $X_n \xrightarrow{e_n} \dots \xrightarrow{e_1} X_0$, $e_{[n]} = e_1 \dots e_n$, and fixed (λ_0, x_0) , put

$$\begin{aligned} \text{LawProv}_{e_\bullet}(\lambda_0) &:= \{(\lambda_1, \dots, \lambda_n) : (e_k)_* \lambda_k = \lambda_{k-1}\}, \\ \text{PtProv}_{e_\bullet}(x_0) &:= \{(x_1, \dots, x_n) : e_k x_k = x_{k-1}\}, \\ \text{LOProv}_{e_\bullet}(\lambda_0, x_0) &:= \text{LawProv}_{e_\bullet}(\lambda_0) \times \text{PtProv}_{e_\bullet}(x_0). \end{aligned}$$

Iterating (PLF/RFE) and projecting to the top coordinate gives mutually inverse maps (affine on the law factor):

$$[\text{LPT}] \quad \text{LawProv}_{e_\bullet}(\lambda_0) \cong_{\text{aff}} \text{Lift}_{e_{[n]}}(\lambda_0) \cong_{\text{aff}} \text{LKer}_{e_{[n]}}(\lambda_0), \quad (224)$$

$$\text{LOProv}_{e_\bullet}(\lambda_0, x_0) \cong \text{LOLift}_{e_{[n]}}(\lambda_0, x_0) = \text{Lift}_{e_{[n]}}(\lambda_0) \times e_{[n]}^{-1}\{x_0\}, \quad (225)$$

$$[\text{LPT}_{\text{tot}}] \quad \text{LawProv}_{e_\bullet}(\lambda_0) \cong \int_{\lambda_1 \in \text{Lift}_{e_1}(\lambda_0)} \dots \text{Lift}_{e_n}(\lambda_{n-1}), \quad (226)$$

$$[\text{LOPC}_0] \quad \text{LOLift}_{VU}(\lambda, z) \cong \int_{(\eta, y) \in \text{LOLift}_V(\lambda, z)} \text{LOLift}_U(\eta, y). \quad (227)$$

The law carrier is convex; no convex structure is imposed on the point factor. Applying (ELR) to the composite and one more (PLF/RFE) step gives

$$[\text{LPT}_{\text{ex}}] \quad \text{ex LawProv}_{e_\bullet}(\lambda_0) \cong \text{Sec}_{e_{[n]}}(\lambda_0), \quad (228)$$

$$[\text{LPSR}] \quad \text{LawProv}_{e_\bullet, e_{n+1}}(\lambda_0) \cong \int_{\lambda \in \text{LawProv}_{e_\bullet}(\lambda_0)} \text{Lift}_{e_{n+1}}(\lambda_n). \quad (229)$$

The first is deterministic conditional provenance, not occurrence; the second is singleton exactly when one previous branch has singleton final lift and all others have empty lift. Thus law settlement need not select an old ambiguous point, the Set/probability face of (RFE/SRSL).

Pointing into the paired carrier is natural, but reverse deterministic law-pointing is forbidden by the transposition-fixed uniform binary law (ANB):

$$[\text{LOPC}_\delta] \quad \Delta_X^\delta : X \rightarrow \text{LawOcc}(X), \quad x \mapsto (\delta_x, x), \quad \text{LawOcc}(e) \Delta_X^\delta = \Delta_Y^\delta e, \quad (230)$$

$$[\text{LOPC}_0] \quad \text{Nat}_{\text{FinBij}_+}(\mathbb{D}, \text{LawOcc})_{\pi_{\text{law}}=1} = \emptyset. \quad (231)$$

Conversely a law naturally determines the law of pointed occurrences. With $\iota_\nu(x) = (\nu, x)$,

$$[\text{LOPC}_{\text{sam}}] \quad \text{Sam}_X(\nu) := (\iota_\nu)_* \nu, \quad (\pi_{\text{law}})_* \text{Sam}_X(\nu) = \delta_\nu, \quad (\pi_{\text{occ}})_* \text{Sam}_X(\nu) = \nu, \quad (232)$$

$$[\text{LOPC}_{\text{nat}}] \quad \text{LawOcc}(e)_* \text{Sam}_X(\nu) = \text{Sam}_Y(e_* \nu). \quad (233)$$

Hence stochastic, not deterministic, pointing is natural. On finite spaces $\text{LawOcc}_+(I) = \{(p, i) : p_i > 0\}$ is preserved by surjective read-outs with fibre $\text{supp } p$, the (SRTK/SQAR) face. The sharp terminal-law/nonterminal-point witness is

$$[\text{LPT}_0] \quad \text{LawProv}_{I_2}(u_2) = \{u_2\}, \quad u_2 \notin \delta(2), \quad (234)$$

so (ANB/DLG/OINB) still forbids natural occurrence. If the (ECP) increment law ν is reached through $X_n \rightarrow \dots \rightarrow E \xrightarrow{a} G$, ambient implementations are $\text{Lift}_{ae_1 \dots e_n}(\nu)$ while (CHA/HAG) depends only on ν : provenance is an implementation residual, not a point source. This standard-Borel statement does not interpret a mixed quantum environment as an epistemic mixture; (OINB/MDNB) remains the quantum firewall.

Countable law-occurrence completion and ghost boundary. For a countable tower of Borel maps between standard-Borel spaces $\dots \xrightarrow{e_3} X_2 \xrightarrow{e_2} X_1 \xrightarrow{e_1} X_0$, let $X_\infty = \varprojlim_n X_n$ (the Borel graph-equation subset of $\prod_n X_n$), with projections π_n , and

$$\text{InfLawProv}_{e_\bullet}(\lambda_0) := \{(\lambda_n)_{n \geq 0} : \lambda_0 \text{ fixed}, (e_n)_* \lambda_n = \lambda_{n-1}\}, \quad L_n := \text{LawProv}_{e_1, \dots, e_n}(\lambda_0).$$

By (CPR), prefix compatibility, and the independent point coordinate,

$$[\text{ILPT}] \quad \text{Lift}_{\pi_0}(\lambda_0) \xrightarrow{\text{aff}} \text{InfLawProv}_{e_\bullet}(\lambda_0), \quad \Gamma \mapsto ((\pi_n)_* \Gamma)_{n \geq 0}, \quad (235)$$

$$[\text{ILPT}_{\text{fin}}] \quad \text{InfLawProv}_{e_\bullet}(\lambda_0) \cong \varprojlim_n L_n \cong_{\text{aff}} \text{Lift}_{\pi_0}(\lambda_0), \quad (236)$$

$$[\text{PLOC}] \quad \text{LawOcc}(X_\infty) \xrightarrow{\sim} \varprojlim_n \text{LawOcc}(X_n), \quad (\Gamma, x) \mapsto ((\pi_n)_* \Gamma, \pi_n x)_{n \geq 0}. \quad (237)$$

This assembles compatible coordinates, not a law-to-point lift; (LOPC_{nat}/LPA/ELR) retains the distinction, and the constant binary tower has a singleton law but two constant point towers.

Finite-horizon solvability need not globalize. Define

$$[\text{CLPG}] \quad \text{LawGhost}_{e_\bullet}(\lambda_0) := [L_n \neq \emptyset \forall n < \omega] \wedge [\text{InfLawProv}_{e_\bullet}(\lambda_0) = \emptyset]. \quad (238)$$

It occurs for $X_0 = *$, $X_n = \{n, n+1, \dots\} \subset \mathbb{N}$ with inclusion bonds above level one: every finite horizon has a Dirac realization, while compatibility would force $\lambda_1(X_n) = 1$ for all n and continuity from above gives $1 = \lambda_1(\cap_n X_n) = 0$. If all X_n are instead nonempty compact metrizable and the bonds continuous, compactness of $\prod_n \mathcal{P}(X_n)$ plus the finite-intersection property gives

$$[\text{ILPT}_{\text{cpt}}] \quad \text{InfLawProv}_{e_\bullet}(\lambda_0) \neq \emptyset. \quad (239)$$

Compactness closes only this finite-to-global gate. Occurrence remains separate even for terminal global law provenance:

$$[\text{ILPT}_0] \quad \text{InfLawProv}_{e_\bullet}(u_2) = \{(u_2, u_2, \dots)\}, \quad u_2 \notin \delta(2). \quad (240)$$

Thus finite-horizon law, global law and occurrence are distinct: (CLPG) and (ILPT₀) refute the two converses, (PI) closes only the first on the compact branch, and (LOPC_δ) maps occurred limit points to Dirac towers only in the forward direction; this is the (PFR/GTS) law face.

For the existing infinite quantum-history consumer, reachable (SQAR) history sets H_n have onto truncations $\tau_{n+1, n}$ because every reachable history has a positive-probability continuation, and exact cylinder laws Γ_n with $(\tau_{n+1, n})_* \Gamma_{n+1} = \Gamma_n$; (CPR) gives

$$[\text{PLOC}_Q] \quad \Gamma_\infty \in \mathcal{P}(H_\infty), \quad H_\infty := \varprojlim_n H_n, \quad (\pi_n)_* \Gamma_\infty = \Gamma_n. \quad (241)$$

For each finite reachable h with active support S_h , put $H_{\text{seed}} := \prod_h \text{RaceKey}_{S_h}$ and $\mathbb{P}_{\text{seed}} := \bigotimes_h \text{Dirichlet}_{S_h}(1)$. If K_h° is the Borel conull unique-winner locus of (SQAR), countability gives $\mathbb{P}_{\text{seed}}(H_{\text{seed}}^\circ) = 1$ for $H_{\text{seed}}^\circ := \prod_h K_h^\circ$. Write $\mathbb{P}_{\text{seed}}^\circ := \mathbb{P}_{\text{seed}}|_{H_{\text{seed}}^\circ}$; coordinatewise Borel

winner maps define a Borel $R : H_{\text{seed}}^{\circ} \rightarrow H_{\infty}$. Its finite prefixes have laws Γ_n , so projective uniqueness gives

$$[\text{PLOC}_{\text{seed}}] \quad R_* \mathbb{P}_{\text{seed}}^{\circ} = \Gamma_{\infty}, \quad (V \mapsto (\Gamma_{\infty}, R(V)))_* \mathbb{P}_{\text{seed}}^{\circ} = \text{Sam}_{H_{\infty}}(\Gamma_{\infty}). \quad (242)$$

Thus the seed-field law generates the history law but occurrence requires a realized seed field. For the symmetric binary instrument $\mathcal{J}_0 = \mathcal{J}_1 = \frac{1}{2} \text{id}$, $H_{\infty} = 2^{\mathbb{N}}$, $\Gamma_{\infty} = u_2^{\otimes \mathbb{N}}$ and global bit flip C satisfy

$$[\text{PLOC}_0] \quad C_* \Gamma_{\infty} = \Gamma_{\infty}, \quad \text{Fix}(C) = \emptyset. \quad (243)$$

Hence even the complete path law has no equivariant realized-history selector, the infinite (ANB/OINB/RNB) law–occurrence face; (QRS/QCAF) counterfactual gluing remains orthogonal.

Recursive law–occurrence re-entry and future-law residual. For a Borel Markov kernel $K : X \rightsquigarrow Y$ between standard–Borel spaces, with $\mathcal{P}(Y)$ carrying the evaluation Borel structure, define

$$[\text{RLOP}] \quad \text{OccStep}_K(x) := \text{Sam}_Y(K_x) \in \mathcal{P}(\text{LawOcc}(Y)), \quad \text{OccStep}_K(x, A) := \int_Y \mathbf{1}_A(K_x, y) K_x(dy) \quad (A \subseteq \text{LawOcc}(Y) \text{ Borel}). \quad (244)$$

Rectangle testing proves this is a Borel Markov kernel, and (LOPC_{Sam}) gives

$$[\text{RLOP}_0] \quad (\pi_{\text{law}})_* \text{OccStep}_K(x) = \delta_{K_x}, \quad (\pi_{\text{occ}})_* \text{OccStep}_K(x) = K_x. \quad (245)$$

Thus the present point may determine a supplied next law, but the law does not select its next point.

For a Borel Markov kernel $L : Y \rightsquigarrow Z$ put

$$[\text{RLOP}_\circ] \quad \text{ReOcc}_{K,L}(x) := \int_Y \text{Sam}_Z(L_y) K_x(dy) \in \mathcal{P}(\text{LawOcc}(Z)), \quad (246)$$

and let $(K \star L)_x(B) = \int_Y L_y(B) K_x(dy)$. Then direct marginalization yields the non-scalar future-law residual

$$[\text{FLV}] \quad (\pi_{\text{law}})_* \text{ReOcc}_{K,L}(x) = \text{FutLaw}_{K,L}(x) = (y \mapsto L_y)_* K_x, \quad (\pi_{\text{occ}})_* \text{ReOcc}_{K,L}(x) = (K \star L)_x. \quad (247)$$

For $\mu_x := (K \star L)_x$ and $c_x : \text{LawOcc}(Z) \rightarrow \text{LawOcc}(Z)$, $c_x(\eta, z) := (\mu_x, z)$,

$$[\text{RLOP}_{\text{end}}] \quad (c_x)_* \text{ReOcc}_{K,L}(x) = \text{Sam}_Z(\mu_x), \quad (248)$$

because both sides push μ_x forward by $z \mapsto (\mu_x, z)$. Hence Kleisli composition is precisely the quotient forgetting the activated conditional future law. Its metric-free zero locus is

$$[\text{FLV}_0] \quad \text{ReOcc}_{K,L}(x) = \text{Sam}_Z(\mu_x) \iff \text{FutLaw}_{K,L}(x) = \delta_{\mu_x} \iff L_y = \mu_x \quad K_x\text{-a.e. } y. \quad (249)$$

The first implication is the law marginal; a Dirac law of the random measure L_y is equivalent to a.e. constancy, which substituted in (RLOP₀) gives the converse. Thus occurrence changes the next possibility law exactly off this locus.

Strictness already occurs for $X = *$, $Y = Z = \mathbf{2}$, $K_* = u_2$, with $L_0^A = \delta_0$, $L_1^A = \delta_1$ and $L_0^B = L_1^B = u_2$: both endpoints are u_2 , whereas

$$[\text{FLV}_{\text{wit}}] \quad \text{ReOcc}_{K,L^A}(*) = \frac{1}{2} \delta_{(\delta_0,0)} + \frac{1}{2} \delta_{(\delta_1,1)}, \quad \text{ReOcc}_{K,L^B}(*) = \frac{1}{2} \delta_{(u_2,0)} + \frac{1}{2} \delta_{(u_2,1)} = \text{Sam}_2(u_2). \quad (250)$$

Hence the Kleisli endpoint does not recover future-law re-entry. Iterating OccStep for history-dependent kernels $K_r(x_0, \dots, x_r; dx_{r+1})$ yields the usual finite history law; a realized prefix h_r selects only the next law $K_r(h_r; \cdot)$. Endpoint marginalization forgets the conditional-law profile, while (LOPC₀/ANB) still forbids choosing x_{r+1} from it.

For a two-step finite-dimensional instrument with first outcome law $p(i)$ and second Born laws $q_i \in \Delta_J$,

$$[\text{RLOP}_Q] \quad \text{ReOcc}_{p,q} = \sum_i p(i) \text{Sam}_J(q_i), \quad \text{FutLaw}_{p,q} = \sum_i p(i) \delta_{q_i}, \quad q_{\text{end}} = \sum_i p(i) q_i. \quad (251)$$

Endpoint forgetting gives $\text{Sam}_J(q_{\text{end}})$, so the first occurrence changes the next law iff the q_i are nonconstant on $\text{supp } p$. The richer (SQAR) consumer sends i to p_i and thence to the next Born law/support, but still supplies no next outcome without a newly realized actualization resource. Thus occurrence can generate future law without selecting future occurrence. The kernel/iterated-integral identities are standard; SCO content is the typed (LOPC/PLOC) factorization and exact (FLV₀/FLV_{wit}) non-collapse, the standard–Borel (OCV/SQAR) face, with history, update and realized seeds retained as attachments.

Future-relevant determination key and barycentric forgetting. The future-law residual has a canonical determination key and an exact endpoint-forgetting map. For a standard–Borel kernel $L : Y \rightsquigarrow Z$, write

$$\ell_L : Y \longrightarrow \mathcal{P}(Z), \quad \ell_L(y) := L_y.$$

The evaluation Borel structure on $\mathcal{P}(Z)$ makes ℓ_L measurable. On underlying sets corestrict it as

$$[\text{FLK}] \quad q_L : Y \rightsquigarrow \text{FutKey}(L) := \text{im } \ell_L, \quad R_{q_L} = R_{\ell_L} = \{(y, y') : L_y = L_{y'}\}. \quad (252)$$

This is the (RKD/FSB_{core}/SDO) determination quotient for ℓ_L ; by (KGC_{cons}), for every corestricted $d_{\text{FLK}_{\text{mm}}} D$, $\ell_L = \bar{\ell} d \iff R_d \subseteq R_{\ell_L} \iff q_L = \phi d$ uniquely, so q_L is the coarsest sufficient quotient. This is Set-valued in general: a Borel image of a Borel map between standard–Borel spaces need only be analytic. On the *Borel-key locus* where $\text{im } \ell_L \subseteq \mathcal{P}(Z)$ is Borel—in particular for finite/countable discrete consumers—the same corestriction is a standard–Borel quotient and can itself carry the pushed-forward law.

The endpoint map is the standard barycenter of a law of laws. Define

$$[\text{BFL}] \quad \beta_Z : \mathcal{P}(\mathcal{P}(Z)) \longrightarrow \mathcal{P}(Z), \quad \beta_Z(M)(B) := \int_{\mathcal{P}(Z)} \eta(B) M(d\eta). \quad (253)$$

Evaluation measurability makes β_Z Borel. For every Borel $f : Z \rightarrow W$, Tonelli gives

$$[\text{BFL}_{\text{nat}}] \quad f_* \beta_Z(M) = \beta_W((f_*)_* M), \quad (254)$$

and a second Tonelli application gives the associative flattening identity

$$[\text{BFL}_{\text{assoc}}] \quad \beta_Z((\beta_Z)_* \mathfrak{M}) = \beta_Z(\beta_{\mathcal{P}(Z)}(\mathfrak{M})) \quad (\mathfrak{M} \in \mathcal{P}(\mathcal{P}(\mathcal{P}(Z)))). \quad (255)$$

These are standard probability-monad identities and are not counted as SCO novelty; they are written only to expose the exact consumer interface.

For $K : X \rightsquigarrow Y$, the future-law profile of (FLV) and the Kleisli endpoint satisfy the exact factorization

$$[\text{BFL}_{\text{RLOP}}] \quad M_x := \text{FutLaw}_{K,L}(x) = (\ell_L)_* K_x, \quad (K \star L)_x = \beta_Z(M_x). \quad (256)$$

The second identity is just $\int_{\mathcal{P}(Z)} \eta(B) d((\ell_L)_* K_x)(\eta) = \int_Y L_y(B) K_x(dy)$, but it identifies the exact map that erases the future-law profile. For $\mu \in \mathcal{P}(Z)$, put

$$[\text{BFL}_{\text{fib}}] \quad \boxed{\text{BarFib}_Z(\mu) := \beta_Z^{-1}\{\mu\}.} \quad (257)$$

Then $M_x \in \text{BarFib}_Z((K \star L)_x)$, and the zero locus of the (FLV) residual is precisely the constant-law point in that fibre:

$$[\text{BFL}_0] \quad \boxed{\text{ReOcc}_{K,L}(x) = \text{Sam}_Z((K \star L)_x) \iff M_x = \delta_{(K \star L)_x}.} \quad (258)$$

The barycenter fibre has an exact terminality classification. Let $\delta_Z : Z \rightarrow \mathcal{P}(Z)$, $z \mapsto \delta_z$, and put $\text{DirMix}_Z(\mu) := (\delta_Z)_* \mu$. Then

$$[\text{BFL}_{\text{term}}] \quad \boxed{\begin{aligned} \beta_Z(\delta_\mu) = \mu &= \beta_Z(\text{DirMix}_Z(\mu)), \\ \text{BarFib}_Z(\mu) = \{\delta_\mu\} &\iff \text{DirMix}_Z(\mu) = \delta_\mu \iff \mu \text{ is Dirac.} \end{aligned}} \quad (259)$$

The first row follows directly from the barycenter definition. If μ is non-Dirac, choose a Borel $B \subseteq Z$ with $0 < \mu(B) < 1$ (standard-Borel spaces have a countable separating generator, so a probability taking only values 0, 1 on all Borel sets is Dirac). Under δ_μ , the measurable evaluation $\eta \mapsto \eta(B)$ is constantly $\mu(B)$, whereas under $\text{DirMix}_Z(\mu)$ it takes only the values 0, 1. Thus the two measures are distinct members of the same barycenter fibre. Conversely, if $\mu = \delta_z$ and $M \in \text{BarFib}_Z(\mu)$, then $1 = \int \eta(\{z\}) M(d\eta)$, hence $\eta = \delta_z$ for M -almost every η and $M = \delta_{\delta_z}$. Therefore every non-Dirac endpoint has irreducible barycentric nonidentifiability, whereas a deterministic endpoint is the unique terminal barycenter fibre. For $Z = \mathbf{2}$, $\mu = u_2$, the two canonical points are exactly

$$[\text{BFL}_{\text{wit}}] \quad \boxed{\delta_{u_2}, \quad \text{DirMix}_2(u_2) = \frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1} \in \text{BarFib}_2(u_2).} \quad (260)$$

This is the L^B/L^A witness of $(\text{FLV}_{\text{wit}})$ after taking the law marginal. Hence endpoint law is the barycentric quotient of a structured future-law fibre and, unless the endpoint law is Dirac, cannot classify that fibre.

The future key is the minimal occurrence information required by that declared future consumer. On the Borel-key locus, if $Q_x := (q_L)_* K_x$, then the injective inclusion $\bar{\ell} : \text{FutKey}(L) \hookrightarrow \mathcal{P}(Z)$ gives

$$[\text{FLK}_{\text{law}}] \quad \boxed{M_x = (\bar{\ell})_* Q_x, \quad \text{ReOcc}_{K,L}(x) = \text{Sam}_Z((K \star L)_x) \iff Q_x \text{ is Dirac.}} \quad (261)$$

Thus an intermediate occurrence is predictively relevant exactly through its q_L -class; any finer detail of y is invisible to the next-law target L , while any coarser quotient loses some distinction between conditional future laws. This minimality is relative to the declared future consumer: two points with the same next Born law may still have different post-states or later operational behavior, in which case the richer SQAR/path-law target has a strictly finer future key. No one-step key is promoted to a complete physical state.

For the two-step finite quantum consumer of (RLOP_Q) , put $\ell(i) = q_i$. Then

$$[\text{FLK}_Q] \quad \boxed{i \sim_{\text{Fut}} j \iff q_i = q_j, \quad M = \sum_i p(i) \delta_{q_i}, \quad q_{\text{end}} = \beta_J(M) = \sum_i p(i) q_i.} \quad (262)$$

The quotient of first outcomes by $\sim_{\text{Fut}}$ is therefore the unique coarsest future-relevant event key for that second instrument. The same endpoint Born law can arise from many different M 's, so averaging cannot reconstruct which future-law classes were opened by the first occurrence. Whenever the declared target is upgraded from one next instrument to a measurable complete conditional future path-law kernel, the same (FLK) construction gives the corresponding history-relevant quotient without changing the theorem.

The terminality statement $(\text{BFL}_{\text{term}})$ is a Set/probability-fibre result and is not identified with categorical settlement (SRSL). The two share the general grammar “totalize first, test terminality second” but have independent carriers: attaching a non-Dirac endpoint law to an already terminal residual category makes BarFib nonterminal, while attaching a Dirac endpoint law to an independently nonterminal residual category does the reverse. Thus no cross-type equivalence is inferred from their matching terminal language.

Accordingly the deeper re-entry chain is not “event $\rightarrow$ endpoint law” but

$$\boxed{\text{occurred point} \longrightarrow \text{future-relevant determination class} \longrightarrow \text{conditional future law} \longrightarrow \text{law of pointed future occurrences},}$$

while barycentric/Kleisli forgetting quotients the middle profile and still supplies no next point. This is a MIXED_BRIDGE: barycenter/monad multiplication is standard, whereas the SCO content is the identification of its fibre with the future-law residual and the RKD-minimal occurrence key, together with the quantum non-collapse consumer.

Predictive branch law, complete-future quotient, and actuality normal form. Let A be finite or countable discrete, $X_A = A^{\mathbb{N}}$, $\sigma : X_A \rightarrow X_A$ the left shift, and $\iota_a(x) = ax$. On $\mathbf{FinBij}_+$ write $\text{PathLaw}(I) := \mathcal{P}(I^{\mathbb{N}})$, with relabeling by $(f^{\mathbb{N}})_*$; this fixes the functor used below in (PNB/PBCF). For a finite positive Borel measure ν put

$$[\text{PLBD}] \quad \boxed{\partial_a \nu := \sigma_*(\nu|_{[a]}).} \quad (263)$$

Since $\sigma|_{[a]}$ and ι_a are inverse Borel isomorphisms and the cylinders $[a]$ partition X_A ,

$$[\text{PLBD}_0] \quad \boxed{|\partial_a \nu| = \nu([a]), \quad \nu = \sum_{a \in A} (\iota_a)_* \partial_a \nu, \quad \sum_a |\partial_a \nu| = |\nu|.} \quad (264)$$

Conversely every summable family of finite positive measures $(\nu_a)_a$ reconstructs $\nu = \sum_a (\iota_a)_* \nu_a$ and has $\partial_a \nu = \nu_a$. Thus $\nu \mapsto (\partial_a \nu)_a$ is an isomorphism onto the cone of summable branch families. For a word $w = a_0 \cdots a_{n-1}$, $\partial_w := \partial_{a_{n-1}} \cdots \partial_{a_0}$ satisfies

$$[\text{PLBD}_{\text{cyl}}] \quad \boxed{|\partial_w \nu| = \nu([w]), \quad \nu([w]) > 0 \implies \partial_w \nu / |\partial_w \nu| = C_w \nu.} \quad (265)$$

Zero-mass branches remain zero and are never normalized.

Now let $H \subseteq A^{<\omega}$ be countable and prefix-closed, with declared complete future laws $\Gamma_h \in \mathcal{P}(X_A)$. The branch isomorphism gives

$$[\text{PLBD}_{\text{min}}] \quad \boxed{(\partial_a \Gamma_h)_{a \in A} = (\partial_a \Gamma_{h'})_{a \in A} \iff \Gamma_h = \Gamma_{h'},} \quad (266)$$

so the complete predictive quotient is equivalently the law image or the branch-family image:

$$[\text{PHQ}] \quad \boxed{q_\infty : H \rightarrow \text{PredKey}(\Gamma) := \text{im}(h \mapsto \Gamma_h), \quad h \sim_\infty h' \iff \Gamma_h = \Gamma_{h'}.} \quad (267)$$

This is the standard predictive-equivalence/causal-state relation⁸⁸. By (KGC_{cons}), every corestricted $d : H \rightarrow D$ obeys $\Gamma = \bar{\Gamma}d \iff R_d \subseteq R_{q_\infty} \iff q_\infty = \phi d$ uniquely; hence q_∞ is the coarsest quotient sufficient for the complete future law, equivalently for all branch measures.

Let q_n corestrict $h \mapsto (\pi_n)_* \Gamma_h$ for the first- n -symbol projection. Cylinder uniqueness and (PLBD_{cyl}) give

$$[\text{PHQ}_\omega] \quad R_{q_\infty} = \bigcap_{n \geq 1} R_{q_n}, \quad h \sim_\infty h' \iff (\pi_n)_* \Gamma_h = (\pi_n)_* \Gamma_{h'} \quad \forall n, \quad (268)$$

and (KGC_{sep}/KGC_{ord}) gives $[q_\infty] = \bigvee_{n \geq 1} [q_n]$. No fixed horizon is sufficient in general: for

$$\Gamma^A = \frac{1}{2} \delta_{000\dots} + \frac{1}{2} \delta_{1100\dots}, \quad \Gamma^B = \frac{1}{2} \delta_{0100\dots} + \frac{1}{2} \delta_{1000\dots},$$

$$[\text{PHQ}_{\text{wit}}] \quad q_1(h_A) = q_1(h_B), \quad q_2(h_A) \neq q_2(h_B), \quad q_\infty(h_A) \neq q_\infty(h_B). \quad (269)$$

Moving the correlation bit beyond any prescribed horizon gives the same refuter at arbitrary finite depth.

Assume history consistency: for every positive branch, $ha \in H$ and $\Gamma_{ha} = C_a \Gamma_h$. Then equality of complete laws makes equal branch masses and normalized branch laws, so

$$[\text{PUP}] \quad h \sim_\infty h', \quad \Gamma_h([a]) > 0 \implies \Gamma_{h'}([a]) = \Gamma_h([a]) > 0, \quad ha \sim_\infty h'a. \quad (270)$$

Thus the occurred symbol defines the partial predictive update. Writing $S = \text{PredKey}(\Gamma)$, the same branch operator gives its full event-labelled normal form

$$[\text{PLOT}] \quad p_s(a) = |\partial_a s|, \quad \text{OccFib}(s) = \{a : \partial_a s \neq 0\}, \quad U_a(s) = \frac{\partial_a s}{|\partial_a s|}, \quad E_s = \sum_{a: \partial_a s \neq 0} |\partial_a s| \delta_{(a, U_a(s))}. \quad (271)$$

Its state-transition marginal is

$$[\text{PLOT}_0] \quad T_s = (\pi_S)_* E_s = \sum_{a: \partial_a s \neq 0} |\partial_a s| \delta_{U_a(s)}, \quad (272)$$

and iterating the branch identity reconstructs every cylinder:

$$[\text{PLOT}_{\text{cyl}}] \quad s([a_0 \cdots a_{n-1}]) = |\partial_{a_0 \cdots a_{n-1}} s| = \prod_{k=0}^{n-1} p_{s_k}(a_k), \quad s_{k+1} = U_{a_k}(s_k), \quad (273)$$

with both sides zero after the first zero branch. Hence E is a lossless generative presentation of the declared future law, while occurrence remains the label needed to choose the next state.

If a corestricted $d : H \rightarrow D$ reconstructs next-symbol probabilities and an event-labelled update whose generated cylinders equal every Γ_h , then equal d -values give equal complete future laws, so

$$[\text{PLOT}_{\text{min}}] \quad R_d \subseteq R_{q_\infty}, \quad q_\infty = \phi d \quad (274)$$

uniquely. Occurrence terminality is instead

$$[\text{RTH}] \quad \text{OccFib}(s) \simeq \mathbf{1} \iff \#\{a : \partial_a s \neq 0\} = 1 \iff p_s \text{ is Dirac}. \quad (275)$$

It does not follow from deterministic state transition: for the fair iid binary law,

$$[\text{PLOT}_{\text{iid}}] \quad \partial_0 s = \partial_1 s = \frac{1}{2} s, \quad U_0(s) = U_1(s) = s, \quad T_s = \delta_s, \quad E_s = \frac{1}{2} \delta_{(0,s)} + \frac{1}{2} \delta_{(1,s)}. \quad (276)$$

Complete branch data still has no natural point selector:

$$[\text{PNB}] \quad \text{Nat}_{\text{FinBij}_*}(\text{PathLaw}, \text{Id}) = \emptyset, \quad (277)$$

by the transposition-fixed fair binary path law.

Resolution acts linearly before branch normalization. For a surjection $f : A \twoheadrightarrow B$, $F = f^{\mathbb{N}}$,

$$[\text{PLBD}_{\text{nat}}] \quad \partial_b(F_* \nu) = F_* \sum_{f(a)=b} \partial_a \nu, \quad (278)$$

$$[\text{PLBD}_{\text{int}}] \quad M_b := \sum_{\substack{a \in S_s \\ f(a)=b}} \frac{p_s(a)}{(f_* p_s)(b)} \delta_{F_* U_a(s)}, \quad U_b(F_* s) = \beta_{B^{\mathbb{N}}}(M_b).$$

For $S_s = \{a : p_s(a) > 0\}$ and $\ell(a) = F_* U_a(s)$,

$$R_{f|_{S_s}} \subseteq R_\ell \iff (\forall b) M_b \text{ is Dirac} \iff F_* U_a(s) = U_{f(a)}(F_* s) \quad (a \in S_s).$$

Thus unnormalized branching always commutes by fibre summation, while normalized re-entry descends pointwise exactly at this zero locus. Otherwise (RKD/KGC) gives the strict least repair and (BFL) its non-Dirac residual. Sharpness is the old $s = \frac{1}{2}(\delta_{0^\infty} + \delta_{12^\infty})$, $f(0) = f(1) = x$, $f(2) = y$ witness; at finite truncation the zero locus is $I(A_0; F(\text{tail}) \mid f(A_0)) = 0$, only a scalar shadow of dynamic sufficiency.

The one-step future-law key remains a strict projection: if the complete future law factors through the one-step law, q_∞ factors through (FLK), but (PHQ_{wit}) refutes the converse. Hence (RLOP/PUP) is the occurrence-driven one-step face, while the branch operator carries the complete path law.

The quantum consumer retains both strictness and branch reconstruction. Let histories h_0, h_1 have qubit post-states $|0\rangle\langle 0|, |1\rangle\langle 1|$, apply the symmetric instrument $\mathcal{J}_0 = \mathcal{J}_1 = \frac{1}{2} \text{id}$, then a sharp Z measurement. The next laws coincide but the subsequent futures differ:

$$[\text{PHQ}_Q] \quad q_1^Q(h_0) = q_1^Q(h_1) = u_2, \quad q_2^Q(h_0) \neq q_2^Q(h_1), \quad q_\infty^Q(h_0) \neq q_\infty^Q(h_1). \quad (279)$$

For the conditional future law Γ_h of a declared history-dependent instrument policy in the (SQAR/PHQ) perimeter,

$$[\text{PLBDQ}] \quad \boxed{|\partial_i \Gamma_h| = p_h(i), \quad p_h(i) > 0 \implies \frac{\partial_i \Gamma_h}{|\partial_i \Gamma_h|} = \Gamma_{hi}, \quad |\partial_{i_{m-1}} \cdots \partial_{i_0} \Gamma_\emptyset| = \text{Tr} \left[I_{i_{m-1}}^{h_{m-1}} \circ \cdots \circ I_{i_0}^\emptyset (\rho_\emptyset) \right].} \quad (280)$$

Thus (SQAR/PLOC) supplies path laws, the branch operator recovers their event-labelled future laws, and (QRS/QCAF) still blocks counterfactual point gluing. Different post-states may remain in one predictive class when the declared policy cannot distinguish their future statistics; adjoining declared policy/read-out families intersects the corresponding predictive kernels, so enlargement can only refine the predictive quotient. Still q_∞ is not the full quantum state and is not the operator-algebraic reflector Pred_T of (PCE/PGA) without an additional cylinder-generation theorem.

The head–tail measure decomposition and causal-state equivalence are `STANDARD_ABI`. The `MIXED_BRIDGE` is the identification

$$\boxed{s \mapsto (\partial_a s)_a \mapsto p_s \mapsto \text{Sam}_A(p_s) \xrightarrow{\text{realized } a} U_a(s) = \partial_a s / |\partial_a s|.$$

The branch family regenerates (PHQ/PUP/PLOT/RTH), lands under resolution in (BFL/KGC), retains (PNB), and projects to the old pointing law (LOPC_{sam}). Together with ($\text{KGC}_{\text{ref}}/\text{RRCX}$), actuality and read-out refinement remain independent axes: fair iid has occurrence with $U_a(s) = s$, while strict KGC refinement needs no occurrence. No endogenous first-point source is asserted.

Predictive orbit-resolution and actuality factorization. For finite I , the PHQ horizon tower has the presentation-invariant stabilizer shadow $H_n = \text{Stab}((\pi_n)_* s) \searrow H_\infty = \text{Stab}(s)$. Each law stabilizes at finite symmetry depth but no uniform depth exists. The orbit maps form an (RRCX) path whose weights telescope to p_s : depth only resolves symmetry. (OIL/GAD) splits orbit/residual-symmetry entropy, with Δ_n the Shannon valuation of one grade, not new law-information. The existing race realizes the same split on one clock field across horizons. (AKI/ECM/PBCF) identifies branch success/exactness by $K_{s,r} = H_\infty \cap G_r$ versus $K_a = H_\infty \cap G_a$; the positive over-resolution fibre is $K_a/K_{s,r}$, with log-cardinality only its scalar shadow after the subgroup gate. (RNB/FDSC_{br}) keeps strict law-generated symmetry loss relatively exogenous, while the (PHQ_Q/PLBD_Q) consumer transfers $\Delta_1 = \log 2$ from symmetry to quotient information without selecting the non-Dirac occurrence. Hence `PORS/OAF` is a `MIXED_BRIDGE` projection of `KGC–PHQ–AKF/PBCF` plus the existing race, not a new state, clock, scalar criterion or occurrence primitive. The exact identities, chain/race coherence, positive fibre, sharp witnesses and proof are retained in Supplementary Material, Sec. S11.

Predictive branch implementation and calibrated provenance. Work on $C = \mathbf{FinBij}_{+,sk}$ with $D = \text{PathLaw}$, $T = U_{\text{set}}$, a resource functor $R : C \rightarrow \mathbf{Set}$, $s \in D(I)$ and $a \in \text{OccFib}(s)$. The target-relative implementation classes are exactly the (AKI/ECM) specializations

$$[\text{PBCF}] \quad \boxed{\begin{aligned} \text{BrSucc}_R(s; a) &:= \{[q_r] : r \in R(I), R_{\hat{s}} \cap R_{\hat{r}} \subseteq R_{\hat{s}} \cap R_{\hat{a}}\}, \\ \text{BrEx}_R(s; a) &:= \{[q_r] : r \in R(I), R_{\hat{s}} \cap R_{\hat{r}} = R_{\hat{s}} \cap R_{\hat{a}}\}. \end{aligned}} \quad (281)$$

Thus success means natural generation of the supplied branch and exactness means realization of $\text{ActKey}(s; a)$ without joint over-resolution. By (PLBD_Q/PLBD_{nat}), for $\text{BrLaw}(I) = \{(\nu_a)_a : \nu_a \in \mathcal{M}_+(I^{\mathbb{N}}), \sum_a |\nu_a| = 1\}$ the map $\mathfrak{B}_I(s) = (\partial_a s)_a$ is a natural isomorphism $\text{PathLaw} \simeq \text{BrLaw}$, hence

$$[\text{PBCF}_0] \quad \boxed{R_{\mathfrak{B}_I(s)} = R_{\hat{s}}.} \quad (282)$$

Branch presentation therefore changes neither actuality keys nor complement fibres.

Let PathOcc^+ be the groupoid of triples (I, s, a) with $a \in \text{OccFib}(s)$, put $F_R^{\text{succ}} = \text{BrSucc}_R$, $F_R^{\text{ex}} = \text{BrEx}_R$, $\tilde{F}_R^*(s; a) = \{r : [q_r] \in F_R^*(s; a)\}$ and $\text{RawFib}_R(\alpha) = \{r : [q_r] = \alpha\}$. Kernel conjugation under bijections and $r \mapsto [q_r]$ give the functorial surjections and the (RFE/URT) totalization

$$[\text{PBCF}_{\text{tot}}] \quad \boxed{\begin{array}{ccccc} \int \tilde{F}_R^{\text{ex}} & \hookrightarrow & \int \tilde{F}_R^{\text{succ}} & & \\ \downarrow & & \downarrow & & \\ \int F_R^{\text{ex}} & \hookrightarrow & \int F_R^{\text{succ}} & \longrightarrow & \text{PathOcc}^+ \longrightarrow \int \text{PathLaw}. \end{array}} \quad (283)$$

The four layers—complete law, occurred branch, effective implementation class, and raw realizer—therefore remain typed. Once (s, a) is supplied, (PUP/PLBD) fixes $U_a(s)$; these fibres classify implementation, not occurrence. A target-blind section would project to the forbidden (PNB) map $\text{PathLaw} \Rightarrow U_{\text{set}}$, and (RNB) makes every strict actuality-enlarging resource relatively exogenous.

On the symmetric iid face $s_I = u_I^{\otimes \mathbb{N}}$, (OAF_R) has $H_\infty = \text{Sym}(I)$ and gives

$$[\text{PBCF}_G] \quad \boxed{[q_r] \in \text{BrSucc}_R(s_I; a) \iff G_r \leq G_a, \quad [q_r] \in \text{BrEx}_R(s_I; a) \iff G_r = G_a.} \quad (284)$$

For $|I| = n$, (`PORScost`) gives branch price $\log n$, while a generic unique-minimum seed has trivial stabilizer and price $\log n!$, hence excess $\log(n-1)!$; this is zero exactly at $n = 2$, recovering (RSL).

For calibrated law provenance put $G_R^*(s) = \coprod_{a \in \text{OccFib}(s)} F_R^*(s; a) \times \{a\}$ and likewise $\tilde{G}_R^*$ with $\tilde{F}$; let $\chi_s^*(r, a) = ([q_r], a)$ and $e_s^*(\alpha, a) = a$. With $\text{Lift}_f^{\mathbb{D}}(p) = \{\gamma : f_* \gamma = p\}$, the finite-support Set calculation underlying (LPA/PLF/LOPC) gives, without any measurable structure on $R(I)$,

$$[\text{PBCF}_{\text{cal}}] \quad \text{Cal}_R^*(s) := \text{Lift}_{e_s^*}^{\mathbb{D}}(p_s), \quad (285)$$

$$[\text{PBCF}_{\text{dis}}] \quad \text{Cal}_R^*(s) \cong_{\text{aff}} \prod_{a \in \text{OccFib}(s)} \mathbb{D}(F_R^*(s; a)), \quad (286)$$

$$[\text{PBCF}_{\text{raw}}] \quad \widetilde{\text{Cal}}_R^*(s) := \text{Lift}_{e_s^* \chi_s^*}^{\mathbb{D}}(p_s) \cong \int_{\gamma \in \text{Cal}_R^*(s)} \prod_{(\alpha, a) \in \text{supp } \gamma} \mathbb{D}(\text{RawFib}_R(\alpha)), \quad \widetilde{\text{Cal}}_R^*(s) \cong_{\text{aff}} \prod_{a \in \text{OccFib}(s)} \mathbb{D}(\tilde{F}_R^*(s; a)). \quad (287)$$

Indeed one first conditions the calibrated law on each positive branch mass $p_s(a)$ and, on the raw side, then conditions on each effective class;

this proves the displayed set-totalization and the direct affine product. The same decomposition proves the exact gate

$$\begin{aligned}
 & \boxed{\begin{aligned}
 & \text{Cal}_R^*(s) \neq \emptyset \iff \widetilde{\text{Cal}}_R^*(s) \neq \emptyset \iff (\forall a \in \text{OccFib}(s)) \text{F}_R^*(s; a) \neq \emptyset, \\
 & \text{Cal}_R^*(s) \simeq \mathbf{1} \iff (\forall a) \text{F}_R^*(s; a) \simeq \mathbf{1}, \\
 & \widetilde{\text{Cal}}_R^*(s) \simeq \mathbf{1} \iff (\forall a) \widetilde{\text{F}}_R^*(s; a) \simeq \mathbf{1}, \\
 & p_s \text{ non-Dirac} \implies \text{every calibrated effective/raw law is non-Dirac.}
 \end{aligned}}
 \end{aligned}
 \tag{PBCF}_{\text{gate}} \tag{288}$$

Equivalently, raw terminality means one effective class α_a and a singleton $\text{RawFib}_R(\alpha_a)$ on every active branch. Hence feasibility, effective-law uniqueness, raw uniqueness and occurrence are distinct. Sharp witnesses remain: for $I = \{0, 1, 2\}$, $s = (1/2, 1/4, 1/4)^{\otimes \mathbb{N}}$ and $R = \mathbf{1}$, (LWP) makes branch 0 uniquely readable with one exact implementation while branches 1, 2 have none, so the uncalibrated successful total is nonempty but branch-complete calibration fails. For $s = u_2^{\otimes \mathbb{N}}$ and $R = U_{\text{set}}$, both raw points represent one exact class α and

$$\boxed{\begin{aligned}
 & \text{Cal}_R^{\text{ex}}(s) = \left\{ \frac{1}{2} \delta_{(\alpha, 0)} + \frac{1}{2} \delta_{(\alpha, 1)} \right\} \simeq \mathbf{1}, \quad \widetilde{\text{Cal}}_R^{\text{ex}}(s) \simeq_{\text{aff}} \mathbb{D}(\mathbf{2})^2 \neq \mathbf{1}.
 \end{aligned}}
 \tag{PBCF}_{\text{bin}} \tag{289}$$

For $|I| = 3$ the point stabilizers are distinct, so the raw exact lift is the unique non-Dirac law $\frac{1}{3} \sum_a \delta_{(a, a)}$; conversely $s = \delta_{000\dots}$ with binary $R = U_{\text{set}}$ has terminal occurrence but $\widetilde{\text{Cal}}_R^{\text{ex}}(s) \simeq \mathbb{D}(\mathbf{2})$ nonterminal. Thus occurrence and raw-provenance terminality are independent, while raw terminality implies effective terminality by push-forward.

The exponential race is the positive effective-calibration consumer. For full-support finite s , take $R(I) = \Omega_I = (0, \infty)^I$ with product unit-exponential law ν_I . The class map $c_I(\xi) = [q_\xi]$ is fixed by the coordinate-equality stabilizer, hence has finite image and is Borel. Let $\nu_{I,s}^{\text{um}}$ be the restriction of ν_I to the full-measure unique-minimum locus and there set $A_s(\xi) = \arg \min_i \xi_i / p_s(i)$. Equivariance plus (AKI/ERD) gives $[q_\xi] \in \text{BrSucc}_R(s; A_s(\xi))$ and $(A_s)_* \nu_{I,s}^{\text{um}} = p_s$, hence

$$\boxed{\begin{aligned}
 & \Gamma_s^{\text{race}} := (\xi \mapsto (c_I(\xi), A_s(\xi)))_* \nu_{I,s}^{\text{um}} \in \text{Cal}_R^{\text{succ}}(s), \quad (e_s^{\text{succ}})_* \Gamma_s^{\text{race}} = p_s.
 \end{aligned}}
 \tag{PBCF}_{\text{race}} \tag{290}$$

Exactness is $G_s \cap G_\xi = G_s \cap G_a$: on symmetric iid laws (RSL) makes binary seeds exact a.s., whereas for $n \geq 3$ generic $G_\xi = \mathbf{1} \neq G_a$ and the exact locus (equal nonwinner coordinates) is nonempty but ν_I -null; if $G_s = \mathbf{1}$, every successful seed is exact. Thus over-resolution is symmetry-relative, not an arity law. Only the finite determination quotient of the continuous race lands in the displayed $\mathbb{D}$ -calibration fibre.

This is a MIXED_BRIDGE: (PLBD/PNB) supplies law/occurrence, (AKF/ECM) implementation, (LPA/PLF/LOPC) calibrated provenance, and (OAF/PORS) the symmetry price. The asymmetric, binary, ternary and deterministic witnesses preserve all non-collapse boundaries; any effective/raw natural point section still projects to (PNB).

Supplementary package S4: Standard modular/contextual structure and stable classical chart. The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S4; every use below invokes that exact proof-bearing package, not an additional assumption.

c. *Half-sided orbit bridge (Borchers–Wiesbrock theory).* On the modular singleton branch of the temporal lift-fiber result VIII B 0 a, put $M_U(a) := U(a)MU(-a)$ and $N_U := M_U(1)$. The Borchers/Araki–Zsidó half-sided modular theorem⁷⁴⁷⁵ (Thm. 2.1) gives (BW) $\Delta^{it}U(s)\Delta^{-it} = U(e^{-2\pi i t} s)$, $\Delta_{N_U}^{it}\Delta^{-it} = U(1 - e^{-2\pi i t})$, $N_U = U(1)MU(-1)$. If $M_U(b) = M_U(a)$ for $b > a$, $c = b - a$ and $0 \leq s \leq c$, then $M = U(s)[U(c - s)MU(s - c)]U(-s) \subseteq U(s)MU(-s) \subseteq M$. Thus $U(s)$ normalizes M ; $U(s)\Omega = \Omega$ gives $[U(s), \Delta^{it}] = 0$ by Tomita uniqueness, and (BW) with $t \rightarrow \infty$ forces $U(s) = \mathbf{1}$, hence $U = \mathbf{1}$. Therefore for $U \neq \mathbf{1}$

$$\begin{aligned}
 & b > a \geq 0 \implies M_U(b) \subsetneq M_U(a), \quad \sigma_t^\omega(M_U(a)) = M_U(e^{-2\pi i t} a), \\
 & [BO] \quad q_U(M_U(a)) = -\frac{1}{2\pi} \log a \quad (a > 0), \quad q_U \circ \sigma_t^\omega = q_U + t.
 \end{aligned}
 \tag{291}$$

Thus q_U is an orbit coordinate, not a defect.

d. *Modularly generated chiral geometry.* Assume the proper half-sided inclusion $(N_U \subset M, \Omega)$ is standard (equivalently Ω is cyclic for $N'_U \cap M$), put $\alpha_a = \text{Ad } U(a)$ and $\mathcal{A}_U(a, b) = \alpha_a(M) \cap \alpha_b(M)' = M_U(a) \cap M_U(b)'$, $\mathcal{A}_U(a, \infty) = M_U(a)$. Guido–Longo–Wiesbrock⁷⁶ (Cor. 1.7 and Cor. 1.9) give the unique isotone, local, strongly additive conformal precosheaf with vacuum Ω and half-lines M, N_U , with its isomorphism-class correspondence to Borchers triples and translation–dilation covariant Haag-dual Bisognano–Wichmann nets. SCO uses only (MGA) $X_q = \mathcal{A}_U(e^{-2\pi i q}, \infty)$, $\sigma_t^\omega X_q = X_{q+t}$, $a = e^{-2\pi i q}$. Thus U, Δ^{it} supply translations/dilations and q_U is the intrinsic logarithmic affine half-line coordinate; no spacetime localization is assumed, and without standardness only (BO) is asserted.

e. *Modular Jordan causal carrier.* The graded carrier has both a rank-one modular-incidence face and an effective cross-composition core. For standard K, L write $\text{ML}_\pm(K, L)$ when $K \cap L$ is standard, both inclusions $K \cap L \subseteq K, L$ are $\pm$ half-sided modular, and $S_{K,L}^\pm := s\text{-lim}_{t \rightarrow \pm\infty} \Delta_K^{it} \Delta_L^{-it}$ satisfies $J_K S_{K,L}^\pm J_K = (S_{K,L}^\pm)^{-1}$; this is exactly the modular-intersection datum of⁷⁷ (Def. 3.20). If standard H_1, H_2, H_3 satisfy $\text{ML}_-(H_1, H_2)$, $\text{ML}_-(H_3, H'_1)$ and $\text{ML}_+(H_2, H_3)$, Wiesbrock’s theorem in the one-particle form⁷⁷ (Thm. 3.23) makes their modular objects generate an antiunitary $\text{PGL}_2(\mathbb{R})$ representation and its connected effective image is trivial or has Lie algebra $\mathfrak{sl}_2(\mathbb{R})$.

For the finite-dimensional Lie-wedge branch pass to the 1-connected cover of the connected even carrier and lift the grading involution; the covering differential identifies the Lie algebra. Write G for this graded extension, let U be an antiunitary representation with discrete kernel, and choose $\gamma \in \text{Hom}_{\text{gr}}(\mathbb{R}^\times, G)$ with $U \circ \gamma = U_V$, where $U_V(e^t) = \Delta_V^{-it/2\pi}$ and $U_V(-1) = J_V$. Put $h := \gamma'(1)$, $\tau := \text{Ad}(\gamma(-1))$, $\mathfrak{h} := \ker(\tau - \mathbf{1})$, $\mathfrak{q} := \ker(\tau + \mathbf{1})$, $\mathfrak{h}_0(h) := \mathfrak{h} \cap \ker \text{ad } h$, $\mathfrak{q}_\lambda(h) := \mathfrak{q} \cap \ker(\text{ad } h - \lambda \mathbf{1})$, $C_U := \{x \in \mathfrak{g} : -i \partial U(x) \geq 0\}$, and $C_\pm := \pm C_U \cap \mathfrak{q}_{\pm 1}(h)$, $S_V := \{g \in G_+ : U(g)V \subseteq V\}$. Bind the standard carrier interfaces once: the Lie-wedge theorem⁷⁸ (Thm. 4.4), the simple 3-graded Jordan/conformal identification⁷⁹ (Lem. 2.14), and the Euclidean-Jordan spectral and rank-two classification⁸⁰ (Thm. III.1.2 and Cor. IV.1.5) give

$$\begin{aligned}
 & \mathbf{L}(S_V) = C_- \oplus \mathfrak{h}_0(h) \oplus C_+, \quad \mathfrak{e}_\pm = C_\pm - C_\pm; \\
 & \mathfrak{g}_{\text{red}} = \mathfrak{e}_- \oplus \mathfrak{h}_0(h) \oplus \mathfrak{e}_+, \quad [\mathfrak{e}_\pm, \mathfrak{e}_\pm] = 0; \\
 & \mathfrak{s} \text{ simple active, } C_\mathfrak{s} := C_U \cap \mathfrak{s} \text{ maximal proper invariant} \\
 & [MJS] \implies E = \mathfrak{s}_1(h) \text{ Euclidean Jordan, } C_{+, \mathfrak{s}} := C_\mathfrak{s} \cap \mathfrak{s}_1(h) = E_+, \quad r_J = \text{rank}_J E; \\
 & r_J = 2 \implies C_{+, \mathfrak{s}} = \{ae + z : a \geq \|z\|_J\}, \quad Q_J(ae + z) = a^2 - \|z\|_J^2, \\
 & \partial C_{+, \mathfrak{s}} = \{Q_J = 0, a \geq 0\}.
 \end{aligned}
 \tag{292}$$

Here $C_{\pm}$ are closed pointed, $e_{\pm}$ are their linear grade spans, and E_+ denotes only the square cone of the simple Euclidean Jordan algebra E , with $\text{tr}_J(e) = 2$, $E_0 = \ker \text{tr}_J$ and $\|z\|_J^2 = \frac{1}{2} \text{tr}_J(z^2)$. The metric appears only at rank two, where the cone fixes the Lorentz conformal class and positive sheet; $\text{Sym}_3(\mathbb{R})_+$ supplies a non-ray proper-face witness for rank > 2 , and $\mathbb{R} \oplus \mathbb{R}$ is the reducible $1 + 1$ boundary. On the Neeb–Ólafsson net regime, the primary net theorem and Jordan-space-time landing⁸¹ (Thm. 5.3; §5.2) realize the BGL datum as a covariant cyclic/standard-subspace net on the relevant simple-Jordan/tube-type causal homogeneous space, with the Bisognano–Wichmann property. Their §1 and the discussion immediately after Theorem 5.3 separately invoke the standard second-quantization functor to obtain the corresponding free quantum field in the Haag–Kastler sense⁸¹ (§1; discussion after Thm. 5.3). This export requires the additional standard-subspace net realization; (MJS) supplies its Lie/Jordan carrier.

Rank-two affine extraction. On the simple active rank-two face of (MJS), the imported carrier interfaces identify $\mathfrak{s} \cong \text{conf}(E)$, with $\mathfrak{s}_1 = E$ constant and $\mathfrak{s}_0$ linear, and identify $E = \mathbb{R}e \oplus V$ as a spin factor. For $B_u = L_{(0,u)}$ and $R \in \mathfrak{so}(V) \subset \text{Der}(E)$,

$$[\text{MAP}] \quad B_u = \begin{pmatrix} 0 & u^\top \\ u & 0 \end{pmatrix}, \quad [B_u, B_v] = \text{diag}(0, uv^\top - vu^\top), \quad [R, B_w] = B_{Rw}, \quad \mathfrak{p}_s := [\mathfrak{s}_0, \mathfrak{s}_0] \ltimes \mathfrak{s}_1 \cong \mathfrak{so}(Q_J) \ltimes E. \quad (293)$$

Its nilradical is E . For unit $v \in V$, $P = e + v$ and $D = -B_v$ satisfy $Q_J(P) = 0$, $[D, P] = -P$, and the connected Lorentz orbit of P generates E_+ ; hence (MSG) holds with $d = \dim_{\mathbb{R}} E$.

For the same grades set $\mathfrak{g}^\# = \text{Lie}(e_- \cup e_+)$, $\mathfrak{g}_0^\# = \text{span}[e_+, e_-]$, $Z_0 = \{a \in \mathfrak{g}_0^\# : [a, e_+ \oplus e_-] = 0\}$ and $\{x, y, z\}_\sigma = [[x, y], z]$. Since $[e_\sigma, e_\sigma] = 0$, Jacobi gives outer symmetry and $[D(x, y), D(u, v)] = D(D(x, y)u, v) - D(u, D(y, x)v)$, so $\mathcal{J}_{\text{mod}} = (e_+, e_-, \{\cdot, \cdot, \cdot\}_\pm)$ is a Jordan pair. The action of $\mathfrak{g}_0^\#$ on $e_+ \oplus e_-$ has kernel exactly Z_0 and image $\text{instr}(\mathcal{J}_{\text{mod}})$; because $e_{\pm}$ generate $\mathfrak{g}^\#$, Z_0 is central. Therefore⁸² $\mathfrak{g}^\# / Z_0 \cong e_- \oplus \text{instr}(\mathcal{J}_{\text{mod}}) \oplus e_+ = \text{TKK}(\mathcal{J}_{\text{mod}})$, the coarsest grade-preserving quotient injective on $e_{\pm}$ and preserving all triple products: every admissible kernel lies in Z_0 .

This cross law is not fixed by grading and positive rays. On the same $\mathbb{R}h \oplus \mathbb{R}e_+ \oplus \mathbb{R}e_-$ with rays $\mathbb{R}_+e_{\pm}$, the brackets $[h, e_{\pm}] = \pm e_{\pm}$ with respectively $[e_+, e_-] = 0$ and $2h$ give $\{e_+, e_-, e_+\} = 0$ and $2e_+$; the carriers are solvable and $\mathfrak{sl}_2(\mathbb{R})$. Thus $\text{instr}(\mathcal{J}_{\text{mod}})$ is the exact missing coordinate at fixed one-sided data; the flat carrier is only the algebraic counterexample. On the nontrivial signed-incidence branch the effective PGL_2 image has $e_{\pm} = \mathbb{R}e_{\pm}$ after graded normalization and recovers this same $\mathfrak{sl}_2$ envelope, while the Euclidean-Jordan clauses are its ordered simple higher-rank continuation. The imported (MJS) carrier and the TKK interface above supply the carriers; the effective cross-law factorization is the intrinsic bridge. The full conformal factor is semisimple, but (MAP) shows that this does not obstruct the affine target: the canonical derived affine subcarrier inside the standard parabolic has abelian nilradical E . What remains independent is the associative/dynamical orientation of the Jordan carrier, classified below.

f. Dimensional obstruction and intrinsic Lorentz recognition. Normalize the Borchers–Wiesbrock Lie carrier by $\mathfrak{a}_{\text{BW}} = \mathbb{R}D \ltimes \mathbb{R}P$, $[D, P] = -P$. For every $d \geq 2$, with $V_d = \mathbb{R}^{1,d-1}$, $\mathfrak{p}_d = \mathfrak{so}(1, d-1) \ltimes V_d$, standard boost B_{01} and $p_+ = e_0 + e_1$, (MDN) $j_d : \mathfrak{a}_{\text{BW}} \hookrightarrow \mathfrak{p}_d$, $D \mapsto B_{01}$, $P \mapsto p_+$, is injective. Hence the abstract BW carrier is dimensionally non-rigid.

Let $\mathfrak{g}$ be a finite-dimensional real Lie algebra and put $\mathfrak{t} = \text{Nilrad}(\mathfrak{g})$, $d = \dim \mathfrak{t}$, $\mathfrak{l} = \mathfrak{g}/\mathfrak{t}$; let $\rho : \mathfrak{l} \rightarrow \mathfrak{gl}(\mathfrak{t})$ be the bracket action. Assume $\mathfrak{t}$ abelian, $d \geq 2$, ρ faithful, $\dim \mathfrak{l} = d(d-1)/2$,

$$[\text{MSG}] \quad Q_{\mathfrak{g}} := \{Q \in \text{Sym}^2(\mathfrak{t}^*) : Q(\rho(X)v, w) + Q(v, \rho(X)w) = 0 \forall X \in \mathfrak{l}, v, w \in \mathfrak{t}\} \text{ contains a nondegenerate } Q, \quad (294)$$

$$\exists(D, P) : D \in \mathfrak{g}, 0 \neq P \in \mathfrak{t}, [D, P] = -P, \quad \mathcal{C}_P := \text{con}(L_{\mathfrak{g}} \cdot P) \text{ pointed,}$$

where $L_{\mathfrak{g}} \leq \text{GL}(\mathfrak{t})$ is connected with Lie algebra $\rho(\mathfrak{l})$. Then, without assuming Poincaré type, (MCO) $\rho(\mathfrak{l}) = \mathfrak{so}(Q)$, $\text{sign } Q \in \{(1, d-1), (d-1, 1)\}$, $\dim Q_{\mathfrak{g}} = 1$, $d(\mathfrak{g}) := \dim \text{Nilrad}(\mathfrak{g}) = d$, $\mathcal{N}_{\mathfrak{g}} := \{v \in \mathfrak{t} : Q(v, v) = 0\}$ intrinsic. For $d \geq 3$, $\mathcal{C}_P$ is one closed causal cone and the positive BW ray chooses its time orientation; for $d = 2$ one BW orbit is one null ray, so a second linearly independent positive null incidence is required for the full causal cone.

Indeed $[B_{01}, p_+] = -p_+$ gives (MDN). Under (MSG), $\rho(\mathfrak{l}) \subseteq \mathfrak{so}(Q)$ and the dimensions agree, hence equality. The BW eigenvector is Q -null. If the signature (p, q) had $p, q \geq 2$, connected π -rotations in positive and negative two-planes would send P to $-P$, contradicting pointedness of $\mathcal{C}_P$; thus Q is Lorentzian. The defining Lorentz module has a one-dimensional space of invariant symmetric forms (also in the $d = 2$ lightlike basis), proving the remaining intrinsic claims. The $SO_0(Q)$ orbit of a positive null ray gives the stated cone boundary. Consequently every Lie carrier satisfying (MSG) has the intrinsic Lorentz reading (MCO). When $\mathfrak{g}$ is produced from modular incidence data, write $d_{\text{mod}}(\mathfrak{g}) := d(\mathfrak{g})$. In particular (MAP) gives $d_{\text{mod}}(\mathfrak{p}_s) = \dim_{\mathbb{R}} E$ for every simple active rank-two factor.

For every simple active E of (MJS) with $r := \text{rank}_J E > 1$, let $J_E : E \otimes_{\mathbb{R}} E \rightarrow E \otimes E$ be the canonical map into the Hanche–Olsen universal EJA tensor and set $\text{LTBr}(E) := \{I \subseteq E \otimes E : I \text{ is a simple Jordan ideal and } c_I := p_{IJ_E} : E \otimes_{\mathbb{R}} E \xrightarrow{\sim} I\}$. Give each EJA its standard $x \& y := Q_{\sqrt{x}}y$. For $I \in \text{LTBr}(E)$ pull I 's unit, cone and $\&$ through c_I . Since $c_I(e \otimes e) = e_I$, pure positive tensors stay positive, and universal-envelope quadratic factorization with Jordan functional calculus for p_I gives $c_I(a \otimes b) \& c_I(c \otimes d) = c_I((a \& c) \otimes (b \& d))$. Thus van de Wetering Definition 4 and Theorem 3 give $E \cong A_{\text{sa}}$; finite-dimensional simplicity and rank force $A \cong M_r(\mathbb{C})$ ⁸³ (Def. 4, Prop. 40 and Thm. 3). Conversely BGW Table 2 gives $M_r(\mathbb{C})_{\text{sa}} \otimes M_r(\mathbb{C})_{\text{sa}} \cong M_{r^2}(\mathbb{C})_{\text{sa}}^{\oplus 2}$, and its standard/transpose branch maps are isomorphisms on Hermitian tensor bases. Hence (CTS) $\text{LTBr}(E) \neq \emptyset \iff E \cong M_r(\mathbb{C})_{\text{sa}}$, with $|\text{LTBr}(E)| = 2$. BGW's envelope is universal for every EJA (exceptional case 0), so complex type is output, not input⁸⁴ (Defs. 3.5, 3.8 and Table 2). On the rank-two face, for right K -torsors F_i put $\text{Rel}_K(F_1, F_2) := (F_1 \times F_2)/\Delta K \cong \text{Iso}_K(F_1, F_2)$; the self-class $[x, x]$ is canonical. For $K = C_2$ write RelOr ; since sign reversal makes $\text{Or}(E)$ a torsor, $\text{LTBr}(E) \cong \text{RelOr}(\text{Or}(E), \text{Or}(E))$, empty off the oriented stratum and two-point same/opposite on it. For $\chi_E(\alpha) = \det(\alpha|_V)$, transpose has $\chi_E = -1$; (α, β) fixes a branch iff $\chi_E(\alpha)\chi_E(\beta) = 1$, so one-leg reversal swaps branches and diagonal reversal does not. For the (ORI) torsor $P_{\text{or}} \rightarrow B$, the same class $o_{\text{or}} = [P_{\text{or}}]$ can be nonzero, yet $(P_{\text{or}} \times_B P_{\text{or}})/\Delta C_2$ has canonical section $[I, I]$; hence the LT relative branch globalizes even when absolute orientation does not. Still on this rank-two face, let $S_{\iota} := \{[\iota(a)] : a \in E\} \subset t_{\iota}^p(M)$ for a unital Jordan monomorphism $\iota : E \rightarrow M_{\text{sa}}$ into a finite-dimensional von Neumann algebra with faithful trace τ and $M = W^*(\iota(E))$. Then (MAP), (ORI) and Alfsen–Shultz give $d_{\text{mod}}(\mathfrak{p}_s) = 4 \iff \text{Or}(E) \neq \emptyset \iff \text{Ax}(V) \neq \emptyset \iff \text{LTBr}(E) \neq \emptyset \iff \exists(M, \tau, \iota) \text{ JLCI}_{\tau}(S_{\iota}) = S_{\iota}$. As τ is tracial, directly $\text{Alg}_{\tau}(S_{\iota}) = W^*(\iota(E)) = M$ and $\text{JLCI}_{\tau}(S_{\iota}) = [M_{\text{sa}}]$. Thus fixedness and $1 \in \iota(E)$ give $M_{\text{sa}} = \iota(E)$, hence $\text{Or}(E) \neq \emptyset$ by Alfsen–Shultz Corollary 2; conversely that corollary supplies a von Neumann product and its trace, giving fixedness with $\iota = \text{id}$. The residual is genuine: the Neeb–Ólafsson introductory table gives $\mathfrak{s} = \mathfrak{so}(2, n)$, $E = \mathbb{R}^{1, n-1}$ simple rank two for every $n > 2$, while Theorem 5.3 and §5.2 supply the corresponding standard-subspace nets in the declared net regime⁸¹ (table on pp. 3–4; Thm. 5.3; §5.2); hence (MJS/MAP) plus net realization forces neither the self-composite nor this fixed point.

Theorem X.1 (Atomic causal boundary and zero-composition selection). *Let $E = \mathfrak{s}_1(h)$ be the Euclidean Jordan algebra of a simple active (MJS) factor and assume its symmetric cone E_+ is not a ray. Write*

$$\text{FaceAt}(E) \iff \text{every nonzero proper face of } E_+ \text{ is a ray.}$$

For every simple Jordan ideal $I \triangleleft E \otimes E$ put $c_I := p_{I|E} : E \otimes_{\mathbb{R}} E \rightarrow I$, view c_I as a morphism of underlying abelian groups, and define the additive self-composition residual by

$$\text{CompRes}(E; I) := \mathfrak{R}_\bullet(c_I) = \text{Cone}(c_I)[-1].$$

Then

$$[\text{ACRS}] \quad \text{FaceAt}(E) \iff \text{rank}_J E = 2, \quad \{I : \text{CompRes}(E; I) \simeq 0\} = \text{LTBr}(E). \quad (295)$$

Independently (CTS) gives $\text{LTBr}(E) \neq \emptyset \iff E \cong M_{\text{rank}_J E}(\mathbb{C})_{\text{sa}}$; hence

$$[\text{M4D}] \quad \text{FaceAt}(E) \ \& \ (\exists I) \ \text{CompRes}(E; I) \simeq 0 \iff E \cong M_2(\mathbb{C})_{\text{sa}}. \quad (296)$$

Sharpness: $\text{FaceAt}(\text{Sym}_2 \mathbb{R}) = 1$ but $\text{LTBr} = \emptyset$, while $\text{LTBr}(M_3(\mathbb{C})_{\text{sa}}) \neq \emptyset$ but $\text{FaceAt} = 0$. On this locus

$$\dim_{\mathbb{R}} E = 4, \quad Q_J(ae + z) = a^2 - \|z\|_J^2 \text{ has signature } (1, 3), \quad \mathfrak{p}_s \cong \mathfrak{so}(1, 3) \otimes \mathbb{R}^{1,3}.$$

The zero-composition branch does not choose an absolute orientation:

$$\{I : \text{CompRes}(E; I) \simeq 0\} \cong \text{RelOr}(\text{Or}(E), \text{Or}(E)),$$

so on the nonempty locus the same/opposite relative branch remains a genuine two-point residual. In particular the dimension 4, Lorentz signature and complex matrix carrier are conclusions of the order/composition conditions above; no spacetime dimension or field choice occurs in their hypotheses.

Proof. By (MJS), rank two gives a strictly convex Lorentz cone, hence every nonzero proper face is a ray. Conversely, for rank $r \geq 3$ a Jordan frame $e_1, \dots, e_r$ and $d = e - e_1 - e_2$ give, by self-duality of the symmetric cone, the exposed proper face $F_d = \{x \in E_+ : \langle x, d \rangle = 0\}$ containing the independent rays $\mathbb{R}_+ e_1, \mathbb{R}_+ e_2$; the non-ray hypothesis excludes rank one, proving the first (ACRS) equivalence. For c_I , the additive residual theorem gives $H_1 \mathfrak{R}_\bullet(c_I) = \ker c_I$ and $H_0 \mathfrak{R}_\bullet(c_I) = \text{coker } c_I$, so finite-dimensional concentration yields $\mathfrak{R}_\bullet(c_I) \simeq 0 \iff c_I$ is an isomorphism $\iff I \in \text{LTBr}(E)$. Intersecting with (CTS) proves (M4D). Finally $E = \mathbb{R}e \oplus V$, $\dim V = 3$, and (MJS/MAP) give the displayed (1, 3) form and $\mathfrak{p}_s \cong \mathfrak{p}_{1,3}$; the relative-orientation clause is the established $\text{LTBr}(E) \cong \text{RelOr}$ identification. $\square$

g. Finite-family realization landing in the affine recognizer. As an independent explicit realization, use the Schroer–Wiesbrock Section 5 finite-family construction: $0 \leq i < j \leq 4$, common standard vector Ω , $M_{ji} = M'_{ij}$, modular-intersection and index-symmetry conditions (56)–(58), and the local composition hypothesis (c') $\text{Ad } J_{12}(\mathcal{P}_\varepsilon^{1,1}) \subseteq \mathcal{P}_\delta^{1,1} \mathcal{L}_{12}$. These hypotheses give the $\text{SL}_2(\mathbb{C})$ Lorentz part⁸⁵ (Thm. 3, Eqs. (56)–(60)); the translation inclusion (61), $T_{12} = J_{N_{12}} J_{12}$, and the CPT/commutation, index-symmetry and covariance assumptions (62)–(65) give the full Poincaré group with spectrum condition⁸⁵ (§5, Eqs. (61)–(65)), with proof details in⁸⁶. Only sufficiency is imported: c' is neither derived from (56)–(58) nor identified with the Jordan cross-derivation. If translations are nontrivial, let $K \triangleleft \mathfrak{p}_{1,3}$ be the infinitesimal kernel and $V = \mathbb{R}^{1,3}$. Irreducibility gives $K \cap V = 0$. The projection of K to the simple Lorentz algebra is an ideal; if nonzero it is all, but then bracketing a lifted nonzero Lorentz generator with V gives a nonzero element of $K \cap V$. Thus $K = 0$, so the effective carrier is $\mathfrak{p}_{1,3}$. Hence it satisfies (MSG) and (MCO) gives $d_{\text{mod}}(\mathfrak{p}_{1,3}) = 4$, independently of the intrinsic selector (ORI).

h. Operational modular ray. To compare the intrinsic ray with a declared locally-covariant spacetime, let $\mathcal{A}$ lie on the dynamically-local regime of (ROG), let π be a fixed faithful $*$ -representation, put $\mathcal{M} = \pi(\mathcal{A}(M))''$, $\omega = \omega_\Omega$, and $D_M^\pi(K) = \pi(D_M(K))''$. Assume $U \neq 1$ and every proper orbit algebra $\mathcal{M}_U(a)$, $a > 0$, lies in $\text{OpReg}_\mathcal{A}^\pi(M) := \text{im } D_M^\pi$. Then (BO/MGA) identifies $\text{Ray}_U^{\text{op}} = \{\mathcal{M}_U(a) : a > 0\}$ with $(\mathbb{R}, \leq)$ by $q_U(\mathcal{M}_U(a)) = -(2\pi)^{-1} \log a$ and carries modular flow to $q \mapsto q + t$. Thus the unpointed ray is an ordered $\mathbb{R}$ -torsor; the mark $X_0 = \mathcal{M}_U(1)$ uniquely fixes this affine coordinate because $X_q = \sigma_q^\omega(X_0)$. For the bare realization fiber put $\text{GeoRay}_U := \{(q, K) \in \mathbb{R} \times \mathcal{K}(M) : D_M^\pi(K) = X_q\}$, $p_U(q, K) = q$, $E_U := R_{p_U}$, $\mathcal{T}_t := p_U^\circ \tau_t p_U$, $\tau_t(q) = q + t$. Operational realization makes p_U onto, so relation calculus gives $\mathcal{T}_s \mathcal{T}_t = \mathcal{T}_{s+t}$, $\mathcal{T}_0 = E_U$, $E_U \mathcal{T}_t = \mathcal{T}_t E_U = \mathcal{T}_t$, $\mathcal{T}_t^\circ = \mathcal{T}_{-t}$. Since $\mathcal{T}_t(q, K) = p_U^{-1}(q + t)$, it is functional for one/every t iff $E_U = \Delta$, equivalently every p_U -fiber is a singleton. Order GeoRay_U by bare inclusion. ROG isotony makes p_U monotone; on $\text{RefGap}_{p_U}(\subseteq, \leq) = 0$, p_U is an order isomorphism (hence has singleton fibers), $q \leq r \iff K_q \subseteq K_r$, and lifts translation by $K_q \mapsto K_{q+t}$. The (ROG) witness blocks this without faithfulness. Absent operational realization no identification with bare spacetime regions is made, while this construction remains intrinsic on the standard branch.

i. Common modular-time bridge. On an overlap specified by a state-preserving normal isomorphism $\Phi : (\mathcal{M}, \omega) \xrightarrow{\sim} (\mathcal{M}_{\text{time}}, \omega_\Xi)$ from the ambient (BO/MGA) system with $U \neq 1$ to a marked $\text{MEnv}(\Xi)$, set $X_q^\Phi := \Phi(\mathcal{M}_U(e^{-2\pi q}))$, $\text{Ray}_U^\Phi := \Phi(\text{Ray}_U^{\text{op}})$, $q_U^\Phi(X_q^\Phi) := q$ and $\tau_t^\Phi X_q^\Phi := X_{q+t}^\Phi$. Since $\Phi \sigma_t^\omega = \sigma_t^{\omega_\Xi} \Phi$, for $X \in \text{Ray}_U^\Phi$ (MC/MGA) gives $u^* \sigma_t^{\omega_\Xi}(u) = e^{it\Theta_\Xi}$, $q_U^\Phi(\sigma_t^{\omega_\Xi} X) = q_U^\Phi(X) + t$, $\Xi_{\Xi, U}^\Phi(t) := (e^{it\Theta_\Xi}, \tau_t^\Phi)$, with $\Xi_{\Xi, U}^\Phi : \mathbb{R} \rightarrow \mathcal{U}(\mathcal{A}) \times \text{Aut}_\leq(\text{Ray}_U^\Phi)$ a one-parameter group; at $X_0^\Phi = \Phi(\mathcal{M}_U(1))$, $q_U^\Phi(\sigma_t^{\omega_\Xi} X_0^\Phi) = t$ and $U \neq 1 \implies \Theta_\Xi \neq 0$ by (MC/BW). Converse: nonzero clock may coexist with trivial Borchers group; only t , not the sorts, is shared.

For $N \in \text{Lat}_\omega(M)$ let e_N project onto $\overline{N\Omega_\omega}$. By (TEXP) and bimodularity, (EWC) $N = M \iff e_N = 1 \iff \Omega_\omega$ cyclic for N , $e_N x e_N = E_N(x) e_N$, $e_N M e_N = N e_N$. Since $U(a)\Omega_\omega = \Omega_\omega$, the vector Ω_ω is cyclic for every $\mathcal{M}_U(a)$. Hence (EWC) gives $\mathcal{M}_U(a) \in \text{Lat}_\omega(M) \implies \mathcal{M}_U(a) = M$; for $U \neq 1$ and $a > 0$ this contradicts the strict orbit in (BO). Stone uniqueness applied to (BW) also gives $\Delta^{it} P \Delta^{-it} = e^{-2\pi t} P$. For $U \neq 1$, $P \neq 0$, and this scaling forces P to be unbounded. Thus a proper half-sided orbit is not an expected-subalgebra compression on the same pointed standard form. Changing only a scalar-defect attachment leaves (M, ω) and its expected-subalgebra lattice fixed, so it cannot create this orbit; the translation generator is additional typed data.

j. Tagged non-collapse and derived factorization. The marked representation strictly exceeds its active reduct: on a one-point base, residuals (0, 1), (1, 1) give trajectories $0, 1 - e^{-t}$ at $R_0 = 0$, and constant costs 0, 1 are likewise erased. Operator-algebraic descendants are generated by expected inclusions, fixed accesses/Petz, (M, ω) , MEnv , temporal CMLSI/current and the Borchers carrier, with boundary $\text{XA } 0$ i; evaluation, time-zero marks, general residuals and impact/extendability remain exactly the witnessed fibers.

k. Marked-envelope reflection-positive consequence. On the finite reversible branch MEnv reconstructs T_0 . The temporal lift-fiber result VIII B 0 a supplies the full reflection-positive/OS supported-semigroup boundary, with $\ker T_0$ the one-time spectral null after the prediction-null quotient and the full-space obstruction. When $T_0 > 0$, (FRM) decides whether the singleton $\text{OSFib}(T_0)$ is also the unique reversible Markov lift: this holds exactly when $(\log T_0)_{ij} \geq 0$ for $i \neq j$. If $\ker T_0 \neq 0$, no finite-state continuous-time Markov embedding exists. The ambient TimeFib remains independent and may be non-singleton.

B. Completion and structural closure

Presentation closure acts on the joint read-out of Definition II.2; operator-algebraic, interface, entropic and current closures keep their native carriers from Theorem VII.1(4) and descend only through (DI) fiber constancy. The residual-profile envelope of Theorem II A 0 a is a typed envelope external only to the local presentation, not a field or silent operator-algebraic enlargement of $\mathfrak{F}_{\text{acc}}$; its data remain stage-internal through the routed residual carriers.

a. Scope of closure. Local closure combines an admissible access section with record saturation, sufficiency and modular invariance; support impact/extendability and the gluing, entropic, clock and residual carriers remain typed separately.

b. Integrative product-order map. Fix a declared active comparison support S inside one comparison presentation: S lists the scalar-defect and predicate-capability coordinates on which comparison is licensed. For effective data D, E carrying that same support, write $E \succeq_{\text{int}}^S D$ when every scalar defect in S is no larger for E , while every predicate-valued capability in S is preserved or gained; strict dominance means at least one strict improvement. This is the product preorder on the declared typed joint read-out at the active regime. Data with different active supports are not compared until a common support is explicitly supplied; absent coordinates are therefore outside, rather than silently neutralized inside, the comparison. This support condition is necessary: with one scalar defect c , data $D_0(c) = 1$, D_1 carrying no c , and $D_2(c) = 0$ would make $D_0 \succeq D_1$ and $D_1 \succeq D_2$ vacuous under shared-coordinate comparison but not $D_0 \succeq D_2$. When S is fixed we suppress it and write $\succeq_{\text{int}}$. On the presentation/content sort that read-out is Definition II.2, while additional operator-algebraic coordinates enter only through their declared constructions. Consequently it is invariant under presentation isomorphism, strictly increasing reparametrization $\delta \mapsto \varphi(\delta)$ of a fixed gluing scale with $\varphi(0) = 0$, deterministic Blackwell equivalence after the canonical identification of the same visible von Neumann algebra and reference, and conjugacy of the dynamic–residual data. Replacing a declared metric by an inequivalent metric is additional comparison data and need not preserve the quantitative order; (QGR) is the finite-dimensional quantum witness, with zero state fiber compatible with a nonzero metric residual.

For shared scalar coordinates $\Delta_a \in C_a$, any declared family of order-preserving exchange maps $\Xi_a : C_a \rightarrow C$ into one ordered commutative cone defines the monotone aggregate $\mathcal{F}_{\Xi} = \sum_a \Xi_a(\Delta_a)$. For a feasible fiber $\mathcal{F}$ inside the same active support, put $\text{Par}(\mathcal{F}) = \{D \in \mathcal{F} : \nexists E \in \mathcal{F}, E \succ_{\text{int}} D\}$. Any objective strictly decreasing along $\succ_{\text{int}}$ has $\text{Argmin} \subseteq \text{Par}(\mathcal{F})$. No canonical point or aggregate is forced: the witness has $\delta(D) < \delta(E)$, $\varepsilon(D) > \varepsilon(E)$; on $\mathcal{F} = \{D, E\}$, both are Pareto-maximal and the two coordinate projections select opposite points. If all coordinate projections are allowed, their scalar preorders intersect in exactly the scalar part of $\succeq_{\text{int}}$. Thus the product preorder is the exchange-rate-free core; on an (EGC) fixed fiber it selects no point, and any objective/scalarization remains attachment data, with clock, type and reconstruction typed separately. On a finite access family, (JOD) sharpens this attachment: the reduced incidence objective embeds in the source-objective quotient with independent quotient $\text{Hom}(\mathbb{K}(H_F), \mathbb{Z})$, nonzero already for the constant two-point read-out.

Pareto dual tangent fiber. For a finite differentiable defect chart $\Delta = (\Delta_i)_{i=1}^m$, closed convex control cone $K = T_0 U$, and $x(u) = x + Bu + o(\|u\|)$, put $L_x v = (d\Delta_i(x)Bv)_i$, $\mathcal{E}_x = \overline{L_x K}$, $\mathfrak{W}_x = \mathcal{E}_x^* \cap \mathbb{R}_{++}^m$, $\mathbb{W}_x = \mathfrak{W}_x / \mathbb{R}_{>0}$. The cone-image duality displayed above gives $\mathcal{E}_x \cap (-\mathbb{R}_{++}^m) = \{0\} \iff \mathfrak{W}_x \neq \emptyset$. If $L_x K$ is closed this is precisely absence of exact first-order integrative improvement. Closedness is sharp: for $K = S_+^2$ and $L(X) = (x_{11}, x_{12})$, $L(K) = \{(a, b) : a > 0, b \in \mathbb{R}\} \cup \{0\}$ has no exact negative rate, while its closure contains $(0, -1)$ and admits no strictly positive dual. Positive diagonal tangent reparametrization sends $\mathfrak{W}_x \mapsto A^{-T} \mathfrak{W}_x$, so $\mathbb{W}_x$ is the invariant no-descent scalarization-ray fiber. For block-independent products, $\mathfrak{W}_{12} = \mathfrak{W}_1 \times \mathfrak{W}_2$, $[\mathbb{W}_{12}] = \{(t\lambda, (1-t)\mu) : 0 < t < 1, \lambda \in \Sigma_1, \mu \in \Sigma_2\}$, where $\Sigma_i = \{\lambda \in \mathfrak{W}_i : \mathbf{1} \cdot \lambda = 1\}$. Thus even singleton leg certificates leave a free relative product weight; there is no canonical strongly monoidal scalarization. For $m = 1$, the displayed certificate is the tangent-descent clause of Theorem IV.2; on attained objective branches $\mathbb{W}_x$ precedes the RF residual V_q^d , coordinating rather than absorbing objective choice.

On the operator-algebraic entropy branch, a chosen locally-BV relative-entropy trajectory has a canonical scalar source/sink reading: the Hahn–Jordan split of dR . This does not select the trajectory from the preceding static order; the scalar residual equation is the absolutely continuous constitutive specialization of this measure-valued law.

Theorem X.2 (Residual current and asymptotic forgetting). *Let V be a finite nonempty set, put*

$$C_0(V) := \mathbb{R}^V, \quad \varepsilon : C_0(V) \rightarrow \mathbb{R}, \quad \varepsilon(x) := \sum_{v \in V} x_v, \quad C_0^0(V) := \ker \varepsilon,$$

and let $R : I \rightarrow C_0(V)$ be locally of bounded variation. Suppose its measure derivative admits a typed balance $dR = d\Pi - d\Sigma - d\mathcal{T}$, where $d\Pi, d\Sigma$ are componentwise non-negative Radon measures and $d\mathcal{T}$ is a $C_0^0(V)$ -valued signed Radon measure. Then:

1. *conservative transfer disappears exactly in the augmentation quotient, $d\varepsilon(R) = \varepsilon(d\Pi) - \varepsilon(d\Sigma)$, $C_0(V)/C_0^0(V) \xrightarrow[\cong]{\varepsilon} \mathbb{R}$; for $q_V : V \twoheadrightarrow \{*\}$, $\mathbb{K}(q_V) \otimes \mathbb{R} \cong C_0^0(V)$ is the constant-quotient (UAR) face, universal for zero-sum-invariant linear observables;*
2. *if a finite oriented graph $G = (V, E)$ is attached, (ICL) makes graph-conservative transfer precisely a boundary. Whenever $d\mathcal{T} = B_G dJ$, its edge-current lifts form the affine cycle fiber $dJ \sim dJ + dZ$, $dZ \in \ker B_G$. For connected G , $q_G = q_V$, so every zero-sum transfer has such a lift and (ICL) gives its exact integral/real cycle carrier. Let the additive monoid $C_1(G)$ act on $C_0(G)$ by $x \cdot J = x + B_G J$. For $w \in C_1(G)^*$, $c_x^w(J) := w(J)$ is an action cocycle and $\text{Hist}_x^w(J, J') = w(J - J')$ whenever $B_G J = B_G J'$. Since e translates by $B_G J$ and g_x is injective on represented translations, $R_{T_x} = R_e = \{(J, J') : B_G J = B_G J'\}$; Proposition S2 yields $H_{\text{act}}^1(x; \mathbb{R}) \cong Z_{\text{rc}}^1(R_e; \mathbb{R})$, so all current history is structural. (BDT/KSP) identifies (HCD) as its history–current duality. Its zero class is the potential branch $w = B_G^* f$, i.e. the responses that factor through the visible nodal transfer $B_G J$; thus (HCD) is the $V_{B_G}^*$ face of Theorem S2 and (ICL). More generally, for*

a chain complex $C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$ over its native coefficient ring and admissible $\mathcal{A} \subseteq C_1$, $\text{BF}_{\mathcal{A}}(b) = \mathcal{A} \cap (c_0 + \ker \partial_1)$; equal-boundary lifts satisfy $c - c' \in Z_1 := \ker \partial_1$ and hence define $[c - c'] \in H_1 := Z_1 / \text{im } \partial_2$. Let R_{∂_1} be equal boundary on the additive right action of C_1 on itself. For every abelian G , $(\text{HRC}) \text{Hom}(H_1, G) \cong \{h \in Z_{\text{rc}}^1(R_{\partial_1}; G) : h(c + \partial_2 b, c) = 0 \forall b, c\}$, $\lambda \mapsto [(c, c') \mapsto \lambda([c - c'])]$. For the canonical forgetting complex $\mathfrak{C}_{\bullet}^{\text{forg}}(r : s)$ of (RGH), with quotient differentials $\bar{\partial}_i$, (RGH/UAR/HRC) gives the typed bridge $Z^1(R_p; G) \cong \{h \in Z_{\text{rc}}^1(R_{\bar{\partial}_1}; G) : h(c + \partial_2 b, c) = 0 \forall b, c\}$, using $\mathcal{O}_{p,G} : \text{Hom}(\mathbb{K}(p), G) \cong Z^1(R_p; G)$. Thus (HRC) is the 2-boundary-blind bidegree- $(1, 0)$ d_C -closed row of Proposition S2, realized by (RGH) forgetting. Outside that chain model homology and continuation curvature stay distinct: the right-zero monoid $S = \{1, a, b\}$ with $q(a) = q(b)$, $u(a) = 1, u(b) = 0$ has $H_1^{\text{sing}} = 0$ but $\partial_{\text{rc}} d_C u(a, b; a) = -1$, whereas the diagonal circle relation has $H_1^{\text{sing}}(S^1; \mathbb{Z}) = \mathbb{Z}$ and zero curvature. Current is the real graph-chain branch and (TF/HJTF) its nonnegative cut. A TimeFib path has endpoint boundary $T - I$, while (TFC/TRP/TCR) makes its root tower the diagonal (R2C/R2F) decomposition pro-object and retains C_0 -landing/TempRes $_{\mathcal{E}}(T)$: circle rotations at $T = I$ share zero boundary but have distinct landing towers, and the coherent scalar-unitary tower of Proposition VIII B 0 a lies in TempRes $_{\mathcal{E}}(I)$. Hence temporal lift data are not homological current coordinates.

- for $|V| = 1$ there is no internal transfer. Every scalar signed current dR has the unique mutually singular Hahn–Jordan reading $dR = d\Pi - d\Sigma$, $d\Pi \perp d\Sigma$, which is therefore the canonical one-vertex source/sink realization of the same current complex. Under source/sink exchange it is the real two-leg face: $\frac{1}{2}(d\Pi + d\Sigma) = \frac{1}{2}|dR|$ is even and $\frac{1}{2}(d\Pi - d\Sigma) = \frac{1}{2}dR$ is odd;
- measure Hahn–Jordan coupling fiber. Let S be standard Borel, write $\text{M}_{\text{sa}}(S)$ for its finite signed measures, and let $r \in \text{M}_{\text{sa}}(S)$ have Hahn–Jordan split $r = r^+ - r^-$, masses $P = r^+(S)$, $N = r^-(S)$, and $e_+ = (P - N)_+$, $e_- = (N - P)_+$. On $S_- = S \sqcup \{\partial_-\}$ and $S_+ = S \sqcup \{\partial_+\}$ put $a_r = r^+ + e_- \delta_{\partial_-}$, $b_r = r^- + e_+ \delta_{\partial_+}$, and

$$\begin{aligned} \text{HJMeasFib}_S(r) &:= \text{Cpl}(a_r, b_r) \xrightarrow{\cong} \{(\pi, \sigma, J) \geq 0 : r^+ = \pi + (\text{pr}_1)_* J, r^- = \sigma + (\text{pr}_2)_* J\}, \\ \text{[MHJ]} \quad \Gamma &\mapsto (\Gamma|_{S \times \{\partial_+\}}, \Gamma|_{\{\partial_-\} \times S}, \Gamma|_{S \times S}), \quad r = \pi - \sigma - BJ, \quad BJ = (\text{pr}_2)_* J - (\text{pr}_1)_* J, \\ &\quad \pi(S) = e_+, \quad \sigma(S) = e_-, \quad J(S \times S) = \min\{P, N\}. \end{aligned} \quad (297)$$

The inverse is $(\pi, \sigma, J) \mapsto J + \pi \otimes \delta_{\partial_+} + \delta_{\partial_-} \otimes \sigma$. Hence the fiber is always nonempty; if $r \neq 0$, (CPL) makes it singleton exactly when $a_r / \max(P, N)$ or $b_r / \max(P, N)$ is Dirac, and otherwise it has cardinal $2^{\aleph_0}$. Its graph is the typed relation-valued span $\text{M}_{\text{sa}}(S) \leftarrow \{(r, \Gamma) : \Gamma \in \text{HJMeasFib}_S(r)\} \rightarrow \text{M}_+(S_- \times S_+)$, with sets read discretely. For finite discrete S this is exactly the augmented transportation matrix of the next item. For equal-mass α on X and β on Y , $S = X \sqcup Y$ and $r = \iota_{X*} \alpha - \iota_{Y*} \beta$ give the full two-leg (CPL) fibre $\text{HJMeasFib}_S(r) \cong \text{Cpl}(\alpha, \beta)$; current, joining and interface data remain distinct refinements.

- finite-tag Hahn–Jordan face. For locally BV R_i put $\lambda = \sum_i |dR_i|$, $r_i = dR_i / d\lambda$, $p_i = (r_i)_+$, $s_i = (-r_i)_+$, $P = \sum_i p_i$, $S = \sum_i s_i$, $e_+ = (P - S)_+$, $e_- = (S - P)_+$, $a = (p_1, \dots, p_n, e_-)$ and $b = (s_1, \dots, s_n, e_+)$. The finite-discrete specialization of (MHJ) is the transportation polytope $\mathfrak{T}_r = \Pi(a, b)$. Writing its exterior legs as π, σ and its interior matrix as J gives exactly (HJTF) $r = \pi - \sigma - BJ$, $\sum_i \pi_i = e_+$, $\sum_i \sigma_i = e_-$, $\sum_{i,j} J_{ij} = \min\{P, S\}$. The inverse in (MHJ) is the unique reconstruction of every admissible (π, σ, J) . Writing $n_a = |\text{supp } a|$, $n_b = |\text{supp } b|$, one has $n_a, n_b \geq 1$ λ -a.e.; there, after deleting zero marginals, (TF/ICL) gives affine difference space $\Lambda_1(K_{n_a, n_b}) \otimes \mathbb{R}$ and dimension $(n_a - 1)(n_b - 1)$; after common-mass normalization, (MCM/CTH) makes a transport point extreme exactly when its positive support graph is a forest. The rate $(1, 1, -1)$ retains the one-dimensional $K_{2,2}$ witness.
- expected-quotient and flag faces. For an ω -preserving expectation E , assume $H_t = D(\rho_t E \| \omega)$ and $V_t = D(\rho_t \| \rho_t E)$ are locally BV. Then expected Pythagoras gives $R_t = D(\rho_t \| \omega) = H_t + V_t$. Put $\lambda = |dH| + |dV|$, $h = dH / d\lambda$, $v = dV / d\lambda$ and $c = \min\{h_+ + v_+, h_- + v_-\}$; then $d\Pi_R = dH^+ + dV^+ - c \, d\lambda$, $d\Sigma_R = dH^- + dV^- - c \, d\lambda$. This is the two-component (HJTF) face. Sharpness: for $u = (1/2, 1/2)$, $\omega = u \otimes u$, first-coordinate expectation and $p_t = (1/2 + t, 1/2 - t)$, the paths $p_t \otimes u$ and $u \otimes p_t$ have respectively $(H, V) = (D(p_t \| u), 0)$ and $(0, D(p_t \| u))$ at the same scalar $R_t = D(p_t \| u)$.

For a finite increasing ω -expected flag $N_0 \subseteq \dots \subseteq N_m = M$, let $t \mapsto \rho_t$ lie on the finite-entropy perimeter and assume each $R_i(t) := e_{i:i+1}(\rho_t)$ is locally BV. Put $\lambda_F = \sum_{i < m} |dR_i|$, $r_i = dR_i / d\lambda_F$, $p_i = (r_i)_+$, $s_i = (-r_i)_+$, $P = \sum_i p_i$, $S = \sum_i s_i$, $e_+ = (P - S)_+$, $e_- = (S - P)_+$, $a = (p_0, \dots, p_{m-1}, e_-)$ and $b = (s_0, \dots, s_{m-1}, e_+)$, with $n_a = |\text{supp } a|$, $n_b = |\text{supp } b|$. Telescoping and (MHJ/HJTF) give $\text{FlagCurFib}_{N, \rho}(t) = \Pi(a(t), b(t))$, $\dim_{\text{aff}} \text{FlagCurFib} = (n_a - 1)(n_b - 1)$. The lift is unique exactly when $n_a = 1$ or $n_b = 1$. Strictness is realized on three independent bits: for $u = (1/2, 1/2)$, $\omega = u^{\otimes 3}$ and the successive-coordinate flag, put $p_x = (1/2 + x, 1/2 - x)$, choose $0 < x_0 < 1/2$ and $\gamma = \log((1 + 2x_0)/(1 - 2x_0))$, and for small t set $\rho_t = p_{x_0+t/\gamma} \otimes p_{x_0+t/\gamma} \otimes p_{x_0-t/\gamma}$. At $t = 0$ the shell rates are $(1, 1, -1)$, so the active $K_{2,2}$ transport graph gives a one-dimensional (FCL) fibre at fixed scalar current. For every $s < r$ the pairwise shell is the partial-sum face

$$G_{s:r} = \sum_{s \leq i < r} e_{i:i+1} = V_s - V_r = H_r - H_s, \quad \Delta G_{s:r} = \Delta V_s - \Delta V_r = \Delta H_r - \Delta H_s.$$

If an ω -preserving state channel commutes with P_s, P_r , then $G_{s:r}(\Phi \rho) \leq G_{s:r}(\rho)$; the coordinate-swap witness still shows covariance is necessary.

Hence total load, nodal redistribution, edge circulation, scalar source/sink and expected-quotient location are typed readings of one balance carrier, with deterministic quotients and lift fibers kept distinct rather than treated as independent residual primitives.

Proof. Graph exactness gives scalar/chain/cycle rows; covariance and ∂_2 -blindness factor curvature through H_1 , while cohomology–homology/Banach duality plus residual profile gives the potential branch. Marginal equations give the unique $J + \pi + \sigma$ split/multiplicity; tangent-cycle law gives affine transport and the nonnegative lift relation the dimension/strict $(1, 1, -1)$ face. Pythagoras gives fibre cost; DPI/examples give separation. $\square$

c. Entropic calibration map. Let $t \mapsto \rho_t \in S_\omega^{\infty}(\mathcal{M})$ be càdlàg, absolutely continuous off a locally finite jump set, and suppose $R_t = D(\rho_t \parallel \omega)$ has locally bounded variation. The one-vertex case of Theorem X.2 gives the canonical Hahn–Jordan reading $dR = d\Pi - d\Sigma$. Standard entropy monotonicity and the declared MLSI calibrate it as follows:

1. on an ω -preserving Markov interval, $d\Pi_{ac} = 0$, and under $\text{MLSI}(\lambda)$, $d\Sigma_{ac} \geq 2\lambda R_t dt$;
2. at a jump $\rho_{t_j} = \Phi_j \rho_{t_j^-}$ generated by a normal state channel Φ_j , with $\Delta_j R = D(\Phi_j \rho_{t_j^-} \parallel \omega) - D(\rho_{t_j^-} \parallel \omega)$, $d\Pi(\{t_j\}) = (\Delta_j R)_+$, $d\Sigma(\{t_j\}) = (-\Delta_j R)_+$;
3. (RMC) on the finite-action domain put $\text{RMC}_\sigma(\omega_0 \rightarrow \omega_1) := D(\sigma \parallel \omega_0) - D(\sigma \parallel \omega_1)$. It is a signed normal-isomorphism-invariant endpoint cocycle, $\text{RMC}_{02} = \text{RMC}_{01} + \text{RMC}_{12}$. With $C_j(\rho) := D(\rho \parallel \omega) - D(\Phi_j \rho \parallel \Phi_j \omega) \geq 0$ by (RFP) and $B_j(\rho) := \text{RMC}_{\Phi_j \rho}(\omega \rightarrow \Phi_j \omega)$, telescoping gives $\Delta_j R = -C_j(\rho_{t_j^-}) + B_j(\rho_{t_j^-})$ and $B_j(\omega) = P_{\Phi_j}$ when finite. On faithful matrix references $B_j(\rho) = \text{Tr}[\Phi_j \rho (\log \Phi_j \omega - \log \omega)] = -i \partial_t(\Phi_j \rho)([D\Phi_j \omega : D\omega]_t)|_0$ by the finite-dimensional (CRN) specialization; RMC may be negative. Thus forcing is its diagonal face, not the positive jump mass;
4. on the constitutive branch $d\Pi = P_t dt$, $d\Sigma = \Lambda_t R_t dt$, $\dot{R}_t = P_t - \Lambda_t R_t$, $R = \text{Sol}_{P,\Lambda}(R_0)$. Hence $P \equiv 0$ gives extinction when $\int_0^\infty \Lambda = \infty$; more generally extinction holds when the source convolution in $\text{Sol}_{P,\Lambda}$ tends to zero, in particular for $P_t \rightarrow 0$ and an eventual floor $\Lambda_t \geq \lambda_0 > 0$; this is Nirodha($P, \Lambda; R_0$). If the sink integral is finite and $R_0 > 0$ on the unforced branch, then $R_t \rightarrow R_0 e^{-\int_0^\infty \Lambda} > 0$. On the convergent asymptotic branch $P_t \rightarrow P_0$, $\Lambda_t \rightarrow \lambda > 0$, variation of constants gives the exact stratification $R_t \rightarrow R_\infty = \frac{P_0}{\lambda}$, $\text{Nirodha}(P, \Lambda; R_0) \iff P_0 = 0$, $\text{Samsara}(P, \Lambda; R_0) \iff P_0 > 0$. At fixed λ , R_∞ is continuous and non-decreasing in P_0 . Since $\mathbb{C}1, \mathcal{M} \in \text{Lat}_\omega(\mathcal{M})$, expected Pythagoras gives for every finite-entropy state $0 = \delta_{\mathcal{M}}(\rho) \leq \delta_{\mathcal{N}}(\rho) \leq \delta_{\mathbb{C}1}(\rho) = D(\rho \parallel \omega)$, $D(\rho \parallel \omega) = \sup_{\mathcal{N} \in \text{Lat}_\omega(\mathcal{M})} \delta_{\mathcal{N}}(\rho)$. Hence on the convergent constitutive branch Nirodha and Samsara are respectively the zero and positive asymptotic sectors of the full expected-access Avidya envelope; vanishing at one access, including $\delta_{\mathcal{M}} = 0$, does not decide them. For $R_t = D(\rho_t \parallel \omega)$, extinction also implies $\|\rho_t - \omega\|_1 \rightarrow 0$.

The first three clauses are respectively entropy monotonicity/MLSI, Hahn–Jordan, and the (RMC)/DPI entropy square; the last is variation of constants. Thus they calibrate the universal current complex rather than define a second residual theorem. Presentation isomorphisms preserve relative entropy, so the sign of $\dot{R}$ and the zero/floor sectors are frame-invariant.

d. Asymptotic residual sectors. For a fixed attached finite oriented graph $G = (V, E)$, put $dE := d\Pi - d\Sigma$. The deterministic current-forgetting spine is (RFC) $(d\Pi, d\Sigma, dT) \xrightarrow{F_0} (dE, dT) \xrightarrow{F_1} dR = dE - dT \xrightarrow{F_2} S_G(dR)$, $dJ \xrightarrow{B_G} dT$. Its fibers are exact: at fixed (dE, dT) , $(d\Pi, d\Sigma) = (dE^+ + \chi, dE^- + \chi)$ with arbitrary componentwise non-negative χ is precisely gross-turnover freedom; at fixed dR , $(dE, dT) = (dR + t, t)$ with arbitrary $C_0^0(V)$ -valued signed Radon measure t is transfer-allocation freedom; and the difference fiber of F_2 is the space of $\ker S_G = \text{im } B_G$ -valued signed measures. On the graph-liftable branch $dT \in \text{im } B_G$, the side cover has fiber $\ker B_G$. The nested kernel relations of the effective-range corestrictions of F_0, F_1, F_2 induce the (KSP) loss filtration for every abelian cocycle coefficient group. Finite source tags lie one level above it: (HJTF) is the relation-valued non-negative lift fiber into the balance carrier, with transportation/cycle freedom, not another deterministic arrow. By (RMC), forcing is diagonal moving-reference data $P_\Phi = \text{RMC}_{\Phi\omega}(\omega \rightarrow \Phi\omega)$, not another (RFC) arrow; reference preservation kills it.

1. Relative globality and terminal closure

a. Relative-global stage map. For every non-empty directed $J \subseteq I$, put $\mathcal{G}_J := \lim_{\leftarrow j \in J} \mathcal{T}_j$, $\rho_{JJ'} : \mathcal{G}_{J'} \rightarrow \mathcal{G}_J$ ($J \subseteq J'$). By the global compatibility/closure result III 0a this is the stage read-out of C_J ; a greatest stage gives the local object, $\mathcal{G}_I = C_*^{\text{rec}}$, and directed exhaustion gives $\mathcal{G}_I \cong \varprojlim_n \mathcal{G}_{J_n}$. This is the common law/residual construction underlying (SPB) and (RCF); on the latter face (PGA) identifies adaptive law, predictive subalgebra and predictive hosted-closure class losslessly. Exogenous stage, policy/priority and realized path remain their typed fibers.

b. Outward stabilization. This is exactly the Takesaki-duality/two-periodicity clause in the classification above; no additional invariant is introduced.

c. Unforced envelope-integration map. The unforced branch combines Theorem VIII.1 with the current complex. Its gauge bound is uniform in cocycle, matrix level and presentation; on the faithful-base branch $R_t = D(\rho_t \parallel \omega_\varphi) = \mathcal{R}_t^g(\varphi)$ and $d\Sigma_{ac} \geq 2c_{S^1} R_t dt$. Pinsker gives norm convergence to the base read-out; cut transfer is quotient-invisible. Reference-preserving failure and forcing remain separate.

Supplementary package S5: Dilation forcing and GNS normal support. The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S5; every use below invokes that exact proof-bearing package, not an additional assumption. *Supplementary package S6: Interface completion, normal support, and terminality.* The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S6; every use below invokes that exact proof-bearing package, not an additional assumption. *Supplementary package S7: Structural hull, adaptive closure, perspective straightening, and provenance.* The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S7; every use below invokes that exact proof-bearing package, not an additional assumption.

XI. RELATIVE-GLOBAL RECONSTRUCTION AND AMBIENT FIBERS

We now pass to the minimal normal core and its residual ambient realization fibre.

On Theorem VII.1(4), fix $\mathbf{P} = (\mathcal{M}, \omega, (\alpha_r : \mathcal{N}_r \rightarrow \mathcal{M})_{r \in F})$, with $\mathcal{M}$ σ -finite, ω faithful normal and the accesses normal UCP, and write $\text{Core}(\mathbf{P}) = \mathcal{G}_F^\omega$, with ω -expectation E_F^{core} . By (BKT) this is a **PAcc** point. The realization regime is external only to the bounded $\mathfrak{F}_{\text{acc}}$ presentation; it remains a typed stage-germ realization fibre, and the theorem classifies its core and residual fibers.

Theorem XI.1 (Minimal sufficient core and irreducible ambient fibers). *For $\mathcal{L}_F = \{(h, t) : h \text{ finitely supported}, t \in \mathbb{R}\}$ let $\mathbb{F}(\mathcal{L}_F)$ be free, put $u_{(h,t)}^F = [DQ_F^\omega(h) : D\omega]_t$, $\chi_F^\omega(\ell_1^{\varepsilon_1} \cdots \ell_m^{\varepsilon_m}) = \omega((u_{\ell_1}^F)^{\varepsilon_1} \cdots (u_{\ell_m}^F)^{\varepsilon_m})$, and $\mathcal{G}_F^\omega = \Gamma_\omega(\mathcal{V}_F, \{Q_F^\omega(h) : h \text{ finitely supported}\})$, with output transport $(h, t) \mapsto (\kappa h, t)$. Scalar price, tracial marginals, common Petz core and Krieger–Connes type remain coarser read-outs, never converse invariants. Resolved pointed core. For every sigma-finite normal access system, $\mathcal{G}_F^\omega = \text{Gen}_\omega\{Q_F^\omega(h) : h \text{ finitely supported}\} = W^*([DQ_F^\omega(h) : D\omega]_t : h, t) = W^*(\sigma_s^\omega(x) : x \in \mathcal{V}_F, s \in \mathbb{R})$. With $y_h = N_F Q_F^\omega(h)$, pressure makes y_h the unique maximizer of $\langle h, y \rangle - J_F^\omega(y)$ and $Q_F^\omega(h) = s_F(y_h)$. For every $\varepsilon > 0$,*

$$\begin{aligned} \mathcal{Y}_{F,\varepsilon}^{\text{con}} &:= \{y : J_F^\omega(y) < \varepsilon, s_F(y) \text{ faithful}, J_F^\omega(y) = D(y_r \| \omega_r) \forall r \in F\}, \\ \text{[ECG]} \quad \{s_F(y) : y \in \mathcal{Y}_{F,\varepsilon}^{\text{con}}\} &= \text{Host}_\omega(\mathcal{A}_F^\omega) \cap \{D(\cdot \| \omega) < \varepsilon\}, \\ \mathcal{A}_F^\omega &= \text{Gen}_\omega\{s_F(y) : y \in \mathcal{Y}_{F,\varepsilon}^{\text{con}}\}. \end{aligned} \quad (298)$$

By (RFP), common U1 contact is $M_F(s_F(y)) = 0$: pressure-exposed lifts generate $\mathcal{G}_F^\omega$, while arbitrarily small finite-action zero-contact lifts (equivalently, zero middle Petz shell) generate $\mathcal{A}_F^\omega$. The product-marginal example and Theorem V.2(E) keep these expected subalgebras distinct. Hence $\mathcal{G}_F^\omega$ is the unique minimal expected-sufficient subalgebra for all finite-support $Q_F^\omega(h)$, with unique ω -preserving E_F^{core} ; (MSA/RRP/NAT) makes $\mathcal{R}_{\mathcal{G}_F^\omega}^n$ greatest among restriction relations of such sufficient subalgebras, while the section carrier admits no scalar replacement.

The characteristic functional and the ancillary/support towers of Section IXB are complete coordinates of the pointed labeled core. For normal reference-preserving output isomorphisms $\kappa_r : \mathcal{N}_r \rightarrow \mathcal{N}'_r$, let $T : \mathcal{V}_F^\mathbb{C} \rightarrow \mathcal{V}'_F^\mathbb{C}$ be unital, $*$ -linear and weak- $*$ homeomorphic, with $\mathcal{V}_F^\mathbb{C} = \mathcal{V}_F + i\mathcal{V}_F$, $T\alpha_r = \alpha'_r \kappa_r$, and $T_n = T \otimes \text{id}_{M_n}$. Set $(\kappa h)_r := \kappa_r(h_r)$ and let $\kappa_\# := \mathbb{F}((h, t) \mapsto (\kappa h, t)) : \mathbb{F}(\mathcal{L}_F) \rightarrow \mathbb{F}(\mathcal{L}'_F)$; write $\kappa w := \kappa_\#(w)$. Write $\mathfrak{J}_{F,\text{nrmr}}^\omega \approx_\kappa \mathfrak{J}'_{F,\text{nrmr}}^\omega$ and $\mathfrak{P}_{F,\text{nrmr}}^\omega \approx_\kappa \mathfrak{P}'_{F,\text{nrmr}}^\omega$ when, for all $n \geq 1$, faithful densities γ with $\text{Tr} \gamma = 1$, and $c \geq 0$, $\mathcal{J}_{F,n}^{\omega,\gamma}(\ell \circ T_n^{-1}) = \mathcal{J}_{F,n}^{\omega,\gamma}(\ell)$ and $\psi_{F,n,c}^{\omega,\gamma}(T_n V) = \psi_{F,n,c}^{\omega,\gamma}(V)$; no multiplicativity of T is assumed. Then $\mathfrak{J}_{F,\text{nrmr}}^\omega \approx_\kappa \mathfrak{J}'_{F,\text{nrmr}}^\omega \iff \mathfrak{P}_{F,\text{nrmr}}^\omega \approx_\kappa \mathfrak{P}'_{F,\text{nrmr}}^\omega \iff \chi_F^\omega(w) = \chi_F^{\omega'}(\kappa w) \forall w \iff \exists! \theta : \mathcal{G}_F^\omega \xrightarrow{\sim} \mathcal{G}_F^{\omega'}$, where θ is normal, state-preserving and label-intertwining, $\theta(u_{h,t}^F) = u_{\kappa h,t}^{\omega'}$, $\theta \alpha_r = \alpha'_r \kappa_r$. For $a = a^* \in \mathcal{N}_r$, let $h^{r,a}$ be the finitely supported tuple with $h_r^{r,a} = a$ and $h_s^{r,a} = 0$ for $s \neq r$; then $X_{r,a} := \frac{1}{i} \frac{d}{dt} \Big|_0 u_{h^{r,a},t}^F = \alpha_r(a) - \Lambda_F^\omega(h^{r,a}) \mathbf{1}$, $\alpha_r(a) = X_{r,a} + [\omega_r(a) - \omega(X_{r,a})] \mathbf{1}$. Hence $\Phi = \iota'_F \theta E_F^{\text{core}}$ and $\Psi = \iota_F \theta^{-1} E_F^{\text{core}}$ are reference-preserving normal UCP simulations inverse on the cores, uniformly across normal type I–III summands.

Lost orientation. By (TR), the tracial slice $\gamma = \tau_n$ can identify transpose-related distinct pointed cores. Full faithful marginals restore $\mathbf{1} \otimes D$ and, in finite dimension, every labeled word moment $C_i(t) = \text{Tr}(\rho^{t_1} X_{i_1} \cdots \rho^{t_q} X_{i_q})$, $t_r \geq 0$, $\sum_r t_r = 1$, and all simplex-vertex jets. Thus the resolved tower is complete, the tracial slice is not, and simultaneous transpose retains the declared $\mathbb{Z}_2$ opposite-product/Jordan-orientation action.

For a normal presentation $\mathbf{P}$ set $\mathfrak{C}_F(\mathbf{P}) = \mathcal{G}_F^\omega$ with restricted reference/labels. Core reduction is split idempotent, (RGI) $E_F^{\text{core}} \iota_F = \text{id}_{\mathcal{G}_F^\omega}$, $\text{Core}^2(\mathbf{P}) = \text{Core}(\mathbf{P})$. Write $\mathbf{P} \sim_{\text{rg}} \mathbf{P}'$ for pointed label-intertwining normal core isomorphism and let $c_F : [\mathbf{P}] \mapsto [\text{Core}(\mathbf{P})]$ be the class-valued ambient-to-core reduction. For each set-sized Core-stable A , put $c_{F,A} := c_F|_A : A \rightarrow c_F(A)$; only these slices enter set-valued residual profiles. For abelian G , $\mathcal{R}_G(c_{F,A})_{\mathbb{C}} = G_{F,A}^{c_{F,A}^{-1}([\mathbb{C}])} / \Delta_{\mathbb{C}} G \cong Z^1(R_{c_{F,A}}|_{c_{F,A}^{-1}([\mathbb{C}])}; G)$, $Z^1(R_{c_{F,A}}; G) \cong \text{coker } c_{F,A}^*$ by (KSP), and the crossed-product witness below makes it nonzero for $G = \mathbb{R}$. For $\Pi_F \rho = (\rho|_{\mathcal{G}_F^\omega}) \circ E_F^{\text{core}}$ and finite-entropy ρ , $D_M(\rho \| \omega) = D_M(\rho \| \Pi_F \rho) + D_{\mathcal{G}_F^\omega}(\rho|_{\mathcal{G}_F^\omega} \| \omega|_{\mathcal{G}_F^\omega})$. Since $\mathcal{V}_F \subseteq \mathcal{G}_F^\omega$, the record readout and, on each fixed- γ ancillary face, $Q_{F,n}$ factor through core restriction; $E_F^{\text{core}} \otimes \text{id}_{M_n}$ preserves that face. Hence (EVD) gives core invariance of J_F^ω and, by U0 support duality, for all $n \geq 1$, faithful γ , $c \geq 0$, on the common $\mathcal{V}_F \odot M_n$ carrier, (ACE) $\mathcal{J}_{F,n}^{\omega,\gamma} = \mathcal{J}_{F,n}^{\omega|_{\mathcal{G}_F^\omega},\gamma}$, $\psi_{F,n,c}^{\omega,\gamma} = \psi_{F,n,c}^{\omega|_{\mathcal{G}_F^\omega},\gamma}$. Hence the NMR/U0/Pareto–Legendre price presentations are core-invariant and classify $\sim_{\text{rg}}$; the core is full iff $\mathcal{G}_F^\omega = \mathcal{M}$. Put $\text{Wit}_F^\omega(\mathbf{P}) := [\chi_F^\omega]_{(T,\kappa)}$ and $\text{Saksin}_F^\omega := \text{Wit}_F^\omega$. At fixed reference, the preceding equivalence and (RGI) give

$$\begin{aligned} \mathbf{P} &\mapsto (\mathcal{M}, \omega, \mathcal{G}_F^\omega) \in \mathbf{PAcc}, \\ \text{[PSF]} \quad [\mathbf{P}] &\xrightarrow{c_F} [\mathfrak{C}_F(\mathbf{P})] \xrightarrow[\cong]{\bar{\chi}_F^\omega} \text{Im Wit}_F^\omega, \end{aligned} \quad (299)$$

Here $\bar{\chi}_F^\omega([\mathbb{C}])$ is the characteristic-functional class: Saksin is the complete core invariant, not another pointing coordinate; this places noncanonicity before pointed-core selection and the c_F -fiber retains post-classification ambient freedom. Also (MSA/RGI) give $\text{Wit}_F^\omega(\text{Core } \mathbf{P}) = \text{Wit}_F^\omega(\mathbf{P})$ and $\text{Gen}_\omega \text{Host}_\omega(\mathcal{G}_F^\omega) = \mathcal{G}_F^\omega$. For $\emptyset \neq E \subseteq F$, let $\iota_{E,F} : \mathcal{L}_E \hookrightarrow \mathcal{L}_F$ be zero-extension in the label coordinates. Then $\chi_F^\omega \circ \mathbb{F}(\iota_{E,F}) = \chi_E^\omega$ and $\mathcal{G}_E^\omega \subseteq \mathcal{G}_F^\omega$: label extension is the vertical **Pres** arrow $(\text{id}, \mathcal{G}_E^\omega \subseteq \mathcal{G}_F^\omega)$ and Saksin restricts compatibly. The arrow has the Avidya/Host/zero-Samskara charts.

Saksin label-lift relation. Let $\mathbf{P}_F, \mathbf{P}'_F$ be resolved pointed systems in this perimeter and let $u = (\theta, \theta \mathcal{G}_F^\omega \subseteq \mathcal{G}_{F'}^{\omega'})$ be the underlying **Pres** arrow. Define

$$\begin{aligned} \text{Lab}(u) &:= \{(\lambda, \kappa) : \lambda : F \hookrightarrow F', \kappa_r : \mathcal{N}_r \xrightarrow{\sim} \mathcal{N}'_{\lambda(r)} \text{ normal reference-preserving } * \text{-isomorphisms}\}, \\ \text{SakLift}(u) &:= \{(\lambda, \kappa) \in \text{Lab}(u) : \theta \alpha_r = \alpha'_{\lambda(r)} \kappa_r \forall r\}. \end{aligned}$$

For $(\lambda, \kappa) \in \text{Lab}(u)$ let $j_{\lambda,\kappa} h \in \bigoplus_{r' \in F'} \mathcal{N}'_{r',\text{sa}}$ be supported on $\lambda(F)$ with $(j_{\lambda,\kappa} h)_{\lambda(r)} = \kappa_r(h_r)$. Then

$$\begin{aligned} (\lambda, \kappa) \in \text{SakLift}(u) &\iff \theta H_F(h) = H_{F'}(j_{\lambda,\kappa} h) \quad (\forall h) \\ \text{[SLF]} \quad &\implies \theta_\# Q_F^\omega(h) = Q_{F'}^{\omega'}(j_{\lambda,\kappa} h), \quad \theta(u_{h,t}^F) = u_{j_{\lambda,\kappa} h,t}^{F'} \end{aligned} \quad (300)$$

$$\chi_F^\omega(w) = \chi_{F'}^{\omega'}(\mathbb{F}(j_{\lambda,\kappa})(w)) \quad (\forall w), \quad \text{SakLift}(v) \circ \text{SakLift}(u) \subseteq \text{SakLift}(vu).$$

Thus bare **Pres** refinement is the base of a relation-valued labeled-transport fiber, not a labeled Saksin transport. Both failure modes occur at fixed core. On (M_2, τ) , the source with one label $(\mathcal{N}_1, \alpha_1) = (M_2, I)$ and the target with two labels the transverse MASAs $\mathcal{D}_z, \mathcal{D}_x \hookrightarrow M_2$ have

the same full core and the identity **Pres** arrow, but **SakLift** = $\emptyset$ because $M_2 \not\equiv \mathcal{D}_z, \mathcal{D}_x$. Conversely, for the single-label system on (M_2, τ) with $N_1 = M_2$ and $\alpha_1(a) = \tau(a)I$, the core is $\mathbb{C}1$ and the identity base arrow admits $(\text{id}, \text{Ad } U)$ for every unitary $U \in M_2$; hence **SakLift** is non-singleton without duplicating a label. The label-extension arrow is the declared canonical point of this fiber. Hence the exact extra datum needed to transport Saksin along a general **Pres** refinement is a label lift; neither existence nor selection follows from the core refinement alone.

For faithful normal ω, ν and $N \in \text{Lat}_\nu(\mathcal{M})$ set $u_t = [D\nu : D\omega]_t$ and $\Lambda_\nu^\omega(N) := \Gamma_\omega(N, \{\nu\}) = W^*(N, u_t : t \in \mathbb{R})$. (CRN) gives $\sigma_s^\omega(\Lambda_\nu^\omega(N)) = \Lambda_\nu^\omega(N)$, so (TEXP) types it in Lat_ω ; on $\text{Lat}_\omega \cap \text{Lat}_\nu$ it is $\text{Br}_{\omega, N}(\nu)$. Since $\mathcal{G}_F^\nu = W^*(\sigma_s^\nu(\mathcal{V}_F) : s \in \mathbb{R})$, $\Lambda_\nu^\omega(\mathcal{G}_F^\nu) = \Gamma_\omega(\mathcal{V}_F, \{\nu\}) = W^*(\mathcal{G}_F^\omega, u_t : t \in \mathbb{R})$, $\Lambda_\nu^\omega(\mathcal{G}_F^\nu) = \mathcal{G}_F^\omega \iff \nu \in \text{Host}_\omega(\mathcal{G}_F^\omega)$. Together (SLF) and this reference transport give the exact two-axis Saksin perimeter: label change requires a point of **SakLift**, while reference change inside $\text{Host}_\omega(\mathcal{G}_F^\omega)$ is cocycle-internal; bare **Pres** refinement supplies neither missing datum. Thus witness resolution is idempotent, access-coherent and reference-indexed; the tracial/ambient non-collapse witnesses remain.

1. History/biflag core and ambient fiber. For expected accesses $P_\alpha = E_{N_\alpha^*}$, (HSD) recovers $N_\alpha = W^*(\text{Fix } \Lambda_{P_\alpha})$; commuting flat words recover intersections, while generator marking gives $\mathcal{W}_P(J) = W^*(N_\alpha : \alpha \in J)$ and $(\text{CTF}_{\text{fib}}/\text{SDR}/\text{FLG})$ gives $\text{Lat}_\omega(\mathcal{M})_{\geq \mathcal{W}_P(J)} \cong \text{Fix}(\text{Cl}_{\omega, \mathcal{W}_P(J)})$, so this branch has no independent range attachment. For $P_r = \Pi_{\{r\}}^\Delta$, $\prod_{r \in S} P_r = \Pi_S^\Delta$, $\text{Ran}(\Pi_S^\Delta)^* = \mathcal{A}_S^\Delta = W^*(\text{Fix } \Lambda_{\Pi_S^\Delta})$. Thus, for $\emptyset \neq F \subseteq \mathcal{R}$, the (DI) joint quotient and $\mathcal{V}_F = (\mathcal{K}_F^{\text{vis}})^\perp$ are the native biflag's recoverability/visibility legs, (VRI/BCG) their lossless pair, and on faithful finite-dimensional constant-rank Araki charts (VAG) its BKM associated graded (not a global tangent splitting). The symmetric I, T, D_λ examples separate the legs; $M_{\text{sa}}/\mathcal{V}_F$ is dual to the invisible predual kernel and modular closure of $\mathcal{V}_F$ generates $\mathcal{G}_F^\omega$.

On full expected flags, record descent is triangular, (FLR) fixed-base covariance shell-diagonal and (RIF) global expectation covariance diagonal; Theorem IX.1 and its identity/ $\mathcal{K}$ pair make the final upgrade a strict fiber. For selected $\mathcal{K}$ and endpoint ρ , with $\phi = \rho|_{\mathcal{K}}$, $p = s(\phi)$, $z = c_{\mathcal{K}}(p)$, the (NSG) central quotient gives $\ker \pi_\phi = \mathcal{K}(1 - z)$ and $M_\phi \cong \mathcal{K}z$; hence support loss $p < 1$ and central-support loss $z < 1$ are distinct, including $p < 1, z = 1$ on a factor.

For product charts $\mathbf{P}_\mathcal{R} = (\mathcal{G} \otimes \mathcal{R}, \omega_\mathcal{G} \otimes \tau, a \mapsto a \otimes 1)$, $Z = Z(\mathcal{G})$, state-preserving over-core isomorphisms are exactly pointed Z -fixed isomorphisms $\vartheta : (Z \otimes \mathcal{R}, \omega_\mathcal{G}|_Z \otimes \tau) \rightarrow (Z \otimes \mathcal{R}', \omega_\mathcal{G}|_Z \otimes \tau')$, via (WFM) $\Phi \mapsto \vartheta = \Phi|_{Z \otimes \mathcal{R}}$, $\vartheta \mapsto \Phi_\vartheta$, $\Phi_\vartheta(g \otimes r) = (g \otimes 1)\vartheta(1 \otimes r)$. For $\tau_p = \text{diag}(p, 1 - p)$, $0 < p < 1/2$, $\mathbf{P}_{2,p} = \mathbf{P}_{(M_2, \tau_p)}$ gives, over every pointed core $\mathbf{C}$, (AIF) $[\mathbf{P}_{2,p}] = [\mathbf{P}_{2,q}] \iff p = q$, $(0, 1/2) \hookrightarrow c_F^{-1}([\mathbf{C}])$, $p \mapsto [\mathbf{P}_{2,p}]$. If $\mathbf{C}$ has separable predual and $\text{Fib}_{\text{sep}}(\mathbf{C})$ denotes the classes in $c_F^{-1}([\mathbf{C}])$ with separable-predual ambient algebra, this continuum injection is sharp there (but not on the unrestricted sigma-finite regime): $(\text{ADF}) |\text{Fib}_{\text{sep}}(\mathbf{C})| = 2^{\aleph_0}$. With $d_2(\mathbf{P}) = \dim_{\text{Hilb}} L^2(\mathcal{M}, \omega)$ and $d_2(\mathbf{C}) = \dim_{\text{Hilb}} L^2(\mathcal{G}, \omega_\mathcal{G})$, the unrestricted σ -finite spectrum is (ADS)

$$\{d_2(\mathbf{P}) : [\mathbf{P}] \in c_F^{-1}([\mathbf{C}]), d_2(\mathbf{P}) \geq \aleph_0\} = \{\kappa : \kappa \text{ infinite}, \kappa \geq \max(\aleph_0, d_2(\mathbf{C}))\}.$$

Indeed the Bernoulli algebra M_U^{Ber} with $|U| = \kappa$ has a finite-coordinate Walsh basis of cardinal κ , while the core inclusion gives the lower bound; tensoring with (M_2, τ_p) preserves κ and WFM separates the continuum family by modular ratio $p/(1 - p)$. Thus every displayed κ -stratum contains $2^{\aleph_0}$ inequivalent charts, density is intrinsic but incomplete, no set-cardinal bound depends only on $[\mathbf{C}]$, and the unrestricted c_F -fibre is proper-class-valued; equivalently (ADS) lies outside every set-sized $c_{F,A}$ slice.

The remaining ambient strata are independently sharp. For $E_0 : M_n \rightarrow \mathbb{C}1$, $E_0(x) = \tau_n(x)1$, one has $I_{E_0} = n$, $H_{E_0} = \log n$ and $E_0(x) \geq n^{-1}x$, sharply on rank-one projections, so order index separates matrix size without identifying the minimal-index interface. For finite G , $d = |G|$, let $R = \bigotimes_{n \geq 1} (M_d, \tau)$ and $\alpha_g = \bigotimes_n \text{Ad } \lambda_g$. If $g \neq e$, a rank-one p with $\lambda_g p \lambda_g^* \perp p$ gives a central sequence p_n with $\|\alpha_g(p_n) - p_n\|_2 = \sqrt{2/d}$, whereas inner automorphisms move bounded central sequences by $o(1)$ in L^2 ; hence α_g is outer. In $M = R \rtimes_\alpha G$,

$$E\left(\sum_g x_g u_g\right) = x_e, \quad x = \sum_g u_g E(u_g^* x) = \sum_g E(x u_g) u_g^*,$$

so $\{(u_g, u_g^*)\}$ is a quasi-basis. If $z = \sum_g z_g u_g \in R' \cap M$, then $z_g \alpha_g(a) = a z_g$ and polar decomposition/outerness force $z_g = 0$ for $g \neq e$, $z_e \in \mathbb{C}$. If $v = \sum_g v_g u_g \in N_M(R)$ and $\beta = \text{Ad } v|_R$, then $v_g \alpha_g(a) = \beta(a) v_g$; two nonzero coefficients would make a nontrivial $\alpha_{hg^{-1}}$ inner, hence $v \in \mathcal{U}(R) u_g$ uniquely. Therefore $R' \cap M = \mathbb{C}$, $N_M(R)/\mathcal{U}(R) \cong G$, and $[M : R] = |G|^{\aleph_1}$ (Sec. 0). Thus $\mathbb{Z}_2, \mathbb{Z}_3$ fix the commutant but change index, while $\mathbb{Z}_4, \mathbb{Z}_2^2$ fix both and change over-core class. The A_0/A_1 factor independently fixes the core/expected-subalgebra lattice while changing the one-seed hull; WFM remains a product chart and the c_F -fibre remains genuinely fibre-valued.

2. Enlargement and exhaustion. For typed pullback j_u , let $I_t = [j_u N_{r'}', j_u \mathcal{A}']$ and $I_s = [N_{\rho(r')}, \mathcal{A}]$. Then

$$\text{WorldCmp}_u(r') = I_t \cap I_s = \begin{cases} [N_{\rho(r')}, j_u \mathcal{A}'], & N_{\rho(r')} \subseteq j_u \mathcal{A}', \\ \emptyset, & \text{otherwise.} \end{cases} \quad (301)$$

On the nonempty Takesaki branch let $N_0 = j_u N_{r'}' \subseteq N_1 = N_{\rho(r')} \subseteq N_2 = j_u \mathcal{A}' \subseteq N_3 = \mathcal{A}$, with middle band $q_{N_1} \geq q_{\mathcal{L}} \geq q_{N_2}$. Its three BKM/entropic shells are lower-transport, intermediate-algebra width and ambient coverage. Along finite-entropy paths with locally BV shell loads, (FCL) is their exact HJTF current lift, so scalar current forgets allocation without collapsing these static coordinates. Order-faithfulness gives $\mathbf{b}_{0:1} = 0 \iff N_0 = N_1$, $\mathbf{b}_{1:2} = 0 \iff N_1 = N_2 \iff |\text{WorldCmp}_u(r')| = 1$, $\mathbf{b}_{2:3} = 0 \iff N_2 = \mathcal{A}$, and the two outer zeros iff $I_t = I_s$. U_0 activation and the c_F -fiber remain independent. Finally $F_j \uparrow F_\infty$ gives $\mathcal{G}_{F_\infty}^\omega = W^*(\bigcup_j \mathcal{G}_{F_j}^\omega)$, full exactly for core-generating terminal access.

Proof. Araki Pythagoras/residual profile gives contact and Host; entropy/cocycle derivatives plus multi-sufficiency reconstruct the core. Equality of χ fixes the labelled cyclic GNS representation and bicommutant map. Disintegration/Takesaki/Connes gives shells/core transport; entropy naturality the state lift; relative-commutant slicing word factorization. Product charts give the continuum type- I_2 lower bound. For the separable-predual upper bound, put every pointed ambient extension in faithful standard form on a separable Hilbert space: the SOT unit ball is

Polish, its von Neumann unit ball is closed, and the normal state and fixed-core inclusion/finite label data are determined on countable strong-* dense sets, so there are at most $2^{\aleph_0}$ codes. Thus AIF gives ADF exactly; Bernoulli gives the full infinite Hilbert-density spectrum/core-only counterexample. Matrix/group examples give factor strata; Hiai⁶⁶ (Thm. 1) the minimum-index expectation on finite-index subfactor legs. Splitting, represented-incidence, BKM/fibre-cost and directed exhaustion give remaining coordinates/current lift independently of c_F . $\square$

a. Core-realization commutation and dynamic attachments. For $\mathfrak{S} \in \mathbf{Acc}_{\text{sb}}$ and finite $F \subset \mathcal{R}_{\text{adm}}$, let $\mathbf{P}_F^{\text{MEnv}}(\mathbf{P})$ be the MEnv system and put $\mathbf{C}_F^{\text{ab}}(\mathfrak{S}) := W^*(j_0 \alpha_r^{\text{ab}}(\mathcal{Y}_r) : r \in F) \subseteq \mathcal{A}_0$, $\mathbf{C}_{\mathfrak{S}}(j_0 f) = j_0((T - I)f)$. Then

$$\begin{aligned} \text{[CRC]} \quad & \text{Cor}(\mathbf{P}_F^{\text{MEnv}}(\mathbf{P})) = \mathbf{C}_F^{\text{ab}}(\mathfrak{S}), \\ & \mathbf{C}_{\mathfrak{S}} = E_0 \text{Ad}(u^*)|_{\mathcal{A}_0} - I_{\mathcal{A}_0}. \end{aligned} \quad (302)$$

On the fixed pointed skeleton $\mathfrak{S}_0^{\#}$, put $m(\mathbf{P}) = [\text{MEnv}(\mathbf{P}, \mu)]_{\text{mark}/\mathfrak{S}_0^{\#}}$ and $c(\mathbf{P}) = T_{\mathbf{P}} - I$, where marked isomorphisms fix $\mathcal{A}_0$, its accesses/records and the crossed-product unitary. Modulo μ -a.e. equality, $R_m = R_c = \Delta$: $T = I + \mathbf{C}_{\mathfrak{S}}$ recovers $\mathbf{P}$ on a countable determining Boolean algebra and hence MEnv, while (NC) recovers $\mathbf{C}_{\mathfrak{S}}$. Modular triviality on $\hat{\mathcal{A}}$ proves the first CRC line, and $\mathbf{C}_{\mathfrak{S}} = 0 \iff T = I \iff \mathbf{P}_x = \delta_x$ a.e. $\iff \mathcal{M}_{\text{time}}$ is abelian. The stationary two-point identity/flip pair fixes the uniform reference, identity access/record skeleton and full core $\mathcal{A}_0$, while changing $\mathbf{C}_{\mathfrak{S}}$ and abelian/nonabelian MEnv; the dynamic datum is therefore a strict core fiber.

Supplementary package S8: Marked-to-ambient realization fibre. The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S8; every use below invokes that exact proof-bearing package, not an additional assumption.

Theorem XI.2 (Projective-temporal recollection). *On the projectively adapted compact-Radon Feller branch, relative to a supplied (FTRF) point, fix compatible (μ_n) , global μ from (PI), pointed $\mathfrak{S}_0^{\#}$ and finite nonempty F . Write $\mathcal{D}_{\text{pr}}^{\text{UI}}$ for the bond-compatible Feller-kernel families $(\mathbf{P}_n)$ of (PKR) compatible with $\mathfrak{S}_0^{\#}$, with $\mu_n \mathbf{P}_n \sim \mu_n$ and both (PN) likelihood-ratio martingales UI; write $\mathcal{K}_{\text{pr}}^{\text{UI}}(\mathfrak{S}_0^{\#})$ for their unique global (PKR) reconstructions and put $\text{MEnvCls}_{\text{pr}} := m(\mathcal{K}_{\text{pr}}^{\text{UI}}(\mathfrak{S}_0^{\#}))$. By (PN) these global kernels satisfy $\mu \mathbf{P} \sim \mu$; the bondwise (PKR) equations, not UI, are the zero record-descent DepRes condition. Then*

$$\mathcal{D}_{\text{pr}}^{\text{UI}} \xrightarrow[\sim]{\text{PI/PKR/PN}} \mathcal{K}_{\text{pr}}^{\text{UI}}(\mathfrak{S}_0^{\#}) \xrightarrow[\sim]{m} \text{MEnvCls}_{\text{pr}} \xrightarrow[\sim]{\text{Tot}} \coprod_{a \in q_F(\text{MEnvCls}_{\text{pr}})} (q_F^{-1}(a) \cap \text{MEnvCls}_{\text{pr}}).$$

For $X_{\mu} = \text{supp } \mu$, quasi-invariance and Feller positivity make restriction to $C(X_{\mu})$ extension-independent and place T_{ξ}^{Fcl} in the strong-closed contraction monoid $\mathcal{E}_{\mu}^{\text{Fcl}} = \{A \in B(C(X_{\mu})) : A \geq 0, A1 = 1\}$. Indeed positive unital maps on $C(X_{\mu})$ are contractions and are closed under composition and strong limits; if $f|_{X_{\mu}} = 0$, then $(\mu \mathbf{P})(|f|) = 0$, so Feller continuity and full support on X_{μ} force $\mathbf{P}|f| = 0$ there. For $d \in \mathcal{D}_{\text{pr}}^{\text{UI}}$ let $\mathbf{P}_d$ be its global (PKR) reconstruction and $\xi_d = m(\mathbf{P}_d)$, and put $\mathcal{D}_{\text{pr}}^{\text{UI,flow}} := \coprod_{d \in \mathcal{D}_{\text{pr}}^{\text{UI}}} \text{TimeFib}_{\mathcal{E}_{\mu}^{\text{Fcl}}}^0(T_{\xi_d}^{\text{Fcl}})$. Base change, totalization and (TRP) give

$$\text{[FTR]} \quad \mathcal{D}_{\text{pr}}^{\text{UI,flow}} \cong \coprod_{\xi \in \text{MEnvCls}_{\text{pr}}} \text{TimeFib}_{\mathcal{E}_{\mu}^{\text{Fcl}}}^0(T_{\xi}^{\text{Fcl}}) \cong \coprod_{a \in q_F(\text{MEnvCls}_{\text{pr}})} \coprod_{\xi \in q_F^{-1}(a) \cap \text{MEnvCls}_{\text{pr}}} \text{Land}_{\mathcal{E}_{\mu}^{\text{Fcl}}}(T_{\xi}^{\text{Fcl}}). \quad (303)$$

For $\xi \in \text{MEnvCls}_{\text{pr}}$ let $\omega_{\xi}, \mathcal{M}_{\xi}$ be its reconstructed temporal state/envelope and $\text{ExpPair}(\xi) = \{\mathcal{N} \subseteq \mathcal{K} \in \text{Lat}_{\omega_{\xi}}(\mathcal{M}_{\xi})\}$. Then

$$\text{ModTemp}_{F,\text{pr}} := \coprod_a \coprod_{\xi \in q_F^{-1}(a) \cap \text{MEnvCls}_{\text{pr}}} \text{ExpPair}(\xi) \times \text{TimeFib}_{\mathcal{E}_{\mu}^{\text{Fcl}}}^0(T_{\xi}^{\text{Fcl}}), \quad (a, \xi, \mathcal{N} \subseteq \mathcal{K}, U) \mapsto (a, \Theta_{\xi}, \mathbf{b}_{\mathcal{N}:\mathcal{K}}, U),$$

where $a \in A_{\text{pr}} := q_F(\text{MEnvCls}_{\text{pr}})$. Put $q_{\text{pr}} := q_F|_{\text{MEnvCls}_{\text{pr}}}$, let $p_{\text{pr}} : \text{ModTemp}_{F,\text{pr}} \rightarrow \text{MEnvCls}_{\text{pr}}$ forget $(\mathcal{N} \subseteq \mathcal{K}, U)$, and set $\rho_{\text{pr}} := q_{\text{pr}} p_{\text{pr}}$. Then (RFE/ROA) gives the exact occurrence refinement

$$\text{Occ}_{\rho_{\text{pr}}}(\xi) \simeq \text{ExpPair}(\xi) \times \text{TimeFib}_{\mathcal{E}_{\mu}^{\text{Fcl}}}^0(T_{\xi}^{\text{Fcl}}), \quad \text{Occ}_{\rho_{\text{pr}}}(a) \simeq \int_{\xi \in \text{Occ}_{q_{\text{pr}}}(a)} \text{Occ}_{\rho_{\text{pr}}}(\xi), \quad \text{Occ}_{q_{\text{pr}}}(a) = q_F^{-1}(a) \cap \text{MEnvCls}_{\text{pr}}.$$

Thus (MOR) is precisely the projectively adapted dependent refinement of the realized (AOS) occurrence fibre; it changes neither the ambient H_0 grade nor any unrealized ambient point. Forgetting ExpPair is (FTR), (MC) derives Θ_{ξ} , and (MSE) is the modular action on every displayed shell. The layers are strict: (MGF/MAF-W) varies ξ at fixed core; on the constant two-point branch take $L = \lambda \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ with $\lambda > 0$, $T = e^L$ and $U_1 = e^{tL}$: for the uniform reference, $U \in \text{TimeFib}_{\mathcal{E}_{\mu}^{\text{Fcl}}}^0(T)$, (MC) gives zero clock and the proper expected $\hat{\mathcal{A}} \subseteq \mathcal{M}_{\text{time}}$ shell is nonzero; for a full-support nonuniform reference $\mu_p = (p, 1 - p)$, $p \neq 1/2$, the same U lands and $\mu_p T \neq \mu_p$, so (MC) gives nonzero clock and $\mathcal{N} = \mathcal{K} = \mathcal{M}_{\text{time}}$ gives zero shell; the tracial/nontracial M_2 fixtures of (MSE) remain global cross-checks; and the one-stage circle gives distinct (TCR) lifts at fixed ξ . Thus ambient status and projective realized dynamics/occurrence form the outer tower, while each realized ξ carries the product fibre $\text{ExpPair}(\xi) \times \text{TimeFib}_{\mathcal{E}_{\mu}^{\text{Fcl}}}^0(T_{\xi}^{\text{Fcl}})$; clock is derived from ξ , and the total produces no canonical section. (SLF) label transport, (SCX/PEL) pre-compact landing, (TFC/TCR) diagonal residual and (TSB) terminal pair remain separate.

Supplementary package S9: Double Grothendieck assembly and reference-cut double carrier. The complete statement(s), hypotheses, tagged identities, boundary cases, witnesses/refuters, and proof(s) are retained in Supplementary Material, Sec. S9; every use below invokes that exact proof-bearing package, not an additional assumption.

Theorem XI.3 (Residual transport, degree transmutation, and strict shadow atlas). *Selector/history and topology/lift/occurrence remain typed attachments. For $\mathbf{D}_{\bullet}^{[s]} = \bigoplus_{|A|=s} \mathbf{D}_{\bullet}(A)$, (DCHD/RDXT) gives (AHT) $H_{s-1-r}(\mathbf{DRes}_C \mathbf{D}_{\bullet}^{[s]}) \cong \text{Hist}_C^r(P_F^s/P_F^{s+1})$ ($2 \leq s \leq |F|$), so endpoint/history may occupy different degrees. Strictness is Theorems S2, S2, IIA 0 a-IIA 0 a.*

For a finite restriction tower $r_p = g_{p,q} r_q$, $p \leq q$, and $\lambda \in \{0, \neq, \rightarrow\}$, let $d_{p,q}^{\lambda}, g_{p,q}^{\lambda}$ be the object/isomorphism-Hom/ordinary-Hom component maps and $I_q^{\lambda} = \text{Im } d_q^{\lambda}$, with complements in $\text{dom } g_{p,q}^{\lambda}$. The raw strict-image mix is $X_{p,q}^{\lambda} := g_{p,q}^{\lambda}(I_q^{\lambda}) \cap g_{p,q}^{\lambda}((I_q^{\lambda})^c)$. Whenever the corresponding p -level strict landing/fullness property makes $g_{p,q}^{\lambda}(I_q^{\lambda})$ exhaust the relevant p -level target, $X_{p,q}^{\lambda} = \emptyset$ iff that same strict image

property holds at q ; this raw-image criterion must not be promoted to categorical equivalence. If r_p is an equivalence, $X_{p,q}^\rightarrow = \emptyset$ is sufficient for fullness of r_q , while faithfulness of r_q is inherited from the faithful composite $r_p = g_{p,q}r_q$. The raw mixed cells are neither necessary nor sufficient for r_q to be an equivalence: for insufficiency take $\mathbf{1} \xrightarrow{r_q} \mathbf{2}_{\text{disc}} \xrightarrow{g} E_2$, with r_q selecting one object and E_2 the chaotic two-object groupoid; then gr_q is an equivalence and $X^0 = X^\rightarrow = \emptyset$, but r_q is not essentially surjective. For nonnecessity take $\mathbf{1} \xrightarrow{r_q} E_2 \xrightarrow{g} \mathbf{1}$ with the same selection; both r_q and gr_q are equivalences while both raw mixed cells are nonempty. Thus categorical equivalence retains its replete essential-image/full-faithfulness coordinate; (MRS) remains the strict-image collision face, consistently with the $\mathbf{2}_{\text{disc}}/BC_2/\text{noninvertible-arrow}$ boundaries and independently of global outer-kernel-zero.

For a locally posetal comparison cell η with finite cuts $(\ell_h(x) \leq u_h(x))_{h \in H_\eta}$, (RCS/GCS/CFR) gives $\text{ICP}_\eta(x) = ([\ell_h, u_h])_{h \in H_\eta}$. Its endpoint map satisfies $\partial \text{ICP}_\eta = \mathfrak{A}_\square$ and $\mathfrak{A}_\square = 0 \iff \text{ICP}_\eta(x) \simeq \mathbf{1}^{H_\eta}$; (RCP) consumes it componentwise. By (HCF), every component has zero reduced homotopy/stable shadow even when $\ell_h < u_h$, so the endpoint profile is strictly pre-homotopy; supplied non-posetal cells retain full (CFR). It is still not determined by horizontal completion: on $P = \{0 < 1\}$ the same span $\mathbf{1} \leftarrow P \rightarrow \mathbf{1}$ admits intervals $[1, 1]$ for $I \Rightarrow I$ and $[0, 1]$ for $c_0 \Rightarrow I$.

On the fixed pointed-core (AUS) face set $C_A = \text{disc}(A)$, $A \in \mathcal{A}_{\text{ASC}}$. Then $A \subseteq A' \Rightarrow C_A \hookrightarrow C_{A'}$, $\text{Ob } C_{A'} \setminus \text{Ob } C_A = A' \setminus A$. (AOS/AUS/MGT) identifies each set-sized ambient enlargement with H_0 extension growth and fixed-image occurrence/multiplicity with H_1 ; (ADS) retains the no-maximal-stage boundary.

For finite $\mathcal{B}, F$, (RFE) gives $\text{Sec}_{\text{cart}}(\text{Tot disc } F) \cong P_{\text{coh}}$, $\text{Sec}_{\text{cart}}(\text{Tot disc } (\mathcal{P}F)) \cong \text{CohLaw}_{\mathcal{B}}(F)$, $\text{PtLaw}_{\mathcal{B}}(F) = \text{conv}(\delta_{F,*} P_{\text{coh}})$. (RCD/CRX/JKC/GTS/TDO) gives completed target-support H_0/H_1 , zero-tail GTS, independent derived tails and post-fixation $\mathbb{K}(j_{J,a})$; (BDI/BDP/BGC, CRT/CRM/CSE/CRH, DCE/DCHD, RHTX/AHT) gives the mixed cubical total with (CRK), (DOD/DOH), (CRM/CSE/DANF) faces. The independent ambient/generation/pre-homotopy/mate/interface boundaries remain typed. (FCR/DFCR/BDR/HBD, IDS/HDF/CDS/CCT/CSM/CEX/CVBS/HVS/PVS, CCI/CHI/PPH, DOB/ICG/AHR/OBF/DOH) is the finite/directed exact-transport boundary: finite cubical exact transport commutes automatically; on a countable tower a covariant exact functor leaves the $(1 - \tilde{T})$ -fibre of its product residual, while a contravariant exact functor leaves the $(1 - \tilde{S})$ -cofiber of its sum-to-dual-product residual. Coefficients give (CCI/CHI), and derived integral duality gives (DOB/OBF), whose degree-one objective birth has the exact unsplit arithmetic-history/saturation/global-character filtration; Hist^1 is the path-component quotient on the free-lattice prophase. On the finite probability face, (DPH/AIPG, WFR_{int}/WFR_z/MTE/MID) gives the compact phase $\text{IntPh}_{I,R}$ with Lie algebra V_R , fundamental lattice $\mathbb{I}_{I,\gamma}$, component group $\text{SatRes}_I(R)$ and character group $\text{Obj}_{I,R}^{\text{inc}}$; EFD is its dual-Lie chart with coordinate $[\log(\pi/q_R)]$. The $K_{2,2}$, E_n and mixed-block witnesses keep its connected/discrete faces mutually nonclassifying. On the two-leg subface, (LRF) gives unbounded LatRank at fixed $(C, \text{SuppDep}_C, \text{MargRest}_C, \text{FibDep}_R) = (1, 0, 0, t)$ for $0 < t < \log 2$, while (DLCF) gives continuum many minimal latent realizations with no least element even at one fixed law γ_p and therefore at one fixed WFR coordinate. Thus even the full compact phase/character carrier does not select a latent implementation point. Modular ray/witness envelope are Theorems XA 0 h, XI.1.

Corollary XI.4 (Residual reconstruction, cofinal closure, and stage-germ terminality). *The bounded grammar has the seven routed typed faces above. Let $\mathfrak{S}_{\text{SCO}}$ be the (DMR)-bounded stages (small typed diagram plus legal set probes; (AOS/AUS): $A \in \mathcal{A}_F$), ordered by code-preserving enlargement. Write $\text{Code}(h)$ for the tagged set of object/arrow/cell/probe codes carried by h and $j_{hh'} : \text{Code}(h) \hookrightarrow \text{Code}(h')$ for $h \preceq h'$, with $j_{hh} = 1$ and $j_{h'h''} j_{hh'} = j_{hh''}$.*

(CRTC) Every set-sized stage family has an upper stage; every set-ordinal legal typed re-entry/residual chain lies in one stage; and every set probe of a declared class-valued face over $x \in \text{Code}(h)$ lies at some $h' \succeq h$. Set unions with (DMR/RTD/RTO/RVN) give the first two clauses; the routed fixed-input faces are sets. The sole unrestricted proper-class face is $c_F^{-1}([\mathbf{C}])$, where lm_F^{MGI} is set-coded and (AUS/ADS) gives the third clause with no last stage.

(CGE) Let Γ_0 be the set-coded tagged native-sort/route signature of the seven faces above. Let CGr_{SCO} be the class of set-coded code-conservative set-ary extensions $\Gamma \supseteq \Gamma_0$: inherited sort/operator tags, arities, source/target typing and fixed-input output laws are literal, while every new tag is a fresh declaration-rigid atom from a proper-class reservoir (equality of tags fixes kind and, when applicable, arity, source/target typing and fixed-input output-law code), so incompatible declarations necessarily use distinct tags. Every new routed operation/constructor code has set-sized arity and, at fixed input, either set-valued output or a declared class-valued output used only through specified set probes. This tag convention makes every set family jointly compatible and its code-conservative embeddings literal inclusions. This CGr_{SCO} is a meta-signature/route carrier, not the internal closure-law grammar $\text{Gr}(L)$ of (GCL/RGW): the latter varies laws inside a fixed native vocabulary, whereas CGr_{SCO} enlarges that vocabulary. Every set family $(\Gamma_i)_{i \in I}$ has the set-coded upper grammar $\Gamma_* := \bigcup_i \Gamma_i$; there is no maximal member, since adjoining one fresh inert sort tag is code-conservative and, by (SGO), asserts no realization.

For $\Gamma \in \text{CGr}_{\text{SCO}}$ let $\mathfrak{S}_\Gamma$ be the (CRTC) bounded-stage construction with Γ -typed codes and legal set probes, and write $\text{Code}_\Gamma(h)$. If $\Gamma \subseteq \Gamma'$, an old stage has the empty extension $h^{\Gamma'} \in \mathfrak{S}_{\Gamma'}$ using no new tag, with a literal code injection $j_h^{\Gamma'}$. The proof of (CRTC) is signature-parametric: set unions of small typed diagrams/probes remain sets, set-ordinal traces remain set-sized, the routed fixed-input outputs are sets and unrestricted class output is only probed setwise. Hence for every set family $(\Gamma_i, h_i)_{i \in I}$,

$$[\text{CGE}] \quad \Gamma_* := \bigcup_i \Gamma_i \in \text{CGr}_{\text{SCO}}, \quad \exists h_* \in \mathfrak{S}_{\Gamma_*} \quad h_i^{\Gamma_*} \preceq h_* \quad (i \in I). \quad (304)$$

Define eventual equality across grammar extensions by

$$(\Gamma, h, x) \sim_\uparrow (\Gamma', h', y) \iff \exists \Lambda \supseteq \Gamma, \Gamma', \exists \ell \in \mathfrak{S}_\Lambda : h^\Lambda, h'^\Lambda \preceq \ell, \quad j_{h^\Lambda \ell} j_h^\Lambda x = j_{h'^\Lambda \ell} j_{h'}^\Lambda y.$$

Set-directedness in (CGE) makes this an equivalence relation, while literal code preservation makes the stage maps below injective:

$$[\text{CGG}] \quad \begin{aligned} \mathcal{P}_{\text{SCO}}^\uparrow &:= \prod_{\Gamma \in \text{CGr}_{\text{SCO}}} \prod_{h \in \mathfrak{S}_\Gamma} \{(\Gamma, h)\} \times \text{Code}_\Gamma(h), & q_{\text{gr}}^\uparrow : \mathcal{P}_{\text{SCO}}^\uparrow &\rightarrow \mathfrak{U}_{\text{SCO}}^\uparrow := \mathcal{P}_{\text{SCO}}^\uparrow / \sim_\uparrow, \\ \iota_{\Gamma, h}(x) &:= q_{\text{gr}}^\uparrow(\Gamma, h, x), & \text{GrProv}^\uparrow(u) &:= (q_{\text{gr}}^\uparrow)^{-1}(u), & \mathcal{P}_{\text{SCO}}^\uparrow &\cong \prod_{u \in \mathfrak{U}_{\text{SCO}}^\uparrow} \text{GrProv}^\uparrow(u), \\ S &\subseteq \mathfrak{U}_{\text{SCO}}^\uparrow, S \text{ a set} \implies \exists \Gamma \in \text{CGr}_{\text{SCO}}, \exists h \in \mathfrak{S}_\Gamma, \exists \tilde{S} \subseteq \text{Code}_\Gamma(h) : S = \iota_{\Gamma, h}(\tilde{S}), \\ & & \forall (\Gamma, h) : & \iota_{\Gamma, h}(\text{Code}_\Gamma(h)) \subseteq \mathfrak{U}_{\text{SCO}}^\uparrow. \end{aligned} \quad (305)$$

The third row follows by taking one upper grammar for the set of representative grammars and then one upper stage by (CGE). The strict last row follows directly by fresh-sort enlargement. Grammar presentation is not erased: if Γ^+ adjoins one fresh inert sort and $x \in \text{Code}_\Gamma(h)$, then $(\Gamma, h, x) \neq (\Gamma^+, h^{\Gamma^+}, x)$ in $\mathcal{P}_{\text{SCO}}^\uparrow$ but both have the same $q_{\text{gr}}^\uparrow$ -image, so the corresponding $\text{GrProv}^\uparrow$ fibre is nontrivial. Every specified set probe of $q_{\text{gr}}^\uparrow$ lies in one grammar-stage by the third row, hence its restriction is a native set map and (YFF/RFD/RTD/RVN) recovers its probed grammar-provenance fibre. Thus the terminal construction re-enters the original residual doctrine instead of introducing a new primitive.

Moreover, for every declared tagged target Z , any sort-preserving family $f_{\Gamma, h} : \text{Code}_\Gamma(h) \rightarrow Z$ compatible with both stage enlargement and empty grammar extension descends uniquely to a class map $\bar{f} : \mathfrak{U}_{\text{SCO}}^\uparrow \rightarrow Z$ with $\bar{f}|_{\Gamma, h} = f_{\Gamma, h}$. Thus (CGG) is the class-colimit/germ universal property, not a selector.

At the base grammar Γ_0 , put $(h, x) \sim_g (k, y)$ iff some $\ell \in \mathfrak{H}_{\Gamma_0}$ receives h, k and satisfies $j_{h\ell}x = j_{k\ell}y$. Code-conservativity gives the fixed-grammar horizon face

$$[\text{CGC}] \quad \mathfrak{U}_{\text{SCO}}^g := \left(\coprod_{h \in \mathfrak{H}_{\Gamma_0}} \{h\} \times \text{Code}_{\Gamma_0}(h) \right) / \sim_g \hookrightarrow \mathfrak{U}_{\text{SCO}}^\uparrow, \quad [h, x]_g \mapsto [\Gamma_0, h, x]_\uparrow. \quad (306)$$

Indeed, all stage maps and empty grammar extensions preserve inherited code tokens literally. If two base-grammar classes agree in an upper grammar, their inherited code tokens are therefore already equal; a common upper Γ_0 -stage supplied by (CRTC) witnesses the same equality, so the map is injective. Thus (CGC) removes horizon presentation only. Its universal property and set-cofinality are the fixed-grammar restrictions of (CGG/CGE), and (ADS) keeps every fixed stage proper; a fresh-sort code lies outside this fixed face, so grammar enlargement remains a distinct coordinate.

Let $\text{Cl}_{\text{SCO}}^\uparrow$ allow set-ordinal traces alternating code-conservative grammar enlargement with the operations declared by the current Γ (whose Γ_0 restriction is exactly the seven routed operations), retaining native sort tags and all hypotheses and taking specified set probes at class-valued outputs. A set of such traces uses only set many new tags, so its dependency/tag union lies in one $\Gamma \in \text{CGr}_{\text{SCO}}$ and then in one stage by (CGE/CRTC). Therefore

$$[\text{CGT}] \quad \begin{aligned} S, T \text{ sets, } T \subseteq \text{Cl}_{\text{SCO}}^\uparrow(S) &\implies \exists(\Gamma, h) : S \cup T \subseteq \iota_{\Gamma, h}(\text{Code}_\Gamma(h)), \\ \text{Cl}_{\text{SCO}}^\uparrow(\mathfrak{U}_{\text{SCO}}^\uparrow) &= \mathfrak{U}_{\text{SCO}}^\uparrow. \end{aligned} \quad (307)$$

The same statement is probe-wise cofinal for a class-sized set-ary tagged extension: if $\text{dep}(P)$ denotes adjoining to P the set of sort/operator tags referenced by the set-coded declarations in P , then the dependency closure of any specified set of tags is $\bigcup_{n < \omega} \text{dep}^n(P)$, hence set-coded. The routed producer families, including (UAL/ULR) and (GCL/RGW), are re-instantiated fibrewise only where their printed hypotheses are typed; (CGE) never transports a theorem to a new law point by fiat. The discrete (CVE) signature-extension totalization is a direct consumer: signature growth is code-conservative, whereas theory/law growth may extinguish its extension fibre and therefore remains a law coordinate. The active set-sized finitary $\text{C}_\emptyset$ constructor hull is a constructor-face within-grammar consumer. Conversely, retyping an inherited symbol is not a (CGE) morphism: a tag cannot be both nullary and unary in a common code-preserving upper grammar; such a change must be freshly tagged or supplied as a distinct law/root.

(CGF) Write $\text{Cl}_{\text{SCO}}^{\text{pr}}$ for the restriction of (CGT) to traces that remain in Γ_0 . Then every specified fixed-grammar set-generated family lies in one later Γ_0 stage and $\text{Cl}_{\text{SCO}}^{\text{pr}}(\mathfrak{U}_{\text{SCO}}^g) = \mathfrak{U}_{\text{SCO}}^g$.

(CIE) The integrative direction is itself invariant under lawful grammar enlargement. If $\Gamma \subseteq \Gamma'$ is a (CGE) extension and $(\mathcal{P}; D, E)$ is a comparison instance already typed in Γ whose active support S and coordinate orders are retained unchanged by the empty extension, let $(\mathcal{P}; D, E)^{\Gamma'}$ denote that extension. Code conservation preserves every inherited defect/capability tag and value, so

$$[\text{CIE}] \quad E \succeq_{\text{int}}^S D \iff E^{\Gamma'} \succeq_{\text{int}}^S D^{\Gamma'}. \quad (308)$$

Hence every old integrative judgment, and every feasible-fibre Pareto or differentiable dual-ray consumer whose printed hypotheses are carried by that instance, survives unchanged along the cumulative grammar germ. This produces direction without selection: it does not create comparisons between distinct supports or presentation types. The existing two-coordinate Pareto witness remains inside one fixed support and still forbids a canonical scalar objective. Thus comparison remains immanent to its typed carrier rather than being imposed by a terminal selector.

(GDI) The cumulative germ quotient is the finest quotient compatible with every lawful re-entry, equivalently the least-collapse intrinsicization of grammar/stage presentation. Write $q = q_{\text{gr}}^\uparrow : \mathcal{P}_{\text{SCO}}^\uparrow \Rightarrow \mathfrak{U}_{\text{SCO}}^\uparrow$. Its kernel relation is exactly the equivalence closure of the pairs identifying a code with its stage enlargement and with its empty code-conservative grammar extension: one inclusion is immediate because q coequalizes those embeddings, while for $p \sim_\uparrow p'$ the defining common upper grammar/stage identifies both with one literal upper code. Hence any quotient or read-out compatible with every legal re-entry edge has kernel containing R_q and factors uniquely through q ; a strictly finer quotient cannot coequalize all such edges.

For every set probe $A \subseteq \mathcal{P}_{\text{SCO}}^\uparrow$ let $q_A : A \Rightarrow q(A)$ be the corestriction. By (CGE/CGG) the probe lies in one grammar-stage, so q_A is an ordinary native Set map and (FSB_{core}/RPL) gives, for every set-valued read-out $a : A \rightarrow Z$,

$$[\text{GDI}] \quad \text{DepRes}_{q_A}(a) = 0 \iff R_{q_A} \subseteq R_a \iff a = \bar{a} q_A \text{ for a unique } \bar{a} : q(A) \rightarrow Z. \quad (309)$$

For a sort-preserving family $f_{\Gamma, h}$ on the full presentation class, compatibility with stage enlargement and empty grammar extension is therefore equivalent to vanishing of this residual on every set probe (already every two-point probe suffices), and hence to the unique class descent $\bar{f} : \mathfrak{U}_{\text{SCO}}^\uparrow \rightarrow Z$ of (CGG). Thus presentation dependence is not removed by decree: it is exactly the nonzero residual along q , while every intrinsic re-entry-invariant read-out is precisely a zero-residual descendant. In particular (CIE) supplies a nontrivial positive-direction consumer of this criterion without adding a scalar objective.

(ICR) On each cumulative Set probe, every jointly separating lawful intrinsic family $(g_i : q(A) \rightarrow Z_i)_i$ reconstructs q_A by (KGC_{sep}). Boolean descendants give a canonical witness: with $\text{Bool}_{\text{int}}(A) = \{b : A \rightarrow \mathbf{2} : \text{DepRes}_{q_A}(b) = 0\}$,

$$[\text{ICR}] \quad \boxed{\text{Pull}_{q_A} : \mathcal{P}(q(A)) \xrightarrow{\sim} \text{Bool}_{\text{int}}(A), \quad U \mapsto \mathbf{1}_U q_A, \quad R_{q_A} = \bigcap_{b \in \text{Bool}_{\text{int}}(A)} R_b.} \quad (310)$$

Thus intrinsic identity is independent of the separating lawful vocabulary; Boolean atoms merely exhibit the germ fibres on Set. Null-quotiented, contextual and operatorial carriers retain their typed non-point residuals.

(GFC) The same germ has no residual set-sized coherence obstruction after lawful re-entry, even when its presentation fibre is nontrivial. For $u \in \mathcal{U}_{\text{SCO}}^\uparrow$ let $\text{ReEnt}(u)$ be the thin re-entry class whose objects are $p = (\Gamma, h, x) \in \text{GrProv}^\uparrow(u)$ and whose relation $p \preceq_{\text{re}} p'$ means that p' is a literal composite of stage enlargement and empty code-conservative grammar extension of p , carrying the inherited code x to the code of p' . For a nonempty set $A \subseteq \text{Ob ReEnt}(u)$ and a common re-entry upper bound r , let $\text{ReEnt}_A(u; r)$ be the full thin category on $A \cup \{r\}$. Then

$$\begin{aligned} \text{[GFC]} \quad & \emptyset \neq A \subseteq \text{Ob ReEnt}(u), A \text{ a set} \implies \exists r \in \text{GrProv}^\uparrow(u) : (\forall p \in A) p \preceq_{\text{re}} r, \quad \text{ReEnt}_A(u; r)^{\text{gpd}} \simeq \mathbf{1}, \\ & r, r' \text{ common upper bounds of } A \implies \exists r'' \in \text{GrProv}^\uparrow(u) : r \preceq_{\text{re}} r'', r' \preceq_{\text{re}} r'', \\ & \forall r \in \text{GrProv}^\uparrow(u) \quad \exists r^+ \in \text{GrProv}^\uparrow(u) : r \prec_{\text{re}} r^+ \text{ and } r^+ \not\preceq_{\text{re}} r. \end{aligned} \quad (311)$$

Indeed fix $p_0 \in A$. For every $p \in A$, equality $q(p) = q(p_0)$ supplies by (CGG) a common grammar/stage witness on which their inherited codes are literally equal. This is a set of proof-local witnesses, so (CGE) gives one common upper grammar/stage for all of them; code preservation carries all equalities to a single upper code z , and the resulting representative $r = (\Lambda, \ell, z)$ lies in the same q -fibre and receives every $p \in A$. Hence r is terminal in the small thin category $\text{ReEnt}_A(u; r)$; the inclusion $\{r\} \hookrightarrow \text{ReEnt}_A(u; r)$ is fully faithful reflective, so (CHL) gives the displayed groupoid contraction. Applying the first clause to $\{r, r'\}$ gives apex-independence after further lawful re-entry. Conversely adjoining one fresh inert sort to the grammar of r gives the strict extension r^+ by (CGE/SGO), so the full presentation fibre has no final representative. Thus presentation multiplicity survives while incompatibility among any specified set of same-germ presentations is absorbed by lawful re-entry; no apex selector is generated. This statement concerns the cumulative presentation germ only and does not contract independent history, duration, law, interface or realization fibres.

(GGL) This coherence identifies the proof-relevant presentation shape of every set probe exactly with its intrinsic germ set. Let ReEnt be the thin re-entry class on $\mathcal{P}_{\text{SCO}}^\uparrow$, with $p \rightarrow p'$ exactly when $p \preceq_{\text{re}} p'$; its q -fibre is $\text{ReEnt}(u)$. For every set $A \subseteq \mathcal{P}_{\text{SCO}}^\uparrow$ there exists a set B with

$$\text{[GGL]} \quad A \subseteq B \subseteq q^{-1}(q(A)), \quad \boxed{(\text{ReEnt}|_B)^{\text{gpd}} \simeq \text{disc}(q(A))}. \quad (312)$$

Indeed, for each $u \in q(A)$ apply (GFC) to the nonempty set $A_u = A \cap q^{-1}(u)$ and choose one common lawful upper representative r_u ; set $B = A \cup \{r_u : u \in q(A)\}$ (for $A = \emptyset$, take $B = \emptyset$). Every re-entry arrow stays in one q -fibre by (GDI), and r_u is terminal in the u -component of $\text{ReEnt}|_B$. Hence the nerve of that component is contractible; the nerve of $\text{ReEnt}|_B$ is therefore the coproduct of contractible components indexed by $q(A)$. In particular (CHL) gives its groupoidal 1-truncation $(\text{ReEnt}|_B)^{\text{gpd}} \simeq \coprod_{u \in q(A)} \mathbf{1} \simeq \text{disc}(q(A))$. Different proof-local choices of the r_u admit componentwise common refinements by (GFC), hence induce the same germ indexing after further lawful re-entry; fresh inert sorts ensure that none is a final representative of its full presentation fibre, so no canonical/functorial final-representative selector is produced. Thus, on every set probe, lawful groupoidification erases exactly grammar/stage presentation inside each germ and no intrinsic germ distinction: strict presentation multiplicity and discrete intrinsic multiplicity are simultaneously retained at their native levels.

(GRC) The terminal germ re-enters the founding proarrow calculus as a direct (PKF) consumer. For a set probe $A \subseteq \mathcal{P}_{\text{SCO}}^\uparrow$, put $q_A : A \rightarrow U_A := q(A)$; by (ICR) this is already the coarsest boundary retaining every Boolean zero-residual re-entry consequence. Let $a : A \rightarrow Z$ be set-valued. Let $j_a : A \rightarrow K_a := \text{Key}_{a|q_A}$, $\pi_a : K_a \rightarrow U_A$ and $\alpha_a : K_a \rightarrow Z$ be the (RKD/PKF) factorization data. Then

$$\begin{aligned} \text{[GRC]} \quad & T_{A,a}^{\text{germ}} := a_* \odot q_A^* \simeq (\alpha_a)_* \odot N_{j_a} \odot \pi_a^*, \quad R_{A,a}^{\text{germ}} := \text{supp}_0 T_{A,a}^{\text{germ}} = \text{im}(q_A, a), \\ & \text{DepRes}_{q_A}(a) \simeq \mathbb{K}(\pi_a), \quad R_{A,a}^{\text{germ}} \text{ functional} \iff a = \bar{a}q_A \iff \text{DepRes}_{q_A}(a) = 0. \end{aligned} \quad (313)$$

The support is the least relation in $U_A \times Z$ containing every $(q(p), a(p))$; PKF now separates its two residual layers literally: N_{j_a} retains multiplicity at fixed (q_A, a) , while π_a^* retains equal- q_A /different- a branching. If $\bar{a}$ is a declared sort-preserving re-entry-compatible family, (GDI/PKF) gives already on every probe $T_{A,\bar{a}|_A}^{\text{germ}} \simeq (\bar{a}|_{U_A})_* \odot N_{q_A}$ and $R_{A,\bar{a}|_A}^{\text{germ}} = \text{graph}(\bar{a}|_{U_A})$. For a (GGL) enlargement $A \subseteq B \subseteq q^{-1}(q(A))$,

$$(\text{ReEnt}|_B)^{\text{gpd}} \simeq \text{disc}(q(A)), \quad T_{B,\bar{a}|_B}^{\text{germ}} \simeq (\bar{a}|_{q(A)})_* \odot N_{q_B}, \quad N_{q_B}(u, u') \simeq \begin{cases} q_B^{-1}\{u\}, & u = u', \\ \emptyset, & u \neq u'. \end{cases}$$

Hence $R_{B,\bar{a}|_B}^{\text{germ}} = \text{graph}(\bar{a}|_{q(A)})$ and the nonempty Prof fibre is $T_{B,\bar{a}|_B}^{\text{germ, Prof}}(u, \bar{a}(u)) \simeq q_B^{-1}(u)$, recovering the previous GRC clauses from one producer. Taking $A = \{p, p^+\}$ with p^+ a fresh-inert extension and intrinsic a makes π_a bijective while N_{j_a} has at least two elements; a same-germ two-point probe with unequal a -values makes N_{j_a} singleton while π_a branches. Thus provenance multiplicity, non-descent and re-entry coherence remain distinct, with no master scalar, world or Prof.

(DTC) Terminality is cumulative under every set-coded code-conservative set-ary enlargement of the declared native-sort/route grammar, not maximal inventory. Since (UAL/ULR) remains typed and (CGG) is a class germ rather than an object of one grammar, (SGO) and the Cantor/Tarski boundary refute

$$\boxed{\text{cumulative typed terminality} \implies \text{absolute same-level self-classification}} \quad \text{as false.}$$

Thus code-conservative extensions preserve inherited distinctions while fresh tags may add new ones; nonconservative retyping lies outside the embedding relation. (CIE/GDI/ICR/GFC/GGL/GRC) preserves typed integrative direction, identifies intrinsic content by zero germ residual and amalgamates every specified set of same-germ presentations without a final representative or canonical objective. (CGC/CGF) and (CGG/CGT) keep fixed-grammar horizon and cumulative grammar presentation distinct, while law, interface/joining, completion and realization variation remain in their native fibres.

XII. CONCLUSION

The determination core is consumer-reflective: (KGC/KGC_{cons}/KGC_{ord}/KGC_{sep}) gives presentation-independent least sufficient objects, (RRCX/BDRF) retains route residuals beyond endpoint or scalar shadows, and (GDI/ICR) reconstructs the Set probe. Thus object identity is quotient stability; (CBR/RCC) separates realized compatibility, least common-base completion and common invariant, while route, interaction, provenance, occurrence and law remain typed attachments. The actuality/predictive spine is selector-free: (IOC/BEH/RRCX;

PLBD_{int}/BFL/KGC) makes a cut stable exactly when declared continuations descend, with failure recorded by a refinement residual; unpredictability, occurrence and structural novelty remain distinct. (CGE/CGG/CGT; GDI/ICR/GFC/GGL/GRC; DTC) preserves this under lawful code-conservative enlargement without a final representative or same-level self-classification; terminality is invariance under lawful enlargement, not maximal inventory. **Conflict of Interest.** None. **Data Availability.** No new data were created or analyzed. **Author Contributions.** Fulvio Bennato: conceptualization, formal analysis, investigation, methodology, drafting, review and editing. **AI Use.** OpenAI ChatGPT (GPT-5.6; OpenAI) assisted analysis and preparation; the author reviewed all arguments and conclusions.

Appendix A: Conservative semantic dictionary

The assumption-free body-readout set is $\mathcal{A}_{\text{atl}} = \{\text{Rta, Satya, Maya, Pramana, Avidya, Upadana, Kala, Smrti, Svapna, Karma, Samskara, Samsara, Nirodha, Nirvana}\}$.

Corollary A.1 (Conservative semantic-atlas determinability). *For a realization $(\Omega, (p_u)_{u \in \mathcal{A}_{\text{atl}}})$ put $\widehat{u}_\Omega = up_u$ (effective-image corestricted), let $\mathcal{V}_{\text{atl}}^\times(\Omega)$ be the finite joint-readout closure including its constant bottom, and set $\mathbb{D}_\Omega(v \mid w) = \text{DepRes}_w(v)$. Then (SDO/SDK/SDC) identifies the resulting finite-join semilattice anti-isomorphically with realized kernel pairs, with $v \preceq_\Omega w \iff \mathbb{D}_\Omega(v \mid w) = 0$; determination closure, novelty, crossed-front synergy and minimal coalitions therefore descend, while the SDC frontier remains non-nullary. Moreover*

$$Q_\Omega : \Omega \rightarrow \Omega / \bigcap_u R_{\widehat{u}_\Omega}$$

is the unique coarsest joint quotient by (DCC/KGC/YFF), with the (RFF/RKD) minimal functionalizing keys and residual package. Hence $\mathfrak{D}_{\text{atl}}(\Omega) = \langle [\widehat{u}_\Omega] \rangle_v$ has at most 2^{14} elements, sharply on the fourteen-coordinate cube; the Banach lift is Theorem IIA 0 a, and (SDO/SDK) gives the unique functorial finite-join transport. For a declared monoid action,

$$[\text{SDS}] \quad R_{\text{atl}}^M(\Omega) = I_M(R_{Q_\Omega}) = \bigcap_{m,u} R_{\widehat{u}_\Omega T_m}, \quad Q_\Omega = c_\Omega^M Q_\Omega^M.$$

The first claims are the cited finite consumer-reflection/kernel-order faces; (IOC/BEH) gives the displayed dynamic closure and orbit/quotient descent recovers the action, orbit/corestricted and residual readouts. Hence the atlas is a conservative finite RTD/GSSF read-out; the three-point witness still separates synergy, additive interaction and ordered residual history, with native domains distinct.

Supplementary Material. S1–S9 and S11 retain the unabsorbed proof-bearing hypotheses, identities, witnesses/refuters and proofs; omission neither strengthens a theorem nor replaces proof by assertion.

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